

# **Asymptotic Transseries and Inversion of Eight Sequences and Special Functions**

Rooted trees, the double factorial, the partition numbers, the swing factorial,  
Gamma and Barnes  $G$ , the hyperfactorial  $K$ , the subfactorial, and a  
real-argument Fibonacci function:  
Lambert-cored reversion and its limits

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## Abstract

Eight subjects are studied here to all orders and then inverted. The first four are combinatorial sequences: the numbers  $a_n$  of unlabelled rooted trees on  $n$  vertices (OEIS A000081, the trees underlying Butcher series), the double factorial  $n!!$ , the partition numbers  $p(n)$  (A000041), and the swing factorial  $\text{sw}(n) = n!/\lfloor n/2 \rfloor!$  (A056040). The next three are special functions whose logarithms have a power–logarithmic phase: Euler’s  $\Gamma$ , the Barnes  $G$ -function, and the hyperfactorial  $K$ -function. The last two are outliers, included because they mark the boundary of the method: the subfactorial  $!n$ , whose inverse borrows its carrier from  $\Gamma^{-1}$  and whose own content is superexponentially small, and a real-argument Fibonacci function, whose inverse has no Lambert core at all and whose expansion converges. For each we give the complete forward asymptotic transseries and then the asymptotic inverse: given a value, recover the index, to all orders and with an explicit error theory.

The subjects look unrelated and their forward analyses genuinely are — a Pólya-tree singularity analysis, two unrelated pairs of gamma-ratio branches, Rademacher’s convergent root-of-unity series, three Stirling-type expansions, a Bell-number tail, and a Binet formula. Their *inversions* are not. Six of the eight, after a monomial change of variable, are the inversion of an exponential–power model

$$F(x) = C \exp(ax^\alpha) x^b \exp Q(x^{-\delta}),$$

or of a power–logarithmic block  $ax^p(\log x - \beta)$ , whose dominant phase is inverted exactly by the Lambert  $W$ -function and whose remaining corrections are generated by one triangular recurrence. Chapter 0 develops that apparatus once, with complete proofs; Chapters 1–8 supply the parameter dictionary for each subject and the mathematics genuinely specific to it. The synthesis chapter compares all eight and records what the common frame does and does not explain — including four square roots, arising in two apparently different ways, that are shown to be one lemma applied four times.

This volume is a consolidation of twenty-four independently written articles, three on each subject, filed in this repository on 3 September 2026. The second twelve were written without knowledge of the first twelve and reproduce much of their apparatus independently; the duplication is quantified in the synthesis chapter. Where the three treatments of a subject agreed, one presentation was kept; where they disagreed, the disagreement is stated in the text and resolved; where a result was asserted without proof, a proof is supplied or the statement is demoted to a remark that says what is actually established. Corrections to the sources are boxed in the text and listed in Appendix B; provenance is in Appendix A.

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# Chapter 1

## The common apparatus

All of the first twelve filed articles carry a private copy of the material of this chapter. The presentation below follows, in the main, the two sections *A universal coefficient calculus for exponential–power inversion* and *Lambert-normalized inverse transseries* of `Swing_Factorial_Transseries_Article`, which state the reversion in the greatest generality of the first twelve. Four things have been changed. (i) The Lagrange fixed-point lemma is reproved: the swing article’s proof appeals to Lagrange–Bürmann inversion in a setting where the required invertibility hypothesis fails, and the gap is closed here by a universality argument (lemma 1.15). (ii) The exponential–power model is separated into a *formal* and an *analytic* reading, and every asymptotic claim is given an explicit remainder hypothesis; the sources routinely pass from a coefficient recurrence to a Poincaré statement without one. (iii) The dominant core is axiomatized (definition 1.8) so that the factorial-type phase of Part 3 — which does *not* belong to the exponential–power model — is covered by the same reversion theorem. (iv) The discrete-inverse material of `Butcher_Tree_Transseries-2`, `double_factorial_transseries-2` and the three partition articles is merged into one statement with a residue-class lattice.

### 1.1 Standing conventions

Throughout the volume  $R$  denotes a commutative  $\mathbb{Q}$ -algebra, and  $R[[t]]$ ,  $R[[t, u]]$  formal power series rings over it. For a formal series  $f$  we write  $[t^n]f$  for the coefficient of  $t^n$ . The symbols  $e, i, d$  are the upright constants and differential;  $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$  are the usual number systems, and  $\mathbb{N} = \{0, 1, 2, \dots\}$ . Asymptotic statements use  $\mathcal{O}$  and  $o$  in the Landau sense as the stated variable tends to  $+\infty$ ; an implied constant is always allowed to depend on every quantity held fixed in the statement (in particular on the truncation index  $N$ ) and on nothing else.

The sources disagree on three points of bookkeeping. We fix them once.

*Remark 1.1* (Conventions adopted, and where the sources differ).

1. *Ordinary versus exponential Bell polynomials.* The partition articles and `Butcher_Tree_Transseries-2` take the *exponential* partial Bell polynomials  $B_{n,k}$  as primitive and convert; `Swing_Factorial_Transseries_Article` and `double_factorial_transseries-2` take the *ordinary* ones  $\hat{B}_{n,k}$  as primitive. The volume defines both (definitions 1.9 and 1.10), records the conversion (lemma 1.11), and uses  $\hat{B}$  in every reversion formula and  $B$  in every exponentiation or logarithm of a *unit* series. The reason is not taste: reversion multiplies a correction series whose coefficients carry no factorials, and  $\hat{B}$  is the transform that leaves them alone.
2. *Sign of the logarithmic phase.* We write the dominant phase as

$$L = aX + b \log X,$$

with  $b$  a *signed* constant. Butcher\_Tree\_Transseries-2 writes  $Z = X - p \log X$  and the partition articles write  $\rho - 2 \log \rho = L$ ; both are the case  $b = -p < 0$ . Every formula below is stated in the signed convention, and the dictionary of table 1.1 gives  $b$  for each chapter. One dividend of the signed convention is corollary 1.21: the Lambert branch is  $W_0$  when  $b > 0$  and  $W_{-1}$  when  $b < 0$ , with no case analysis beyond the sign.

3. *Truncation index.* A truncation “of order  $N$ ” retains the terms with index  $n = 1, \dots, N$  *inclusive*, together with the leading term. Swing\_Factorial\_Transseries\_Article uses the exclusive convention for its forward table (“the truncation labelled  $N$  retains  $A_0, \dots, A_{N-1}$ ”) and the inclusive one for its inverse table (“the first  $N$  correction terms are retained”); the two tables of that article are therefore not on a common scale. The volume is inclusive throughout.

## 1.2 The exponential–power model

### 1.2.1 Formal and analytic readings

**Definition 1.2** (The exponential–power model). Fix constants  $C > 0$ ,  $a > 0$ ,  $\alpha \in (0, 1]$ ,  $\delta > 0$ ,  $b \in \mathbb{R}$ , and a formal series without constant term,

$$Q(t) = \sum_{j \geq 1} q_j t^j \in t \mathbb{R}[[t]].$$

The associated *exponential–power model* is the expression

$$F(x) = C \exp(ax^\alpha) x^b \exp Q(x^{-\delta}), \quad (1.1)$$

understood through its logarithm

$$\log F(x) = \log C + ax^\alpha + b \log x + Q(x^{-\delta}). \quad (1.2)$$

We call  $(C, a, \alpha, b, \delta, Q)$  the *model data*.

A positive function  $F \in C^1([x_0, \infty))$  is said to be of *exponential–power type with data*  $(C, a, \alpha, b, \delta, Q)$  when, for every  $M \in \mathbb{N}$ , there is a constant  $K_M$  with

$$\left| \log F(x) - \log C - ax^\alpha - b \log x - \sum_{j=1}^M q_j x^{-j\delta} \right| \leq K_M x^{-(M+1)\delta} \quad (x \geq x_0), \quad (1.3)$$

and *differentiably* of that type when in addition

$$(\log F)'(x) = a\alpha x^{\alpha-1} + \frac{b}{x} + \mathcal{O}(x^{-1-\delta}) \quad (x \rightarrow \infty). \quad (1.4)$$

*Remark 1.3* (Why the two readings are kept apart). The series  $Q$  is *not* assumed convergent. In Parts 2, 3 and 5 it is divergent of Gevrey type, which is what makes theorem 1.46 bite there. It is *not* divergent in Part 4: for the partition numbers  $Q(t) = \log(1-t)$  has radius of convergence 1, every algebraic expansion of that chapter converges, and theorem 1.46 correspondingly has no content there — the error floor of Part 4 is set by the exponentially small Rademacher remainder and not by a least term. This is precisely why the two readings are kept apart: convergence of  $Q$  is a property of the subject, not of the frame. Every purely algebraic assertion below — coefficient recurrences, closed coefficient formulae, degree bounds — is a statement about (1.2) as a formal object and needs no analytic hypothesis at all. Every assertion with an error term uses (1.3), and every assertion that inverts uses (1.4) as well: an asymptotic expansion may not be differentiated without a hypothesis, and the inversion of  $F$  is controlled by  $(\log F)'$ . In each chapter (1.4) is verified directly from a digamma, Laplace or Rademacher representation, never inferred from (1.3).

### 1.2.2 Reduction to a linear phase

**Lemma 1.4** (Monomial  $\alpha$ -reduction). *Let  $F$  be of exponential–power type with data  $(C, a, \alpha, b, \delta, Q)$  on  $[x_0, \infty)$ , and put*

$$\xi = x^\alpha, \quad x = \xi^{1/\alpha}, \quad G(\xi) := F(\xi^{1/\alpha}). \quad (1.5)$$

*Then  $G$  is of exponential–power type on  $[x_0^\alpha, \infty)$  with data*

$$\left(C, a, 1, \frac{b}{\alpha}, \frac{\delta}{\alpha}, Q\right), \quad (1.6)$$

*that is,*

$$\log G(\xi) = \log C + a\xi + \frac{b}{\alpha} \log \xi + Q(\xi^{-\delta/\alpha}). \quad (1.7)$$

*The coefficients  $q_j$  are unchanged. If moreover  $F$  is differentiable of its type then  $G$  is differentiable of its type. Finally  $x \mapsto x^\alpha$  is an increasing bijection of  $[x_0, \infty)$  onto  $[x_0^\alpha, \infty)$ , so  $F$  is strictly increasing if and only if  $G$  is, and then*

$$F^{-1}(y) = (G^{-1}(y))^{1/\alpha}. \quad (1.8)$$

*Proof.* Substituting  $x = \xi^{1/\alpha}$  into (1.2) and using  $x^\alpha = \xi$ ,  $\log x = \alpha^{-1} \log \xi$  and  $x^{-j\delta} = \xi^{-j\delta/\alpha}$  gives (1.7) term by term; the truncated inequality (1.3) transforms into the corresponding inequality for  $G$  with the same constants  $K_M$ , because  $x^{-(M+1)\delta} = \xi^{-(M+1)\delta/\alpha}$ . For the derivative,  $dx/d\xi = \alpha^{-1} \xi^{1/\alpha-1}$ , so

$$(\log G)'(\xi) = (\log F)'(\xi^{1/\alpha}) \cdot \frac{\xi^{1/\alpha-1}}{\alpha} = \left(a\alpha \xi^{1-1/\alpha} + \frac{b}{\xi^{1/\alpha}} + \mathcal{O}(\xi^{-(1+\delta)/\alpha})\right) \frac{\xi^{1/\alpha-1}}{\alpha} = a + \frac{b/\alpha}{\xi} + \mathcal{O}(\xi^{-1-\delta/\alpha}),$$

which is (1.4) for  $G$  with  $\alpha$  replaced by 1. The last two assertions are immediate from strict monotonicity of  $x \mapsto x^\alpha$  on  $(0, \infty)$ .  $\square$

**Remark 1.5** (Normalization used from here on). By lemma 1.4 every statement about the model (1.1) reduces to the case  $\alpha = 1$ , and we assume  $\alpha = 1$  from section 1.4 onwards. We assume in addition  $\delta = 1$ ; this is the value of  $\delta/\alpha$  in every chapter of the volume (table 1.1). Proposition 1.27 states the general- $\delta$  reversion, on the two-generator monomial grid it requires.

**Remark 1.6** (The affine shift is taken *before* the reduction). Chapters 3 and 4 apply the model not to the combinatorial index  $n$  but to a shifted variable  $x = n - x_*$  ( $x_* = 1/24$  for the partition numbers,  $x_* = -1$  for the double factorial). This is not a loss of generality but a genuine choice of normal form, and the order of operations matters. If  $F$  has data  $(C, a, \alpha, b, \delta, Q)$  in  $x$  and one re-expands in  $n = x + x_*$ , then for  $\alpha < 1$

$$a(n - x_*)^\alpha = an^\alpha + a \sum_{k \geq 1} \binom{\alpha}{k} (-x_*)^k n^{\alpha-k},$$

so the phase acquires an entire new tower of powers  $n^{\alpha-k}$ : the re-expanded object no longer has model data with a single exponent grid. The partition articles say this explicitly — their infinite algebraic coefficient arrays in the variable  $n^{-1/2}$  “arise entirely from re-expanding the shift  $N = n - 1/24$ ”. The volume therefore normalizes the shift first, inverts in  $x$ , and returns to the index by  $n = x + x_*$ , an exact identity that costs nothing.

**Remark 1.7** (Which chapters the model covers, and which it does not). Chapters 2, 4 and 5 are governed by definition 1.2: the rooted-tree counts and the swing factorial with  $\alpha = 1$ , the partition numbers with  $\alpha = 1/2$  (table 1.1). Part 3 is not.

**Claim repaired.** It is natural, and it is asserted in the merge brief for this volume, that the case  $\alpha = 1$  of definition 1.2 covers the double factorial alongside the swing factorial and the rooted-tree counts. It does not. For the swing factorial  $n!/[n/2]!$ <sup>2</sup> the growth is  $2^n n^{\pm 1/2}$  and the model applies with

$a = \log 2$ . For the double factorial,

$$\log((x+1)!!) = \frac{x+1}{2}(\log(x+1) - 1) + \mathcal{O}(\log x),$$

so the dominant phase is  $\frac{1}{2}s \log s$  with  $s = x + 1$ , not  $ax$ : no choice of the data  $(C, a, \alpha, b, \delta, Q)$  with  $\alpha \leq 1$  reproduces it, since  $s \log s / s^\alpha \rightarrow \infty$  for every  $\alpha \leq 1$ . The three double factorial articles are consistent with this; they never claim membership of the exponential–power family, and they invert the phase  $\frac{s}{2}(\log s - 1)$  directly. **Correction.** The chapter states the reversion for an axiomatized *dominant core* (definition 1.8 and theorem 1.29) of which both the linear-plus-logarithmic phase  $aX + b \log X$  and the factorial phase  $X(\kappa \log X + d)$  are instances, both exactly invertible by Lambert  $W$  (theorem 1.20 and proposition 1.23). The theorem named in the Part 0 interface, theorem 1.25, is stated for the exponential–power model as specified, and theorem 1.29 is the version Part 3 cites.

**Definition 1.8** (Admissible dominant core). Let  $X_* \geq 0$ . A function  $\Phi_0 \in C^2((X_*, \infty))$  is an *admissible core* when

1.  $\Phi'_0(X) > 0$  for  $X > X_*$  and  $\Phi_0(X) \rightarrow +\infty$  as  $X \rightarrow +\infty$ ;
2.  $\Phi'_0$  is eventually monotone and  $\Phi'_0(X) \rightarrow \Lambda_\infty \in (0, +\infty]$ ;
3.  $X\Phi'_0(X) \rightarrow +\infty$ .

Its *core slope* is  $\Phi'_0$ . Given a target  $L$  in the range of  $\Phi_0$ , the *core solution*  $X = X(L)$  is the unique large solution of  $\Phi_0(X) = L$ .

The two cores used in this volume are

$$\Phi_0(X) = aX + b \log X \quad (\text{Part 2, 4, 5}), \quad \Phi_0(X) = X(\kappa \log X + d) \quad (\text{Part 3}), \quad (1.9)$$

with  $a, \kappa > 0$ . Both are admissible on a suitable half-line, and both are inverted exactly by  $W$  in section 1.4.

## 1.3 The coefficient calculus

### 1.3.1 Two families of partial Bell polynomials

**Definition 1.9** (Ordinary partial Bell polynomials). For a sequence  $\gamma = (\gamma_1, \gamma_2, \dots)$  in  $R$  and  $n, k \in \mathbb{N}$  put

$$\widehat{B}_{n,k}(\gamma) := [t^n] \left( \sum_{r \geq 1} \gamma_r t^r \right)^k. \quad (1.10)$$

Explicitly,

$$\widehat{B}_{n,k}(\gamma) = \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_r m_r = k, \sum_r r m_r = n}} \frac{k!}{\prod_{r \geq 1} m_r!} \prod_{r \geq 1} \gamma_r^{m_r}. \quad (1.11)$$

In particular  $\widehat{B}_{0,0} = 1$ ,  $\widehat{B}_{n,0} = 0$  for  $n \geq 1$ ,  $\widehat{B}_{n,k} = 0$  for  $k > n$ ,  $\widehat{B}_{n,n} = \gamma_1^n$ , and  $\widehat{B}_{n,k}$  depends only on  $\gamma_1, \dots, \gamma_{n-k+1}$ .

**Definition 1.10** (Exponential partial Bell polynomials). For  $x = (x_1, x_2, \dots)$  in  $R$  and  $n \geq k \geq 0$  put

$$B_{n,k}(x_1, x_2, \dots) := n! \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_j m_j = k, \sum_j j m_j = n}} \prod_{j \geq 1} \frac{1}{m_j!} \left( \frac{x_j}{j!} \right)^{m_j}, \quad (1.12)$$

and let  $B_n := \sum_{k=0}^n B_{n,k}$  be the complete polynomial. These are characterized by the generating identities

$$\frac{1}{k!} \left( \sum_{j \geq 1} x_j \frac{t^j}{j!} \right)^k = \sum_{n \geq k} B_{n,k}(x_1, x_2, \dots) \frac{t^n}{n!}, \quad (1.13)$$

$$\exp \left( \sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (1.14)$$

**Lemma 1.11** (Conversion). *For all  $n \geq k \geq 0$ ,*

$$\widehat{B}_{n,k}(\gamma_1, \gamma_2, \dots) = \frac{k!}{n!} B_{n,k}(1! \gamma_1, 2! \gamma_2, \dots). \quad (1.15)$$

*Proof.* Set  $x_j = j! \gamma_j$  in (1.13). The left-hand side becomes  $(k!)^{-1} (\sum_j \gamma_j t^j)^k$ , whose coefficient of  $t^n$  is  $\widehat{B}_{n,k}(\gamma)/k!$ ; the right-hand side has coefficient  $B_{n,k}(1! \gamma_1, \dots)/n!$ .  $\square$

### 1.3.2 Powers and logarithms of a unit series

**Lemma 1.12** (Powers and logarithms). *Let  $w = \sum_{r \geq 1} \gamma_r t^r \in tR[[t]]$  and  $n \geq 1$ .*

1. *For every  $\beta$  in  $R$  (or in any commutative  $\mathbb{Q}$ -algebra containing  $R$ ),*

$$[t^n] (1+w)^\beta = \sum_{k=1}^n \binom{\beta}{k} \widehat{B}_{n,k}(\gamma), \quad \binom{\beta}{k} = \frac{\beta(\beta-1) \cdots (\beta-k+1)}{k!}. \quad (1.16)$$

2. *For an integer  $j \geq 1$ ,*

$$[t^n] (1+w)^{-j} = \sum_{k=0}^n (-1)^k \binom{j+k-1}{k} \widehat{B}_{n,k}(\gamma). \quad (1.17)$$

3.

$$[t^n] \log(1+w) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma). \quad (1.18)$$

*Equivalently, in exponential form, for  $U(t) = 1 + \sum_{j \geq 1} u_j t^j$ ,*

$$[t^N] U(t)^\beta = \frac{1}{N!} \sum_{k=0}^N \beta^{\underline{k}} B_{N,k}(1! u_1, 2! u_2, \dots), \quad \beta^{\underline{k}} = \beta(\beta-1) \cdots (\beta-k+1), \quad (1.19)$$

$$[t^N] \log U(t) = \frac{1}{N!} \sum_{k=1}^N (-1)^{k-1} (k-1)! B_{N,k}(1! u_1, 2! u_2, \dots). \quad (1.20)$$

*Proof.* All four identities are the substitution of  $w$  into a formal series in one variable. Since  $w \in tR[[t]]$ , the family  $(w^k)_{k \geq 0}$  is summable in the  $t$ -adic topology and  $[t^n] w^k = 0$  for  $k > n$ , so every sum displayed is finite and the substitution is legitimate. Now  $(1+w)^\beta = \sum_{k \geq 0} \binom{\beta}{k} w^k$  gives (1.16) on applying definition 1.9;  $\binom{-j}{k} = (-1)^k \binom{j+k-1}{k}$  gives (1.17); and  $\log(1+w) = \sum_{k \geq 1} (-1)^{k+1} w^k / k$  gives (1.18). For the exponential forms write  $U = 1 + w$  with  $\gamma_j = u_j$  and substitute (1.15) into (1.16) and (1.18), using  $\binom{\beta}{k} = \beta^{\underline{k}} / k!$  and  $(-1)^{k+1} / k = (-1)^{k-1} (k-1)! / k!$ .  $\square$

**Corollary 1.13** (Exponential and logarithmic jets). *Let  $D(t) = \sum_{j \geq 1} d_j t^j \in tR[[t]]$  and  $A(t) = \exp D(t) = \sum_{n \geq 0} A_n t^n$ . Then  $A_0 = 1$ ,*

$$A_n = \frac{1}{n!} B_n(1! d_1, 2! d_2, \dots, n! d_n) = \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_j j k_j = n}} \prod_{j \geq 1} \frac{d_j^{k_j}}{k_j!}, \quad (1.21)$$

and the coefficients obey the triangular recurrence

$$nA_n = \sum_{j=1}^n j d_j A_{n-j} \quad (n \geq 1). \quad (1.22)$$

Conversely, if  $A = 1 + \sum_{n \geq 1} A_n t^n$  is a unit series then  $d_n = [t^n] \log A$  is given by (1.20).

*Proof.* The first equality is (1.14) with  $x_j = j!d_j$ ; the second is the expansion of  $\exp$  as a product over  $j$ . For (1.22) differentiate  $A = \exp D$  to get  $A' = D'A$  and compare coefficients of  $t^{n-1}$ . The converse is (1.20).  $\square$

### 1.3.3 Lagrange–Bürmann inversion and the fixed-point form

**Theorem 1.14** (Lagrange–Bürmann). *Let  $\varphi \in R[[w]]$  with  $\varphi(0)$  invertible in  $R$ . There is a unique  $W \in tR[[t]]$  with*

$$W = t\varphi(W), \quad (1.23)$$

and for every  $G \in R[[w]]$  and every  $n \geq 1$ ,

$$[t^n] G(W(t)) = \frac{1}{n} [w^{n-1}] \left( G'(w) \varphi(w)^n \right). \quad (1.24)$$

In particular

$$[t^n] W(t) = \frac{1}{n} [w^{n-1}] \varphi(w)^n. \quad (1.25)$$

*Proof.* Existence and uniqueness: (1.23) determines  $[t^n] W$  from  $[t^{<n}] W$ , because  $t\varphi(W)$  modulo  $t^{n+1}$  depends only on  $W$  modulo  $t^n$ .

For (1.24) work in the field of formal Laurent series  $R((w))$  and let  $\text{Res}$  denote the coefficient of  $w^{-1}$ . Two facts are needed.

(a)  $\text{Res } f' = 0$  for every  $f \in R((w))$ , since  $w^{-1}$  has no antiderivative among the monomials:  $(w^k)' = kw^{k-1}$  is a multiple of  $w^{-1}$  only for  $k = 0$ , when it vanishes.

(b) *Change of variables.* Put  $\psi(w) := w/\varphi(w)$ , a series in  $wR[[w]]$  with  $\psi'(0) = \varphi(0)^{-1}$  invertible; then  $W$  is the compositional inverse of  $\psi$ , so  $\psi(W(t)) = t$ . For  $f \in R((w))$ ,

$$\text{Res}_w(f(w)) = \text{Res}_t(f(W(t)) W'(t)). \quad (1.26)$$

Indeed both sides are  $R$ -linear and continuous, so it suffices to check  $f = w^k$ . For  $k \neq -1$ ,  $W^k W' = (W^{k+1}/(k+1))'$  and (a) gives  $0 = \text{Res}_w w^k$ . For  $k = -1$ , write  $W = t\vartheta(t)$  with  $\vartheta$  a unit; then  $W'/W = t^{-1} + \vartheta'/\vartheta$  and the second summand lies in  $R[[t]]$ , so  $\text{Res}_t(W'/W) = 1 = \text{Res}_w w^{-1}$ .

Now apply (1.26) to  $f(w) := G(w)\psi(w)^{-n-1}\psi'(w)$ , noting that  $f(W(t))W'(t) = G(W(t))t^{-n-1}\frac{d}{dt}\psi(W(t)) = G(W(t))t^{-n-1}$  because  $\psi(W(t)) = t$ :

$$[t^n] G(W(t)) = \text{Res}_t(G(W(t)) t^{-n-1}) = \text{Res}_w(G(w)\psi(w)^{-n-1}\psi'(w)).$$

Since  $\psi^{-n-1}\psi' = -\frac{1}{n}(\psi^{-n})'$ , fact (a) and the product rule give

$$\text{Res}_w(G\psi^{-n-1}\psi') = -\frac{1}{n} \text{Res}_w(G \cdot (\psi^{-n})') = \frac{1}{n} \text{Res}_w(G' \psi^{-n}) = \frac{1}{n} \text{Res}_w(G'(w) \varphi(w)^n w^{-n}),$$

which is (1.24). Taking  $G(w) = w$  gives (1.25).  $\square$

The reversions of this volume all take the form  $U = \Psi(t, U)$  in which the rôle of  $\varphi(0)$  is played by  $\Psi(t, 0)$ , an element of  $tR[[t]]$ , which is *never* invertible. Theorem 1.14 therefore does not apply directly, and the following universality lemma is what makes the passage legitimate.

**Lemma 1.15** (Lagrange’s formula without an invertibility hypothesis). *Let  $S$  be any commutative  $\mathbb{Q}$ -algebra and  $\varphi \in S[[w]]$  arbitrary (in particular  $\varphi(0)$  need not be invertible). Let  $W \in zS[[z]]$  be the unique solution of  $W = z\varphi(W)$ . Then for every  $m \geq 1$ ,*

$$[z^m] W = \frac{1}{m} [w^{m-1}] \varphi(w)^m. \quad (1.27)$$

*Proof.* Existence and uniqueness of  $W$  hold as in theorem 1.14 (the recursion is  $z$ -adic and uses no invertibility). Fix  $m$ . Iterating  $W = z\varphi(W)$  shows that  $[z^m] W$  is a universal polynomial with integer coefficients in  $\varphi_0, \varphi_1, \dots, \varphi_{m-1}$ , where  $\varphi = \sum_i \varphi_i w^i$ ; the right-hand side of (1.27) is a universal polynomial with rational coefficients in the same variables. Both sides are therefore elements  $P_m, \tilde{P}_m$  of the polynomial ring  $\mathcal{P} := \mathbb{Q}[\varphi_0, \dots, \varphi_{m-1}]$ , and they are obtained from  $\varphi$  by the unique  $\mathbb{Q}$ -algebra homomorphism  $\mathcal{P} \rightarrow S$  sending  $\varphi_i \mapsto \varphi_i$ . It thus suffices to prove  $P_m = \tilde{P}_m$  in  $\mathcal{P}$ . The localization map  $\mathcal{P} \rightarrow \mathcal{P}[\varphi_0^{-1}]$  is injective ( $\mathcal{P}$  is an integral domain), and in  $\mathcal{P}[\varphi_0^{-1}]$  the element  $\varphi_0$  is invertible, so theorem 1.14 applies and gives  $P_m = \tilde{P}_m$  there. Injectivity transports the equality back to  $\mathcal{P}$ , and the homomorphism transports it to  $S$ .  $\square$

**Lemma 1.16** (Lagrange fixed-point formula). *Let  $\Psi \in tR[[t, u]]$ , that is, a formal series in  $t$  and  $u$  each of whose monomials has  $t$ -degree at least one. Then the equation*

$$U(t) = \Psi(t, U(t)) \quad (1.28)$$

*has a unique solution  $U \in tR[[t]]$ , and for every  $n \geq 1$*

$$[t^n] U = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi(t, u)^m. \quad (1.29)$$

*Proof.* Uniqueness and existence:  $\Psi(t, U)$  modulo  $t^{n+1}$  depends only on  $U$  modulo  $t^n$  because  $\Psi$  is divisible by  $t$ ; so (1.28) determines the coefficients of  $U$  recursively, and the resulting series solves the equation by  $t$ -adic continuity.

For (1.29) apply lemma 1.15 over the ring  $S := R[[t]]$  with  $\varphi(w) := \Psi(t, w) \in S[[w]]$  and marker  $z$ : the unique  $\tilde{U} \in zS[[z]]$  with  $\tilde{U} = z\varphi(\tilde{U})$  satisfies

$$[z^m] \tilde{U} = \frac{1}{m} [w^{m-1}] \Psi(t, w)^m \in S.$$

Since  $\Psi \in tR[[t, u]]$ , we have  $\Psi^m \in t^m R[[t, u]]$ , whence  $[t^n] [w^{m-1}] \Psi(t, w)^m = 0$  for  $m > n$ . Consequently the family  $([z^m] \tilde{U})_{m \geq 1}$  is summable in the  $t$ -adic topology of  $S$ , the specialization  $z = 1$  is defined coefficientwise in  $t$ , and the resulting series  $U := \sum_{m \geq 1} [z^m] \tilde{U}$  satisfies  $U = \Psi(t, U)$  by the same summability. Uniqueness identifies it with the solution of (1.28), and extracting  $[t^n]$  gives (1.29) with the sum truncated at  $m = n$ .  $\square$

**Source claim.** `Swing_Factorial_Transseries_Article`, Lemma *Lagrange fixed-point formula*, proves (1.29) by writing “Ordinary Lagrange–Bürmann inversion in  $z$  gives  $U = \sum_{m \geq 1} z^m m^{-1} [u^{m-1}] \Psi(t, u)^m$ ”. The same step, with the same silence, appears in `Partition_Number_Transseries_and_Inverse` (“Introduce  $u = z\Phi_\rho(u)$ . Lagrange–Bürmann gives ...”), in `Partition_Numbers_Transseries_and_Inverse`, and in `double_factorial_transseries-2`. **Defect.** The classical Lagrange–Bürmann theorem in the form (1.25) requires  $\varphi(0)$  to be invertible in the coefficient ring. Here  $\varphi(0) = \Psi(t, 0) \in tR[[t]]$ , a non-unit, and the residue proof of theorem 1.14 breaks at the change of variables:  $\psi = w/\varphi$  is not a well-defined element of  $wR[[t]][[w]]$ . **Repair.** Lemma 1.15: the identity is a polynomial identity in the coefficients of  $\varphi$  over  $\mathbb{Q}$ , hence holds in the polynomial ring by injectivity of localization at  $\varphi_0$ , hence in every  $\mathbb{Q}$ -algebra. `double_factorial_transseries-2` comes closest to noticing the issue — “Although  $A$  is a divergent Gevrey series, it is a valid formal power series with nonzero linear coefficient” — but that remark addresses convergence, not invertibility, and



its own  $G_\Lambda$  does have an invertible linear coefficient, so the gap there is only in the inherited lemma.

**Lemma 1.17** (Fixed-point formula with a quadratic self-interaction). *Let  $\Psi = h + f$  with  $h \in u^2 R[[u]]$  and  $f \in tR[[t, u]]$ . Then  $U = \Psi(t, U)$  has a unique solution  $U \in tR[[t]]$ , and for every  $n \geq 1$*

$$[t^n] U = \sum_{m=1}^{2n-1} \frac{1}{m} [t^n u^{m-1}] \Psi(t, u)^m. \quad (1.30)$$

*Proof. Existence and uniqueness.* Write  $U = U_{<n} + e_n t^n + \dots$ . Then  $h(U) = h(U_{<n}) + h'(U_{<n}) e_n t^n + \dots$ , and  $h'(U_{<n}) \in tR[[t]]$  because  $h' \in uR[[u]]$  and  $U_{<n} \in tR[[t]]$ ; so  $e_n$  does not occur in  $[t^n] h(U)$ . It does not occur in  $[t^n] f(t, U)$  either, since  $f$  is divisible by  $t$ . Hence the equation determines  $e_n$ , as  $[t^n] \Psi(t, U_{<n})$ , from  $U_{<n}$ ; the resulting series solves the equation by  $t$ -adic continuity.

*The closed formula.* Introduce a marker  $z$  and let  $\tilde{U} \in zS[[z]]$ ,  $S := R[[t]]$ , solve  $\tilde{U} = z\Psi(t, \tilde{U})$ . Lemma 1.15 gives  $[z^m] \tilde{U} = m^{-1} [u^{m-1}] \Psi(t, u)^m$ . We claim

$$[t^n u^{m-1}] \Psi(t, u)^m = 0 \quad \text{whenever } m > 2n - 1. \quad (1.31)$$

Expand  $\Psi^m = (h + f)^m$  into  $2^m$  products; a product using  $h$  in  $\sigma$  of the  $m$  slots and  $f$  in the remaining  $m - \sigma$  has  $u$ -degree at least  $2\sigma$  and  $t$ -degree at least  $m - \sigma$ . For the  $u$ -degree to equal  $m - 1$  we need  $2\sigma \leq m - 1$ , hence the  $t$ -degree is at least  $m - \frac{m-1}{2} = \frac{m+1}{2}$ ; so  $[t^n]$  of that product vanishes unless  $n \geq (m + 1)/2$ , which is (1.31). Consequently the family  $([z^m] \tilde{U})_{m \geq 1}$  is  $t$ -adically summable, the specialization  $z = 1$  is defined coefficientwise, and, again by summability, the resulting series solves  $U = \Psi(t, U)$ . Uniqueness identifies it with  $U$ , and extracting  $[t^n]$  gives (1.30).  $\square$

Lemma 1.16 is the case  $h = 0$  of lemma 1.17, in which the argument above sharpens the range of  $m$  from  $2n - 1$  to  $n$ .

### 1.3.4 Perturbed inversion around an exactly invertible core

The reversions carried out in Chapters 1–4 all reduce to one step: a core  $F$  that can be inverted exactly is perturbed by a correction  $E$ , and the inverse of  $F + \varepsilon E$  is required to all orders in  $\varepsilon$ . The answer is a single differential formula, stated here once. It is the tool behind theorem 4.62 in the ordinate of the partition chapter and behind lemma 5.39 in the phase coordinate of the swing chapter, and those two are the same identity in different letters.

**Theorem 1.18** (Perturbed inversion). *Let  $F$  and  $E$  be analytic in a neighbourhood of  $\mu_0 \in \mathbb{C}$ , let  $F'(\mu_0) \neq 0$ , and put  $y := F(\mu_0)$ . Write  $\mu_0(\cdot)$  for the local inverse of  $F$ , so that  $\mu_0(y) = \mu_0$  and  $\mu'_0(y) = 1/F'(\mu_0(y))$ , and let*

$$\mathcal{D} := \frac{1}{F'(\mu)} \frac{d}{d\mu}, \quad (1.32)$$

*which is the derivation  $d/dy$  read in the coordinate  $y = F(\mu)$ . Then there are a neighbourhood  $\mathcal{U}$  of  $\mu_0$  and an  $\varepsilon_0 > 0$  such that for  $|\varepsilon| < \varepsilon_0$  the equation*

$$F(\mu) + \varepsilon E(\mu) = y \quad (1.33)$$

*has exactly one root  $\mu(y; \varepsilon)$  in  $\mathcal{U}$ ; it is analytic in  $\varepsilon$  there, and*

$$\mu(y; \varepsilon) = \mu_0(y) + \sum_{M \geq 1} \frac{(-\varepsilon)^M}{M!} \mathcal{D}^{M-1} \left( \frac{E(\mu_0(y))^M}{F'(\mu_0(y))} \right), \quad (1.34)$$

*the series converging for  $|\varepsilon| < \varepsilon_0$ , with  $M$ th coefficient*

$$\frac{(-1)^M}{M!} \mathcal{D}^{M-1} \left( \frac{E(\mu_0)^M}{F'(\mu_0)} \right) = \frac{(-1)^M}{M} \operatorname{Res}_{\mu=\mu_0} \frac{E(\mu)^M}{(F(\mu) - y)^M}. \quad (1.35)$$

If moreover  $E = \sum_j \varepsilon_j E_j$  with finitely many  $E_j$  analytic near  $\mu_0$ , then for every multi-index  $\mathbf{m} \neq \mathbf{0}$  with  $M := |\mathbf{m}|$  the coefficient of  $\varepsilon^{\mathbf{m}}$  in  $\mu - \mu_0$  is

$$V_{\mathbf{m}} = \frac{(-1)^M}{\mathbf{m}!} \mathcal{D}^{M-1} \left( \frac{E^{\mathbf{m}}(\mu_0)}{F'(\mu_0)} \right) = \frac{(-1)^M (M-1)!}{\mathbf{m}!} \operatorname{Res}_{\mu=\mu_0} \frac{E^{\mathbf{m}}(\mu)}{(F(\mu) - F(\mu_0))^M}, \quad (1.36)$$

where  $E^{\mathbf{m}} := \prod_j E_j^{m_j}$  and  $\mathbf{m}! := \prod_j m_j!$ .

*Proof.* Let  $\gamma$  be a small positively oriented circle about  $\mu_0$ , contained with its interior  $\mathcal{U}$  in the common domain of analyticity and enclosing no zero of  $F(\cdot) - y$  other than  $\mu_0$ ; such a  $\gamma$  exists because  $F'(\mu_0) \neq 0$  makes  $\mu_0$  a simple zero, hence isolated. Put  $m_\gamma := \min_\gamma |F - y| > 0$  and  $M_\gamma := \max_\gamma |E|$ , and let  $\varepsilon_0 := m_\gamma / (M_\gamma + 1)$ . For  $|\varepsilon| < \varepsilon_0$  we have  $|\varepsilon E| < |F - y|$  on  $\gamma$ , so Rouché gives exactly one zero  $\mu(y; \varepsilon)$  of  $F - y + \varepsilon E$  inside  $\gamma$ , and it is simple. The argument principle, applied to that function, expresses it as

$$\mu(y; \varepsilon) = \frac{1}{2\pi i} \int_\gamma \zeta \, d\zeta \log(F(\zeta) - y + \varepsilon E(\zeta)), \quad (1.37)$$

which is analytic in  $\varepsilon$  on  $|\varepsilon| < \varepsilon_0$  because the integrand is and  $\gamma$  is compact.

Subtract from (1.37) the same expression at  $\varepsilon = 0$ , whose value is  $\mu_0$ , and expand

$$\log\left(1 + \varepsilon \frac{E}{F - y}\right) = \sum_{M \geq 1} \frac{(-1)^{M-1}}{M} \varepsilon^M \frac{E^M}{(F - y)^M},$$

uniformly convergent on  $\gamma$  since  $|\varepsilon E / (F - y)| < 1$  there. Integrating by parts once turns  $\zeta \, d\zeta(\cdot)$  into  $-(\cdot) \, d\zeta$  (the boundary term vanishes on a closed contour), so the coefficient of  $\varepsilon^M$  is

$$\frac{(-1)^M}{M} \operatorname{Res}_{\zeta=\mu_0} \frac{E(\zeta)^M}{(F(\zeta) - F(\mu_0))^M},$$

which is the right-hand side of (1.35).

It remains to identify that residue with the differential expression. Change coordinate by  $w = F(\zeta)$ ,  $\zeta = \mu_0(w)$ ,  $d\zeta = \mu'_0(w) \, dw$ , legitimate on a neighbourhood of  $\mu_0$  because  $F'(\mu_0) \neq 0$ :

$$\operatorname{Res}_{\zeta=\mu_0} \frac{E(\zeta)^M}{(F(\zeta) - y)^M} d\zeta = \operatorname{Res}_{w=y} \frac{E(\mu_0(w))^M \mu'_0(w)}{(w - y)^M} dw = \frac{1}{(M-1)!} \frac{d^{M-1}}{dy^{M-1}} \left( \frac{E(\mu_0(y))^M}{F'(\mu_0(y))} \right),$$

the last step being the residue of a pole of order  $M$ . Multiplying by  $(-1)^M / M$  and using  $d/dy = \mathcal{D}$  gives (1.34) and (1.35).

For (1.36) replace  $E$  by  $\sum_j \varepsilon_j E_j$  and expand the  $M$ th power by the multinomial theorem: the monomial  $\varepsilon^{\mathbf{m}}$  occurs with coefficient  $M! / \mathbf{m}!$ , which cancels the  $M!$  in (1.34).  $\square$

The identity (1.34) is needed in Chapter 4 with a correction  $E$  that is a *divergent* formal series, where no contour is available. What survives is the coefficientwise identity, and it survives for the reason the Lagrange fixed-point formula does.

**Corollary 1.19** (Formal form of the reversion). *Fix  $M \geq 1$  and let  $\mathcal{P} := \mathbb{Q}[e_0, \dots, e_{M-1}, f_2, \dots, f_M][f_1^{-1}]$ . Both of the following are elements of  $\mathcal{P}$ , obtained by writing out the indicated recursions with  $e_i$  and  $f_i$  standing for  $E^{(i)}(\mu_0)$  and  $F^{(i)}(\mu_0)$ :*

1. the coefficient  $\mu_M$  of  $\varepsilon^M$  produced by solving (1.33) by formal Taylor iteration, i.e. by substituting  $\mu = \mu_0 + \sum_{k \geq 1} \mu_k \varepsilon^k$ , expanding  $F$  and  $E$  in their Taylor jets at  $\mu_0$ , and equating coefficients of  $\varepsilon^M$ ;
2. the expression  $\frac{(-1)^M}{M!} \mathcal{D}^{M-1} (E(\mu_0)^M / F'(\mu_0))$  of (1.34), expanded by the Leibniz rule.

They are equal in  $\mathcal{P}$ . Consequently (1.34) and (1.36) hold coefficientwise in any commutative  $\mathbb{Q}$ -algebra carrying jets  $e_i, f_i$  with  $f_1$  invertible — in particular when  $F$  and  $E$  are formal, possibly divergent, series, and when  $\mathcal{D}$  is the derivation (1.32) written in any coordinate.

*Proof.* Each recursion visibly terminates and involves only  $e_0, \dots, e_{M-1}, f_2, \dots, f_M$  and divisions by  $f_1$ : in (i) the coefficient of  $\varepsilon^M$  in  $F(\mu_0 + \sum_{k \geq 1} \mu_k \varepsilon^k) + \varepsilon E(\mu_0 + \sum_{k \geq 1} \mu_k \varepsilon^k)$  equals  $f_1 \mu_M$  plus a polynomial in  $f_2, \dots, f_M, e_0, \dots, e_{M-1}$  and  $\mu_1, \dots, \mu_{M-1}$ , so  $\mu_M$  is determined by one division by  $f_1$ ; in (ii) each application of  $\mathcal{D} = f_1^{-1} d/d\mu$  lowers the available jet order by one and contributes one factor  $f_1^{-1}$ , and  $M - 1$  applications to a function of  $e_0, \dots$  and  $f_1, \dots$  reach at most  $e_{M-1}$  and  $f_M$ . So both lie in  $\mathcal{P}$ .

Choose  $N$  with  $f_1^N \mu_M$  and  $f_1^N \cdot$  (ii) both in  $\mathcal{A} := \mathbb{Q}[e_0, \dots, e_{M-1}, f_1, \dots, f_M]$ , and regard them as polynomial functions on  $\mathbb{C}^{2M}$ . Every point of  $\mathbb{C}^{2M}$  with  $f_1 \neq 0$  is realized by a pair of *polynomials*: take  $\mu_0 = 0$ ,  $E(\mu) := \sum_{i=0}^{M-1} e_i \mu^i / i!$  and  $F(\mu) := \sum_{i=1}^M f_i \mu^i / i!$ , which are entire with  $F'(0) = f_1 \neq 0$  and with the prescribed jets. For such a pair theorem 1.18 applies, and the Taylor coefficients in  $\varepsilon$  of the analytic root  $\mu(y; \varepsilon)$  satisfy the recursion of (i) — substituting the convergent expansion into (1.33) and equating coefficients is legitimate — so (i) and (ii) agree at that point. Two elements of  $\mathcal{A}$  agreeing on the Zariski-dense open set  $\{f_1 \neq 0\}$  of  $\mathbb{C}^{2M}$  are equal in  $\mathbb{C}[e_\bullet, f_\bullet]$ , hence equal in  $\mathcal{A}$  and, after dividing by  $f_1^N$ , in  $\mathcal{P}$ . The final assertion follows by applying the unique  $\mathbb{Q}$ -algebra homomorphism  $\mathcal{P} \rightarrow S$  determined by the given jets, exactly as in lemma 1.15.  $\square$

## 1.4 The Lambert core

### 1.4.1 Exact inversion of the linear–logarithmic phase

**Theorem 1.20** (Exact solution of the dominant block). *Let  $a > 0$ ,  $b \in \mathbb{R}$ , and consider, for  $X > 0$ ,*

$$\Phi_0(X) := aX + b \log X = L. \quad (1.38)$$

1. *If  $b = 0$  the unique solution is  $X = L/a$ .*
2. *If  $b \neq 0$ , put  $W := (a/b)X$ . Then (1.38) holds if and only if*

$$W e^W = z(L), \quad z(L) := \frac{a}{b} e^{L/b}, \quad (1.39)$$

*so that the solutions of (1.38) are exactly*

$$X = \frac{b}{a} W_k \left( \frac{a}{b} e^{L/b} \right) \quad (1.40)$$

*as  $k$  ranges over the branches of the Lambert function for which  $W_k(z(L))$  is defined.*

3. *Branch rule. Suppose  $b \neq 0$ .*

- *If  $b > 0$  then  $z(L) > 0$ , equation (1.38) has exactly one positive solution for every real  $L$ ,  $\Phi_0$  is strictly increasing on  $(0, \infty)$ , and  $X = (b/a)W_0(z(L))$ .*
- *If  $b < 0$  then  $z(L) = -(a/|b|)e^{-L/|b|}$  lies in  $(-e^{-1}, 0)$  precisely when*

$$L > L_c := |b| \left( 1 - \log \frac{|b|}{a} \right) = \Phi_0 \left( \frac{|b|}{a} \right), \quad (1.41)$$

*and then (1.38) has exactly two positive solutions, namely  $(b/a)W_0(z(L)) < |b|/a < (b/a)W_{-1}(z(L))$ .  $\Phi_0$  is strictly increasing on  $(|b|/a, \infty)$  and the large solution is*

$$X = \frac{b}{a} W_{-1} \left( \frac{a}{b} e^{L/b} \right) = \frac{|b|}{a} \left( -W_{-1}(z(L)) \right). \quad (1.42)$$

*In both cases the large solution satisfies  $X \rightarrow +\infty$  and  $X = L/a + \mathcal{O}(\log L)$  as  $L \rightarrow +\infty$ .*

4. Slope identities. With  $W = (a/b)X$  and  $b \neq 0$ ,

$$\Phi'_0(X) = a + \frac{b}{X} = a \frac{1+W}{W}, \quad \frac{dX}{dL} = \frac{1}{\Phi'_0(X)} = \frac{1}{a} \frac{W}{1+W}. \quad (1.43)$$

*Proof.* (1) is immediate. For (2), multiply (1.38) by  $1/b$  and exponentiate:

$$e^{L/b} = e^{(a/b)X} X = e^W \cdot \frac{b}{a} W,$$

because  $X = (b/a)W$ . Multiplying by  $a/b$  gives (1.39); the step is reversible, and no logarithm of a negative number occurs, which is what makes the argument uniform in the sign of  $b$ . By definition, the solutions of  $We^W = z$  are the values  $W_k(z)$ , whence (1.40).

(3) We have  $\Phi'_0(X) = a + b/X$ . For  $b > 0$  this is positive on  $(0, \infty)$  and  $\Phi_0$  increases from  $-\infty$  to  $+\infty$ , so there is exactly one solution; correspondingly  $W = (a/b)X > 0$  and  $W_0$  is the unique real branch on  $(0, \infty)$ . For  $b < 0$ ,  $\Phi'_0$  vanishes only at  $X_* = |b|/a$ , is negative on  $(0, X_*)$  and positive on  $(X_*, \infty)$ ;  $\Phi_0$  decreases from  $+\infty$  to  $\Phi_0(X_*) = |b| - |b| \log(|b|/a) = L_c$  and then increases back to  $+\infty$ . Hence there are two positive solutions exactly when  $L > L_c$ , one in each monotonicity interval. On the  $W$  side  $W = (a/b)X < 0$ , and the two real branches of  $W$  on  $(-e^{-1}, 0)$  are  $W_0$ , with values in  $(-1, 0)$ , and  $W_{-1}$ , with values in  $(-\infty, -1)$ ; the map  $W \mapsto (b/a)W$  reverses order, so  $W_{-1}$  produces the larger  $X$ , namely the one exceeding  $|b|/a$ . Also  $z(L) > -e^{-1}$  is equivalent to  $(a/|b|)e^{-L/|b|} < e^{-1}$ , that is to (1.41). Finally  $aX = L - b \log X$  with  $X \rightarrow \infty$  gives  $aX = L + \mathcal{O}(\log L)$ .

(4)  $a + b/X = a + b \cdot \frac{a}{bW} = a(1 + W^{-1})$ , and  $dX/dL$  is the reciprocal of  $\Phi'_0$  by the inverse function theorem, applicable because  $\Phi'_0 \neq 0$  on the relevant interval.  $\square$

**Corollary 1.21** (Branch rule). *For  $b \neq 0$  the branch of the Lambert function carrying the large solution of  $L = aX + b \log X$  is determined by the sign of  $b$  alone:  $W_0$  if  $b > 0$ ,  $W_{-1}$  if  $b < 0$ .*

*Remark 1.22* (Why the branch is not a detail). Three of the four chapters have  $b < 0$  and therefore use  $W_{-1}$ ; the odd phase of the swing factorial has  $b = +1/2$  and uses  $W_0$ . *Swing\_Factorial\_Transseries\_Article* puts the point sharply: the principal branch “would produce the small root of the simplified equation and is irrelevant to the large-index inverse”. For  $b < 0$  the two roots are the two sides of a genuine fold at  $X_* = |b|/a$ , and choosing wrongly yields an object all of whose asymptotic corrections are then computed correctly about the wrong point.

## 1.4.2 The factorial core

**Proposition 1.23** (Exact inversion of a log-linear phase). *Let  $\kappa > 0$  and  $d \in \mathbb{R}$ , and consider*

$$\Phi_0(X) := X(\kappa \log X + d) = L, \quad X > 0. \quad (1.44)$$

*For  $L > 0$  put*

$$v := W_0\left(\frac{L}{\kappa} e^{d/\kappa}\right), \quad X := \exp\left(v - \frac{d}{\kappa}\right). \quad (1.45)$$

*Then  $X$  is the unique solution of (1.44) with  $\kappa \log X + d > 0$ , and*

$$\Phi'_0(X) = \kappa \log X + d + \kappa = \kappa(v + 1), \quad \frac{dX}{dL} = \frac{1}{\kappa(v + 1)}. \quad (1.46)$$

*Moreover  $\Phi_0$  is an admissible core in the sense of definition 1.8 on  $(e^{-(d+\kappa)/\kappa}, \infty)$ , with  $\Phi'_0(X) \rightarrow +\infty$ .*

*Proof.* Put  $s = \log X$ . Then (1.44) reads  $e^s(\kappa s + d) = L$ , that is  $(s + \frac{d}{\kappa})e^{s+d/\kappa} = (L/\kappa)e^{d/\kappa}$ . The right-hand side is positive, so  $W_0$  is the unique real branch and  $s + d/\kappa = v$ , which is (1.45). Uniqueness under the stated side condition holds because  $\kappa s + d > 0$  forces  $s + d/\kappa > 0$ , and  $\sigma \mapsto \sigma e^\sigma$  is injective on  $(0, \infty)$ . Differentiating,  $\Phi'_0(X) = \kappa \log X + d + \kappa = \kappa(s + d/\kappa) + \kappa = \kappa(v + 1)$ . Admissibility:  $\Phi'_0$  is increasing, tends to  $+\infty$ , is positive exactly for  $\log X > -(d + \kappa)/\kappa$ , and  $X\Phi'_0(X) \rightarrow \infty$ .  $\square$

*Remark 1.24* (The double factorial normalization). Part 3 uses  $\kappa = \frac{1}{2}$ ,  $d = -\frac{1}{2}$  and the centred variable  $s = x + 1$ , so that  $\Phi_0(s) = \frac{s}{2}(\log s - 1)$ ,  $(L/\kappa)e^{d/\kappa} = 2L/e$ ,  $X = e^{v+1}$  with  $v = W_0(2L/e)$ , and  $\Phi'_0(X) = \frac{1}{2}(v+1) = \frac{1}{2}\log X$ . This reproduces the core of all three double factorial articles; the quantity they call  $\Lambda$  is  $v+1 = \log X$ .

## 1.5 All-orders reversion around the core

### 1.5.1 The Lambert-centred inverse

**Theorem 1.25** (All-orders reversion around the Lambert core). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra, let  $a \in R$  be invertible, let  $b \in R$ , and let  $Q = \sum_{j \geq 1} q_j t^j \in tR[[t]]$ . Define*

$$\Psi(t, u) := -\frac{1}{a} \left[ b \log(1 + tu) + Q\left(\frac{t}{1 + tu}\right) \right]. \quad (1.47)$$

1.  $\Psi \in tR[[t, u]]$ .

2. There is a unique  $U = \sum_{n \geq 1} c_n t^n \in tR[[t]]$  with  $U = \Psi(t, U)$ , and

$$c_n = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi(t, u)^m \quad (n \geq 1). \quad (1.48)$$

3. Triangular Bell recurrence. Put  $\gamma_1 := 0$  and  $\gamma_r := c_{r-1}$  for  $r \geq 2$ . Then, for  $n \geq 1$ ,

$$c_n = -\frac{1}{a} \left[ b \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma) + \sum_{j=1}^n q_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma) \right], \quad (1.49)$$

and only  $c_1, \dots, c_{n-1}$  occur on the right-hand side. In particular  $c_n$  lies in the  $\mathbb{Q}$ -subalgebra of  $R$  generated by  $a^{-1}$ ,  $b$  and  $q_1, \dots, q_n$ ; no logarithm appears.

4. Formal exactness. In  $t^{-1}R[[t]]$  put

$$x := t^{-1} + U(t) = t^{-1} + \sum_{n \geq 1} c_n t^n. \quad (1.50)$$

Then  $tx = 1 + tU \in 1 + tR[[t]]$  and  $1/x = t/(1 + tU) \in tR[[t]]$ , and the following identity holds in  $R[[t]]$ :

$$a(x - t^{-1}) + b \log(tx) + Q(1/x) = 0. \quad (1.51)$$

Equivalently: writing  $t = X^{-1}$  and  $x = X + \sum_{n \geq 1} c_n X^{-n}$ , the quantity  $ax + b \log x + Q(1/x)$  equals  $aX + b \log X$  to all orders.

5. Analytic version. Let  $F$  be differentiable of exponential-power type with data  $(C, a, 1, b, 1, Q)$  on  $[x_0, \infty)$  (definition 1.2), strictly increasing there, with  $F(x) \rightarrow \infty$ . For  $y$  in the range of  $F$  set  $L = \log(y/C)$  and let  $X = X(y)$  be the large core solution of  $L = aX + b \log X$  furnished by theorem 1.20. Then  $X(y) \rightarrow \infty$  and, for every fixed  $N \geq 0$ ,

$$F^{-1}(y) = X + \sum_{n=1}^N \frac{c_n}{X^n} + \mathcal{O}(X^{-N-1}) \quad (y \rightarrow \infty). \quad (1.52)$$

*Proof.* (1)  $\log(1 + tu) = \sum_{k \geq 1} (-1)^{k+1} (tu)^k / k \in tR[[t, u]]$ , and  $t/(1 + tu) = t \sum_{k \geq 0} (-tu)^k \in tR[[t, u]]$ , so  $(t/(1 + tu))^j \in t^j R[[t, u]]$  and the sum over  $j$  is summable; hence  $Q(t/(1 + tu)) \in tR[[t, u]]$ .

(2) is lemma 1.16.

(3) Put  $w := tU = \sum_{r \geq 1} \gamma_r t^r$  with  $\gamma$  as stated;  $\gamma_1 = 0$  because  $U \in tR[[t]]$ . Then  $[t^n] \log(1+w)$  is given by (1.18), while

$$[t^n] Q\left(\frac{t}{1+w}\right) = \sum_{j \geq 1} q_j [t^{n-j}] (1+w)^{-j} = \sum_{j=1}^n q_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma)$$

by (1.17). Since  $c_n = [t^n] \Psi(t, U)$ , this is (1.49). For triangularity,  $\widehat{B}_{n,k}(\gamma)$  depends only on  $\gamma_1, \dots, \gamma_{n-k+1}$ , that is on  $c_1, \dots, c_{n-k}$ ; in the first sum  $k \geq 1$  and in the second  $j \geq 1$ , so  $c_n$  never occurs. (In fact  $\widehat{B}_{n,k}(\gamma) = 0$  for  $2k > n$ , because  $\gamma_1 = 0$ .)

(4) With  $x = t^{-1}(1+w)$ ,  $w = tU$ , one has  $a(x-t^{-1}) = aU$ ,  $\log(tx) = \log(1+w)$  and  $1/x = t/(1+w)$ , so the left-hand side of (1.51) equals  $aU + b \log(1+tU) + Q(t/(1+tU)) = a(U - \Psi(t, U)) = 0$ .

(5) Fix  $N \geq 0$ , set  $t := 1/X$ ,  $U_N(t) := \sum_{n=1}^N c_n t^n$  and  $x_N := X + U_N(t) = X(1 + tU_N(t))$ . By theorem 1.20(3),  $X \rightarrow \infty$  and  $X \sim L/a$ ; since  $tU_N(t) = \mathcal{O}(t^2)$  we get  $x_N = X(1 + \mathcal{O}(X^{-2}))$ , so  $x_N \rightarrow \infty$  and  $x_N \geq x_0$  for all large  $y$ .

*Residual.* Apply (1.3) with  $M = N$ :

$$\log F(x_N) = \log C + ax_N + b \log x_N + \sum_{j=1}^N q_j x_N^{-j} + E_N, \quad |E_N| \leq K_N x_N^{-N-1}.$$

Since  $\log y = \log C + aX + b \log X$  and  $\log x_N = \log X + \log(1 + tU_N(t))$ ,

$$r_N := \log F(x_N) - \log y = g(t) + E_N, \quad g(t) := aU_N(t) + b \log(1 + tU_N(t)) + \sum_{j=1}^N q_j \left( \frac{t}{1 + tU_N(t)} \right)^j.$$

Here  $g$  is real-analytic on a neighbourhood of  $t = 0$ , being a finite composition of analytic maps with  $1 + tU_N(t) \rightarrow 1$ ; its Taylor series at  $t = 0$  is therefore the corresponding formal series. By (4) the formal series  $aU + b \log(1 + tU) + Q(t/(1 + tU))$  vanishes identically. Replacing  $U$  by  $U_N$  changes it by an element of  $t^{N+1}R[[t]]$ , because  $U - U_N \in t^{N+1}R[[t]]$  and the maps  $u \mapsto \log(1 + tu)$ ,  $u \mapsto (t/(1 + tu))^j$  are  $t$ -adically continuous; and discarding the terms of  $Q$  with  $j > N$  changes it by an element of  $t^{N+1}R[[t]]$  as well. Hence the Taylor series of  $g$  vanishes through order  $t^N$ , so  $g(t) = \mathcal{O}(t^{N+1})$  and

$$|r_N| \leq |g(1/X)| + K_N x_N^{-N-1} = \mathcal{O}(X^{-N-1}).$$

*Slope.* By (1.4) with  $\alpha = 1$ ,  $(\log F)'(x) = a + \mathcal{O}(x^{-1})$ , so there is  $x_1 \geq x_0$  with  $(\log F)' \geq a/2$  on  $[x_1, \infty)$ . For all large  $y$  both  $x_N$  and  $F^{-1}(y)$  lie in  $[x_1, \infty)$ . Theorem 1.42, applied to  $\log F$  on that half-line, gives

$$|x_N - F^{-1}(y)| \leq \frac{|r_N|}{a/2} = \mathcal{O}(X^{-N-1}),$$

which is (1.52). □

**Remark 1.26** (No logarithms in the coefficients). Part (3) is the structural reason for centring at the Lambert core: the whole logarithmic tower has been resummed into  $X$ , so the  $c_n$  are constants of the coefficient ring. The alternative expansion of section 1.6 has coefficients that are polynomials in  $\log L$  whose degree grows with the order. This is an algebraic statement and by itself says nothing about numerical accuracy; the numerical comparison is a separate, chapterwise matter (see remark 1.36).

**Proposition 1.27** (General exponent grid). *Let  $\delta > 0$  and keep the remaining hypotheses of theorem 1.25. Let  $s$  be a second formal variable and put*

$$\Psi_\delta(t, s, u) := -\frac{1}{a} \left[ b \log(1 + tu) + Q\left(\frac{s}{(1 + tu)^\delta}\right) \right], \quad (1.53)$$

where  $(1 + tu)^{-\delta}$  is the formal binomial series (1.16). Then  $\Psi_\delta$  lies in the ideal  $(t, s)$  of  $R[[t, s, u]]$ , the equation  $U = \Psi_\delta(t, s, U)$  has a unique solution  $U = \sum_{i+j \geq 1} c_{i,j} t^i s^j$  in the ideal  $(t, s)$  of  $R[[t, s]]$ , and the substitution  $t = X^{-1}$ ,  $s = X^{-\delta}$  reverts (1.2): formally

$$ax + b \log x + Q(x^{-\delta}) = aX + b \log X, \quad x = X + \sum_{i+j \geq 1} c_{i,j} X^{-i-j\delta}. \quad (1.54)$$

For  $\delta = 1$  the two markers coincide and  $\Psi_\delta(t, t, u) = \Psi(t, u)$  of (1.47).

*Proof.* The membership  $\Psi_\delta \in (t, s)$  and the existence and uniqueness of  $U$  follow exactly as in parts (1) and (2) of theorem 1.25, with the  $t$ -adic filtration replaced by the  $(t, s)$ -adic one:  $\Psi_\delta$  lies in  $(t, s)$ , so  $\Psi_\delta(t, s, U)$  modulo  $(t, s)^{d+1}$  depends only on  $U$  modulo  $(t, s)^d$ . The identity (1.54) is the computation of part (4) with  $1/x = t/(1 + tU)$  replaced by  $x^{-\delta} = s(1 + tU)^{-\delta}$ . Finally the substitution is legitimate because, for each real exponent  $\theta$ , only finitely many pairs  $(i, j)$  with  $i, j \geq 0$  satisfy  $i + j\delta = \theta$  when  $\delta > 0$ .  $\square$

*Remark 1.28* (When the grid degenerates). If  $\delta$  is irrational the exponents  $i + j\delta$  are pairwise distinct and the monomial scale is a free monoid on two generators. If  $\delta = p/q$  in lowest terms, different pairs  $(i, j)$  can give the same exponent and the correct statement is that the reversion lives in  $R[[X^{-1/q}]]$ . Every chapter of this volume has  $\delta/\alpha = 1$  after lemma 1.4, so the degeneracy is never met here.

### 1.5.2 Reversion about an arbitrary admissible core

**Theorem 1.29** (Reversion about an arbitrary core). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and  $\Lambda \in R$  invertible. Consider*

$$\Lambda E + h(E) + f(t, E) = 0, \quad h \in u^2 R[[u]], \quad f \in tR[[t, u]]. \quad (1.55)$$

*Then (1.55) has a unique solution  $E = \sum_{n \geq 1} e_n t^n \in tR[[t]]$ ; it satisfies the triangular recurrence*

$$e_n = -\frac{1}{\Lambda} [t^n] \left[ h(E_{<n}) + f(t, E_{<n}) \right], \quad E_{<n} := \sum_{j=1}^{n-1} e_j t^j, \quad (1.56)$$

*and the closed formula*

$$e_n = \sum_{m=1}^{2n-1} \frac{1}{m} [t^n u^{m-1}] \left( -\Lambda^{-1} [h(u) + f(t, u)] \right)^m. \quad (1.57)$$

*If moreover  $h(u) = \sum_{m \geq 2} h_m u^m$  and  $f(t, u) = \sum_{k \geq 1} \phi_k(t) (1 + u)^{-\mu_k}$  with  $\phi_k \in tR[[t]]$  and  $\mu_k \in \mathbb{N}$ , then*

$$e_n = -\frac{1}{\Lambda} \left[ \sum_{m=2}^n h_m \widehat{B}_{n,m}(e) + \sum_{k \geq 1} \sum_{i=1}^n ([t^i] \phi_k) \sum_{m=0}^{n-i} (-1)^m \binom{\mu_k + m - 1}{m} \widehat{B}_{n-i,m}(e) \right], \quad (1.58)$$

*in which only  $e_1, \dots, e_{n-1}$  occur.*

*Proof.* Existence, uniqueness and (1.57) are lemma 1.17 applied to  $\Psi := -\Lambda^{-1}[h + f]$ . For (1.56), note as in the proof of that lemma that  $e_n$  occurs neither in  $[t^n] h(E)$  nor in  $[t^n] f(t, E)$ , so both may be computed from  $E_{<n}$ . For (1.58) apply definition 1.9 to  $h$  and (1.17) to each  $(1 + E)^{-\mu_k}$ , then convolve with  $\phi_k$ ; the displayed ranges are those outside which the ordinary Bell polynomials vanish.  $\square$

*Remark 1.30* (How the chapters instantiate it). For the linear-logarithmic core the shorter route of theorem 1.25 is available, because measuring the displacement in units of  $X$  makes the entire perturbation divisible by  $t$ . Part 3 needs the general form: there  $\Lambda$  is the normalized core slope  $\log X$  of proposition 1.23,  $h(E) = (1 + E) \log(1 + E) - E$  and  $f(t, E) = \sum_{k \geq 1} \alpha_k t^{2k} (1 + E)^{-(2k-1)}$ , and (1.58) is exactly the Bell recurrence of `double_factorial_transseries-2`. The coefficient ring there is  $\mathbb{Q}[\Lambda^{-1}]$ : the  $e_n$  are *not* constants but polynomials in  $\Lambda^{-1}$ , of degree at most  $2n - 1$  and with no constant term.

### 1.5.3 The perturbation-operator form

**Proposition 1.31** (Inverse perturbation operator). *Let  $\Phi_0$  be a phase with  $\Phi'_0$  nonvanishing near the point considered, let  $X$  solve  $\Phi_0(X) = L$ , and consider*

$$L = \Phi_0(x) + \varepsilon \varrho(x). \quad (1.59)$$

Put  $\mathcal{D}_0 := \Phi'_0(X)^{-1} d/dX$ . Then, as a formal power series in  $\varepsilon$ ,

$$x = X + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} \mathcal{D}_0^{m-1} \left[ \frac{\varrho(X)^m}{\Phi'_0(X)} \right]. \quad (1.60)$$

*Proof.* Change coordinate to  $\lambda := \Phi_0(x)$ , so that  $x = \Phi_0^{-1}(\lambda)$  and (1.59) becomes  $L = \lambda + \varepsilon \varrho_0(\lambda)$  with  $\varrho_0 := \varrho \circ \Phi_0^{-1}$ . This is a near-identity reversion in  $\lambda$  with small parameter  $\varepsilon$ ; Lagrange–Bürmann in the form (1.24), with the observable  $G = \Phi_0^{-1}$ , gives

$$\Phi_0^{-1}(\lambda) = \Phi_0^{-1}(L) + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} \left( \frac{d}{dL} \right)^{m-1} \left[ (\Phi_0^{-1})'(L) \varrho_0(L)^m \right].$$

Finally  $(\Phi_0^{-1})'(L) = 1/\Phi'_0(X)$ ,  $\varrho_0(L) = \varrho(X)$  and  $d/dL = \mathcal{D}_0$  by the chain rule.  $\square$

*Remark 1.32* (What the operator form is for). This is the form in which an *exponentially small* forward correction is transported through the inverse map: with  $\varrho$  a Stokes discontinuity or a terminant remainder rather than a power series,  $-\varepsilon \varrho(X)/\Phi'_0(X)$  is the leading displacement and the higher terms generate the nonlinear sectors. As a statement about a formal series in  $\varepsilon$  it is exact; as an asymptotic statement about a particular  $\varrho$  it needs the hypotheses of theorem 1.44. Expanding  $\varrho$  in inverse powers of  $X$  reproduces (1.48).

## 1.6 Flattening the core

Theorem 1.25 hides the explicit scale of the answer inside  $X$ . Unfolding  $X$  produces the polynomial–logarithmic transseries in which the four chapters state their headline formulae. Throughout this section  $H$  is an indeterminate; in the analytic application it is specialized to  $\log(L/a)$ .

**Theorem 1.33** (Direct polynomial–logarithmic reversion). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra,  $a \in R$  invertible,  $b \in R$ ,  $Q = \sum_{j \geq 1} q_j t^j \in tR[[t]]$ , and let  $H$  be an indeterminate. Set*

$$\Phi(t, u; H) := -b \log(1 + t(u - bH)) - Q \left( \frac{at}{1 + t(u - bH)} \right) \in tR[H][[t, u]]. \quad (1.61)$$

1. *There is a unique  $V = \sum_{n \geq 1} P_n(H) t^n \in tR[H][[t]]$  with  $V = \Phi(t, V; H)$ , and*

$$P_n(H) = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Phi(t, u; H)^m. \quad (1.62)$$

2. *Triangular Bell recurrence. Put  $\gamma_1 := -bH$  and  $\gamma_r := P_{r-1}(H)$  for  $r \geq 2$ . Then*

$$P_n(H) = -b \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma) - \sum_{j=1}^n q_j a^j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma), \quad (1.63)$$

*in which only  $P_1, \dots, P_{n-1}$  occur.*



3. Shape.  $P_n \in R[H]$  with

$$\deg_H P_n \leq n, \quad [H^n] P_n = \frac{b^{n+1}}{n}, \quad P_n(0) = a^{n+1} c_n, \quad (1.64)$$

where  $c_n$  are the Lambert-centred coefficients of theorem 1.25. In particular  $\deg_H P_n = n$  exactly when  $b$  is not a zero divisor, and the coefficients of  $P_n$  lie in the  $\mathbb{Q}$ -subalgebra generated by  $a, a^{-1}, b, q_1, \dots, q_n$ .

4. Formal exactness and uniqueness. Put  $t = L^{-1}$  and  $H = \log(L/a)$ , and

$$x := \frac{1}{a} \left[ L - bH + \sum_{n \geq 1} \frac{P_n(H)}{L^n} \right]. \quad (1.65)$$

Then  $ax + b \log x + Q(1/x) = L$  as an identity of formal series in  $L^{-1}$  with polynomial coefficients in  $H$ . Conversely, (1.65) is the only expansion of the form  $x = a^{-1} [L - bH + \sum_{n \geq 1} \Pi_n(H) L^{-n}]$ , with  $\Pi_n \in R[H]$ , that solves the equation coefficientwise.

5. Analytic version. Under the hypotheses of theorem 1.25(5), with  $L = \log(y/C)$  and  $H = \log(L/a)$ , for every fixed  $N \geq 0$ ,

$$F^{-1}(y) = \frac{1}{a} \left[ L - bH + \sum_{n=1}^N \frac{P_n(H)}{L^n} \right] + \mathcal{O}\left(\frac{(1+H)^{N+1}}{L^{N+1}}\right) \quad (y \rightarrow \infty). \quad (1.66)$$

*Proof.* (1) Membership of  $\Phi$  in  $tR[H][[t, u]]$  is checked as in theorem 1.25(1), the extra additive term  $-bH$  inside  $1 + t(u - bH)$  being harmless because it is multiplied by  $t$ . Lemma 1.16 over the ring  $R[H]$  then gives existence, uniqueness and (1.62).

(2) Put  $w := t(V - bH) = \sum_{r \geq 1} \gamma_r t^r$  with  $\gamma$  as stated; apply (1.18) to  $\log(1+w)$  and (1.17) to  $(1+w)^{-j}$ , and convolve the latter with the factor  $(at)^j$ . Only  $\gamma_1, \dots, \gamma_{n-k+1}$ , that is  $H$  and  $P_1, \dots, P_{n-k}$ , occur.

(4) Set  $t = L^{-1}$ . With  $x$  as in (1.65),

$$x = \frac{1}{at} (1 + t(V - bH)) = \frac{1+w}{at}, \quad ax = \frac{1}{t} + (V - bH), \quad \log x = H + \log(1+w), \quad \frac{1}{x} = \frac{at}{1+w},$$

where  $\log x$  uses  $\log(1/(at)) = \log(L/a) = H$ . Hence

$$ax + b \log x + Q(1/x) - L = (V - bH) + b(H + \log(1+w)) + Q\left(\frac{at}{1+w}\right) = V - \Phi(t, V; H) = 0.$$

For uniqueness, substituting a general  $\Pi = \sum \Pi_n(H) t^n$  in place of  $V$  turns the equation into  $\Pi = \Phi(t, \Pi; H)$ , and lemma 1.16 has a unique solution.

(3) We show by strong induction that  $\deg_H P_n \leq n$  and, in addition, that  $\deg_H \gamma_r \leq r - 1$  for  $r \geq 2$ . The base is  $\gamma_1 = -bH$ , of degree 1. Assume  $\deg_H P_j \leq j$  for all  $j < n$ , so  $\deg_H \gamma_r = \deg_H P_{r-1} \leq r - 1$  for  $2 \leq r \leq n$ . A monomial of  $\widehat{B}_{n,k}(\gamma)$  is a product  $\gamma_{r_1} \cdots \gamma_{r_k}$  with  $r_1 + \cdots + r_k = n$  and  $r_\nu \geq 1$ , hence of  $H$ -degree at most  $\sum_\nu r_\nu = n$ ; so the first sum of (1.63) has  $H$ -degree at most  $n$ . In the second sum the ordinary Bell polynomial has first index  $n - j$  with  $j \geq 1$ , so its  $H$ -degree is at most  $n - 1$ . Hence  $\deg_H P_n \leq n$ , completing the induction.

A monomial of  $H$ -degree exactly  $n$  requires  $\deg_H \gamma_{r_\nu} = r_\nu$  for every  $\nu$ , which forces  $r_\nu = 1$  for all  $\nu$  because  $\deg_H \gamma_r \leq r - 1$  for  $r \geq 2$ . Thus  $k = n$ , the monomial is  $\gamma_1^n = (-bH)^n$ , and  $[H^n] \widehat{B}_{n,k}(\gamma) = 0$  for  $k < n$  while  $\widehat{B}_{n,n}(\gamma) = (-bH)^n$ . Only the term  $k = n$  of the first sum of (1.63) therefore survives, and

$$[H^n] P_n = -b \cdot \frac{(-1)^{n+1}}{n} \cdot (-b)^n = \frac{(-1)^{2n+2} b^{n+1}}{n} = \frac{b^{n+1}}{n}.$$

For the constant term put  $H = 0$  in (1.61):

$$\Phi(t, u; 0) = -b \log(1 + tu) - Q\left(\frac{at}{1 + tu}\right).$$

If  $U(t) = \sum c_n t^n$  solves  $U = \Psi(t, U)$  with  $\Psi$  of (1.47), put  $\tilde{V}(t) := a U(at)$ . Then  $t\tilde{V}(t) = (at)U(at)$ , so

$$\Phi(t, \tilde{V}(t); 0) = -b \log(1 + (at)U(at)) - Q\left(\frac{at}{1 + (at)U(at)}\right) = a \Psi(at, U(at)) = a U(at) = \tilde{V}(t),$$

so  $\tilde{V}$  solves the  $H = 0$  fixed-point equation; by uniqueness  $P_n(0) = [t^n] \tilde{V} = a \cdot a^n c_n = a^{n+1} c_n$ .

(5) Fix  $N$ . Put  $x_N$  for the truncation displayed in (1.66). Exactly as in the proof of theorem 1.25(5), the residual  $r_N := \log F(x_N) - \log y$  splits as  $g(t) + E_N$  with  $|E_N| \leq K_N x_N^{-N-1}$  and  $g$  analytic in  $t = 1/L$  with coefficients polynomial in  $H$ ; by (4) the Taylor coefficients of  $g$  vanish through order  $N$ , and the coefficient of  $t^{N+1}$  is a polynomial in  $H$  of degree at most  $N+1$ . Hence  $|r_N| = \mathcal{O}((1+H)^{N+1} L^{-N-1})$ . The slope bound  $(\log F)' \geq a/2$  and theorem 1.42 finish the proof, after noting  $L \asymp aX$  and  $H = \log(L/a)$  so that  $x_N \rightarrow \infty$ .  $\square$

**Theorem 1.34** (The pure Lambert block in closed form). *Let  $\begin{bmatrix} n \\ m \end{bmatrix}$  be the unsigned Stirling numbers of the first kind, so that the signed ones are  $(-1)^{n-m} \begin{bmatrix} n \\ m \end{bmatrix}$  and  $\log(1+z)^m = m! \sum_{n \geq m} (-1)^{n-m} \begin{bmatrix} n \\ m \end{bmatrix} z^n / n!$ . When  $Q = 0$  the polynomials of theorem 1.33 are*

$$P_n^{(0)}(H) = b^{n+1} \sum_{m=1}^n (-1)^{m+1} \frac{\begin{bmatrix} n \\ m \end{bmatrix}}{(n-m+1)!} H^{n-m+1} \quad (1.67)$$

for  $n \geq 1$ . Equivalently, clearing the binomial,

$$P_n^{(0)}(H) = \frac{b^{n+1}}{n!} \sum_{m=1}^n (-1)^{m+1} (m-1)! \begin{bmatrix} n \\ m \end{bmatrix} \binom{n}{m-1} H^{n-m+1}. \quad (1.68)$$

In particular

$$[H^n] P_n^{(0)} = \frac{b^{n+1}}{n}, \quad P_n^{(0)}(0) = 0, \quad (1.69)$$

and the first four are

$$P_1^{(0)} = b^2 H, \quad P_2^{(0)} = \frac{b^3}{2} (H^2 - 2H), \quad P_3^{(0)} = \frac{b^4}{6} (2H^3 - 9H^2 + 6H), \quad P_4^{(0)} = \frac{b^5}{12} (3H^4 - 22H^3 + 36H^2 - 12H). \quad (1.70)$$

Consequently the Lambert core itself unfolds as

$$X = \frac{1}{a} \left[ L - bH + \sum_{n \geq 1} \frac{P_n^{(0)}(H)}{L^n} \right], \quad H = \log \frac{L}{a}, \quad (1.71)$$

formally, and with remainder  $\mathcal{O}((1+H)^{N+1} L^{-N-1})$  after truncation at  $n = N$ .

*Proof.* With  $Q = 0$ , (1.62) reads

$$P_n^{(0)}(H) = \sum_{m=1}^n \frac{(-b)^m}{m} [t^n u^{m-1}] \log(1 + t(u - bH))^m.$$

Insert  $\log(1+z)^m = m! \sum_{r \geq m} (-1)^{r-m} \begin{bmatrix} r \\ m \end{bmatrix} z^r / r!$  with  $z = t(u - bH)$ ; the coefficient of  $t^n$  is  $m! (-1)^{n-m} \begin{bmatrix} n \\ m \end{bmatrix} (u - bH)^n / n!$ , and  $[u^{m-1}] (u - bH)^n = \binom{n}{m-1} (-bH)^{n-m+1}$ . Hence

$$P_n^{(0)}(H) = \sum_{m=1}^n \frac{(-b)^m}{m} \cdot \frac{m! (-1)^{n-m} \begin{bmatrix} n \\ m \end{bmatrix}}{n!} \binom{n}{m-1} (-b)^{n-m+1} H^{n-m+1},$$

and  $(-b)^m(-b)^{n-m+1}(-1)^{n-m} = (-1)^{n+1}(-1)^{n-m}b^{n+1} = (-1)^{m+1}b^{n+1}$ , which gives (1.68). Since  $(m-1)!\binom{n}{m-1}/n! = 1/(n-m+1)!$ , the two displayed forms agree and (1.67) follows. Every term carries  $H^{n-m+1}$  with  $m \leq n$ , so no constant term occurs and  $P_n^{(0)}(0) = 0$ ; the term  $m = 1$  has  $\binom{n}{1} = (n-1)!$  and gives  $[H^n] P_n^{(0)} = b^{n+1}(n-1)!/n! = b^{n+1}/n$ . Finally (1.71) is theorem 1.33(4) and (5) with  $Q = 0$ , in which case the unique solution of  $ax + b \log x = L$  is the core  $X$  of theorem 1.20.  $\square$

*Remark 1.35* (Three source formulae, one identity). The closed form (1.67) appears in the sources in three different dresses, all with the signed Stirling numbers  $s(n, m) = (-1)^{n-m}\binom{n}{m}$ . `Swing_Factorial_Transseries_Article` states (1.68); `Partition_Number_Transseries_and_Asymptotic_Inverse` states  $P_j(L) = 2^{j+1} \sum_{q \leq j} s(j, q) L^{j+1-q} / (j+1-q)!$ ; `Partition_Number_Transseries_and_Inverse` states  $P_m(\ell) = \frac{1}{m!} \sum_j 2^j (j-1)! s(m, j) \binom{m}{j-1} (2\ell)^{m-j+1}$ . The last two are the specialization  $a = 1, b = -2$  of (1.67), and all three agree with it as exact rational polynomials for  $1 \leq n \leq 8$ , which was checked in exact rational arithmetic (remark 1.48). The volume uses (1.67): it is the shortest of the four, it avoids the letter  $s$ , which Part 4 needs for Dedekind sums, and it makes the leading coefficient (1.69) a one-line consequence.

*Remark 1.36* (Both normalizations, and what is and is not claimed). The Lambert-centred expansion (1.52) and the flattened expansion (1.66) are two truncations of the same object: by (1.71), expanding  $X$  and then each power  $X^{-n}$  in the flattened scale returns (1.66), and conversely collecting in the flattened series all terms generated by  $b \log x$  alone rebuilds  $X$ . What differs is the coefficient ring: the  $c_n$  are constants, the  $P_n$  are polynomials of degree  $n$  in  $H = \log(L/a)$ . Several of the sources add that Lambert centring is numerically more accurate at equal truncation order and support this with tables. Part 0 makes no such claim: the comparison depends on the size of  $H$  relative to  $L$  and on the particular data, it is an empirical statement about specific truncations, and the supporting tables belong to the chapters, where they are reported with their provenance. What Part 0 does assert is the algebraic statement of remark 1.26, and the remainder bounds (1.52) and (1.66), whose right-hand sides differ by the factor  $(1 + H)^{N+1}$  — and that factor is the only rigorous difference between the two at fixed order.

## 1.7 The three inverses of a discrete sequence

**Definition 1.37** (The three inverse objects). Let  $(A_n)_{n \geq n_0}$  be real numbers, strictly increasing for  $n \geq n_1 \geq n_0$ , with  $A_n \rightarrow +\infty$ .

1. The *integer staircase*, or generalized inverse, is

$$N_*(y) := \min\{n \geq n_1 : A_n \geq y\}, \quad y \leq \sup_n A_n. \quad (1.72)$$

It is an integer-valued, nondecreasing step function, constant on each interval  $(A_{n-1}, A_n]$ .

2. Given a continuous, strictly increasing  $\mathcal{A} : [n_1, \infty) \rightarrow \mathbb{R}$  with  $\mathcal{A}(n) = A_n$  for every integer  $n \geq n_1$  — an *admissible interpolation* — the *interpolated inverse* is  $\nu_{\mathcal{A}} := \mathcal{A}^{-1}$ , defined on  $[A_{n_1}, \infty)$ .
3. The *asymptotic inverse* is the formal transseries produced by theorem 1.25 or theorem 1.33 from a model  $F$  for which the chapter in question proves an asymptotic identity along the sequence. It is attached to no particular interpolation.

*Remark 1.38* (The three are genuinely different). Object (1) is canonical but not smooth; object (2) is smooth but not canonical — distinct admissible interpolations agree at the integers and differ in between, hence have different inverses; object (3) is canonical as a formal series but determines a function only after a summation convention is fixed. The three coincide only in the following precise senses:  $N_* = \lceil \nu_{\mathcal{A}} \rceil$  for every admissible  $\mathcal{A}$  (theorem 1.39(1)), and  $\nu_{\mathcal{A}}(A_n) = n$  for every admissible  $\mathcal{A}$  and every integer  $n \geq n_1$ . The asymptotic series (3) is independent of the choice in  $\mathcal{A}$  at every algebraic order, but the exponentially small layers are not: the partition articles make this explicit for the Rademacher amplitudes, which are periodic in the index and therefore genuinely depend on how the sequence is continued off the lattice.

**Theorem 1.39** (Staircase recovery and the separation condition). *Notation as in definition 1.37, with  $\mathcal{A}$  admissible and  $\nu := \nu_{\mathcal{A}}$ .*

1. Exact rounding. For every  $y$  in the range of  $\mathcal{A}$ , that is for  $A_{n_1} \leq y < \sup_n A_n$ ,

$$N_*(y) = \lceil \nu(y) \rceil, \quad (1.73)$$

with  $\lceil m \rceil = m$  for integers  $m$ . The identity holds for every admissible interpolation.

2. Separation condition. Let  $x_* \in \mathbb{R}$  and  $\eta \geq 0$  satisfy  $|x_* - \nu(y)| \leq \eta$ . If

$$\lceil x_* - \eta \rceil = \lceil x_* + \eta \rceil, \quad (1.74)$$

then  $N_*(y)$  equals that common value. Condition (1.74) holds if and only if the interval  $[x_* - \eta, x_* + \eta]$  is contained in a single half-open interval  $(m - 1, m]$ ; it can fail for arbitrarily small  $\eta > 0$ , namely when  $\nu(y)$  is within  $\eta$  of an integer from above.

3. Inverse along the range. If  $y = A_n$  for an integer  $n \geq n_1$  and  $|x_* - \nu(y)| \leq \eta < \frac{1}{2}$ , then  $n = \lfloor x_* + \frac{1}{2} \rfloor$ . This case is unconditional: no separation hypothesis is needed, because  $\nu(A_n) = n$  exactly.
4. Residue classes. Suppose  $r \geq 1$  and that for each  $\rho \in \{0, 1, \dots, r - 1\}$  the subsequence  $(A_n)_{n \equiv \rho \pmod{r}}$  is strictly increasing for  $n \geq n_1$  and admits an admissible interpolation  $\mathcal{A}_\rho$  on the class, with inverse  $\nu_\rho$ . Then, for  $y$  large enough,

$$N_*(y) = \min_{0 \leq \rho < r} \left( \rho + r \left\lceil \frac{\nu_\rho(y) - \rho}{r} \right\rceil \right). \quad (1.75)$$

5. No uniform smooth representation. For every  $Y$  in the range and every continuous  $g : [Y, \infty) \rightarrow \mathbb{R}$ ,

$$\sup_{y \geq Y} |N_*(y) - g(y)| \geq \frac{1}{2}. \quad (1.76)$$

Equivalently, the  $\mathcal{O}(1)$  periodic layer  $\lceil x \rceil - x - \frac{1}{2} = \pi^{-1} \sum_{m \geq 1} m^{-1} \sin(2\pi m x)$  (valid for  $x \notin \mathbb{Z}$ ) is not a refinement that can be dropped: it is the leading discrete transmonomial of the generalized inverse.

*Proof.* (1) For an integer  $n \geq n_1$ ,  $A_n \geq y$  is equivalent to  $\mathcal{A}(n) \geq y$ , hence to  $n \geq \nu(y)$  by strict monotonicity of  $\mathcal{A}$ . The least such integer is  $\lceil \nu(y) \rceil$ .

(2) The function  $\lceil \cdot \rceil$  is constant on each  $(m - 1, m]$  and nondecreasing; since  $\nu(y) \in [x_* - \eta, x_* + \eta]$ , (1.74) forces  $\lceil \nu(y) \rceil$  to equal the common value, and (1) applies. The characterization of (1.74) and the failure mode are immediate from the same description of  $\lceil \cdot \rceil$ .

(3)  $\nu(A_n) = n$  because  $\mathcal{A}(n) = A_n$ ; hence  $|x_* - n| \leq \eta < 1/2$  and rounding to the nearest integer returns  $n$ .

(4) Fix  $\rho$ . Among the integers  $n \equiv \rho \pmod{r}$  with  $n \geq n_1$ , the condition  $A_n \geq y$  is, by (1) applied to the class, equivalent to  $n \geq \nu_\rho(y)$ . Writing  $n = \rho + rk$  with  $k \in \mathbb{Z}$ , the least admissible  $k$  is  $\lceil (\nu_\rho(y) - \rho)/r \rceil$ , so the least admissible  $n$  in the class is  $\rho + r \lceil (\nu_\rho(y) - \rho)/r \rceil$ . Taking the minimum over  $\rho$  gives the least admissible  $n$  without a congruence restriction, which is  $N_*(y)$ .

(5) Fix  $n$  with  $A_n > Y$ . On  $(A_{n-1}, A_n]$  we have  $N_* = n$  and on  $(A_n, A_{n+1}]$  we have  $N_* = n + 1$ . For  $\epsilon > 0$  small,

$$1 = |N_*(A_n + \epsilon) - N_*(A_n)| \leq |N_*(A_n + \epsilon) - g(A_n + \epsilon)| + |g(A_n + \epsilon) - g(A_n)| + |g(A_n) - N_*(A_n)|.$$

Letting  $\epsilon \downarrow 0$  and using continuity of  $g$  gives  $1 \leq 2 \sup_{y \geq Y} |N_* - g|$ . For the Fourier identity, the function  $x \mapsto \lceil x \rceil - x - \frac{1}{2}$  is 1-periodic, equal to  $-(x - \lfloor x \rfloor) + \frac{1}{2}$  shifted, i.e. minus the standard sawtooth, whose Fourier series is the displayed one and converges pointwise off  $\mathbb{Z}$ .  $\square$

**Proposition 1.40** (Certified discrete inversion). *In the setting of theorem 1.39(4), suppose that for each  $\rho$  an approximation  $x_\rho$  with  $|x_\rho - \nu_\rho(y)| \leq \eta_\rho$  is available. Then  $N_*(y)$  is determined by at most  $\sum_\rho (\lfloor 2\eta_\rho/r \rfloor + 2)$  exact evaluations of the sequence: for each  $\rho$  list the integers  $n \equiv \rho \pmod{r}$  in  $[x_\rho - \eta_\rho, x_\rho + \eta_\rho + r]$ , evaluate  $A_n$ , and take the least  $n$  over all classes with  $A_n \geq y$ .*

*Proof.* By theorem 1.39(4) the class- $\rho$  candidate is  $\rho + r \lceil (\nu_\rho(y) - \rho)/r \rceil$ , and  $\nu_\rho(y)$  lies in  $[x_\rho - \eta_\rho, x_\rho + \eta_\rho]$ ; the least integer of the class that is  $\geq \nu_\rho(y)$  therefore lies in  $[x_\rho - \eta_\rho, x_\rho + \eta_\rho + r]$ . Since the class subsequence is increasing, the test  $A_n \geq y$  identifies it, and the outer minimum over  $\rho$  is (1.75).  $\square$

**Remark 1.41** (The parity instances). Part 5 and Part 3 are the case  $r = 2$ . The formula  $\rho + r \lceil (\nu_\rho - \rho)/r \rceil$  specializes to  $2 \lceil \nu_0/2 \rceil$  and  $2 \lceil (\nu_1 - 1)/2 \rceil + 1$ , which is exactly the pair of parity candidates of `double_factorial_transseries-2`, and  $N_* = \min$  of the two is its “exact generalized inverse” theorem. `Swing_Factorial_Transseries_Article` states the corresponding *floor* formulae for the threshold query “the greatest index in the class  $\rho$  whose sequence value is at most  $y$ ”; those are  $\rho + r \lfloor (\nu_\rho - \rho)/r \rfloor$ , obtained from (1.75) by replacing  $\lceil \cdot \rceil$  with  $\lfloor \cdot \rfloor$  and  $\min$  with  $\max$ , with the same proof. The partition chapter is the case  $r = 1$ , where (1.75) reduces to (1.73).

**Source claim.** Several sources write, or invite the reader to write,  $N_*(y) = \nu(y) + o(1)$ . **Defect.** This is false uniformly in  $y$ , by theorem 1.39(5). `Butcher_Tree_Transseries-2` already flags it (“An expansion such as  $N_*(y) = \nu(y) + o(1)$  is false uniformly in  $y$ ”), and `Partition_Number_Transseries_and_Asymptotic_Inverse` likewise separates the three objects; but the qualification is absent from the summary statements of the other articles. **Correction.** The volume uses only the three statements theorem 1.39(1), (3), (4): an exact ceiling identity valid for every admissible interpolation, an unconditional nearest-integer recovery *along the range*  $y = A_n$ , and a residue-class refinement. Nothing else about  $N_*$  is asserted.

## 1.8 Error control, transport, and optimal truncation

### 1.8.1 Backward error certifies forward error

**Theorem 1.42** (Residual-to-root conversion). *Let  $I \subseteq \mathbb{R}$  be an interval and  $f : I \rightarrow \mathbb{R}$  differentiable with*

$$0 < m \leq f'(\xi) \leq M \quad (\xi \in I). \quad (1.77)$$

*Let  $x_\dagger \in I$  satisfy  $f(x_\dagger) = \lambda$ , let  $x_* \in I$ , and set the residual  $r_* := f(x_*) - \lambda$ . Then*

$$\frac{|r_*|}{M} \leq |x_* - x_\dagger| \leq \frac{|r_*|}{m}. \quad (1.78)$$

*Moreover, if  $f$  is continuous on  $I$  and  $[x_* - \varrho, x_* + \varrho] \subseteq I$  with  $\varrho := |r_*|/m$ , then  $f$  takes the value  $\lambda$  somewhere in that interval; the residual is therefore a certificate, not merely a bound conditional on the existence of a nearby root.*

*Proof.* By the mean value theorem,  $r_* = f(x_*) - f(x_\dagger) = f'(\xi)(x_* - x_\dagger)$  for some  $\xi$  between the two points, and (1.77) gives (1.78). For the certificate, suppose  $r_* > 0$  (the case  $r_* < 0$  is symmetric and  $r_* = 0$  is trivial). Then  $f(x_*) = \lambda + r_* > \lambda$ , while

$$f(x_* - \varrho) \leq f(x_*) - m\varrho = \lambda + r_* - |r_*| = \lambda,$$

using  $f' \geq m$ . The intermediate value theorem supplies a point of  $[x_* - \varrho, x_*]$  where  $f = \lambda$ .  $\square$

**Remark 1.43** (Use in the logarithmic variable). In every application  $f = \log F$  and  $\lambda = \log y$ :  $F$  itself overflows any fixed-precision arithmetic long before the asymptotic regime begins, whereas  $\log F$  is computable from a digamma or Laplace representation throughout. The lower bound in (1.78) makes the residual test sharp — a small residual is necessary as well as sufficient.

### 1.8.2 Transport of a controlled forward remainder

**Theorem 1.44** (Remainder transport). *Let  $I = [x_1, \infty)$ , let  $f_0 : I \rightarrow \mathbb{R}$  be differentiable with  $f'_0 \geq m > 0$ , and let  $\varepsilon : I \rightarrow \mathbb{R}$  be differentiable with  $\sup_I |\varepsilon'| \leq \theta < m$ . Put  $f := f_0 + \varepsilon$ . For  $\lambda$  in the common range let  $x_0 = x_0(\lambda) \in I$  solve  $f_0(x_0) = \lambda$  and  $x = x(\lambda) \in I$  solve  $f(x) = \lambda$ . Then*

1. Rigorous bound.

$$|x(\lambda) - x_0(\lambda)| \leq \frac{|\varepsilon(x_0(\lambda))|}{m - \theta}. \quad (1.79)$$

2. First-order law with explicit error. *If moreover  $f_0 \in C^2(I)$  with  $\sup_I |f''_0| \leq M_2$ , then*

$$x(\lambda) - x_0(\lambda) = -\frac{\varepsilon(x_0)}{f'_0(x_0)} + E, \quad |E| \leq \frac{|\varepsilon(x_0)|}{m(m - \theta)} \left( \frac{M_2 |\varepsilon(x_0)|}{m - \theta} + \theta \right). \quad (1.80)$$

*In particular, if  $\varepsilon(x) = \mathcal{O}(x^{-M-1})$  and  $\varepsilon'(x) = o(1)$  as  $x \rightarrow \infty$ , then  $x(\lambda) - x_0(\lambda) = -\varepsilon(x_0)/f'_0(x_0) (1 + o(1)) = \mathcal{O}(x_0^{-M-1})$ .*

*Proof.* Since  $f' = f'_0 + \varepsilon' \geq m - \theta > 0$ ,  $f$  is strictly increasing and  $x(\lambda)$  is unique. The residual of  $x_0$  in  $f$  is  $f(x_0) - \lambda = f_0(x_0) + \varepsilon(x_0) - \lambda = \varepsilon(x_0)$ , so theorem 1.42 applied to  $f$  with  $m$  replaced by  $m - \theta$  gives (1.79).

For (2), write  $d := x - x_0$ . By the mean value theorem there is  $\xi$  between  $x_0$  and  $x$  with  $0 = f(x) - \lambda = \varepsilon(x_0) + f'(\xi)d$ , so  $d = -\varepsilon(x_0)/f'(\xi)$  and

$$d + \frac{\varepsilon(x_0)}{f'_0(x_0)} = \varepsilon(x_0) \left( \frac{1}{f'_0(x_0)} - \frac{1}{f'(\xi)} \right) = \varepsilon(x_0) \frac{f'(\xi) - f'_0(x_0)}{f'_0(x_0) f'(\xi)}.$$

Now  $|f'(\xi) - f'_0(x_0)| \leq |f'_0(\xi) - f'_0(x_0)| + |\varepsilon'(\xi)| \leq M_2|d| + \theta \leq M_2|\varepsilon(x_0)|/(m - \theta) + \theta$  by (1.79), while  $f'_0(x_0) \geq m$  and  $f'(\xi) \geq m - \theta$ . Substituting gives (1.80). The final assertion follows since then  $\theta \rightarrow 0$  and  $|\varepsilon(x_0)| \rightarrow 0$ .  $\square$

**Corollary 1.45** (Forward truncation to inverse accuracy). *Let  $F$  be differentiable of exponential–power type with data  $(C, a, 1, b, 1, Q)$ , strictly increasing on  $[x_1, \infty)$  with  $(\log F)' \geq m > 0$  there. Let  $x_*$  solve exactly the truncated equation*

$$\log C + ax_* + b \log x_* + \sum_{j=1}^M q_j x_*^{-j} = \log y.$$

*Then  $|x_* - F^{-1}(y)| \leq K_M x_*^{-M-1}/m$  with  $K_M$  the constant of (1.3).*

*Proof.* The residual of  $x_*$  in  $\log F$  is exactly the forward remainder, of modulus at most  $K_M x_*^{-M-1}$ ; apply theorem 1.42.  $\square$

### 1.8.3 Least-term truncation and the beyond-all-orders floor

**Theorem 1.46** (Optimal truncation and the exponential floor). *Let  $A > 0$ ,  $\beta \in \mathbb{R}$ ,  $K > 0$ ,  $x_0 \geq 1$ , and let  $\varphi : [x_0, \infty) \rightarrow \mathbb{R}$  satisfy the Gevrey-1 remainder bound*

$$\left| \varphi(x) - \sum_{j=1}^M q_j x^{-j} \right| \leq K \Gamma(M + \beta) A^{-M} x^{-M} \quad (M \geq 1, x \geq x_0). \quad (1.81)$$

1. Least-term envelope. *There are  $x_1 \geq x_0$  and  $K' = K'(K, A, \beta)$  such that, choosing  $M = M(x) := \lfloor Ax \rfloor$ ,*

$$\left| \varphi(x) - \sum_{j=1}^{M(x)} q_j x^{-j} \right| \leq K' x^{\beta-1/2} e^{-Ax} \quad (x \geq x_1). \quad (1.82)$$

2. Transported inverse error. Let  $F$  be as in corollary 1.45, with  $\log F(x) = \log C + ax + b \log x + \varphi(x)$  and  $(\log F)' \geq m > 0$  on  $[x_1, \infty)$ . If  $x_*$  solves the optimally truncated equation exactly, then

$$|x_* - F^{-1}(y)| \leq \frac{K'}{m} x_*^{\beta-1/2} e^{-Ax_*}. \quad (1.83)$$

3. The floor is real. Suppose in addition that there are  $M_0 \geq 1$  and  $0 < k \leq K''$  with

$$k \Gamma(M + \beta) A^{-M} \leq |q_M| \leq K'' \Gamma(M + \beta) A^{-M} \quad (M \geq M_0). \quad (1.84)$$

Then

$$\log \min_{M \geq M_0} |q_M| x^{-M} = -Ax + \mathcal{O}(\log x) \quad (x \rightarrow \infty), \quad (1.85)$$

so that no truncation of the algebraic series achieves an error of smaller exponential order than  $e^{-Ax}$ .

*Proof.* (1) Stirling's formula gives constants  $C_\beta$  and  $M_\beta \geq 1$  with  $\Gamma(M + \beta) \leq C_\beta M^{M+\beta-1/2} e^{-M}$  for all  $M \geq M_\beta$ . With  $M = \lfloor Ax \rfloor$  and  $x$  so large that  $M \geq M_\beta$ ,

$$\Gamma(M + \beta) A^{-M} x^{-M} \leq C_\beta M^{\beta-1/2} e^{-M} \left( \frac{M}{Ax} \right)^M \leq C_\beta M^{\beta-1/2} e^{-M},$$

because  $M \leq Ax$ . Since  $M > Ax - 1$ , we have  $e^{-M} < e \cdot e^{-Ax}$ , and  $M^{\beta-1/2} \leq C'_\beta x^{\beta-1/2}$  for  $x \geq x_1$  (with  $C'_\beta$  depending only on  $A$  and  $\beta$ , since  $Ax - 1 \leq M \leq Ax$ ). Combining with (1.81) gives (1.82).

(2) The residual of  $x_*$  in  $\log F$  is exactly the left-hand side of (1.82) evaluated at  $x_*$ ; apply theorem 1.42.

(3) Enlarging  $M_0$  we may assume  $M_0 \geq M_\beta$ . Write  $u := Ax$  and  $g(M) := (M/u)^M e^{-M}$ , so that by Stirling there are  $0 < c_\beta \leq C_\beta$  with

$$c_\beta M^{\beta-1/2} g(M) \leq \Gamma(M + \beta) (Ax)^{-M} \leq C_\beta M^{\beta-1/2} g(M) \quad (M \geq M_\beta). \quad (1.86)$$

*Upper bound.* Take  $M = \lfloor Ax \rfloor$  in the right-hand inequality of (1.84) and repeat the estimate of part (1); this gives  $\min_{M \geq M_0} |q_M| x^{-M} \leq K''' x^{\beta-1/2} e^{-Ax}$  once  $\lfloor Ax \rfloor \geq M_0$ .

*Lower bound.* Note first that  $\log g(M) = M \log(M/u) - M$  has derivative  $\log(M/u)$ , so  $g$  decreases on  $(0, u)$ , increases on  $(u, \infty)$ , and

$$g(M) \geq g(u) = e^{-u} = e^{-Ax} \quad (M > 0). \quad (1.87)$$

Set  $h(M) := M^{\beta-1/2} g(M)$ ; then  $(\log h)'(M) = (\beta - \frac{1}{2})/M + \log(M/u)$ , which is positive as soon as  $M \geq 2u$  and  $2u \geq 2|\beta - \frac{1}{2}|/\log 2$ , so  $h$  is increasing on  $[2u, \infty)$  for all large  $x$ . Consequently

$$\min_{M \geq M_0} h(M) = \min_{M_0 \leq M \leq 2u} h(M) \geq \left( \min_{M_0 \leq M \leq 2u} M^{\beta-1/2} \right) e^{-Ax} \geq c x^{-|\beta-1/2|} e^{-Ax}$$

for  $x \geq x_1$ , using (1.87) and  $\min_{M_0 \leq M \leq 2u} M^{\beta-1/2} \geq \min(M_0^{\beta-1/2}, (2Ax)^{\beta-1/2})$ . Combining with the left-hand inequalities of (1.84) and (1.86),  $|q_M| x^{-M} \geq k c_\beta c x^{-|\beta-1/2|} e^{-Ax}$  for every  $M \geq M_0$ . Taking logarithms of the two bounds gives (1.85).  $\square$

*Remark 1.47* (What may and may not be appended). Theorem 1.46 bounds the size of the optimal remainder; it does not produce it. In particular, adding a term  $c e^{-Ax}$  with an arbitrarily chosen constant  $c$  to the divergent series is *not* a valid transseries completion. The double factorial and swing sources make the point explicitly: on the positive real axis the Borel singularities of these phases lie at  $\pm i\pi m$ , so the exponentially small sectors are terminant-weighted combinations of  $e^{\mp i\pi m x}$ , whose optimally truncated magnitude is  $e^{-\pi x}$  but which are not real exponentials. The unambiguous completion on the positive axis is the exact Borel–Laplace representation, when the chapter provides one. Where a chapter does supply a rigorously normalized exponential sector, its transport through the inverse map is proposition 1.31 at first order and theorem 1.44 with an explicit error.

Table 1.1: Model data of the four chapters, in the normal form of definition 1.2 and after the  $\alpha$ -reduction of lemma 1.4. “Core” names the admissible core of definition 1.8 and the Lambert branch selected by corollary 1.21. The row for Part 3 is *not* an instance of definition 1.2; see the repair box in section 1.2.

| Chapter | sequence                  | variable $x$       | $\alpha$ | $b$        | core, branch                            |
|---------|---------------------------|--------------------|----------|------------|-----------------------------------------|
| 2       | rooted trees, A000081     | $n$                | 1        | $-3/2$     | $ax + b \log x, W_{-1}$                 |
| 3       | double factorial, A006882 | $n + 1$            | —        | —          | $\frac{x}{2}(\log x - 1), W_0$          |
| 4       | partitions, A000041       | $n - \frac{1}{24}$ | $1/2$    | $-1$       | $ax^{1/2} + b \log x, W_{-1}$           |
| 5       | swing factorial, A056040  | $n$                | 1        | $\sigma/2$ | $ax + b \log x, W_0 \text{ or } W_{-1}$ |

## 1.9 The parameter dictionary

The constants  $a$  are:  $a = \log \alpha_{\text{Otter}}$  for Part 2;  $a = \pi\sqrt{2/3}$  for Part 4, in which  $\alpha = 1/2$  and lemma 1.4 substitutes  $\xi = (n - 1/24)^{1/2}$ , after which  $b/\alpha = -2$  and the core equation is  $\xi \mapsto a\xi - 2 \log \xi$  — the equation the three partition articles write as  $\rho - 2 \log \rho = L$  after rescaling  $a$  to 1; and  $a = \log 2$  for Part 5, with  $\sigma = (-1)^{n+1}$  the parity transmonomial. Each chapter verifies (1.4) directly and states its own  $q_j$ .

*Remark 1.48* (Reproducibility of the Part 0 identities). Every coefficient identity of this chapter was checked in exact rational arithmetic, with no floating-point step, over six independent choices of the data  $(a, b, q_1, \dots, q_N)$  with  $a, b, q_j \in \mathbb{Q}$ ,  $N \in \{10, 12\}$ , using a truncated-series engine over  $\mathbb{Q}[H]$  built on Python’s `Fractions.Fraction`. The identities checked were: the Lagrange closed formula (1.48) against the defining fixed point; the ordinary-Bell recurrence (1.49); the vanishing of the residual  $ax + b \log x + Q(1/x) - L$  under the flattened ansatz (1.65) through order  $L^{-N}$ ; the shape statements (1.64); and the closed form (1.67) together with its agreement with the three source formulae listed in remark 1.35. All checks passed. No numerical constant is asserted anywhere in this chapter.



## Chapter 2

# Butcher–Pólya rooted trees (A000081)

This chapter merges the three independently written treatments *Butcher\_Tree\_Transseries*, *Butcher\_Tree\_Transseries-2* and *Butcher\_Tree\_Transseries-3* of the special-function-inversion draft set. Below they are called S1, S2 and S3. S2 is a book-length treatment and is the most complete on the inverse and on the inherited singularity at  $-\sqrt{\rho}$ ; S3 is the most careful about what is and is not proved beyond the dominant sector; S1 has the shortest route to the universal Cayley coefficients and the only treatment of the compositional inverse of the generating function. Where they disagree the disagreement is stated in the text. Every numerical constant retained here has been recomputed from the exact divisor recurrence at 60–110 working decimal digits; see section 2.11.3.

The subject of this chapter is the sequence

$$0, 1, 1, 2, 4, 9, 20, 48, 115, 286, 719, 1842, 4766, 12486, 32973, 87811, \dots \quad (2.1)$$

(OEIS A000081), where for  $n \geq 1$  the term  $a_n$  counts isomorphism classes of rooted, unlabelled, non-plane trees on  $n$  vertices. These are the *Pólya trees*; in numerical analysis the same shapes index Butcher series, elementary differentials and Runge–Kutta order conditions, which is the origin of the name *Butcher tree numbers*. We set  $a_0 = 0$  throughout.

The sequence grows like  $C\alpha^n n^{-3/2}$  with  $\alpha = \rho^{-1}$  and  $\rho$  the Otter singularity. Thus, in the language of definition 1.2, its asymptotic interpolation is an exponential–power model with

|                              |                                               |                                    |               |
|------------------------------|-----------------------------------------------|------------------------------------|---------------|
| $\alpha_{\text{model}} = 1,$ | $a_{\text{model}} = \lambda := \log(1/\rho),$ | $b_{\text{model}} = -\frac{3}{2},$ | $\delta = 1.$ |
|------------------------------|-----------------------------------------------|------------------------------------|---------------|

(2.2)

This chapter is therefore the *linear-phase* instance of the Part 0 apparatus: no  $\alpha$ -reduction (lemma 1.4) is needed, the Lambert core (theorem 1.20) applies verbatim, and the entire reversion is a substitution into theorem 1.25. Everything specific to rooted trees is the computation of the model’s data  $C$ ,  $\lambda$  and  $Q$ ; that computation is the substance of sections 2.1 to 2.5, and the inversion in section 2.7 is short precisely because Part 0 has already been proved.

## 2.1 The functional equation and the exact recurrence

### 2.1.1 Pólya’s multiset equation

Let  $\mathcal{T}$  be the class of unlabelled non-plane rooted trees and let  $\mathcal{Z}$  be one atom (one vertex). Deleting the root of a tree leaves an unordered multiset of rooted trees, and grafting a multiset of rooted trees onto a fresh root inverts this. Hence

$$\mathcal{T} = \mathcal{Z} \times \text{MSET}(\mathcal{T}). \quad (2.3)$$

**Proposition 2.1** (Pólya’s equation and the Euler product). *Let  $T(z) = \sum_{n \geq 1} a_n z^n \in \mathbb{Z}[[z]]$ . Then, as identities of formal power series,*

$$T(z) = z \prod_{r \geq 1} (1 - z^r)^{-a_r}, \quad (2.4)$$

$$T(z) = z \exp \left( \sum_{k \geq 1} \frac{T(z^k)}{k} \right). \quad (2.5)$$

*Proof.* A multiset of rooted trees that contains  $m_j$  copies of the  $j$ th shape has size  $\sum_j m_j |j|$  and contributes  $z^{\sum_j m_j |j|}$ . For one fixed shape of size  $r$ , unbounded multiplicity contributes  $\sum_{m \geq 0} z^{rm} = (1 - z^r)^{-1}$ ; there are  $a_r$  shapes of size  $r$ , and distinct shapes are chosen independently, so the forest hanging from the root has generating function  $\prod_{r \geq 1} (1 - z^r)^{-a_r}$ . Every factor is a unit of  $\mathbb{Z}[[z]]$  and the product is  $z$ -adically convergent because the  $r$ th factor is  $1 + \mathcal{O}(z^r)$ . Multiplying by the root gives eq. (2.4). Taking the formal logarithm and using  $-\log(1 - w) = \sum_{k \geq 1} w^k/k$ ,

$$\log \frac{T(z)}{z} = \sum_{r \geq 1} a_r \sum_{k \geq 1} \frac{z^{rk}}{k} = \sum_{k \geq 1} \frac{1}{k} \sum_{r \geq 1} a_r (z^k)^r = \sum_{k \geq 1} \frac{T(z^k)}{k},$$

the rearrangement being legitimate because for each fixed power of  $z$  only finitely many pairs  $(r, k)$  contribute. Exponentiating gives eq. (2.5).  $\square$

*Remark 2.2.* Equation (2.5) is purely formal; no analytic hypothesis is used, and none is available at this stage. All three siblings state it correctly, but only S2 says explicitly that the identity holds in  $\mathbb{Z}[[z]]$  before any radius of convergence has been produced. We keep that discipline: section 2.2 is the first place where analysis enters.

### 2.1.2 The divisor recurrence

**Definition 2.3** (Divisor transform). For  $m \geq 1$  put

$$b_m := \sum_{d|m} d a_d. \quad (2.6)$$

**Theorem 2.4** (Otter–Pólya recurrence). *With  $a_1 = 1$  one has, for every  $n \geq 2$ ,*

$$(n - 1) a_n = \sum_{j=1}^{n-1} b_j a_{n-j}. \quad (2.7)$$

*Proof.* Write  $G(z) = T(z)/z = \sum_{n \geq 0} a_{n+1} z^n$ , a unit of  $\mathbb{Q}[[z]]$  with  $G(0) = 1$ . From eq. (2.4) and the computation in the proof of proposition 2.1,

$$\log G(z) = \sum_{r \geq 1} a_r \sum_{k \geq 1} \frac{z^{rk}}{k} = \sum_{m \geq 1} \frac{z^m}{m} \sum_{d|m} d a_d = \sum_{m \geq 1} \frac{b_m}{m} z^m,$$

where the inner rearrangement groups the pairs  $(r, k)$  with  $rk = m$  and uses  $1/k = d/m$  for  $d = r$ . Formal logarithmic differentiation gives  $G'(z) = G(z) \sum_{m \geq 1} b_m z^{m-1}$ . Extracting  $[z^{n-1}]$ ,

$$n a_{n+1} = \sum_{j=1}^n b_j a_{n-j+1},$$

and replacing  $n$  by  $n - 1$  yields eq. (2.7).  $\square$

*Remark 2.5* (Why the recurrence is triangular, and its cost). Although  $b_n$  contains the summand  $n a_n$ , the right-hand side of eq. (2.7) uses only  $b_j$  with  $j < n$ , so  $a_n$  is determined by  $a_1, \dots, a_{n-1}$ . Implemented as a sieve – after computing  $a_n$ , add  $n a_n$  to  $b_n, b_{2n}, b_{3n}, \dots$  – the divisor transform costs  $\mathcal{O}(N \log N)$  additions and the convolutions cost  $\mathcal{O}(N^2)$  integer operations for  $a_1, \dots, a_N$ . The integrality assertion  $(n-1) \mid \sum_j b_j a_{n-j}$  is a free exact self-test on every step, and was used as such in the computations of section 2.11.3.

### 2.1.3 The self-interaction and its coefficients

Separating the  $k = 1$  term of eq. (2.5) is the structural move on which everything else rests.

**Definition 2.6** (Disturbance and Cayley argument). Put

$$R(z) := \sum_{k \geq 2} \frac{T(z^k)}{k}, \quad \zeta(z) := z e^{R(z)}. \quad (2.8)$$

With this notation eq. (2.5) reads

$$T(z) = \zeta(z) e^{T(z)}. \quad (2.9)$$

**Lemma 2.7** (Coefficients of the disturbance).  $R(z) = \sum_{r \geq 2} \varrho_r z^r$  with

$$\varrho_r = \frac{1}{r} \sum_{\substack{d \mid r \\ d < r}} d a_d = \frac{b_r - r a_r}{r} \in \mathbb{Q}_{\geq 0}. \quad (2.10)$$

In particular  $\varrho_r$  depends only on  $a_d$  with  $d \leq r/2$ , and  $\varrho_2 = \frac{1}{2}$ ,  $\varrho_3 = \frac{1}{3}$ ,  $\varrho_4 = \frac{3}{4}$ ,  $\varrho_5 = \frac{1}{5}$ ,  $\varrho_6 = \frac{3}{2}$ ,  $\varrho_7 = \frac{1}{7}$ ,  $\varrho_8 = \frac{19}{8}$ .

*Proof.*  $R(z) = \sum_{k \geq 2} \sum_{d \geq 1} \frac{a_d}{k} z^{kd}$ ; collecting the terms with  $kd = r$  gives  $\varrho_r = \sum_{d \mid r, d < r} a_d / (r/d) = \frac{1}{r} \sum_{d \mid r, d < r} d a_d$ , which is  $(b_r - r a_r) / r$  by eq. (2.6). Every divisor  $d < r$  of  $r$  satisfies  $d \leq r/2$ . The listed values follow from  $a_1 = a_2 = 1$ ,  $a_3 = 2$ ,  $a_4 = 4$ : e.g.  $\varrho_4 = (1 \cdot 1 + 2 \cdot 1) / 4 = \frac{3}{4}$  and  $\varrho_6 = (1 + 2 \cdot 1 + 3 \cdot 2) / 6 = \frac{3}{2}$ .  $\square$

Lemma 2.7 is due to S2; S1 and S3 carry  $R$  only through the double sum  $\sum_{k \geq 2} \sum_n a_n \rho^{kn}(\dots)$ . The single-sum form is used throughout this chapter because it turns every quantity below into one geometrically convergent series in the exact integers  $a_r$  (lemma 2.20).

## 2.2 The Cayley–Lambert normal form and the Otter singularity

### 2.2.1 Factorisation through the Cayley tree function

**Definition 2.8** (Cayley tree function). Let  $\mathcal{C}(w)$  be the unique formal solution of  $\mathcal{C} = w e^{\mathcal{C}}$  with  $\mathcal{C}(0) = 0$ . It is analytic on  $|w| < e^{-1}$  and

$$\mathcal{C}(w) = -W_0(-w), \quad (2.11)$$

where  $W_0$  is the principal Lambert branch.

Comparing eq. (2.9) with definition 2.8,

$$T(z) = \mathcal{C}(\zeta(z)) = -W_0(-\zeta(z)). \quad (2.12)$$

This holds as formal series, and as an identity of analytic functions on the common domain of the branches selected at the origin. The point of the factorisation is a clean separation of labour: *all* of the singular mechanism is the universal square-root branch point of  $\mathcal{C}$  at  $w = e^{-1}$ , and *all* of the tree-specific information is packed into the map  $\zeta$ , which – as we now show – is analytic on a strictly larger disc than  $T$  itself.

**Lemma 2.9** (Elementary bounds on the radius). *The radius of convergence  $\rho$  of  $T$  satisfies  $\frac{1}{4} \leq \rho \leq e^{-1}$ .*

*Proof.* Since  $a_n \geq 1$  for  $n \geq 1$  we have  $\rho \leq 1$ , so for  $0 < x < \rho$  every  $x^k$  with  $k \geq 2$  lies in  $(0, \rho)$  as well. Upper bound: for such  $x$  the series  $T(x)$  and  $R(x)$  converge, all terms are nonnegative, and  $\zeta(x) = xe^{R(x)} \geq x$ ; so eq. (2.9) gives  $T(x) \geq xe^{T(x)}$ , i.e.  $x \leq T(x)e^{-T(x)} \leq \sup_{u \geq 0} ue^{-u} = e^{-1}$ . Letting  $x \uparrow \rho$  gives  $\rho \leq e^{-1}$ . Lower bound: forgetting the identification of isomorphic children embeds the shapes counted by  $a_n$  into the rooted *plane* trees on  $n$  vertices, of which there are  $\frac{1}{n} \binom{2n-2}{n-1} \leq 4^n$ ; hence  $\limsup a_n^{1/n} \leq 4$  and  $\rho \geq \frac{1}{4}$ .  $\square$

**Proposition 2.10** (Analyticity of the disturbance).  *$R$ , and hence  $\zeta$ , is analytic on  $|z| < \sqrt{\rho}$ , and the series  $\sum_{k \geq 2} T(z^k)/k$  converges normally on compact subsets together with all its derivatives. In particular  $R$  and  $\zeta$  are analytic at  $z = \rho$ , since  $\rho < \sqrt{\rho}$  by lemma 2.9.*

*Proof.* Fix  $r < \sqrt{\rho}$ , so  $r^2 < \rho$ . Since  $T$  has radius  $\rho$  and  $T(w) = w + \mathcal{O}(w^2)$ , there are  $M > 0$  and  $r_0 \in (0, 1)$  with  $|T(w)| \leq M|w|$  for  $|w| \leq r_0$ . For  $|z| \leq r$ , only finitely many  $k$  have  $r^k > r_0$ , and for each such  $k$  we have  $r^k \leq r^2 < \rho$ , so  $T(z^k)$  is defined and bounded there. For the remaining  $k$ ,  $|T(z^k)/k| \leq Mr^k/k$ , which is summable. Differentiating  $j$  times multiplies the  $k$ th tail term by a factor  $\mathcal{O}(k^j r^{-j})$ , still summable. Weierstrass’ theorem gives analyticity and termwise differentiation.  $\square$

**Proposition 2.11** (Critical characterisation of  $\rho$ ). *With  $\rho \in [\frac{1}{4}, e^{-1}]$  the radius of convergence of  $T$ ,*

$$T(\rho) = 1, \quad \zeta(\rho) = e^{-1}, \quad \log \rho + 1 + R(\rho) = 0. \quad (2.13)$$

*Moreover  $\zeta$  is strictly increasing on  $(0, \sqrt{\rho})$  and*

$$\theta_1 := e\rho\zeta'(\rho) = 1 + \rho R'(\rho) > 0. \quad (2.14)$$

*Proof.* All coefficients of  $T$ , hence of  $R$  and of  $\zeta = ze^R$ , are nonnegative, and  $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \dots$  has positive linear coefficient; therefore  $\zeta'(x) > 0$  on  $(0, \sqrt{\rho})$ , and  $\zeta$  is strictly increasing there.

*Step 1:*  $T(x) < 1$  on  $[0, \rho)$ , and  $T = \mathcal{C} \circ \zeta$  there. On  $[0, \rho)$  the function  $T$  is continuous, increasing, with  $T(0) = 0$ , and eq. (2.9) gives  $\zeta(x) = T(x)e^{-T(x)} \leq e^{-1}$ . Let  $\rho_1 := \sup\{x \in [0, \rho) : T(x) < 1\}$ . On  $[0, \rho_1)$  we have  $\zeta(x) < e^{-1}$ , so  $\mathcal{C}$  is analytic at  $\zeta(x)$  and  $\mathcal{C} \circ \zeta$  is an analytic solution of eq. (2.9) agreeing with  $T$  at 0; by uniqueness of that germ and analytic continuation,  $T = \mathcal{C} \circ \zeta$  on  $[0, \rho_1)$ . If  $\rho_1 < \rho$  then  $T(\rho_1) = 1$  and  $\zeta(\rho_1) = e^{-1}$ ; but  $\zeta$  is strictly increasing, so  $\zeta(x) > e^{-1}$  for  $x \in (\rho_1, \rho)$ , contradicting  $\zeta(x) \leq e^{-1}$ . Hence  $\rho_1 = \rho$ .

*Step 2:*  $\zeta(\rho) = e^{-1}$ . By proposition 2.10 the function  $\zeta$  is analytic at  $\rho$ , and by Step 1 and monotonicity  $\zeta(\rho) = \lim_{x \uparrow \rho} \zeta(x) \leq e^{-1}$ . If the inequality were strict,  $\mathcal{C}$  would be analytic at  $\zeta(\rho)$  and  $\mathcal{C} \circ \zeta$  would be analytic in a full neighbourhood of  $\rho$  and would agree with  $T$  on  $[0, \rho)$ ; then  $T$  would continue analytically through  $\rho$ , contradicting Pringsheim’s theorem, which makes the positive point  $z = \rho$  a singularity of a series with nonnegative coefficients. Hence  $\zeta(\rho) = e^{-1}$  and, by definition 2.8,  $T(\rho) = \mathcal{C}(e^{-1}) = 1$ .

*Step 3.* Taking logarithms in  $\zeta(\rho) = \rho e^{R(\rho)} = e^{-1}$  gives the scalar equation; equivalently  $R(\rho) = \lambda - 1$  with  $\lambda = -\log \rho$ , and  $R(\rho) > 0$  independently re-proves  $\rho < e^{-1}$ . Finally  $\zeta' = (1/z + R')\zeta$ , so  $e\rho\zeta'(\rho) = e\zeta(\rho)(1 + \rho R'(\rho)) = 1 + \rho R'(\rho)$ , positive because  $R$  has nonnegative coefficients.  $\square$

**Proposition 2.12** (Uniqueness of the dominant singularity).  *$z = \rho$  is the only singularity of  $T$  on the circle  $|z| = \rho$ .*

*Proof.* Take  $z = \rho e^{i\vartheta}$  with  $\vartheta \neq 0$ . The series  $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \dots$  has nonnegative coefficients, so  $|\zeta(z)| \leq \zeta(\rho) = e^{-1}$ , with equality only if the three displayed nonzero terms  $z, \frac{1}{2}z^3, \frac{1}{3}z^4$  share one argument, i.e.  $e^{2i\vartheta} = e^{3i\vartheta} = 1$ , hence  $e^{i\vartheta} = 1$  and  $\vartheta \equiv 0$ . So  $|\zeta(z)| < e^{-1}$  strictly,  $\mathcal{C}$  is analytic at  $\zeta(z)$ , and  $\zeta$  is analytic at  $z$  because  $\rho < \sqrt{\rho}$  (proposition 2.10). Hence  $T = \mathcal{C} \circ \zeta$  is analytic at  $z$ .  $\square$

**Remark 2.13.** This is S3’s argument, and it is the only one of the three that is self-contained: S1 and S2 appeal to “positivity and aperiodicity” without exhibiting the aperiodicity witness. The three coefficients  $z, z^3, z^4$  of  $\zeta$  furnish that witness explicitly, because  $\gcd(3-1, 4-1) = 1$ .

### 2.2.2 The constants

**Convention 2.14** (Growth constants). Throughout the chapter

$$\alpha := \rho^{-1}, \quad \lambda := \log \alpha = -\log \rho, \quad p := \frac{3}{2}. \quad (2.15)$$

**Notation collision between the siblings.** S2 and S3 write  $\alpha = \rho^{-1}$  and  $\lambda = \log \alpha$ ; S1 writes exactly the opposite,  $\lambda = \rho^{-1}$  and  $\alpha = -\log \rho$ . The symbols therefore cross-denote between the drafts, and S1's  $\Omega_-/\alpha$ ,  $e^{\alpha n}$  and  $d_m$  formulas must be read with  $\alpha = \lambda_{\text{here}}$ . This volume adopts the two-to-one majority reading eq. (2.15); every formula imported from S1 has been translated.

Solving  $\log \rho + 1 + R(\rho) = 0$  from eq. (2.10) with 700 exact tree numbers at 110 working digits (see section 2.11.3) gives

$$\rho = 0.338321856899207695196112625717017053183774607532967795572303776 \dots, \quad (2.16)$$

$$\alpha = 2.955765285651994974714817524123194588375492304663596595350472478 \dots, \quad (2.17)$$

$$\lambda = 1.083757597244624660482711980896174279049662124202672234914427065 \dots, \quad (2.18)$$

$$R(\rho) = 0.083757597244624660482711980896174279049662124202672234914427065 \dots, \quad (2.19)$$

and, as an independent consistency check on the whole pipeline,

$$\lambda = 1 + R(\rho), \quad (2.20)$$

which is nothing but eq. (2.13) rearranged; the two computed values agree to all 70 digits carried. Two further constants are recorded here because they are quoted repeatedly:

$$\theta_1 = 1.216004561824048922157212619228187444 \dots, \quad \zeta'(\rho) = 1.322241142696930264451284889 \dots \quad (2.21)$$

## 2.3 The universal Cayley branch germ

The next result is universal: it depends on the Cayley function alone, not on rooted trees, and applies verbatim to *any* class satisfying  $T = \zeta e^T$  with analytic  $\zeta$ .

**Definition 2.15** (Cayley kernel). Put

$$\Phi(v) := \frac{2(1 - (1 - v)e^v)}{v^2} = \sum_{m \geq 0} \frac{2(m+1)}{(m+2)!} v^m = \sum_{m \geq 0} \frac{2}{(m+2)m!} v^m = 1 + \frac{2}{3}v + \frac{1}{4}v^2 + \frac{1}{15}v^3 + \dots \quad (2.22)$$

The two displayed coefficient forms agree because  $2(m+1)/(m+2)! = 2/((m+2)m!)$ ; S1 and S3 use the first, S2 the second.

**Theorem 2.16** (Universal Cayley coefficients). Set  $w = (1 - v)/e$  and  $\mathcal{C}(w) = 1 - v$  on the branch reached from  $0 < w < e^{-1}$ . Then  $v = \frac{1}{2}v^2\Phi(v)$ , the inverse germ is  $v = \sum_{m \geq 1} \omega_m(2v)^{m/2}$ , and for every  $m \geq 1$

$$\omega_m = \frac{1}{m} [v^{m-1}] \Phi(v)^{-m/2} = \frac{1}{m} \sum_{k=0}^{m-1} \binom{-m/2}{k} \hat{B}_{m-1,k} \left( \frac{2}{3}, \frac{1}{4}, \frac{1}{15}, \dots \right) \in \mathbb{Q}. \quad (2.23)$$

Consequently, on that branch,

$$\mathcal{C}\left(\frac{1-v}{e}\right) = 1 + \sum_{r \geq 1} c_r v^{r/2}, \quad c_r = -2^{r/2} \omega_r. \quad (2.24)$$

*Proof.* From  $\mathcal{C} = we^{\mathcal{C}}$  we get  $ew = \mathcal{C}e^{1-\mathcal{C}}$ , so with  $\mathcal{C} = 1 - v$ ,

$$1 - v = ew = (1 - v)e^v, \quad \text{hence} \quad v = 1 - (1 - v)e^v = \sum_{m \geq 2} \frac{m-1}{m!} v^m = \frac{v^2}{2} \Phi(v),$$

the middle identity by expanding  $e^v$  and collecting. Put  $\varsigma = \sqrt{2v}$  with the determination  $\varsigma/v \rightarrow 1$ ; then  $\varsigma = v \Phi(v)^{1/2}$  is an ordinary power series in  $v$  with nonzero linear coefficient, so theorem 1.14 applies with  $\phi(v) = \Phi(v)^{1/2}$  and yields

$$[\varsigma^m] v(\varsigma) = \frac{1}{m} [v^{m-1}] \phi(v)^{-m} = \frac{1}{m} [v^{m-1}] \Phi(v)^{-m/2} = \omega_m.$$

The Bell form is lemma 1.12 applied to the unit series  $\Phi$ , whose coefficients are  $2(m+1)/(m+2)!$  for  $m \geq 1$ . Rationality follows because  $\binom{-m/2}{k} \in \mathbb{Q}$  and  $\widehat{B}_{m-1,k}$  is a polynomial with rational coefficients in rational arguments. Finally  $\mathcal{C} = 1 - v = 1 - \sum_m \omega_m \varsigma^m = 1 - \sum_m \omega_m 2^{m/2} v^{m/2}$ , which is eq. (2.24).  $\square$

*Remark 2.17* (Two normalisations, one object). S2 works with  $\omega_m$  (expanding in  $\varsigma = \sqrt{2v}$ ) while S1 and S3 work with  $c_r$  (expanding in  $v^{1/2}$ ). The dictionary is  $c_r = -2^{r/2} \omega_r$ , verified here for  $1 \leq r \leq 18$  by exact rational arithmetic. The  $\omega_m$  are rational, the  $c_r$  alternate between rationals and rational multiples of  $\sqrt{2}$ ; this is the only reason to prefer  $\omega$  for tables.

Table 2.1: Exact universal Cayley coefficients  $\omega_m$  of eq. (2.23), recomputed by rational arithmetic from eq. (2.22).

| $m$ | $\omega_m$              | $m$ | $\omega_m$                                      |
|-----|-------------------------|-----|-------------------------------------------------|
| 1   | 1                       | 10  | $-5776369/1515591000$                           |
| 2   | $-1/3$                  | 11  | $169709463197/69528040243200$                   |
| 3   | $11/72$                 | 12  | $-1118511313/709296588000$                      |
| 4   | $-43/540$               | 13  | $667874164916771/650782456676352000$            |
| 5   | $769/17280$             | 14  | $-500525573/744761417400$                       |
| 6   | $-221/8505$             | 15  | $103663334225097487/234281684403486720000$      |
| 7   | $680863/43545600$       | 16  | $-466901817532379/1595278956070800000$          |
| 8   | $-1963/204120$          | 17  | $21235294185086305043/109242202556140093440000$ |
| 9   | $226287557/37623398400$ | 18  | $-106040742894306601/818378104464320400000$     |

The corresponding  $c_r$  are

$$\begin{aligned} c_1 &= -\sqrt{2}, & c_2 &= \frac{2}{3}, & c_3 &= -\frac{11\sqrt{2}}{36}, & c_4 &= \frac{43}{135}, \\ c_5 &= -\frac{769\sqrt{2}}{4320}, & c_6 &= \frac{1768}{8505}, & c_7 &= -\frac{680863\sqrt{2}}{5443200}, & c_8 &= \frac{3926}{25515}. \end{aligned} \quad (2.25)$$

*Remark 2.18* (On the observed sign pattern). All three siblings display  $\omega_m$  with strictly alternating signs and none proves it. The data in table 2.1 are consistent with  $\text{sgn } \omega_m = (-1)^{m+1}$  for  $1 \leq m \leq 18$ , and this is what one expects from the fact that the inverse function  $v(\varsigma)$  of  $\varsigma = v \Phi(v)^{1/2}$  has its nearest singularity on the negative  $\varsigma$ -axis; but no proof is offered here and the statement is recorded as an observation, not as a result.

## 2.4 The Puiseux expansion at the Otter singularity

### 2.4.1 The local disturbance series

**Definition 2.19** (Local coordinate and disturbance). Write

$$s := 1 - \frac{z}{\rho}, \quad \theta(s) := 1 - e \zeta(\rho(1-s)) = \sum_{m \geq 1} \theta_m s^m. \quad (2.26)$$

The constant term vanishes by eq. (2.13). By proposition 2.10,  $\theta$  is analytic on the image of  $|z| < \sqrt{\rho}$  under  $z \mapsto s$ , namely the disc  $|s - 1| < \rho^{-1/2}$ ; in particular its Taylor series at  $s = 0$  converges for  $|s| < \rho^{-1/2} - 1 = 0.71923\dots$

Taylor's formula gives the derivative form

$$\theta_m = \frac{(-1)^{m+1} e \rho^m}{m!} \zeta^{(m)}(\rho), \quad m \geq 1, \quad (2.27)$$

consistent with eq. (2.14):  $\theta_1 = e \rho \zeta'(\rho)$ . For computation the following coefficient form is preferable, since it needs nothing but the integers  $a_r$  and the single scalar  $\rho$ .

**Lemma 2.20** (Local coefficients of  $R$  from the tree numbers). *Let  $\hat{R}_m := [s^m] R(\rho(1-s))$ . Then*

$$\hat{R}_m = (-1)^m \sum_{r \geq 2} \varrho_r \binom{r}{m} \rho^r = (-1)^m \frac{\rho^m R^{(m)}(\rho)}{m!}, \quad (2.28)$$

with  $\varrho_r$  as in eq. (2.10). The series converges geometrically: for fixed  $m$  the general term is  $\mathcal{O}(r^m \rho^{r/2})$ .

*Proof.* By lemma 2.7,  $R(\rho(1-s)) = \sum_{r \geq 2} \varrho_r \rho^r (1-s)^r$ , and  $[s^m] (1-s)^r = (-1)^m \binom{r}{m}$ . The middle expression is a Taylor coefficient of  $R$  at  $\rho$  in the variable  $\rho s$ , giving the right-hand form. For the convergence rate, every divisor  $d < r$  of  $r$  satisfies  $d \leq r/2$ , so  $\varrho_r \leq a_{\lfloor r/2 \rfloor} d(r)$  with  $d(r)$  the number of divisors of  $r$ ; since  $\limsup_n a_n^{1/n} = \rho^{-1}$ , for every  $\epsilon > 0$  there is  $K_\epsilon$  with  $\varrho_r \rho^r \leq K_\epsilon (\sqrt{\rho} + \epsilon)^r$ . With  $\binom{r}{m} \leq r^m$  the claim follows.  $\square$

**Remark 2.21** (A dedup). S1 (its  $\ell_m$ ) and S3 (its  $[s^m] R(\rho(1-s))$ ) both give the double sum  $(-1)^m \sum_{k \geq 2} \frac{1}{k} \sum_{n \geq 1} a_n \binom{kn}{m} \rho^{kn}$ . Equation (2.28) is the same number after the inner divisor collection of lemma 2.7, and is one sum rather than two. The two forms were checked against each other numerically to 70 digits for  $0 \leq m \leq 30$ .

**Lemma 2.22** (The disturbance coefficients). *With  $\hat{R}_m$  as in lemma 2.20, put  $\hat{R}_+(s) := \sum_{m \geq 1} \hat{R}_m s^m$  and  $E(s) := \exp \hat{R}_+(s) = \sum_{m \geq 0} E_m s^m$ . Then  $E_0 = 1$ ,*

$$E_m = \sum_{k=0}^m \frac{1}{k!} \hat{B}_{m,k}(\hat{R}_1, \hat{R}_2, \dots) = \frac{1}{m} \sum_{k=1}^m k \hat{R}_k E_{m-k}, \quad (2.29)$$

and

$$\theta_m = E_{m-1} - E_m, \quad m \geq 1. \quad (2.30)$$

*Proof.* By eq. (2.13),  $e \zeta(\rho) = 1$ , so

$$e \zeta(\rho(1-s)) = \frac{\rho(1-s) e^{R(\rho(1-s))}}{\rho e^{R(\rho)}} = (1-s) e^{\hat{R}_+(s)} = (1-s) E(s),$$

because  $\hat{R}_0 = R(\rho)$ . Hence  $\theta(s) = 1 - (1-s)E(s)$  and  $\theta_m = -(E_m - E_{m-1})$  for  $m \geq 1$ , while the constant term is  $1 - E_0 = 0$ . The two expressions for  $E_m$  are the ordinary Bell expansion of an exponential (definition 1.9) and the standard triangular exponential recurrence obtained from  $E' = \hat{R}'_+ E$ .  $\square$

**Remark 2.23** (The jet route). S2 computes  $\theta_m$  instead from the *scaled logarithmic jets*

$$\ell_j := \rho^j \left. \frac{d^j}{dz^j} \log \zeta(z) \right|_{z=\rho} = (-1)^{j-1} (j-1)! + \sum_{r \geq j} \varrho_r r^j \rho^r, \quad (2.31)$$

via  $\theta_m = \frac{(-1)^{m+1}}{m!} B_m(\ell_1, \dots, \ell_m)$ , where  $B_m$  is the complete exponential Bell polynomial and  $r^j = r(r-1)\dots(r-j+1)$ . This is equivalent to lemma 2.22 through  $\hat{R}_m = (-1)^m \rho^m R^{(m)}(\rho)/m!$  and

$\ell_j = (-1)^j j! \widehat{R}_j$  for  $j \geq 1$  up to the  $\log z$  term; both routes were implemented and agree. Lemma 2.22 is preferred because it avoids the factorial rescaling and its intermediate quantities stay  $\mathcal{O}(1)$ . S2's normalisation  $s_j = 2\theta_j$  (it expands  $2(1 - e\zeta)$  rather than  $1 - e\zeta$ ) accounts for every factor of 2 that differs between the siblings' formulas for  $t_r$ .

The first disturbance coefficients, recomputed, are

$$\begin{aligned} \theta_1 &= 1.216004561824048922157212619 \dots, & \theta_2 &= -0.458415729856391719180847717 \dots, \\ \theta_3 &= 0.448104774389535043665734122 \dots, & \theta_4 &= -0.399561679956209677406231980 \dots, \\ \theta_5 &= 0.385251341946613000719546396 \dots, & \theta_6 &= -0.389806561959203341949164640 \dots \end{aligned} \quad (2.32)$$

### 2.4.2 All local coefficients

**Theorem 2.24** (Every Puiseux coefficient at  $\rho$ ). *Factor  $\theta(s) = \theta_1 s(1 + U(s))$  with  $U(s) = \sum_{j \geq 1} u_j s^j$ ,  $u_j = \theta_{j+1}/\theta_1$ . Then in a slit neighbourhood of  $z = \rho$  there is a convergent Puiseux expansion*

$$T(z) = 1 + \sum_{n \geq 1} t_n s^{n/2}, \quad s = 1 - \frac{z}{\rho}, \quad (2.33)$$

and for every  $n \geq 1$

$$t_n = \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r \theta_1^{r/2} [s^{(n-r)/2}] (1 + U(s))^{r/2} = \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r \theta_1^{r/2} \sum_{k=0}^{(n-r)/2} \binom{r/2}{k} \widehat{B}_{\frac{n-r}{2}, k}(u_1, u_2, \dots). \quad (2.34)$$

Thus  $t_n$  is a finite expression in  $\omega_1, \dots, \omega_n, \theta_1^{\pm 1/2}$  and  $\theta_2, \dots, \theta_{\lceil (n+1)/2 \rceil}$ .

*Proof.* Near  $z = \rho$  we have  $e\zeta = 1 - \theta(s)$  with  $\theta(0) = 0$ , so by eq. (2.12) and eq. (2.24),  $T = 1 + \sum_{r \geq 1} c_r \theta(s)^{r/2}$ , the branch being the one with  $T \rightarrow 1^-$  as  $s \rightarrow 0^+$ . Substituting  $\theta(s)^{r/2} = \theta_1^{r/2} s^{r/2} (1 + U(s))^{r/2}$ , a term of index  $r$  can contribute to  $s^{n/2}$  only when  $n - r$  is a nonnegative even integer. Extracting  $[s^{(n-r)/2}]$  and applying lemma 1.12 to the unit series  $1 + U$  gives eq. (2.34). Convergence:  $\theta$  is analytic and has a simple zero at  $s = 0$  with  $\theta_1 > 0$  (eq. (2.14)), so  $U$  is analytic with  $U(0) = 0$ ; choosing a slit neighbourhood on which  $|U| < 1$  and a consistent square root, the convergent series eq. (2.24) may be composed with the analytic map  $s \mapsto \theta(s)$ .  $\square$

**Corollary 2.25.**  $t_1 = -\sqrt{2\theta_1}$ ,  $t_2 = \frac{2}{3}\theta_1$ , and  $t_3 = -\frac{11\sqrt{2}}{36}\theta_1^{3/2} - \frac{\sqrt{2}}{2}\theta_2\theta_1^{-1/2}$ .

*Proof.* Take  $n = 1, 2, 3$  in eq. (2.34) with  $c_1 = -\sqrt{2}$ ,  $c_2 = \frac{2}{3}$ ,  $c_3 = -\frac{11\sqrt{2}}{36}$  and  $\widehat{B}_{0,0} = 1$ ,  $\widehat{B}_{1,1}(u) = u_1$ ,  $\binom{1/2}{1} = \frac{1}{2}$ .  $\square$

**Remark 2.26** (Sibling cross-check). In S2's normalisation  $s_1 = 2\theta_1$ ,  $s_2 = 2\theta_2$ , the same coefficient reads  $t_3 = -(11s_1^2 + 36s_2)/(72\sqrt{s_1})$ ; substituting  $s_j = 2\theta_j$  reproduces corollary 2.25 exactly. Likewise  $t_1 = -\sqrt{s_1}$  and  $t_2 = s_1/3$ . Both were checked numerically.

**Remark 2.27** (Analytic versus singular local terms). The even part  $\sum_j t_{2j} s^j$  of eq. (2.33) is analytic at  $z = \rho$ . A fixed analytic monomial  $(1 - z/\rho)^j$  is a polynomial in  $z$  and contributes nothing to  $[z^n]$  once  $n > j$ ; hence only the odd coefficients  $t_{2j+1}$  enter the algebraic  $\rho^{-n} n^{-\bullet}$  asymptotics of section 2.5. This is not a discarding of information: the analytic germ continues through  $\rho$  and reappears in the exponentially smaller sectors of section 2.9. The sign pattern of table 2.2 is not monotone and should not be over-read; these are local analytic data, not the final coefficient constants.



Table 2.2: Puiseux coefficients in  $T(z) = 1 + \sum_{n \geq 1} t_n(1 - z/\rho)^{n/2}$ , recomputed from eq. (2.34) at 90 working digits.

| $n$ | $t_n$                    | $n$ | $t_n$                    |
|-----|--------------------------|-----|--------------------------|
| 1   | -1.559490020374640885542 | 10  | 0.205984831277907418736  |
| 2   | 0.810669707882699281438  | 11  | -0.052724708498190562315 |
| 3   | -0.285487021612845605304 | 12  | 0.017026568754953669446  |
| 4   | 0.165372365712083893477  | 13  | -0.152370624366325396286 |
| 5   | -0.342459970402154201042 | 14  | 0.173702883299850462894  |
| 6   | 0.317407225946528563568  | 15  | -0.014473703739527044857 |
| 7   | -0.107778800291631009182 | 16  | -0.021899517611215562233 |
| 8   | 0.061384957055835104689  | 17  | -0.144547193570909705477 |
| 9   | -0.195212383597556463613 | 18  | 0.176077108885017779062  |

In Otter’s coordinate  $\rho - z = \rho s$  the leading behaviour reads  $T(z) = 1 - b(\rho - z)^{1/2} + \mathcal{O}(\rho - z)$  with

$$b = -\frac{t_1}{\sqrt{\rho}} = 2.68112814726711223857732878 \dots \quad (2.35)$$

**Theorem 2.28** (Local expansion with an explicit remainder). *For every  $N \geq 1$  there are a dented (“ $\Delta$ -domain”) neighbourhood  $\Delta$  of  $\rho$  and a constant  $K_N$  such that*

$$\left| T(z) - 1 - \sum_{n=1}^N t_n \left(1 - \frac{z}{\rho}\right)^{n/2} \right| \leq K_N \left|1 - \frac{z}{\rho}\right|^{(N+1)/2}, \quad z \in \Delta. \quad (2.36)$$

*Proof.* By proposition 2.10 the map  $s \mapsto \theta(s)$  is analytic on a disc  $|s| < \varepsilon_0$ ; by eq. (2.14) it has a simple zero at  $s = 0$  with  $\theta_1 > 0$ . Shrinking  $\varepsilon_0$ ,  $|U(s)| < \frac{1}{2}$  there. The series eq. (2.24) converges for  $|v|$  small; a Cauchy estimate on the analytic function  $v \mapsto \mathcal{C}((1-v)/e)$  restricted to a slit disc gives  $|c_r| \leq K\varepsilon_1^{-r/2}$ . Composing convergent expansions and using Taylor’s theorem with remainder on  $(1+U)^{r/2}$  gives eq. (2.36) on the slit disc  $\{|s| < \varepsilon, |\arg s| < \pi - \eta\}$ , which is a  $\Delta$ -domain at  $\rho$  in the  $z$ -variable.  $\square$

## 2.5 Transfer to the all-orders coefficient asymptotics

### 2.5.1 Exact transfer of a singular monomial

For  $\beta \notin \mathbb{N}_0$  and  $n \geq 0$ ,

$$[z^n] \left(1 - \frac{z}{\rho}\right)^\beta = \rho^{-n} (-1)^n \binom{\beta}{n} = \rho^{-n} \frac{\Gamma(n - \beta)}{\Gamma(-\beta)\Gamma(n + 1)}. \quad (2.37)$$

For  $\beta \in \mathbb{N}_0$  the left-hand side vanishes for  $n > \beta$  – the analytic-germ statement of remark 2.27. It remains to expand the gamma ratio to all orders.

**Lemma 2.29** (Gamma-ratio coefficients). *Let  $B_m(x)$  be the Bernoulli polynomials, and for  $\beta \in \mathbb{C}$  set*

$$\varkappa_m(\beta) := \frac{(-1)^{m+1}}{m(m+1)} (B_{m+1}(-\beta) - B_{m+1}(1)), \quad m \geq 1, \quad (2.38)$$

and define  $g_r(\beta)$  by  $\exp(\sum_{m \geq 1} \varkappa_m(\beta) v^m) = \sum_{r \geq 0} g_r(\beta) v^r$ , equivalently

$$g_0(\beta) = 1, \quad g_r(\beta) = \frac{1}{r} \sum_{m=1}^r m \varkappa_m(\beta) g_{r-m}(\beta) = \sum_{k=0}^r \frac{1}{k!} \widehat{B}_{r,k}(\varkappa_1(\beta), \varkappa_2(\beta), \dots). \quad (2.39)$$

Then, for fixed  $\beta$  and  $n \rightarrow \infty$  in any sector avoiding the negative real axis,

$$\frac{\Gamma(n - \beta)}{\Gamma(n + 1)} \sim n^{-\beta-1} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r}. \quad (2.40)$$

Each  $g_r$  is a polynomial of degree  $2r$  in  $\beta$  with rational coefficients; the first five are

$$\begin{aligned} g_0 &= 1, & g_1(\beta) &= \frac{\beta(\beta + 1)}{2}, & g_2(\beta) &= \frac{\beta(\beta + 1)(\beta + 2)(3\beta + 1)}{24}, \\ g_3(\beta) &= \frac{\beta^2(\beta + 1)^2(\beta + 2)(\beta + 3)}{48}, \end{aligned} \quad (2.41)$$

$$g_4(\beta) = \frac{\beta(\beta + 1)(\beta + 2)(\beta + 3)(\beta + 4)(15\beta^3 + 30\beta^2 + 5\beta - 2)}{5760}. \quad (2.42)$$

In particular  $(g_r(\frac{1}{2}))_{r \geq 0} = 1, \frac{3}{8}, \frac{25}{128}, \frac{105}{1024}, \frac{1659}{32768}, \frac{6237}{262144}, \dots$

*Proof.* The Euler–Maclaurin/Bernoulli form of Stirling’s expansion gives, for fixed  $a, b$ ,

$$\log \frac{\Gamma(n + a)}{\Gamma(n + b) n^{a-b}} \sim \sum_{m \geq 1} \frac{(-1)^{m+1}}{m(m+1)} \frac{B_{m+1}(a) - B_{m+1}(b)}{n^m}$$

as  $n \rightarrow \infty$  in the stated sectors. Taking  $a = -\beta, b = 1$  yields  $\log(\Gamma(n - \beta)/\Gamma(n + 1)) \sim -(\beta + 1) \log n + \sum_m \varkappa_m(\beta) n^{-m}$ , and exponentiating an asymptotic series term by term is legitimate, producing exactly the coefficients of  $\exp(\sum \varkappa_m v^m)$ . The recurrence follows from differentiating that generating identity; the Bell form is definition 1.9. Degree:  $\varkappa_m$  has degree  $m + 1$  in  $\beta$  and vanishes at  $\beta = 0$ , so a monomial  $\prod \varkappa_{m_i}$  with  $\sum m_i = r$  has degree  $\leq r + \#\{i\} \leq 2r$ ; the displayed  $g_1, \dots, g_4$  were recomputed here from eq. (2.38) by exact rational arithmetic.  $\square$

*Remark 2.30* (Part 0 candidate). Lemma 2.29 is not specific to rooted trees and is used verbatim wherever a chapter transfers an algebraic singularity to coefficient asymptotics. It is listed in the report as a candidate for promotion into Part 0. All three siblings state it; S2 and S3 give the same recurrence, S1 the same object under the name  $\gamma_m(\beta)$  with  $\eta_r(\beta)$  for  $\varkappa_r(\beta)$ .

## 2.5.2 The master coefficient formula

**Theorem 2.31** (All-orders dominant expansion of A000081). *For every fixed  $M \geq 0$ ,*

$$a_n = \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \left( \sum_{m=0}^M \frac{\tau_m}{n^m} + \mathcal{O}(n^{-M-1}) \right), \quad n \rightarrow \infty, \quad (2.43)$$

where

$$\tau_m = \sum_{j=0}^m \Delta_j t_{2j+1} g_{m-j}(j + \tfrac{1}{2}), \quad \Delta_j := \frac{(-1)^{j+1} (2j+1)!!}{2^{j+1}} = \frac{\sqrt{\pi}}{\Gamma(-j - \frac{1}{2})}. \quad (2.44)$$

Every ingredient is a finite coefficient extraction:  $t_{2j+1}$  by eq. (2.34),  $g_r$  by eq. (2.39),  $\theta_m$  by eq. (2.30),  $\omega_m$  by eq. (2.23),  $\varrho_r$  by eq. (2.10).

*Proof.* Fix  $M$  and apply theorem 2.28 with  $N = 2M + 2$ : on a  $\Delta$ -domain at  $\rho$ ,

$$T(z) = A(z) + \sum_{j=0}^M t_{2j+1} \left( 1 - \frac{z}{\rho} \right)^{j+1/2} + \mathcal{O} \left( \left| 1 - \frac{z}{\rho} \right|^{M+3/2} \right),$$

with  $A$  the polynomial collecting the even (analytic) terms. By proposition 2.12 the only singularity on  $|z| = \rho$  is  $\rho$  itself, so the Cauchy contour may be deformed into the standard Flajolet–Odlyzko  $\Delta$ -contour. The polynomial  $A$  contributes nothing to  $[z^n]$  for  $n > \deg A$ . Each retained monomial contributes exactly  $\rho^{-n} t_{2j+1} \Gamma(n - j - \frac{1}{2}) / (\Gamma(-j - \frac{1}{2}) \Gamma(n + 1))$  by eq. (2.37), whose asymptotic expansion is eq. (2.40); the  $\mathcal{O}$ -term transfers to  $\mathcal{O}(\rho^{-n} n^{-M-5/2})$  by the standard Hankel-contour estimate. Collecting the coefficient of  $\rho^{-n} n^{-m-3/2}$  from the pairs  $j + r = m$  and multiplying through by  $\sqrt{\pi}$  gives eq. (2.44), the identity  $\Gamma(-j - \frac{1}{2}) = (-4)^{j+1} (j+1)! \sqrt{\pi} / (2j+2)! = (-1)^{j+1} 2^{j+1} \sqrt{\pi} / (2j+1)!!$  supplying  $\Delta_j$ .  $\square$

**Corollary 2.32** (Otter’s constant). *With  $C := \tau_0 / \sqrt{\pi}$ ,*

$$\tau_0 = -\frac{t_1}{2} = \sqrt{\frac{\theta_1}{2}}, \quad C = \sqrt{\frac{\theta_1}{2\pi}} = \sqrt{\frac{e\rho \zeta'(\rho)}{2\pi}} = \sqrt{\frac{1 + \rho R'(\rho)}{2\pi}}, \quad a_n \sim C \alpha^n n^{-3/2}. \quad (2.45)$$

*Numerically*

$$\tau_0 = 0.7797450101873204427711036979 \dots, \quad C = 0.4399240125710253040409033914345447648 \dots \quad (2.46)$$

*Proof.* Set  $m = 0$  in eq. (2.44):  $\Delta_0 = -\frac{1}{2}$ ,  $g_0 = 1$ , so  $\tau_0 = -t_1/2 = \sqrt{2\theta_1}/2 = \sqrt{\theta_1}/2$  by corollary 2.25. Divide by  $\sqrt{\pi}$  and use eq. (2.14).  $\square$

The identity  $C^2 = \theta_1/(2\pi)$  is a genuinely independent check: the left side comes through the whole Puiseux/transfer chain, the right side directly from one derivative of  $\zeta$ . The two agree to all 55 digits carried.

Expanding eq. (2.44) for small  $m$  gives the transfer identities

$$\begin{aligned} \tau_0 &= -\frac{1}{2}t_1, \\ \tau_1 &= -\frac{3}{16}(t_1 - 4t_3) = -\frac{3}{16}t_1 + \frac{3}{4}t_3, \\ \tau_2 &= -\frac{5}{256}(5t_1 - 72t_3 + 96t_5) = -\frac{25}{256}t_1 + \frac{45}{32}t_3 - \frac{15}{8}t_5, \\ \tau_3 &= -\frac{105}{2048}(t_1 - 44t_3 + 160t_5 - 128t_7) = -\frac{105}{2048}t_1 + \frac{1155}{512}t_3 - \frac{525}{64}t_5 + \frac{105}{16}t_7, \\ \tau_4 &= -\frac{21}{65536}(79t_1 - 10800t_3 + 81600t_5 - 161280t_7 + 92160t_9). \end{aligned} \quad (2.47)$$

S2 states the right-hand forms, S3 the factored left-hand forms; they are identical after clearing denominators, and both were re-derived here symbolically from eq. (2.44). For any  $m$  beyond 4 the compact convolution is far preferable to the expanded form.

Table 2.3: Coefficients of the dominant expansion  $a_n \sim \rho^{-n}(\pi n^3)^{-1/2} \sum_{m \geq 0} \tau_m n^{-m}$  and their normalisations  $f_m = \tau_m/\tau_0$ , recomputed here.

| $m$ | $\tau_m$                  | $f_m = \tau_m/\tau_0$    |
|-----|---------------------------|--------------------------|
| 0   | 0.7797450101873204427711  | 1                        |
| 1   | 0.07828911261061096206128 | 0.1004034800964014347637 |
| 2   | 0.3929402676631860184743  | 0.5039343151023035331022 |
| 3   | 1.537879315978838091102   | 1.972284908382278469206  |
| 4   | 8.200844090435596159220   | 10.51734090413157399274  |
| 5   | 57.29291473494343787740   | 73.47647498402047106417  |
| 6   | 503.0445050262735818248   | 645.1397552456612928153  |
| 7   | 5359.600933884326033403   | 6873.530274463408619941  |
| 8   | 67342.06920114653048341   | 86364.21948371140797700  |
| 9   | 975425.4970695924732213   | 1250954.458605978293018  |
| 10  | 15996932.49293173312961   | 20515594.56480361813667  |
| 11  | 292822531.3353392231146   | —                        |
| 12  | 5914523441.293932519117   | —                        |

Higher values continue smoothly; for reference  $\tau_{15} = 8.078305401468884289 \times 10^{13}$ ,  $\tau_{18} = 2.073828085209893749 \times 10^{18}$ .

Writing

$$F(v) := 1 + \sum_{m \geq 1} f_m v^m, \quad f_m = \frac{\tau_m}{\tau_0}, \quad (2.48)$$

the inversion-ready normal form is

$$a_n \sim C e^{\lambda n} n^{-p} F(1/n), \quad p = \frac{3}{2}. \quad (2.49)$$

*Remark 2.33* (Divergence and least-term truncation). Equation (2.43) asserts a Poincaré expansion at every fixed order; it does *not* assert convergence, and table 2.3 shows the coefficients growing rapidly. The optimal truncation and the size of the resulting floor are governed by theorem 1.46; the relevant nearest competing exponential scale is identified in section 2.9 as  $\rho^{-n/2}$ , so the least-term floor at index  $n$  is  $\mathcal{O}(\rho^{n/2})$  relative to the leading term — consistent with the error tables of section 2.11.2, where the improvement stops at  $n = 20$  around  $M = 6$ . S3 states this correctly as a numerical strategy rather than a theorem; S1 and S2 call the late-term growth “the perturbative shadow of farther singularities”, which is the right heuristic but is not proved anywhere in the three drafts and is not proved here.

## 2.6 The multiplicative reciprocal

If “inverse” is read as the reciprocal sequence, the answer is immediate from eq. (2.49) and requires no inversion theory at all.

**Proposition 2.34** (All-orders reciprocal). *Write  $1/F(v) = 1 + \sum_{m \geq 1} \bar{f}_m v^m$ . Then*

$$\bar{f}_m = \sum_{k=1}^m (-1)^k \hat{B}_{m,k}(f_1, f_2, \dots) = - \sum_{k=1}^m f_k \bar{f}_{m-k} \quad (\bar{f}_0 = 1), \quad (2.50)$$

and for every fixed  $M \geq 0$

$$\frac{1}{a_n} = \frac{\sqrt{\pi} \rho^n n^{3/2}}{\tau_0} \left( \sum_{m=0}^M \frac{\bar{f}_m}{n^m} + \mathcal{O}(n^{-M-1}) \right). \quad (2.51)$$

*Proof.* The first equality in eq. (2.50) is lemma 1.12 with exponent  $-1$ ; the second is the Cauchy product  $F \cdot F^{-1} = 1$ . For eq. (2.51), note that  $a_n > 0$  and, by eq. (2.43),  $a_n = C \alpha^n n^{-p} (1 + \varepsilon_n)$  with  $\varepsilon_n \rightarrow 0$ ; hence  $1/a_n$  is defined and  $1/(1 + \varepsilon_n)$  has the asymptotic expansion obtained by formally inverting the (asymptotic) series for  $1 + \varepsilon_n$  — a legitimate operation because the leading term is  $1 \neq 0$  and each truncation error is  $\mathcal{O}(n^{-M-1})$ .  $\square$

The first reciprocal coefficients, recomputed, are

$$\begin{aligned} \bar{f}_1 &= -0.1004034800964014347637, & \bar{f}_2 &= -0.4938534562868350540384, \\ \bar{f}_3 &= -1.872103543737176348103, & \bar{f}_4 &= -9.882481221441551170916, \\ \bar{f}_5 &= -69.51082491336461883211, & \bar{f}_6 &= -616.9168645978433816315. \end{aligned} \quad (2.52)$$

*Remark 2.35* (Reciprocal of the full transseries). Writing  $a_n = \mathcal{M}_n(1 + \mathcal{E}_n)$  with  $\mathcal{M}_n$  the complete dominant sector and  $\mathcal{E}_n$  the sum of relative exponentially small sectors of section 2.9, one has  $1/a_n = \mathcal{M}_n^{-1} \sum_{q \geq 0} (-\mathcal{E}_n)^q$ ; the products generate exactly the additive semigroup of the singular actions, with coefficients given by finite Cauchy products. This is a formal statement about the transseries, and inherits whatever status theorem 2.55 has on the chosen continuation domain.

*Warning 2.36.* The reciprocal is *not* a functional inverse. Its exponential scale is  $\rho^n$ , while the functional inverse of section 2.7 grows only logarithmically in its argument. All three siblings make this point; it is repeated here because the ambiguity of the word “inverse” is exactly what definition 1.37 exists to remove.

## 2.7 The asymptotic inverse: the Part 0 dictionary

### 2.7.1 The parameter dictionary

Everything needed to invert eq. (2.49) has been proved in Part 0. What this chapter must supply is only the dictionary and the verification of Part 0's hypotheses.

**Proposition 2.37** (A000081 as an instance of the exponential–power model). *Let*

$$\mathcal{A}(x) := C e^{\lambda x} x^{-p} F(x^{-1}), \quad p = \frac{3}{2}, \quad (2.53)$$

with  $C$  from corollary 2.32 and  $F$  from eq. (2.48). Then  $\mathcal{A}$  is an instance of definition 1.2 under

|        |                              |                          |          |                |          |                    |
|--------|------------------------------|--------------------------|----------|----------------|----------|--------------------|
| Part 0 | $C$                          | $a$                      | $\alpha$ | $b$            | $\delta$ | $Q(t)$             |
| here   | $C = \sqrt{\theta_1/(2\pi)}$ | $\lambda = \log(1/\rho)$ | 1        | $-\frac{3}{2}$ | 1        | $H(t) = \log F(t)$ |

(2.54)

with  $H \in t\mathbb{R}[[t]]$ . In particular  $a = \lambda > 0$  and  $\alpha = 1$ , so  $\mathcal{A}$  is already in the linear-phase normal form and lemma 1.4 is the identity substitution.

*Proof.* Equation (2.49) is definition 1.2 with the stated substitutions;  $\lambda > 0$  because  $\rho < 1$  (proposition 2.11),  $C > 0$  because  $\theta_1 > 0$ , and  $H(0) = \log F(0) = \log 1 = 0$  so  $H \in t\mathbb{R}[[t]]$ .  $\square$

**Definition 2.38** (Additive phase coefficients). Write  $H(v) = \log F(v) = \sum_{m \geq 1} \varphi_m v^m$ . By lemma 1.12,

$$\varphi_m = \sum_{k=1}^m \frac{(-1)^{k-1}}{k} \widehat{B}_{m,k}(f_1, f_2, \dots) = f_m - \frac{1}{m} \sum_{k=1}^{m-1} k \varphi_k f_{m-k}. \quad (2.55)$$

Taking logarithms in eq. (2.43) gives the form used for inversion,

$$\log a_n = \lambda n - \frac{3}{2} \log n + \log C + \sum_{m=1}^M \frac{\varphi_m}{n^m} + \mathcal{O}(n^{-M-1}), \quad (2.56)$$

with, recomputed here,

$$\begin{aligned} \varphi_1 &= 0.1004034800964014347637, & \varphi_2 &= 0.4988938856945692935703, \\ \varphi_3 &= 1.922025533841821821591, & \varphi_4 &= 10.19739642337376887053, \\ \varphi_5 &= 71.47146706207073604412, & \varphi_6 &= 630.8599376073372478899. \end{aligned} \quad (2.57)$$

### 2.7.2 The exact Lambert carrier

**Corollary 2.39** (Carrier; specialisation of theorem 1.20). *Let  $y > y_* := C(e\lambda/p)^p = 1.2108232025837572603\dots$ . Then the equation  $y = C e^{\lambda x} x^{-p}$  has exactly one solution  $x_0 = x_0(y) > p/\lambda$ , namely*

$$x_0(y) = -\frac{p}{\lambda} W_{-1} \left( -\frac{\lambda}{p} \left( \frac{C}{y} \right)^{1/p} \right) = -\frac{3}{2\lambda} W_{-1} \left( -\frac{2\lambda}{3} \left( \frac{C}{y} \right)^{2/3} \right). \quad (2.58)$$

*Proof.* Raise  $y = C e^{\lambda x} x^{-p}$  to the power  $-1/p$  and multiply by  $-\lambda/p$ :

$$-\frac{\lambda}{p} \left( \frac{C}{y} \right)^{1/p} = -\frac{\lambda x}{p} e^{-\lambda x/p} = \xi e^{\xi}, \quad \xi := -\frac{\lambda x}{p}.$$

Write  $\epsilon := -\frac{\lambda}{p} (C/y)^{1/p}$ . The map  $x \mapsto C e^{\lambda x} x^{-p}$  is strictly increasing on  $(p/\lambda, \infty)$  (its logarithmic derivative is  $\lambda - p/x > 0$  there) with infimum  $y_*$  attained as  $x \downarrow p/\lambda$ , so  $y > y_*$  gives a unique root  $x_0 > p/\lambda$ , i.e.  $\xi < -1$ ; and  $y > y_*$  is exactly  $\epsilon \in (-e^{-1}, 0)$ . On that range  $\xi \mapsto \xi e^{\xi}$  has the two real inverse branches  $W_0 \in (-1, 0)$  and  $W_{-1} < -1$ , and the required root is the second. This is theorem 1.20 with  $L = \log(y/C)$ ,  $a = \lambda$ ,  $b = -p$  and branch index  $k = -1$ ; the branch rule there selects  $k = -1$  under exactly the condition  $x > -b/a$ .  $\square$

*Remark 2.40.* The principal branch  $W_0$  returns the other, small root of the same equation, which is the spurious solution: all three siblings say so, and the threshold  $y_*$  makes the dichotomy quantitative. For A000081 the threshold lies below  $a_3 = 2$ , so eq. (2.58) may be used at every sequence value from  $n = 3$  on.

### 2.7.3 All algebraic corrections

**Theorem 2.41** (Corrections; specialisation of theorem 1.25). *Let*

$$\Psi(t, u) := \frac{p}{\lambda} \log(1 + tu) - \frac{1}{\lambda} H\left(\frac{t}{1 + tu}\right) \in t\mathbb{R}[[t, u]]. \quad (2.59)$$

*Then there is a unique  $D(t) = \sum_{m \geq 1} d_m t^m \in t\mathbb{R}[[t]]$  with  $D(t) = \Psi(t, D(t))$ , its coefficients are given by lemma 1.16,*

$$d_m = \sum_{k=1}^m \frac{1}{k} [t^m u^{k-1}] \Psi(t, u)^k, \quad (2.60)$$

*equivalently by the triangular extraction*

$$d_m = [t^m] \left\{ \frac{p}{\lambda} \log(1 + tD_{<m}(t)) - \frac{1}{\lambda} H\left(\frac{t}{1 + tD_{<m}(t)}\right) \right\}, \quad D_{<m} = \sum_{j < m} d_j t^j, \quad (2.61)$$

*or by the completely explicit ordinary-Bell form*

$$d_m = \frac{p}{\lambda} \sum_{r=1}^m \frac{(-1)^{r-1}}{r} \widehat{B}_{m-r,r}(d_1, d_2, \dots) - \frac{1}{\lambda} \sum_{k=1}^m \varphi_k \sum_{r=0}^{m-k} (-1)^r \binom{k+r-1}{r} \widehat{B}_{m-k-r,r}(d_1, d_2, \dots), \quad (2.62)$$

*in which the index constraints force only  $d_1, \dots, d_{m-1}$  to occur on the right. With  $x_0$  from eq. (2.58) the complete algebraic inverse is*

$$\nu(y) \sim x_0(y) + \sum_{m \geq 1} \frac{d_m}{x_0(y)^m}. \quad (2.63)$$

*Proof.* Theorem 1.25 states exactly this for a model  $Ce^{ax}x^be^{Q(x^{-\delta})}$  with  $\Psi(t, u) = -\frac{1}{a}[b \log(1 + tu) + Q(t/(1 + tu))]$ ; substituting the dictionary eq. (2.54) ( $a = \lambda$ ,  $b = -p$ ,  $\delta = 1$ ,  $Q = H$ ) gives eq. (2.59). Concretely: putting  $x = x_0 + D$  into  $\log(y/C) = \lambda x - p \log x + H(x^{-1})$  and subtracting the carrier equation  $\log(y/C) = \lambda x_0 - p \log x_0$  gives, with  $t = x_0^{-1}$ ,

$$\lambda D - p \log(1 + tD) + H\left(\frac{t}{1 + tD}\right) = 0,$$

which is  $D = \Psi(t, D)$ . Since  $\Psi \in t\mathbb{R}[[t, u]]$ , the formal implicit-function theorem gives a unique solution in  $t\mathbb{R}[[t]]$ ; lemma 1.16 converts it into eq. (2.60), and eq. (2.61) is the same statement read as a recursion. Equation (2.62) follows by expanding  $\log(1 + tD) = \sum_r \frac{(-1)^{r-1}}{r} t^r D^r$  and  $H(t/(1 + tD)) = \sum_k \varphi_k t^k \sum_r (-1)^r \binom{k+r-1}{r} t^r D^r$  and applying definition 1.9 to the powers  $D^r$ .  $\square$

**A formal-to-analytic slide in S1.** S1’s “General coefficient formula for the inverse” is stated as  $x(y) \sim X(y) + \sum_m d_m X(y)^{-m}$ , an asymptotic assertion, but its proof ends with “substitution ... proves formal inversion order by order” and supplies no remainder. As stated, theorem 2.41 is a theorem about formal series only; the asymptotic reading requires theorem 2.42 below, whose argument is S3’s – the only one of the three drafts that separates the two statements. The volume keeps them separate.

**Theorem 2.42** (Remainder; specialisation of theorem 1.44). *Fix  $M \geq 1$  and set  $\nu_M(y) := x_0(y) + \sum_{m=1}^{M-1} d_m x_0(y)^{-m}$ . Let  $f(x) = \log C + \lambda x - p \log x + H(x^{-1})$  be the (asymptotic) logarithm of the forward interpolant. Then*

$$f(\nu_M(y)) - \log y = \mathcal{O}(x_0(y)^{-M}), \quad \nu(y) - \nu_M(y) = \mathcal{O}(x_0(y)^{-M}) = \mathcal{O}((\log y)^{-M}). \quad (2.64)$$

*Proof.* By theorem 2.41 the coefficients of  $t^1, \dots, t^{M-1}$  in  $D - \Psi(t, D)$  vanish, so truncating leaves a residual  $\mathcal{O}(t^M)$ ; this is the first claim. Since  $f'(x) = \lambda - p/x + \mathcal{O}(x^{-2})$  is bounded away from 0 for large  $x$ , theorem 1.42 converts the residual into a forward error of the same order divided by  $f'$ ; this is theorem 1.44 with  $a = \lambda$ . Finally  $x_0(y) = \lambda^{-1} \log y (1 + o(1))$  by eq. (2.58).  $\square$

### 2.7.4 Coefficients

Expanding eq. (2.61) through  $t^5$  – and verifying the result symbolically – gives the closed forms

$$d_1 = -\frac{\varphi_1}{\lambda}, \quad (2.65)$$

$$d_2 = -\frac{p\varphi_1 + \lambda\varphi_2}{\lambda^2}, \quad (2.66)$$

$$d_3 = -\frac{\lambda\varphi_1^2 + p^2\varphi_1 + p\lambda\varphi_2 + \lambda^2\varphi_3}{\lambda^3}, \quad (2.67)$$

$$d_4 = -\frac{1}{2\lambda^4} \left( 5p\lambda\varphi_1^2 + 6\lambda^2\varphi_1\varphi_2 + 2p^3\varphi_1 + 2p^2\lambda\varphi_2 + 2p\lambda^2\varphi_3 + 2\lambda^3\varphi_4 \right), \quad (2.68)$$

$$d_5 = -\frac{1}{2\lambda^5} \left( 4\lambda^2\varphi_1^3 + 9p^2\lambda\varphi_1^2 + 14p\lambda^2\varphi_1\varphi_2 + 8\lambda^3\varphi_1\varphi_3 + 4\lambda^3\varphi_2^2 + 2p^4\varphi_1 + 2p^3\lambda\varphi_2 + 2p^2\lambda^2\varphi_3 + 2p\lambda^3\varphi_4 + 2\lambda^4\varphi_5 \right). \quad (2.69)$$

Beyond  $m = 5$  the expanded expressions grow fast, and eqs. (2.60) and (2.61) are both shorter and less error-prone.

Table 2.4: Additive phase coefficients  $\varphi_m$  and Lambert-carrier corrections  $d_m$ , recomputed here. The last column is S2’s normalisation  $D_m = \lambda^{m+1}d_m$  (see remark 2.43).

| $m$ | $\varphi_m$           | $d_m$                   | $D_m = \lambda^{m+1}d_m$ |
|-----|-----------------------|-------------------------|--------------------------|
| 1   | 0.1004034800964014348 | −0.09264385352561312984 | −0.1088130343442745145   |
| 2   | 0.4988938856945692936 | −0.5885630399313472628  | −0.7491856512881932495   |
| 3   | 1.922025533841821822  | −2.596680167407531021   | −3.582177326832277789    |
| 4   | 10.19739642337376887  | −13.14905339996706569   | −19.65872121138776004    |
| 5   | 71.47146706207073604  | −85.49870151462556652   | −138.5327478518560861    |
| 6   | 630.8599376073372479  | −711.2629148329465609   | −1248.979326371692854    |
| 7   | 6753.629059387619229  | −7306.788270284967257   | −13905.40884474814835    |
| 8   | 85171.04383262353790  | −89561.76088263724860   | −184719.1911265080905    |
| 9   | 1236987.135642947528  | −1274829.503083708903   | —                        |
| 10  | 20325777.10708850337  | −20641099.00032138544   | —                        |

**Numerical error in S2’s audit appendix.** S2’s appendix table of  $\eta_j$  and  $D_j$  reports  $D_3 = -3.582177326832123\dots$ , disagreeing with its own main text ( $-3.582177326832277789103009844$ ) from the fourteenth significant digit. Recomputation confirms the main-text value,  $D_3 = -3.58217732683227778910301\dots$ ; the appendix entry is wrong. Every other entry of that table was checked and is correct to the digits shown.

*Remark 2.43* (Three normalisations of the same inverse). The siblings work in three different variables, which is why their coefficient lists look different:

- S1 and S3 expand in the index variable  $x$  itself and obtain the  $d_m$  of table 2.4; their carriers agree with eq. (2.58).
- S2 rescales to  $X = \lambda x$ , so that its carrier is  $X_0 = -pW_{-1}(-\frac{1}{p}e^{-Z/p})$  with  $Z = \log(y/(C\lambda^p))$ , its corrections are  $D_m = \lambda^{m+1}d_m$ , and its phase coefficients are  $\eta_m^{S2} = \lambda^m\varphi_m$ . Both dictionaries were verified numerically for  $1 \leq m \leq 8$ , and  $X_0 = \lambda x_0$  holds identically because the two carrier equations differ by that substitution.
- S3 also uses  $x$  but writes the polynomial–logarithmic form in terms of  $\mathfrak{q} = p/\lambda$  and  $\eta_m^{S3} = \varphi_m/\lambda$ .

The constant appearing in S2's  $Z$  is  $C\lambda^p = 0.4963361416595997474727369464\dots$

### 2.7.5 Polynomial–logarithmic form

Eliminating the Lambert function gives the expansion in the canonical logarithmic scale. That is theorem 1.33; only the dictionary is recorded here.

**Corollary 2.44** (Flattened inverse). *Put*

$$\mathcal{X} := \frac{1}{\lambda} \log \frac{y}{C}, \quad L := \log \mathcal{X}, \quad \mathfrak{q} := \frac{p}{\lambda}, \quad \eta_k := \frac{\varphi_k}{\lambda}. \quad (2.70)$$

Then the inverse problem is  $x - \mathfrak{q} \log x + \sum_{k \geq 1} \eta_k x^{-k} = \mathcal{X}$ , and theorem 1.33 gives

$$\boxed{\nu(y) \sim \mathcal{X} + \mathfrak{q}L + \sum_{m \geq 1} \frac{P_m(L)}{\mathcal{X}^m},} \quad (2.71)$$

where each  $P_m$  is a polynomial of degree  $m$  over  $\mathbb{Q}[\mathfrak{q}, \eta_1, \dots, \eta_m]$  with

$$[L^m] P_m(L) = \frac{(-1)^{m+1}}{m} \mathfrak{q}^{m+1}, \quad P_m(0) = d_m. \quad (2.72)$$

The identity  $P_m(0) = d_m$  is not an accident: setting  $L = 0$  in the flattening recursion reduces it to the fixed-point equation solved by the Lambert carrier, and formal uniqueness in theorem 1.25 identifies the two. The first two polynomials are

$$P_1(L) = \mathfrak{q}^2 L - \eta_1 = 1.9156590172195452616 L - 0.0926438535256131298, \quad (2.73)$$

$$\begin{aligned} P_2(L) &= -\frac{1}{2} \mathfrak{q}^3 L^2 + (\mathfrak{q}^3 + \mathfrak{q}\eta_1) L - \mathfrak{q}\eta_1 - \eta_2 \\ &= -1.3257062894576032164 L^2 + 2.7896427837264004647 L - 0.5885630399313472628. \end{aligned} \quad (2.74)$$

*Remark 2.45* (The two flattenings are not the same polynomials). S2 flattens in  $Z = \log(y/(C\lambda^p))$  and  $\log Z$ ; S3 flattens in  $\mathcal{X} = \lambda^{-1} \log(y/C)$  and  $\log \mathcal{X}$ . The two recursions are formally the *same* recursion with different parameters – S3's is S2's under  $(p, \eta_k^{S2}) \mapsto (\mathfrak{q}, \eta_k^{S3})$  – but they expand the same function in *different* variables, so their coefficient polynomials genuinely differ. Neither draft is in error; the caution is only that the two lists must not be compared term by term. We adopt S3's normalisation because it keeps the index variable  $x$  throughout and so matches eq. (2.63).

Retaining only the first two scales,

$$\nu(y) = \frac{\log y}{\lambda} + \frac{p}{\lambda} \log \log y - \frac{1}{\lambda} \log(C\lambda^p) + o(1), \quad (2.75)$$

that is

$$\nu(y) = 0.9227155616186015248 \log y + 1.3840733424279022872 \log \log y + 0.6463639827002088073 + o(1). \quad (2.76)$$

Equation (2.63) is preferable numerically because  $W_{-1}$  resums the entire logarithmic tower exactly before any truncation is made.



## 2.8 The discrete inverse

The objects distinguished in definition 1.37 are, for this sequence: the smooth inverse  $\nu$  of section 2.7; the integer staircase

$$N(y) := \min\{n \geq 1 : a_n \geq y\}; \quad (2.77)$$

and the round-to-nearest generalized inverse. Applying theorem 1.39 requires its separation hypothesis, which for A000081 is supplied by the next two lemmas.

**Lemma 2.46** (Strict monotonicity).  $a_1 = a_2 = 1$  and  $a_{n+1} > a_n$  for every  $n \geq 2$ .

*Proof.* Monotonicity: the map sending a rooted tree  $t$  on  $n$  vertices to the rooted tree whose root has the single child subtree  $t$  is an injection from the  $n$ -vertex shapes into the  $(n+1)$ -vertex shapes, so  $a_{n+1} \geq a_n$ . Strictness for  $n \geq 2$ : when  $n+1 \geq 3$ , the star  $K_{1,n}$  rooted at its centre has  $n \geq 2$  children and is therefore not in the image of that injection, whence  $a_{n+1} \geq a_n + 1$ . The values  $a_1 = a_2 = 1$  are immediate.  $\square$

**Lemma 2.47** (Separation).  $a_{n+1}/a_n \rightarrow \alpha = 2.9557\dots > 1$ ; more precisely

$$\log \frac{a_{n+1}}{a_n} = \lambda - \frac{3}{2n} + \frac{\frac{3}{4} - \varphi_1}{n^2} + \frac{-\frac{1}{2} + \varphi_1 - 2\varphi_2}{n^3} + \mathcal{O}(n^{-4}). \quad (2.78)$$

*Proof.* Subtract eq. (2.56) at  $n$  from the same relation at  $n+1$ , and expand  $\log(n+1) = \log n + n^{-1} - \frac{1}{2}n^{-2} + \frac{1}{3}n^{-3} - \dots$  and  $(n+1)^{-m} = n^{-m}(1 - mn^{-1} + \dots)$ . Collecting powers of  $n^{-1}$  gives eq. (2.78); the limit statement is its leading term.  $\square$

**Corollary 2.48** (Staircase recovery). Let  $\nu_M$  be as in theorem 2.42. For every  $M \geq 2$  and all sufficiently large  $n$ ,  $|\nu_M(a_n) - n| < \frac{1}{2}$ , so  $n$  is recovered from  $a_n$  by rounding. For a general real target  $y > 1$ ,  $N(y) = \nu(y) + \mathcal{O}(1)$ , and certifying  $N(y)$  exactly requires the two-sided test  $a_{m-1} < y \leq a_m$ , which eq. (2.63) reduces to a bounded neighbourhood of  $m \approx \nu(y)$ .

*Proof.* The first claim is theorem 2.42 with  $M \geq 2$ , since  $\mathcal{O}((\log a_n)^{-2}) = \mathcal{O}(n^{-2})$ . For the second, apply theorem 1.39, whose separation hypothesis holds by lemmas 2.46 and 2.47: the logarithmic derivative of the interpolation tends to  $\lambda > 0$ , so multiplying  $y$  by a bounded factor moves  $\nu(y)$  by a bounded amount, while consecutive sequence values differ by the asymptotically fixed factor  $\alpha$ . The final sentence is the definition of  $N$ .  $\square$

**Warning 2.49** (The  $\mathcal{O}(1)$  rounding layer is not a small correction). An assertion of the form  $N(y) = \nu(y) + o(1)$  uniformly in real  $y$  is false, and no descending power–logarithmic series can be true uniformly: arbitrarily small changes of  $y$  across a value  $a_n$  change  $N(y)$  by one. Once a monotone interpolation  $\mathcal{A}_{\text{ex}}$  with  $\mathcal{A}_{\text{ex}}(n) = a_n$  has been fixed, one has exactly  $N(y) = \lceil \nu_{\text{ex}}(y) \rceil$  with  $\nu_{\text{ex}}$  its ordinary inverse, and for non-integral  $\chi$

$$\lceil \chi \rceil = \chi + \frac{1}{2} + \frac{1}{\pi} \sum_{m \geq 1} \frac{\sin(2\pi m \chi)}{m}, \quad (2.79)$$

an  $\mathcal{O}(1)$  periodic layer that never tends to zero. The correct statements are: about  $\nu$ ; or at the special values  $y = a_n$ , where  $\nu(a_n) = n + \mathcal{O}(n^{-M})$  for every fixed  $M$ ; or about a rounded exact interpolation. S2 states this sharply and S3 restates it; S1’s phrase “can never be resolved more accurately than one unit” is the same point stated loosely, and is corrected here. The layer eq. (2.79) must not be confused with the exponentially small oscillation of section 2.9: the latter is genuinely  $o(1)$  and has an entirely different origin.

**Remark 2.50** (A definitional disagreement). S1 and S2 define the staircase as  $\min\{n \geq 2 : a_n \geq y\}$ , S3 as  $\min\{n \geq 1 : a_n \geq y\}$ . The two differ only at  $y \leq 1$ , where the tie  $a_1 = a_2 = 1$  makes the choice a pure convention. This volume uses eq. (2.77) and confines all statements to  $y > 1$ , where the definitions agree.

## 2.9 Exponentially small sectors

### 2.9.1 Why one exponential scale is not the whole story

Equation (2.43) is complete in inverse powers of  $n$  at the scale  $\rho^{-n}$ . It cannot see a term  $\sigma^{-n}$  with  $|\sigma| > \rho$ , because  $(\rho/|\sigma|)^n$  decays faster than every power of  $n^{-1}$ . Such terms come from singularities of the analytic continuation of  $T$ , and the functional equation exposes exactly two mechanisms for producing them:

1. **inherited preimages:** if  $T$  is singular at  $\tau$  and  $\sigma^k = \tau$  for some  $k \geq 2$ , the summand  $T(z^k)/k$  of  $R$  can make  $R$ , hence  $T$ , singular at  $\sigma$ ;
2. **new Cayley critical points:** on a continued sheet where  $R$  is analytic, any  $\sigma$  with  $\zeta(\sigma) = e^{-1}$  and  $\zeta'(\sigma) \neq 0$  is a square-root branch point of  $T$  by theorem 2.16.

Iterating the first mechanism from the dominant point produces the *root-lift skeleton*

$$\mathcal{R} := \{\rho^{1/m} e^{2\pi i j/m} : m \geq 1, 0 \leq j < m\}, \quad (2.80)$$

whose radii increase to 1 and whose arguments become dense. Relative to the dominant sector the level- $m$  points carry the exponential weight

$$\sigma_{m,j}^{-n} = \rho^{-n/m} e^{-2\pi i j n/m}, \quad \text{i.e.} \quad \exp\left[-\lambda \left(1 - \frac{1}{m}\right) n\right] \text{ relative to } \rho^{-n}. \quad (2.81)$$

**Warning 2.51** (What the skeleton is and is not). Equation (2.80) is a list of *candidate* inherited germs generated by the recursion. It is not a theorem that every listed point is singular on a given sheet, that every singularity is listed, or that a listed point is reachable along a prescribed path. Continuation can also meet solutions of  $\zeta(z) = e^{-1}$  that are not in  $\mathcal{R}$ , and a critical point can coalesce with an inherited germ (which, by eq. (2.89) below, halves an exponent and can create quarter powers).

**Remark 2.52** (A disagreement about the unit circle). S1 and S2 assert that  $|z| = 1$  is a natural boundary for  $T$  and that the continued singularities accumulate at every point of it, citing Drmota–Gittenberger. S3 makes no such assertion and states only the candidate skeleton with the caveat of warning 2.51. This volume records the natural-boundary claim as *reported* and does not use it: every statement in this section is conditioned on the explicit finite-radius hypothesis  $H(\mathcal{R})$  of definition 2.54, and nothing below depends on what happens as  $\mathcal{R} \uparrow 1$ .

### 2.9.2 Transfer from a general singular germ

**Theorem 2.53** (Universal sector coefficients). *Let  $\sigma \neq 0$  be an accessible singularity on a specified sheet, put  $u = 1 - z/\sigma$ , and suppose that in a  $\Delta$ -domain at  $\sigma$*

$$T(z) = A_\sigma(u) + \sum_{\beta \in E_\sigma} c_{\sigma,\beta} u^\beta + \mathcal{O}(|u|^{\beta_{\max}}), \quad (2.82)$$

with  $A_\sigma$  analytic,  $E_\sigma \subset \mathbb{Q}_{>0} \setminus \mathbb{N}_0$  finite and  $\beta_{\max} > \max E_\sigma$ . Then the contribution of eq. (2.82) to  $[z^n] T$  is

$$\boxed{\mathfrak{S}_\sigma(n) \sim \sigma^{-n} \sum_{\beta \in E_\sigma} \frac{c_{\sigma,\beta}}{\Gamma(-\beta)} n^{-\beta-1} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r}}, \quad (2.83)$$

so the coefficient of  $\sigma^{-n} n^{-\beta-1-r}$  is exactly  $c_{\sigma,\beta} g_r(\beta)/\Gamma(-\beta)$ .

*Proof.* Apply eq. (2.37) with  $\rho$  replaced by  $\sigma$  to each displayed monomial and expand the gamma ratio by lemma 2.29. The analytic part  $A_\sigma$  has no Hankel jump and contributes no large- $n$  tail; the  $\mathcal{O}$ -term transfers by the usual Hankel estimate.  $\square$

Equation (2.83) is the exact analogue of eq. (2.44) and allows arbitrary ramification exponents, not only half-integers. Relative to the dominant sector one introduces the *action*

$$\Omega_\sigma := \operatorname{Log} \frac{\sigma}{\rho}, \quad \text{so that} \quad \sigma^{-n} = \rho^{-n} e^{-n\Omega_\sigma}. \quad (2.84)$$

**Definition 2.54** (Continuation hypothesis  $H(\mathcal{R})$ ). Fix  $\rho < \mathcal{R} < 1$ . We say  $H(\mathcal{R})$  holds for a chosen continuation of  $T$  if that continuation is analytic on  $|z| < \mathcal{R}$  after removing finitely many pairwise disjoint cuts issuing from a finite set  $\Sigma_{\mathcal{R}}$ ; every  $\sigma \in \Sigma_{\mathcal{R}}$  carries a germ of the form eq. (2.82) in a  $\Delta$ -domain; and the deformed coefficient contour may be taken on a circle of radius at least  $\mathcal{R}$ , up to harmless indentations, on which  $T$  has at most polynomial growth.

**Theorem 2.55** (Finite-level coefficient transseries). Assume  $H(\mathcal{R})$  and let  $\mathfrak{S}_\sigma$  be the multiplier-weighted sector of eq. (2.83) attached to the local Hankel cycle actually traversed. Then, for any finite truncation of the exponent and inverse-power sums,

$$a_n = \sum_{\sigma \in \Sigma_{\mathcal{R}}} \mathfrak{S}_\sigma(n) + \mathcal{O}(\mathcal{R}^{-n} n^K) \quad (2.85)$$

for a finite  $K$  determined by the boundary growth.

*Proof.* Deform the Cauchy contour from a small circle about 0 into the union of small Hankel contours around the cuts issuing from  $\Sigma_{\mathcal{R}}$  plus an outer contour of radius  $\mathcal{R}$ . On each Hankel contour apply theorem 2.53; the outer contour is bounded by  $\mathcal{R}^{-n}$  times the allowed polynomial factor. Only finitely many singularities and finitely many local terms are retained, so the deformation and the term-by-term summation are legitimate. Passing to a different sheet changes the homology coefficients of the local cycles; those coefficients are the Stokes/intersection multipliers absorbed into  $\mathfrak{S}_\sigma$ .  $\square$

**S1’s global transseries is not a theorem.** S1 displays  $a_n \sim \mathcal{A}_\rho(n) + \sum_{\sigma \in \mathfrak{S} \setminus \{\rho\}} S_\sigma \mathcal{A}_\sigma(n)$  as a numbered displayed result over “the singular set encountered by the chosen contour continuation”, with no hypothesis attached, while conceding in its introduction that globally the formula “is best understood as a formal resurgent/Darboux transseries”. The volume replaces it by theorem 2.55, which is S3’s formulation and the only one of the three carrying an explicit hypothesis; the unrestricted sum over an infinite  $\mathfrak{S}$  is not asserted anywhere here. The correct reading of a “global” transseries in the presence of accumulating singularities is the compatible family of the finite-radius expansions eq. (2.85) as  $\mathcal{R}$  grows, which is exactly what S2’s “projective meaning” remark says.

### 2.9.3 The local recursion produces every germ

**Lemma 2.56** (Pull-back coordinates). Let  $z = \sigma(1 - u)$  and  $\tau = \sigma^k$ . Then

$$1 - \frac{z^k}{\tau} = 1 - (1 - u)^k = k u \left( 1 + \sum_{r \geq 1} p_{k,r} u^r \right), \quad p_{k,r} = \frac{(-1)^r}{k} \binom{k}{r+1}, \quad (2.86)$$

and consequently, for a parent monomial,

$$\left( 1 - \frac{z^k}{\tau} \right)^\beta = (ku)^\beta \sum_{N \geq 0} u^N \sum_{j=0}^N \binom{\beta}{j} \widehat{B}_{N,j}(p_{k,1}, p_{k,2}, \dots). \quad (2.87)$$

*Proof.*  $1 - (1 - u)^k = -\sum_{r \geq 1} \binom{k}{r} (-u)^r = ku - \sum_{r \geq 2} \binom{k}{r} (-1)^r u^r$ ; dividing by  $ku$  and re-indexing  $r \mapsto r+1$  gives  $p_{k,r}$ . Equation (2.87) is lemma 1.12 applied to the unit series  $1 + \sum_r p_{k,r} u^r$ .  $\square$

**Theorem 2.57** (Recursive completeness of the local calculus). *Fix a sheet and a point  $\sigma \neq 0$ , and choose a local uniformiser  $z = \sigma(1 - v^L)$  in which every parent germ needed in  $R(z)$  is an ordinary power series in  $v$ . Then the local germ of  $T$  at  $\sigma$  is computable to any finite order by the following finite operations, and only these:*

1. *pull back each singular summand  $T(z^k)/k$  by eq. (2.87);*
2. *add the regular Taylor parts to obtain  $R$ , and exponentiate by eq. (2.29) to obtain  $\zeta = ze^R$ ;*
3. *if  $\zeta_0 := \zeta(\sigma) \neq e^{-1}$  on the reached branch, so that the selected Cayley branch  $C_b$  is analytic at  $\zeta_0$ , compose regularly:*

$$T = \sum_{r \geq 0} \frac{C_b^{(r)}(\zeta_0)}{r!} (\zeta - \zeta_0)^r, \quad C_b'(w) = \frac{C_b(w)}{w(1 - C_b(w))}, \quad (2.88)$$

*the higher derivatives following by Faà di Bruno;*

4. *if  $\zeta_0 = e^{-1}$  and the reached branch has  $T(\sigma) = 1$ , compose critically with the universal germ of theorem 2.16:*

$$T = 1 + \sum_{r \geq 1} c_r \theta_\sigma^{r/2}, \quad \theta_\sigma := 1 - e\zeta. \quad (2.89)$$

*If  $\theta_\sigma$  vanishes to order  $\varsigma$  in  $v$ , then  $\theta_\sigma^{1/2}$  begins at  $v^{\varsigma/2}$  and the ramification index doubles when  $\varsigma$  is odd; this is the only new ramification mechanism.*

*Proof.* Induction on the depth of  $\sigma$  in the pull-back graph. At depth 0, theorem 2.24 gives the dominant germ. At the induction step,  $z \mapsto z^k$  converts each already-known germ into a Puiseux series in a common uniformiser by eq. (2.87); the ring  $\mathbb{C}[[v]]$  is closed under the sum defining  $R$ , under exponentiation and under multiplication by  $z = \sigma(1 - v^L)$ ; and the outer composition is either eq. (2.88) or eq. (2.89), both of which determine every coefficient at finite order by definitions 1.9 and 1.10. The derivative formula in eq. (2.88) follows by differentiating  $C = we^C$ .  $\square$

**Remark 2.58** (A dedup). S1 introduces a “generalized Bell algebra”  $\mathfrak{B}_{\gamma, m}$  on a semigroup-graded power-series ring for this step. Theorem 2.57 avoids it: after a common uniformiser  $v$  has been chosen, everything is an *ordinary* power series in  $v$  and the standard  $\hat{B}$ ,  $B$  of definitions 1.9 and 1.10 suffice. That is S3’s organisation and is adopted here; nothing is lost, because the choice of  $L$  is exactly what the semigroup grading was encoding.

### 2.9.4 The first inherited sector, at $-\sqrt{\rho}$

The point

$$\sigma_- := -\sqrt{\rho} = -0.5816544136333942538101 \dots \quad (2.90)$$

is reached by continuing the germ at 0 along the negative real axis, which never meets the dominant positive singularity. At  $z = \sigma_-$  the summand  $T(z^2)/2$  of  $R$  reaches  $T(\rho)/2$  and inherits the dominant square-root branch, while every  $T(z^k)$  with  $k \geq 3$  is evaluated at  $|\sigma_-|^k \leq \rho^{3/2} < \rho$ , strictly inside the original disc. Take the uniformiser

$$z = \sigma_-(1 - v^2), \quad v = \left(1 - \frac{z}{\sigma_-}\right)^{1/2}, \quad \text{so} \quad 1 - \frac{z^2}{\rho} = v^2(2 - v^2). \quad (2.91)$$

**Proposition 2.59** (Complete local data at  $\sigma_-$ ). *Write  $R(\sigma_-(1 - v^2)) = \sum_{m \geq 0} r_m^- v^m$  and  $\zeta(\sigma_-(1 - v^2)) = \sum_{m \geq 0} \zeta_m^- v^m$ . Then, with  $t_0 := 1$ ,*

$$r_m^- = \frac{1}{2} \sum_{\substack{0 \leq r \leq m \\ r \equiv m(2)}} t_r (-1)^{\frac{m-r}{2}} 2^{\frac{r}{2} - \frac{m-r}{2}} \binom{r/2}{\frac{m-r}{2}} + \mathbf{1}_{2|m} (-1)^{m/2} \sum_{k \geq 3} \frac{1}{k} \sum_{n \geq 1} a_n \sigma_-^{kn} \binom{kn}{m/2}, \quad (2.92)$$

both parts converging geometrically. Defining  $\mathcal{E}_m^-$  by  $\sum_{m \geq 0} \mathcal{E}_m^- v^m = \exp(\sum_{m \geq 0} r_m^- v^m)$ , so that  $\mathcal{E}_m^- = e^{r_0^-} \sum_{k \geq 0} \frac{1}{k!} \hat{B}_{m,k}(r_1^-, r_2^-, \dots)$ , one has

$$\zeta_m^- = \sigma_- (\mathcal{E}_m^- - \mathcal{E}_{m-2}^-), \quad \mathcal{E}_{<0}^- := 0. \quad (2.93)$$

Setting  $y_- := \mathcal{C}_0(\zeta_0^-) = -W_0(-\zeta_0^-)$ , the outer composition is regular because  $y_- \neq 1$ , and writing  $T(\sigma_-(1-v^2)) = y_- + \sum_{m \geq 1} t_m^- v^m$  one has the closed form

$$\boxed{r_1^- = \frac{t_1}{\sqrt{2}}, \quad t_1^- = \frac{y_-}{1-y_-} \cdot \frac{t_1}{\sqrt{2}}.} \quad (2.94)$$

*Proof.* The  $k = 2$  contribution to  $R$  is  $\frac{1}{2}T(\rho(1-v^2(2-v^2)))$ ; inserting eq. (2.33) with  $s = v^2(2-v^2)$ , so  $s^{r/2} = v^r 2^{r/2} (1-v^2/2)^{r/2}$ , and expanding  $(1-v^2/2)^{r/2}$  binomially gives the first sum. The  $k \geq 3$  contributions are  $\sum_{k \geq 3} \frac{1}{k} \sum_n a_n \sigma_-^{kn} (1-v^2)^{kn}$ , whose  $v^{2j}$  coefficient is the second sum; convergence is geometric already at  $k = 3$  because  $|\sigma_-|^3 < \rho$ . Equation (2.93) is  $\zeta = \sigma_-(1-v^2)e^R$ . For eq. (2.94): only  $r = 1$  contributes to  $v^1$  in the first sum, giving  $r_1^- = \frac{1}{2}t_1\sqrt{2} = t_1/\sqrt{2}$ ; hence  $\zeta_1^- = \zeta_0^- r_1^-$  and, by eq. (2.88) with  $\mathcal{C}'_0(w) = \mathcal{C}_0(w)/(w(1-\mathcal{C}_0(w)))$ ,  $t_1^- = \mathcal{C}'_0(\zeta_0^-)\zeta_1^- = \frac{y_-}{\zeta_0^-(1-y_-)}\zeta_0^- r_1^-$ , which is the claim.  $\square$

Recomputed at 80 working digits (see the repair below),

$$\begin{aligned} r_0^- &= 0.4686366279248455480872, & \zeta_0^- &= -0.9293757352012050888501, \\ y_- &= -0.5410329414204534926205, & t_1^- &= 0.3871501110302429034772, \end{aligned} \quad (2.95)$$

and the first local coefficients are

$$\begin{aligned} t_2^- &= -0.03256491851564763414731, & t_3^- &= 0.01720042160312209148301, \\ t_4^- &= 0.10407703951838443852546, & t_5^- &= 0.16194736077270236118952, \\ t_6^- &= -0.06292599818799308689694, & t_7^- &= -0.24403103923142135135510. \end{aligned} \quad (2.96)$$

**S2's minus-sector constants are printed past their accuracy.** S2 prints  $r_0^-$ ,  $\zeta_0^-$ ,  $y_-$  and  $b_1^-$  to 40 significant digits, e.g.  $r_0^- = 0.4686366279248455480872026033545827113234$ . Two independent recomputations here – one through the germ recursion at 90 digits, one by direct summation of  $\frac{1}{2} + \sum_{k \geq 3} T(\sigma_-^k)/k$  at 80 digits with 1400 exact tree numbers and no truncation of the inner sums – agree with each other and give  $r_0^- = 0.468636627924845548087202621931819314\dots$ , which departs from S2's value at the *twenty-sixth* significant digit. The same 25-digit agreement, and no more, is found for  $\zeta_0^-$ ,  $y_-$ ,  $t_1^-$  and every derived quantity, consistent with a single truncation of the  $k \geq 3$  tail in S2's computation. All minus-sector constants are therefore quoted here to at most 22 digits.

Only the odd  $t_{2j+1}^-$  are singular in  $z$  at  $\sigma_-$ , so theorem 2.53 with  $\beta = j + \frac{1}{2}$  gives

$$\boxed{a_n^{(-)} \sim \mathfrak{s}_- \sigma_-^{-n} n^{-3/2} \sum_{k \geq 0} \frac{B_k^-}{n^k}, \quad B_k^- = \sum_{j=0}^k \frac{t_{2j+1}^-}{\Gamma(-j-\frac{1}{2})} g_{k-j}(j+\frac{1}{2}),} \quad (2.97)$$

where  $\mathfrak{s}_-$  is the multiplier of the local cycle; for the direct negative-axis continuation it is 1, and it is displayed only to keep the contour dependence visible in larger deformations. Since  $\sigma_-^{-n} = (-1)^n \rho^{-n/2}$ , the sector is alternating, with

$$\begin{aligned} B_0^- &= -0.10921302995631017570549, & B_1^- &= -0.03367666220778285336719, \\ B_2^- &= -0.17900090117304631638903, & B_3^- &= -1.64234200495327998300, \\ B_4^- &= -18.9266458632920839673, & B_5^- &= -277.723089846678995649. \end{aligned} \quad (2.98)$$

Relative to the leading dominant term its amplitude is

$$\frac{B_0^-}{C} = -0.2482543049151649529916 \dots, \quad \frac{a_n^{(-)}}{a_n^{(0)}} \sim -0.2482543049 \dots (-\sqrt{\rho})^n, \quad (2.99)$$

which is smaller than every power of  $1/n$ : this is the first explicitly computed genuinely beyond-all-orders correction to A000081. Its action is

$$\Omega_- = \text{Log} \frac{\sigma_-}{\rho} = \frac{\lambda}{2} + i\pi \pmod{2\pi i}, \quad \frac{\Omega_-}{\lambda} = \frac{1}{2} + i\frac{\pi}{\lambda}, \quad \frac{\pi}{\lambda} = 2.8987964297339787377 \dots \quad (2.100)$$

*Remark 2.60.* The point  $+\sqrt{\rho}$  lies behind the dominant branch point on real continuation and belongs to sheet-dependent continuations; complex critical points and overlapping root lifts may occur farther out. Their local data are produced by theorem 2.57 and their multipliers depend on the chosen contour. Equation (2.97) is one explicit sector, not an assertion that it exhausts modulus  $\sqrt{\rho}$ .

### 2.9.5 Transport of the sectors through the inverse

Write the completed forward interpolation as

$$\mathfrak{a}(x) = \mathcal{A}(x)(1 + \mathcal{E}(x)), \quad \mathcal{E}(x) = \sum_{\Omega \neq 0} e^{-\Omega x} x^{\mu_\Omega} \sum_{r \geq 0} e_{\Omega, r} x^{-r}, \quad (2.101)$$

with  $\mathcal{A}$  from eq. (2.53) and the support of  $\mathcal{E}$  contained in the additive semigroup generated by the actions eq. (2.84). For non-integer  $x$  a factor  $\sigma^{-x}$  requires a choice of  $\text{Log} \sigma$ ; different choices agree at integers and differ between them, so the exponentially small part of a *functional* inverse is not canonical until a continuation and pairing convention has been fixed. Fix one, pairing conjugate branches so that  $\mathfrak{a}$  is real on the positive axis.

**Proposition 2.61** (Sector transport). *Let  $\nu_0$  be the complete algebraic inverse of  $\mathcal{A}$  (eq. (2.63)) and put  $\nu_{\text{full}} = \nu_0 + \Delta$ . Then  $\Delta$  is the unique solution, in the transseries field ordered by the semigroup of actions, of*

$$\Delta = \Psi_{\nu_0}(\Delta), \quad \Psi_{\nu_0}(w) := -\frac{\log(1 + \mathcal{E}(\nu_0 + w))}{\mathcal{Q}_{\nu_0}(w)}, \quad \mathcal{Q}_x(w) := \frac{f(x+w) - f(x)}{w}, \quad (2.102)$$

with  $f = \log \mathcal{A}$  and  $\mathcal{Q}_x(0) = f'(x)$ ; its coefficients are given by lemma 1.16, so that in particular

$$\Delta = -\frac{\mathcal{E}(\nu_0)}{f'(\nu_0)} + \mathcal{O}(\mathcal{E}^2) \sim -\frac{\mathcal{E}(\nu_0)}{\lambda}, \quad (2.103)$$

and a single relative forward sector  $\mathcal{E}_\Omega \sim e_{\Omega,0} e^{-\Omega x} x^{\mu_\Omega}$  maps to

$$\Delta_\Omega(y) \sim -\frac{e_{\Omega,0}}{\lambda} C^{\Omega/\lambda} y^{-\Omega/\lambda} \nu_0^{\mu_\Omega - p\Omega/\lambda}. \quad (2.104)$$

*Proof.*  $f(\nu_0 + \Delta) - f(\nu_0) + \log(1 + \mathcal{E}(\nu_0 + \Delta)) = 0$  is the exact equation, and  $f(\nu_0 + w) - f(\nu_0) = w \mathcal{Q}_{\nu_0}(w)$  by definition; dividing gives eq. (2.102). Since  $f'(\nu_0)$  has leading term  $\lambda \neq 0$  it is invertible in the coefficient field, and every occurrence of  $\Delta$  on the right is either nonlinear or multiplied by a positive transmonomial, so the support is well ordered and each requested transmonomial receives finitely many contributions; lemma 1.16 then gives the coefficients. Equation (2.103) is the first term. For eq. (2.104), use  $y \sim C e^{\lambda \nu_0} \nu_0^{-p}$  to write  $e^{-\Omega \nu_0} = (y/C)^{-\Omega/\lambda} \nu_0^{-p\Omega/\lambda}$ .  $\square$

For the first inherited sector,  $\Omega_- = \frac{\lambda}{2} + i\pi$  and  $e_{-,0} = B_0^-/C$ . Pairing the two lateral determinations  $\pm i\pi$  replaces  $e^{-i\pi x}$  by  $\cos(\pi x)$ , and  $e^{-\lambda \nu_0/2} = C^{1/2} y^{-1/2} \nu_0^{-3/4}$ , so

$$\Delta_-(y) \sim 0.15193347365953098826 \dots y^{-1/2} \nu_0(y)^{-3/4} \cos(\pi \nu_0(y)), \quad (2.105)$$

the amplitude being  $\sqrt{C} \cdot (-B_0^-/(\lambda C))$  with  $-B_0^-/(\lambda C) = 0.22906811038403197857\dots$ . The phase is

$$\pi\nu_0(y) = \frac{\pi}{\lambda} \log y + \frac{3\pi}{2\lambda} \log \log y + \mathcal{O}(1), \quad \frac{\pi}{\lambda} = 2.89879642973397873771\dots, \quad \frac{3\pi}{2\lambda} = 4.34819464460096810657\dots \quad (2.106)$$

so the first nonperturbative correction to the inverse has envelope  $y^{-1/2}(\log y)^{-3/4}$  and a logarithmic oscillation with a slower  $\log y$  drift.

**Warning 2.62** (Two different oscillations). Equation (2.105) tends to zero and comes from a complex singular action. The sawtooth of eq. (2.79) is  $\mathcal{O}(1)$ , never tends to zero, and comes from integer rounding. They have different origins, different scales, and must not be conflated.

## 2.10 Other inverse notions: the compositional inverse of $T$

Since  $T(z) = z + \mathcal{O}(z^2)$  with integer coefficients and unit linear coefficient,  $T$  has a compositional inverse  $Z := T^{\langle -1 \rangle} \in z\mathbb{Z}[[z]]$ , unrelated to the growth inversion of section 2.7.

**Proposition 2.63** (Coefficients of  $Z$ ). For  $n \geq 1$ ,

$$[w^n] Z(w) = \frac{1}{n} [z^{n-1}] \left( \frac{z}{T(z)} \right)^n = \frac{1}{n} [z^{n-1}] \exp \left( -n \sum_{k \geq 1} \frac{T(z^k)}{k} \right), \quad (2.107)$$

a finite ordinary-Bell expression in  $a_1, \dots, a_{n-1}$ . The first sixteen values, recomputed by exact rational arithmetic, are

$$1, -1, 0, 1, -1, 1, -4, 11, -18, 18, 0, -60, 189, -360, 453, -373. \quad (2.108)$$

*Proof.* The first equality is theorem 1.14 with  $\phi(z) = T(z)/z$ , a unit series. The second is eq. (2.5) rearranged, and definition 1.9 converts the exponential into a finite Bell sum. Integrality is automatic for the compositional inverse of a series in  $z + z^2\mathbb{Z}[[z]]$ .  $\square$

**Remark 2.64.** S1 displays  $Z(w) = w - w^2 + w^4 - w^5 + w^6 - 4w^7 + 11w^8 - \dots$ , which agrees with eq. (2.108), and identifies the coefficients with OEIS A050395. The expansion is confirmed here; the OEIS identification is reported from S1 and has not been re-checked in this volume.

Near the dominant value  $w = 1$  the branch of  $Z$  through  $Z(1) = \rho$  is *analytic* in  $1 - w$ , because it undoes the square root: from  $T = \zeta e^T$  one has  $\zeta = T e^{-T}$ , hence

$$Z(w) = \zeta^{\langle -1 \rangle} (w e^{-w}) \quad (2.109)$$

on that branch,  $\zeta$  being injective on  $(0, \sqrt{\rho})$  by proposition 2.11. Since  $w e^{-w} - e^{-1}$  has a double zero at  $w = 1$ , the local expansion of  $Z$  at  $w = 1$  has no linear term. Its coefficients follow from one more Lagrange extraction: writing  $\zeta(\rho + x) - e^{-1} = x \psi(x)$  with  $\psi(0) = \zeta'(\rho) \neq 0$ ,

$$[v^m] (\zeta^{\langle -1 \rangle} (e^{-1} + v) - \rho) = \frac{1}{m} [x^{m-1}] \psi(x)^{-m}, \quad (2.110)$$

followed by composition with  $(1 - \varsigma) e^{-(1-\varsigma)} - e^{-1} = -e^{-1} \sum_{m \geq 2} \frac{m-1}{m!} \varsigma^m = -e^{-1} \frac{\varsigma^2}{2} \Phi(\varsigma)$ , the same kernel eq. (2.22) that governed the forward branch.

## 2.11 Algorithms, validation and audit

### 2.11.1 The pipeline

**Algorithm 2.65** (All dominant and inverse coefficients through order  $M$ ).

1. Generate  $a_1, \dots, a_N$  and  $b_1, \dots, b_N$  exactly by eq. (2.7) with a divisor sieve, checking the divisibility assertion at each step; form  $\varrho_r$  by eq. (2.10).
2. Solve  $\log x + 1 + \sum_{r=2}^N \varrho_r x^r = 0$  for  $\rho$  by safeguarded Newton or interval bisection. Choose  $N$  so that  $\rho^{N/2}$  is below the target precision; the omitted tail is geometrically small because  $R$  is analytic on  $|z| < \sqrt{\rho}$  (proposition 2.10).
3. Compute  $\widehat{R}_0, \dots, \widehat{R}_{2M+1}$  by eq. (2.28); exponentiate by eq. (2.29) and form  $\theta_1, \dots, \theta_{2M+2}$  by eq. (2.30), then  $u_j = \theta_{j+1}/\theta_1$ .
4. Compute  $\omega_1, \dots, \omega_{2M+1}$  exactly by eq. (2.23) (rationals), set  $c_r = -2^{r/2}\omega_r$ , and form  $t_1, t_3, \dots, t_{2M+1}$  by eq. (2.34).
5. Compute  $g_r(j + \frac{1}{2})$  exactly for  $j + r \leq M$  by eq. (2.39), then  $\tau_0, \dots, \tau_M$  by eq. (2.44), then  $C = \tau_0/\sqrt{\pi}$  and  $f_m = \tau_m/\tau_0$ .
6. Reciprocal:  $\bar{f}_m$  by eq. (2.50).
7. Inverse:  $\varphi_m$  by eq. (2.55), then  $d_m$  by eq. (2.61), or  $P_m(L)$  by the flattening recursion of theorem 1.33.
8. Sectors: apply theorem 2.57 outward in modulus, transfer each germ by eq. (2.83), and invert by proposition 2.61.

At every stage the coefficient of order  $m$  depends only on data of order at most  $m$ ; no nonlinear global solve occurs after  $\rho$  is known. The useful power-series primitives are the three dense recurrences

$$[v^n] F^\gamma = \frac{1}{n} \sum_{k=1}^n ((\gamma+1)k - n) F_k [v^{n-k}] F^\gamma, \quad E_n = \frac{1}{n} \sum_{k=1}^n k A_k E_{n-k}, \quad H_n = F_n - \frac{1}{n} \sum_{k=1}^{n-1} k H_k F_{n-k}, \quad (2.111)$$

for  $F^\gamma$  with  $F_0 = 1$ , for  $E = e^A$  with  $A_0 = 0$ , and for  $H = \log F$  with  $F_0 = 1$ . They are equivalent to the Bell forms used in the statements and are what an implementation should actually run.

*Warning 2.66* (Precision discipline). The coefficients grow rapidly and eq. (2.44) combines alternating contributions from several Puiseux coefficients, so obtaining  $d$  reliable digits of  $\tau_M$  can require substantially more than  $d$  working digits. The tables in this chapter were produced with 90–110 decimal working digits, with universal objects  $(\omega_m, g_r(\beta), a_n)$  held in exact rational or integer arithmetic. The repair recorded at section 2.9.4 is exactly what happens when this discipline lapses.

### 2.11.2 Numerical validation

Let  $A_M(n)$  denote eq. (2.43) truncated after  $\tau_M n^{-M}$ . Table 2.5 reports the signed relative error  $(A_M(n) - a_n)/a_n$  against exact integers from eq. (2.7); table 2.6 reports the signed index error  $\nu_M(a_n) - n$  for the Lambert-normalised inverse  $\nu_M = x_0 + \sum_{m \leq M} d_m x_0^{-m}$ . Both tables were regenerated here and agree entry by entry with the corresponding tables of S1 and S2 at the displayed precision.

Table 2.5: Signed relative error  $(A_M(n) - a_n)/a_n$  of the forward expansion.

| $n$ | $M = 0$               | $M = 2$               | $M = 4$                | $M = 6$                | $M = 8$                | $M = 10$               |
|-----|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
| 10  | $-1.52 \cdot 10^{-2}$ | $-3.49 \cdot 10^{-4}$ | $2.63 \cdot 10^{-3}$   | $3.99 \cdot 10^{-3}$   | $5.52 \cdot 10^{-3}$   | $8.77 \cdot 10^{-3}$   |
| 20  | $-6.58 \cdot 10^{-3}$ | $-3.42 \cdot 10^{-4}$ | $-3.18 \cdot 10^{-5}$  | $1.04 \cdot 10^{-6}$   | $9.73 \cdot 10^{-6}$   | $1.41 \cdot 10^{-5}$   |
| 50  | $-2.22 \cdot 10^{-3}$ | $-1.77 \cdot 10^{-5}$ | $-2.88 \cdot 10^{-7}$  | $-1.20 \cdot 10^{-8}$  | $-9.84 \cdot 10^{-10}$ | $-1.36 \cdot 10^{-10}$ |
| 100 | $-1.06 \cdot 10^{-3}$ | $-2.08 \cdot 10^{-6}$ | $-8.06 \cdot 10^{-9}$  | $-7.88 \cdot 10^{-11}$ | $-1.50 \cdot 10^{-12}$ | $-4.73 \cdot 10^{-14}$ |
| 200 | $-5.15 \cdot 10^{-4}$ | $-2.53 \cdot 10^{-7}$ | $-2.40 \cdot 10^{-10}$ | $-5.73 \cdot 10^{-13}$ | $-2.66 \cdot 10^{-15}$ | $-2.04 \cdot 10^{-17}$ |
| 500 | $-2.03 \cdot 10^{-4}$ | $-1.59 \cdot 10^{-8}$ | $-2.39 \cdot 10^{-12}$ | $-9.02 \cdot 10^{-16}$ | $-6.62 \cdot 10^{-19}$ | $-8.01 \cdot 10^{-22}$ |

Two features are worth naming. First, at  $n = 10$  and  $n = 20$  the error stops improving and then worsens: the expansion is asymptotic, and the least-term truncation of theorem 1.46 has been passed.



Table 2.6: Signed index error  $\nu_M(a_n) - n$  of the Lambert-normalised inverse;  $M = 0$  is the bare carrier  $x_0$ .

| $n$ | $M = 0$              | $M = 1$              | $M = 2$              | $M = 4$               | $M = 6$               | $M = 8$               |
|-----|----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
| 10  | $1.64 \cdot 10^{-2}$ | $7.15 \cdot 10^{-3}$ | $1.29 \cdot 10^{-3}$ | $-2.61 \cdot 10^{-3}$ | $-4.16 \cdot 10^{-3}$ | $-5.76 \cdot 10^{-3}$ |
| 20  | $6.55 \cdot 10^{-3}$ | $1.91 \cdot 10^{-3}$ | $4.44 \cdot 10^{-4}$ | $3.76 \cdot 10^{-5}$  | $-2.02 \cdot 10^{-7}$ | $-9.39 \cdot 10^{-6}$ |
| 50  | $2.11 \cdot 10^{-3}$ | $2.59 \cdot 10^{-4}$ | $2.32 \cdot 10^{-5}$ | $3.32 \cdot 10^{-7}$  | $1.26 \cdot 10^{-8}$  | $9.98 \cdot 10^{-10}$ |
| 100 | $9.88 \cdot 10^{-4}$ | $6.16 \cdot 10^{-5}$ | $2.74 \cdot 10^{-6}$ | $9.34 \cdot 10^{-9}$  | $8.35 \cdot 10^{-11}$ | $1.53 \cdot 10^{-12}$ |
| 200 | $4.78 \cdot 10^{-4}$ | $1.50 \cdot 10^{-5}$ | $3.33 \cdot 10^{-7}$ | $2.79 \cdot 10^{-10}$ | $6.09 \cdot 10^{-13}$ | $2.71 \cdot 10^{-15}$ |
| 500 | $1.88 \cdot 10^{-4}$ | $2.38 \cdot 10^{-6}$ | $2.10 \cdot 10^{-8}$ | $2.78 \cdot 10^{-12}$ | $9.59 \cdot 10^{-16}$ | $6.75 \cdot 10^{-19}$ |

Second, the inverse errors track the forward relative errors divided by  $f' \approx \lambda$ , exactly as theorem 2.42 predicts; this is why tables 2.5 and 2.6 are so nearly proportional.

*Remark 2.67* (On comparisons with the literature). S1 states that its first five coefficients “agree with Genitrini’s independently computed values” and S3 that its Puiseux table “agrees with the independent table in [Genitrini 2016]”. Those comparisons are reported from the drafts and have *not* been re-performed here; what has been done is a full independent recomputation of every constant from the exact recurrence, described in section 2.11.3.

### 2.11.3 Audit

Every numerical value printed in this chapter was produced by an independent implementation of algorithm 2.65 written for this volume: exact integer generation of  $a_1, \dots, a_N$  with  $N$  between 700 and 1400; exact rational  $\varrho_r, \omega_m$  and  $g_r(\beta)$  (the last from an exact Bernoulli-number table); and 60–110 decimal working digits for everything else. The following independent identities were checked and hold to all digits carried.

1.  $(n - 1) \mid \sum_{j < n} b_j a_{n-j}$  at every step, for  $n \leq 1400$ ;  $a_{10} = 719$ ,  $a_{20} = 12\,826\,228$ ,  $a_{50} = 425\,976\,989\,835\,141\,038\,353$ .
2.  $\lambda = 1 + R(\rho)$ , i.e. eq. (2.20).
3.  $\theta_0 = 0$ , the vanishing of the constant term in eq. (2.26) predicted by  $e\zeta(\rho) = 1$ .
4.  $C^2 = \theta_1/(2\pi)$ , i.e. eq. (2.45) – the two sides travel through disjoint parts of the pipeline.
5.  $c_r = -2^{r/2}\omega_r$  for  $1 \leq r \leq 18$  in exact arithmetic, and  $t_1 = -\sqrt{2\theta_1}$ ,  $t_2 = \frac{2}{3}\theta_1$ .
6. Equation (2.47) re-derived symbolically from eq. (2.44), and the  $g_r(\beta)$  of eqs. (2.41) and (2.42) re-derived from eq. (2.38).
7.  $D_m = \lambda^{m+1}d_m$  and  $\eta_m^{S2} = \lambda^m\varphi_m$  for  $1 \leq m \leq 8$ , reconciling S1/S3 with S2.
8.  $P_1(0) = d_1$ ,  $P_2(0) = d_2$  (a case of eq. (2.72)).
9.  $r_1^- = t_1/\sqrt{2}$  and the closed form eq. (2.94), each against the germ recursion; and  $r_0^-$  by a second, wholly independent summation (see the repair in section 2.9.4).
10. The single-sum eq. (2.28) against the siblings’ double sum, for  $0 \leq m \leq 30$ .

## 2.12 Summary

The Pólya equation becomes transparent after the factorisation  $T = \mathcal{C} \circ \zeta$  with  $\mathcal{C}(w) = -W_0(-w)$ : the universal layer is one Lagrange coefficient  $[v^{r-1}] \Phi(v)^{-r/2}$ , the problem-specific layer is ordinary Bell

Table 2.7: Principal symbols of this chapter.

| Symbol                                  | Meaning                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $a_n, T(z)$                             | A000081 and its ordinary generating function.                                                                |
| $b_m, \varrho_r$                        | Divisor transform $\sum_{d m} da_d$ ; coefficients of $R$ , $\varrho_r = (b_r - ra_r)/r$ .                   |
| $R, \zeta$                              | Disturbance $\sum_{k \geq 2} T(z^k)/k$ and Cayley argument $ze^R$ ; $T = -W_0(-\zeta)$ .                     |
| $\rho, \alpha, \lambda, p$              | Otter singularity; $\alpha = \rho^{-1}$ , $\lambda = \log \alpha$ , $p = \frac{3}{2}$ .                      |
| $\Phi, \omega_m, c_r$                   | Universal Cayley kernel and its two coefficient normalisations, $c_r = -2^{r/2}\omega_r$ .                   |
| $s, \theta, \theta_m, u_j$              | Local coordinate $1 - z/\rho$ ; disturbance $1 - e\zeta$ ; its coefficients; $u_j = \theta_{j+1}/\theta_1$ . |
| $\hat{R}_m, \ell_j$                     | $[s^m] R(\rho(1 - s))$ ; scaled logarithmic jets $\rho^j (\log \zeta)^{(j)}(\rho)$ .                         |
| $t_n$                                   | Puiseux coefficients, $T = 1 + \sum t_n s^{n/2}$ .                                                           |
| $\varkappa_m(\beta), g_r(\beta)$        | Bernoulli increments and gamma-ratio transfer coefficients.                                                  |
| $\tau_m, C, f_m, \bar{f}_m$             | Forward coefficients; $C = \tau_0/\sqrt{\pi}$ ; $f_m = \tau_m/\tau_0$ ; reciprocal coefficients.             |
| $F, H, \varphi_m$                       | $F = 1 + \sum f_m v^m$ ; $H = \log F = \sum \varphi_m v^m$ ; this $H$ is the Part 0 datum $Q$ .              |
| $x_0, d_m, \nu$                         | Lambert carrier; its corrections; the smooth asymptotic inverse.                                             |
| $\mathcal{X}, L, q, \eta_k, P_m$        | Flattening variables and coefficient polynomials.                                                            |
| $N(y)$                                  | Integer staircase $\min\{n \geq 1 : a_n \geq y\}$ .                                                          |
| $\sigma_-, t_m^-, B_k^-, \Omega_\sigma$ | The inherited singularity $-\sqrt{\rho}$ , its local and sector coefficients, and the actions.               |

composition of the analytic disturbance  $\zeta$ , and coefficient transfer is Bernoulli polynomials in the gamma ratio. The chain is

$$a_n \rightarrow \varrho_r \rightarrow \rho \rightarrow \hat{R}_m \rightarrow \theta_m \rightarrow t_n \rightarrow \tau_m \rightarrow f_m \rightarrow \varphi_m \rightarrow d_m \text{ or } P_m(L), \quad (2.112)$$

every arrow a finite Bell- or Bernoulli-polynomial formula. Inversion adds nothing new: A000081 is the  $\alpha = 1$  case of definition 1.2 with  $a = \lambda$ ,  $b = -\frac{3}{2}$ ,  $Q = H$ , so theorem 1.20 supplies the carrier exactly, theorem 1.25 every algebraic correction, theorem 1.33 the logarithmic form, theorem 1.44 the error, and theorem 1.39 the passage to the integer inverse. Beyond all orders, the recursion supplies a singularity grammar whose first explicit instance is the alternating sector of relative size  $0.2483 \dots (-\sqrt{\rho})^n$  at  $\sigma_- = -\sqrt{\rho}$ , transported by proposition 2.61 into an inverse correction of envelope  $y^{-1/2}(\log y)^{-3/4}$  with a  $\log y$ -periodic phase. What remains open is not coefficient algebra but analysis: the classification of accessible sheets, of critical preimages, and of their Stokes connections – and, prior to that, a proof of  $H(\mathcal{R})$  for  $\mathcal{R}$  close to 1.

## Chapter 3

# The double factorial

This chapter merges three independent treatments of  $n!!$ : *Double\_Factorial\_Transseries* (henceforth S1), *Double\_Factorial\_Transseries-3* (S2), and *double\_factorial\_transseries-2* (S3). All three reach the same forward and inverse coefficients; they differ in organisation, in what is proved, and – decisively – in notation, where the letter  $C$  carries three incompatible meanings. The chapter adopts S3’s *exact* centred normal form as its spine, S2’s Borel kernel and sharp remainder, S2’s graded closed coefficient formula, and S1/S2’s parity-mean centring of the periodic interpolation. Every displayed rational number below was regenerated with exact rational arithmetic (sympy) from the defining recurrences, and every numerical table was recomputed at 220-digit precision (mpmath); see section 3.14.

### 3.1 Orientation: which inverse?

For  $n \in \mathbb{N}$  the double factorial is

$$0!! = 1!! = 1, \quad n!! = n(n-2)!! \quad (n \geq 2), \quad (3.1)$$

so that  $n!!$  is the product of the positive integers not exceeding  $n$  of the same parity as  $n$ . The sequence  $1, 1, 2, 3, 8, 15, 48, 105, 384, 945, \dots$  is OEIS A006882. Its elementary closed forms already contain the whole difficulty:

$$(2m)!! = 2^m m!, \quad (2m+1)!! = \frac{2^{m+1} \Gamma(m + \frac{3}{2})}{\sqrt{\pi}}. \quad (3.2)$$

The even subsequence is a gamma function of the halved argument; the odd subsequence is the *same* gamma function of the halved argument times a different constant. The two constants do not agree, and no amount of reindexing makes them agree.

*Warning 3.1.*  $n!!$  is not the restriction to  $\mathbb{N}$  of one canonical slowly varying function. Its even and odd subsequences are two gamma-type branches whose quotient is a nonzero constant  $\sqrt{2/\pi} \neq 1$ . Consequently *there is no canonical real inverse of  $n!!$  until an interpolation is fixed*, and different admissible interpolations give inverses that differ at an order *larger* than the entire Bernoulli correction grid (theorem 3.8). Every statement in this chapter that begins “the inverse” names its interpolation first.

Following definition 1.37 we therefore distinguish, for a target  $y \rightarrow +\infty$ :

- (i) *the two parity-resolved functional inverses*, obtained by inverting the canonical gamma continuation of one parity subsequence (section 3.6);
- (ii) *the functional inverse of a single all-integer interpolation*, for instance the periodic one recorded by OEIS (section 3.8);
- (iii) *the integer staircase inverses*, the least  $n$  with  $n!! \geq y$  and the greatest  $n$  with  $n!! \leq y$  (section 3.9).

Objects (i) and (iii) are intrinsic to the sequence; object (ii) is not. This is the chapter's spine, and theorem 3.8 makes it quantitative.

Two asymptotic scales must also be kept apart. Taking logarithms converts the growth of  $n!!$  into a phase  $\frac{1}{2}u(\log u - 1)$  whose inversion is *exactly* a Lambert expression (proposition 1.23); expanding that Lambert function produces an *outer* layer in powers of  $1/\log \log y$  with polynomial dependence on  $\log \log \log y$  (theorem 1.33). The Bernoulli corrections generate a far finer *inner* grid in odd inverse powers of the Lambert core. The fixed shift  $-1$  lies strictly between the two levels. Writing all of this as one informal series in  $1/\log y$  destroys the ordering; see proposition 3.42.

## 3.2 Exact continuations and the parity split

### 3.2.1 The two parity branches

**Definition 3.2** (Parity branches). For the parity label  $\delta \in \{0, 1\}$  put

$$\kappa_\delta = \left(\frac{2}{\pi}\right)^{\delta/2}, \quad \mathcal{D}_\delta(x) = \kappa_\delta 2^{x/2} \Gamma\left(\frac{x}{2} + 1\right) \quad (x > -2), \quad (3.3)$$

and write

$$A_\delta = \log(\kappa_\delta \sqrt{\pi}) = \begin{cases} \frac{1}{2} \log \pi, & \delta = 0, \\ \frac{1}{2} \log 2, & \delta = 1, \end{cases} \quad e^{A_0} = \sqrt{\pi}, \quad e^{A_1} = \sqrt{2}. \quad (3.4)$$

Here  $\delta = 0$  labels the even subsequence and  $\delta = 1$  the odd one.

**Proposition 3.3** (Exact parity interpolation and monotonicity). *For every integer  $n \geq 0$  with  $n \equiv \delta \pmod{2}$  one has  $\mathcal{D}_\delta(n) = n!!$ . Moreover each  $\mathcal{D}_\delta$  is positive and strictly increasing on  $[0, \infty)$ , so it has a single-valued inverse  $\mathcal{D}_\delta^{-1}$  on its range; and*

$$\mathcal{D}_1(x) = \sqrt{\frac{2}{\pi}} \mathcal{D}_0(x), \quad \mathcal{D}_1^{-1}(y) = \mathcal{D}_0^{-1}\left(\sqrt{\frac{\pi}{2}} y\right). \quad (3.5)$$

*Proof.* If  $n = 2m$  then  $\mathcal{D}_0(2m) = 2^m \Gamma(m+1) = 2^m m! = (2m)!!$  by eq. (3.2). If  $n = 2m+1$  then

$$\mathcal{D}_1(2m+1) = \sqrt{\frac{2}{\pi}} 2^{m+1/2} \Gamma\left(m + \frac{3}{2}\right) = \frac{2^{m+1}}{\sqrt{\pi}} \Gamma\left(m + \frac{3}{2}\right) = (2m+1)!!.$$

Positivity is clear because  $\Gamma > 0$  on  $(0, \infty)$ . For monotonicity,

$$\frac{\mathcal{D}'_\delta(x)}{\mathcal{D}_\delta(x)} = \frac{1}{2} \left( \log 2 + \psi\left(\frac{x}{2} + 1\right) \right), \quad \psi = \Gamma'/\Gamma, \quad (3.6)$$

and  $\psi$  is strictly increasing on  $(0, \infty)$  because  $\psi'(z) = \sum_{m \geq 0} (z+m)^{-2} > 0$ . At  $x = 0$  the bracket equals  $\log 2 - \gamma = 0.11593 \dots > 0$ , hence it is positive for all  $x \geq 0$ . Equation (3.5) is immediate from eq. (3.3), since  $\kappa_1/\kappa_0 = \sqrt{2/\pi}$ .  $\square$

**Remark 3.4.** The constant  $\kappa_\delta$  disappears from eq. (3.6). Both branches therefore have the same local conditioning, the same forward correction series, and — as theorem 3.29 shows — literally the same inverse coefficient polynomials. Parity survives only as the additive constant  $A_\delta$  subtracted from  $\log y$ .

### 3.2.2 The periodic OEIS interpolation, and why it is not canonical

**Definition 3.5** (Periodic interpolation). Put

$$\mathcal{D}(x) = 2^{x/2} \Gamma\left(\frac{x}{2} + 1\right) \left(\frac{2}{\pi}\right)^{(1-\cos \pi x)/4}. \quad (3.7)$$

Because  $\frac{1}{2} \sin^2(\pi x/2) = \frac{1}{4}(1 - \cos \pi x)$ , this is the interpolation recorded in the OEIS entry for A006882, written there with the half-argument sine.

**Proposition 3.6** (Interpolation and eventual monotonicity).  $\mathcal{D}(n) = n!!$  for every integer  $n \geq 0$ , and  $\mathcal{D}$  is strictly increasing on  $[1.04, \infty)$ . Its inverse on that ray, denoted  $\mathcal{D}^{-1}$ , is what “the inverse of  $\mathcal{D}$ ” means below; it is not a globally single-valued inverse near the origin.

*Proof.* The periodic factor equals 1 at even integers and  $\sqrt{2/\pi}$  at odd integers, so  $\mathcal{D}$  agrees with  $\mathcal{D}_0$  at even and with  $\mathcal{D}_1$  at odd integers; proposition 3.3 finishes the first claim. Next,

$$\frac{\mathcal{D}'(x)}{\mathcal{D}(x)} = \frac{1}{2} \left( \log 2 + \psi\left(\frac{x}{2} + 1\right) \right) + \frac{\pi}{4} \log\left(\frac{2}{\pi}\right) \sin \pi x. \quad (3.8)$$

For  $z > \frac{1}{2}$  one has  $\psi(z) > \log(z - \frac{1}{2})$ , hence  $\psi(x/2 + 1) > \log((x+1)/2)$  and the first summand exceeds  $\frac{1}{2} \log(x+1)$ . The second summand has modulus at most  $\pi a$ , where

$$a := \frac{1}{4} \log \frac{\pi}{2} = 0.112895676322 \dots > 0. \quad (3.9)$$

Thus  $\mathcal{D}'/\mathcal{D} > \frac{1}{2} \log(x+1) - \pi a$ , which is positive as soon as  $x+1 > e^{2\pi a} = 2.03265 \dots$ , i.e. for  $x \geq 1.04$ .  $\square$

*Remark 3.7.* The bound in the proof is not sharp: numerically the last zero of eq. (3.8) is at  $x = 0.6970947 \dots$ , so  $\mathcal{D}$  is in fact strictly increasing on  $[0.70, \infty)$ . Only the proved threshold is used below.

The next theorem is the precise form of warning 3.1. None of the three sources states it; all three assert the phenomenon informally.

**Theorem 3.8** (The inverse transseries is not determined by the sequence). For  $\lambda \in \mathbb{R}$  set  $\mathcal{D}_\lambda(x) = \mathcal{D}(x) e^{\lambda \sin \pi x}$ . Then:

- (a)  $\mathcal{D}_\lambda(n) = n!!$  for every integer  $n \geq 0$ , and  $\mathcal{D}_\lambda$  is real-analytic on  $\mathbb{R}$ ;
- (b)  $\mathcal{D}_\lambda$  is strictly increasing on  $[e^{2\pi(a+|\lambda|)} - 1, \infty)$ , so it has a large real inverse;
- (c) with  $T$  and  $\Lambda$  the Lambert data of eq. (3.32) built from  $R_* = \log y - \frac{1}{4} \log(2\pi)$ , the leading displacement of that inverse beyond the core is

$$\mathcal{D}_\lambda^{-1}(y) - (T - 1) = \frac{2\sqrt{a^2 + \lambda^2}}{\Lambda} \cos\left(\pi T - \arctan \frac{\lambda}{a}\right) + \mathcal{O}(\Lambda^{-2}). \quad (3.10)$$

In particular the amplitude  $2\sqrt{a^2 + \lambda^2}/\Lambda$  of the leading oscillatory sector is an unbounded function of the interpolation and is larger, by the factor  $T$ , than the entire Bernoulli grid  $\mathcal{O}(T^{-1}\Lambda^{-1})$  of theorem 3.29.

*Proof.* (a)  $\sin \pi n = 0$  for integer  $n$ , and  $e^{\lambda \sin \pi x}$  is entire.

(b) Differentiating logarithmically adds  $\lambda \pi \cos \pi x$  to eq. (3.8); repeating the estimate in the proof of proposition 3.6 gives  $\mathcal{D}'_\lambda/\mathcal{D}_\lambda > \frac{1}{2} \log(x+1) - \pi(a + |\lambda|)$ .

(c) By lemma 3.44 below, with  $u = x + 1$ ,

$$\log \mathcal{D}_\lambda(u - 1) = \frac{u}{2} (\log u - 1) + \frac{1}{4} \log(2\pi) - a \cos \pi u - \lambda \sin \pi u + \Omega(u),$$

because  $\cos \pi x = -\cos \pi u$  and  $\sin \pi x = -\sin \pi u$ . Subtracting the core identity  $\frac{1}{2} T (\log T - 1) = R_*$ , putting  $u = T + \Delta$  and multiplying by 2 exactly as in lemma 3.45 gives

$$\Lambda \Delta + T \varphi(\Delta/T) + 2\Omega(T + \Delta) - 2a \cos(\theta + \pi \Delta) - 2\lambda \sin(\theta + \pi \Delta) = 0, \quad \theta = \pi T.$$

All terms except the last two are  $\mathcal{O}(T^{-1})$  once  $\Delta = \mathcal{O}(\Lambda^{-1})$ , so at leading order  $\Lambda \Delta_0 = 2a \cos(\theta + \pi \Delta_0) + 2\lambda \sin(\theta + \pi \Delta_0) = 2\sqrt{a^2 + \lambda^2} \cos(\theta + \pi \Delta_0 - \arctan(\lambda/a))$ . Since  $\Delta_0 = \mathcal{O}(\Lambda^{-1})$ , replacing  $\theta + \pi \Delta_0$  by  $\theta$  costs  $\mathcal{O}(\Lambda^{-2})$ , which is eq. (3.10).  $\square$

*Remark 3.9.* The family  $e^{\lambda \sin \pi x}$  is only the simplest witness. Any real-analytic  $g$  vanishing on  $\mathbb{N}$  and with  $g'$  bounded produces the same conclusion, with  $2a \cos \theta$  replaced by the value of  $-2g$  transported to the core. What is intrinsic is the pair of parity-branch transseries and the integer staircase; the periodic sector is a property of the chosen interpolation and nothing else.

### 3.2.3 The centred coordinate and the exact normal form

**Definition 3.10** (Centred coordinate). Throughout,  $u := x + 1$  and

$$\Phi(u) := \frac{u}{2}(\log u - 1), \quad \Omega(u) := \log \Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2} \log \frac{u}{2} + \frac{u}{2} - \frac{1}{2} \log(2\pi). \quad (3.11)$$

**Theorem 3.11** (Exact centred normal form). For every  $x > -1$  and every  $\delta \in \{0, 1\}$ , with  $u = x + 1$ ,

$$\mathcal{D}_\delta(x) = e^{A_\delta} \left(\frac{u}{e}\right)^{u/2} e^{\Omega(u)}, \quad \text{equivalently} \quad \log \mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \Omega(u). \quad (3.12)$$

This is an identity, not an asymptotic statement, and  $\Omega$  does not depend on  $\delta$ .

*Proof.* Insert  $x = u - 1$  into eq. (3.3):

$$\log \mathcal{D}_\delta(x) = \log \kappa_\delta + \frac{u-1}{2} \log 2 + \log \Gamma\left(\frac{u+1}{2}\right).$$

Subtract  $A_\delta + \Phi(u)$  and use  $A_\delta = \log \kappa_\delta + \frac{1}{2} \log \pi$ :

$$\begin{aligned} \log \mathcal{D}_\delta(x) - A_\delta - \Phi(u) &= \frac{u-1}{2} \log 2 + \log \Gamma\left(\frac{u+1}{2}\right) - \frac{1}{2} \log \pi - \frac{u}{2} \log u + \frac{u}{2} \\ &= \log \Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2} (\log u - \log 2) + \frac{u}{2} - \frac{1}{2} \log 2 - \frac{1}{2} \log \pi, \end{aligned}$$

which is exactly  $\Omega(u)$  as defined in eq. (3.11).  $\square$

*Remark 3.12* (Why  $u = x + 1$ ). The shift is not cosmetic. It turns the gamma argument into  $\frac{1}{2}(u+1) = \frac{u}{2} + \frac{1}{2}$ , and Bernoulli polynomials evaluated at  $\frac{1}{2}$  annihilate every odd index; consequently  $\Omega$  has only *odd* inverse powers of  $u$  (theorem 3.16) and the explicit  $\frac{1}{2} \log x$  of the uncentred Stirling form (proposition 3.23) is absorbed into the dominant phase. This sparse support is precisely why the inverse lives on the grid  $T^{-1}, T^{-3}, T^{-5}, \dots$

At an integer  $n$  the two constants  $A_0, A_1$  combine into one parity transmonomial: writing  $\chi_n = (-1)^n$ ,

$$A(n) = \frac{1}{4} \log(2\pi) + \frac{\chi_n}{4} \log \frac{\pi}{2}, \quad e^{A(n)} = (2\pi)^{1/4} \left(\frac{\pi}{2}\right)^{\chi_n/4}, \quad (3.13)$$

so that for every integer  $n \geq 0$

$$n!! = \left(\frac{n+1}{e}\right)^{(n+1)/2} \exp(A(n) + \Omega(n+1)). \quad (3.14)$$

Equation (3.14) is exact. The factor  $(-1)^n$  is a genuine transmonomial of the sequence's leading behaviour, not an error term.

*Remark 3.13* (Notation across the three sources). The letter  $C$  means three different things in the siblings, and two Greek letters collide:

| this chapter                                                | S1              | S2         | S3              |
|-------------------------------------------------------------|-----------------|------------|-----------------|
| $\kappa_\delta = (2/\pi)^{\delta/2}$                        | $\kappa_\delta$ | $\kappa_r$ | $C_\varepsilon$ |
| $e^{A_\delta} \in \{\sqrt{\pi}, \sqrt{2}\}$                 | $C_\delta$      | $A_r$      | —               |
| $A_\delta \in \{\frac{1}{2} \log \pi, \frac{1}{2} \log 2\}$ | $\log C_\delta$ | $C_r$      | $A_\varepsilon$ |
| $c_k$ (centred log. coefficient)                            | $c_k$           | $\beta_k$  | $c_k$           |
| $a_k = 2c_k$                                                | $2c_k$          | $\alpha_k$ | $a_k$           |
| $a = \frac{1}{4} \log(\pi/2)$                               | $\rho/2$        | $a$        | —               |
| $P_r(L_2)$ (Lambert polynomial)                             | $P_r$           | $u_{r+1}$  | $\alpha_{r+1}$  |
| $\tau_m(L_2)$ (reciprocal poly.)                            | $h_m$           | $\tau_m$   | $\beta_m$       |

Part 0 adds one further collision: definition 1.8 writes the factorial core as  $\Phi_0(X) = X(\kappa \log X + d)$ , and that  $\kappa$  is *not* the branch prefactor  $\kappa_\delta$  of S1 and S2. This chapter therefore never abbreviates the core parameters, always writing  $\Phi_0(u) = u(\frac{1}{2} \log u - \frac{1}{2})$  in full; see remark 1.24. In particular S2's  $\alpha_k, \beta_k$  are Bernoulli data while S3's  $\alpha_j, \beta_n$  are outer Lambert polynomials. Readers comparing the sources should translate before comparing formulae; the numbers themselves agree everywhere (section 3.14).

### 3.3 Parameter dictionary for the Part 0 apparatus

The chapter uses the Part 0 toolkit rather than re-deriving it. Its dominant core is the *factorial* one of definition 1.8, not the exponential–power model definition 1.2; see warning 3.14. The specialisation is the following.

| Part 0 datum                                       | even branch $\delta = 0$                          | odd branch $\delta = 1$                        |
|----------------------------------------------------|---------------------------------------------------|------------------------------------------------|
| interpolation                                      | $2^{x/2} \Gamma(\frac{x}{2} + 1)$                 | $\sqrt{2/\pi} 2^{x/2} \Gamma(\frac{x}{2} + 1)$ |
| affine shift $x_*$ of definition 1.8               | $x_* = -1$ , taken first                          | $x_* = -1$ , taken first                       |
| centred coordinate $u = x - x_*$                   | $u = x + 1$                                       | $u = x + 1$                                    |
| admissible core $\Phi_0$ of definition 1.8         | $\Phi_0(u) = u(\frac{1}{2} \log u - \frac{1}{2})$ | same                                           |
| additive constant                                  | $A_0 = \frac{1}{2} \log \pi$                      | $A_1 = \frac{1}{2} \log 2$                     |
| perturbation $Q(s) = \sum_{k \geq 1} a_k s^{2k-1}$ | $a_k = 2c_k$ , eq. (3.19)                         | same                                           |
| target after constant removal                      | $R_0 = \log y - \frac{1}{2} \log \pi$             | $R_1 = \log y - \frac{1}{2} \log 2$            |
| core (proposition 1.23)                            | $T_\delta = 2R_\delta/W_0(2R_\delta/e)$           | same formula                                   |
| core slope parameter                               | $\Lambda_\delta = \log T_\delta$                  | same formula                                   |
| inner grid                                         | $q = T_\delta^{-1}, z = q^2$                      | same                                           |

Concretely: definitions 1.9 and 1.10 supply the Bell polynomials used throughout; lemma 1.12 supplies the coefficients of  $(1+w)^{-j}$  and  $\log(1+w)$ ; proposition 1.23 inverts the dominant phase exactly by  $W_0$ ; theorem 1.29, whose formal engine is lemma 1.16, generates the correction blocks and is specialised once in lemma 3.28; theorem 1.33 unfolds  $W_0$  into the outer layer of section 3.7; definition 1.37 and theorem 1.39 govern section 3.9; theorems 1.42 and 1.44 convert forward residuals into inverse errors; and theorem 1.46 fixes the least-term scale of section 3.10.

*Warning 3.14* (The double factorial is not of exponential–power type). It is tempting — and wrong — to file  $\mathcal{D}_\delta$  under definition 1.2,  $F(x) = C \exp(ax^\alpha) x^b \exp Q(x^{-\delta})$  with  $\alpha \in (0, 1]$ . By theorem 3.11,  $\log \mathcal{D}_\delta(x) = \frac{u}{2}(\log u - 1) + A_\delta + \Omega(u)$  with  $u = x + 1$ , so the dominant phase is  $\frac{1}{2}u \log u$ . For every  $\alpha \leq 1$ ,

$$\frac{u \log u}{u^\alpha} \longrightarrow \infty \quad (u \rightarrow +\infty),$$

so no admissible choice of  $(C, a, \alpha, b, \delta, Q)$  reproduces it: the double factorial lies outside the exponential–power family, in the *factorial* class of definition 1.8, with the parameters that remark 1.24 records for this chapter. This is the structural difference from Parts 2, 4 and 5, whose sequences have  $\log a_n = an^\alpha + b \log n + \dots$  with  $\alpha \leq 1$ . What survives from the exponential–power theory is the *shape* of the reversion, not the model: theorem 1.29 covers both cores, and the resulting fixed-point map is eq. (3.37). Note also that the shift  $x_* = -1$  must be applied *before* any further reduction: it is what removes the

$\frac{1}{2} \log x$  of eq. (3.30) and produces the sparse odd-power perturbation (remark 3.12). None of S1, S2, S3 claims membership of the exponential–power family; all three invert  $\frac{1}{2}u(\log u - 1)$  directly.

### 3.4 Borel structure of the centred correction

The exact Borel kernel is due to S2 and S3, independently and in the same form; S1 has only the formal Bernoulli series. The sharp first-omitted-term bound of theorem 3.17 is S2's; S3 proves the weaker bound with  $\eta(2N + 2)$  replaced by 1, and that weaker form is not used here.

**Theorem 3.15** (Borel–Laplace representation). *For  $\Re u > 0$ ,*

$$\Omega(u) = \int_0^\infty e^{-us} \widehat{\Omega}(s) \, ds, \quad \widehat{\Omega}(s) = \frac{1}{2s^2} \left( \frac{s}{\sinh s} - 1 \right), \quad (3.15)$$

*the singularity of  $\widehat{\Omega}$  at  $s = 0$  being removable. Moreover*

$$\widehat{\Omega}(s) = \sum_{m \geq 1} \frac{(-1)^m}{s^2 + \pi^2 m^2}, \quad (3.16)$$

*so the Borel singularities of  $\Omega$  are exactly the simple poles  $s = \pm i\pi m$ ,  $m \geq 1$ , with residues*

$$\text{Res}_{s=i\pi m} \widehat{\Omega}(s) = \frac{(-1)^{m+1}i}{2\pi m}, \quad \text{Res}_{s=-i\pi m} \widehat{\Omega}(s) = \frac{(-1)^m i}{2\pi m}. \quad (3.17)$$

*Proof.* Put  $t = u/2$  and  $\widetilde{\Omega}(t) = \log \Gamma(t + \frac{1}{2}) - t \log t + t - \frac{1}{2} \log(2\pi)$ , so that  $\Omega(u) = \widetilde{\Omega}(u/2)$  by eq. (3.11). Binet's first integral and Frullani's formula give, for  $t > 0$ ,

$$\psi(t + \tfrac{1}{2}) - \log t = \int_0^\infty e^{-tr} \left( \frac{1}{r} - \frac{1}{2 \sinh(r/2)} \right) \, dr,$$

the integrand being  $\mathcal{O}(r)$  at the origin and exponentially damped at infinity. On the other hand

$$\frac{d}{dt} \int_0^\infty e^{-tr} \frac{r/(2 \sinh(r/2)) - 1}{r^2} \, dr = \int_0^\infty e^{-tr} \left( \frac{1}{r} - \frac{1}{2 \sinh(r/2)} \right) \, dr,$$

differentiation under the integral being justified by the same bounds. Since  $\widetilde{\Omega}'(t) = \psi(t + \frac{1}{2}) - \log t$  and both  $\widetilde{\Omega}(t)$  and the displayed integral tend to 0 as  $t \rightarrow +\infty$ , the two functions coincide. Substituting  $r = 2s$  and  $t = u/2$  turns the integral into eq. (3.15); analytic continuation in  $u$  extends it to  $\Re u > 0$ . The expansion  $s/\sinh s = 1 - s^2/6 + \mathcal{O}(s^4)$  removes the origin.

For eq. (3.16), the elementary Mittag–Leffler expansion  $\frac{s}{\sinh s} = 1 + 2s^2 \sum_{m \geq 1} \frac{(-1)^m}{s^2 + \pi^2 m^2}$  gives the claim after division by  $2s^2$ . The residues follow from  $1/(s^2 + \pi^2 m^2) = [(s - i\pi m)(s + i\pi m)]^{-1}$ .  $\square$

**Theorem 3.16** (The centred logarithmic coefficients). *As  $u \rightarrow \infty$  in any closed subsector of  $|\arg u| < \pi/2$ ,*

$$\Omega(u) \sim \sum_{k \geq 1} \frac{c_k}{u^{2k-1}}, \quad (3.18)$$

*where the following three expressions coincide:*

$$c_k = \frac{2^{2k-1} B_{2k}(\frac{1}{2})}{2k(2k-1)} = \frac{(1 - 2^{2k-1}) B_{2k}}{2k(2k-1)} = (-1)^k \frac{(2k-2)! \eta(2k)}{\pi^{2k}} \quad (3.19)$$

*with  $B_m(z)$  the Bernoulli polynomials,  $B_m = B_m(0)$ , and  $\eta(s) = (1 - 2^{1-s})\zeta(s)$  the Dirichlet eta function. There are no even inverse powers.*



*Proof.* Expand each summand of eq. (3.16) by

$$\frac{1}{s^2 + A^2} = \sum_{j=0}^{N-1} \frac{(-1)^j s^{2j}}{A^{2j+2}} + \frac{(-1)^N s^{2N}}{A^{2N}(s^2 + A^2)}, \quad A > 0, \quad (3.20)$$

with  $A = \pi m$ , sum over  $m$ , and integrate term by term against  $e^{-us}$ , using  $\int_0^\infty e^{-us} s^{2j} ds = (2j)! u^{-2j-1}$ . The coefficient of  $u^{-(2j+1)}$  is

$$(2j)! \sum_{m \geq 1} \frac{(-1)^m (-1)^j}{(\pi m)^{2j+2}} = (-1)^{j+1} \frac{(2j)! \eta(2j+2)}{\pi^{2j+2}},$$

which is the third expression in eq. (3.19) with  $k = j+1$ . Euler's evaluation  $\zeta(2k) = (-1)^{k+1} (2\pi)^{2k} B_{2k} / (2(2k)!)$  together with  $\eta(2k) = (1 - 2^{1-2k})\zeta(2k)$  and  $B_{2k}(\frac{1}{2}) = (2^{1-2k} - 1)B_{2k}$  converts it into the first two. Only odd powers  $u^{-(2k-1)}$  occur because  $\widehat{\Omega}$  is even.  $\square$

**Theorem 3.17** (Sharp alternating remainder). *For real  $u > 0$  and  $N \geq 0$  put  $R_N(u) = \Omega(u) - \sum_{k=1}^N c_k u^{-(2k-1)}$  (empty sum for  $N = 0$ ). Then  $R_N(u)$  has the sign of the first omitted term and is smaller than it in modulus:*

$$0 < (-1)^{N+1} R_N(u) < \frac{|c_{N+1}|}{u^{2N+1}} = \frac{(2N)! \eta(2N+2)}{\pi^{2N+2} u^{2N+1}}. \quad (3.21)$$

*Proof.* Insert eq. (3.20) with  $A = \pi m$  into eq. (3.16) and Laplace-transform. The residual kernel is  $(-1)^{N+1} s^{2N} S_N(s)$  with

$$S_N(s) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{(\pi m)^{2N} ((\pi m)^2 + s^2)}.$$

Writing  $((\pi m)^2 + s^2)^{-1} = \int_0^\infty e^{-s^2 v} e^{-\pi^2 m^2 v} dv$  gives

$$S_N(s) = \int_0^\infty e^{-s^2 v} \sum_{m \geq 1} \frac{(-1)^{m-1} e^{-\pi^2 m^2 v}}{(\pi m)^{2N}} dv.$$

For each  $v \geq 0$  the inner series is alternating with strictly decreasing positive terms, hence strictly positive. Therefore  $S_N(s) > 0$  and  $S_N$  is strictly decreasing in  $s^2$ , so  $0 < S_N(s) \leq S_N(0) = \eta(2N+2)/\pi^{2N+2}$ . Multiplying by  $s^{2N}$ , integrating against  $e^{-us}$  and using  $\int_0^\infty e^{-us} s^{2N} ds = (2N)! u^{-(2N+1)}$  gives eq. (3.21).  $\square$

**Corollary 3.18** (Certified forward logarithm). *For  $x \rightarrow +\infty$ ,  $u = x + 1$  and either parity,*

$$\log \mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \sum_{k=1}^N \frac{c_k}{u^{2k-1}} + \theta_N \frac{c_{N+1}}{u^{2N+1}} \quad \text{for some } \theta_N = \theta_N(u) \in (0, 1). \quad (3.22)$$

*Thus truncating after any number of terms overshoots or undershoots by strictly less than the first omitted term, with the sign of that term. For the integer sequence replace  $A_\delta$  by  $A(n)$  of eq. (3.13).*

*Proof.* Combine theorems 3.11 and 3.17.  $\square$

Table 3.1: The centred logarithmic coefficients  $c_k$  of eq. (3.19) and the doubled coefficients  $a_k = 2c_k$  used in the inverse problem. Regenerated in exact rational arithmetic from  $B_{2k}$ .

| $k$ | $c_k$   | $a_k = 2c_k$ |
|-----|---------|--------------|
| 1   | $-1/12$ | $-1/6$       |

| $k$ | $c_k$              | $a_k = 2c_k$       |
|-----|--------------------|--------------------|
| 2   | 7/360              | 7/180              |
| 3   | -31/1260           | -31/630            |
| 4   | 127/1680           | 127/840            |
| 5   | -511/1188          | -511/594           |
| 6   | 1414477/360360     | 1414477/180180     |
| 7   | -8191/156          | -8191/78           |
| 8   | 118518239/122400   | 118518239/61200    |
| 9   | -5749691557/244188 | -5749691557/122094 |

*Remark 3.19* (Late coefficients). From the third form in eq. (3.19) and  $1 < \eta(2k) < 1 + 2^{-2k} \cdot \frac{4}{3}$ ,

$$c_k = (-1)^k \frac{(2k-2)!}{\pi^{2k}} (1 + \mathcal{O}(4^{-k})), \quad (3.23)$$

so eq. (3.18) is Gevrey-1 in the variable  $u^{-2}$ . The ratio of consecutive term moduli is  $\sim 2k(2k-1)/(\pi u)^2$ ; the least term occurs near  $2k = \pi u$  and has size  $\asymp u^{-1/2} e^{-\pi u}$ , matching the distance  $\pi$  from the origin to the nearest Borel poles  $\pm i\pi$ . This is the instance of theorem 1.46 relevant here; see section 3.10.

## 3.5 The forward transseries

### 3.5.1 Multiplicative, reciprocal, and arbitrary-power coefficients

Exponentiation of eq. (3.18) is an instance of the exponential formula, so definition 1.10 does all the work at once for every power.

**Definition 3.20.** Embed the  $c_k$  sparsely on the odd indices,

$$\gamma_r = \begin{cases} c_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases} \quad r \geq 1, \quad (3.24)$$

and for a scalar  $\alpha$  define  $s_m^{(\alpha)}$  by

$$\exp\left(\alpha \sum_{r \geq 1} \gamma_r u^{-r}\right) = \sum_{m \geq 0} \frac{s_m^{(\alpha)}}{u^m}, \quad s_m := s_m^{(1)}. \quad (3.25)$$

**Proposition 3.21** (All multiplicative coefficients at once). *For every  $\alpha$  and every  $m \geq 0$ ,*

$$s_m^{(\alpha)} = \frac{1}{m!} B_m(\alpha 1! \gamma_1, \alpha 2! \gamma_2, \dots, \alpha m! \gamma_m) \quad (3.26)$$

$$= \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_{r \geq 1} r k_r = m}} \prod_{r \geq 1} \frac{(\alpha \gamma_r)^{k_r}}{k_r!} = \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_{j \geq 1} (2j-1) k_j = m}} \prod_{j \geq 1} \frac{(\alpha c_j)^{k_j}}{k_j!}, \quad (3.27)$$

and equivalently

$$s_0^{(\alpha)} = 1, \quad m s_m^{(\alpha)} = \alpha \sum_{1 \leq j \leq (m+1)/2} (2j-1) c_j s_{m-2j+1}^{(\alpha)} \quad (m \geq 1). \quad (3.28)$$

Consequently, for  $x \rightarrow +\infty$ ,  $u = x + 1$  and every fixed  $M \geq 0$ ,

$$\mathcal{D}_\delta(x)^\alpha = e^{\alpha A_\delta} \left(\frac{u}{e}\right)^{\alpha u/2} \left(\sum_{m=0}^M \frac{s_m^{(\alpha)}}{u^m} + \mathcal{O}(u^{-(M+1)})\right). \quad (3.29)$$

The cases  $\alpha = 1$  and  $\alpha = -1$  give  $\mathcal{D}_\delta$  itself and its reciprocal.

*Proof.* Put  $t = u^{-1}$  and  $S_\alpha(t) = \exp(\alpha \sum_r \gamma_r t^r)$ . Equation (3.26) is the complete-Bell generating identity of definition 1.10; multinomial expansion of the exponential gives eq. (3.27), and the second form there uses  $\gamma_{2j-1} = c_j$ ,  $\gamma_{2j} = 0$ . Logarithmic differentiation  $tS'_\alpha = \alpha(\sum_r r\gamma_r t^r)S_\alpha$  and comparison of coefficients give eq. (3.28). For eq. (3.29), take  $N$  with  $2N - 1 \geq M$ ; by theorems 3.11 and 3.17,  $\alpha\Omega(u) = \alpha \sum_{k \leq N} c_k u^{1-2k} + \mathcal{O}(u^{-(2N+1)})$ , and exp of a finite sum of  $\mathcal{O}(u^{-1})$  terms plus a  $\mathcal{O}(u^{-(2N+1)})$  error equals  $\sum_{m \leq M} s_m^{(\alpha)} u^{-m} + \mathcal{O}(u^{-(M+1)})$ , the exponential being Lipschitz on bounded sets.  $\square$

**Corollary 3.22** (Reciprocal sign symmetry).  $s_m^{(-\alpha)} = (-1)^m s_m^{(\alpha)}$  for all  $m \geq 0$ ; in particular the reciprocal coefficients are  $s_m^{(-1)} = (-1)^m s_m$ .

*Proof.* Only odd  $r$  carry a nonzero  $\gamma_r$ , so  $\sum_r \gamma_r (-t)^r = -\sum_r \gamma_r t^r$  and hence  $S_\alpha(-t) = S_{-\alpha}(t)$ . Reading off  $[t^m]$  gives the claim.  $\square$

Table 3.2: Centred multiplicative coefficients  $s_m = s_m^{(1)}$  of eq. (3.29) and the uncentred coefficients  $g_m$  of proposition 3.23. Regenerated from eq. (3.28) in exact rational arithmetic.

| $m$ | $s_m$ (centred)             | $g_m$ (uncentred)        |
|-----|-----------------------------|--------------------------|
| 0   | 1                           | 1                        |
| 1   | -1/12                       | 1/6                      |
| 2   | 1/288                       | 1/72                     |
| 3   | 1003/51840                  | -139/6480                |
| 4   | -4027/2488320               | -571/155520              |
| 5   | -5128423/209018880          | 163879/6531840           |
| 6   | 168359651/75246796800       | 5246819/1175731200       |
| 7   | 68168266699/902961561600    | -534703531/7054387200    |
| 8   | -58728355451/86684309913600 | -4483131259/338610585600 |

### 3.5.2 The uncentred normal form and the centre-shift identity

**Proposition 3.23** (Uncentred expansion). As  $x \rightarrow +\infty$ ,

$$\log \mathcal{D}_\delta(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2} \log x + A_\delta + \sum_{k \geq 1} \frac{b_k}{x^{2k-1}}, \quad b_k = \frac{2^{2k-1} B_{2k}}{2k(2k-1)} = (-1)^{k-1} \frac{(2k-2)! \zeta(2k)}{\pi^{2k}}, \quad (3.30)$$

with  $b_1 = \frac{1}{6}$ ,  $b_2 = -\frac{1}{45}$ ,  $b_3 = \frac{8}{315}$ ,  $b_4 = -\frac{8}{105}$ ,  $b_5 = \frac{128}{297}$ . Exponentiating,  $\mathcal{D}_\delta(x) \sim e^{A_\delta} \sqrt{x} (x/e)^{x/2} \sum_m g_m x^{-m}$ , where  $g_m$  is obtained from eq. (3.28) on replacing  $c_j$  by  $b_j$ ; this is the familiar form, with amplitudes  $\sqrt{\pi}$  (even) and  $\sqrt{2}$  (odd). The exact Borel kernel of the uncentred correction is  $\hat{\mathcal{U}}(s) = \frac{\coth s}{2s} - \frac{1}{2s^2} = \sum_{m \geq 1} (s^2 + \pi^2 m^2)^{-1}$ , with the same pole lattice  $\pm i\pi m$  but all residues of one sign pattern.

*Proof.* Apply the standard Stirling expansion to  $\log \Gamma(z+1)$  with  $z = x/2$  and add  $(x/2) \log 2 + \log \kappa_\delta$ ; the elementary terms give the first three summands of eq. (3.30), and the corrections scale as  $z^{-(2k-1)} = 2^{2k-1} x^{-(2k-1)}$ . Euler's evaluation of  $\zeta(2k)$  gives the second form of  $b_k$ . For the kernel, the uncentred correction is  $\mathcal{U}(x) = \Omega_\Gamma(x/2)$  with  $\Omega_\Gamma$  Binet's function; substituting  $t = 2s$  in  $\Omega_\Gamma(z) = \int_0^\infty e^{-zt} \left( \frac{\coth(t/2)}{2t} - \frac{1}{t^2} \right) dt$  gives the stated kernel, and  $\frac{\coth s}{s} = \frac{1}{s^2} + 2 \sum_{m \geq 1} (s^2 + \pi^2 m^2)^{-1}$  gives the partial fractions.  $\square$

**Warning 3.24.** The two statements  $n!! \sim e^{A(n)} \sqrt{n} (n/e)^{n/2}$  and  $n!! \sim e^{A(n)} ((n+1)/e)^{(n+1)/2}$  are both correct and are not in conflict: they use different normal forms. The first leaves a factor  $\sqrt{n}$  and an *uncentred* correction series; the second absorbs that factor into the phase. Only the centred form leads to the sparse inverse grid, and it is the one used from section 3.6 onwards.

**Proposition 3.25** (Centre-shift identity). *For every  $r \geq 1$ ,*

$$\frac{(-1)^{r+1}}{2r(r+1)} + (-1)^{r+1} \sum_{k=1}^{\lfloor (r+1)/2 \rfloor} \binom{r-1}{2k-2} c_k = \begin{cases} b_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases} \quad (3.31)$$

*Proof.* Substitute  $u = x + 1 = x(1 + w)$ ,  $w = x^{-1}$ , into  $\Phi(u) + \sum_k c_k u^{1-2k}$  and compare with eq. (3.30). First, with  $\log u = \log x + \log(1 + w)$ ,

$$\Phi(u) = \frac{x}{2}(\log x - 1) + \frac{1}{2}(\log x - 1) + \frac{x+1}{2} \log(1 + w).$$

Expanding  $\log(1 + w) = \sum_{j \geq 1} (-1)^{j-1} w^j / j$  and splitting the prefactor  $\frac{x+1}{2} = \frac{x}{2} + \frac{1}{2}$  gives

$$\frac{x+1}{2} \log(1 + w) = \frac{1}{2} + \frac{1}{2} \sum_{r \geq 1} (-1)^r x^{-r} \left( \frac{1}{r+1} - \frac{1}{r} \right) = \frac{1}{2} + \sum_{r \geq 1} \frac{(-1)^{r+1}}{2r(r+1)} x^{-r},$$

so that  $\Phi(u) = \frac{x}{2}(\log x - 1) + \frac{1}{2} \log x + \sum_{r \geq 1} (-1)^{r+1} (2r(r+1))^{-1} x^{-r}$ ; the two halves  $\pm \frac{1}{2}$  cancel. Second,

$$\sum_{k \geq 1} \frac{c_k}{u^{2k-1}} = \sum_{k \geq 1} c_k x^{-(2k-1)} (1+w)^{-(2k-1)} = \sum_{r \geq 1} x^{-r} \sum_{k \leq (r+1)/2} c_k \binom{-(2k-1)}{r-2k+1},$$

and  $\binom{-(2k-1)}{r-2k+1} = (-1)^{r-2k+1} \binom{r-1}{r-2k+1} = (-1)^{r+1} \binom{r-1}{2k-2}$ . Adding the two coefficient lists and matching against eq. (3.30) – where the coefficient of  $x^{-r}$  is  $b_{(r+1)/2}$  for odd  $r$  and 0 for even  $r$  – gives eq. (3.31).  $\square$

*Remark 3.26.* Equation (3.31) was verified symbolically for  $1 \leq r \leq 9$ : the even cases return 0 and the odd cases return  $\frac{1}{6}, -\frac{1}{45}, \frac{8}{315}, -\frac{8}{105}, \frac{128}{297}$ . It is the cheapest available certificate that the centring is consistent, because it fails immediately under any sign or index slip in eq. (3.19). S3 states the identity without proof; the proof above is supplied here.

## 3.6 The branchwise functional inverse

All three siblings reach the same objects in this section. The presentation follows S3 for the exact normal form and S2 for the graded closed formula and the sign law; S1’s “scaling from the half-shifted inverse Gamma problem” is subsumed by theorem 3.61. The proofs of corollary 3.35 and theorem 3.37 are new: the corresponding source arguments are sketches (see the chapter report).

### 3.6.1 The Lambert core

**Lemma 3.27** (Exact inversion of the dominant phase). *Let  $R > 0$ . The equation  $\Phi(T) = \frac{1}{2}T(\log T - 1) = R$  has a unique solution  $T > e$ , namely*

$$w = W_0\left(\frac{2R}{e}\right), \quad T = \frac{2R}{w} = e^{w+1}, \quad \Lambda := w + 1 = \log T, \quad (3.32)$$

and then

$$\Phi'(T) = \frac{\Lambda}{2}, \quad \frac{dT}{dR} = \frac{2}{\Lambda}, \quad T(\Lambda - 1) = 2R. \quad (3.33)$$

*Proof.* This is proposition 1.23 for the parameters of remark 1.24; for completeness, setting  $w = \log T - 1$  turns  $\Phi(T) = R$  into  $we^{w+1} = 2R$ , i.e.  $we^w = 2R/e > 0$ , whose unique real solution is  $w = W_0(2R/e) > 0$ , the principal branch being the only real branch on  $(0, \infty)$ .  $\Phi$  is strictly increasing on  $(1, \infty)$  with range  $(-\frac{1}{2}, \infty)$ , which gives uniqueness of  $T$ . The identities in eq. (3.33) follow from  $\Phi'(T) = \frac{1}{2} \log T$ , implicit differentiation of  $\Phi(T) = R$ , and  $T = 2R/w = 2R/(\Lambda - 1)$ .  $\square$

For the parity branch  $\delta$  one takes

$$R_\delta = \log y - A_\delta, \quad w_\delta, T_\delta, \Lambda_\delta \text{ as in eq. (3.32).} \quad (3.34)$$

The subscript  $\delta$  is suppressed inside coefficient calculations, where it never appears.

### 3.6.2 The normalised displacement equation

**Lemma 3.28** (The factorial-core reversion kernel). *Write  $u = T(1 + E)$ ,  $q = T^{-1}$ ,  $z = q^2$ , and*

$$\varphi(v) = (1 + v) \log(1 + v) - v = \sum_{m \geq 2} \frac{(-1)^m}{m(m-1)} v^m, \quad a_k = 2c_k. \quad (3.35)$$

*Then the equation  $\Phi(u) + \Omega(u) = R$ , with  $\Omega$  replaced by its asymptotic series, is equivalent to the formal normal form*

$$\boxed{\Lambda E + \varphi(E) + \sum_{k \geq 1} a_k z^k (1 + E)^{-(2k-1)} = 0.} \quad (3.36)$$

*Equivalently, in absolute displacement  $U = TE$  with  $t = q$ ,*

$$U = \Psi(t, U), \quad \Psi(t, v) = -\frac{1}{\Lambda} \left[ \frac{\varphi(tv)}{t} + \sum_{k \geq 1} a_k t^{2k-1} (1 + tv)^{-(2k-1)} \right] \in t \mathbb{Q}(\Lambda)[[t, v]], \quad (3.37)$$

*which is the specialisation of theorem 1.29 to the factorial core, so that lemma 1.16 applies verbatim. There is a unique formal solution, and it has the form*

$$E = \sum_{n \geq 1} e_n(\Lambda) z^n, \quad \text{hence} \quad u - T = \sum_{n \geq 1} \frac{e_n(\Lambda)}{T^{2n-1}} : \quad (3.38)$$

*only odd inverse powers of the Lambert core occur.*

*Proof.* Because  $\Phi(T) = R$ ,

$$\Phi(T(1 + E)) - \Phi(T) = \frac{T(1 + E)}{2} (\log T + \log(1 + E) - 1) - \frac{T}{2} (\log T - 1) = \frac{T}{2} [\Lambda E + \varphi(E)],$$

using  $\Lambda = \log T$  and  $(1 + E) \log(1 + E) = \varphi(E) + E$ . Also  $\Omega(u) = \sum_k c_k T^{-(2k-1)} (1 + E)^{-(2k-1)}$  formally. Multiplying the whole equation by  $2/T = 2q$  and using  $a_k = 2c_k$  and  $q \cdot q^{2k-1} = z^k$  gives eq. (3.36). Dividing by  $T$  and solving for  $U = TE$  gives eq. (3.37); the bracket lies in  $t \mathbb{Q}[[t, v]]$  because  $\varphi(tv) = t^2 v^2 / 2 + \mathcal{O}(t^3)$ , so  $\varphi(tv)/t = \mathcal{O}(t)$ , and every Bernoulli term carries  $t^{2k-1}$  with  $k \geq 1$ . Existence and uniqueness of the fixed point are lemma 1.16. Finally, if  $E$  contains only even powers of  $q$  then so does every term of eq. (3.36); an induction on the  $q$ -valuation therefore forces  $E \in z \mathbb{Q}(\Lambda)[[z]]$ .  $\square$

### 3.6.3 The all-orders inverse and its remainder

**Theorem 3.29** (Complete branchwise inverse transseries). *Fix  $\delta \in \{0, 1\}$  and let  $R_\delta, T_\delta, \Lambda_\delta$  be as in eq. (3.34). Write  $d_{2n-1} := e_n$  for the coefficients of eq. (3.38). Then for every fixed  $N \geq 0$ , as  $y \rightarrow +\infty$ ,*

$$\boxed{\mathcal{D}_\delta^{-1}(y) = T_\delta - 1 + \sum_{n=1}^N \frac{d_{2n-1}(\Lambda_\delta)}{T_\delta^{2n-1}} + \mathcal{O}\left(\frac{1}{T_\delta^{2N+1} \Lambda_\delta}\right),} \quad (3.39)$$

*with an implied constant that may be chosen uniformly in  $\delta \in \{0, 1\}$ . All parity dependence sits in  $R_\delta = \log y - A_\delta$ ; the coefficient functions  $d_{2n-1}$  are universal.*

*Proof.* Let  $u_N = T + \sum_{n=1}^N e_n(\Lambda) T^{1-2n}$  and  $E_N = (u_N - T)/T$ . By construction, substituting  $E_N$  into eq. (3.36) annihilates the coefficients of  $z^1, \dots, z^N$ , so the formal residual is  $\mathcal{O}(z^{N+1})$  with coefficients rational in  $\Lambda$  and bounded as  $\Lambda \rightarrow \infty$ . Undoing the normalisation (multiplying by  $T/2$ ) therefore bounds the exact forward residual

$$\rho_N := \Phi(u_N) + A_\delta + \Omega(u_N) - \log y$$

by  $\mathcal{O}(T \cdot z^{N+1}) = \mathcal{O}(T^{-(2N+1)})$ , where replacing  $\Omega$  by its  $N$ -term truncation costs at most  $|c_{N+1}| u_N^{-(2N+1)}$  by theorem 3.17, of the same order. This is the hypothesis of theorem 1.42. For the slope, on the interval between  $u_N$  and the exact root,

$$\frac{d}{du} [\Phi(u) + \Omega(u)] = \frac{1}{2} \log u + \Omega'(u) = \frac{\Lambda}{2} (1 + o(1)),$$

because  $u = T(1 + \mathcal{O}(T^{-2}))$  and, differentiating eq. (3.15) under the integral sign,  $\Omega'(u) = -\int_0^\infty s e^{-us} \widehat{\Omega}(s) ds = \mathcal{O}(u^{-2})$ . The mean-value theorem now divides  $\rho_N$  by a quantity asymptotic to  $\Lambda/2$ , which is the instance of theorem 1.44 for the factorial core and gives eq. (3.39). The parity constant  $A_\delta$  disappears on differentiation, so the estimate is uniform in  $\delta$ .  $\square$

### 3.6.4 Two generators for the coefficients

Equation (3.36) is exactly eq. (1.55), the reversion problem of theorem 1.29, with the instance data

$$h(v) = \varphi(v), \quad h_m = \frac{(-1)^m}{m(m-1)}, \quad f(z, v) = \sum_{k \geq 1} \phi_k(z) (1+v)^{-\mu_k}, \quad \phi_k(z) = a_k z^k, \quad \mu_k = 2k-1. \quad (3.40)$$

Its triangular recurrence eq. (1.56) and its Bell form eq. (1.58) therefore apply verbatim; written out, the former is

$$e_n(\Lambda) = -\frac{1}{\Lambda} [z^n] \left\{ \varphi(E_{n-1}) + \sum_{k=1}^n a_k z^k (1 + E_{n-1})^{-(2k-1)} \right\}, \quad E_{n-1} = \sum_{j < n} e_j z^j. \quad (3.41)$$

Only the coefficient *structure* is subject-specific.

**Theorem 3.30** (Structure of the inverse blocks). *For every  $n \geq 1$ ,*

$$e_n(\Lambda) \in \Lambda^{-1} \mathbb{Q}[\Lambda^{-1}], \quad \deg_{\Lambda^{-1}} e_n \leq 2n-1, \quad e_n(\Lambda) = -\frac{a_n}{\Lambda} + \mathcal{O}(\Lambda^{-2}). \quad (3.42)$$

*Proof.* Specialising eq. (1.58) to eq. (3.40) gives

$$e_n = -\frac{1}{\Lambda} \left[ \sum_{m=2}^n \frac{(-1)^m}{m(m-1)} \widehat{B}_{n,m}(e) + \sum_{k=1}^n a_k \sum_{m=0}^{n-k} (-1)^m \binom{2k+m-2}{m} \widehat{B}_{n-k,m}(e) \right], \quad (3.43)$$

in which every occurring  $e_j$  has  $j < n$ . A monomial of  $\widehat{B}_{r,m}$  is a product  $e_{j_1} \cdots e_{j_m}$  with  $j_1 + \cdots + j_m = r$ ; by induction its  $\Lambda^{-1}$ -degree lies between  $m$  and  $\sum_{\nu} (2j_\nu - 1) = 2r - m$ , and the outer factor  $\Lambda^{-1}$  raises this by one. In the first sum  $r = n$  and  $m \geq 2$ , giving degree at most  $2n-1$ ; in the second  $r = n-k$ , giving at most  $2(n-k) - m + 1 \leq 2n-1$ . Rationality is preserved throughout. Finally the only contribution of  $\Lambda^{-1}$ -degree exactly 1 is the term  $k = n, m = 0$ , namely  $-a_n/\Lambda$ , because every Bell monomial contains at least one  $e_j = \mathcal{O}(\Lambda^{-1})$ .  $\square$

**Theorem 3.31** (Graded closed formula). *Let  $A(z) = \sum_{k \geq 1} a_k z^k$  and put*

$$A_{n,p} := [z^n] A(z)^p = \widehat{B}_{n,p}(a), \quad K_{n,j,p} := [v^{j-1}] \varphi(v)^{j-p} (1+v)^{-(2n-p)}. \quad (3.44)$$

Expand  $d_{2n-1}(\Lambda) = \sum_{j \geq 1} c_{n,j} \Lambda^{-j}$ . Then

$$c_{n,j} = \frac{(-1)^j}{j} \sum_{p=\lceil (j+1)/2 \rceil}^{\min(j,n)} \binom{j}{p} A_{n,p} K_{n;j,p} \quad (1 \leq j \leq 2n-1), \quad (3.45)$$

and  $c_{n,j} = 0$  for  $j > 2n-1$ . The right-hand side is a finite expression in Bernoulli numbers, binomial coefficients and ordinary partial Bell polynomials; it computes any single scalar coefficient without generating the lower blocks.

*Proof.* Put  $H(z, v) = \varphi(v) + \sum_{k \geq 1} a_k z^k (1+v)^{-(2k-1)}$ , so that eq. (3.36) reads  $E = -\Lambda^{-1} H(z, E)$ . The closed formula eq. (1.57) of theorem 1.29 gives

$$e_n = \sum_{j \geq 1} \frac{1}{j} [z^n v^{j-1}] \left( -\frac{H(z, v)}{\Lambda} \right)^j = \sum_{j \geq 1} \frac{(-1)^j}{j \Lambda^j} [z^n v^{j-1}] H(z, v)^j,$$

which already exhibits the grading by powers of  $\Lambda^{-1}$ . It remains to evaluate  $[z^n v^{j-1}] H^j$ . Expand  $H^j$  binomially, taking  $p$  of the  $j$  factors from the Bernoulli sector:

$$H^j = \sum_{p=0}^j \binom{j}{p} \varphi(v)^{j-p} \left( \sum_{k \geq 1} a_k z^k (1+v)^{-(2k-1)} \right)^p.$$

Extracting  $z^n$  from the  $p$ -th power, the surviving terms have  $k_1 + \dots + k_p = n$ ; their coefficients sum to  $A_{n,p}$  and they share the factor  $(1+v)^{-\sum_i (2k_i-1)} = (1+v)^{-(2n-p)}$ . Hence  $[z^n] H^j = \sum_p \binom{j}{p} A_{n,p} \varphi(v)^{j-p} (1+v)^{-(2n-p)}$ , and extracting  $v^{j-1}$  gives eq. (3.45). (Equation (1.57) caps the outer sum at  $j \leq 2n-1$ ; the support argument below recovers that bound independently.)

For the summation range:  $A_{n,p} = 0$  unless  $1 \leq p \leq n$ , while  $\varphi(v)^{j-p}$  has  $v$ -valuation  $2(j-p)$ , so a nonzero  $v^{j-1}$ -coefficient forces  $2(j-p) \leq j-1$ , i.e.  $p \geq \lceil (j+1)/2 \rceil$ . Combining,  $j \leq 2p-1 \leq 2n-1$ ; this re-proves the degree bound of eq. (3.42) and shows the expansion terminates.  $\square$

*Remark 3.32.* Equation (3.45) was checked against eq. (3.41) for all  $1 \leq n \leq 5$  and all  $1 \leq j \leq 2n-1$  in exact rational arithmetic: all 25 scalars agree. For explicit evaluation,  $K_{n;j,p}$  unfolds by definition 1.9 and lemma 1.12 into

$$K_{n;j,p} = \sum_{h=2(j-p)}^{j-1} \widehat{B}_{h,j-p}(\nu) (-1)^{j-1-h} \binom{2n-p+j-2-h}{j-1-h}, \quad \nu_1 = 0, \quad \nu_m = \frac{(-1)^m}{m(m-1)} \quad (m \geq 2),$$

with the conventions  $\widehat{B}_{0,0} = 1$  and  $\widehat{B}_{h,0} = 0$  for  $h > 0$ .

**Proposition 3.33** (A second, ungraded closed formula). *Let  $S = A^{(-1)}$  be the formal compositional inverse of  $A(z) = \sum_k a_k z^k$ , which exists because  $a_1 = -\frac{1}{6} \neq 0$ , with*

$$[w^m] S(w) = \frac{1}{m} [t^{m-1}] \left( \frac{t}{A(t)} \right)^m. \quad (3.46)$$

Put

$$G_\Lambda(E) = (1+E)^2 S\left(-\frac{\Lambda E + \varphi(E)}{1+E}\right) \in E \mathbb{Q}(\Lambda)[[E]], \quad G'_\Lambda(0) = -\frac{\Lambda}{a_1} = 6\Lambda \neq 0. \quad (3.47)$$

Then

$$e_n(\Lambda) = \frac{1}{n} [E^{n-1}] \left( \frac{E}{G_\Lambda(E)} \right)^n. \quad (3.48)$$

*Proof.* Since  $\sum_k a_k z^k (1+E)^{-(2k-1)} = (1+E) A(z/(1+E)^2)$ , eq. (3.36) reads  $A(z/(1+E)^2) = -(\Lambda E + \varphi(E))/(1+E)$ . Applying  $S$  and multiplying by  $(1+E)^2$  gives  $z = G_\Lambda(E)$ ; differentiating at  $E = 0$ , where  $\varphi(0) = \varphi'(0) = 0$ , gives  $G'_\Lambda(0) = S'(0) \cdot (-\Lambda) = -\Lambda/a_1$ . Equations (3.46) and (3.48) are then two applications of theorem 1.14. Although  $A$  is a divergent Gevrey series, every finite jet uses only finitely many  $a_k$ , so this is a formal identity requiring no convergence hypothesis.  $\square$

*Remark 3.34.* Equation (3.48) was verified against eq. (3.41) for  $n \leq 4$ . It is less useful than eq. (3.45) in practice: it does not separate the powers of  $\Lambda^{-1}$ , and it requires reverting  $A$  first. Its interest is conceptual — it exhibits the inverse blocks as a *double* Lagrange–Bürmann extraction, the inner one reverting the pure Bernoulli series.

### 3.6.5 Boundary coefficients and the global sign law

**Corollary 3.35** (The two edges of the coefficient triangle). *For every  $n \geq 1$ ,*

$$c_{n,1} = -a_n = -2c_n, \quad c_{n,2n-1} = \frac{1}{3^n} \binom{1/2}{n} = -\frac{1}{2^{n-1}(2n-1)} \binom{2n-1}{n} a_1^n. \quad (3.49)$$

*In particular the two closed forms recorded separately by S1 and S2 for the deepest coefficient are the same number.*

*Proof.* For  $j = 1$  the range in eq. (3.45) forces  $p = 1$ ; then  $A_{n,1} = a_n$ ,  $K_{n;1,1} = [v^0] (1+v)^{-(2n-1)} = 1$ , and the prefactor is  $-1$ .

For  $j = 2n - 1$ , the range forces  $p = n$ ; then  $A_{n,n} = a_1^n$  and

$$K_{n;2n-1,n} = [v^{2n-2}] \varphi(v)^{n-1} (1+v)^{-n} = [v^{2n-2}] \left( \frac{v^2}{2} \right)^{n-1} = 2^{1-n},$$

because  $\varphi(v)^{n-1}$  has valuation  $2n - 2$ . With  $\frac{(-1)^{2n-1}}{2^{n-1}} \binom{2n-1}{n}$  as prefactor this is the second expression in eq. (3.49).

Equality of the two expressions is elementary. Using  $\binom{1/2}{n} = (-1)^{n-1} \binom{2n}{n} (4^n (2n-1))^{-1}$ ,

$$\frac{1}{3^n} \binom{1/2}{n} = (-1)^{n-1} \frac{\binom{2n}{n}}{12^n (2n-1)},$$

while  $a_1 = -\frac{1}{6}$  and  $\binom{2n-1}{n} = \frac{1}{2} \binom{2n}{n}$  give

$$-\frac{\binom{2n-1}{n} a_1^n}{2^{n-1} (2n-1)} = (-1)^{n+1} \frac{\binom{2n}{n}}{2 \cdot 2^{n-1} 6^n (2n-1)} = (-1)^{n+1} \frac{\binom{2n}{n}}{12^n (2n-1)}.$$

$\square$

*Remark 3.36* (The deepest edge as a square root). Let  $t_n = c_{n,2n-1}$ . Only the quadratic part of  $\varphi$  and the single coefficient  $a_1$  can reach  $\Lambda^{-(2n-1)}$ : in eq. (3.43) the degree bound  $2r - m + 1$  attains  $2n - 1$  only for  $r = n$ ,  $m = 2$  in the first sum, since in the second sum  $2(n-k) - m + 1 = 2n - 1$  forces  $k = 1$ ,  $m = 0$ , where  $\widehat{B}_{n-1,0} = 0$  for  $n \geq 2$ . Hence  $t_1 = -a_1 = \frac{1}{6}$  and  $t_n = -\frac{1}{2} \sum_{j=1}^{n-1} t_j t_{n-j}$  for  $n \geq 2$ , i.e.  $T(z) = \sum_n t_n z^n$  satisfies  $T + \frac{1}{2} T^2 = \frac{1}{6} z$ , whence  $T(z) = -1 + \sqrt{1 + z/3}$  and  $t_n = 3^{-n} \binom{1/2}{n}$ . Equivalently, the maximal-pole part of eq. (3.36) is  $\Lambda E + \frac{1}{2} E^2 - \frac{1}{6} z = 0$  with root  $E = -\Lambda + \sqrt{\Lambda^2 + z/3}$ . S1 asserts this without proof; the degree bookkeeping above supplies it.

**Theorem 3.37** (Coefficientwise sign law). *For every  $n \geq 1$ ,*

$$(-1)^{n+1} d_{2n-1}(\Lambda) \in \Lambda^{-1} \mathbb{Q}_{>0}[\Lambda^{-1}], \quad \deg_{\Lambda^{-1}} d_{2n-1} = 2n - 1 \text{ exactly.} \quad (3.50)$$

*That is,  $c_{n,j} \neq 0$  with sign  $(-1)^{n+1}$  for every  $1 \leq j \leq 2n - 1$ , and  $c_{n,j} = 0$  beyond.*



*Proof.* Write  $a_k = (-1)^k a_k^+$  with  $a_k^+ = 2(2k-2)! \eta(2k) \pi^{-2k} > 0$  by eq. (3.19). Substituting  $z = -\zeta$  and  $E = -P$  into eq. (3.36) and using  $\varphi(-P) = \sum_{m \geq 2} P^m / (m(m-1))$  turns it into

$$\Lambda P = \sum_{m \geq 2} \frac{P^m}{m(m-1)} + \sum_{k \geq 1} a_k^+ \zeta^k (1-P)^{-(2k-1)}. \quad (3.51)$$

Every coefficient on the right is *strictly positive*:  $1/(m(m-1)) > 0$ ,  $a_k^+ > 0$ , and  $(1-P)^{-(2k-1)} = \sum_{m \geq 0} \binom{2k+m-2}{m} P^m$  has positive coefficients. Writing  $P = \sum_n p_n(\Lambda) \zeta^n$ , the triangular recurrence obtained from eq. (3.51) expresses  $p_n$  as  $\Lambda^{-1}$  times a positive-coefficient polynomial in  $p_1, \dots, p_{n-1}$ ; by induction each  $p_n \in \Lambda^{-1} \mathbb{Q}_{\geq 0}[\Lambda^{-1}]$ , with no cancellation possible anywhere. Since  $E(z) = -P(-z)$ , we get  $d_{2n-1} = e_n = (-1)^{n+1} p_n$ , which gives the sign statement once we show that  $p_n$  has a *strictly positive* coefficient at every  $\Lambda^{-j}$  with  $1 \leq j \leq 2n-1$ .

We prove that by induction, exhibiting one contributing configuration for each  $j$ ; because all contributions are nonnegative, one suffices. For  $n = 1$ ,  $p_1 = a_1^+ / \Lambda$ , and  $2n-1 = 1$ . Let  $n \geq 2$  and assume the claim for all smaller indices.

- $j = 1$ : take the forcing term  $k = n$ ,  $m = 0$  in eq. (3.51), contributing  $a_n^+ \Lambda^{-1}$ .
- $2 \leq j \leq 2n-2$ : take the forcing term  $k = 1$  together with the single factor  $m = 1$  of  $(1-P)^{-1}$ , so that the contribution is  $\Lambda^{-1} a_1^+ \binom{1}{1}$  times a coefficient of  $p_{n-1}$ . By induction  $p_{n-1}$  has strictly positive coefficients at  $\Lambda^{-j'}$  for every  $1 \leq j' \leq 2n-3$ , and  $j-1$  runs over exactly that range.
- $j = 2n-1$ : take the nonlinear term  $m = 2$ , i.e. a product  $p_{n_1} p_{n_2}$  with  $n_1 + n_2 = n$ ,  $n_i \geq 1$ , at their top degrees  $2n_i - 1$ ; the resulting  $\Lambda^{-1}$ -degree is  $1 + (2n_1 - 1) + (2n_2 - 1) = 2n - 1$ .

Hence every  $c_{n,j}$ ,  $1 \leq j \leq 2n-1$ , is nonzero of sign  $(-1)^{n+1}$ ; theorem 3.31 already showed  $c_{n,j} = 0$  for  $j > 2n-1$ .  $\square$

*Remark 3.38.* The sign law was confirmed symbolically for  $n \leq 7$ : after multiplication by  $(-1)^{n+1}$  all 1, 3, 5, 7, 9, 11, 13 coefficients of the respective blocks are positive rationals, and the degree is exactly  $2n-1$  in each case. S2 states the theorem but its proof gestures at “a positive rooted-tree contribution” without exhibiting one; the attainability induction above replaces that step. The law is also the cheapest runtime invariant for a symbolic implementation (algorithm 3.64).

### 3.6.6 The explicit blocks

Repeated use of eq. (3.41) gives, in exact rational arithmetic,

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \quad (3.52)$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \quad (3.53)$$

$$d_5(\Lambda) = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\Lambda^5}, \quad (3.54)$$

$$d_7(\Lambda) = -\frac{1}{5443200\Lambda^7} (822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 + 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625), \quad (3.55)$$

$$d_9(\Lambda) = \frac{1}{359251200\Lambda^9} (309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 + 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 + 242550\Lambda + 40425), \quad (3.56)$$

$$d_{11}(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}} (46195687238400\Lambda^{10} + 8854370366400\Lambda^9$$

$$\begin{aligned}
& + 2171643318000\Lambda^8 + 605515022112\Lambda^7 + 212869408752\Lambda^6 \\
& + 84136852800\Lambda^5 + 35589153600\Lambda^4 + 14234019800\Lambda^3 \\
& + 4868113250\Lambda^2 + 1213962750\Lambda + 165540375). \tag{3.57}
\end{aligned}$$

The rapidly growing numerators are an argument for regenerating blocks from eq. (3.41) rather than storing a table;  $d_{13}$ , which S3 also displays, is omitted here for that reason. The alternating block signs and the coefficientwise positivity after multiplication by  $(-1)^{n+1}$  illustrate theorem 3.37; the leading and trailing coefficients illustrate corollary 3.35. In separated form,

$$\begin{aligned}
d_1 &= \frac{1}{6\Lambda}, & d_3 &= -\frac{7}{180\Lambda} - \frac{1}{36\Lambda^2} - \frac{1}{72\Lambda^3}, \\
d_5 &= \frac{31}{630\Lambda} + \frac{7}{270\Lambda^2} + \frac{17}{1080\Lambda^3} + \frac{5}{648\Lambda^4} + \frac{1}{432\Lambda^5}, \tag{3.58}
\end{aligned}$$

$$d_7 = -\frac{127}{840\Lambda} - \frac{4063}{75600\Lambda^2} - \frac{11413}{453600\Lambda^3} - \frac{1}{80\Lambda^4} - \frac{157}{25920\Lambda^5} - \frac{35}{15552\Lambda^6} - \frac{5}{10368\Lambda^7}. \tag{3.59}$$

The common-denominator forms are less prone to transcription error; the separated form makes  $c_{n,1} = -a_n$  visible. Through three blocks,

$$\begin{aligned}
\mathcal{D}_\delta^{-1}(y) &= T - 1 + \frac{1}{6T\Lambda} - \frac{14\Lambda^2 + 10\Lambda + 5}{360T^3\Lambda^3} \\
&+ \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360T^5\Lambda^5} + \mathcal{O}\left(\frac{1}{T^7\Lambda}\right). \tag{3.60}
\end{aligned}$$

*Remark 3.39* (Lambert-only form, and a correction to the usual first guess). Because  $T = 2R_\delta/w$  and  $\Lambda = w + 1$ , the expansion can be written without  $T$ . The first three terms are

$$\begin{aligned}
\mathcal{D}_\delta^{-1}(y) &\sim \frac{2R_\delta}{w} - 1 + \frac{w}{12R_\delta(w+1)} - \frac{w^3(14(w+1)^2 + 10(w+1) + 5)}{2880R_\delta^3(w+1)^3} \\
&+ \frac{w^5(2232(w+1)^4 + 1176(w+1)^3 + 714(w+1)^2 + 350(w+1) + 105)}{1451520R_\delta^5(w+1)^5} + \dots, \tag{3.61}
\end{aligned}$$

with  $w = W_0(2R_\delta/e)$ . A frequently used rough inversion is  $x \approx 2R_\delta/W_0(2R_\delta/e) - 1$ ; its leading correction is *not* of order  $T^{-1}$  alone but

$$\frac{1}{6\Lambda T} = \frac{w}{12R_\delta(w+1)} = \frac{1}{12R_\delta} \left(1 - \frac{1}{w+1}\right) \sim \frac{1}{12\log y}.$$

The factor  $\Lambda^{-1}$  is forced by the slope  $\Phi'(T) = \Lambda/2$  of eq. (3.33).

### 3.7 The outer iterated-logarithmic layer

Keeping  $W_0$  unexpanded is both more accurate and algebraically cleaner. To place the result inside a conventional polynomial–logarithmic transseries one applies theorem 1.33. Set

$$Z = \frac{2R_\delta}{e}, \quad L_1 = \log Z, \quad L_2 = \log L_1. \tag{3.62}$$

**Proposition 3.40** (Outer data). *With  $\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]$  the unsigned Stirling numbers of the first kind, theorem 1.33 gives, for  $Z \rightarrow +\infty$ ,*

$$W_0(Z) \sim L_1 - L_2 + \sum_{r \geq 1} \frac{P_r(L_2)}{L_1^r}, \quad P_r(t) = \sum_{m=1}^r (-1)^{r-m} \left[ \begin{smallmatrix} r \\ r-m+1 \end{smallmatrix} \right] \frac{t^m}{m!}. \tag{3.63}$$

Write  $W_0(Z)/L_1 = 1 + \sum_{j \geq 1} v_j(L_2)L_1^{-j}$  with  $v_1 = -L_2$  and  $v_j = P_{j-1}$  for  $j \geq 2$ , and define the reciprocal polynomials

$$\tau_0 = 1, \quad \tau_m = - \sum_{j=1}^m v_j \tau_{m-j} \quad (m \geq 1). \quad (3.64)$$

Then

$$T \sim \frac{2R_\delta}{L_1} \sum_{m \geq 0} \frac{\tau_m(L_2)}{L_1^m}, \quad \Lambda = 1 + W_0(Z). \quad (3.65)$$

The first data, recomputed from eqs. (3.63) and (3.64), are

$$P_1 = t, \quad P_2 = \frac{1}{2}t^2 - t, \quad P_3 = \frac{1}{3}t^3 - \frac{3}{2}t^2 + t, \quad (3.66)$$

$$\begin{aligned} \tau_0 &= 1, & \tau_1 &= L_2, & \tau_2 &= L_2^2 - L_2, \\ \tau_3 &= L_2^3 - \frac{5}{2}L_2^2 + L_2, & \tau_4 &= L_2^4 - \frac{13}{3}L_2^3 + \frac{9}{2}L_2^2 - L_2, \\ \tau_5 &= L_2^5 - \frac{77}{12}L_2^4 + \frac{71}{6}L_2^3 - 7L_2^2 + L_2. \end{aligned} \quad (3.67)$$

**Theorem 3.41** (Every coefficient of the fully expanded transseries). Put  $t = L_1^{-1}$ ,

$$\mathcal{W}(t; L_2) = 1 + \sum_{j \geq 1} v_j(L_2)t^j, \quad \mathcal{L}(t; L_2) = \mathcal{W}(t; L_2) + t, \quad (3.68)$$

so that  $W_0/L_1 = \mathcal{W}$ ,  $\Lambda/L_1 = \mathcal{L}$  and  $T = (2R_\delta/L_1)\mathcal{W}^{-1}$ . With  $e_n(\Lambda) = \sum_{p=1}^{2n-1} c_{n,p}\Lambda^{-p}$ ,

$$\mathcal{D}_\delta^{-1}(y) \sim \frac{2R_\delta}{L_1} \sum_{r \geq 0} \frac{\tau_r(L_2)}{L_1^r} - 1 + \sum_{n \geq 1} \left( \frac{L_1}{2R_\delta} \right)^{2n-1} \sum_{r \geq 1} \frac{D_{n,r}(L_2)}{L_1^r}, \quad (3.69)$$

where every  $D_{n,r} \in \mathbb{Q}[L_2]$  is given by the finite extraction

$$D_{n,r}(L_2) = [t^r] \left\{ \mathcal{W}(t; L_2)^{2n-1} \sum_{p=1}^{2n-1} c_{n,p} t^p \mathcal{L}(t; L_2)^{-p} \right\}. \quad (3.70)$$

Together with eqs. (3.45), (3.63) and (3.64) this is a general formula for every coefficient at every level. The leading term of the  $n$ -th inverse-power sector is  $-a_n L_1^{2n-2} (2R_\delta)^{-(2n-1)}$ .

*Proof.* The core term is eq. (3.65), and eq. (3.64) is reciprocal series division. For the  $n$ -th block,

$$e_n(\Lambda) T^{-(2n-1)} = \left( \frac{L_1}{2R_\delta} \right)^{2n-1} \mathcal{W}^{2n-1} \sum_{p=1}^{2n-1} c_{n,p} L_1^{-p} \mathcal{L}^{-p},$$

and setting  $t = L_1^{-1}$  and extracting  $[t^r]$  gives eq. (3.70); at fixed  $r$  only finitely many  $v_j$  and  $c_{n,p}$  occur, so the extraction is finite. The first nonzero power is  $r = 1$ , from  $p = 1$  and the constant terms  $\mathcal{W}(0) = \mathcal{L}(0) = 1$ , with  $c_{n,1} = -a_n$  by corollary 3.35.  $\square$

**Proposition 3.42** (Ordering of the scales). For every fixed  $M$ ,  $T/L_1^M \rightarrow \infty$  as  $y \rightarrow \infty$ . Consequently

$$\frac{T}{L_1^M} \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \frac{1}{T^5\Lambda} \gg \cdots, \quad (3.71)$$

so the constant shift  $-1$  is beyond every fixed order of the outer  $L_1^{-1}$ -expansion, yet dominates the entire inner grid. In particular the first Bernoulli displacement satisfies  $1/(6T\Lambda) \asymp 1/(12R_\delta) \asymp 1/(12 \log y)$ .

*Proof.*  $T = e^\Lambda$  with  $\Lambda = 1 + W_0(Z) \sim L_1$ , while  $L_1^M = e^{ML_2}$  with  $L_2 = \log L_1 = o(L_1)$ ; hence  $\log(T/L_1^M) = \Lambda - ML_2 \rightarrow \infty$ . The remaining comparisons are immediate from  $T \rightarrow \infty$  and eq. (3.33), which give  $T\Lambda = 2R_\delta\Lambda/(\Lambda - 1) \sim 2R_\delta$ .  $\square$

**Remark 3.43.** This ordering is destroyed if all terms are written as one informal series in  $1/\log y$ : the shift  $-1$ , the outer polynomials  $\tau_m(L_2)/L_1^m$  and the inner blocks  $d_{2n-1}(\Lambda)T^{1-2n}$  then become indistinguishable. S1, S2 and S3 all make this point; the quantitative version is eq. (3.71).

### 3.8 The periodic interpolation and its oscillatory inverse

S1 and S2 centre the periodic problem at the parity-mean constant  $\frac{1}{4} \log(2\pi)$ ; S3 centres it at the even-branch constant  $\frac{1}{2} \log \pi$  and consequently reports a leading displacement  $2a(1 + \cos \theta)/\Lambda$  rather than  $2a \cos \theta/\Lambda$ . The two are the same function expanded about different cores (they differ by  $2a/\Lambda + \mathcal{O}(\Lambda^{-2})$ , which is exactly  $h' \cdot a$  for the core map  $h$ ). This chapter adopts the mean centring, which makes the carrier a mean-zero oscillation with the cleanest harmonic content.

**Lemma 3.44** (Centred exact form of  $\mathcal{D}$ ). *With  $u = x + 1$ ,  $a = \frac{1}{4} \log(\pi/2)$  as in eq. (3.9) and*

$$A_* := \frac{1}{4} \log(2\pi) = 0.4594692666 \dots, \quad e^{A_*} = (2\pi)^{1/4} = 1.5832334871 \dots, \quad (3.72)$$

*one has, exactly and for all  $x > -1$ ,*

$$\log \mathcal{D}(x) = \Phi(u) + A_* - a \cos(\pi u) + \Omega(u). \quad (3.73)$$

*At an integer  $x = n$ ,  $\cos \pi u = (-1)^{n+1}$  and  $A_* - a \cos \pi u = A(n)$  of eq. (3.13): the periodic factor freezes to the correct parity constant.*

*Proof.* By eq. (3.7) and theorem 3.11 with  $\delta = 0$ ,  $\log \mathcal{D}(x) = \Phi(u) + A_0 + \Omega(u) + P(x)$  where  $P(x) = \frac{1}{4}(1 - \cos \pi x) \log(2/\pi) = -a(1 - \cos \pi x)$ . Since  $\cos \pi x = \cos \pi(u - 1) = -\cos \pi u$ ,  $P = -a(1 + \cos \pi u) = -a - a \cos \pi u$ , and  $A_0 - a = \frac{1}{2} \log \pi - \frac{1}{4} \log(\pi/2) = \frac{1}{4} \log(2\pi) = A_*$ .  $\square$

**Lemma 3.45** (Exact displacement equation). *Let  $R_* = \log y - A_*$  and let  $T, \Lambda$  be the core data of eq. (3.32) built from  $R_*$ ; put  $q = T^{-1}$ ,  $\theta = \pi T$ . Then  $x = \mathcal{D}^{-1}(y)$  satisfies  $x = T - 1 + \Delta$  where  $\Delta$  is the unique small root of*

$$\Lambda \Delta + \frac{\varphi(q\Delta)}{q} + 2\Omega(T + \Delta) - 2a \cos(\theta + \pi \Delta) = 0. \quad (3.74)$$

*Replacing  $\Omega$  by its asymptotic series turns eq. (3.74) into the formal equation  $\mathcal{F}(q, \Delta) = 0$  with*

$$\mathcal{F}(q, v) = \Lambda v + \sum_{m \geq 2} \frac{(-1)^m}{m(m-1)} v^m q^{m-1} - 2a \cos(\theta + \pi v) + 2 \sum_{k \geq 1} c_k q^{2k-1} (1 + qv)^{-(2k-1)}. \quad (3.75)$$

*Since the periodic term is  $\mathcal{O}(1)$  while every other nonlinear term carries a positive power of  $q$ , one has  $\Delta = \mathcal{O}(\Lambda^{-1})$ , not  $\mathcal{O}(q\Lambda^{-1})$ .*

*Proof.* Evaluate eq. (3.73) at  $u = T + \Delta$ , set it equal to  $\log y = R_* + A_*$ , subtract  $\Phi(T) = R_*$ , and multiply by 2. By the computation in lemma 3.28 with  $E = q\Delta$ ,  $2[\Phi(T + \Delta) - \Phi(T)] = T[\Lambda q\Delta + \varphi(q\Delta)] = \Lambda \Delta + \varphi(q\Delta)/q$ , since  $T = q^{-1}$ . This gives eq. (3.74). The last sum in eq. (3.75) is  $2\Omega(T(1 + qv))$  expanded. Finally  $\varphi(q\Delta)/q = q\Delta^2/2 + \mathcal{O}(q^2)$ , so at  $q = 0$  the equation reduces to  $\Lambda \Delta_0 = 2a \cos(\theta + \pi \Delta_0)$ , forcing  $\Delta_0 \asymp \Lambda^{-1}$ .  $\square$

#### 3.8.1 The periodic carrier

**Theorem 3.46** (Existence, uniqueness, and the closed carrier expansion). *Fix  $\theta \in \mathbb{R}$  and put  $\mu = 2a/\Lambda$ .*

(a) *If  $\Lambda > 2\pi a = 0.70934 \dots$ , the equation*

$$\Delta_0 = \mu \cos(\theta + \pi \Delta_0) \quad (3.76)$$

*has exactly one real solution, and  $|\Delta_0| \leq \mu$ .*

- (b) If moreover  $\mu \leq \frac{1}{5}$ , equivalently  $\Lambda \geq 10a = 1.12896 \dots$ , that solution is the sum of the convergent Lagrange–Bürmann series

$$\Delta_0 = \sum_{m \geq 1} \frac{\mu^m}{m!} \frac{d^{m-1}}{dv^{m-1}} \cos^m(\theta + \pi v) \Big|_{v=0} = \sum_{m \geq 1} \frac{a^m}{m! \Lambda^m} \sum_{j=0}^m \binom{m}{j} (i\pi(m-2j))^{m-1} e^{i(m-2j)\theta}. \quad (3.77)$$

The second expression is real. At inverse-logarithmic order  $m$  only the Fourier modes  $m, m-2, \dots, -m$  occur.

- (c) The first three terms are

$$\Delta_0 = \mu \cos \theta - \frac{\pi}{2} \mu^2 \sin 2\theta + \frac{\pi^2}{2} \mu^3 \cos \theta (2 - 3 \cos^2 \theta) + \mathcal{O}(\mu^4). \quad (3.78)$$

*Proof.* (a) The map  $g(v) = \mu \cos(\theta + \pi v)$  satisfies  $|g'| \leq \pi\mu = 2\pi a/\Lambda < 1$  and  $|g| \leq \mu$ , so it is a contraction of  $\mathbb{R}$  into  $[-\mu, \mu]$ ; Banach's theorem gives a unique fixed point there, and any real fixed point lies in  $[-\mu, \mu]$ .

(b) Equation (3.76) is  $v = \mu\phi(v)$  with  $\phi(v) = \cos(\theta + \pi v)$  entire. By the classical convergence criterion for Lagrange's series, if there is  $r > 0$  with  $|\mu| \max_{|v|=r} |\phi(v)| < r$ , then the series  $\sum_m \frac{\mu^m}{m!} \left(\frac{d}{dv}\right)^{m-1} \phi(v)^m \Big|_{v=0}$  converges and its sum is the unique root of  $v = \mu\phi(v)$  in  $|v| < r$ . Here  $|\cos(\theta + \pi v)| \leq \cosh(\pi|\Im v|) \leq \cosh(\pi r)$  on  $|v| = r$ ; with  $r = 0.38$  and  $\mu \leq \frac{1}{5}$ ,  $\mu \cosh(0.38\pi) \leq 0.3603 < 0.38$ . (The optimal choice  $\max_{r>0} r/\cosh(\pi r) = 0.21095 \dots$  at  $r = 0.382$  shows  $\mu < 0.2109$  already suffices.) For the Fourier form, insert  $\cos^m v = 2^{-m} \sum_j \binom{m}{j} e^{i(m-2j)v}$  into the derivative:

$$\frac{d^{m-1}}{dv^{m-1}} \cos^m(\theta + \pi v) \Big|_{v=0} = 2^{-m} \sum_{j=0}^m \binom{m}{j} (i\pi(m-2j))^{m-1} e^{i(m-2j)\theta},$$

and  $\mu^m 2^{-m} = a^m \Lambda^{-m}$ . Pairing  $j$  with  $m-j$  replaces  $m-2j$  by its negative, conjugating each summand, so the sum is real. Only  $|m-2j| \leq m$  occurs, with the same parity as  $m$ .

(c) For  $m = 1$  the derivative is  $\cos \theta$ . For  $m = 2$ ,  $\frac{d}{dv} \cos^2 = -2\pi \cos \sin = -\pi \sin 2(\cdot)$ , giving  $-\frac{\pi}{2} \mu^2 \sin 2\theta$ . For  $m = 3$ ,  $\frac{d^2}{dv^2} \cos^3(\theta + \pi v)|_0 = 3\pi^2 \cos \theta (2 - 3 \cos^2 \theta)$ , and division by  $3!$  gives the third term.  $\square$

S1 states the carrier expansion as “convergent for sufficiently large  $\Lambda$ ” with no threshold and no criterion, and S2 proves only the contraction estimate, from which convergence of the Lagrange series does not follow. Theorem 3.46 separates the two claims and supplies an explicit sufficient condition,  $\mu \leq \frac{1}{5}$ . Numerically at  $\Lambda = 3$ ,  $\theta = 2$ , the partial sums  $m \leq 1, \dots, 5$  of eq. (3.77) approximate the exact fixed point  $-0.0256993556$  with errors  $5.6 \cdot 10^{-3}, 1.1 \cdot 10^{-3}, 1.8 \cdot 10^{-4}, 1.9 \cdot 10^{-5}, 1.9 \cdot 10^{-6}$ , consistent with a convergence ratio  $\approx \mu/0.211$ .

### 3.8.2 Algebraic corrections around the carrier

**Theorem 3.47** (Triangular periodic-grid recurrence). *Let*

$$M := \partial_v \mathcal{F}(0, \Delta_0) = \Lambda + 2\pi a \sin(\theta + \pi \Delta_0), \quad (3.79)$$

which satisfies  $M \geq \Lambda - 2\pi a > 0$  for  $\Lambda > 2\pi a$ . Seek  $\Delta = \Delta_0 + \sum_{n \geq 1} \Delta_n q^n$  with each  $\Delta_n$  a formal series in  $\Lambda^{-1}$  with trigonometric-polynomial coefficients, and put  $\Delta_{<n} = \Delta_0 + \sum_{j < n} \Delta_j q^j$ . Then the solution of  $\mathcal{F} = 0$  is unique and generated by

$$\Delta_n = -\frac{1}{M} [q^n] \mathcal{F}(q, \Delta_{<n}(q)), \quad n \geq 1. \quad (3.80)$$

The first two are

$$\Delta_1 = \frac{\frac{1}{6} - \frac{1}{2}\Delta_0^2}{M}, \quad \Delta_2 = -\frac{1}{M} \left[ a\pi^2 \cos(\theta + \pi\Delta_0) \Delta_1^2 + \Delta_0 \Delta_1 + \frac{\Delta_0 - \Delta_0^3}{6} \right]. \quad (3.81)$$

*Proof.* In  $[q^n] \mathcal{F}(q, \Delta)$  the unknown  $\Delta_n$  appears only linearly, through the  $v$ -derivative of the  $q$ -free part  $\Lambda v - 2a \cos(\theta + \pi v)$  evaluated at  $v = \Delta_0$ , which is  $M$ ; every other contribution involves only  $\Delta_j$  with  $j < n$ . This is lemma 1.16 applied in the variable  $q$  over the coefficient ring generated by  $\Lambda^{-1}$  and  $e^{\pm i\theta}$ , and  $M$  is invertible there because  $M = \Lambda(1 + \mathcal{O}(\Lambda^{-1}))$ .

For  $n = 1$ : the  $m = 2$  term of eq. (3.75) gives  $\frac{1}{2}\Delta_0^2$ , the cosine gives  $2a\pi \sin(\theta + \pi\Delta_0)\Delta_1$ , and the  $k = 1$  Bernoulli term gives  $2c_1 = -\frac{1}{6}$ ; collecting gives  $M\Delta_1 + \frac{1}{2}\Delta_0^2 - \frac{1}{6} = 0$ . For  $n = 2$ : the  $m = 2$  term gives  $\Delta_0\Delta_1$ , the  $m = 3$  term gives  $-\frac{1}{6}\Delta_0^3$ , the second-order Taylor term of the cosine gives  $a\pi^2 \cos(\theta + \pi\Delta_0)\Delta_1^2$ , and expanding  $2c_1 q(1 + q\Delta)^{-1}$  gives  $-2c_1\Delta_0 = \frac{1}{6}\Delta_0$ ; collecting gives eq. (3.81).  $\square$

**Corollary 3.48** (Asymptotic meaning). *Fix  $N, K \geq 0$ . Truncate  $\Delta$  after  $q^N$  and expand each  $\Delta_n$  in  $\Lambda^{-1}$  through  $\Lambda^{-K}$ , obtaining  $\tilde{\Delta}$ . Then, as  $y \rightarrow +\infty$ ,*

$$\mathcal{D}^{-1}(y) = T - 1 + \tilde{\Delta} + \mathcal{O}\left(\frac{1}{T^{N+1}\Lambda}\right) + \mathcal{O}\left(\frac{1}{\Lambda^{K+2}}\right), \quad (3.82)$$

uniformly in the phase  $\theta = \pi T$ .

*Proof.* By construction eq. (3.80) cancels the residual of  $\mathcal{F}$  through  $q^N$ , and theorem 3.46 cancels it through  $\Lambda^{-K}$  at each fixed  $q$ -order; the replacement of  $\Omega$  by its truncation is controlled by theorem 3.17. The exact  $v$ -derivative of the left side of eq. (3.74) equals  $M + \mathcal{O}(q)$  and is bounded below by  $\Lambda - 2\pi a - \mathcal{O}(q)$ , a positive multiple of  $\Lambda$  for large  $T$ ; all trigonometric coefficients and their derivatives are bounded, so theorem 1.42 converts the residual into a root error after division by a quantity  $\asymp \Lambda$ , uniformly in  $\theta$ .  $\square$

**Proposition 3.49** (The parity branches resum the periodic sector). *At an integer  $u = n + 1$  one has  $A_* - a \cos \pi u = A_\delta$  with  $\delta \equiv n \pmod{2}$ . The branchwise normalisation therefore absorbs the whole  $\mathcal{O}(1)$  periodic constant into the target  $R_\delta$  before the core is formed, so the  $\mathcal{O}(\Lambda^{-1})$  carrier disappears and the first correction is the much smaller  $1/(6T\Lambda)$ . In this precise sense the two parity transseries are subsequence resummations of the periodic one: they build the parity phase into the core instead of expanding it perturbatively.*

*Proof.* Immediate from lemma 3.44 and theorems 3.29 and 3.46.  $\square$

**Proposition 3.50** (Phase-coordinate reversion, formal). *Let  $h(R) = 2R/W_0(2R/e)$  be the inverse of  $\Phi$  (lemma 3.27), and set  $\mathcal{A}(R) = -a \cos(\pi h(R)) + \Omega(h(R))$ . Then the equation  $\Phi(u) + \mathcal{A}(\Phi(u)) = R_*$  — which is eq. (3.73) rewritten in the phase coordinate  $\sigma = \Phi(u)$  — has the formal reversion*

$$\sigma = R_* + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left( \frac{d}{dR_*} \right)^{m-1} \mathcal{A}(R_*)^m, \quad \mathcal{D}^{-1}(y) = h(\sigma) - 1. \quad (3.83)$$

*Proof.* Write  $\sigma = R_* + \varsigma$ ; then  $\varsigma = -\mathcal{A}(R_* + \varsigma)$ , and Lagrange's formula for  $\varsigma = \tau \phi(\varsigma)$  with  $\phi(\varsigma) = -\mathcal{A}(R_* + \varsigma)$ ,  $\tau = 1$ , together with the translation invariance of  $d/d\varsigma$ , gives eq. (3.83).  $\square$

S3 states eq. (3.83) as a boxed asymptotic expansion, writing “setting  $\tau = 1$  as an asymptotic expansion gives”. That is a formal-to-analytic slide:  $\mathcal{A}$  is not small in any norm that makes the reversion series converge, and  $m$ -fold differentiation with respect to  $R_*$  produces factors  $h'(R_*) = 2/\Lambda$  whose accumulated effect is exactly what needs to be estimated. The statement is demoted here to a *formal* proposition. An asymptotic claim would require, at each order  $m$ , a bound on  $(d/dR_*)^{m-1} \mathcal{A}^m$  uniform in the oscillatory phase — available for the truncated grid of corollary 3.48, which is therefore

the statement to use. Comparing the two routes at leading order: the  $m = 1$  term of eq. (3.83) contributes  $h'(R_*) \cdot (-\mathcal{A}) = 2a \cos(\pi T)/\Lambda + \mathcal{O}(T^{-1}\Lambda^{-1})$ , in agreement with eq. (3.78).

**Warning 3.51.** The carrier of theorem 3.46 records the choice eq. (3.7), not a property of  $n!!$ . By theorem 3.8 another interpolation, agreeing with  $n!!$  at every nonnegative integer, changes the amplitude of this sector from  $2a$  to  $2\sqrt{a^2 + \lambda^2}$  and shifts its phase, while leaving every integer value and both parity transseries untouched.

### 3.9 The discrete inverses

Objects (i) and (iii) of section 3.1 are intrinsic to the sequence. They are the residue-class staircase of definition 1.37 and theorem 1.39 at modulus  $r = 2$ , and nothing new has to be proved: remark 1.41 already records this chapter as that instance. Only the dictionary and the arithmetic caveat are subject-specific.

**Proposition 3.52** (Specialisation of the staircase). *Take  $r = 2$ , residue classes  $\rho \in \{0, 1\}$ ,  $A_n = n!!$ , and  $\nu_\rho(y) = \mathcal{D}_\rho^{-1}(y)$ , which is legitimate because  $\mathcal{D}_\rho$  interpolates the class- $\rho$  subsequence and is strictly increasing (proposition 3.3). Then eq. (1.75) reads*

$$N(y) := \min\{n \in \mathbb{N} : n!! \geq y\} = \min \left\{ 2 \left\lceil \frac{\nu_0(y)}{2} \right\rceil, 2 \left\lceil \frac{\nu_1(y) - 1}{2} \right\rceil + 1 \right\} \quad (y > 1), \quad (3.84)$$

and its floor counterpart, obtained from theorem 1.39 by replacing  $\lceil \cdot \rceil$  with  $\lfloor \cdot \rfloor$  and  $\min$  with  $\max$ , is

$$J(y) := \max\{n \in \mathbb{N} : n!! \leq y\} = \max_{\rho \in \{0, 1\}} \left\{ \rho + 2 \left\lfloor \frac{\nu_\rho(y) - \rho}{2} \right\rfloor \right\}. \quad (3.85)$$

Replacing  $\nu_\rho$  by the truncation of eq. (3.39) with a certified error bound  $\eta_\rho$  (for instance from eq. (3.100)) determines  $N(y)$  and  $J(y)$  exactly under the separation condition of theorem 1.39, and proposition 1.40 bounds the number of exact evaluations of  $n!!$  needed by  $\sum_\rho (\lfloor \eta_\rho \rfloor + 2)$ . Those evaluations should be made as  $\log \mathcal{D}_\rho(n) = \log \kappa_\rho + \frac{n}{2} \log 2 + \log \Gamma(\frac{n}{2} + 1)$ , or by eq. (3.1) in exact integers.

*Proof.* Everything is theorem 1.39 and proposition 1.40 with  $r = 2$ ; the hypotheses hold by proposition 3.3 and, for the asymptotic input, by theorem 3.29.  $\square$

**Warning 3.53** (Never form  $y$ ). Equation (3.84), evaluated with four inverse blocks, was tested at  $y = n!!$  for every  $6 \leq n \leq 199$  and returned the correct  $N(y) = n$  in all 194 cases – but only when the comparison  $A_n \geq y$  is made on exact integers or on  $\log y$ . Converting  $n!!$  to a fixed-precision float first destroys the comparison long before the asymptotics fail: at 80 working digits this already produces spurious off-by-one answers from  $n \approx 120$  onwards, because  $120!!$  has more than 100 digits. The failure is arithmetic, not asymptotic, and it is easy to misdiagnose as a defect of the expansion.

### 3.10 Beyond all algebraic orders

**Proposition 3.54** (Optimal truncation, forward and inverse). *Let  $u > 0$ . The terms of eq. (3.18) have moduli  $|c_k|u^{1-2k}$  with consecutive ratio  $\sim 2k(2k-1)/(\pi u)^2$ ; the least term occurs at  $2k \approx \pi u$  and*

$$\min_k \left| \frac{c_k}{u^{2k-1}} \right| = \mathcal{O}\left(u^{-1/2} e^{-\pi u}\right). \quad (3.86)$$

Consequently, by theorem 1.46 and the slope  $\Phi'(T) = \Lambda/2$ , the optimally truncated branchwise inverse has error

$$\mathcal{O}\left(\frac{e^{-\pi T}}{\sqrt{T} \Lambda}\right). \quad (3.87)$$

*Proof.* Equation (3.23) gives  $|c_k| = (2k-2)!\pi^{-2k}(1 + \mathcal{O}(4^{-k}))$ , whence the ratio. At  $2k = \pi u$ , Stirling's formula for  $(2k-2)!$  gives eq. (3.86). For the inverse, theorem 3.17 bounds the forward logarithmic residual at optimal truncation by eq. (3.86) with  $u = T(1 + \mathcal{O}(T^{-2}))$ ; dividing by the slope  $\frac{1}{2}\Lambda + \mathcal{O}(T^{-2})$  via theorems 1.42 and 1.44 gives eq. (3.87).  $\square$

**Proposition 3.55** (Transport of a forward exponential sector). *Suppose an exponentially improved representation of the centred forward phase retains the algebraic terms  $c_1, \dots, c_K$  and has an additional residual  $\mathcal{R}$ , and let  $u_{\text{alg}}$  solve the correspondingly truncated algebraic inverse problem. Then*

$$u - u_{\text{alg}} = -\frac{2\mathcal{R}(u_{\text{alg}})}{\log u_{\text{alg}} - 2\sum_{k=1}^K (2k-1)c_k u_{\text{alg}}^{-2k}} + \mathcal{O}\left(\frac{\sup |F''| \mathcal{R}^2}{\Lambda^3} + \frac{|\mathcal{R} \mathcal{R}'|}{\Lambda^2}\right), \quad (3.88)$$

where  $F$  is the doubled algebraic phase and the supremum is over the interval between  $u_{\text{alg}}$  and the exact root; the estimate assumes the displayed denominator is  $\asymp \Lambda$  and  $|\mathcal{R}'| = o(\Lambda)$ . At leading order this is  $-2\mathcal{R}(T)/\Lambda$ .

*Proof.* Let  $F$  be the doubled algebraic phase minus  $2\mathcal{R}$ , so  $F(u_{\text{alg}}) = 0$  and the exact phase equals  $F + 2\mathcal{R}$ . Taylor expansion of  $0 = F(u_{\text{alg}} + \Delta) + 2\mathcal{R}(u_{\text{alg}} + \Delta)$  about  $u_{\text{alg}}$ , with  $F'(u_{\text{alg}})$  the stated denominator, gives the first Newton correction; replacing  $F' + 2\mathcal{R}'$  by  $F'$  and iterating once produces the two displayed error terms.  $\square$

**Warning 3.56** (What the exponential sector is, and is not). The Borel poles of eq. (3.16) lie at  $s = \pm i\pi m$ , so the exponentials associated with them are  $e^{\mp i\pi m u}$  together with direction-dependent terminant multipliers, not real exponentials. On the positive axis the optimally truncated remainder has modulus  $\asymp u^{-1/2}e^{-\pi u}$ , but this is the size of a *terminant-weighted combination*: appending a bare term  $\text{const} \cdot e^{-\pi u}$  to the divergent Bernoulli series is not a valid transseries completion, and conflating the two gives incorrect Stokes statements. The unambiguous positive-real completion is the exact integral eq. (3.15) itself, which by theorem 3.17 even provides two-sided bounds at every order; eq. (3.88) then carries any rigorously normalised forward completion through the inverse map. All three sources make this point; S3 states it most sharply.

### 3.11 Complex inverse sheets

**Proposition 3.57** (Local sheets). *Fix a branch of  $\log y$  and integers  $\ell$  (logarithmic winding) and  $k$  (Lambert branch). Put*

$$R_{\delta,\ell} = \log y + 2\pi i\ell - A_\delta, \quad w_{k,\ell} = W_k\left(\frac{2R_{\delta,\ell}}{e}\right), \quad T_{k,\ell} = \frac{2R_{\delta,\ell}}{w_{k,\ell}}, \quad \Lambda_{k,\ell} = w_{k,\ell} + 1. \quad (3.89)$$

*In any sector in which  $T_{k,\ell} \rightarrow \infty$ , the Stirling expansion for  $\log \Gamma$  is uniform, and the corresponding solution branch of  $\mathcal{D}_\delta(x) = y$  is simple, the same universal polynomials give*

$$x_{\delta;k,\ell}(y) \sim T_{k,\ell} - 1 + \sum_{n \geq 1} \frac{d_{2n-1}(\Lambda_{k,\ell})}{T_{k,\ell}^{2n-1}}. \quad (3.90)$$

*Proof.* Every step of lemma 3.28 and theorem 3.30 is an identity of formal series over  $\mathbb{C}(\Lambda)$  and uses only  $\Phi(T) = R$ ,  $\Lambda = \log T$ ; eq. (3.89) secures those. The analytic statement is the sectorial version of theorem 3.29, with the mean-value step replaced by the standard implicit-function estimate in the sector.  $\square$

**Remark 3.58** (Branch caveats). The pair  $(k, \ell)$  is an asymptotic label, not a global classification: distinct labels can be joined by analytic continuation around the poles of  $\Gamma$  or the logarithmic cut. Branch collisions occur exactly where  $\mathcal{D}'_\delta = 0$ , i.e. where  $\log 2 + \psi(x/2 + 1) = 0$ ; by proposition 3.3 this never happens for real  $x \geq 0$ , but it does for complex  $x$ , and near such points eq. (3.90) must be replaced by a Puiseux-type local analysis.



### 3.12 Multifactorials

Only S3 treats multifactorials. Its normal form is reproduced here with a proof, and theorem 3.61 is new: it subsumes the “half-shifted inverse Gamma” scaling that S1 and S2 record separately, by identifying the Gamma case as the member  $p = 1$ .

**Proposition 3.59** (Residue-class branches). *Let  $p \geq 1$  and  $r \in \{1, \dots, p\}$ . For integers  $x \equiv r \pmod{p}$ ,  $x \geq r$ , set  $x!_{(p)} = x(x-p)(x-2p) \cdots r$ . Then, with*

$$\mathcal{D}_{p,r}(x) = C_{p,r} p^{x/p} \Gamma\left(\frac{x}{p} + 1\right), \quad C_{p,r} = \frac{p^{1-r/p}}{\Gamma(r/p)}, \quad (3.91)$$

one has  $\mathcal{D}_{p,r}(x) = x!_{(p)}$  for every such  $x$ . For  $p = 2$ ,  $C_{2,2} = 1 = \kappa_0$  and  $C_{2,1} = \sqrt{2/\pi} = \kappa_1$ ; for  $p = 1$ ,  $C_{1,1} = 1$  and  $\mathcal{D}_{1,1}(x) = \Gamma(x+1)$ .

*Proof.* Write  $x = r + mp$ . Then  $x!_{(p)} = \prod_{i=0}^m (r + ip) = p^{m+1} \prod_{i=0}^m (\frac{r}{p} + i) = p^{m+1} \Gamma(\frac{r}{p} + m + 1) / \Gamma(\frac{r}{p})$ , while  $\mathcal{D}_{p,r}(x) = C_{p,r} p^{r/p+m} \Gamma(\frac{r}{p} + m + 1)$ . Equating gives  $C_{p,r} = p^{1-r/p} / \Gamma(r/p)$ .  $\square$

**Theorem 3.60** (Centred multifactorial normal form). *Put  $u = x + \frac{p}{2}$  and  $A_{p,r} = \log C_{p,r} + \frac{1}{2} \log(2\pi/p)$ . Then, as  $x \rightarrow +\infty$ ,*

$$\log \mathcal{D}_{p,r}(x) = \frac{u}{p} (\log u - 1) + A_{p,r} + \Omega_p(u), \quad \Omega_p(u) \sim \sum_{k \geq 1} \frac{c_k^{(p)}}{u^{2k-1}}, \quad c_k^{(p)} = \frac{p^{2k-1} B_{2k}(\frac{1}{2})}{2k(2k-1)}, \quad (3.92)$$

and  $\Omega_p$  has the exact Borel–Laplace representation

$$\Omega_p(u) = \int_0^\infty e^{-us} \frac{\frac{ps}{2 \sinh(ps/2)} - 1}{ps^2} ds, \quad (3.93)$$

whose singularities are the simple poles  $s = \pm 2\pi im/p$ . Hence the optimally truncated forward remainder has scale  $e^{-2\pi u/p}$ . For  $p = 2$  this is theorems 3.11 and 3.15; for  $p = 1$  it is the half-shifted Stirling expansion of  $\log \Gamma(x+1) = \log \Gamma(u + \frac{1}{2})$ .

*Proof.* Put  $t = u/p$ , so  $x = pt - \frac{p}{2}$  and  $\log \mathcal{D}_{p,r} = \log C_{p,r} + (t - \frac{1}{2}) \log p + \log \Gamma(t + \frac{1}{2})$ . The half-shifted Stirling expansion  $\log \Gamma(t + \frac{1}{2}) \sim t \log t - t + \frac{1}{2} \log(2\pi) + \sum_k B_{2k}(\frac{1}{2}) (2k(2k-1))^{-1} t^{1-2k}$  – the case  $a = \frac{1}{2}$  of  $\log \Gamma(t+a)$ , in which every odd-index term vanishes because  $B_{2j+1}(\frac{1}{2}) = 0$  – gives

$$\log \mathcal{D}_{p,r} \sim \log C_{p,r} - \frac{1}{2} \log p + \frac{1}{2} \log(2\pi) + \frac{u}{p} (\log u - 1) + \sum_k \frac{p^{2k-1} B_{2k}(\frac{1}{2})}{2k(2k-1) u^{2k-1}},$$

since  $(t - \frac{1}{2}) \log p + t \log t = (u/p) \log u - \frac{1}{2} \log p$  and  $t^{1-2k} = p^{2k-1} u^{1-2k}$ . The constant is  $A_{p,r}$ . For eq. (3.93),  $\Omega_p(u) = \tilde{\Omega}(u/p)$  with  $\tilde{\Omega}$  as in the proof of theorem 3.15, where  $\tilde{\Omega}(t) = \int_0^\infty e^{-tr} (r/(2 \sinh(r/2)) - 1) r^{-2} dr$ ; substituting  $r = ps$  gives eq. (3.93), and  $\sinh(ps/2) = 0$  at  $s = 2\pi im/p$ .  $\square$

**Theorem 3.61** (Universal multifactorial inverse and its scaling). *Let  $R = \log y - A_{p,r}$ ,  $v = W_0(pR/e)$ ,  $X = pR/v$ ,  $\Lambda = v + 1 = \log X$ , so that  $\frac{X}{p} (\log X - 1) = R$ . With  $u = X(1+E)$ ,  $q = X^{-1}$ ,  $z = q^2$ , the normalised equation is*

$$\Lambda E + \varphi(E) + \sum_{k \geq 1} a_k^{(p)} z^k (1+E)^{-(2k-1)} = 0, \quad a_k^{(p)} := p c_k^{(p)} = p^{2k} \frac{B_{2k}(\frac{1}{2})}{2k(2k-1)}. \quad (3.94)$$

Every statement of theorems 3.30, 3.31 and 3.37, proposition 3.33, and corollary 3.35 holds verbatim with  $a_k$  replaced by  $a_k^{(p)}$ , and for every fixed  $N$

$$\mathcal{D}_{p,r}^{-1}(y) = X - \frac{p}{2} + \sum_{n=1}^N \frac{d_{2n-1}^{(p)}(\Lambda)}{X^{2n-1}} + \mathcal{O}\left(\frac{1}{X^{2N+1}\Lambda}\right). \quad (3.95)$$

Moreover the coefficient polynomials of all multifactorials are one family:

$$d_{2n-1}^{(p)}(\Lambda) = p^{2n} d_{2n-1}^{(1)}(\Lambda) \quad \text{for every } p \geq 1, n \geq 1, \quad (3.96)$$

where  $d^{(1)}$  are the blocks of the half-shifted inverse-Gamma problem. In particular  $d_{2n-1} = d_{2n-1}^{(2)} = 4^n d_{2n-1}^{(1)}$ .

*Proof.* The derivation of eq. (3.94) repeats lemma 3.28 with  $\frac{2}{T}$  replaced by  $\frac{p}{X}$ : the phase difference is  $\frac{X}{p}[\Lambda E + \varphi(E)]$ , and  $\frac{p}{X}\Omega_p(u) = \sum_k p c_k^{(p)} q^{2k} (1+E)^{-(2k-1)}$ . Since eq. (3.94) differs from eq. (3.36) only in the values of the forcing coefficients, all structural statements transfer. Now  $a_k^{(p)} = p^{2k} a_k^{(1)}$  by eq. (3.92), so substituting  $q \mapsto pq$  (equivalently  $z \mapsto p^2 z$ ) carries the  $p = 1$  equation into the general one with  $\Lambda$  untouched. Uniqueness of the formal solution gives  $E^{(p)}(q) = E^{(1)}(pq)$ , i.e.  $e_n^{(p)} = p^{2n} e_n^{(1)}$ , which is eq. (3.96). The remainder in eq. (3.95) follows from theorems 1.42 and 1.44 exactly as in theorem 3.29, the slope now being  $\frac{1}{p} \log X = \Lambda/p$ .  $\square$

*Remark 3.62.* Equation (3.96) explains and replaces the “scaling from the half-shifted inverse Gamma problem” of S1 ( $p_n = 4^n d_{2n-1}$ ) and the identity  $d_{2n-1}^{!!} = 4^n d_{2n-1}^{\Gamma}$  of S2: both are the case  $(p, 1) = (2, 1)$ . It was verified symbolically for  $n \leq 4$  and  $p \in \{2, 3\}$ . As a coefficient-polynomial identity it says nothing about the numerical Lambert data, which differ between the two problems; the identity is in  $\Lambda$ , not in  $y$ . It also supplies a strong regression test: an implementation storing only  $d^{(1)}$  reproduces every multifactorial.

| $p$ | object        | centre $u$        | phase                     | $a_1^{(p)}$       | Borel poles              |
|-----|---------------|-------------------|---------------------------|-------------------|--------------------------|
| 1   | $\Gamma(x+1)$ | $x + \frac{1}{2}$ | $u(\log u - 1)$           | $-\frac{1}{24}$   | $\pm 2\pi i m$           |
| 2   | $n!!$         | $x + 1$           | $\frac{u}{2}(\log u - 1)$ | $-\frac{1}{6}$    | $\pm \pi i m$            |
| 3   | $n!!!$        | $x + \frac{3}{2}$ | $\frac{u}{3}(\log u - 1)$ | $-\frac{3}{8}$    | $\pm \frac{2\pi}{3} i m$ |
| $p$ | $x!_{(p)}$    | $x + \frac{p}{2}$ | $\frac{u}{p}(\log u - 1)$ | $-\frac{p^2}{24}$ | $\pm \frac{2\pi}{p} i m$ |

### 3.13 Algorithms and certification

**Algorithm 3.63** (Forward coefficient table). Given  $M$  and a power  $\alpha \in \mathbb{Q}$ : compute  $c_k$  from eq. (3.19) for  $1 \leq k \leq \lceil M/2 \rceil$ ; set  $\gamma_{2k-1} = c_k$ ,  $\gamma_{2k} = 0$ ; run eq. (3.28). This costs  $\mathcal{O}(M^2)$  rational operations and never forms a symbolic exponential.

**Algorithm 3.64** (Inverse blocks). On the variable  $z = q^2$ , with  $\Lambda$  symbolic:

```

input: N
E := 0
for n := 1, ..., N:
    H := (1+E)*log(1+E) - E
    for k := 1, ..., n:
        H := H + a[k]*z^k*(1+E)^(-(2*k-1))
    H := truncate(H, z^(n+1))
    e[n] := - coefficient(H, z^n) / Lambda
    E := E + e[n]*z^n
    assert truncate(residual(E), z^(n+1)) = 0      # (2.residual)
    assert sign_invariant(e[n], n)                 # Theorem sign law
return e[1], ..., e[N]
```

Every product, logarithm and negative power may be truncated above  $z^n$  at step  $n$ ; no full symbolic logarithm is ever needed. Representing each  $e_n$  as a dense polynomial in  $\Lambda^{-1}$  of degree exactly  $2n - 1$  (theorem 3.37) gives an exact allocation size.

The two assertions are worth more than a comparison against a stored table. The first is the *formal residual certificate*: with  $E_N = \sum_{n \leq N} e_n z^n$ ,

$$\Lambda E_N + \varphi(E_N) + \sum_{k=1}^{N+1} a_k z^k (1 + E_N)^{-(2k-1)} = \mathcal{O}(z^{N+1}), \quad (3.97)$$

and the coefficient of  $z^{N+1}$ , divided by  $-\Lambda$ , is the next block. This single identity catches sign, indexing and centring mistakes at once. The second is theorem 3.37: after multiplication by  $(-1)^{n+1}$  every one of the  $2n - 1$  coefficients must be a positive rational. Independent checks available in addition:

- (1) expanding the exact gamma ratio of theorem 3.11 reproduces  $c_k$ ;
- (2) expanding the Borel kernel eq. (3.16) reproduces  $c_k$  from  $\eta$ -values, and proposition 3.23 reproduces  $b_k$  from  $\zeta$ -values;
- (3) the centre-shift identity eq. (3.31) links the two;
- (4) eq. (3.45) and eq. (3.48) must agree with eq. (3.41) coefficient by coefficient;
- (5) eq. (3.96) compares every block with the independently generated  $p = 1$  family.

Every displayed block  $d_1, \dots, d_{13}$  of this chapter passes all five.

**Algorithm 3.65** (Stable evaluation from a logarithmic target). Never form  $y$ ; accept  $\ell = \log y$ . For branch  $\delta$ :

$$R = \ell - A_\delta, \quad w = W_0(2R/e), \quad T = 2R/w, \quad \Lambda = w + 1, \quad x \leftarrow T - 1 + \sum_{n=1}^N \frac{d_{2n-1}(\Lambda)}{T^{2n-1}}. \quad (3.98)$$

Refine by Newton on the logarithmic residual  $F_\delta(x) = \log \kappa_\delta + \frac{x}{2} \log 2 + \log \Gamma(\frac{x}{2} + 1) - \ell$ ,

$$x_{\text{new}} = x - \frac{F_\delta(x)}{\frac{1}{2}(\log 2 + \psi(\frac{x}{2} + 1))}, \quad (3.99)$$

whose denominator is positive by eq. (3.6); convergence is quadratic once the asymptotic value is in the basin, which a few blocks always achieve in the ranges tested. For  $\mathcal{D}$ , add  $\frac{1}{4}(1 - \cos \pi x) \log(2/\pi)$  to  $F_0$  and its derivative to the denominator, and start from eq. (3.82).

**Proposition 3.66** (A posteriori certificate). Let  $\tilde{x}$  be any approximation and  $\rho = F_\delta(\tilde{x})$ . If  $I$  is an interval containing  $\tilde{x}$  and  $\mathcal{D}_\delta^{-1}(y)$ , then

$$|\tilde{x} - \mathcal{D}_\delta^{-1}(y)| \leq \frac{2|\rho|}{\inf_{\xi \in I} (\log 2 + \psi(\xi/2 + 1))}, \quad (3.100)$$

and since  $\psi$  is increasing the infimum is attained at the left endpoint. At large arguments the denominator is  $\sim \Lambda$ , matching eq. (3.39).

*Proof.* This is theorem 1.42 with the explicit slope eq. (3.6); the mean-value theorem applied to  $F_\delta$  on  $I$  gives eq. (3.100).  $\square$

Table 3.3: Forward: centred logarithmic expansion,  $\Phi(u) + A_\delta + \sum_{k \leq N} c_k u^{1-2k} - \log \mathcal{D}_\delta(n)$ ,  $u = n + 1$ . Signs alternate, confirming theorem 3.17.

| $\delta$ | $n$ | $N = 0$                 | $N = 1$                  | $N = 2$                  | $N = 3$                   | $N = 4$                  |
|----------|-----|-------------------------|--------------------------|--------------------------|---------------------------|--------------------------|
| 0        | 20  | $3.96616 \cdot 10^{-3}$ | $-2.09362 \cdot 10^{-6}$ | $5.98269 \cdot 10^{-9}$  | $-4.14413 \cdot 10^{-11}$ | $5.30661 \cdot 10^{-13}$ |
| 1        | 21  | $3.78606 \cdot 10^{-3}$ | $-1.82137 \cdot 10^{-6}$ | $4.74399 \cdot 10^{-9}$  | $-2.99567 \cdot 10^{-11}$ | $3.49749 \cdot 10^{-13}$ |
| 0        | 100 | $8.25064 \cdot 10^{-4}$ | $-1.88702 \cdot 10^{-8}$ | $2.34020 \cdot 10^{-12}$ | $-7.04697 \cdot 10^{-16}$ | $3.92938 \cdot 10^{-19}$ |
| 1        | 101 | $8.16975 \cdot 10^{-4}$ | $-1.83207 \cdot 10^{-8}$ | $2.22773 \cdot 10^{-12}$ | $-6.57742 \cdot 10^{-16}$ | $3.59602 \cdot 10^{-19}$ |

Table 3.4: Forward: relative error  $|F_M/\mathcal{D}_\delta - 1|$  of the centred multiplicative expansion eq. (3.29) truncated after  $s_M u^{-M}$ , with  $\delta \equiv x \pmod{2}$ .

| $x$ | $\delta$ | $M = 0$               | $M = 1$               | $M = 3$                | $M = 5$                | $M = 9$                |
|-----|----------|-----------------------|-----------------------|------------------------|------------------------|------------------------|
| 10  | 0        | $7.590 \cdot 10^{-3}$ | $4.330 \cdot 10^{-5}$ | $2.599 \cdot 10^{-7}$  | $4.974 \cdot 10^{-9}$  | $1.388 \cdot 10^{-11}$ |
| 11  | 1        | $6.957 \cdot 10^{-3}$ | $3.538 \cdot 10^{-5}$ | $1.751 \cdot 10^{-7}$  | $2.782 \cdot 10^{-9}$  | $5.454 \cdot 10^{-12}$ |
| 20  | 0        | $3.974 \cdot 10^{-3}$ | $9.988 \cdot 10^{-6}$ | $1.432 \cdot 10^{-8}$  | $6.756 \cdot 10^{-11}$ | $1.314 \cdot 10^{-14}$ |
| 21  | 1        | $3.793 \cdot 10^{-3}$ | $9.014 \cdot 10^{-6}$ | $1.166 \cdot 10^{-8}$  | $4.972 \cdot 10^{-11}$ | $7.956 \cdot 10^{-15}$ |
| 50  | 0        | $1.635 \cdot 10^{-3}$ | $1.483 \cdot 10^{-6}$ | $3.106 \cdot 10^{-10}$ | $2.113 \cdot 10^{-13}$ | $9.594 \cdot 10^{-19}$ |
| 51  | 1        | $1.604 \cdot 10^{-3}$ | $1.424 \cdot 10^{-6}$ | $2.861 \cdot 10^{-10}$ | $1.866 \cdot 10^{-13}$ | $7.800 \cdot 10^{-19}$ |

Table 3.5: Branchwise inverse:  $x_N - x$  with  $y = \mathcal{D}_\delta(x)$  exact and  $x_N$  the truncation of eq. (3.39) after  $N$  blocks. Signs alternate, as theorem 3.37 predicts for the dominant term.

| $\delta$ | $x$ | $N = 0$                 | $N = 1$                | $N = 2$                  | $N = 3$                 | $N = 5$                 |
|----------|-----|-------------------------|------------------------|--------------------------|-------------------------|-------------------------|
| 0        | 10  | $-6.3074 \cdot 10^{-3}$ | $1.6444 \cdot 10^{-5}$ | $-1.6097 \cdot 10^{-7}$  | $3.6890 \cdot 10^{-9}$  | $1.1386 \cdot 10^{-11}$ |
| 1        | 11  | $-5.5808 \cdot 10^{-3}$ | $1.2104 \cdot 10^{-5}$ | $-9.9815 \cdot 10^{-8}$  | $1.9338 \cdot 10^{-9}$  | $4.2632 \cdot 10^{-12}$ |
| 0        | 20  | $-2.6055 \cdot 10^{-3}$ | $1.7520 \cdot 10^{-6}$ | $-4.7751 \cdot 10^{-9}$  | $3.1029 \cdot 10^{-11}$ | $7.6506 \cdot 10^{-15}$ |
| 1        | 21  | $-2.4497 \cdot 10^{-3}$ | $1.4956 \cdot 10^{-6}$ | $-3.7170 \cdot 10^{-9}$  | $2.2041 \cdot 10^{-11}$ | $4.5233 \cdot 10^{-15}$ |
| 0        | 50  | $-8.3109 \cdot 10^{-4}$ | $8.9794 \cdot 10^{-8}$ | $-4.1941 \cdot 10^{-11}$ | $4.7156 \cdot 10^{-14}$ | $3.4429 \cdot 10^{-19}$ |
| 1        | 51  | $-8.1110 \cdot 10^{-4}$ | $8.4219 \cdot 10^{-8}$ | $-3.7845 \cdot 10^{-11}$ | $4.0942 \cdot 10^{-14}$ | $2.7669 \cdot 10^{-19}$ |
| 0        | 100 | $-3.5755 \cdot 10^{-4}$ | $9.5805 \cdot 10^{-9}$ | $-1.1469 \cdot 10^{-12}$ | $3.3159 \cdot 10^{-16}$ | $1.5896 \cdot 10^{-22}$ |
| 1        | 101 | $-3.5329 \cdot 10^{-4}$ | $9.2785 \cdot 10^{-9}$ | $-1.0892 \cdot 10^{-12}$ | $3.0878 \cdot 10^{-16}$ | $1.4232 \cdot 10^{-22}$ |

Table 3.6: Inverse of the periodic interpolation:  $(T - 1 + \tilde{\Delta}) - x$  with  $y = \mathcal{D}(x)$  exact, successively adding the carrier  $\Delta_0$  (computed as the exact fixed point of eq. (3.76)),  $\Delta_1/T$  and  $\Delta_2/T^2$  from eq. (3.81). The core error matches the predicted carrier amplitude  $2a/\Lambda$ : at  $x = 20$ ,  $2a/\Lambda = 0.074080$  against an observed 0.071518; at  $x = 100$ , 0.048919 against 0.048564. No source validates this sector numerically.

| $x$    | core                    | $T - 1$                 | $+\Delta_0$             | $+\Delta_1/T$          | $+\Delta_2/T^2$ |
|--------|-------------------------|-------------------------|-------------------------|------------------------|-----------------|
| 20     | $7.1518 \cdot 10^{-2}$  | $-2.5603 \cdot 10^{-3}$ | $-1.2693 \cdot 10^{-5}$ | $1.6637 \cdot 10^{-6}$ |                 |
| 21     | $-7.5539 \cdot 10^{-2}$ | $-2.4124 \cdot 10^{-3}$ | $1.4359 \cdot 10^{-5}$  | $1.4221 \cdot 10^{-6}$ |                 |
| 20.5   | $-2.5254 \cdot 10^{-3}$ | $-3.2850 \cdot 10^{-3}$ | $2.0627 \cdot 10^{-6}$  | $2.2808 \cdot 10^{-6}$ |                 |
| 100    | $4.8564 \cdot 10^{-2}$  | $-3.5495 \cdot 10^{-4}$ | $-2.3071 \cdot 10^{-7}$ | $9.4103 \cdot 10^{-9}$ |                 |
| 100.25 | $3.4218 \cdot 10^{-2}$  | $-3.9845 \cdot 10^{-4}$ | $-1.8912 \cdot 10^{-7}$ | $1.0779 \cdot 10^{-8}$ |                 |

Table 3.7: Multifactorial inverse eq. (3.95), using only the  $p = 1$  blocks rescaled by eq. (3.96). Entries are  $x_N - x$  with  $y = \mathcal{D}_{p,r}(x)$  exact; the  $p = 2$  row reproduces table 3.5.

| $p$ | $r$ | $x$ | $N = 0$                  | $N = 1$                 | $N = 2$                  | $N = 3$                  |
|-----|-----|-----|--------------------------|-------------------------|--------------------------|--------------------------|
| 2   | 2   | 20  | $-2.60549 \cdot 10^{-3}$ | $1.75197 \cdot 10^{-6}$ | $-4.77514 \cdot 10^{-9}$ | $3.10287 \cdot 10^{-11}$ |
| 3   | 1   | 22  | $-5.05001 \cdot 10^{-3}$ | $6.03949 \cdot 10^{-6}$ | $-2.95265 \cdot 10^{-8}$ | $3.44131 \cdot 10^{-10}$ |
| 3   | 2   | 20  | $-5.67882 \cdot 10^{-3}$ | $8.16471 \cdot 10^{-6}$ | $-4.75923 \cdot 10^{-8}$ | $6.60178 \cdot 10^{-10}$ |
| 4   | 3   | 23  | $-8.27262 \cdot 10^{-3}$ | $1.54424 \cdot 10^{-5}$ | $-1.18248 \cdot 10^{-7}$ | $2.15627 \cdot 10^{-9}$  |
| 5   | 2   | 27  | $-1.04166 \cdot 10^{-2}$ | $2.15596 \cdot 10^{-5}$ | $-1.85311 \cdot 10^{-7}$ | $3.79755 \cdot 10^{-9}$  |

### 3.14 Numerical validation

All tables in this section were recomputed at 220-digit precision with `mpmath`: the exact target  $\ell = \log \mathcal{D}_\delta(n)$  is formed from  $\log \Gamma$ , the Lambert data from eq. (3.98), and the blocks from eq. (3.41) in exact rational arithmetic. No rounded value of  $y$  is ever formed. Entries are *signed* (approximation minus exact), which the source tables partly suppress.

*Remark 3.67.* The near-geometric improvement at these moderate arguments is not a claim of convergence: at fixed  $x$  the Bernoulli-driven expansion diverges, and by proposition 3.54 the attainable accuracy is bounded below by  $\asymp e^{-\pi T}/(\sqrt{T}\Lambda)$ . In high-precision work, truncate at the least term or finish with eq. (3.99). Note also that the parity dependence is confined to the core: once the correct  $A_\delta$  is used, the two subsequences show nearly identical error patterns, as remark 3.4 predicts.

### 3.15 Logical status

Three kinds of statement have been used, and the chapter keeps them apart.

**Exact identities.** Equations (3.3), (3.7) and (3.91) and the interpolation statements propositions 3.3, 3.6 and 3.59; the centred normal forms eqs. (3.12), (3.14) and (3.73); the Lambert identities eq. (3.33); the Borel–Laplace representations eqs. (3.15) and (3.93) and the Mittag–Leffler decomposition eq. (3.16). These hold wherever the functions are defined.

**Formal coefficient identities.** The Bell formulae eqs. (3.26) and (3.43), the recurrences eqs. (3.28), (3.41) and (3.80), the closed formulae eqs. (3.45), (3.48) and (3.70), the boundary and sign statements corollary 3.35 and theorem 3.37, the centre-shift identity eq. (3.31), the scaling eq. (3.96), and proposition 3.50. These are identities of formal series over  $\mathbb{Q}(\Lambda)$ , or over the ring generated by  $\Lambda^{-1}$  and  $e^{\pm i\theta}$  in the periodic case. No convergence of the Bernoulli series is asserted anywhere.

**Poincaré asymptotics with remainders.** Theorems 3.16 and 3.17 and corollary 3.18 for the forward direction — where the remainder is in fact two-sided and sharp; theorem 3.29, corollary 3.48, and eq. (3.95) for the inverse, each obtained from a forward residual by theorems 1.42 and 1.44 together with the explicit slope eq. (3.6); and propositions 3.54 and 3.55 for the beyond-all-orders layer. Theorem 3.46(b) is a genuine convergence statement, with an explicit threshold.

*Remark 3.68* (Practical summary). For symbolic work, centre at  $u = x + 1$ , keep  $W_0$  exact, generate  $c_k$  from Bernoulli numbers, exponentiate with complete Bell polynomials, and invert with the triangular  $z = T^{-2}$  recurrence, checking eq. (3.97) and the sign law at every step. Expand  $W_0$  into iterated logarithms only as a final presentation layer. When the parity of  $n$  is known, use the branch inverse; when it is not, state the interpolation — by theorem 3.8 its choice changes the leading sector of the answer.

## Chapter 4

# The partition numbers (A000041)

This chapter merges three independent treatments of the partition numbers filed in the special-function-inversion collection: *Partition\_Number\_Transseries\_and\_Asymptotic\_Inverse* (below: **S1**), *Partition\_Number\_Transseries\_and\_Inverse* (**S2**), and *Partition\_Numbers\_Transseries\_and\_Inverse* (**S3**). All three take Rademacher's convergent series as analytic input and build a coefficient calculus on top of it; they differ in notation, in how the non-integral arithmetic phase is defined, and in which tail estimate is proved. Everything below that is not attributed is common to all three, restated once. Numerical constants, coefficient tables and error tables have been recomputed here (exact rationals through `fractions/sympy`; 140-digit `mpmath` for the tables); the recomputation is described in section 4.12.

### 4.1 Orientation: what is specific to $p(n)$

The partition function  $p(n)$  is defined by Euler's product

$$\sum_{n \geq 0} p(n) q^n = \prod_{m \geq 1} \frac{1}{1 - q^m}, \quad |q| < 1, \quad (4.1)$$

and is OEIS A000041. Hardy and Ramanujan proved

$$p(n) \sim \frac{1}{4\sqrt{3}n} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \quad (4.2)$$

and Rademacher replaced the asymptotic formula by an exactly convergent series. That single fact makes this chapter structurally different from Parts 2, 3 and 5: the exponentially small sectors are *given*, not reconstructed from the late behaviour of a divergent perturbative series. Three features are genuinely specific to  $p(n)$  and are the subject of this chapter.

1. **An arithmetic sector index.** The sectors are labelled by a positive integer  $k$  (a root of unity  $e^{2\pi i h/k}$ ) and a sign  $\sigma$ , and their amplitudes are the Rademacher sums  $A_k(n)$  built from Dedekind sums. These amplitudes are *periodic* in  $n$  with period  $k$ ; they are not constants, and after inversion they become bounded oscillations whose phase is quadratic in  $\log y$ .
2. **Non-well-ordered actions.** Relative to the dominant sector, the exponential actions are  $1 - 1/k$  (increasing to 1) and  $1 + 1/k$  (decreasing to 1). The second family has no least element. The support of the completed object is therefore not a locally finite Hahn grid, and the two signs may not be separated before summing over  $k$ .

3. **Convergence where one expects divergence.** Every fixed sector's algebraic series in  $n^{-1/2}$  has radius of convergence exactly  $\sqrt{24}$  in the variable  $n^{-1/2}$ , and the reversion series of the dominant sector converges as well. Nothing in this chapter is a divergent asymptotic series; consequently the least-term machinery of theorem 1.46 degenerates here, as recorded in remark 4.37.

Everything else — the Lambert core, the reversion around it, the flattening into logarithms, the passage from a smooth inverse to an integer staircase, and the transport of remainders — is Part 1 material and is *cited*, never re-proved.

#### 4.1.1 The exponential–power reduction and the Part 0 dictionary

Fix throughout

$$N = n - \frac{1}{24}, \quad \lambda = \pi\sqrt{\frac{2}{3}} = \frac{\pi\sqrt{6}}{3}, \quad \mu = \lambda\sqrt{N} = \frac{\pi}{6}\sqrt{24n-1}, \quad \kappa = \frac{\pi^2}{6\sqrt{3}}. \quad (4.3)$$

The dominant Rademacher sector, derived in theorem 4.14, is

$$P_{1,+}(n) = \frac{e^{\lambda\sqrt{N}}}{4\sqrt{3}N} \left(1 - \frac{1}{\lambda\sqrt{N}}\right). \quad (4.4)$$

Read as a function of  $N$ , eq. (4.4) is exactly an instance of the exponential–power model definition 1.2,  $F(x) = C \exp(ax^\alpha)x^b \exp Q(x^{-\delta})$ , with

$$x = N, \quad C = \frac{1}{4\sqrt{3}}, \quad a = \lambda, \quad \alpha = \frac{1}{2}, \quad b = -1, \quad \delta = \frac{1}{2}, \quad Q(t) = \log\left(1 - \frac{t}{\lambda}\right) = -\sum_{r \geq 1} \frac{t^r}{r\lambda^r}. \quad (4.5)$$

So the partition problem is the case  $\alpha = 1/2$ : a *square-root* phase, not a linear one. It is reduced to the linear case in two steps: the monomial substitution  $\xi = x^\alpha = \sqrt{N}$  of lemma 1.4, followed by the elementary linear rescaling  $\mu = \lambda\xi$  that normalises the exponential rate to 1.

**Lemma 4.1** (The  $\alpha$ -reduction for  $p(n)$ ). (i) *Lemma 1.4 applied to eq. (4.5) with  $\xi = \sqrt{N}$  gives an  $\alpha = 1$  model with data  $(\frac{1}{4\sqrt{3}}, \lambda, 1, -2, 1, Q)$ ,  $Q(t) = \log(1 - t/\lambda)$ ; note  $b/\alpha = -2$  and  $\delta/\alpha = 1$ , so this is the normalisation  $\delta = 1$  assumed from Part 1 onwards.* (ii) *The further substitution  $\mu = \lambda\xi$  rescales any  $\alpha = 1$  model with data  $(C, a, 1, b, 1, Q)$  to one with data  $(Ca^{-b}, 1, 1, b, 1, Q(a \cdot))$ . Here the two steps give*

$$P_{1,+} = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right) = C' e^{a'\mu} \mu^{b'} \exp Q'(\mu^{-\delta'}), \quad (4.6)$$

with the dictionary

$$C' = \kappa = \frac{\pi^2}{6\sqrt{3}}, \quad a' = 1, \quad \alpha' = 1, \quad b' = -2, \quad \delta' = 1, \quad Q'(t) = \log(1 - t).$$

The inverse map back to the index is exact and quadratic:

$$n = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}. \quad (4.7)$$

*Proof.* (i) is lemma 1.4 at  $\alpha = \delta = 1/2$ ,  $b = -1$ , which leaves  $C$ ,  $a$  and the coefficients of  $Q$  unchanged and replaces  $(b, \delta)$  by  $(b/\alpha, \delta/\alpha)$ . For (ii), substituting  $\xi = \mu/a$  in  $Ce^{a\xi}\xi^b \exp Q(\xi^{-1})$  gives  $Ca^{-b}e^\mu \mu^b \exp Q(a/\mu)$ , and  $t \mapsto Q(at)$  again lies in  $t\mathbb{R}[[t]]$ . Here  $a = \lambda$ ,  $b = -2$ , so  $Ca^{-b} = \lambda^2/(4\sqrt{3}) = \pi^2/(6\sqrt{3}) = \kappa$  (using  $\lambda^2 = 2\pi^2/3$ ) and  $Q'(t) = Q(\lambda t) = \log(1 - t)$ . Composing,  $\mu = \lambda\sqrt{N}$  and eq. (4.4) become eq. (4.6). Inverting  $\mu = \lambda\sqrt{n - 1/24}$  gives eq. (4.7), again by  $\lambda^2 = 2\pi^2/3$ .  $\square$

*Remark 4.2* (The model hypotheses hold exactly, and  $Q$  converges). Definition 1.2 asks for the analytic estimates eq. (1.3) and eq. (1.4) in addition to the formal shape. For the partition envelope both hold *exactly and trivially*:  $F(\mu) = \kappa e^\mu \mu^{-2} (1 - 1/\mu)$  is an elementary closed form, so  $\log F(\mu) - \log \kappa - \mu + 2 \log \mu = \log(1 - 1/\mu)$  is literally the sum of the series  $Q'$ , and  $(\log F)' = \Phi'$  is computed in closed form in lemma 4.49; no asymptotic estimate is being differentiated. In particular  $Q'(t) = \log(1 - t)$  is *convergent*, with radius 1. Remark 1.3 states that  $Q$  is divergent of Gevrey type in all four chapters of this volume; that is not so here, and the discrepancy should be read as a further instance of remark 4.37 — the partition chapter is the one place in the volume where the whole apparatus applies to a convergent object.

*Remark 4.3* (Why this is the point of the whole construction). After lemma 4.1 nothing in the inversion of the dominant sector is partition-specific. The three sources each rediscover, in their own notation, the Lambert core of theorem 1.20, the centred reversion of theorem 1.25 and the logarithmic flattening of theorem 1.33. We record the dictionary once (table 4.1) and cite. What is *not* generic — and occupies the rest of the chapter — is the arithmetic: the sector index, the Dedekind-sum amplitudes, the accumulation of actions at 1, and the interaction between the periodic amplitudes and the  $\mathcal{O}(1)$  shift produced by the reversion.

Table 4.1: Parameter dictionary from Part 1 to the partition chapter. All Part 0 statements are used at these values.

| Part 0 object                | Partition value                                                     | Reference        |
|------------------------------|---------------------------------------------------------------------|------------------|
| model variable $x$           | $\mu = \pi \sqrt{2(n - 1/24)/3}$                                    | eq. (4.3)        |
| $C$                          | $\kappa = \pi^2/(6\sqrt{3})$                                        | lemma 4.1        |
| $a, \alpha$                  | 1, 1 (after reduction and rescaling; $\lambda, 1/2$ before)         | lemma 4.1        |
| $b$                          | -2                                                                  | lemma 4.1        |
| $Q(t), \delta$               | $\log(1 - t), 1$                                                    | lemma 4.1        |
| Lambert core $X$             | $-2W_{-1}\left(-\frac{1}{2}\sqrt{\kappa/y}\right)$                  | proposition 4.46 |
| $\Psi(t, u)$ of theorem 1.25 | $3 \log(1 + tu) - \log(1 - t + tu)$                                 | theorem 4.50     |
| reversion coefficients $c_m$ | $1, \frac{5}{2}, \frac{13}{3}, \frac{53}{12}, -\frac{14}{5}, \dots$ | table 4.2        |
| index map $x \mapsto n$      | $n = \frac{1}{24} + 3\mu^2/(2\pi^2)$                                | eq. (4.7)        |
| staircase separation         | unit spacing of $\mathbb{Z}$ ; see theorem 4.83                     | theorem 1.39     |

*Remark 4.4* (Notation across the three sources). The sources disagree on symbols in a way that has to be stated once, because two of the clashes are dangerous.

1. **The letter  $x$ .** S1 writes  $N = n - 1/24$  for the shifted index and  $x$  for the phase  $\pi \sqrt{2N/3}$ ; S2 writes  $x = n - 1/24$  for the shifted index and  $\mu$  for the phase; S3 writes  $N$  and  $\mu$ . This volume uses  $N$  for the shifted index and  $\mu$  for the phase, i.e. the S3 convention, which is the intersection of the other two.
2. **The shift itself.** Every displayed *exact* identity in this chapter is in the shifted variable  $N = n - 1/24$ . Expansions in  $n^{-1/2}$  (section 4.5) re-expand the shift and are the only place where unshifted  $n$  appears inside a coefficient. Confusing the two is the single most common error in this subject: e.g. replacing  $N$  by  $n$  in eq. (4.4) costs a relative  $\mathcal{O}(n^{-1/2})$  and destroys every statement about exponentially small sectors.
3. **The normalising constant** is  $\mathcal{C}$  in S1,  $\kappa$  in S2,  $D$  in S3; all equal  $\pi^2/(6\sqrt{3})$ . We write  $\kappa$ , reserving  $D$  for nothing and  $\Phi'$  for the slope.
4. **The logarithmic variable.** S1 and S2 expand in  $R = \log(y/\kappa)$ ; S3 expands in  $H = \log(4y/\kappa) = R + 2 \log 2$ . The two expansions have the *same* coefficient polynomials (see remark 4.48) but are different series and must not be mixed.



## 4.2 Dedekind sums and the arithmetic amplitudes

### 4.2.1 Definitions

**Definition 4.5** (Sawtooth and Dedekind sum). For  $u \in \mathbb{R}$  put

$$((u)) = \begin{cases} u - \lfloor u \rfloor - \frac{1}{2}, & u \notin \mathbb{Z}, \\ 0, & u \in \mathbb{Z}, \end{cases} \quad (4.8)$$

and for  $k \geq 1, h \in \mathbb{Z}$ ,

$$s(h, k) = \sum_{r=1}^{k-1} \left( \left( \frac{r}{k} \right) \right) \left( \left( \frac{hr}{k} \right) \right), \quad (4.9)$$

with the empty-sum convention  $s(h, 1) = 0$ .

Throughout,  $U_k = \{h : 0 \leq h < k, \gcd(h, k) = 1\}$ , so that  $U_1 = \{0\}$ .

**Definition 4.6** (Rademacher amplitude). For  $k \geq 1$  and  $n \in \mathbb{Z}$ ,

$$A_k(n) = \sum_{h \in U_k} \exp \left( \pi i s(h, k) - \frac{2\pi i h n}{k} \right). \quad (4.10)$$

**Lemma 4.7** (Elementary properties). For every  $k \geq 1$  and  $n \in \mathbb{Z}$ :

1.  $A_k(n) \in \mathbb{R}$ , and  $A_k(n) = \sum_{h \in U_k} \cos(\pi s(h, k) - 2\pi h n / k)$ ;
2.  $A_k(n + k) = A_k(n)$ ;
3.  $|A_k(n)| \leq \varphi(k) \leq k$ ;
4.  $A_1(n) = 1$  and  $A_2(n) = (-1)^n$ .

*Proof.* (i) For  $k \geq 2$  the involution  $h \mapsto k - h$  is a bijection of  $U_k$  (note  $k - h \neq h$  unless  $k = 2h$ , impossible for  $\gcd(h, k) = 1, k \geq 3$ ; for  $k = 2$  the set is  $\{1\}$  and  $s(1, 2) = 0$  makes the single term real). The reciprocity  $s(-h, k) = -s(h, k)$ , immediate from  $((-u)) = -((u))$ , shows that the terms at  $h$  and  $k - h$  are complex conjugates. Hence the sum is real and equals the sum of the real parts, which is the stated cosine sum. (ii) Replacing  $n$  by  $n + k$  changes each exponent by  $-2\pi i h$ , an integer multiple of  $2\pi i$ . (iii) Triangle inequality on  $\varphi(k)$  unimodular terms. (iv)  $U_1 = \{0\}$  and  $s(0, 1) = 0$  give  $A_1 \equiv 1$ ;  $U_2 = \{1\}$ ,  $s(1, 2) = 0$  give  $A_2(n) = e^{-\pi i n} = (-1)^n$ .  $\square$

Only the trivial bound (iii) is used anywhere in this chapter. Weil-type bounds for these Kloosterman-like sums would sharpen the constants in theorems 4.23 and 4.78 but change no structure; see remark 4.26.

**Proposition 4.8** (The first amplitudes). With  $s(1, k) = \frac{(k-1)(k-2)}{12k}$ ,

$$\begin{aligned} A_1(n) &= 1, & A_2(n) &= (-1)^n, \\ A_3(n) &= 2 \cos \left( \frac{\pi}{18} - \frac{2\pi n}{3} \right), & A_4(n) &= 2 \cos \left( \frac{\pi}{8} - \frac{\pi n}{2} \right), \\ A_5(n) &= 2 \cos \left( \frac{\pi}{5} - \frac{2\pi n}{5} \right) + 2 \cos \frac{4\pi n}{5}, & A_6(n) &= 2 \cos \left( \frac{5\pi}{18} - \frac{\pi n}{3} \right). \end{aligned} \quad (4.11)$$

*Proof.*  $s(1, k) = \sum_{r=1}^{k-1} ((r/k))^2 = \sum_{r=1}^{k-1} (r/k - 1/2)^2 = \frac{(k-1)(2k-1)}{6k} - \frac{k-1}{2} + \frac{k-1}{4} = \frac{(k-1)(k-2)}{12k}$ . Hence  $s(1, 3) = \frac{1}{18}$ ,  $s(1, 4) = \frac{1}{8}$ ,  $s(1, 5) = \frac{1}{5}$ ,  $s(1, 6) = \frac{5}{18}$ , and  $s(k-1, k) = -s(1, k)$ . For  $k = 3, 4, 6$  the set  $U_k$  is  $\{1, k-1\}$ , and the two terms are conjugate, giving twice the real part of the  $h = 1$  term after reducing  $2\pi(k-1)n/k \equiv -2\pi n/k \pmod{2\pi}$ . For  $k = 5$ ,  $U_5 = \{1, 2, 3, 4\}$  and a direct evaluation of eq. (4.9) gives  $s(2, 5) = s(3, 5) = 0$ : with  $((1/5), (2/5), (3/5), (4/5)) = (-\frac{3}{10}, -\frac{1}{10}, \frac{1}{10}, \frac{3}{10})$ , the four products in  $s(2, 5)$  are  $\frac{3}{100}, -\frac{3}{100}, -\frac{3}{100}, \frac{3}{100}$ . Pairing  $h = 1$  with  $h = 4$  and  $h = 2$  with  $h = 3$  gives the stated two cosines.  $\square$

## 4.2.2 The real-analytic interpolation of the amplitudes

The inverse problem needs the amplitudes at non-integral arguments: the smooth inverse of a smooth interpolation evaluates  $A_k$  at a real point. The finite Fourier shape of eq. (4.10) supplies a canonical choice.

**Definition 4.9** (Entire amplitude interpolation). For  $k \geq 1$  and  $\nu \in \mathbb{C}$  set

$$\tilde{A}_k(\nu) = \sum_{h \in U_k} \cos\left(\pi s(h, k) - \frac{2\pi h\nu}{k}\right). \quad (4.12)$$

**Lemma 4.10.**  $\tilde{A}_k$  is entire,  $k$ -periodic, real on  $\mathbb{R}$ , satisfies  $\tilde{A}_k(n) = A_k(n)$  for  $n \in \mathbb{Z}$ , and  $|\tilde{A}_k^{(r)}(\nu)| \leq \varphi(k)(2\pi/k)^r \cosh(2\pi|\Im \nu|) \leq k(2\pi)^r e^{2\pi|\Im \nu|}$  for every  $r \geq 0$ . Moreover  $\tilde{A}_1 \equiv 1$  and  $\tilde{A}_2(\nu) = \cos \pi\nu$ , while

$$\tilde{A}_3(\nu) = 2 \cos(\pi\nu) \cos\left(\frac{\pi}{18} + \frac{\pi\nu}{3}\right). \quad (4.13)$$

*Proof.* Each summand is entire and  $k$ -periodic in  $\nu$ ; realness on  $\mathbb{R}$  is manifest; agreement at integers is lemma 4.7(i). The derivative bound is termwise, using  $|h/k| \leq 1$  and  $|\cos^{(r)}(z)| \leq \cosh(\Im z)$ . For  $k = 1$ ,  $U_1 = \{0\}$  makes the single term  $\cos 0 = 1$  identically. For  $k = 3$ , sum-to-product on  $A = \frac{\pi}{18} - \frac{2\pi\nu}{3}$  and  $B = -\frac{\pi}{18} - \frac{4\pi\nu}{3}$  gives  $\frac{A+B}{2} = -\pi\nu$  and  $\frac{A-B}{2} = \frac{\pi}{18} + \frac{\pi\nu}{3}$ , whence eq. (4.13).  $\square$

**S2, Definition “Real Rademacher amplitude”.** S2 defines the interpolation as a sum over  $1 \leq h \leq k$  with  $\gcd(h, k) = 1$ . For  $k \geq 2$  this is the same index set as  $U_k$ , but for  $k = 1$  it produces the single term  $\cos(2\pi\nu)$  instead of the constant 1. S2 nevertheless uses  $\tilde{A}_1 \equiv 1$  throughout (its dominant sector carries no amplitude, and its relative-correction formula separates the  $k = 1$  terms with amplitude 1), so the definition contradicts its own use: with S2’s index set the interpolating function  $\mathcal{P}$  would not be an interpolation of  $p$  – its dominant term would oscillate off-lattice with period 1 at full relative strength, and the “eventual monotonicity” proposition would be false. **Fix.** Take  $h$  over  $U_k = \{0 \leq h < k : \gcd(h, k) = 1\}$  with  $U_1 = \{0\}$ , i.e. definition 4.9; equivalently, decree  $\tilde{A}_1 \equiv 1$  separately, as S3 does. All of S2’s later formulas are then correct as printed.

*Remark 4.11* (Non-uniqueness, honestly stated). definition 4.9 is *a* choice, not *the* choice. Any strictly increasing smooth  $\mathcal{P}$  with  $\mathcal{P}(n) = p(n)$  has an inverse, and the algebraic (Lambert–logarithmic) part of that inverse is the same for all of them, because it is determined by the dominant sector, which is interpolation-independent once  $N$  is treated as a real variable. The exponentially small sectors are *not* interpolation-independent. definition 4.9 is singled out only because it is entire, finite-Fourier, real on  $\mathbb{R}$ , and literally the analytic continuation of the amplitudes already present in Rademacher’s theorem. section 4.10 gives the alternative that avoids interpolation altogether.

## 4.3 Rademacher’s formula in exact transseries normal form

### 4.3.1 The analytic input

**Theorem 4.12** (Hardy–Ramanujan–Rademacher; quoted). For every integer  $n \geq 1$ ,

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \frac{d}{dn} \left[ \frac{\sinh\left(\frac{\lambda}{k} \sqrt{n - \frac{1}{24}}\right)}{\sqrt{n - \frac{1}{24}}} \right], \quad (4.14)$$

the series converging when summed in increasing  $k$ .

This is the only unproved input of the chapter. It is Rademacher's theorem (1937, 1938, 1943); a textbook proof is in Apostol, and a machine-checked proof exists in Isabelle/HOL (Eberl–Paulson, 2026); see remark 4.100. All three sources take it as given and so do we. Note the factor  $\sqrt{k}$ : some normalisations absorb it into the definition of the arithmetic sum, and mixing conventions is a standard source of a spurious factor  $k$ . Remark 4.87 records the cross-checks that detect such a slip.

### 4.3.2 Exact two-exponential sectors

**Lemma 4.13** (Elementary derivative). *For  $a > 0$  and  $v > 0$ ,*

$$\frac{d}{dv} \left( v^{-1/2} \sinh(a\sqrt{v}) \right) = \frac{1}{4v^{3/2}} \left[ (a\sqrt{v} - 1)e^{a\sqrt{v}} + (a\sqrt{v} + 1)e^{-a\sqrt{v}} \right]. \quad (4.15)$$

*Proof.* With  $w = a\sqrt{v}$  one has  $dw/dv = w/(2v)$  and  $v^{-1/2} \sinh w = a w^{-1} \sinh w$ , so

$$\frac{d}{dv} \left( v^{-1/2} \sinh(a\sqrt{v}) \right) = a \frac{d}{dw} \left( \frac{\sinh w}{w} \right) \frac{w}{2v} = \frac{a}{2v} \left( \cosh w - \frac{\sinh w}{w} \right) = \frac{1}{2v^{3/2}} (w \cosh w - \sinh w),$$

using  $a = w/\sqrt{v}$  in the last step. Finally  $2(w \cosh w - \sinh w) = (w - 1)e^w + (w + 1)e^{-w}$ .  $\square$

**Theorem 4.14** (Exact sector decomposition). *Let  $N, \lambda, \mu, \kappa$  be as in eq. (4.3) and set, for  $k \geq 1$  and  $\sigma \in \{+1, -1\}$ ,*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k} N} e^{\sigma\mu/k} \left( 1 - \frac{\sigma k}{\mu} \right) = \frac{\kappa}{\mu^2} \frac{A_k(n)}{\sqrt{k}} e^{\sigma\mu/k} \left( 1 - \frac{\sigma k}{\mu} \right). \quad (4.16)$$

*Then, for every integer  $n \geq 1$ ,*

$$p(n) = \sum_{k \geq 1} (P_{k,+}(n) + P_{k,-}(n)), \quad (4.17)$$

*the two signs being paired before the sum over  $k$  is formed. Equivalently, with the entire paired kernel*

$$g(z) = \cosh z - \frac{\sinh z}{z} = \sum_{m \geq 1} \frac{2m}{(2m+1)!} z^{2m}, \quad g(0) = 0, \quad (4.18)$$

*one has the manifestly convergent form*

$$p(n) = \frac{2\kappa}{\mu^2} \sum_{k \geq 1} \frac{A_k(n)}{\sqrt{k}} g\left(\frac{\mu}{k}\right). \quad (4.19)$$

*Proof.* Apply lemma 4.13 with  $v = N$  and  $a = \lambda/k$ , so that  $a\sqrt{v} = \mu/k$ , inside eq. (4.14). The  $k$ th summand becomes

$$\frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}} \cdot \frac{1}{4N^{3/2}} \left[ \left(\frac{\mu}{k} - 1\right) e^{\mu/k} + \left(\frac{\mu}{k} + 1\right) e^{-\mu/k} \right] = \frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}} \cdot \frac{\mu}{4kN^{3/2}} \left[ e^{\mu/k} \left(1 - \frac{k}{\mu}\right) + e^{-\mu/k} \left(1 + \frac{k}{\mu}\right) \right].$$

Now  $\mu/(\pi\sqrt{2}\sqrt{N}) = \lambda/(\pi\sqrt{2}) = 1/\sqrt{3}$ , so the prefactor is  $A_k(n)/(4\sqrt{3k}N)$ , which is eq. (4.16) summed over  $\sigma$ . Since  $\mu^2 = \lambda^2 N = (2\pi^2/3)N$ , we get  $1/(4\sqrt{3k}N) = \kappa/(\sqrt{k}\mu^2)$ , the second form of eq. (4.16). Adding the two signs and using  $e^z(1 - 1/z) + e^{-z}(1 + 1/z) = 2g(z)$  with  $z = \mu/k$  gives eq. (4.19). All manipulations are termwise algebra, so the sum is unchanged.  $\square$

**Remark 4.15** (The three sources' normal forms are the same). S1 writes  $p(n) = \frac{\zeta}{x^2} \sum_k \frac{A_k}{\sqrt{k}} \{e^{x/k}(1 - k/x) + e^{-x/k}(1 + k/x)\}$  (its  $x$  is our  $\mu$ ); S2 writes the sectors  $P_{k,\sigma} = \frac{A_k}{4\sqrt{3k}x} e^{\sigma\mu/k} (1 - \sigma k/\mu)$  (its  $x$  is our  $N$ ); S3 writes  $p(n) = D\mu^{-3} \sum_k A_k \sqrt{k} [(\frac{\mu}{k} - 1)e^{\mu/k} + (\frac{\mu}{k} + 1)e^{-\mu/k}]$ . Factoring  $\mu/k$  out of S3's bracket and using  $\sqrt{k} \cdot (1/k) = 1/\sqrt{k}$  turns S3 into S1, and  $\kappa/\mu^2 = 1/(4\sqrt{3k}N) \cdot \sqrt{k}$  turns S1 into S2. All three are eq. (4.16). We adopt the second form of eq. (4.16) because it displays the Part 0 model  $\kappa e^{\mu} \mu^{-2} (1 - 1/\mu)$  at  $k = 1, \sigma = +$  verbatim.

### 4.3.3 The dominant envelope and the exact relative factorisation

**Definition 4.16** (Dominant envelope, actions). Put

$$F(\mu) = P_{1,+} = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right) = \kappa e^\mu \frac{\mu - 1}{\mu^3}, \quad (4.20)$$

and for  $k \geq 1$ ,  $\sigma = \pm 1$  (excluding  $(1, +)$ ) the *action*

$$\alpha_{k,\sigma} = 1 - \frac{\sigma}{k} \quad \text{so that} \quad \alpha_{1,-} = 2, \quad \alpha_{k,+} = 1 - \frac{1}{k}, \quad \alpha_{k,-} = 1 + \frac{1}{k}. \quad (4.21)$$

**Proposition 4.17** (Exact relative factorisation). For  $\mu > 1$ ,

$$p(n) = F(\mu) (1 + \mathcal{R}(\mu; n)), \quad (4.22)$$

where, with the amplitude ratios

$$r_{k,\sigma}(\mu; n) = \frac{A_k(n)}{\sqrt{k}} \frac{1 - \sigma k/\mu}{1 - 1/\mu}, \quad (4.23)$$

one has the finite-level resolved form, valid for each  $K \geq 1$ ,

$$\mathcal{R} = r_{1,-} e^{-2\mu} + \sum_{k=2}^K \sum_{\sigma=\pm 1} r_{k,\sigma}(\mu; n) e^{-\alpha_{k,\sigma}\mu} + \mathcal{R}_{>K}(\mu; n), \quad (4.24)$$

and the exact paired form

$$\mathcal{R}(\mu; n) = \frac{1 + \mu^{-1}}{1 - \mu^{-1}} e^{-2\mu} + \frac{2e^{-\mu}}{1 - \mu^{-1}} \sum_{k \geq 2} \frac{A_k(n)}{\sqrt{k}} g\left(\frac{\mu}{k}\right), \quad \mathcal{R}_{>K} = \frac{2e^{-\mu}}{1 - \mu^{-1}} \sum_{k > K} \frac{A_k(n)}{\sqrt{k}} g\left(\frac{\mu}{k}\right). \quad (4.25)$$

*Proof.* Divide eq. (4.16) by eq. (4.20):  $P_{k,\sigma}/F = \frac{A_k}{\sqrt{k}} \frac{1 - \sigma k/\mu}{1 - 1/\mu} e^{(\sigma/k - 1)\mu}$ , which is eq. (4.23) times  $e^{-\alpha_{k,\sigma}\mu}$ ; the case  $k = 1, \sigma = -1$  has  $A_1 = 1$  and gives the first term of eq. (4.25). For  $k \geq 2$  the paired sum  $P_{k,+} + P_{k,-} = \frac{2\kappa}{\mu^2} \frac{A_k}{\sqrt{k}} g(\mu/k)$  from eq. (4.19), divided by  $F = \kappa e^\mu \mu^{-2} (1 - 1/\mu)$ , gives the second term.  $\square$

*Remark 4.18* (What the Hardy–Ramanujan formula omits). Equation (4.2) is  $F(\mu)$  after replacing  $N$  by  $n$  and discarding  $1 - 1/\mu$ . Each of those two simplifications costs a relative  $\mathcal{O}(n^{-1/2})$ ; together they account for the entire algebraic half-power series of section 4.5. Equation (4.22) loses nothing: it records every Ford-circle arc, in both signs, with its exact algebraic amplitude.

## 4.4 Ordering of the sectors and tail estimates

### 4.4.1 The action-one accumulation

The relative actions of eq. (4.21) are

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \uparrow 1 \quad \text{and} \quad 2, \frac{3}{2}, \frac{4}{3}, \dots \downarrow 1.$$

The first family is increasing with supremum 1; the second is decreasing with infimum 1 and *has no least element*. Two consequences must be stated before any transseries language is used.

**Definition 4.19** (Rademacher transseries). A *Rademacher transseries* for  $p$  is the datum of

1. for each  $K \geq 1$ , the finite family of resolved sectors  $\{(k, \sigma) : k \leq K\}$  with the coefficients of section 4.5, treated as independent formal exponential variables (so that coefficient extraction takes place in a finitely generated multivariate power-series ring), together with

2. the exact paired remainder  $\mathcal{R}_{>K}$  of eq. (4.25), retained as one analytic block.

The limit  $K \rightarrow \infty$  is taken analytically, through the convergence in theorem 4.12, and never by claiming that the whole support is a locally finite Hahn grid.

*Remark 4.20* (Why the pairing is mandatory). For  $k \gg \mu$  each of  $P_{k,+}$  and  $P_{k,-}$  separately contains a term proportional to  $\sqrt{k}/\mu$ , and  $\sum_k \sqrt{k}/\mu$  diverges. Only in the sum do those pieces cancel:  $2g(\mu/k) = \frac{2}{3}(\mu/k)^2 + O((\mu/k)^4)$ , which makes the  $k$ th paired term  $\mathcal{O}(k^{-3/2})$ . Thus eq. (4.17) prescribes an order of summation; it is not an absolutely rearrangeable double series. All three sources say this; S2 and S3 say it most sharply, and definition 4.19 follows S2.

*Remark 4.21* (Truncating “through  $e^{-\mu}$ ” is not a prescription). Because the positive actions accumulate at 1, an instruction such as “retain all terms down to relative order  $e^{-\mu}$ ” selects infinitely many sectors and is meaningless. A legitimate truncation names either a Rademacher level  $K$ , or an action cutoff  $\eta < 1$  (which then names finitely many positive sectors, namely  $k \leq (1 - \eta)^{-1}$ ), or a target accuracy together with theorem 4.23.

#### 4.4.2 The tail bound

**Lemma 4.22** (Kernel bound). *For  $z \geq 0$ ,*

$$0 \leq g(z) \leq \frac{z^2}{3} \cosh z \leq \frac{z^2}{3} e^z, \quad \text{hence} \quad g(z) \leq \frac{\cosh 1}{3} z^2 \quad \text{for } 0 \leq z \leq 1. \quad (4.26)$$

*The factor  $\cosh z$  may not be dropped:  $g(1) = e^{-1} = 0.3679 \dots > \frac{1}{3}$ .*

*Proof.* By eq. (4.18),  $g(z) = \sum_{m \geq 1} \frac{2m}{(2m+1)!} z^{2m}$  has non-negative coefficients, whence  $g \geq 0$ . For the upper bound compare with  $\frac{z^2}{3} \cosh z = \sum_{m \geq 1} \frac{z^{2m}}{3(2m-2)!}$  termwise: the claim is  $\frac{2m}{(2m+1)!} \leq \frac{1}{3(2m-2)!}$ , i.e.  $6m(2m-2)! \leq (2m+1)!$ , i.e.  $6m \leq (2m+1)2m(2m-1)$ , which holds for all  $m \geq 1$  with equality at  $m = 1$ . Monotonicity of  $\cosh$  gives the last inequality on  $[0, 1]$ , and  $g(1) = \cosh 1 - \sinh 1 = e^{-1}$  shows the factor cannot be dropped.  $\square$

**Theorem 4.23** (Explicit finite- $K$  tail bound). *Let  $\mu > 1$  and  $K \geq 1$ . Then*

$$|\mathcal{R}_{>K}(\mu; n)| \leq \frac{4\mu^2}{3(1 - \mu^{-1})\sqrt{K}} \exp\left[-\left(1 - \frac{1}{K+1}\right)\mu\right]. \quad (4.27)$$

*The same bound holds with  $A_k(n)$  replaced by  $\tilde{A}_k(\nu)$  for any real  $\nu$ , since  $|\tilde{A}_k| \leq \varphi(k) \leq k$  as well.*

*Proof.* From eq. (4.25),  $|A_k| \leq k$  and lemma 4.22,

$$|\mathcal{R}_{>K}| \leq \frac{2e^{-\mu}}{1 - \mu^{-1}} \sum_{k > K} \sqrt{k} g\left(\frac{\mu}{k}\right) \leq \frac{2e^{-\mu}}{1 - \mu^{-1}} \cdot \frac{\mu^2}{3} \sum_{k > K} k^{-3/2} e^{\mu/k} \leq \frac{2\mu^2 e^{-\mu + \mu/(K+1)}}{3(1 - \mu^{-1})} \sum_{k > K} k^{-3/2},$$

and  $\sum_{k > K} k^{-3/2} \leq \int_K^\infty t^{-3/2} dt = 2K^{-1/2}$ .  $\square$

**Theorem 4.24** (Sharper fixed- $K$  tail). *For each fixed  $K \geq 1$ , as  $\mu \rightarrow \infty$ ,*

$$\sum_{k > K} (P_{k,+} + P_{k,-}) = \mathcal{O}_K\left(\mu^{-1/2} e^{\mu/(K+1)}\right), \quad \mathcal{R}_{>K} = \mathcal{O}_K\left(\mu^{3/2} e^{-(1-1/(K+1))\mu}\right). \quad (4.28)$$

*Proof.* Split at  $k = \mu$ . For  $K < k \leq \mu$  use  $|A_k| \leq k$  and  $2g(\mu/k) \leq 2 \cosh(\mu/k) \leq 2e^{\mu/(K+1)}$  in eq. (4.19), so those terms contribute at most  $\frac{2\kappa}{\mu^2} \cdot 2e^{\mu/(K+1)} \sum_{k \leq \mu} \sqrt{k} = \mathcal{O}(\mu^{-1/2} e^{\mu/(K+1)})$  since  $\sum_{k \leq \mu} \sqrt{k} = \mathcal{O}(\mu^{3/2})$ . For  $k > \mu$  put  $z = \mu/k \in (0, 1)$ ; then  $g(z) \leq \frac{1}{3} z^2 \cosh 1$  by lemma 4.22, and the  $k$ th term is  $\mathcal{O}(\mu^{-2} \sqrt{k} \cdot \mu^2 k^{-2}) = \mathcal{O}(k^{-3/2})$ , whose tail is  $\mathcal{O}(\mu^{-1/2})$ . Dividing by  $F \asymp \kappa \mu^{-2} e^\mu$  gives the relative statement.  $\square$

*Remark 4.25* (Which bound to use). Theorem 4.23 (from S2) is explicit and uniform in  $K$ ; theorem 4.24 (from S1 and S3) is sharper in  $\mu$  by a factor  $\mu^{-1/2}$  but only for fixed  $K$ . Both are needed: the first to certify a computation, the second to state clean asymptotic orders. S1 and S3 state only the second, S2 only the first.

*Remark 4.26* (Sharper amplitude bounds). All constants above come from  $|A_k| \leq k$ . Lehmer's explicit remainder estimates and the effective error bounds of Banerjee–Paule–Radu–Schneider (remark 4.100) are much sharper, and Weil-type bounds for the Kloosterman-like sums  $A_k$  give  $|A_k(n)| \ll_\varepsilon k^{1/2+\varepsilon}$  for many  $k$ . Substituting any such bound in the proofs of theorems 4.23 and 4.24 improves the powers of  $\mu$  but changes no statement of this chapter, all of which are of the form “a fixed Rademacher level is exponentially accurate”. For a *specific* residue class some  $A_k$  vanish, so the first non-zero omitted sector can be much smaller than the generic estimate.

#### 4.4.3 An exact condensation at the accumulation level

Right at the accumulation action 1, an alternative to sector-by-sector bookkeeping is to sum the kernel expansion first. This is S3's contribution and has no counterpart in S1 or S2.

**Definition 4.27** (Rademacher–Kloosterman zeta). For  $\Re s > 2$  and  $n \in \mathbb{Z}$  put  $\mathcal{Z}_n(s) = \sum_{k \geq 1} \frac{A_k(n)}{k^s}$ ; the series converges absolutely there by lemma 4.7(iii). The same definition with  $\tilde{A}_k(\nu)$  defines  $\mathcal{Z}_\nu(s)$  for real  $\nu$ .

**Theorem 4.28** (Exact zeta condensation). *For every integer  $n \geq 1$ ,*

$$p(n) = \kappa \sum_{r \geq 1} \frac{4r}{(2r+1)!} \mu^{2r-2} \mathcal{Z}_n\left(2r + \frac{1}{2}\right), \quad (4.29)$$

*and the double series obtained by expanding  $\mathcal{Z}_n$  converges absolutely.*

*Proof.* Insert  $2g(z) = \sum_{r \geq 1} \frac{4r}{(2r+1)!} z^{2r+1} \cdot z^{-1}$  – more precisely  $2g(z) = \sum_{r \geq 1} \frac{4r}{(2r+1)!} z^{2r}$  – into eq. (4.19) with  $z = \mu/k$ :

$$p(n) = \frac{\kappa}{\mu^2} \sum_{k \geq 1} \frac{A_k(n)}{\sqrt{k}} \sum_{r \geq 1} \frac{4r}{(2r+1)!} \frac{\mu^{2r}}{k^{2r}} = \kappa \sum_{k,r} \frac{4r}{(2r+1)!} \mu^{2r-2} \frac{A_k(n)}{k^{2r+1/2}}.$$

Absolute convergence:  $\sum_k |A_k| k^{-2r-1/2} \leq \sum_k k^{-2r+1/2} < \infty$  for  $r \geq 1$ , and  $\sum_{r \geq 1} \frac{4r}{(2r+1)!} \mu^{2r-2} \zeta(2r - \frac{1}{2})$  converges for every fixed  $\mu$  because  $\zeta(2r - \frac{1}{2}) \rightarrow 1$  and  $\sum_r \frac{4r \mu^{2r-2}}{(2r+1)!} < \infty$ . Tonelli's theorem justifies the interchange, giving eq. (4.29).  $\square$

*Remark 4.29.* Equation (4.29) orders nothing by asymptotic size – it is an exact analytic rearrangement, not a transseries. Its use is precisely where the transseries language fails: it packages the whole action-one accumulation into Dirichlet values  $\mathcal{Z}_n(2r + \frac{1}{2})$ , which can be used as a single block in the inverse formulas of section 4.9.

### 4.5 All-order coefficients in the unshifted variable

The shifted phase  $\mu$  makes each sector elementary; the price is that the conventional scale is  $n^{-1/2}$ . The entire infinite algebraic array of this subject comes from re-expanding the single shift  $N = n - 1/24$ . This section gives all of it, in one formula valid for every sector.

### 4.5.1 The sector generating function

**Theorem 4.30** (Sectorwise convergent half-power expansion). *Fix  $k \geq 1$  and  $\sigma \in \{+1, -1\}$ , and set  $t = n^{-1/2}$ . Then*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}n} \exp\left(\frac{\sigma\lambda\sqrt{n}}{k}\right) \sum_{m \geq 0} c_m^{(k,\sigma)} n^{-m/2}, \quad (4.30)$$

where the  $c_m^{(k,\sigma)}$  are the Taylor coefficients at  $t = 0$  of

$$\Omega_{k,\sigma}(t) = \frac{1}{1 - t^2/24} \exp\left[\frac{\sigma\lambda}{kt} \left(\sqrt{1 - \frac{t^2}{24}} - 1\right)\right] \left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right). \quad (4.31)$$

The function  $\Omega_{k,\sigma}$  is holomorphic on  $|t| < \sqrt{24}$ , so the series in eq. (4.30) converges for every real  $n > 1/24$ ; in particular  $c_0^{(k,\sigma)} = 1$ .

*Proof.* With  $t = n^{-1/2}$  we have  $N = n(1 - t^2/24)$  and  $\mu = \lambda\sqrt{N} = (\lambda/t)\sqrt{1 - t^2/24}$ . Substituting in eq. (4.16),

$$P_{k,\sigma} = \frac{A_k(n)}{4\sqrt{3k}n} \cdot \frac{1}{1 - t^2/24} \cdot e^{\sigma\mu/k} \left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right),$$

and  $e^{\sigma\mu/k} = e^{\sigma\lambda/(kt)} \exp\left[\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1)\right]$  with  $e^{\sigma\lambda/(kt)} = e^{\sigma\lambda\sqrt{n}/k}$ . This is eq. (4.30) with eq. (4.31).

**Holomorphy:** the only candidate singularities of eq. (4.31) in  $|t| < \sqrt{24}$  are  $t = 0$  and the zeros of  $\sqrt{1 - t^2/24}$ , and the latter lie on  $|t| = \sqrt{24}$ . At  $t = 0$  the exponent is  $\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1) = -\frac{\sigma\lambda t}{48k} + \mathcal{O}(t^3)$ , so the singularity is removable, and the last factor is holomorphic there as well. Setting  $t = 0$  gives  $\Omega_{k,\sigma}(0) = 1$ .  $\square$

**S1, Theorem “Universal sector coefficient generator”.** S1 states only that the series “converges for all sufficiently small  $|t|$  and supplies an asymptotic expansion to arbitrary order”. This understates the truth and, read literally, invites the error the next remark forbids: it suggests the half-power series is a merely asymptotic object. S2 and S3 both identify the radius as exactly  $\sqrt{24}$ , i.e. convergence for every real  $n > 1/24$ ; theorem 4.30 adopts their statement with the branch-point argument made explicit. **Consequence.** Nothing in this chapter is a divergent series in  $n^{-1/2}$ , and no least-term truncation is available; see remark 4.37.

### 4.5.2 A finite closed formula for every sector coefficient

Write

$$\beta = \beta_{k,\sigma} = \frac{\sigma\pi}{6k}, \quad \text{so that} \quad \frac{\sigma\lambda}{k} = 2\sqrt{6}\beta, \quad \frac{\sigma k}{\lambda} = \frac{1}{12\beta} \cdot \sqrt{6}, \quad (4.32)$$

the two identities following from  $\lambda = \pi\sqrt{6}/3$  and  $k = \sigma\pi/(6\beta)$ . Thus each sector is the *same* function of the single parameter  $\beta$ ; the sector label enters nowhere else.

**Theorem 4.31** (Finite coefficient formula, every sector). *For every  $k \geq 1$ ,  $\sigma = \pm 1$  and  $m \geq 0$ ,*

$$c_m^{(k,\sigma)} = \frac{1}{(-4\sqrt{6})^m} \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m+1}{j} \frac{m+1-j}{(m+1-2j)!} \beta_{k,\sigma}^{m-2j}, \quad (4.33)$$

terms with a negative factorial index being absent. In particular  $c_m^{(k,\sigma)} \in \mathbb{Q}[\beta, \beta^{-1}] \cdot \sqrt{6}^{m \bmod 2}$ , and the dependence on the sector is only through  $\beta_{k,\sigma} = \sigma\pi/(6k)$ .

*Proof.* Put  $u = -t/(4\sqrt{6})$ , so that  $t^2/24 = 4u^2$  and, by eq. (4.32),  $\frac{\sigma\lambda}{kt} = \frac{2\sqrt{6}\beta}{-4\sqrt{6}u} = -\frac{\beta}{2u}$  and  $\frac{\sigma kt}{\lambda} = -\frac{4\sqrt{6}u\sqrt{6}}{12\beta} = -\frac{2u}{\beta}$ . Writing  $s = \sqrt{1 - 4u^2}$ , eq. (4.31) becomes

$$G(u) := \Omega_{k,\sigma}(-4\sqrt{6}u) = \frac{e^{\beta T}}{s^2} \left(1 + \frac{2u}{\beta s}\right), \quad T = T(u) = \frac{1-s}{2u}, \quad (4.34)$$

where we used  $-\frac{\beta}{2u}(s-1) = \beta\frac{1-s}{2u} = \beta T$  and  $s^2 = 1 - 4u^2$ . The Catalan variable  $T$  satisfies

$$T = u(1 + T^2), \quad u = \frac{T}{1 + T^2}, \quad s = \frac{1 - T^2}{1 + T^2}, \quad du = \frac{1 - T^2}{(1 + T^2)^2} dT. \quad (4.35)$$

(The first identity follows from  $2uT = 1 - s$  and  $s^2 = 1 - 4u^2$ ; the rest are algebra.) Substituting eq. (4.35) into eq. (4.34) gives the exact differential identity

$$G(u) du = e^{\beta T} \left( \frac{1}{1 - T^2} + \frac{2T}{\beta(1 - T^2)^2} \right) dT = \frac{1}{\beta} d\left( \frac{e^{\beta T}}{1 - T^2} \right), \quad (4.36)$$

because  $\frac{d}{dT} \frac{e^{\beta T}}{1 - T^2} = e^{\beta T} \left( \frac{\beta}{1 - T^2} + \frac{2T}{(1 - T^2)^2} \right)$ . Now let  $D_m = [u^m] G(u) = (-4\sqrt{6})^m c_m^{(k,\sigma)}$ . Then

$$D_m = \text{Res}_{u=0} \frac{G(u) du}{u^{m+1}} = \frac{1}{\beta} \text{Res}_{T=0} \frac{(1 + T^2)^{m+1}}{T^{m+1}} d\left( \frac{e^{\beta T}}{1 - T^2} \right),$$

using  $u^{-m-1} = T^{-m-1}(1 + T^2)^{m+1}$ . The residue of an exact differential vanishes, so integrating by parts,

$$D_m = -\frac{1}{\beta} \text{Res}_{T=0} \frac{e^{\beta T}}{1 - T^2} d(T^{-m-1}(1 + T^2)^{m+1}) = \frac{m+1}{\beta} \text{Res}_{T=0} e^{\beta T} (1 + T^2)^m \frac{dT}{T^{m+2}},$$

where we used  $d(T^{-m-1}(1 + T^2)^{m+1}) = (1 + T^2)^m T^{-m-2} (-(m+1)(1 + T^2) + 2(m+1)T^2) dT = -(m+1)(1 + T^2)^m (1 - T^2) T^{-m-2} dT$ , and the factor  $1 - T^2$  cancels. Hence

$$D_m = \frac{m+1}{\beta} [T^{m+1}] e^{\beta T} (1 + T^2)^m = \frac{m+1}{\beta} \sum_{j \geq 0} \binom{m}{j} \frac{\beta^{m+1-2j}}{(m+1-2j)!},$$

the sum running over  $0 \leq j \leq \lfloor (m+1)/2 \rfloor$ . Finally  $(m+1) \binom{m}{j} = (m+1-j) \binom{m+1}{j}$  and division by  $(-4\sqrt{6})^m$  gives eq. (4.33).  $\square$

*Remark 4.32* (Why a finite sum exists). A square-root shift would normally give an infinite convolution at each order. The Catalan coordinate  $T$  absorbs the square root, and the derivative already present in Rademacher's formula turns the integrand into the exact differential eq. (4.36). After the change-of-variable Jacobian cancels the denominator  $1 - T^2$ , only the polynomial  $(1 + T^2)^m$  survives.

*Remark 4.33* (Provenance and the scope of the generalisation). For  $(k, \sigma) = (1, +1)$ , eq. (4.33) at  $\beta = \pi/6$  is O'Sullivan's coefficient formula, also used by Banerjee, Paule, Radu and Schneider in their error analysis. S1 proves only that case, by the same Catalan substitution. S2 and S3 both observe that the proof never uses  $k = 1$  or  $\sigma = +1$  – the sector enters only through  $\beta$  – and state the general formula; S3 gives a second, generating-function proof through the identity  $\sum_{j \geq 0} \binom{m+2j}{j} q^j = C(q)^m / \sqrt{1 - 4q}$  for the Catalan series  $C$ . The residue proof above is S2's, written out. We record the generalisation as the principal coefficient statement of the chapter, since it covers every exponentially small sector at no extra cost.

The first coefficients, as functions of  $\beta$ , are

$$\begin{aligned} c_0 &= 1, & c_1 &= -\frac{\sqrt{6}(\beta^2 + 2)}{24\beta}, & c_2 &= \frac{\beta^2 + 12}{192}, \\ c_3 &= -\frac{\sqrt{6}(\beta^4 + 36\beta^2 + 72)}{13824\beta}, & c_4 &= \frac{\beta^4 + 80\beta^2 + 720}{221184}, & c_5 &= -\frac{\sqrt{6}(\beta^6 + 150\beta^4 + 3600\beta^2 + 7200)}{26542080\beta}, \end{aligned} \quad (4.37)$$



$$c_6 = \frac{\beta^6 + 252\beta^4 + 12600\beta^2 + 100800}{637009920}, \quad c_7 = -\frac{\sqrt{6}(\beta^8 + 392\beta^6 + 35280\beta^4 + 705600\beta^2 + 1411200)}{107017666560\beta}. \quad (4.38)$$

These were recomputed here by expanding eq. (4.31) symbolically to order  $t^9$  and comparing with eq. (4.33) coefficient by coefficient in exact arithmetic; the two agree identically. The parity pattern visible in eq. (4.37) —  $c_m$  is  $\sqrt{6}$  times an odd Laurent polynomial in  $\beta$  for odd  $m$ , and a polynomial in  $\beta^2$  for even  $m$  — is immediate from eq. (4.33), since  $m - 2j \equiv m \pmod{2}$  and the prefactor contributes  $\sqrt{6}^m$ .

### 4.5.3 The Bell-polynomial route

Equation (4.33) is the fastest generator. For symbolic systems, or when the sector coefficients are wanted as a byproduct of a general composition engine, the following route uses only the primitives of definitions 1.9 and 1.10.

**Proposition 4.34** (Logarithmic coefficients of a sector). *Write  $\log \Omega_{k,\sigma}(t) = \sum_{j \geq 1} \ell_j t^j$  (suppressing the sector labels) and put  $a = 1/24$ . Then, for  $j \geq 1$ ,*

$$\ell_j = \mathbf{1}_{2|j} \frac{a^{j/2}}{j/2} + \mathbf{1}_{2 \nmid j} \frac{\sigma \lambda}{k} \binom{1/2}{(j+1)/2} (-a)^{(j+1)/2} - \sum_{\substack{1 \leq q \leq j \\ 2|(j-q)}} \frac{1}{q} \left( \frac{\sigma k}{\lambda} \right)^q \frac{(q/2)_{(j-q)/2}}{((j-q)/2)!} a^{(j-q)/2}, \quad (4.39)$$

where  $(x)_r = x(x+1) \cdots (x+r-1)$ . Consequently the sector coefficients are the complete exponential Bell polynomials

$$c_m = \frac{1}{m!} B_m(1! \ell_1, 2! \ell_2, \dots, m! \ell_m), \quad (4.40)$$

equivalently the triangular recurrence

$$c_0 = 1, \quad c_m = \frac{1}{m} \sum_{j=1}^m j \ell_j c_{m-j} \quad (m \geq 1). \quad (4.41)$$

*Proof.* Taking logarithms in eq. (4.31),

$$\log \Omega = -\log(1 - at^2) + \frac{\sigma \lambda}{kt} (\sqrt{1 - at^2} - 1) + \log \left( 1 - \frac{\sigma k}{\lambda} t(1 - at^2)^{-1/2} \right).$$

The first term contributes  $a^r/r$  at degree  $2r$ . For the second,  $\sqrt{1 - at^2} - 1 = \sum_{r \geq 1} \binom{1/2}{r} (-a)^r t^{2r}$ , so dividing by  $t$  contributes  $\frac{\sigma \lambda}{k} \binom{1/2}{r} (-a)^r$  at degree  $2r - 1$ . For the third, expand  $\log(1 - w) = -\sum_{q \geq 1} w^q/q$  with  $w = \frac{\sigma k}{\lambda} t(1 - at^2)^{-1/2}$  and use  $(1 - at^2)^{-q/2} = \sum_{r \geq 0} \frac{(q/2)_r}{r!} a^r t^{2r}$ ; the term of degree  $j = q + 2r$  is the displayed sum. Equation (4.40) is the exponential formula of definition 1.10, and eq. (4.41) follows by comparing coefficients in  $\Omega' = (\log \Omega)' \Omega$ .  $\square$

For orientation, at general  $(k, \sigma)$  and  $a = 1/24$ ,

$$\begin{aligned} \ell_1 &= -\frac{\sigma \lambda a}{2k} - \frac{\sigma k}{\lambda}, & \ell_2 &= a - \frac{k^2}{2\lambda^2}, \\ \ell_3 &= -\frac{\sigma \lambda a^2}{8k} - \frac{\sigma k a}{2\lambda} - \frac{\sigma k^3}{3\lambda^3}, & \ell_4 &= \frac{a^2}{2} - \frac{k^2 a}{2\lambda^2} - \frac{k^4}{4\lambda^4}. \end{aligned} \quad (4.42)$$

### 4.5.4 The dominant Poincaré expansion, and what it does not say

**Definition 4.35.**  $\omega_m = c_m^{(1,+1)}$ , i.e. eq. (4.33) at  $\beta = \pi/6$ .

**Corollary 4.36.** *For each fixed  $M \geq 1$ ,*

$$p(n) = \frac{e^{\lambda\sqrt{n}}}{4\sqrt{3}n} \left( \sum_{m=0}^{M-1} \omega_m n^{-m/2} + \mathcal{O}_M(n^{-M/2}) \right), \quad (4.43)$$

with

$$\omega_0 = 1, \quad \omega_1 = -\frac{\sqrt{6}(\pi^2 + 72)}{144\pi}, \quad \omega_2 = \frac{\pi^2 + 432}{6912}, \quad \omega_3 = -\frac{\sqrt{6}(\pi^4 + 1296\pi^2 + 93312)}{2985984\pi}, \quad (4.44)$$

$$\omega_4 = \frac{\pi^4 + 2880\pi^2 + 933120}{286654464}, \quad (4.45)$$

numerically  $\omega_1 = -0.443287976873582391\dots$ ,  $\omega_2 = 0.063927894155250197\dots$ ,  $\omega_3 = -0.027730933907464250\dots$ ,  $\omega_4 = 0.003354707463289919\dots$

*Proof.* By theorem 4.30 with  $(k, \sigma) = (1, +1)$ , the displayed sum plus a convergent tail is exactly  $P_{1,+}(n)$  divided by  $e^{\lambda\sqrt{n}}/(4\sqrt{3}n)$ ; the tail after  $M$  terms is  $\mathcal{O}_M(n^{-M/2})$  because  $\Omega_{1,+}$  is holomorphic at 0. The difference  $p(n) - P_{1,+}(n)$  is  $\mathcal{O}(\mu^{3/2}e^{-\mu/2})$  relative to  $P_{1,+}$  by theorem 4.24 with  $K = 1$ , hence beyond every algebraic order. The closed forms follow from eq. (4.33) at  $\beta = \pi/6$ ; they were recomputed here in exact arithmetic and agree with the tables of S1 and S2 (which print the same numbers in two different but equal groupings).  $\square$

**Convergent sector versus asymptotic total.** This is the trap of the subject and all three sources are, at some point, loose about it. The three statements below are *different*:

1.  $\sum_{m \geq 0} \omega_m n^{-m/2}$  converges for every  $n > 1/24$  (theorem 4.30);
2. its sum is  $P_{1,+}(n) \cdot 4\sqrt{3}n e^{-\lambda\sqrt{n}}$ , *not*  $p(n) \cdot 4\sqrt{3}n e^{-\lambda\sqrt{n}}$ ;
3. eq. (4.43) is an *asymptotic* statement about  $p(n)$ , whose error at every fixed order is a sum of two utterly different things: a convergent algebraic tail of size  $\mathcal{O}(n^{-M/2})$  and an exponentially small Rademacher remainder of size  $\mathcal{O}(\mu^{3/2}e^{-\mu/2})$  relative to the dominant sector.

S1's "Dominant Poincaré expansion" subsection writes  $p(n) \sim \dots$  without separating (ii) from (iii) and, in its Theorem on the sector generating function, weakens (i) to "sufficiently small  $|t|$ ". S3's remark after its dominant Poincaré display states the correct trichotomy in one sentence. S2 states it as a standalone remark and is the presentation adopted here. **The practical consequence:** summing the  $\omega_m$  series to convergence does *not* compute  $p(n)$  to arbitrary accuracy; it computes  $P_{1,+}(n)$ , and the residual error stalls at the  $e^{-\mu/2}$  floor.

*Remark 4.37* (Least-term truncation degenerates here). Theorem 1.46 determines the optimal truncation point of a *factorially divergent* algebraic series and the size of the resulting beyond-all-orders floor. Its hypothesis fails for  $p(n)$ : by theorem 4.30 the algebraic series of each sector has a finite positive radius of convergence, so there is no least term and no truncation-induced floor. The floor that does exist — the  $e^{-\mu/2}$  Rademacher gap — is not produced by resummation ambiguity but is supplied explicitly by the next sector. The partition function is thus the degenerate, best possible case of theorem 1.46: the exponentially small sectors are known exactly and independently of the perturbative data, and the optimal algebraic truncation is simply "as many terms as one wishes". This is the structural reason the numerical accuracy in table 4.3 improves so violently with  $K$  but not at all with extra algebraic orders past the  $e^{-\mu/2}$  level.

## 4.6 The multiplicative reciprocal $1/p(n)$

The phrase "inverse of the partition numbers" is ambiguous. The reciprocal is the easy reading and is disposed of first; it needs no reversion at all.

**Proposition 4.38** (Exact reciprocal transseries). *Let  $\mu > 1$  be large enough that  $|\mathcal{R}(\mu; n)| < 1$ , with  $\mathcal{R}$  as in eq. (4.25). Then*

$$\frac{1}{p(n)} = \frac{\mu^2}{\kappa} \frac{e^{-\mu}}{1 - \mu^{-1}} \sum_{r \geq 0} (-1)^r \mathcal{R}(\mu; n)^r, \quad (4.46)$$

*the series converging. As a formal identity in definition 4.19 it holds without the inequality. If  $\mathcal{R}$  is written at finite level as a linear form  $\sum_{j \in J_K} r_j(\mu; n) q_j$  in independent exponential variables  $q_j = e^{-\alpha_j \mu}$ , then for every multi-index  $\alpha = (\alpha_j)_{j \in J_K} \neq \mathbf{0}$ ,*

$$[\mathbf{q}^\alpha] (1 + \mathcal{R})^{-1} = (-1)^{|\alpha|} \frac{|\alpha|!}{\alpha!} \prod_{j \in J_K} r_j(\mu; n)^{\alpha_j}. \quad (4.47)$$

*Proof.* Invert eq. (4.22) and expand  $(1 + \mathcal{R})^{-1}$  geometrically;  $1/F = \mu^2 e^{-\mu} / (\kappa(1 - \mu^{-1}))$ . For eq. (4.47), only the term  $(-1)^{|\alpha|} \mathcal{R}^{|\alpha|}$  of the geometric series can produce  $\mathbf{q}^\alpha$ , and the multinomial theorem supplies the coefficient.  $\square$

**Corollary 4.39** (Reciprocal Poincaré coefficients). *Define  $\eta_m$  by  $(\sum_{m \geq 0} \omega_m t^m)^{-1} = \sum_{m \geq 0} \eta_m t^m$ , i.e.*

$$\eta_0 = 1, \quad \eta_m = - \sum_{j=1}^m \omega_j \eta_{m-j} \quad (m \geq 1), \quad \text{equivalently} \quad \eta_m = \frac{1}{m!} B_m(-1! \ell_1, \dots, -m! \ell_m) \quad (4.48)$$

*with  $\ell_j = \ell_j^{(1,+1)}$  of eq. (4.39). Then for each fixed  $M$ ,*

$$\frac{1}{p(n)} = 4\sqrt{3} n e^{-\lambda\sqrt{n}} \left( \sum_{m=0}^{M-1} \eta_m n^{-m/2} + \mathcal{O}_M(n^{-M/2}) \right), \quad (4.49)$$

*and  $\sum_{m \geq 0} \eta_m t^m$  converges on  $|t| < \sqrt{24}$  minus the zero set of  $\Omega_{1,+}$ , to  $1/\Omega_{1,+}(t)$ .*

*Proof.* The recurrence is Cauchy inversion of a power series with unit constant term; the Bell form is the exponential formula applied to  $\Omega_{1,+}^{-1} = \exp(-\sum \ell_j t^j)$ . Convergence is inherited from theorem 4.30 and  $\Omega_{1,+}(0) = 1$ . The asymptotic statement follows from corollary 4.36 exactly as there; the same caveats of the repair box after corollary 4.36 apply verbatim, with “ $P_{1,+}$ ” replaced by “ $1/P_{1,+}$ ”.  $\square$

**Remark 4.40.** The reciprocal therefore requires no new analytic input and no series reversion. It is recorded because two of the three sources devote a section to it, and because it is the natural place to notice that the multinomial formula eq. (4.47) is the  $|\alpha|$ -linear shadow of the far harder reversion formulas of section 4.9.

## 4.7 What “the inverse” of a discrete sequence means here

**Proposition 4.41** (Strict monotonicity).  *$p(n) < p(n+1)$  for every  $n \geq 1$ .*

*Proof.* Adjoining a part 1 injects the partitions of  $n$  into those of  $n+1$ , and the one-part partition  $(n+1)$  is not in the image.  $\square$

By definition 1.37 there are three distinct objects, and this chapter needs all three.

### 1. The integer staircase (the generalised inverse)

$$p^{\leftarrow}(y) = \min\{n \geq 1 : p(n) \geq y\}, \quad y \geq 1. \quad (4.50)$$

Well defined by proposition 4.41. It is integer valued with unit jumps, so no smooth expansion can represent it to  $o(1)$ .

2. **The functional inverse of a chosen interpolation.** Any strictly increasing smooth  $\mathcal{P}$  with  $\mathcal{P}(n) = p(n)$  has an inverse  $\mathcal{N}$ , and  $p^{\leftarrow}(y) = \lceil \mathcal{N}(y) \rceil$  (theorem 4.44). definition 4.9 supplies the canonical choice used here.
3. **The inverse along the range.** If one knows in advance that  $y = p(n)$  for an integer  $n$ , the target is that integer. A smooth formula that is accurate to  $o(1/2)$  recovers it by rounding (theorem 4.83), and the accuracy available is exponentially small, not merely  $o(1)$ .

**Definition 4.42** (The Rademacher interpolation). For real  $\nu > 1/24$  put  $\mu(\nu) = \frac{\pi}{6}\sqrt{24\nu - 1}$  and

$$\mathcal{P}(\nu) = \frac{2\kappa}{\mu(\nu)^2} \sum_{k \geq 1} \frac{\tilde{A}_k(\nu)}{\sqrt{k}} g\left(\frac{\mu(\nu)}{k}\right), \quad (4.51)$$

with  $g$  as in eq. (4.18) and  $\tilde{A}_k$  as in definition 4.9.

**Proposition 4.43** (Interpolation, regularity, eventual monotonicity). *The series eq. (4.51) and all its derivatives converge locally uniformly on  $(1/24, \infty)$ ;  $\mathcal{P}$  is real analytic there and  $\mathcal{P}(n) = p(n)$  for every integer  $n \geq 1$ . Moreover there is  $\nu_0$  with  $\mathcal{P}'(\nu) > 0$  for  $\nu \geq \nu_0$ , so  $\mathcal{P}$  has a real inverse  $\mathcal{N}$  on  $[\mathcal{P}(\nu_0), \infty)$ .*

*Proof.* Fix a compact  $I \subset (1/24, \infty)$  and a complex neighbourhood of it on which  $\mu$  is holomorphic and bounded. By lemma 4.10 every  $\nu$ -derivative of  $\tilde{A}_k$  is  $\mathcal{O}_r(k)$  uniformly in  $k$  there, while  $g(\mu/k) = \mathcal{O}(k^{-2})$  uniformly with all parameter derivatives (from eq. (4.18) and  $\mu$  bounded). Hence the  $k$ th summand of eq. (4.51) and each of its derivatives is  $\mathcal{O}(k \cdot k^{-1/2} \cdot k^{-2}) = \mathcal{O}(k^{-3/2})$ , and the Weierstrass  $M$ -test gives locally uniform convergence and legitimises termwise differentiation. Equality at integers is lemma 4.10 combined with theorem 4.14.

Monotonicity: write  $\mathcal{P}(\nu) = F(\mu(\nu))(1 + \mathcal{R}(\mu(\nu); \nu))$  as in proposition 4.17 with  $\tilde{A}_k$  in place of  $A_k$ . We have  $d\mu/d\nu = \lambda^2/(2\mu) > 0$  and  $d \log F/d\mu = 1 - 3/\mu + 1/(\mu - 1) \rightarrow 1$ . Differentiating  $\mathcal{R}$  term by term and applying theorem 4.23 with  $K = 1$  together with lemma 4.10 shows  $\mathcal{R} + \partial_\mu \mathcal{R} = \mathcal{O}(\mu^3 e^{-\mu/2})$ ; note that differentiating  $\tilde{A}_k(\nu(\mu))$  costs a factor  $\mathcal{O}(k\mu)$ , absorbed by replacing  $k^{-3/2}$  with  $k^{-1/2}$  in the tail sum and hence by one more power of  $\mu$ . Thus  $\mathcal{P}'(\nu) = F'(\mu)\mu'(\nu)(1 + \mathcal{O}(\mu^3 e^{-\mu/2}))$ , which is positive for large  $\nu$ .  $\square$

**Theorem 4.44** (Staircase from the smooth inverse). *For all sufficiently large real  $y$ ,*

$$p^{\leftarrow}(y) = \lceil \mathcal{N}(y) \rceil, \quad (4.52)$$

*and  $\mathcal{N}(p(n)) = n$  exactly for every integer  $n \geq \nu_0$ .*

*Proof.* This is the specialisation of theorem 1.39 to the unit-spaced lattice  $\mathbb{Z}$ , whose separation hypothesis is proposition 4.41. Concretely: let  $m = p^{\leftarrow}(y)$ , so  $p(m-1) < y \leq p(m)$ . Since  $\mathcal{P}$  is increasing on  $[\nu_0, \infty)$  and  $\mathcal{P}(j) = p(j)$ , we get  $m-1 < \mathcal{N}(y) \leq m$ , whence  $\lceil \mathcal{N}(y) \rceil = m$ .  $\square$

**Remark 4.45** (The unavoidable sawtooth). For general real  $y$  the difference  $p^{\leftarrow}(y) - \mathcal{N}(y)$  is the ceiling defect, of size  $\mathcal{O}(1)$ , with Fourier expansion

$$\lceil v \rceil - v = \frac{1}{2} + \frac{1}{\pi} \sum_{r \geq 1} \frac{\sin(2\pi r v)}{r} \quad (v \notin \mathbb{Z}), \quad (4.53)$$

the series converging to the midpoint  $1/2$  at integers where the true defect is 0. Since the first arithmetic correction to  $\mathcal{N}$  is of size  $y^{-1/2}$  (theorem 4.73), *exponential precision in  $\mathcal{N}$  buys nothing for  $p^{\leftarrow}$  at generic  $y$* : the order-one sawtooth is an intrinsic final sector of the discrete problem, not an error to be removed. It matters only along the range  $y = p(n)$ , where the defect vanishes identically. This warning is S3's and is the sharpest of the three.

## 4.8 The Lambert core and the algebraic inverse

Throughout this section and the next,  $y$  denotes a large ordinate and

$$R = \log \frac{y}{\kappa}, \quad \ell = \log R. \quad (4.54)$$

### 4.8.1 The carrier

**Proposition 4.46** (Lambert carrier; specialisation of theorem 1.20). *The large solution  $X = X(y)$  of  $\kappa e^X X^{-2} = y$ , equivalently of*

$$X - 2 \log X = R, \quad (4.55)$$

is

$$X(y) = -2 W_{-1} \left( -\frac{1}{2} \sqrt{\frac{\kappa}{y}} \right) = -2 W_{-1} \left( -\frac{\pi}{\sqrt{24\sqrt{3}y}} \right). \quad (4.56)$$

*Proof.* This is theorem 1.20 at the dictionary values  $a = 1$ ,  $b = -2$ ,  $C = \kappa$  of table 4.1:  $X = (b/a)W_k((a/b)e^{R/b}) = -2W_k(-\frac{1}{2}e^{-R/2})$  and  $e^{-R/2} = \sqrt{\kappa/y}$ . The branch rule of theorem 1.20 selects  $k = -1$  because  $X \rightarrow +\infty$  forces  $-X/2 \leq -1$ . The second form uses  $\frac{1}{2}\sqrt{\kappa} = \frac{1}{2}\pi/\sqrt{6\sqrt{3}} = \pi/\sqrt{24\sqrt{3}}$ .  $\square$

### 4.8.2 Flattening into logarithms

**Theorem 4.47** (Carrier coefficient polynomials). *As  $y \rightarrow \infty$ ,*

$$X \sim R + 2\ell + \sum_{j \geq 1} \frac{\Pi_j(\ell)}{R^j}, \quad (4.57)$$

in the sense of theorem 1.33, where

$$\Pi_j(\ell) = \frac{2^{j+1}}{j+1} [v^j] (\ell + \log(1+v))^{j+1} = 2^{j+1} \sum_{q=1}^j \frac{s(j,q)}{(j+1-q)!} \ell^{j+1-q}, \quad (4.58)$$

with  $s(j,q)$  the signed Stirling numbers of the first kind,  $v(v-1) \cdots (v-j+1) = \sum_q s(j,q)v^q$ . For each fixed  $J$ ,

$$X = R + 2\ell + \sum_{j=1}^J \frac{\Pi_j(\ell)}{R^j} + \mathcal{O}\left(\frac{\ell^{J+1}}{R^{J+1}}\right). \quad (4.59)$$

Moreover  $\deg \Pi_j = j$  with leading coefficient  $(-1)^{j-1}2^{j+1}/j$ , and  $\Pi_j(0) = 0$ .

*Proof.* Write  $X = R(1+v)$  in eq. (4.55):  $R + Rv - 2 \log R - 2 \log(1+v) = R$ , i.e.

$$v = z(\ell + \log(1+v)), \quad z = \frac{2}{R}, \quad (4.60)$$

a Lagrange fixed point  $v = z\phi(v)$  with  $\phi(v) = \ell + \log(1+v)$  and  $\phi(0) = \ell \neq 0$ . Theorem 1.14 gives  $v = \sum_{m \geq 1} \frac{z^m}{m} [v^{m-1}] \phi(v)^m$ ; multiplying by  $R$  and putting  $m = j+1$  (so  $z^m R = 2^{j+1} R^{-j}$ ) yields the first equality of eq. (4.58). For the second, expand  $(\ell + \log(1+v))^{j+1} = \sum_q \binom{j+1}{q} \ell^{j+1-q} \log(1+v)^q$  and use  $\log(1+v)^q = q! \sum_{m \geq q} s(m,q)v^m/m!$ ; extracting  $[v^j]$  keeps only  $1 \leq q \leq j+1$ , and the term  $q = j+1$  contributes  $s(j, j+1) = 0$ . The surviving coefficient of  $\ell^{j+1-q}$  is  $\frac{2^{j+1}}{j+1} \binom{j+1}{q} q! \frac{s(j,q)}{j!} = 2^{j+1} \frac{s(j,q)}{(j+1-q)!}$ , as claimed. Since  $q \geq 1$ , every monomial carries a positive power of  $\ell$ , so  $\Pi_j(0) = 0$ ; the top power is  $q = 1$ , with  $s(j, 1) = (-1)^{j-1}(j-1)!$ , giving leading coefficient  $2^{j+1}(-1)^{j-1}(j-1)!/j! = (-1)^{j-1}2^{j+1}/j$ . The remainder statement eq. (4.59) is the general estimate of theorem 1.33 applied on the scale  $\ell/R \rightarrow 0$ ; concretely, eq. (4.60) has an analytic solution  $v = v(z, \ell)$  near  $z = 0$  for  $\ell$  in any fixed compact set, and  $\ell/R \rightarrow 0$  lets Taylor's theorem with remainder be applied at order  $J$ .  $\square$

The first polynomials, recomputed here from the Lagrange form and independently from the Stirling form (they agree identically), are

$$\begin{aligned}\Pi_1 &= 4\ell, & \Pi_2 &= -4\ell(\ell - 2), \\ \Pi_3 &= \frac{8}{3}\ell(2\ell^2 - 9\ell + 6), & \Pi_4 &= -\frac{8}{3}\ell(3\ell^3 - 22\ell^2 + 36\ell - 12), \\ \Pi_5 &= \frac{16}{15}\ell(12\ell^4 - 125\ell^3 + 350\ell^2 - 300\ell + 60), & \Pi_6 &= -\frac{16}{15}\ell(20\ell^5 - 274\ell^4 + 1125\ell^3 - 1700\ell^2 + 900\ell - 120),\end{aligned}\quad (4.61)$$

$$\Pi_7 = \frac{32}{105}\ell(120\ell^6 - 2058\ell^5 + 11368\ell^4 - 25725\ell^3 + 24500\ell^2 - 8820\ell + 840), \quad (4.62)$$

so that, explicitly,

$$X = R + 2\ell + \frac{4\ell}{R} + \frac{-4\ell^2 + 8\ell}{R^2} + \frac{\frac{16}{3}\ell^3 - 24\ell^2 + 16\ell}{R^3} + \frac{-8\ell^4 + \frac{176}{3}\ell^3 - 96\ell^2 + 32\ell}{R^4} + \cdots \quad (4.63)$$

*Remark 4.48* (Three printings of one polynomial family). The sources print eq. (4.58) in three different shapes, and it is worth recording once that they are the same family, because a reader comparing them will otherwise suspect an error.

- S1:  $P_j(L) = 2^{j+1} \sum_{q=1}^j \frac{s(j,q)}{(j+1-q)!} L^{j+1-q}$  – identical to eq. (4.58).
- S2:  $P_m(\ell) = \frac{1}{m!} \sum_{q=1}^m 2^q (q-1)! s(m,q) \binom{m}{q-1} (2\ell)^{m-q+1}$ . The coefficient of  $\ell^{m-q+1}$  is  $\frac{2^q (q-1)!}{m!} \binom{m}{q-1} 2^{m-q+1} = \frac{2^{m+1} s(m,q)}{(m-q+1)!}$  after cancelling  $\binom{m}{q-1} = \frac{m!}{(q-1)!(m-q+1)!}$ . Identical.
- S3:  $P_r(\ell) = \frac{1}{r!} \sum_{q=1}^r (q-1)! s(r,q) \binom{r}{q-1} \ell^{r-q+1}$ , appearing in the expansion as  $2^{r+1} P_r(\ell)$ . The same cancellation gives  $2^{r+1} P_r = \Pi_r$ . Identical.

**But the variables differ, and this is a genuine disagreement.** S1 and S2 expand in  $R = \log(y/\kappa)$  with  $\ell = \log R$ ; S3 expands in  $H = \log(4y/\kappa) = R + 2\log 2$  with  $\ell_{S3} = \log(H/2)$ , which is the natural variable for the substitution  $X = 2w$ ,  $w - \log w = H/2$ . The two series

$$X = R + 2\log R + \sum_j \frac{\Pi_j(\log R)}{R^j} \quad \text{and} \quad X = H + 2\log \frac{H}{2} + \sum_j \frac{\Pi_j(\log \frac{H}{2})}{H^j}$$

are *different* asymptotic expansions of the same function, agreeing order by order only after re-expanding  $H = R + 2\log 2$ . Both are correct; mixing them is not. This volume uses the S1/S2 variables  $R, \ell$  throughout, because  $R$  makes the phase equation exactly eq. (4.55) with no additive constant. We verified all of S3's displayed coefficients (its  $u$ - and  $N_0$ -expansions in  $H$ , through  $H^{-4}$  and  $H^{-3}$  respectively) by symbolic re-expansion; they are correct as printed in S3's own variables.

### 4.8.3 Restoring the dominant amplitude: the centred reversion

The carrier  $X$  solves  $y = \kappa e^X X^{-2}$ ; the *dominant core*  $\mu_0 = \mu_0(y)$  solves the full dominant sector,

$$y = F(\mu_0) = \kappa e^{\mu_0} \mu_0^{-2} \left(1 - \frac{1}{\mu_0}\right), \quad \text{i.e.} \quad \Phi(\mu_0) = R, \quad (4.64)$$

where the *dominant phase* is

$$\Phi(\mu) = \mu - 2\log \mu + \log\left(1 - \frac{1}{\mu}\right) = \mu - 3\log \mu + \log(\mu - 1), \quad \mu > 1. \quad (4.65)$$

**Lemma 4.49** (Derivatives of the dominant phase). *For  $\mu > 1$ ,*

$$\Phi'(\mu) = 1 - \frac{3}{\mu} + \frac{1}{\mu - 1} = 1 - \frac{2}{\mu} + \frac{1}{\mu(\mu - 1)} = \frac{\mu^2 - 3\mu + 3}{\mu(\mu - 1)} > 0, \quad (4.66)$$

and for  $m \geq 1$

$$\Phi^{(m)}(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)! \left( \frac{1}{(\mu-1)^m} - \frac{3}{\mu^m} \right). \quad (4.67)$$

In particular  $\Phi'(\mu) = 1 + \mathcal{O}(\mu^{-1})$ , and  $\mu^2 - 3\mu + 3 > 0$  for all real  $\mu$  since its discriminant is  $-3$ . Writing  $\varrho_{\pm} = \frac{1}{2}(3 \pm i\sqrt{3})$  for the roots of  $\mu^2 - 3\mu + 3$ ,

$$F'(\mu) = \kappa e^{\mu} \frac{\mu^2 - 3\mu + 3}{\mu^4}, \quad \frac{d^m}{d\mu^m} \log F'(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)! \left( \frac{1}{(\mu - \varrho_+)^m} + \frac{1}{(\mu - \varrho_-)^m} - \frac{4}{\mu^m} \right). \quad (4.68)$$

*Proof.* Differentiate eq. (4.65) in the form  $\mu - 3 \log \mu + \log(\mu - 1)$  and clear denominators for the three displayed shapes of  $\Phi'$ ; eq. (4.67) is  $m - 1$  further differentiations of  $-3\mu^{-1} + (\mu - 1)^{-1}$ . For eq. (4.68),  $F = \kappa e^{\mu}(\mu - 1)\mu^{-3}$  gives  $F' = F(1 + \frac{1}{\mu-1} - \frac{3}{\mu}) = F\Phi'(\mu)$ , and  $F\Phi' = \kappa e^{\mu} \frac{\mu-1}{\mu^3} \cdot \frac{\mu^2-3\mu+3}{\mu(\mu-1)}$ . Then  $\log F' = \log \kappa + \mu + \log(\mu - \varrho_+) + \log(\mu - \varrho_-) - 4 \log \mu$ , and differentiate.  $\square$

**Theorem 4.50** (The dominant-core correction; specialisation of theorem 1.25). *Put  $z = X^{-1}$ . There is a unique  $U \in z\mathbb{Q}[[z]]$  with*

$$U = 3 \log(1 + zU) - \log(1 - z + zU), \quad (4.69)$$

and  $U$  is analytic in a neighbourhood of  $z = 0$ . The dominant core is

$$\mu_0(y) = X(y) + U(X(y)^{-1}), \quad U(z) = \sum_{m \geq 1} c_m z^m. \quad (4.70)$$

Moreover eq. (4.69) is exactly the fixed-point equation  $U = \Psi(z, U)$  of theorem 1.25 at the dictionary values of table 4.1, so

$$c_m = \sum_{q=1}^m \frac{1}{q} [z^m v^{q-1}] \left( 3 \log(1 + zv) - \log(1 - z + zv) \right)^q \quad (4.71)$$

by lemma 1.16, and equivalently

$$c_m = [z^m] \left\{ 3 \log(1 + zU_{<m}) - \log(1 - z + zU_{<m}) \right\}, \quad U_{<m}(z) = \sum_{j=1}^{m-1} c_j z^j, \quad (4.72)$$

a triangular recurrence over  $\mathbb{Q}$ .

*Proof.* Put  $\mu = X + U$  and divide eq. (4.64) by the carrier identity  $y = \kappa e^X X^{-2}$ :

$$e^U (1 + zU)^{-2} \left( 1 - \frac{z}{1 + zU} \right) = 1, \quad z = \frac{1}{X},$$

since  $\mu = X(1 + zU)$  and  $1/\mu = z/(1 + zU)$ . As  $1 - \frac{z}{1+zU} = \frac{1-z+zU}{1+zU}$ , taking logarithms gives  $U - 3 \log(1 + zU) + \log(1 - z + zU) = 0$ , i.e. eq. (4.69). Write  $G(z, U)$  for the left side:  $G(0, 0) = 0$  and  $\partial_U G(0, 0) = 1$ , so the analytic implicit function theorem gives a unique analytic solution near 0, necessarily with rational Taylor coefficients since  $G$  has rational ones.

For the identification with theorem 1.25: that theorem's fixed-point map at parameters  $a, b, Q$  is  $\Psi(t, u) = -\frac{1}{a} [b \log(1 + tu) + Q(\frac{t}{1+tu})]$ . At  $a = 1, b = -2, Q(s) = \log(1 - s)$  this is

$$\Psi(t, u) = 2 \log(1 + tu) - \log \left( 1 - \frac{t}{1 + tu} \right) = 2 \log(1 + tu) - \log(1 - t + tu) + \log(1 + tu) = 3 \log(1 + tu) - \log(1 - t + tu),$$

which is the right side of eq. (4.69). Equation (4.71) is then lemma 1.16 verbatim. Equation (4.72) holds because in eq. (4.69) every occurrence of  $U$  on the right is multiplied by an explicit  $z$ , so  $[z^m]$  of the right side involves only  $c_1, \dots, c_{m-1}$ .  $\square$

Table 4.2: The centred-reversion coefficients  $c_m$  of eq. (4.70), recomputed here in exact rational arithmetic from eq. (4.72) (truncated power-series arithmetic over  $\mathbb{Q}$ ,  $M = 20$ ). All eighteen agree with the tables printed by the three sources, which call them  $b_m$  (S1),  $a_m$  (S2) and  $d_r$  (S3).

|                                                 |                                                    |                                          |                                           |
|-------------------------------------------------|----------------------------------------------------|------------------------------------------|-------------------------------------------|
| $c_1 = 1$                                       | $c_2 = \frac{5}{2}$                                | $c_3 = \frac{13}{3}$                     | $c_4 = \frac{53}{12}$                     |
| $c_5 = -\frac{14}{5}$                           | $c_6 = -\frac{763}{30}$                            | $c_7 = -\frac{2179}{35}$                 | $c_8 = -\frac{42289}{630}$                |
| $c_9 = \frac{132973}{1260}$                     | $c_{10} = \frac{3449711}{5040}$                    | $c_{11} = \frac{29813143}{18480}$        | $c_{12} = \frac{496569589}{356400}$       |
| $c_{13} = -\frac{4334124833}{926640}$           | $c_{14} = -\frac{509723063851}{21621600}$          | $c_{15} = -\frac{298798016297}{5896800}$ | $c_{16} = -\frac{933244850273}{33633600}$ |
| $c_{17} = \frac{4993013022640963}{23156733600}$ | $c_{18} = \frac{208843210189406957}{231567336000}$ |                                          |                                           |

Thus

$$\mu_0 = X + \frac{1}{X} + \frac{5}{2X^2} + \frac{13}{3X^3} + \frac{53}{12X^4} - \frac{14}{5X^5} - \frac{763}{30X^6} - \frac{2179}{35X^7} + \cdots \quad (4.73)$$

**Remark 4.51** (Implementation check). The cheapest correctness test for a table of  $c_m$  is the residual identity: with  $U_M = \sum_{m \leq M} c_m z^m$ ,

$$U_M - 3 \log(1 + zU_M) + \log(1 - z + zU_M) = \mathcal{O}(z^{M+1}). \quad (4.74)$$

We verified eq. (4.74) at  $M = 12$  symbolically: the residual is  $\mathcal{O}(z^{13})$  exactly. This test is far more reliable than comparing printed digits, and it is the one we used to certify table 4.2.

#### 4.8.4 The algebraic inverse index

**Definition 4.52.**  $N_0(y) = \frac{1}{24} + \frac{3\mu_0(y)^2}{2\pi^2}$ , the *algebraic* (phase-independent) inverse index; and  $N_X(y) = \frac{1}{24} + \frac{3X(y)^2}{2\pi^2}$ , the carrier index.

**Corollary 4.53** (All algebraic coefficients of the inverse index). *With  $c_m$  as in table 4.2, put*

$$\gamma_j = 3c_{j+1} + \frac{3}{2} \sum_{r=1}^{j-1} c_r c_{j-r} \quad (j \geq 1). \quad (4.75)$$

Then

$$N_0(y) = \frac{3X^2}{2\pi^2} + \left( \frac{1}{24} + \frac{3}{\pi^2} \right) + \frac{1}{\pi^2} \sum_{j \geq 1} \frac{\gamma_j}{X^j}, \quad (4.76)$$

with

$$\gamma_1 = \frac{15}{2}, \quad \gamma_2 = \frac{29}{2}, \quad \gamma_3 = \frac{83}{4}, \quad \gamma_4 = \frac{559}{40}, \quad \gamma_5 = -\frac{611}{20}, \quad \gamma_6 = -\frac{112459}{840}, \quad (4.77)$$

$$\gamma_7 = -\frac{101329}{420}, \quad \gamma_8 = -\frac{228677}{3360}, \quad \gamma_9 = \frac{1709167}{1680}, \quad \gamma_{10} = \frac{325098619}{92400}. \quad (4.78)$$

*Proof.* Insert  $\mu_0 = X + U$  into definition 4.52. The cross term  $\frac{3}{2\pi^2} \cdot 2XU = \frac{3}{\pi^2} \sum_{m \geq 1} c_m X^{1-m}$  contributes  $3c_1/\pi^2 = 3/\pi^2$  to the constant and  $3c_{j+1}/(\pi^2 X^j)$  at order  $X^{-j}$ ; the square  $\frac{3}{2\pi^2} U^2$  contributes  $\frac{3}{2\pi^2} (\sum_{r=1}^{j-1} c_r c_{j-r}) X^{-j}$  and nothing at order  $X^0$  or above. The listed values were recomputed from eq. (4.75);  $\gamma_1, \dots, \gamma_8$  agree with the tables of all three sources, and  $\gamma_9, \gamma_{10}$  are new here.  $\square$

**Corollary 4.54** (The carrier bias).  $N_0(y) - N_X(y) = \frac{3}{\pi^2} + \mathcal{O}(X^{-1})$ , and

$$\frac{3}{\pi^2} = 0.30396355092701331433 \dots \quad (4.79)$$

*Proof.* Subtract  $N_X = \frac{1}{24} + 3X^2/(2\pi^2)$  from eq. (4.76): the  $3X^2/(2\pi^2)$  and  $\frac{1}{24}$  cancel, leaving  $3/\pi^2 + \pi^{-2} \sum_{j \geq 1} \gamma_j X^{-j}$ . The constant was evaluated here to 20 digits.  $\square$



*Remark 4.55* (Phase locking; do not expand the arithmetic phase about the carrier). This is the single most consequential structural point of the inverse problem, and only S2 states it explicitly.

The amplitudes  $\tilde{A}_k$  are *periodic on the index scale*, with period  $k$ . The core correction is small in the phase variable,  $\mu_0 - X = \mathcal{O}(X^{-1})$ ; but  $dn/d\mu = 3\mu/\pi^2 \asymp X$ , so the induced change in the index is  $\mathcal{O}(1)$  — exactly  $3/\pi^2$  in the limit, by corollary 4.54. A shift of 0.304 in the index is *not* a small perturbation of  $\cos(\pi\nu)$ . Consequently:

- Taylor-expanding  $\tilde{A}_k(N(\mu_0))$  around  $N_X$  does not begin with a small correction, and produces the wrong phase already at the leading exponentially small order.
- Every arithmetic amplitude must be evaluated at the *full algebraic core*  $N_0(y)$ , not at  $N_X(y)$  and not at any truncation of  $N_0$  coarser than  $o(1)$ . By eq. (4.76), truncating the  $\gamma$ -series after  $M$  terms leaves an index error  $\mathcal{O}(X^{-M})$ , so  $M \geq 1$  already suffices for  $o(1)$ ;  $M = 0$  (i.e. dropping the constant  $3/\pi^2$ ) does not.
- Only the *subsequent* exponentially small displacement may be Taylor-expanded; that is what section 4.9 does.

The numerical signature of the effect is the first column of table 4.4, which tends to  $-3/\pi^2$  and not to 0.

*Remark 4.56* (Elementary-logarithm form). Substituting eq. (4.57) into eq. (4.76) gives the pure logarithmic block

$$N_0(y) = \frac{3R^2}{2\pi^2} + \frac{6R\ell}{\pi^2} + \frac{6\ell^2 + 12\ell + 3}{\pi^2} + \frac{1}{24} + \frac{3(8\ell^2 + 16\ell + 5)}{2\pi^2 R} + \mathcal{O}\left(\frac{\ell^3}{R^2}\right), \quad (4.80)$$

with  $R = \log(y/\kappa)$  and  $\ell = \log R$ . We recomputed this block symbolically; it agrees with S2's display (which stops at the constant) and, after the change of variable  $H = R + 2\log 2$ , with S3's. For *numerical* work eq. (4.80) is markedly inferior to evaluating eq. (4.56) and eq. (4.73) directly, and it is recorded only to exhibit the transseries shape:  $N_0$  is a polynomial of degree 2 in  $R$  with coefficients polynomial in  $\ell$ , plus a descending series in  $R^{-1}$  with  $\deg_\ell$  growing by one per order.

## 4.9 The full arithmetic inverse transseries

Everything so far is phase-independent:  $X$ ,  $\mu_0$ ,  $N_0$  are functions of  $y$  alone. The remaining correction is exponentially small and arithmetically oscillatory, and the oscillation must be frozen before “analytic inverse” has a meaning. The three sources do this in two different ways, both correct.

### 4.9.1 Two ways to freeze the phase

**Definition 4.57** (Residue-frozen finite level; S1). For  $K \geq 1$  let  $\Lambda_K = \text{lcm}(1, 2, \dots, K)$  and fix a residue  $r \in \mathbb{Z}/\Lambda_K\mathbb{Z}$ . Since  $A_k$  has period  $k \mid \Lambda_K$  for  $k \leq K$ , the numbers  $A_k(r)$  are well defined, and

$$P_{K,r}(\mu) = \frac{\kappa}{\mu^2} \sum_{k=1}^K \frac{A_k(r)}{\sqrt{k}} \left[ e^{\mu/k} \left(1 - \frac{k}{\mu}\right) + e^{-\mu/k} \left(1 + \frac{k}{\mu}\right) \right] \quad (4.81)$$

is a genuine analytic function of  $\mu$  with constant coefficients. For large  $\mu$  it is increasing (the  $k = 1$  positive term dominates), and we write  $\mu_{K,r}(y)$  for its large inverse branch.

**Definition 4.58** (Interpolated level; S2, S3).  $\mathcal{P}_K(\mu)$  is eq. (4.81) with  $A_k(r)$  replaced by  $\tilde{A}_k(n(\mu))$ ,  $n(\mu) = \frac{1}{24} + 3\mu^2/(2\pi^2)$ ; its large inverse branch is written  $\mu_K(y)$ .

**Proposition 4.59** (The two conventions agree below action one). *Fix  $K$ , let  $y$  be large, and let  $r \equiv \lfloor N_0(y) \rfloor \pmod{\Lambda_K}$ , where  $\lfloor \cdot \rfloor$  is nearest-integer rounding. Assume  $|N_0(y) - r| \leq \frac{1}{2}$  modulo  $\Lambda_K$  has been resolved (which holds along the range  $y = p(n)$  for large  $n$  by theorem 4.83). Then every inverse transmonomial of total action strictly less than 1 is the same for  $\mu_{K,r}$  and  $\mu_K$ ; the two first differ at total action  $\geq 1$ .*

*Proof.* Both inverses are constructed as  $\mu_0 + v$  with  $v$  exponentially small, and in both the coefficient of the transmonomial  $e^{-\alpha_j \mu_0}$  at first order is  $-r_j(\mu_0)/\Phi'(\mu_0)$  (theorem 4.69 below), where  $r_j$  differs between the two conventions only in the amplitude,  $A_k(r)$  versus  $\tilde{A}_k(n(\mu_0)) = \tilde{A}_k(N_0(y))$ . Now  $\tilde{A}_k$  is entire and  $k$ -periodic with  $\tilde{A}_k(r) = A_k(r)$ , so  $|\tilde{A}_k(N_0) - A_k(r)| \leq \sup |A'_k| \cdot |N_0 - r|$ . Along the range,  $|N_0(p(n)) - n| = \mathcal{O}(\mu^{5/2}e^{-\mu/2})$  by corollary 4.79, so the amplitude discrepancy is  $\mathcal{O}(k\mu^{5/2}e^{-\mu/2})$  and its contribution to the sector carries the extra factor  $e^{-\mu/2}$ , i.e. additional action  $\frac{1}{2}$ . Since the smallest primary action is  $\frac{1}{2}$ , the first affected total action is  $\frac{1}{2} + \frac{1}{2} = 1$ . Higher-order terms of the reversion have total action  $\geq 1$  already.  $\square$

*Remark 4.60* (Which convention to prefer). The residue-frozen convention is intrinsic: it invents no interpolation, and at each finite level it reduces the problem to an ordinary analytic reversion with *constant* coefficients. Its cost is that the inverse is defined only after the residue class is known, which theorem 4.83 supplies. The interpolated convention defines a single function  $\mathcal{N}$  for all real  $y$  and connects directly to the staircase (theorem 4.44); its cost is the interpolation dependence of remark 4.11 and the appearance of the amplitude's own  $\mu$ -derivatives in every higher coefficient. This chapter develops the reversion in a form that covers both: all formulas below are stated for a family of *blocks*  $E_j(\mu)$ , and the two conventions are two choices of block.

### 4.9.2 Blocks, multi-indices, and the reversion equation

**Definition 4.61** (Sector blocks). Fix  $K \geq 1$  and put

$$J_K = \{(1, -1)\} \cup \{(k, \sigma) : 2 \leq k \leq K, \sigma = \pm 1\}. \quad (4.82)$$

For  $j = (k, \sigma) \in J_K$  set  $\alpha_j = 1 - \sigma/k$  as in eq. (4.21), and define the *block*

$$E_j(\mu) = \frac{\kappa}{\mu^2} \frac{a_k(\mu)}{\sqrt{k}} \left(1 - \frac{\sigma k}{\mu}\right) e^{\sigma \mu/k} = F(\mu) r_j(\mu) e^{-\alpha_j \mu}, \quad r_j(\mu) = \frac{a_k(\mu)}{\sqrt{k}} \frac{1 - \sigma k/\mu}{1 - 1/\mu}, \quad (4.83)$$

where  $a_k(\mu)$  is either the constant  $A_k(r)$  (residue-frozen convention, definition 4.57) or  $\tilde{A}_k(n(\mu))$  (interpolated convention, definition 4.58);  $a_1 \equiv 1$  in both. A block may also be the whole paired tail  $\mathcal{R}_{>K} \cdot F$  of eq. (4.25), or the zeta-condensation block of theorem 4.28, treated as one indivisible symbol.

Introduce independent formal variables  $\varepsilon_j$ ,  $j \in J_K$ , and write  $\varepsilon^{\mathbf{m}} = \prod_j \varepsilon_j^{m_j}$ ,  $|\mathbf{m}| = \sum_j m_j$ ,  $\mathbf{m}! = \prod_j m_j!$  for multi-indices  $\mathbf{m} \in \mathbb{N}^{J_K}$ . The forward object at level  $K$  is

$$\mathcal{P}_K(\mu; \varepsilon) = F(\mu) + \sum_{j \in J_K} \varepsilon_j E_j(\mu), \quad (4.84)$$

and the inverse problem is to solve  $\mathcal{P}_K(\mu; \varepsilon) = y$  for  $\mu$  near  $\mu_0 = F^{-1}(y)$ , as a formal series in  $\varepsilon$  which is finally specialised at  $\varepsilon_j = 1$ . Two equivalent engines are available; both are finite at every prescribed multi-index.

### 4.9.3 Engine I: additive Lagrange–Bürmann

**Theorem 4.62** (Additive reversion). Let  $F$  and  $E$  be analytic near  $\mu_0$  with  $F'(\mu_0) \neq 0$  and  $F(\mu_0) = y$ . Write  $\mathcal{D} = \frac{1}{F'(\mu)} \frac{d}{d\mu}$  evaluated along  $\mu = \mu_0(y)$ , i.e.  $\mathcal{D} = d/dy$  in the coordinate  $y$ . Then the solution of  $F(\mu) + \varepsilon E(\mu) = y$  near  $\mu_0$  is

$$\mu(y; \varepsilon) = \mu_0(y) + \sum_{M \geq 1} \frac{(-\varepsilon)^M}{M!} \mathcal{D}^{M-1} \left( \frac{E(\mu_0)^M}{F'(\mu_0)} \right). \quad (4.85)$$

Consequently, for the multivariate problem eq. (4.84), the coefficient of  $\varepsilon^{\mathbf{m}}$  with  $M = |\mathbf{m}| \geq 1$  in  $\mu - \mu_0$  is

$$V_{\mathbf{m}}(\mu_0) = \frac{(-1)^M}{\mathbf{m}!} \mathcal{D}^{M-1} \left( \frac{E^{\mathbf{m}}(\mu_0)}{F'(\mu_0)} \right) = \frac{(-1)^M (M-1)!}{\mathbf{m}!} \operatorname{Res}_{\mu=\mu_0} \frac{E^{\mathbf{m}}(\mu)}{(F(\mu) - F(\mu_0))^M}, \quad (4.86)$$

where  $E^{\mathbf{m}} = \prod_j E_j^{m_j}$ .

*Proof.* Both displays are theorem 1.18 in the same letters: eq. (4.85) is (1.34) and eq. (4.86) is (1.36).

What that theorem supplies is convergence for  $|\varepsilon| < \varepsilon_0$ , with  $\varepsilon_0$  read off a small circle about  $\mu_0$ ; here the specialisation  $\varepsilon_j = 1$  is wanted, and for the partition blocks it is legitimate. With  $E = \sum_j E_j + \mathcal{R}_{>K} F$  we have  $E/F = \mathcal{O}(\mu^C e^{-\mu/2})$  together with the same bound for finitely many derivatives, by theorem 4.23 and lemma 4.10; so on a fixed relative neighbourhood of a large  $\mu_0$  the Rouché comparison made in the proof of theorem 1.18 holds for all  $|\varepsilon| \leq 1 + \eta$  with some  $\eta > 0$ , and the Taylor series in  $\varepsilon$  converges at  $\varepsilon = 1$ .  $\square$

**Remark 4.63** ( $\mathcal{D}$  is iterated, not a plain derivative). In eq. (4.85) the operator  $\mathcal{D}^{M-1}$  is the  $(M-1)$ st power of  $\frac{1}{F'} \frac{d}{d\mu}$ . Replacing it by  $\frac{1}{(F')^{M-1}} \frac{d^{M-1}}{d\mu^{M-1}}$  after “pulling out” the powers of  $F'$  is wrong from  $M = 3$  onwards, because  $F'$  is not constant. S3 flags this explicitly; it is the most common implementation bug in this formula.

**Corollary 4.64** (Bell realisation). *Let  $H_{\mathbf{m}} = E^{\mathbf{m}}/F'$  and  $\Lambda_s^{(\mathbf{m})} = \frac{d^s}{d\mu^s} \log H_{\mathbf{m}}$ , and let  $\mu_0^{(s)} = d^s \mu_0 / dy^s$  be generated by*

$$\mu_0^{(1)} = \frac{1}{F'(\mu_0)}, \quad \mu_0^{(s)} = -\frac{1}{F'(\mu_0)} \sum_{q=2}^s F^{(q)}(\mu_0) B_{s,q}(\mu_0^{(1)}, \dots, \mu_0^{(s-q+1)}), \quad s \geq 2. \quad (4.87)$$

Then, with  $M = |\mathbf{m}| \geq 1$ ,

$$V_{\mathbf{m}} = \frac{(-1)^M H_{\mathbf{m}}(\mu_0)}{\mathbf{m}!} \sum_{s=0}^{M-1} B_{M-1,s}(\mu_0^{(1)}, \mu_0^{(2)}, \dots) B_s(\Lambda_1^{(\mathbf{m})}, \dots, \Lambda_s^{(\mathbf{m})}), \quad (4.88)$$

with the conventions  $B_{0,0} = B_0 = 1$ .

*Proof.* Faà di Bruno (definition 1.10) gives  $\frac{d^{M-1}}{dy^{M-1}} H_{\mathbf{m}}(\mu_0(y)) = \sum_s H_{\mathbf{m}}^{(s)}(\mu_0) B_{M-1,s}(\mu_0^{(1)}, \dots)$ , and writing  $H_{\mathbf{m}} = \exp(\log H_{\mathbf{m}})$  gives  $H_{\mathbf{m}}^{(s)} = H_{\mathbf{m}} B_s(\Lambda_1^{(\mathbf{m})}, \dots)$ . Substitute in eq. (4.86). Equation (4.87) is obtained by differentiating  $F(\mu_0(y)) = y$   $s$  times and solving the Faà di Bruno identity for the term containing  $\mu_0^{(s)}$ .  $\square$

The point of eq. (4.88) is that every ingredient is elementary and closed-form for the partition blocks. Indeed, splitting each aggregate block into its primitive Fourier constituents

$$E_{k,h,\sigma,\epsilon}(\mu) = \frac{\kappa}{2\sqrt{k}} \frac{\mu - \sigma k}{\mu^3} \exp \left[ \frac{\sigma \mu}{k} + i\epsilon \left( \pi s(h, k) - \frac{2\pi h}{k} n(\mu) \right) \right], \quad h \in U_k, \epsilon = \pm 1, \quad (4.89)$$

and using  $n'(\mu) = 3\mu/\pi^2$ ,  $n''(\mu) = 3/\pi^2$ ,  $n^{(s)} = 0$  for  $s \geq 3$ , one gets the closed logarithmic jets

$$\frac{d^s}{d\mu^s} \log E_{k,h,\sigma,\epsilon} = (-1)^{s-1} (s-1)! \left( \frac{1}{(\mu - \sigma k)^s} - \frac{3}{\mu^s} \right) + \mathbf{1}_{s=1} \frac{\sigma}{k} - i\epsilon \frac{2\pi h}{k} n^{(s)}(\mu), \quad (4.90)$$

while  $\frac{d^s}{d\mu^s} \log F'$  is eq. (4.68). Hence  $\Lambda_s^{(\mathbf{m})} = \sum_j m_j$  (eq. (4.90)) – (eq. (4.68)) with no hidden differentiation anywhere. This is S3’s presentation and is the most explicit of the three.

#### 4.9.4 Engine II: the triangular Newton–Hensel recurrence

The reversion may equally be organised in the phase coordinate, which is cheaper when many coefficients are wanted.

**Theorem 4.65** (Triangular recurrence). *Put  $\mu = \mu_0 + v$  and*

$$Q(v; \varepsilon) = \log \left( 1 + \sum_{j \in J_K} \varepsilon_j B_j(v) \right), \quad B_j(v) = r_j(\mu_0 + v) e^{-\alpha_j v}, \quad (4.91)$$

the bookkeeping variable  $\varepsilon_j$  standing for the actual exponential  $e^{-\alpha_j \mu_0}$ , which is substituted only at the very end; the residual  $v$ -dependence of  $e^{-\alpha_j(\mu_0+v)}$  is the factor  $e^{-\alpha_j v}$  inside  $B_j$ . Keeping the  $\varepsilon_j$  independent until the last step is what makes resonant exponent sums ( $\alpha_j + \alpha_{j'} = \alpha_{j''}$ , which does occur here) come out right. Then the reversion equation is

$$\Phi'(\mu_0) v + \sum_{s \geq 2} \frac{\Phi^{(s)}(\mu_0)}{s!} v^s + Q(v; \varepsilon) = 0, \quad (4.92)$$

and its unique formal solution  $v = \sum_{\mathbf{m} \neq \mathbf{0}} V_{\mathbf{m}} \varepsilon^{\mathbf{m}}$  obeys

$$V_{\mathbf{m}} = -\frac{1}{\Phi'(\mu_0)} [\varepsilon^{\mathbf{m}}] \left\{ \sum_{s \geq 2} \frac{\Phi^{(s)}(\mu_0)}{s!} v_{<\mathbf{m}}^s + Q(v_{<\mathbf{m}}; \varepsilon) \right\}, \quad (4.93)$$

where  $v_{<\mathbf{m}}$  collects the already-computed coefficients. Each right-hand side is a finite expression in  $\mu_0$ ,  $(\mu_0 - 1)^{-1}$ , the actions  $\alpha_j$  and the amplitudes  $a_k$ .

*Proof.* Taking logarithms in  $\mathcal{P}_K(\mu; \varepsilon) = y$  and subtracting  $\Phi(\mu_0) = R$  gives  $\Phi(\mu_0 + v) - \Phi(\mu_0) + Q(v; \varepsilon) = 0$ ; Taylor expansion of  $\Phi$  at  $\mu_0$  is eq. (4.92), legitimate as a formal identity because  $v$  has no constant term in  $\varepsilon$ . For triangularity: at a fixed  $\mathbf{m}$ , the unknown  $V_{\mathbf{m}}$  occurs linearly only in the term  $\Phi'(\mu_0)v$ ; each  $v^s$  with  $s \geq 2$  needs a decomposition of  $\mathbf{m}$  into at least two non-zero parts and so uses only strictly smaller multi-indices; and every monomial of  $Q$  carries at least one explicit  $\varepsilon_j$ , so an occurrence of  $V_{\mathbf{m}}$  inside  $Q$  would be multiplied into a strictly larger multi-index. Solving the coefficient equation gives eq. (4.93). Finiteness holds because a fixed multi-index has finitely many additive decompositions.  $\square$

**Corollary 4.66** (Multinomial expansion of  $Q$ ). *For  $\mathbf{m} \neq \mathbf{0}$ ,*

$$[\varepsilon^{\mathbf{m}}] Q(v; \varepsilon) \Big|_{v \text{ fixed}} = (-1)^{|\mathbf{m}|+1} \frac{(|\mathbf{m}| - 1)!}{\mathbf{m}!} \prod_j B_j(v)^{m_j}. \quad (4.94)$$

*Proof.* Only  $S^{|\mathbf{m}|}$  in  $\log(1+S) = \sum_{d \geq 1} (-1)^{d+1} S^d / d$  contributes, with multinomial coefficient  $|\mathbf{m}|! / \mathbf{m}!$ .  $\square$

**Corollary 4.67** (Closed multi-index Lagrange form in the phase). *Let  $G(v) = (\Phi(\mu_0 + v) - \Phi(\mu_0)) / v$  for  $v \neq 0$  and  $G(0) = \Phi'(\mu_0)$ . Then  $v = -Q(v; \varepsilon) / G(v)$ , and by theorem 1.14,*

$$V_{\mathbf{m}} = \sum_{q=1}^{|\mathbf{m}|} \frac{(-1)^q}{q} [v^{q-1}] \frac{1}{G(v)^q} \sum_{\substack{\mathbf{m}_1 + \dots + \mathbf{m}_q = \mathbf{m} \\ \mathbf{m}_i \neq \mathbf{0}}} \prod_{i=1}^q [\varepsilon^{\mathbf{m}_i}] Q(v; \varepsilon), \quad (4.95)$$

both sums being finite.

*Proof.*  $G$  is analytic near 0 with  $G(0) = \Phi'(\mu_0) \neq 0$ , and eq. (4.92) reads  $vG(v) + Q(v; \varepsilon) = 0$ . Apply theorem 1.14 to  $v = z \cdot (-Q(v; \varepsilon) / G(v))$  and set  $z = 1$ ; since every monomial of  $Q$  has positive  $\varepsilon$ -degree, only  $q \leq |\mathbf{m}|$  contributes. Expanding  $Q^q$  and using eq. (4.94) gives eq. (4.95).  $\square$

**Remark 4.68** (Three engines, one answer). Theorem 4.62 works in the ordinate  $y$  and is S3's; corollary 4.67 works in the phase  $\mu$  and is S2's; theorem 4.65 is S1's. They compute the same coefficients: all three solve the same analytic equation with the same normalisation  $V_0 = 0$ , and the solution of a triangular system is unique. In practice eq. (4.93) is preferable when many multi-indices are wanted (it reuses everything), eq. (4.86) when a single high-order multi-index is wanted (it needs no lower ones), and eq. (4.88) when the answer is wanted symbolically in  $\mu_0$ .

#### 4.9.5 First-order sectors and the leading parity chirp

**Theorem 4.69** (The first correction from each sector). *Let  $e_j$  be the unit multi-index at  $j = (k, \sigma) \in J_K$ . Then*

$$V_{e_j}(\mu_0) = -\frac{r_j(\mu_0)}{\Phi'(\mu_0)} = -\frac{a_k(\mu_0)}{\sqrt{k}} \frac{\mu_0(\mu_0 - \sigma k)}{\mu_0^2 - 3\mu_0 + 3}, \quad (4.96)$$

so that the corresponding displacement of the inverse phase is  $V_{e_j}(\mu_0) e^{-\alpha_j \mu_0}$ , and the corresponding displacement of the inverse index is

$$\delta N_{k,\sigma}^{(1)}(y) = -\frac{3 a_k(\mu_0)}{\pi^2 \sqrt{k}} \frac{\mu_0^2(\mu_0 - \sigma k)}{\mu_0^2 - 3\mu_0 + 3} e^{-(1-\sigma/k)\mu_0}. \quad (4.97)$$

*Proof.* Take  $\mathbf{m} = e_j$  in eq. (4.93): the  $s \geq 2$  terms and all higher  $Q$ -monomials vanish, leaving  $V_{e_j} = -[\varepsilon_j] Q(0; \varepsilon)/\Phi'(\mu_0) = -B_j(0)/\Phi'(\mu_0) = -r_j(\mu_0)/\Phi'(\mu_0)$ . Insert  $r_j = \frac{a_k}{\sqrt{k}} \frac{1-\sigma k/\mu}{1-1/\mu} = \frac{a_k}{\sqrt{k}} \frac{\mu-\sigma k}{\mu-1}$  and eq. (4.66): the factors  $\mu_0 - 1$  cancel, giving eq. (4.96). Finally  $n(\mu) = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}$  is exactly quadratic, so

$$n(\mu_0 + v) = N_0(y) + \frac{3\mu_0}{\pi^2} v + \frac{3}{2\pi^2} v^2, \quad (4.98)$$

and the linear term at first order gives eq. (4.97).  $\square$

**Corollary 4.70** (Index coefficient of every transmonomial). *For every  $\mathbf{m} \neq \mathbf{0}$ , the coefficient of  $\varepsilon^{\mathbf{m}}$  in the inverse index is*

$$\mathfrak{N}_{\mathbf{m}}(\mu_0) = \frac{3\mu_0}{\pi^2} V_{\mathbf{m}} + \frac{3}{2\pi^2} \sum_{\substack{\mathbf{a}+\mathbf{b}=\mathbf{m} \\ \mathbf{a}, \mathbf{b} \neq \mathbf{0}}} V_{\mathbf{a}} V_{\mathbf{b}}. \quad (4.99)$$

*Proof.* Expand eq. (4.98) with  $v = \sum V_{\mathbf{m}} \varepsilon^{\mathbf{m}}$  and compare coefficients. No higher powers occur because  $n$  is quadratic.  $\square$

The first two positive sectors, with  $a_2 = \tilde{A}_2(N_0) = \cos(\pi N_0)$  and  $a_3 = \tilde{A}_3(N_0) = 2 \cos(\pi N_0) \cos(\frac{\pi}{18} + \frac{\pi N_0}{3})$  (interpolated convention) or  $a_2 = (-1)^r$ ,  $a_3 = 2 \cos(\frac{\pi}{18} - \frac{2\pi r}{3})$  (residue-frozen), are

$$\delta N_{2,+}^{(1)} = -\frac{3 a_2}{\pi^2 \sqrt{2}} \frac{\mu_0^2(\mu_0 - 2)}{\mu_0^2 - 3\mu_0 + 3} e^{-\mu_0/2}, \quad (4.100)$$

$$\delta N_{3,+}^{(1)} = -\frac{3 a_3}{\pi^2 \sqrt{3}} \frac{\mu_0^2(\mu_0 - 3)}{\mu_0^2 - 3\mu_0 + 3} e^{-2\mu_0/3}. \quad (4.101)$$

Thus *parity of the index appears at relative scale  $e^{-\mu_0/2}$ , the residue modulo 3 at  $e^{-2\mu_0/3}$ , and the residue modulo  $k$  first appears in the growing  $k$ th sector at  $e^{-(1-1/k)\mu_0}$ .*

#### 4.9.6 The fractional-power staircase in the ordinate

The core equation converts exponentials in  $\mu_0$  into fractional powers of  $y$  exactly:

$$e^{-\alpha \mu_0} = \left( \frac{\kappa(\mu_0 - 1)}{y \mu_0^3} \right)^\alpha \quad \text{for every } \alpha, \quad (4.102)$$

directly from  $y = \kappa e^{\mu_0} (\mu_0 - 1) \mu_0^{-3}$ .

**Proposition 4.71** (Every first-order sector in closed  $y$ -form). *With  $\alpha = \alpha_{k,\sigma}$  and  $z = 1/\mu_0$ ,*

$$\delta N_{k,\sigma}^{(1)} = -\frac{3\kappa^\alpha}{\pi^2 \sqrt{k}} a_k(\mu_0) y^{-\alpha} \mu_0^{1-2\alpha} \sum_{s \geq 0} g_s^{(k,\sigma)} \mu_0^{-s}, \quad (4.103)$$

where

$$\sum_{s \geq 0} g_s^{(k, \sigma)} z^s = \frac{(1 - \sigma k z)(1 - z)^\alpha}{1 - 3z + 3z^2}, \quad g_s^{(k, \sigma)} = \sum_{i=0}^s (-1)^i \binom{\alpha}{i} (\chi_{s-i} - \sigma k \chi_{s-i-1}), \quad (4.104)$$

with  $\chi_{-1} = 0$  and  $\chi_s = \frac{\varrho_+^{s+1} - \varrho_-^{s+1}}{\varrho_+ - \varrho_-}$  the coefficients of  $(1 - 3z + 3z^2)^{-1}$  ( $\varrho_{\pm} = \frac{1}{2}(3 \pm i\sqrt{3})$ ;  $\chi_0 = 1, \chi_1 = 3, \chi_2 = 6, \chi_3 = 9, \chi_4 = 9, \chi_5 = 0, \dots$ ).

*Proof.* Insert eq. (4.102) into eq. (4.97) and factor  $\mu_0^{1-2\alpha}$  out of  $\frac{\mu_0^2(\mu_0 - \sigma k)(\mu_0 - 1)^\alpha}{(\mu_0^2 - 3\mu_0 + 3)\mu_0^{3\alpha}}$ ; with  $z = 1/\mu_0$  the remaining factor is  $\frac{(1 - \sigma k z)(1 - z)^\alpha}{1 - 3z + 3z^2}$ . The coefficient formula is the Cauchy product of  $(1 - z)^\alpha = \sum_i (-1)^i \binom{\alpha}{i} z^i$  with  $(1 - \sigma k z)/(1 - 3z + 3z^2)$ , and  $\chi_s$  is the standard solution of  $\chi_s = 3\chi_{s-1} - 3\chi_{s-2}$ ,  $\chi_0 = 1, \chi_1 = 3$ , whose closed form is as stated since  $\varrho_{\pm}$  are the roots of  $\varrho^2 = 3\varrho - 3$ .  $\square$

**Corollary 4.72** (The inverse staircase of scales). *Since  $\mu_0 \sim \log y$ , the primary positive sectors contribute, in order,*

$$y^{-1/2}, \quad y^{-2/3}(\log y)^{-1/3}, \quad y^{-3/4}(\log y)^{-1/2}, \quad \dots, \quad y^{-(1-1/k)}(\log y)^{-1+2/k}, \quad (4.105)$$

*accumulating at the scale  $y^{-1}$ , where they meet the square of the first sector and the whole negative family.*

*Proof.* By eq. (4.103) the  $(k, +)$  sector has size  $y^{-\alpha} \mu_0^{1-2\alpha} (1 + \mathcal{O}(\mu_0^{-1}))$  times a bounded amplitude, with  $\alpha = 1 - 1/k$ , so  $1 - 2\alpha = -1 + 2/k$ ; and  $\mu_0 \sim \log y$  because  $\Phi(\mu_0) = \log(y/\kappa)$  with  $\Phi(\mu) = \mu(1 + o(1))$ . Substituting  $k = 2, 3, 4$  gives the first three displayed scales. Since  $\alpha_{k,+} \uparrow 1$ , the scales accumulate at  $y^{-1}$ ; the square of the  $(2, +)$  sector has action  $\frac{1}{2} + \frac{1}{2} = 1$  and every negative action is  $\geq 1$ .  $\square$

**Theorem 4.73** (The leading parity chirp). *In the interpolated convention,*

$$\mathcal{N}(y) = N_0(y) - \frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} \frac{\cos(\pi N_0(y))}{\sqrt{y}} \left(1 + \mathcal{O}\left(\frac{1}{\log y}\right)\right) + \mathcal{O}\left(y^{-2/3}(\log y)^{-1/3}\right), \quad (4.106)$$

with the constant

$$\frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} = \frac{3^{1/4}}{2\pi} = 0.2094596846361762626265841 \dots \quad (4.107)$$

Since  $N_0(y) = \frac{3}{2\pi^2}(\log y)^2(1 + o(1))$ , the cosine is a quadratic logarithmic chirp, not a log-periodic oscillation.

*Proof.* Take  $k = 2, \sigma = +1, \alpha = \frac{1}{2}$  in eq. (4.103): the power  $\mu_0^{1-2\alpha} = \mu_0^0$  is absent,  $g_0^{(2,+)} = 1$ , and  $a_2 = \cos(\pi N_0)$  by lemma 4.10. The constant is  $3\kappa^{1/2}/(\pi^2\sqrt{2})$ , and  $\sqrt{\kappa} = \pi/\sqrt{6\sqrt{3}}$  gives  $\frac{3\pi}{\pi^2\sqrt{2}\sqrt{6\sqrt{3}}} = \frac{3}{\pi\sqrt{12\sqrt{3}}} = \frac{3}{2\pi \cdot 3^{3/4}} = \frac{3^{1/4}}{2\pi}$ , which we evaluated to 25 digits. The relative  $\mathcal{O}(1/\log y)$  is the  $s \geq 1$  tail of eq. (4.103), and the next omitted contribution is the  $k = 3$  sector at scale  $y^{-2/3}(\log y)^{-1/3}$  by corollary 4.72; all products of two blocks have action  $\geq 1$ .  $\square$

**Remark 4.74** (Constant check). S2 prints this constant as 0.2094596846361762626; the closed form  $3^{1/4}/(2\pi)$  is S3's. The two agree, and both agree with our independent 25-digit evaluation eq. (4.107). S1 does not state the constant, giving instead the equivalent phase-variable form eq. (4.100); the three are algebraically identical, as the cancellation of  $\mu_0 - 1$  in the proof of theorem 4.69 shows.

## 4.10 The profinite arithmetic phase

The residue-frozen convention of definition 4.57 organises itself into a projective system, which is the cleanest way to say what data an “arithmetically complete” inverse of  $p$  actually depends on.

**Theorem 4.75** (Compatibility and the profinite parameter). *Let  $\Lambda_K = \text{lcm}(1, \dots, K)$ , so  $\Lambda_K \mid \Lambda_{K+1}$ . Suppose  $r_K \in \mathbb{Z}/\Lambda_K\mathbb{Z}$  satisfy  $r_{K+1} \equiv r_K \pmod{\Lambda_K}$ , and let  $\mu_{K,r_K}$  be the corresponding inverse branches of definition 4.57. Then for every  $K$ , the transseries of  $\mu_{K+1,r_{K+1}}$  obtained by deleting all monomials containing the variables of level  $K+1$  coincides with the transseries of  $\mu_{K,r_K}$ . Consequently the complete arithmetic inverse is a projective family indexed by the profinite integers  $\hat{r} \in \hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/m\mathbb{Z}$ , and an actual partition index  $n$  supplies its own phase through the diagonal embedding  $n \mapsto \hat{n}$ .*

*Proof.* Deleting the level- $(K+1)$  variables from the defining equation eq. (4.92) at level  $K+1$  produces the defining equation at level  $K$  provided the retained amplitudes agree, i.e. provided  $A_k(r_{K+1}) = A_k(r_K)$  for  $k \leq K$ . That is exactly the compatibility hypothesis, since  $A_k$  has period  $k \mid \Lambda_K$  and  $r_{K+1} \equiv r_K \pmod{\Lambda_K}$ . Both truncated equations are triangular with the same normalisation  $V_0 = 0$ , and a triangular system has a unique solution, so the solutions agree coefficient by coefficient. The projective limit statement is the definition of  $\hat{\mathbb{Z}}$ : compatible systems  $(r_K)$  are exactly the elements of  $\varprojlim \mathbb{Z}/\Lambda_K\mathbb{Z} = \hat{\mathbb{Z}}$ , because  $\{\Lambda_K\}$  is cofinal in the divisibility order on  $\mathbb{N}$ .  $\square$

**Remark 4.76** (Why a profinite parameter and not a real one). At every algebraic order all indices share one expansion. Parity enters at relative order  $e^{-\mu/2}$ ; the residue mod 3 at  $e^{-2\mu/3}$ ; the residue mod  $k$  at  $e^{-(1-1/k)\mu}$ . No finite real-valued smooth parameter records all of these congruence classes simultaneously, and no single real analytic function of  $n$  interpolates all  $A_k$  canonically (remark 4.11). A profinite integer does so canonically and finitely at every exponential resolution: each finite action cutoff  $\eta < 1$  sees only  $A_k$  for  $k \leq (1-\eta)^{-1}$ , i.e. only  $\hat{r}$  modulo  $\Lambda_{\lfloor (1-\eta)^{-1} \rfloor}$ . This is S1’s observation and is absent from S2 and S3.

**Remark 4.77** (A structure suggested, not proved). S1 proposes an “arithmetic transseries sheaf” with base  $\hat{\mathbb{Z}}$  and fibres the finite-level analytic transseries algebras, and calls theorem 4.75 its elementary case. We record this as a suggestion. Nothing beyond theorem 4.75 — namely, projective compatibility of the coefficient systems — is established here or in the sources: no topology on the fibres, no gluing statement, and no continuity of  $\hat{r} \mapsto \mu_{K,\hat{r}}$  in any topology on  $\hat{\mathbb{Z}}$  has been proved. It is stated as conjecture 4.96 in section 4.15, not as a theorem.

## 4.11 Validity, truncation control, and integer recovery

### 4.11.1 Remainder transport

**Theorem 4.78** (Inverse error after  $K$  Rademacher levels). *Fix  $K \geq 1$ . Let  $y = p(n)$ , let  $\mu = \lambda\sqrt{n-1/24}$ , and let  $\mu_K$  denote either inverse branch of definition 4.57 (with  $r \equiv n$ ) or of definition 4.58. Then*

$$\mu_K(y) - \mu = \mathcal{O}_K\left(\mu^{3/2} e^{-(1-\frac{1}{K+1})\mu}\right), \quad n_K(y) - n = \mathcal{O}_K\left(\mu^{5/2} e^{-(1-\frac{1}{K+1})\mu}\right), \quad (4.108)$$

where  $n_K = \frac{1}{24} + 3\mu_K^2/(2\pi^2)$ . Explicitly, with the constant of theorem 4.23, for  $\mu \geq \mu_*$  (any threshold at which  $\Phi' \geq \frac{1}{2}$  and the omitted tail is  $\leq \frac{1}{2}$  in absolute value),

$$|n_K(y) - n| \leq \frac{3\mu}{\pi^2} \cdot \frac{2}{1} \cdot \frac{8\mu^2}{3(1-\mu^{-1})\sqrt{K}} \exp\left[-\left(1 - \frac{1}{K+1}\right)\mu\right] (1 + o(1)). \quad (4.109)$$

*Proof.* This is theorem 1.44 with the forward remainder supplied by theorems 4.23 and 4.24. In detail: let  $\mathcal{F}_K = \log \mathcal{P}_K$  and  $\mathcal{F} = \log p$  regarded as functions of the phase. By proposition 4.17,  $\mathcal{F}(\mu) - \mathcal{F}_K(\mu) = \log(1 + \mathcal{R}_{>K}/(1 + \mathcal{R}_{\leq K}))$ , which by theorem 4.24 and  $|\log(1+u)| \leq 2|u|$  for  $|u| \leq \frac{1}{2}$  is  $\mathcal{O}_K(\mu^{3/2} e^{-(1-1/(K+1))\mu})$ . Since  $\mathcal{F}'_K(\mu) = \Phi'(\mu)(1 + o(1)) \rightarrow 1$ , the mean value theorem converts this

logarithmic residual into the phase error, which is the first half of eq. (4.108). Finally  $dn/d\mu = 3\mu/\pi^2$  gives the index error. The explicit form eq. (4.109) replaces theorem 4.24 by the explicit theorem 4.23 (losing  $\mu^{-1/2}$ , gaining a constant) and  $\mathcal{F}'_K \geq \frac{1}{2}$ .  $\square$

**Corollary 4.79** (Beyond-all-orders accuracy of the core inverse). *For  $y = p(n)$ ,*

$$N_0(y) - n = \mathcal{O}\left(\mu^{5/2}e^{-\mu/2}\right) = \mathcal{O}\left(n^{5/4}\exp\left(-\frac{1}{2}\pi\sqrt{2n/3}\right)\right), \quad (4.110)$$

so  $N_0(p(n)) - n \rightarrow 0$  faster than every power of  $n^{-1}$ .

*Proof.* Theorem 4.78 at  $K = 1$ ; note  $N_0 = n_1$  because the  $K = 1$  level of definition 4.57 contains the sectors  $(1, \pm)$ , and the  $(1, -)$  sector has action 2, hence contributes below the  $K = 1$  error bound.  $\square$

**Proposition 4.80** (A posteriori certificate; specialisation of theorem 1.42). *Let  $\mathcal{F}_K(\mu) = \log \mathcal{P}_K(\mu)$  and let  $\rho(\tilde{\mu}) = \mathcal{F}_K(\tilde{\mu}) - \log y$  be the residual of a candidate  $\tilde{\mu}$ . If  $\mathcal{F}'_K \geq m > 0$  on an interval  $I$  containing both  $\tilde{\mu}$  and the exact root  $\mu_K(y)$ , then*

$$|\tilde{\mu} - \mu_K(y)| \leq \frac{|\rho(\tilde{\mu})|}{m}, \quad |\tilde{n} - n_K(y)| \leq \frac{3}{2\pi^2} (|\tilde{\mu}| + |\mu_K(y)|) \frac{|\rho(\tilde{\mu})|}{m}. \quad (4.111)$$

*Proof.* Mean value theorem applied to  $\mathcal{F}_K$ , then the factorisation  $\tilde{\mu}^2 - \mu_K^2 = (\tilde{\mu} - \mu_K)(\tilde{\mu} + \mu_K)$ . Since  $\mathcal{F}'_K \rightarrow 1$ , one may take  $m = \frac{1}{2}$  beyond an effectively checkable threshold.  $\square$

**Remark 4.81** (Why a residual certificate rather than term counting). Proposition 4.80 separates the *local inversion* error (how well  $\tilde{\mu}$  solves the level- $K$  equation) from the *Rademacher tail* (how far the level- $K$  equation is from the truth). Only the second needs theorem 4.23. This modularity is what makes it possible to certify a truncated symbolic transseries and a numerically refined Newton root by the same inequality.

### 4.11.2 Algebraic truncation

**Proposition 4.82.** *With  $U_M(z) = \sum_{m \leq M} c_m z^m$  as in theorem 4.50,*

$$\mu_0 - (X + U_M(X^{-1})) = \mathcal{O}(X^{-M-1}), \quad N_0 - \left[ \frac{1}{24} + \frac{3}{2\pi^2} (X + U_M(X^{-1}))^2 \right] = \mathcal{O}(X^{-M}), \quad (4.112)$$

*the loss of one order coming from  $dn/d\mu \asymp X$ . In particular  $M \geq 1$  is necessary for the index to be correct to  $o(1)$  and hence for the arithmetic phases of section 4.9 to be evaluated at the right point (remark 4.55).*

*Proof.*  $U$  is analytic at 0 by theorem 4.50, so its Taylor remainder after  $M$  terms is  $\mathcal{O}(z^{M+1})$ ; substitute  $z = X^{-1}$  and use  $N_0 - \tilde{N} = \frac{3}{2\pi^2}(\mu_0 - \tilde{\mu})(\mu_0 + \tilde{\mu})$  with  $\mu_0 + \tilde{\mu} = \mathcal{O}(X)$ .  $\square$

### 4.11.3 Integer recovery

**Theorem 4.83** (Eventual exact recovery by rounding). *There is  $n_0$  such that for every integer  $n \geq n_0$ ,*

$$n = \lfloor N_0(p(n)) \rfloor, \quad (4.113)$$

$\lfloor \cdot \rfloor$  denoting nearest-integer rounding. *Consequently the arithmetic phase  $\hat{n}$  of theorem 4.75 is recoverable from  $y = p(n)$  before any exponentially small sector is evaluated.*

*Proof.* By eq. (4.110),  $|N_0(p(n)) - n| \rightarrow 0$ ; it is therefore  $< \frac{1}{2}$  for all large  $n$ , at which point rounding is exact. The last sentence is the observation that  $A_k(n)$  depends only on  $n \bmod k$ .  $\square$



*Remark 4.84* (The threshold is very small in practice). Theorem 4.83 is asymptotic and claims no optimal  $n_0$ . We computed  $N_0(p(n)) - n$  for all  $1 \leq n \leq 59$  in 140-digit arithmetic; the largest absolute value is  $0.1723 \dots$  at  $n = 1$ , and the sequence decreases in magnitude thereafter (through  $2.8 \times 10^{-2}$  at  $n = 10$ ,  $4.5 \times 10^{-4}$  at  $n = 50$ ,  $1.5 \times 10^{-5}$  at  $n = 100$ ,  $4.3 \times 10^{-17}$  at  $n = 1000$ ). So  $n_0 = 1$  is consistent with the data, though we prove only the asymptotic statement; producing a certified  $n_0$  requires effective constants in theorem 4.78, i.e. effective Rademacher remainder bounds, which is listed as conjecture 4.95.

*Remark 4.85* (Circularity, avoided). The arithmetic transseries needs the residue  $r$  of  $n$ , which appears to require knowing  $n$ . The order of operations resolves this: (i) compute  $X$  from eq. (4.56); (ii) compute  $\mu_0$  and  $N_0$ , which are phase-independent; (iii) round, obtaining  $n$  by theorem 4.83 for  $y$  on the range; (iv) now every  $A_k(n)$  is known, and the exponentially small sectors refine an already identified integer. The arithmetic layer is thus a *refinement* of a known answer, not a device for guessing the phase. For  $y$  off the range, step (iii) yields  $\lceil \mathcal{N}(y) \rceil$  by theorem 4.44 instead, and the sectors describe  $\mathcal{N}$  itself.

## 4.12 Numerical verification

Every number in this chapter was recomputed for this volume. Partition values were generated exactly by Euler's pentagonal recurrence

$$p(n) = \sum_{m \geq 1} (-1)^{m-1} \left[ p\left(n - \frac{m(3m-1)}{2}\right) + p\left(n - \frac{m(3m+1)}{2}\right) \right], \quad p(0) = 1, \quad p(r) = 0 \quad (r < 0), \quad (4.114)$$

in exact integer arithmetic (checked against  $p(10) = 42$ ,  $p(50) = 204226$ ,  $p(100) = 190569292$ ); Dedekind sums exactly as rationals from eq. (4.9); the transcendental factors in 140-digit floating point. Coefficient identities were verified in exact rational or exact symbolic arithmetic, never by floating-point spot checks. The tables check normalisation, signs, phases and the hierarchy of actions; they are used in no proof.

Table 4.3: Relative error  $(S_K(n) - p(n))/p(n)$  of the paired truncation  $S_K = \sum_{k \leq K} (P_{k,+} + P_{k,-})$ , recomputed at 140 digits. Accuracy need not improve monotonically in  $K$  at small  $n$ : the amplitudes  $A_k(n)$  change sign and can suppress or enhance an individual sector.

| $n$  | $p(n)$                 | $K = 1$                  | $K = 2$                 | $K = 3$                  | $K = 5$                  |
|------|------------------------|--------------------------|-------------------------|--------------------------|--------------------------|
| 10   | 42                     | $-8.8621 \cdot 10^{-3}$  | $1.6606 \cdot 10^{-3}$  | $3.7391 \cdot 10^{-4}$   | $3.4722 \cdot 10^{-4}$   |
| 25   | 1 958                  | $1.0717 \cdot 10^{-3}$   | $3.0045 \cdot 10^{-6}$  | $-6.1015 \cdot 10^{-5}$  | $-3.4866 \cdot 10^{-6}$  |
| 50   | 204 226                | $-7.3074 \cdot 10^{-5}$  | $3.9028 \cdot 10^{-6}$  | $2.1050 \cdot 10^{-7}$   | $-1.1350 \cdot 10^{-7}$  |
| 100  | 190 569 292            | $-1.8220 \cdot 10^{-6}$  | $8.6838 \cdot 10^{-9}$  | $-4.9491 \cdot 10^{-9}$  | $3.1454 \cdot 10^{-10}$  |
| 250  | $2.3079 \cdot 10^{14}$ | $-1.0760 \cdot 10^{-9}$  | $7.2267 \cdot 10^{-13}$ | $4.3021 \cdot 10^{-14}$  | $2.1256 \cdot 10^{-15}$  |
| 500  | $2.3002 \cdot 10^{21}$ | $-2.4389 \cdot 10^{-13}$ | $1.7537 \cdot 10^{-17}$ | $-2.0017 \cdot 10^{-19}$ | $-3.5933 \cdot 10^{-22}$ |
| 1000 | $2.4061 \cdot 10^{31}$ | $-1.6997 \cdot 10^{-18}$ | $1.2574 \cdot 10^{-24}$ | $-3.4744 \cdot 10^{-27}$ | $3.9453 \cdot 10^{-30}$  |

Three features of table 4.4 confirm the theory point by point. (i) The first column tends to  $-3/\pi^2 = -0.30396 \dots$ , not to 0: this is corollary 4.54 and the numerical signature of remark 4.55. (S1 and S2 tabulate  $N_X = \frac{1}{24} + 3X^2/(2\pi^2)$  and report  $\rightarrow -0.304$ ; S3 tabulates  $3X^2/(2\pi^2)$  without the  $\frac{1}{24}$  and reports  $\rightarrow -0.346$ . The columns differ by exactly  $1/24$  and each is correct for its own definition; we keep the  $\frac{1}{24}$  so that  $N_X$  and  $N_0$  are directly comparable.) (ii) Restoring the amplitude factor  $1 - 1/\mu$ , i.e. passing from  $X$  to  $\mu_0$ , removes that entire  $\mathcal{O}(1)$  bias and leaves an exponentially small error. (iii) The residual after the core alternates with the parity of  $n$ : at  $n = 49, 50, 51$  it is  $-5.2643 \cdot 10^{-4}$ ,  $+4.5102 \cdot 10^{-4}$ ,  $-4.0920 \cdot 10^{-4}$ , exactly as eq. (4.100) predicts through  $\tilde{A}_2 = \cos \pi N_0$ . Adding the single first-order  $k = 2$  sector reduces these to  $-1.0890 \cdot 10^{-5}$ ,  $-2.4087 \cdot 10^{-5}$ ,  $+2.9029 \cdot 10^{-5}$ ; adding the first-order sectors through  $k = 10$  reduces them to  $-2.24 \cdot 10^{-7}$ ,  $-9.95 \cdot 10^{-8}$ ,  $-5.81 \cdot 10^{-8}$ . This checks both the sign and the phase of the leading chirp eq. (4.106).

Table 4.4: Inverse errors at exact ordinates  $y = p(n)$ ; entries are (approximation)  $-n$ .  $N_X$  is the bare Lambert carrier index,  $N_0$  the dominant core,  $N_K$  the Newton inverse of the level- $K$  interpolated sum (definition 4.58). The residue-frozen inverses (definition 4.57) reproduce the last three columns to all five digits shown for  $n \geq 100$ ; at  $n = 50$  they differ in the fifth digit of the  $K = 3, 5$  entries ( $-1.2992 \cdot 10^{-6}$ ,  $7.0057 \cdot 10^{-7}$ ), at  $n = 25$  in the fourth ( $2.7926 \cdot 10^{-4}$ ,  $1.5958 \cdot 10^{-5}$ ), at  $n = 10$  in the third ( $-5.3338 \cdot 10^{-3}$ ,  $-1.2006 \cdot 10^{-3}$ ,  $-1.1150 \cdot 10^{-3}$ ). That is exactly the pattern proposition 4.59 predicts.

| $n$  | $N_X - n$               | $N_0 - n$               | $N_2 - n$                | $N_3 - n$                | $N_5 - n$                |
|------|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|
| 10   | $-3.9910 \cdot 10^{-1}$ | $2.8447 \cdot 10^{-2}$  | $-5.3290 \cdot 10^{-3}$  | $-1.1863 \cdot 10^{-3}$  | $-1.1015 \cdot 10^{-3}$  |
| 25   | $-3.7877 \cdot 10^{-1}$ | $-4.9052 \cdot 10^{-3}$ | $-1.3751 \cdot 10^{-5}$  | $2.7902 \cdot 10^{-4}$   | $1.5941 \cdot 10^{-5}$   |
| 50   | $-3.5044 \cdot 10^{-1}$ | $4.5102 \cdot 10^{-4}$  | $-2.4089 \cdot 10^{-5}$  | $-1.2993 \cdot 10^{-6}$  | $7.0059 \cdot 10^{-7}$   |
| 100  | $-3.3600 \cdot 10^{-1}$ | $1.5378 \cdot 10^{-5}$  | $-7.3293 \cdot 10^{-8}$  | $4.1772 \cdot 10^{-8}$   | $-2.6548 \cdot 10^{-9}$  |
| 250  | $-3.2364 \cdot 10^{-1}$ | $1.3942 \cdot 10^{-8}$  | $-9.3644 \cdot 10^{-12}$ | $-5.5747 \cdot 10^{-13}$ | $-2.7544 \cdot 10^{-14}$ |
| 500  | $-3.1768 \cdot 10^{-1}$ | $4.4042 \cdot 10^{-12}$ | $-3.1668 \cdot 10^{-16}$ | $3.6146 \cdot 10^{-18}$  | $6.4888 \cdot 10^{-21}$  |
| 1000 | $-3.1356 \cdot 10^{-1}$ | $4.2960 \cdot 10^{-17}$ | $-3.1780 \cdot 10^{-23}$ | $8.7815 \cdot 10^{-26}$  | $-9.9717 \cdot 10^{-29}$ |

*Remark 4.86* (What was verified in exact arithmetic). (1) Equation (4.33) against the Taylor expansion of eq. (4.31), symbolically in  $\beta$  through order  $t^9$ : identical; the closed forms eqs. (4.37) and (4.38) were produced from eq. (4.33) and match S2 (through  $c_5$ ) and S3 (through  $c_8$ ). (2)  $\omega_0, \dots, \omega_4$  against both the S1 and the S2 grouping of the same numbers: identical (they look different only because S1 splits each into partial fractions in  $\pi$ ). (3)  $c_1, \dots, c_{18}$  of table 4.2 from eq. (4.72) in exact rationals, plus the residual test eq. (4.74) at  $M = 12$ ; all eighteen agree with the sources. (4)  $\gamma_1, \dots, \gamma_8$  against all three sources;  $\gamma_9, \gamma_{10}$  are new here. (5)  $\Pi_1, \dots, \Pi_8$  from the Lagrange form against the Stirling form and against each of the three printed shapes, confirming remark 4.48; S1's printed  $P_6, P_7$  are correct. (6) S3's  $u$ -expansion in  $H$  through  $H^{-4}$  and its  $N_0$ -expansion through  $H^{-3}$ , and S2's  $N_0$ -block in  $R$  through the constant term: correct as printed. (7) Equation (4.11) against direct evaluation of eq. (4.10) from exact Dedekind sums. (8) The chirp constant eq. (4.107) to 25 digits in both forms. No discrepancy with any source's arithmetic was found: every repair recorded in this chapter is structural, not numerical.

*Remark 4.87* (Normalisation cross-checks). Because eq. (4.14) carries a factor  $\sqrt{k}$  that some normalisations absorb into  $A_k$ , a missing or spurious power of  $k$  is the easiest error to make and the hardest to see. Three identifications of one quantity detect it: the  $k$ th derivative factor of eq. (4.14) equals  $\frac{\mu}{4kN^{3/2}} [e^{\mu/k}(1 - k/\mu) + e^{-\mu/k}(1 + k/\mu)]$ , and after multiplication by  $\sqrt{k}/(\pi\sqrt{2})$  it may be written either as  $\frac{1}{4\sqrt{3k}N} [\dots]$  or as  $\frac{\kappa}{\sqrt{k}\mu^2} [\dots]$ ; at  $k = 1$  the positive term is  $\frac{e^\mu}{4\sqrt{3}N}(1 - \frac{1}{\mu})$ , which becomes eq. (4.2) after replacing  $N$  by  $n$  and dropping  $1 - 1/\mu$ . Any of these disagreeing with the others by a power of  $k$  localises the slip.

### 4.13 Algorithms

All formulas of this chapter are constructive and need only formal power-series arithmetic over  $\mathbb{Q}(\pi)$  together with exact rational Dedekind sums.

**Algorithm 4.88** (Forward sector coefficients). Given  $k, \sigma, M$ : set  $\beta = \sigma\pi/(6k)$  and evaluate the finite sum eq. (4.33) for  $0 \leq m \leq M$  ( $\mathcal{O}(m)$  field operations each, no series manipulation; keep  $\beta$  symbolic if sector-uniform output is wanted). If the whole array is wanted at once, evaluate  $\ell_1, \dots, \ell_M$  from eq. (4.39) and run the triangular recurrence eq. (4.41) instead. Assemble eq. (4.30), pairing the two signs before summing over large  $k$  (remark 4.20).

**Algorithm 4.89** (Phase-independent inverse). Given a large  $y$  and an order  $M$ : (1) compute  $X$  from eq. (4.56) with a standard  $W_{-1}$  routine — do *not* substitute the logarithmic expansion eq. (4.57), which is numerically far inferior; (2) form  $\mu_0 = X + \sum_{m \leq M} c_m X^{-m}$  from the universal rationals of table 4.2, optionally refined by one Newton step on  $\Phi(\mu) = R$ ; (3) return  $N_0 = \frac{1}{24} + 3\mu_0^2/(2\pi^2)$ ; (4) certify by evaluating  $\rho = \mu_0 - 2 \log \mu_0 + \log(1 - \mu_0^{-1}) - \log(y/\kappa)$  and applying proposition 4.80; (5) round (theorem 4.83) if  $y$  is known to lie on the range, else take  $\lceil \cdot \rceil$  for the staircase (theorem 4.44).

**Algorithm 4.90** (Arithmetic exponential correction). Given  $y$ , a level  $K$  and a finite set of multi-indices: (1) compute  $\mu_0, N_0$  by algorithm 4.89 and use  $N_0$ , never  $N_X$ , to evaluate amplitudes (remark 4.55); (2) compute exact rational  $s(h, k)$  for  $k \leq K$  and hence  $A_k(r)$  (frozen, with  $r$  from theorem 4.83) or  $\tilde{A}_k(N_0)$  (interpolated); (3) build the actions and amplitudes of eq. (4.83); (4) traverse the multi-indices in a graded monomial order using eq. (4.93), or, for one isolated  $\mathbf{m}$ , use eq. (4.86) or the Bell form eq. (4.88) with the closed jets eq. (4.90); (5) substitute  $\varepsilon_j = e^{-\alpha_j \mu_0}$  and convert to the index by eq. (4.99); (6) certify by proposition 4.80 on  $\mathcal{P}_K$ , adding theorem 4.23 if the comparison is with the exact sequence.

*Remark 4.91* (Modular structure of the computation). The layers are independent and should be cached separately: the carrier polynomials  $\Pi_j$  depend only on the exponent data  $(a, b) = (1, -2)$ ; the rationals  $c_m$  only on the amplitude  $1 - 1/\mu$ ; the inverse-derivative polynomials  $\mu_0^{(s)}$  only on  $\Phi$ ; and the arithmetic enters solely through the numbers  $a_k$  in eq. (4.83). Universal tables are computed once and residue-dependent evaluation is then cheap.

## 4.14 A template for Rademacher-type sequences

Nothing above used the partition function except through four properties. S3 has the most careful version of this observation and S1 the most usable one; what follows merges them into a hypothesis list and a theorem, so that the template is a statement one can check rather than a slogan.

**Definition 4.92** (Rademacher-type representation). A sequence  $(a(n))_{n \geq n_1}$  of positive reals is of *Rademacher type* if:

- (T1) there is a real analytic, strictly increasing *phase*  $\phi : (\nu_1, \infty) \rightarrow (0, \infty)$  with  $\phi \rightarrow \infty$  and real analytic inverse, giving the large variable  $\mu = \phi(n)$ ;
- (T2) the *envelope*  $F(\mu) = Ce^{a\mu^\alpha} \mu^b \exp Q(\mu^{-\delta})$  is an exponential–power model definition 1.2 with  $a > 0, \alpha \in (0, 1]$ ;
- (T3) there are actions  $\alpha_j > 0$  ( $j \in J$ ) with  $\inf_j \alpha_j > 0$ , amplitudes  $a_j(n)$  taking values in a fixed finite-dimensional space at each truncation level (e.g. periodic in  $n$ ), and  $g_j$  analytic at  $\infty$ , with

$$a(n) = F(\mu) \left( 1 + \sum_{j \in J'} a_j(n) g_j(\mu) e^{-\alpha_j \mu} + \mathcal{T}_{J'}(\mu; n) \right) \quad \text{for every finite } J' \subset J; \quad (4.115)$$

- (T4)  $\{j : \alpha_j \leq \eta\}$  is finite for every  $\eta < \sup_j \alpha_j$ , and either  $\mathcal{T}_{J'} = \mathcal{O}(\mu^{C'} e^{-\eta' \mu})$  for some  $\eta' > \max_{j \in J'} \alpha_j$ , or — if the actions accumulate — the grouped remainder is retained as one analytic block.

**Theorem 4.93** (Template). *Let  $(a(n))$  be of Rademacher type. Then:*

1. (Reduction.)  $\mu = \xi^{1/\alpha}$  of lemma 1.4 brings (T2) to the linear-phase model with computable  $(C', 1, b', Q')$ .
2. (Carrier.)  $C' e^{\xi} \xi^{b'} = y$  is solved exactly by theorem 1.20; equivalently, for  $y = Ce^{a\mu^\alpha} \mu^{-\beta}$ ,

$$\mu = \left( -\frac{\beta}{a\alpha} W_{-1} \left( -\frac{a\alpha}{\beta} \left( \frac{C}{y} \right)^{\alpha/\beta} \right) \right)^{1/\alpha}. \quad (4.116)$$

3. (Amplitude.) The envelope inverse is  $\xi_0 = X + U(X^{-1})$  with  $U$  the unique solution of the fixed-point equation of theorem 1.25 at  $(1, b', Q')$ ; coefficients by lemma 1.16 or by isolating the linear occurrence of  $U$ .
4. (Phase locking.) If some  $a_j$  is periodic on the index scale and  $dn/d\mu \rightarrow \infty$ , the amplitudes must be evaluated at  $\xi_0$ , not at  $X$ : the carrier bias  $n(\xi_0) - n(X)$  is  $\mathcal{O}(1)$  in general.

5. (Sectors.) *With the phase frozen, every inverse transmonomial is given by eq. (4.86) (in  $y$ ) or eq. (4.95)/eq. (4.93) (in  $\mu$ ), with first-order term*

$$\mu = \mu_0 - \frac{a_j g_j(\mu_0)}{(\log F)'(\mu_0)} e^{-\alpha_j \mu_0} + \mathcal{O}(e^{-2\alpha_{\min} \mu_0}). \quad (4.117)$$

6. (Transport.) *A relative forward remainder  $\epsilon$  transports to a phase error  $\epsilon/(\log F)'(\mu_0)$  and an index error  $\epsilon n'(\mu_0)/(\log F)'(\mu_0)$ , by theorem 1.44.*
7. (Staircase.) *If  $a$  is integer valued with gaps bounded below, the generalised inverse is the ceiling of the smooth one (theorem 1.39), with the order-one sawtooth of remark 4.45 as an irreducible final sector.*

*Proof.* (i)–(iii) are lemma 1.4 and theorems 1.20 and 1.25 at the stated parameters. For the display in (ii): raising  $y = Ce^{a\mu^\alpha} \mu^{-\beta}$  to the power  $\alpha/\beta$  and rearranging gives  $(\frac{a\alpha}{\beta} \mu^\alpha) e^{-(a\alpha/\beta)\mu^\alpha} = \frac{a\alpha}{\beta} (C/y)^{\alpha/\beta}$ , which is the Lambert equation with the stated argument, the branch  $-1$  forced by  $\mu \rightarrow \infty$ . (iv) is the computation of remark 4.55 with  $3\mu/\pi^2$  replaced by  $n'(\mu_0)$ : what matters is only that  $n' \rightarrow \infty$  while  $\xi_0 - X \rightarrow 0$ , so their product need not vanish. (v) is theorems 4.62 and 4.65 and corollary 4.67, whose proofs used no property of  $F$  or of the blocks beyond analyticity and  $F' \neq 0$ ; the first-order display is eq. (4.96) in that generality. (vi) is the mean-value argument of theorem 4.78, and (vii) is theorem 4.44.  $\square$

*Remark 4.94* (The partition case, and the scope of the template). Taking  $\phi(n) = \pi\sqrt{2(n-1/24)/3}$ ,  $(C, a, \alpha, b, Q)$  as in eq. (4.5),  $J = \{(k, \sigma)\}$ ,  $\alpha_{k,\sigma} = 1 - \sigma/k$  and  $a_k = A_k$ , hypotheses (T1)–(T3) hold by theorem 4.14 and proposition 4.17, and (T4) holds in its second form: the actions *do* accumulate at 1, and the grouped block is the paired tail eq. (4.25) controlled by theorem 4.23. With  $(\alpha, \beta) = (1, 2)$  in  $\mu$ , eq. (4.116) reduces to eq. (4.56). The template applies verbatim to coefficient sequences of weakly holomorphic modular forms and eta-quotients with Rademacher-type expansions; the partition function is unusually clean because the Bessel index  $3/2$  makes the kernel two exponentials with amplitudes linear in  $\mu^{-1}$ , so step (iii) is the explicit rational recurrence eq. (4.72) rather than a general implicit solve. *No theorem about any other specific sequence is claimed here: verifying (T1)–(T4) for a given family is exactly the work the template does not do.*

## 4.15 Structural consequences and open questions

### 4.15.1 Convergent blocks inside an exponentially complete object

The exact sector sum eq. (4.19) converges; each sector's algebraic series in  $n^{-1/2}$  converges on  $|n^{-1/2}| < \sqrt{24}$  (theorem 4.30); and the reversion series  $U$  converges near 0 (theorem 4.50). Divergence is therefore *not* what creates the non-perturbative sectors here: they are supplied by modularity and the circle method, independently of the perturbative data. The useful trichotomy is that a *Poincaré expansion* records one dominant sector and forgets all exponentially small information; a *transseries completion* records the other sectors, whether or not the series in each diverges; and an *exact transseries representation*, as here, may itself converge. Compare Parts 2, 3 and 5, where the algebraic series do diverge and theorem 1.46 is the operative tool.

### 4.15.2 Arithmetic amplitudes are not Stokes constants

The  $A_k(n)$  weight distinct exponential sectors and govern exponentially small corrections to the inverse, which is formally the role of Stokes constants. They are not Stokes constants: they are arithmetic, periodic in  $n$  rather than fixed, and no natural analytic continuation in the index produces them canonically (remark 4.11). Theorem 4.75 says what replaces the Stokes data: a profinite integer, of which every finite exponential resolution sees only a finite quotient.

### 4.15.3 Open questions

**Conjecture 4.95** (Effective rounding threshold). *There is an explicit small  $n_0$  — the evidence of remark 4.84 suggests  $n_0 = 1$  — with  $n = \lfloor N_0(p(n)) \rfloor$  for all  $n \geq n_0$ , provable by inserting effective Rademacher remainder bounds (Lehmer; Banerjee–Paule–Radu–Schneider) in place of  $|A_k| \leq k$  in theorem 4.78.*

**Conjecture 4.96** (Arithmetic transseries sheaf). *The projective family of theorem 4.75 extends to a sheaf of transseries algebras over  $\widehat{\mathbb{Z}}$ , with  $\widehat{r} \mapsto (\text{level-}K \text{ inverse transseries})$  continuous for the profinite topology and the finite-level algebras gluing over the standard clopen basis. Only the projective compatibility is proved here (remark 4.77).*

**Conjecture 4.97** (Uniform treatment at the accumulation action). *The zeta condensation eq. (4.29) can serve as a single block in theorem 4.62, yielding a description of the inverse uniform across the accumulation action 1, with error bounds depending on the analytic behaviour of  $\mathcal{Z}_n(s)$  rather than on a sector cutoff. This needs continuation and vertical growth information for the Kloosterman-type Dirichlet series  $\mathcal{Z}_n$ , which is not established here. S3 raises it as an optimisation problem rather than a gap, correctly: the exact Rademacher representation is always available as a fallback.*

**Remark 4.98** (Classifying interpolations). Different strictly increasing smooth interpolations of  $p$  agree at every algebraic order of the inverse and on the integer range, but differ in the exponentially small off-lattice sectors: candidates include definition 4.9, piecewise-logarithmic interpolation through  $(n, \log p(n))$ , Newton or cardinal series with prescribed growth, and continuations built from Poincaré series. Which part of the inverse transseries is interpolation-independent? Proposition 4.59 answers this for one pair of choices — they agree strictly below action 1 — and a general statement is open.

**Remark 4.99** (Formalisation targets). Rademacher’s theorem is now machine-checked (Eberl and Paulson, Archive of Formal Proofs, 2026). The natural next steps, in dependency order, are: (a) the normalisation theorem 4.14; (b) the generating function eq. (4.31) and the recurrence eq. (4.41); (c) the Catalan/residue proof of theorem 4.31; (d) the implicit construction of  $U$  and eq. (4.72); (e) the triangular multivariate inversion eq. (4.93); (f) residual-to-root transfer, proposition 4.80, under an explicit derivative lower bound. Steps (b), (d), (e) are Bell-polynomial infrastructure shared with Part 1.

## 4.16 Summary

1. In the shifted phase  $\mu = \pi\sqrt{2(n - 1/24)}/3$ , Rademacher’s series is the exact sector sum eq. (4.17), each sector having a two-term amplitude linear in  $\mu^{-1}$ . At  $(k, \sigma) = (1, +1)$  it is the exponential-power model definition 1.2 with  $\alpha = 1/2$ , reduced by lemma 1.4 to  $\kappa e^\mu \mu^{-2} (1 - \mu^{-1})$  (lemma 4.1 and table 4.1).
2. Re-expansion in  $n^{-1/2}$  is governed by one analytic function eq. (4.31); its coefficients have the finite closed form eq. (4.33) in every sector, which enters only through  $\beta = \sigma\pi/(6k)$ . Each such series converges while the total is only asymptotic — confusing the two is the characteristic error of the subject.
3. The actions  $1 \mp 1/k$  accumulate at 1 from both sides, so the completed object is not a Hahn series and the two signs may not be separated (definition 4.19 and remark 4.20); theorem 4.23 bounds the tail explicitly and theorem 4.28 packages the accumulation exactly.
4. The inverse carrier is  $-2W_{-1}(-\frac{1}{2}\sqrt{\kappa/y})$  with Stirling flattening polynomials eq. (4.58); the amplitude is restored by theorem 1.25 at  $\Psi(t, u) = 3\log(1 + tu) - \log(1 - t + tu)$ , whose coefficients are table 4.2.
5. Since  $dn/d\mu \asymp \mu$ , that reversion produces an  $\mathcal{O}(1)$  index shift  $3/\pi^2$  which must be resummed before the periodic amplitudes are evaluated (remark 4.55).

6. With the phase frozen — by a residue class mod  $\text{lcm}(1, \dots, K)$  or by the entire interpolation definition 4.9, which agree strictly below action 1 (proposition 4.59) — every inverse transmonomial is given by eq. (4.86), eq. (4.88), eq. (4.95) or eq. (4.93). The leading correction is the parity chirp eq. (4.106) with constant  $3^{1/4}/(2\pi)$ , and the primary sectors form the staircase eq. (4.105).
7. Along the range the core alone recovers  $n$  by rounding (theorem 4.83); for general  $y$  the staircase is the ceiling of the smooth inverse (theorem 4.44), whose order-one sawtooth dominates every arithmetic sector (remark 4.45).

*Remark 4.100* (Works referred to). Analytic input: Rademacher, Proc. Nat. Acad. Sci. USA **23** (1937), 78–84; Proc. London Math. Soc. (2) **43** (1938), 241–254; Ann. of Math. **44** (1943), 416–422; textbook treatment in Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer GTM 41, 1990; machine-checked in Eberl and Paulson, *Rademacher’s series for the partition function*, Archive of Formal Proofs, 2026. Leading asymptotic: Hardy and Ramanujan, Proc. London Math. Soc. (2) **17** (1918), 75–115. Remainder estimates: Lehmer, J. London Math. Soc. **12** (1937), 171–176 and Trans. Amer. Math. Soc. **43** (1938), 271–295, **46** (1939), 362–373; Banerjee, Paule, Radu and Schneider, arXiv:2209.07887. The dominant-sector case of eq. (4.33) is O’Sullivan’s, arXiv:2205.13468 and arXiv:2203.02868. Lambert  $W$ : Corless, Gonnet, Hare, Jeffrey and Knuth, Adv. Comput. Math. **5** (1996), 329–359; Jeffrey, Corless, Hare and Knuth, arXiv:math/9512230. The sequence is OEIS A000041.

## Chapter 5

# The swing factorial (A056040)

This chapter merges three independent treatments of the swinging factorial: *All-Orders Asymptotic Transseries for the Swing Factorial and Its Two Inverse Branches* (cited below as S1), *The Full Asymptotic Transseries of the Swing Factorial and Its Inverse* (S2), and *Parity-Resolved Transseries for the Swing Factorial and Its Inverse* (S3). The general apparatus of S2 — its universal coefficient calculus for exponential–power inversion and its Lambert-normalized reversion theorem — was promoted to Part 1; here it is only *cited* and *specialized*. S1 works in the half index  $m$  with the carrier constant  $\log 4$ ; S2 and S3 work in the original index with  $\log 2$  and differ only in the sign convention for the parity marker. The chapter adopts the S2/Part 1 normalization throughout and gives the translation dictionary in remark 5.12. S1’s genuinely distinct contribution — the centred (translation-symmetric) normal form — is kept, but recast so that it too is a single  $\sigma$ -uniform statement.

Every rational, polynomial and algebraic coefficient displayed in this chapter was recomputed from the stated recurrences in exact arithmetic (sympy, rational Bernoulli input, truncated power series over  $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$ ); every decimal in the validation tables was recomputed at 50 significant digits. Where a recomputation reproduced a source table, that is said; where it did not, the discrepancy is recorded in a repairbox.

## 5.1 The sequence, the parity obstruction, and what “inverse” can mean

### 5.1.1 Definition and exact parity identities

**Definition 5.1** (Swing factorial). For  $n \in \mathbb{N}_0$  put

$$\text{sw}(n) = \frac{n!}{\lfloor n/2 \rfloor!^2}. \quad (5.1)$$

This is OEIS A056040, the *swinging factorial*.

Splitting on the parity of  $n$  gives at once

$$\text{sw}(2m) = \binom{2m}{m}, \quad \text{sw}(2m+1) = (2m+1) \binom{2m}{m}, \quad (5.2)$$

whence the one-step recurrence

$$\text{sw}(0) = 1, \quad \text{sw}(n) = \begin{cases} n \text{ sw}(n-1), & n \text{ odd,} \\ \frac{4}{n} \text{ sw}(n-1), & n \text{ even,} \end{cases} \quad (5.3)$$

and the closed product form

$$\text{sw}(n) = 2^{n-(n \bmod 2)} \prod_{k=1}^n k^{(-1)^{k+1}}. \quad (5.4)$$

The first values are

$$1, 1, 2, 6, 6, 30, 20, 140, 70, 630, 252, 2772, 924, 12012, 3432, 51480, \dots$$

**Proposition 5.2** (Generating functions).

$$\sum_{n \geq 0} \text{sw}(n) \frac{z^n}{n!} = (1+z) I_0(2z), \quad (5.5)$$

$$\sum_{n \geq 0} \text{sw}(n) z^n = \frac{1}{\sqrt{1-4z^2}} + \frac{z}{(1-4z^2)^{3/2}}, \quad (5.6)$$

where  $I_0$  is the modified Bessel function of order zero.

*Proof.* By eq. (5.2) the even exponential part is  $\sum_{m \geq 0} z^{2m}/(m!)^2 = I_0(2z)$  and the odd exponential part is  $z I_0(2z)$ , which is eq. (5.5). For the ordinary series put  $F(v) = \sum_{m \geq 0} \binom{2m}{m} v^m = (1-4v)^{-1/2}$ ; then the even part is  $F(z^2)$  and the odd part is  $z(2vF'(v) + F(v))|_{v=z^2} = z(1-4z^2)^{-3/2}$ .  $\square$

*Remark 5.3* (The generating function already shows the obstruction). The two singularities  $z = \pm \frac{1}{2}$  of eq. (5.6) have equal modulus. Co-dominant singularities of equal modulus at  $\pm \rho$  produce a period-two oscillation in the coefficient asymptotics, and no single non-oscillatory singular expansion can represent the whole sequence. Even/odd projection separates the two contributions exactly. This is the generating-function shadow of everything that follows.

### 5.1.2 The sequence is not eventually monotone

**Proposition 5.4** (Exact adjacent ratios). For  $m \geq 0$ ,

$$\frac{\text{sw}(2m+1)}{\text{sw}(2m)} = 2m+1, \quad \frac{\text{sw}(2m+2)}{\text{sw}(2m+1)} = \frac{2}{m+1}. \quad (5.7)$$

Consequently  $\text{sw}$  rises strictly from every even index to the next odd index, and falls strictly from every odd index to the next even index as soon as  $m \geq 2$ . In particular  $\text{sw}$  is not eventually monotone and has no eventually defined single-valued inverse.

*Proof.* Both ratios are eq. (5.3). The second is  $< 1$  exactly when  $m+1 > 2$ .  $\square$

*Warning 5.5* (There is no “the” inverse swing factorial). Any formula presented as *the* inverse of  $\text{sw}$  without naming a parity sheet or a generalized-inverse convention is incomplete: of the three inverse objects of definition 1.37, none exists globally for  $\text{sw}$  and all three exist branchwise (section 5.10).

### 5.1.3 The two analytic branches

All three sources resolve the obstruction the same way: interpolate the two bisections separately by gamma quotients. We use the parity marker  $\sigma \in \{-1, +1\}$  of S2, with

$$\sigma = -1 \text{ (even phase)}, \quad \sigma = +1 \text{ (odd phase)}, \quad \sigma_n := (-1)^{n+1}, \quad \lambda := \log 2. \quad (5.8)$$

The coefficient ring of the whole chapter is the *parity algebra*

$$\mathcal{R} := \mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1), \quad (5.9)$$

in which every formula below is a single element carrying both sheets. This is not a notational economy only:  $\sigma_n$  is a discrete transmonomial of modulus one, and it must remain visible at every order of every expansion.



**Definition 5.6** (Parity-resolved gamma branches). For  $x > 0$  and  $\sigma \in \{-1, +1\}$  set

$$\mathcal{S}_\sigma(x) := \frac{\Gamma(x+1)}{\Gamma(x/2 + (3-\sigma)/4)^2}. \quad (5.10)$$

Explicitly  $\mathcal{S}_-(x) = \Gamma(x+1)/\Gamma(x/2+1)^2$  and  $\mathcal{S}_+(x) = \Gamma(x+1)/\Gamma((x+1)/2)^2$ , and

$$\mathcal{S}_-(2m) = \text{sw}(2m), \quad \mathcal{S}_+(2m+1) = \text{sw}(2m+1) \quad (m \in \mathbb{N}_0). \quad (5.11)$$

**Proposition 5.7** (Duplication normal form and duality). For  $x > 0$ ,

$$\mathcal{S}_\sigma(x) = \frac{2^x}{\sqrt{\pi}} \left( \frac{\Gamma(x/2+1)}{\Gamma(x/2+1/2)} \right)^\sigma, \quad (5.12)$$

and consequently

$$\mathcal{S}_+(x) \mathcal{S}_-(x) = \frac{4^x}{\pi}, \quad \frac{\mathcal{S}_\sigma(x+2)}{\mathcal{S}_\sigma(x)} = \frac{16(x+1)(x+2)}{(2x+3-\sigma)^2} = \begin{cases} \frac{4(x+1)}{x+2}, & \sigma = -1, \\ \frac{4(x+2)}{x+1}, & \sigma = +1. \end{cases} \quad (5.13)$$

*Proof.* Legendre duplication in the form  $\Gamma(x+1) = 2^x \pi^{-1/2} \Gamma((x+1)/2) \Gamma(x/2+1)$  turns eq. (5.10) into eq. (5.12); multiplying the two values of  $\sigma$  makes the gamma ratio cancel and leaves  $4^x/\pi$ . For the two-step ratio,  $\Gamma(x+3)/\Gamma(x+1) = (x+1)(x+2)$  while the denominator contributes  $(x/2 + (3-\sigma)/4)^{-2} = 16(2x+3-\sigma)^{-2}$ .  $\square$

**Proposition 5.8** (Strict monotonicity and existence of the real inverses). For  $x > 0$ ,

$$\frac{d}{dx} \log \mathcal{S}_\sigma(x) = \lambda + \frac{\sigma}{2} \left[ \psi\left(\frac{x}{2} + 1\right) - \psi\left(\frac{x}{2} + \frac{1}{2}\right) \right], \quad (5.14)$$

where  $\psi = \Gamma'/\Gamma$ . The bracket decreases strictly from  $2\lambda$  at  $x = 0$  to 0 at  $x = \infty$ . Hence  $\mathcal{S}_+$  has strictly positive derivative on  $[0, \infty)$ ,  $\mathcal{S}_-$  has derivative 0 at  $x = 0$  and strictly positive derivative on  $(0, \infty)$ , and both are strictly increasing bijections of  $[0, \infty)$  onto  $[1, \infty)$ . Write

$$\mathcal{I}_\sigma := \mathcal{S}_\sigma^{-1} : [1, \infty) \rightarrow [0, \infty). \quad (5.15)$$

*Proof.* Differentiating eq. (5.12) logarithmically gives eq. (5.14). Put  $g(x) = \psi(x/2+1) - \psi(x/2+1/2)$ . Then  $g'(x) = \frac{1}{2} [\psi'(x/2+1) - \psi'(x/2+1/2)] < 0$  because the trigamma function  $\psi'(t) = \sum_{k \geq 0} (t+k)^{-2}$  is strictly decreasing on  $(0, \infty)$ ; and  $g(x) \rightarrow 0$  as  $x \rightarrow \infty$  by  $\psi(t+1/2) - \psi(t) \rightarrow 0$ . Finally  $g(0) = \psi(1) - \psi(1/2) = 2 \log 2 = 2\lambda$ . Thus  $0 < g(x) < 2\lambda$  for  $x > 0$ . For  $\sigma = +1$ , eq. (5.14) equals  $\lambda + g/2 > 0$  on  $[0, \infty)$ ; for  $\sigma = -1$  it equals  $\lambda - g/2$ , which vanishes only at  $x = 0$  and is positive afterwards. Both branches take the value 1 at  $x = 0$  and tend to  $+\infty$ , so each is a continuous increasing bijection onto  $[1, \infty)$ .  $\square$

### 5.1.4 Why one cannot simply interpolate through both parities

**Proposition 5.9** (Every smooth interpolation of the whole sequence oscillates). With  $R(x) = \Gamma(x/2+1)/\Gamma((x+1)/2)$ , put

$$\widehat{\mathcal{S}}(x) = \frac{2^x}{\sqrt{\pi}} R(x)^{-\cos \pi x}. \quad (5.16)$$

Then  $\widehat{\mathcal{S}}(n) = \text{sw}(n)$  for every  $n \in \mathbb{N}_0$ , yet  $\widehat{\mathcal{S}}$  is not eventually monotone: its logarithmic derivative

$$\frac{d}{dx} \log \widehat{\mathcal{S}}(x) = \lambda + \pi \sin(\pi x) \log R(x) - \cos(\pi x) \frac{R'(x)}{R(x)} \quad (5.17)$$

has an oscillating term of amplitude  $\sim \frac{\pi}{2} \log x$  against remaining terms that are  $O(1)$ , so it changes sign infinitely often. The same holds for the convex interpolation  $\frac{1+\cos \pi x}{2} \mathcal{S}_- + \frac{1-\cos \pi x}{2} \mathcal{S}_+$ . Consequently the phase-resolved pair  $(\mathcal{S}_-, \mathcal{S}_+)$ , and not any single function, is the asymptotically stable continuation of  $\text{sw}$ ; the corresponding inverse object is the pair  $(\mathcal{I}_-, \mathcal{I}_+)$ .

*Proof.* At  $x = n$  one has  $-\cos \pi n = (-1)^{n+1} = \sigma_n$ , so  $\widehat{S}(n) = (2^n/\sqrt{\pi})R(n)^{\sigma_n} = \mathcal{S}_{\sigma_n}(n) = \text{sw}(n)$  by eq. (5.12) and eq. (5.11). Differentiating  $\log \widehat{S} = x\lambda - \frac{1}{2} \log \pi - \cos(\pi x) \log R(x)$  gives eq. (5.17); by corollary 5.15 below,  $\log R(x) = \frac{1}{2} \log(x/2) + O(x^{-1})$  and  $R'/R = O(x^{-1})$ , so the oscillating term has amplitude  $\frac{\pi}{2} \log x + O(1) \rightarrow \infty$  and dominates. The convex interpolation is treated the same way after writing it as  $\mathcal{S}_-(x)$  times a factor whose logarithmic derivative oscillates with amplitude  $\asymp \log x$ .  $\square$

## 5.2 Exact normal form, Binet reduction, and Borel structure

### 5.2.1 The single scalar correction

**Definition 5.10** (The gamma-ratio correction). For  $x > 0$  set

$$D(x) := \log \frac{\Gamma(x/2 + 1)}{\Gamma(x/2 + 1/2)} - \frac{1}{2} \log \frac{x}{2}. \quad (5.18)$$

**Theorem 5.11** (Exact  $\sigma$ -uniform normal form). For  $x > 0$  and  $\sigma \in \{-1, +1\}$ ,

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma e^{\lambda x} x^{\sigma/2} e^{\sigma D(x)}, \quad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}.} \quad (5.19)$$

At integer arguments this is the exact parity formula

$$\log \text{sw}(n) = n\lambda - \frac{1}{2} \log \pi + \sigma_n \left( \frac{1}{2} \log \frac{n}{2} + D(n) \right), \quad n \geq 1. \quad (5.20)$$

*Proof.* By eq. (5.18),  $\Gamma(x/2 + 1)/\Gamma(x/2 + 1/2) = (x/2)^{1/2} e^{D(x)}$ . Substituting into eq. (5.12) gives  $\mathcal{S}_\sigma = (2^x/\sqrt{\pi})(x/2)^{\sigma/2} e^{\sigma D}$ , and  $(x/2)^{\sigma/2} = 2^{-\sigma/2} x^{\sigma/2}$ . For eq. (5.20), take logarithms of eq. (5.19) at  $x = n$ ,  $\sigma = \sigma_n$ , and use  $\frac{1}{2} \log n - \frac{\lambda}{2} = \frac{1}{2} \log \frac{n}{2}$ .  $\square$

Equation (5.19) is exactly definition 1.2 with

$$\boxed{\alpha = 1, \quad a = \lambda = \log 2, \quad b = \frac{\sigma}{2}, \quad C = C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}, \quad q_j = \sigma d_j,} \quad (5.21)$$

where  $d_j$  are the coefficients of the asymptotic series of  $D$  (theorem 5.18). Because  $\alpha = 1$  no monomial substitution is needed and lemma 1.4 is not invoked. We record the dictionary once and use it in every specialization below.

**Remark 5.12** (Dictionary between the three sources). Let  $\varepsilon_{S3} \in \{+1, -1\}$  be the marker of S3 and  $\varepsilon_{S1} \in \{0, 1\}$  that of S1. Then

$$\sigma = -\varepsilon_{S3}, \quad \sigma = 2\varepsilon_{S1} - 1, \quad S3 : Q = -D, \quad q_r^{S3} = -d_{2r-1}, \quad S1 : \alpha_{S1} = \log 4 = 2\lambda.$$

S1 parametrizes each sheet by the half index  $m$  with  $S_\epsilon^{S1}(m) = \text{sw}(2m + \epsilon)$ ; as functions this is the reparametrization  $x = 2m + \epsilon$  of definition 5.6, so

$$S_\epsilon^{S1}(m) = \mathcal{S}_{2\epsilon-1}(2m + \epsilon), \quad \mathcal{I}_\sigma(y) = 2M_\epsilon^{S1}(y) + \epsilon.$$

The reparametrization is invisible in the forward coefficients up to the substitution  $x \mapsto 2m + \epsilon$ , but it is *not* invisible in the inverse coefficients: the two normalizations invert different leading phases ( $\lambda x + \frac{\sigma}{2} \log x$  against  $2\lambda m + \frac{\sigma}{2} \log m$ ) and therefore have different Lambert cores. Only for the even sheet do the two cores agree exactly, through  $X_-(y) = 2w_0^{S1}(y)$ ; see remark 5.37.

### 5.2.2 The Binet kernel collapses to a hyperbolic tangent

**Theorem 5.13** (Exact Borel–Laplace representation). *For  $\Re x > 0$ ,*

$$D(x) = \int_0^\infty e^{-xt} \widehat{D}(t) dt, \quad \widehat{D}(t) = \frac{\tanh(t/2)}{2t}, \quad (5.22)$$

*the singularity of  $\widehat{D}$  at  $t = 0$  being removable with  $\widehat{D}(0) = \frac{1}{4}$ .*

*Proof.* Put  $z = x/2$  and  $h(z) = \log \Gamma(z+1) - \log \Gamma(z + \frac{1}{2}) - \frac{1}{2} \log z$ , so  $h(x/2) = D(x)$ . From the standard integral for a digamma difference,

$$\psi(z+1) - \psi(z + \frac{1}{2}) = \int_0^\infty \frac{e^{-(z+1/2)s} - e^{-(z+1)s}}{1 - e^{-s}} ds = 2 \int_0^\infty \frac{e^{-2zv}}{e^v + 1} dv,$$

the second equality by  $s = 2v$  and  $(e^{-v} - e^{-2v})/(1 - e^{-2v}) = 1/(e^v + 1)$ . Since  $1/(2z) = \int_0^\infty e^{-2zv} dv$ ,

$$h'(z) = \psi(z+1) - \psi(z + \frac{1}{2}) - \frac{1}{2z} = \int_0^\infty e^{-2zv} \left( \frac{2}{e^v + 1} - 1 \right) dv = - \int_0^\infty e^{-2zv} \tanh \frac{v}{2} dv.$$

Both  $h(z)$  and the right-hand side of eq. (5.22) tend to 0 as  $\Re z \rightarrow +\infty$ , so integrating from  $+\infty$  down to  $z$ ,

$$h(z) = \int_z^\infty \int_0^\infty e^{-2\zeta v} \tanh \frac{v}{2} dv d\zeta = \int_0^\infty e^{-2zv} \frac{\tanh(v/2)}{2v} dv,$$

which is eq. (5.22) after  $x = 2z$ . Fubini applies because the double integral is absolutely convergent for  $\Re z > 0$ .  $\square$

S3 reaches eq. (5.22) from Binet's first formula, through the elegant identity  $b(t) - 2b(2t) = -\frac{1}{2} \tanh(t/2)$  for  $b(t) = (e^t - 1)^{-1} - t^{-1} + \frac{1}{2}$ ; S2 reaches it from the digamma integral, as above. S1 keeps Binet's *second* formula (the arctan kernel) instead and never simplifies the difference of the two Binet integrals. The tanh collapse is the decisive simplification: it is what makes the Borel plane elementary, and it is adopted here.

**Corollary 5.14** (Sign, monotonicity, and elementary bounds). *For  $x > 0$ ,*

$$0 < D(x) < \frac{1}{4x}, \quad 0 < -D'(x) < \frac{1}{4x^2}, \quad (-1)^k D^{(k)}(x) > 0 \quad (k \geq 0). \quad (5.23)$$

*In particular  $D$  is completely monotone and strictly decreasing.*

*Proof.*  $\widehat{D} > 0$  on  $(0, \infty)$ , so  $(-1)^k D^{(k)}(x) = \int_0^\infty e^{-xt} t^k \widehat{D}(t) dt > 0$ . From  $0 < \tanh(t/2) < t/2$  we get  $0 < \widehat{D}(t) < \frac{1}{4}$ , and integrating  $e^{-xt} t^k$  gives the two displayed bounds.  $\square$

**Corollary 5.15** (Asymptotics of the gamma ratio). *With  $R(x) = \Gamma(x/2 + 1)/\Gamma((x+1)/2)$  as in eq. (5.16),*

$$\log R(x) = \frac{1}{2} \log(x/2) + D(x) = \frac{1}{2} \log(x/2) + O(x^{-1}), \quad \frac{R'(x)}{R(x)} = \frac{1}{2x} + D'(x) = O(x^{-1}).$$

*Proof.* Immediate from eq. (5.18) and eq. (5.23).  $\square$

### 5.2.3 The Borel singularity lattice

**Proposition 5.16** (Mittag–Leffler expansion and poles). *The Borel transform of eq. (5.22) satisfies*

$$\widehat{D}(t) = 2 \sum_{k \geq 0} \frac{1}{t^2 + \pi^2(2k+1)^2}, \quad (5.24)$$

*is even and meromorphic on  $\mathbb{C}$ , and has exactly the simple poles*

$$t = \pm i\pi(2k+1), \quad k \geq 0, \quad \text{Res}_{t=t_k} \widehat{D}(t) = \frac{1}{t_k}. \quad (5.25)$$

*The positive real ray is free of singularities; the nearest singularities lie at distance  $\pi$ , so the formal series of  $D$  is Gevrey-1 with singulant modulus  $\pi$ .*

*Proof.*  $\tanh(w) = \sum_{k \geq 0} 2w / (w^2 + \pi^2(k + \frac{1}{2})^2)$  is the classical Mittag–Leffler expansion; putting  $w = t/2$  and dividing by  $2t$  gives eq. (5.24), both sides being even, meromorphic, with the same poles and residues and the same value  $\frac{1}{4}$  at  $t = 0$  (using  $\sum_{k \geq 0} (2k+1)^{-2} = \pi^2/8$ ). At  $t = t_k$  the function  $\tanh(t/2)$  has residue 2, so  $\widehat{D}$  has residue  $2/(2t_k) = 1/t_k$ .  $\square$

*Remark 5.17* (What the exponentially small scale is, and is not). The quantity  $e^{-\pi x}$  that governs optimal truncation (section 5.9) is the envelope set by the distance  $\pi$  of the nearest *off-axis* Borel poles; it is not an ambiguity of the positive-axis sum, since the ray  $\arg t = 0$  meets no singularity and eq. (5.22) is exact. Genuine Stokes jumps appear only when the Laplace direction is rotated onto the imaginary rays. S1, retaining an unsimplified Binet integral, makes the weaker but correct statement that no Stokes constant need be assigned on the real axis; the Borel picture explains why.

## 5.3 The logarithmic coefficients and a sharp enveloping remainder

### 5.3.1 All coefficients

We use the Bernoulli convention  $z/(e^z - 1) = \sum_{n \geq 0} B_n z^n / n!$ , so  $B_1 = -\frac{1}{2}$ , and the Bernoulli polynomials  $te^{wt}/(e^t - 1) = \sum_n B_n(w) t^n / n!$ .

**Theorem 5.18** (All logarithmic coefficients). *As  $x \rightarrow \infty$  in any closed subsector of  $|\arg x| < \pi/2$ ,*

$$D(x) \sim \sum_{j \geq 1} \frac{d_j}{x^j}, \quad d_{2r} = 0, \quad d_{2r+1} = \frac{(2^{2r+2} - 1)B_{2r+2}}{(2r+1)(2r+2)}. \quad (5.26)$$

*Equivalently*

$$d_{2r+1} = (-1)^r \frac{2(2r)!}{\pi^{2r+2}} (1 - 2^{-2r-2}) \zeta(2r+2) = (-1)^r \frac{2(2r)!}{\pi^{2r+2}} \sum_{k \geq 0} \frac{1}{(2k+1)^{2r+2}}, \quad (5.27)$$

*and the formal Borel transform of the series is exactly the Taylor expansion of  $\widehat{D}$ :*

$$\widehat{D}(t) = \sum_{r \geq 0} d_{2r+1} \frac{t^{2r}}{(2r)!} = \frac{\tanh(t/2)}{2t}. \quad (5.28)$$

*Proof.* Insert into eq. (5.24) the finite geometric identity

$$\frac{1}{t^2 + a^2} = \sum_{r=0}^{N-1} \frac{(-1)^r t^{2r}}{a^{2r+2}} + \frac{(-1)^N t^{2N}}{a^{2N}(t^2 + a^2)}, \quad a = a_k = \pi(2k+1),$$

and integrate the finite sum against  $e^{-xt}$  using  $\int_0^\infty e^{-xt} t^{2r} dt = (2r)! x^{-2r-1}$ . This gives the second expression in eq. (5.27), hence the first by  $\zeta(2r+2)(1-2^{-2r-2}) = \sum_{k \geq 0} (2k+1)^{-(2r+2)}$ , and hence eq. (5.26) by Euler's evaluation  $B_{2m} = (-1)^{m-1} 2(2m)! \zeta(2m) / (2\pi)^{2m}$ . The unexpanded last term is the remainder controlled in theorem 5.20, which in particular proves the asymptotic statement on  $\Re x > 0$ ; the identity eq. (5.28) is the Taylor expansion of  $\widehat{D}$  at the origin.  $\square$

*Remark 5.19* (Sector of validity: a disagreement between the sources). S2 and S3 state eq. (5.26) on closed subsectors of  $|\arg x| < \pi/2$ , which is what the Laplace representation eq. (5.22) gives. S1 states the corresponding expansion on  $|\arg m| \leq \pi - \delta$ . Both are correct: the wider sector follows not from the Laplace integral but from the Tricomi–Erdélyi expansion of a gamma ratio,

$$\log \frac{\Gamma(z+a)}{\Gamma(z+b)} \sim (a-b) \log z + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n(n+1)z^n} (B_{n+1}(a) - B_{n+1}(b)), \quad |\arg z| \leq \pi - \delta, \quad (5.29)$$

applied with  $a = 1, b = \frac{1}{2}, z = x/2$ . The volume states the Laplace sector where an *exact* representation and a signed remainder are wanted, and the wider sector where only a Poincaré statement is wanted. The two routes give the same coefficients: with  $B_m(\frac{1}{2}) = (2^{1-m} - 1)B_m$  and  $B_m(1) = B_m$  for  $m \neq 1$ , eq. (5.29) yields  $d_n = 2^n \frac{(-1)^{n+1}}{n(n+1)} (2 - 2^{-n}) B_{n+1}$ , which is eq. (5.26).

Table 5.1: The logarithmic coefficients  $d_{2r+1}$  of theorem 5.18, recomputed from eq. (5.26) in exact rational arithmetic and cross-checked against the Taylor expansion of  $\tanh(t/2)/(2t)$ .

| $j$   | 1             | 3               | 5              | 7                 | 9               | 11                | 13                |
|-------|---------------|-----------------|----------------|-------------------|-----------------|-------------------|-------------------|
| $d_j$ | $\frac{1}{4}$ | $-\frac{1}{24}$ | $\frac{1}{20}$ | $-\frac{17}{112}$ | $\frac{31}{36}$ | $-\frac{691}{88}$ | $\frac{5461}{52}$ |

Thus

$$D(x) \sim \frac{1}{4x} - \frac{1}{24x^3} + \frac{1}{20x^5} - \frac{17}{112x^7} + \frac{31}{36x^9} - \frac{691}{88x^{11}} + \frac{5461}{52x^{13}} - \dots \quad (5.30)$$

### 5.3.2 An enveloping remainder, not merely a Poincaré one

**Theorem 5.20** (Exact remainder and strict next-term envelope). *For  $x > 0$  and  $N \geq 0$  write  $D(x) = \sum_{r=0}^{N-1} d_{2r+1} x^{-(2r+1)} + R_N(x)$  (empty sum for  $N = 0$ ). Then, with  $a_k = \pi(2k+1)$ ,*

$$R_N(x) = 2(-1)^N \sum_{k \geq 0} a_k^{-2N} \int_0^\infty \frac{e^{-xt} t^{2N}}{t^2 + a_k^2} dt, \quad (5.31)$$

and consequently

$$0 < (-1)^N R_N(x) < \frac{|d_{2N+1}|}{x^{2N+1}} \quad (x > 0, N \geq 0). \quad (5.32)$$

Every truncation therefore brackets  $D(x)$  together with its successor.

*Proof.* Substitute the finite geometric identity of the previous proof into eq. (5.24) and then into eq. (5.22); Tonelli's theorem justifies the interchange since all terms are positive. This gives eq. (5.31), in which every integral is positive. Replacing  $(t^2 + a_k^2)^{-1}$  by the strict upper bound  $a_k^{-2}$  and using  $\int_0^\infty e^{-xt} t^{2N} dt = (2N)! x^{-2N-1}$  gives

$$0 < (-1)^N R_N(x) < 2(2N)! x^{-(2N+1)} \sum_{k \geq 0} a_k^{-(2N+2)} = \frac{|d_{2N+1}|}{x^{2N+1}},$$

by the last expression in eq. (5.27).  $\square$

*Remark 5.21.* Theorem 5.20 is strictly stronger than the Poincaré statement, and it is what makes every later analytic claim in this chapter effective: all remainder constants below are explicit rational multiples of the first omitted  $|d_{2N+1}|$ . S2 and S3 both prove it; S1 does not have it, because it never simplifies the Binet kernel. It was verified numerically at  $x = 5$  and  $x = 10$  for  $0 \leq N \leq 4$ : the bracketing eq. (5.32) holds with the two sides agreeing to within a factor 1.33 at  $x = 5$  and 1.09 at  $x = 10$ .

### 5.3.3 Large order and least term

**Proposition 5.22** (Least term and its envelope). *As  $r \rightarrow \infty$ ,  $d_{2r+1} \sim (-1)^r 2(2r)! \pi^{-2r-2}$ , so the ratio of consecutive nonzero term magnitudes in eq. (5.30) is  $\sim (2r+2)(2r+1)/(\pi^2 x^2)$ . The least term therefore occurs at  $r \approx \pi x/2$  and has magnitude*

$$\frac{2\sqrt{2}}{\pi\sqrt{x}} e^{-\pi x} (1 + O(x^{-1})). \quad (5.33)$$

*By theorem 5.20, truncation at the least term attains an error of exactly this scale, with no hidden constant.*

*Proof.* The asymptotic form of  $d_{2r+1}$  is eq. (5.27) with  $\zeta(2r+2) \rightarrow 1$ . The ratio statement is immediate. Setting  $2r = \pi x$  and applying Stirling's formula to  $(2r)!$  gives  $2(2r)! \pi^{-2r-2} x^{-2r-1} \sim 2\sqrt{2\pi^2 x} e^{-\pi x} / (\pi^2 x) = 2\sqrt{2} e^{-\pi x} / (\pi\sqrt{x})$ .  $\square$

*Remark 5.23* (Numerical confirmation). Recomputed: at  $x = 10$  the least nonzero term of eq. (5.30) occurs at  $r = 15$  (against  $\pi x/2 = 15.708$ ) with magnitude  $6.5448 \times 10^{-15}$ , while eq. (5.33) predicts  $6.4659 \times 10^{-15}$ ; at  $x = 20$  the least term occurs at  $r = 31$  (against 31.416) with magnitude  $1.03859 \times 10^{-28}$  against a predicted  $1.03837 \times 10^{-28}$ .

## 5.4 The forward transseries

### 5.4.1 Exponentiation

**Theorem 5.24** (All forward coefficients). *Define  $A_0 = 1$  and, for  $n \geq 1$ ,*

$$A_n = [t^n] \exp\left(\sum_{j \geq 1} d_j t^j\right) = \frac{1}{n!} B_n(1! d_1, 2! d_2, \dots, n! d_n) = \sum_{\substack{k_1, k_2, \dots \geq 0 \\ \sum_j j k_j = n}} \prod_{j \geq 1} \frac{d_j^{k_j}}{k_j!}, \quad (5.34)$$

*with  $B_n$  the complete exponential Bell polynomial of definition 1.10. Equivalently and most cheaply,*

$$A_0 = 1, \quad n A_n = \sum_{j=1}^n j d_j A_{n-j} = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} j d_j A_{n-j}. \quad (5.35)$$

*Then, for every fixed  $N \geq 1$  and uniformly on closed subsectors of  $\Re x > 0$ ,*

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma 2^x x^{\sigma/2} \left( \sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n} + O_N(x^{-N}) \right)}. \quad (5.36)$$

*Proof.* The three expressions in eq. (5.34) are the definition of  $B_n$ , its partition form, and the expansion of the exponential as a product; eq. (5.35) follows from  $A'(t) = A(t) \sum_j j d_j t^{j-1}$  for  $A(t) = \exp \sum_j d_j t^j$ , together with  $d_{2r} = 0$ . The parity of the  $\sigma$ -dependence is exact: since  $\sum_j d_j t^j$  is an odd formal series,

$$\exp\left(-\sum_j d_j t^j\right) = \exp\left(\sum_j d_j (-t)^j\right) = \sum_{n \geq 0} (-1)^n A_n t^n,$$

so  $\exp(\sigma D)$  has coefficients  $\sigma^n A_n$  in either sheet. For the remainder, proposition 5.25 below gives an explicit bound.  $\square$

**Proposition 5.25** (Effective forward remainder). *Fix  $N \geq 1$  and let  $x \geq x_N := \max\{1, 2N\}$ . Then*

$$\left| e^{\sigma D(x)} - \sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n} \right| \leq \frac{K_N}{x^N}, \quad K_N := e^{1/4} \left( |d_{2\lceil N/2 \rceil + 1}| + \sum_{n \geq N} \frac{|A_n|}{x_N^{n-N}} \right) < \infty, \quad (5.37)$$

where the tail sum converges because  $A_n = O(C^n n!^{1/2})$  for some  $C$ .

*Proof.* Write  $D_{<M}(x) = \sum_{r < M} d_{2r+1} x^{-(2r+1)}$  with  $M = \lceil N/2 \rceil$ , so that by theorem 5.20  $|D(x) - D_{<M}(x)| \leq |d_{2M+1}| x^{-(2M+1)} \leq |d_{2M+1}| x^{-N}$  for  $2M+1 \geq N$ . Since  $|D(x)| \leq \frac{1}{4}$  and  $|D_{<M}(x)| \leq \frac{1}{4}$  for  $x \geq 1$  by eq. (5.32) with  $N = 0$ , the mean value theorem gives  $|e^{\sigma D} - e^{\sigma D_{<M}}| \leq e^{1/4} |D - D_{<M}|$ . Finally  $e^{\sigma D_{<M}(x)}$  is an absolutely convergent power series in  $1/x$  for  $x \geq x_N$  whose coefficients agree with  $\sigma^n A_n$  for  $n < N$  up to terms already accounted for, and whose tail is bounded by  $\sum_{n \geq N} |A_n| x^{-n}$ . Collecting the two bounds gives eq. (5.37). The growth bound on  $A_n$  follows from  $d_{2r+1} = O((2r)! \pi^{-2r})$  and the Bell-polynomial partition form.  $\square$

**Source claim.** S1 obtains its multiplicative remainder by writing “ $e^{R_N(m)} = 1 + O(m^{-N})$ ”, so formal exponentiation also gives the claimed analytic remainder”. **Defect.** That sentence exponentiates a *formal* truncation error and silently assumes that the tail of the exponentiated series is of the same order; no bound on the coefficients  $A_n$  enters, although the series  $\sum A_n x^{-n}$  diverges. **Repair.** Proposition 5.25 supplies the missing ingredients (a bound on  $D - D_{<M}$  from the enveloping theorem, a uniform bound on the exponential’s Lipschitz constant, and an explicit tail bound), so the constant in eq. (5.36) is effective.

### 5.4.2 Coefficient table and the two sectors

Table 5.2: The universal forward coefficients  $A_n$  of eq. (5.34), recomputed by eq. (5.35) in exact rational arithmetic. The table reproduces the corresponding tables of all three sources (S3 tabulates  $C_m = (-1)^n A_n$ ; S1 tabulates  $A_n/2^n$  in its half index).

| $n$ | $A_n$         | $n$ | $A_n$                           |
|-----|---------------|-----|---------------------------------|
| 0   | 1             | 8   | -334477/8388608                 |
| 1   | 1/4           | 9   | 28717403/33554432               |
| 2   | 1/32          | 10  | 59697183/268435456              |
| 3   | -5/128        | 11  | -8400372435/1073741824          |
| 4   | -21/2048      | 12  | -34429291905/17179869184        |
| 5   | 399/8192      | 13  | 7199255611995/68719476736       |
| 6   | 869/65536     | 14  | 14631594576045/549755813888     |
| 7   | -39325/262144 | 15  | -4251206967062925/2199023255552 |

Explicitly, with  $C_- = \sqrt{2/\pi}$  and  $C_+ = 1/\sqrt{2\pi}$ ,

$$\mathcal{S}_-(x) \sim \sqrt{\frac{2}{\pi}} \frac{2^x}{\sqrt{x}} \left( 1 - \frac{1}{4x} + \frac{1}{32x^2} + \frac{5}{128x^3} - \frac{21}{2048x^4} - \frac{399}{8192x^5} + \frac{869}{65536x^6} + \cdots \right), \quad (5.38)$$

$$\mathcal{S}_+(x) \sim \frac{2^x \sqrt{x}}{\sqrt{2\pi}} \left( 1 + \frac{1}{4x} + \frac{1}{32x^2} - \frac{5}{128x^3} - \frac{21}{2048x^4} + \frac{399}{8192x^5} + \frac{869}{65536x^6} + \cdots \right), \quad (5.39)$$

and for the integer sequence itself,

$$\text{sw}(n) \sim \frac{2^n}{\sqrt{\pi}} 2^{-\sigma_n/2} n^{\sigma_n/2} \sum_{k \geq 0} \frac{\sigma_n^k A_k}{n^k}, \quad \sigma_n = (-1)^{n+1}. \quad (5.40)$$

**Remark 5.26** (Period-two, not oscillatory error). Equation (5.40) is a two-phase (period-two) transseries: the discrete transmonomial  $\sigma_n$  has modulus one, so it is never absorbed into a remainder, and it changes the algebraic scale from  $2^n n^{-1/2}$  to  $2^n n^{+1/2}$ . Equivalently, with the idempotents  $\chi_\sigma(n) = (1 + \sigma\sigma_n)/2$ ,

$$\text{sw}(n) \sim \sum_{\sigma=\pm 1} \chi_\sigma(n) C_\sigma 2^n n^{\sigma/2} \sum_{k \geq 0} \sigma^k A_k n^{-k}.$$

**Corollary 5.27** (Central binomial coefficient and the reciprocal). *Putting  $x = 2m$  in eq. (5.38),*

$$\binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}} \left( 1 - \frac{1}{8m} + \frac{1}{128m^2} + \frac{5}{1024m^3} - \frac{21}{32768m^4} - \frac{399}{262144m^5} + \cdots \right), \quad (5.41)$$

*that is, the half-index coefficients of S1 are  $A_n/2^n$  on the even sheet. Moreover*

$$\frac{1}{\mathcal{S}_\sigma(x)} \sim \frac{1}{C_\sigma} 2^{-x} x^{-\sigma/2} \sum_{n \geq 0} \frac{(-\sigma)^n A_n}{x^n}. \quad (5.42)$$

*Proof.* For eq. (5.41) substitute  $x = 2m$  and  $C_- 2^{2m} (2m)^{-1/2} = 4^m / \sqrt{\pi m}$ . For eq. (5.42) note that the reciprocal of eq. (5.19) carries  $e^{-\sigma D}$ , whose coefficients are  $(-\sigma)^n A_n$  by the oddness argument in the proof of theorem 5.24.  $\square$

## 5.5 The centred normal form

This section is S1's distinctive contribution: a translation of the index that makes the correction series *sparse in a second way* and equalizes the two leading constants. S1 states it as two parallel statements in its half index; it is recast here as a single  $\sigma$ -uniform identity in the original index, and the coefficients are recomputed in that normalization.

**Theorem 5.28** (Centred exact normal form). *Put*

$$u := \frac{x}{2} + \frac{1}{4}. \quad (5.43)$$

*Then, exactly and for both parities,*

$$\boxed{\mathcal{S}_\sigma(x) = \frac{4^u}{\sqrt{2\pi}} \left( \frac{\Gamma(u + 3/4)}{\Gamma(u + 1/4)} \right)^\sigma}, \quad (5.44)$$

*and, as  $u \rightarrow \infty$  in  $|\arg u| \leq \pi - \delta$ ,*

$$\mathcal{S}_\sigma(x) \sim \frac{4^u}{\sqrt{2\pi}} u^{\sigma/2} \exp\left(\sigma \sum_{q \geq 1} \frac{\nu_{2q}}{u^{2q}}\right), \quad \nu_{2q} = \frac{B_{2q+1}(1/4)}{q(2q+1)}. \quad (5.45)$$

*In particular every odd inverse power is absent from the centred correction, and the leading constant  $1/\sqrt{2\pi}$  is the same on both sheets.*

*Proof.* From  $u = x/2 + 1/4$  we get  $x/2 + 1 = u + 3/4$  and  $x/2 + 1/2 = u + 1/4$ , while  $2^x = 2^{2u-1/2} = 4^u / \sqrt{2}$ ; substituting in eq. (5.12) gives eq. (5.44). For eq. (5.45) apply eq. (5.29) with  $a = \frac{3}{4}$ ,  $b = \frac{1}{4}$ ,  $z = u$ : the logarithmic term is  $(a - b) \log u = \frac{1}{2} \log u$ , and the  $n$ -th coefficient is  $\frac{(-1)^{n+1}}{n(n+1)} (B_{n+1}(\frac{3}{4}) - B_{n+1}(\frac{1}{4}))$ . Since  $a + b = 1$ , the reflection formula  $B_m(1 - w) = (-1)^m B_m(w)$  gives  $B_{n+1}(\frac{1}{4}) = (-1)^{n+1} B_{n+1}(\frac{3}{4})$ , so the difference vanishes for odd  $n$  and equals  $2B_{n+1}(\frac{3}{4}) = -2B_{n+1}(\frac{1}{4})$  for even  $n = 2q$ . This yields  $\nu_{2q} = B_{2q+1}(1/4)/(q(2q+1))$ .  $\square$



Table 5.3: Centred logarithmic coefficients  $\nu_{2q}$  of eq. (5.45), recomputed from Bernoulli polynomials at  $1/4$ . They agree with S1's  $\widehat{\lambda}_{2q}^{(0)} = -\nu_{2q}$  after the marker translation of remark 5.12.

| $2q$       | 2              | 4                 | 6                  | 8                       | 10                       |
|------------|----------------|-------------------|--------------------|-------------------------|--------------------------|
| $\nu_{2q}$ | $\frac{1}{64}$ | $-\frac{5}{2048}$ | $\frac{61}{49152}$ | $-\frac{1385}{1048576}$ | $\frac{50521}{20971520}$ |

**Remark 5.29** (Two different sparsities). The uncentred correction  $\sigma D(x)$  contains only *odd* powers of  $1/x$ ; the centred correction contains only *even* powers of  $1/u$ . There is no contradiction: the two statements concern different remainder functions, because the  $\frac{\sigma}{2} \log$  term is attached to  $x$  in one and to  $u$  in the other. Both are instances of one rule, recorded in lemma 5.30 below, and the swing shifts are precisely the optimally centred ones.

**Lemma 5.30** (Optimal centring of a gamma ratio). *Let  $a, b \in \mathbb{R}$  with  $\beta := a - b \neq 0$ , and set  $s = (a + b - 1)/2$  and  $z = w + s$ . Then  $\Gamma(w + a)/\Gamma(w + b) = \Gamma(z + \frac{1+\beta}{2})/\Gamma(z + \frac{1-\beta}{2})$ , the two shifts sum to 1, and*

$$\log \frac{\Gamma(z + \frac{1+\beta}{2})}{\Gamma(z + \frac{1-\beta}{2})} \sim \beta \log z - \sum_{q \geq 1} \frac{B_{2q+1}(\frac{1-\beta}{2})}{q(2q+1)} z^{-2q}, \quad |\arg z| \leq \pi - \delta, \quad (5.46)$$

with no odd inverse powers.

*Proof.* The shift identity is immediate. Apply eq. (5.29) and the argument of theorem 5.28: complementary shifts make the Bernoulli difference vanish in odd degree by  $B_m(1 - w) = (-1)^m B_m(w)$ .  $\square$

For the swing factorial  $a = 1, b = \frac{1}{2}, w = x/2$ , so  $s = \frac{1}{4}$  and  $z = u$ , and  $\beta = \frac{1}{2}$ : eq. (5.45) is exactly eq. (5.46) raised to the power  $\sigma$ . The first centred expansions are

$$\mathcal{S}_-(x) \sim \frac{4^u}{\sqrt{2\pi u}} \left( 1 - \frac{1}{64u^2} + \frac{21}{8192u^4} - \frac{671}{524288u^6} + \cdots \right), \quad (5.47)$$

$$\mathcal{S}_+(x) \sim \frac{4^u \sqrt{u}}{\sqrt{2\pi}} \left( 1 + \frac{1}{64u^2} - \frac{19}{8192u^4} + \frac{631}{524288u^6} + \cdots \right), \quad (5.48)$$

whose coefficients were recomputed from  $\exp(\sigma \sum_q \nu_{2q} z^q)$  and agree with S1.

## 5.6 The two Lambert cores, and why the branch matters

### 5.6.1 Specialization of the core theorem

Write

$$L_\sigma(y) := \log \frac{y}{C_\sigma} = \log y + \frac{1}{2} \log \pi + \frac{\sigma}{2} \lambda. \quad (5.49)$$

Dropping the correction in eq. (5.19) leaves the *core equation*

$$L_\sigma(y) = \lambda X + \frac{\sigma}{2} \log X, \quad \text{equivalently} \quad C_\sigma 2^X X^{\sigma/2} = y. \quad (5.50)$$

**Theorem 5.31** (Explicit Lambert cores). *Theorem 1.20 with the dictionary eq. (5.21) gives*

$$X_{\sigma,k}(y) = \frac{\sigma}{2\lambda} W_k(2\sigma \lambda e^{2\sigma L_\sigma(y)}), \quad (5.51)$$

and the unique large positive real solutions are

$$\boxed{X_+(y) = \frac{W_0(4\pi \lambda y^2)}{2\lambda}, \quad X_-(y) = -\frac{W_{-1}(-4\lambda/(\pi y^2))}{2\lambda}.} \quad (5.52)$$

$X_+$  is defined for all  $y > 0$ ;  $X_-$  is defined exactly for

$$y \geq y_* := 2\sqrt{\frac{e\lambda}{\pi}} = 1.548870223880\dots, \quad (5.53)$$

and satisfies  $X_-(y) \geq 1/(2\lambda) = 0.72134752\dots$  there.

*Proof.* With  $a = \lambda$ ,  $b = \sigma/2$ , theorem 1.20 reads  $X = (b/a)W_k((a/b)e^{L/b})$ , and  $1/\sigma = \sigma$  gives eq. (5.51). For  $\sigma = +1$ ,  $C_+ = 1/\sqrt{2\pi}$  so  $e^{2L_+} = 2\pi y^2$  and the argument is  $4\pi\lambda y^2 > 0$ , on which  $W_0$  is the only real branch. For  $\sigma = -1$ ,  $C_- = \sqrt{2/\pi}$  so  $e^{-2L_-} = 2/(\pi y^2)$  and the argument is  $-4\lambda/(\pi y^2) < 0$ ; it lies in  $[-1/e, 0)$  exactly when  $y^2 \geq 4\lambda e/\pi$ , i.e.  $y \geq y_*$ . On that interval both  $W_0$  and  $W_{-1}$  are real;  $W_{-1} \leq -1$  gives  $X_- \geq 1/(2\lambda)$ , and  $W_{-1}(z) \rightarrow -\infty$  as  $z \rightarrow 0^-$  gives  $X_-(y) \rightarrow \infty$ . Proposition 5.32 identifies the two roots and shows that  $W_0$  gives the wrong one.  $\square$

### 5.6.2 Why $W_0$ is the wrong root on the even sheet

**Proposition 5.32** (The even carrier is not injective). *Let  $g_-(X) = C_- 2^X X^{-1/2}$  be the even carrier of eq. (5.50). Then  $g_-$  is strictly decreasing on  $(0, 1/(2\lambda))$ , strictly increasing on  $(1/(2\lambda), \infty)$ , and*

$$\min_{X>0} g_-(X) = g_-\left(\frac{1}{2\lambda}\right) = 2\sqrt{\frac{e\lambda}{\pi}} = y_*. \quad (5.54)$$

Hence for  $y > y_*$  the equation  $g_-(X) = y$  has exactly two positive roots  $X^{\text{small}}(y) < 1/(2\lambda) < X^{\text{large}}(y)$ ; they coalesce at  $y = y_*$ . In terms of the Lambert branches,

$$X^{\text{large}}(y) = -\frac{W_{-1}(-4\lambda/(\pi y^2))}{2\lambda} = X_-(y) \xrightarrow{y \rightarrow \infty} \infty, \quad X^{\text{small}}(y) = -\frac{W_0(-4\lambda/(\pi y^2))}{2\lambda} = \frac{2}{\pi y^2}(1 + O(y^{-2})). \quad (5.55)$$

The principal branch therefore returns a root that tends to zero; it is not a large-index inverse at any order, and no amount of asymptotic correction can repair the choice. On the odd sheet the carrier  $g_+(X) = C_+ 2^X X^{1/2}$  is a strictly increasing bijection of  $(0, \infty)$  onto  $(0, \infty)$ , its Lambert argument is positive, and  $W_0$  is the only real branch.

*Proof.*  $(\log g_-)'(X) = \lambda - 1/(2X)$  vanishes only at  $X = 1/(2\lambda)$  and changes sign there from  $-$  to  $+$ ; the minimum value is  $C_- e^{\lambda/(2\lambda)} (2\lambda)^{1/2} = \sqrt{2/\pi} \cdot \sqrt{e} \cdot \sqrt{2\lambda} = 2\sqrt{e\lambda/\pi}$ , which is eq. (5.54), and equals  $y_*$  of eq. (5.53). Strict monotonicity on each side gives exactly two roots for  $y > y_*$ . Both are given by eq. (5.51) with  $\sigma = -1$  and a real branch of  $W$ ; since  $W_{-1} \leq -1 \leq W_0 \leq 0$  on  $[-1/e, 0)$ , the branch  $W_{-1}$  gives the root  $\geq 1/(2\lambda)$  and  $W_0$  the root  $\leq 1/(2\lambda)$ . Finally  $W_0(z) = z + O(z^2)$  as  $z \rightarrow 0$ , so  $X^{\text{small}} = -(-4\lambda/(\pi y^2) + O(y^{-4}))/2\lambda = 2/(\pi y^2) + O(y^{-4})$ .  $\square$

**Remark 5.33** (Numerical confirmation and the branch point). Recomputed at 50 digits: for  $y = 10^2, 10^3, 10^5$  the principal-branch root is  $6.366759642 \times 10^{-5}, 6.366203342 \times 10^{-7}, 6.366197724 \times 10^{-11}$ , against  $2/(\pi y^2) = 6.366197724 \times 10^{-5}, 6.366197724 \times 10^{-7}, 6.366197724 \times 10^{-11}$ . The threshold  $y_*$  where the two roots coalesce is exactly the value at which the Lambert argument reaches the branch point  $-1/e$ ; the agreement of these two descriptions is a useful consistency check on the whole normalization. Note also  $\mathcal{S}_-(1/(2\lambda)) = 1.15213\dots < y_*$ , a fact used in proposition 5.35.

**Remark 5.34** (The branch rule is forced by sign  $b$ ). All three sources agree on the assignment:  $W_{-1}$  on the even sheet,  $W_0$  on the odd sheet. In the model definition 1.2 the rule is  $b < 0 \Rightarrow W_{-1}$ ,  $b > 0 \Rightarrow W_0$  for the large real root, and  $b = \sigma/2$ ; so the branch distinction is forced by the sign of the power of  $x$  in the forward asymptotics — that is, by the fact that the even bisection carries  $n^{-1/2}$  and the odd bisection  $n^{+1/2}$ . It is a consequence of the parity split, not an independent convention.

### 5.6.3 Two exact inequalities

The sources deduce the *sign* of the first inverse correction from  $D > 0$ . The following sharpening is exact at all  $y$ , not only to first order, and costs nothing.

**Proposition 5.35** (The core brackets the true inverse). *For every  $y > 1$ ,*

$$\mathcal{I}_+(y) < X_+(y), \quad (5.56)$$

*and for every  $y > y_*$ ,*

$$\mathcal{I}_-(y) > X_-(y). \quad (5.57)$$

*Proof.* By eq. (5.19) and corollary 5.14,  $\mathcal{S}_+(x) = g_+(x)e^{D(x)} > g_+(x)$  for all  $x > 0$ , where  $g_+$  is the odd carrier. Both  $\mathcal{S}_+$  and  $g_+$  are strictly increasing. Put  $x_1 = \mathcal{I}_+(y)$ ; then  $g_+(x_1) < \mathcal{S}_+(x_1) = y = g_+(X_+(y))$ , and  $g_+$  increasing gives  $x_1 < X_+(y)$ .

For  $\sigma = -1$ ,  $\mathcal{S}_-(x) = g_-(x)e^{-D(x)} < g_-(x)$ . Let  $x_0 = \mathcal{I}_-(y)$  with  $y > y_*$ . Since  $\mathcal{S}_-$  is increasing and  $\mathcal{S}_-(1/(2\lambda)) = 1.15213\dots < y_* < y$ , we have  $x_0 > 1/(2\lambda)$ ; also  $X_-(y) \geq 1/(2\lambda)$  by theorem 5.31. On  $[1/(2\lambda), \infty)$  the carrier  $g_-$  is strictly increasing (proposition 5.32), and  $g_-(x_0) > \mathcal{S}_-(x_0) = y = g_-(X_-(y))$ , whence  $x_0 > X_-(y)$ .  $\square$

*Remark 5.36.* This explains without computation the signs  $c_1^{(+1)} = -1/(4\lambda) < 0$  and  $c_1^{(-1)} = +1/(4\lambda) > 0$  found in section 5.7, and supplies a rigorous one-sided bracket at every  $y$ , used as a check in algorithm 5.63.

### 5.6.4 A scaled form, and the half-index cores

Putting  $w_\sigma = 2\lambda X_\sigma$  turns eq. (5.50) into the single scalar equation

$$w + \sigma \log w = \Lambda_\sigma, \quad \Lambda_\sigma := \log(\pi y^2) + \sigma \log(4\lambda), \quad (5.58)$$

so that  $w_+ = W_0(e^{\Lambda_+})$  and  $w_- = -W_{-1}(-e^{-\Lambda_-})$ . This is S3's normalization, with its  $\varepsilon = -\sigma$ ; it is convenient in section 5.8.

*Remark 5.37* (Half-index cores). S1's carriers, in the half index and with  $\alpha = \log 4$ , are  $w_0^{S1} = -\frac{1}{2\alpha}W_{-1}(-2\alpha/(\pi y^2))$  and  $w_1^{S1} = \frac{1}{2\alpha}W_0(\pi\alpha y^2/2)$ . On the even sheet  $n = 2m$  exactly, and one checks directly that  $X_-(y) = 2w_0^{S1}(y)$ ; the two normalizations then agree, and their inverse coefficients are related by  $c_n^{(-1)} = 2^{n+1}d_n^{S1,(0)}$ , which we verified symbolically for  $1 \leq n \leq 8$ . On the odd sheet  $n = 2m + 1$ , so the two carriers solve *different* equations ( $C_+ 2^X X^{1/2} = y$  against  $K_1 4^w w^{1/2} = y$ ) and their coefficient lists are *not* termwise comparable, although both expansions converge asymptotically to the same function  $\mathcal{I}_+$ . Failing to notice this is the easiest way to “discover” a contradiction between S1 and S2/S3 where there is none.

## 5.7 All-orders reversion around the core

### 5.7.1 Specialization of the reversion theorem

**Theorem 5.38** (Swing specialization of theorem 1.25). *Let  $X = X_\sigma(y)$  solve eq. (5.50), let  $t = X^{-1}$ , and put*

$$\Psi_\sigma(t, u) = -\frac{1}{\lambda} \left[ \frac{\sigma}{2} \log(1 + tu) + \sigma \sum_{j \geq 1} d_j \left( \frac{t}{1 + tu} \right)^j \right]. \quad (5.59)$$

*Then the unique  $U \in t\mathcal{R}[[t]]$  with  $U = \Psi_\sigma(t, U)$  has coefficients*

$$c_n^{(\sigma)} = [t^n] U = \sum_{m=1}^n \frac{1}{m} [t^n u^{m-1}] \Psi_\sigma(t, u)^m \quad (5.60)$$

by lemma 1.16, and equivalently, with  $\gamma_1 = 0$  and  $\gamma_r = c_{r-1}^{(\sigma)}$  for  $r \geq 2$ ,

$$c_n^{(\sigma)} = -\frac{1}{\lambda} \left[ \frac{\sigma}{2} \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma) + \sigma \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} d_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma) \right], \quad (5.61)$$

with  $\widehat{B}$  the ordinary partial Bell polynomials of definition 1.9 and the expansions of lemma 1.12. Only  $c_1^{(\sigma)}, \dots, c_{n-1}^{(\sigma)}$  occur on the right. The coefficients lie in the parity algebra  $\mathcal{R}$  of eq. (5.9); no logarithm of  $L_\sigma$  occurs.

*Proof.* This is theorem 1.25 with the dictionary eq. (5.21):  $a = \lambda$ ,  $b = \sigma/2$ ,  $q_j = \sigma d_j$ , which turns the general  $\Psi$  into eq. (5.59). The restriction of the inner sum to odd  $j$  is  $d_{2r} = 0$ .  $\square$

### 5.7.2 Recomputed coefficients

The coefficients below were recomputed from eq. (5.59) by truncated power-series arithmetic over  $\mathbb{Q}[\lambda, \lambda^{-1}]$ , separately for  $\sigma = +1$  and  $\sigma = -1$ , and then recombined into  $\mathcal{R}$  by  $c = \frac{1}{2}(c^+ + c^-) + \sigma \cdot \frac{1}{2}(c^+ - c^-)$ . They agree with S2 for  $1 \leq n \leq 10$  and with S3 for  $1 \leq n \leq 8$  after the marker translation  $\sigma = -\varepsilon_{S3}$ .

$$c_1^{(\sigma)} = -\frac{\sigma}{4\lambda}, \quad c_2^{(\sigma)} = \frac{1}{8\lambda^2}, \quad (5.62)$$

$$c_3^{(\sigma)} = \frac{2\sigma\lambda^2 - 3\lambda - 3\sigma}{48\lambda^3}, \quad c_4^{(\sigma)} = \frac{-4\lambda^2 + 15\sigma\lambda + 6}{192\lambda^4}, \quad (5.63)$$

$$c_5^{(\sigma)} = -\frac{96\sigma\lambda^4 - 80\lambda^3 + 40\sigma\lambda^2 + 135\lambda + 30\sigma}{1920\lambda^5}, \quad (5.64)$$

$$c_6^{(\sigma)} = \frac{96\lambda^4 - 180\sigma\lambda^3 + 215\lambda^2 + 210\sigma\lambda + 30}{3840\lambda^6}, \quad (5.65)$$

$$c_7^{(\sigma)} = \frac{1}{53760\lambda^7} (8160\sigma\lambda^6 - 4312\lambda^5 + 1428\sigma\lambda^4 + 1050\lambda^3 - 4025\sigma\lambda^2 - 2100\lambda - 210\sigma), \quad (5.66)$$

$$c_8^{(\sigma)} = \frac{1}{645120\lambda^8} (-48960\lambda^6 + 56056\sigma\lambda^5 - 40908\lambda^4 + 15015\sigma\lambda^3 + 50610\lambda^2 + 17010\sigma\lambda + 1260), \quad (5.67)$$

$$c_9^{(\sigma)} = -\frac{1}{1290240\lambda^9} (1111040\sigma\lambda^8 - 413184\lambda^7 + 79616\sigma\lambda^6 + 43512\lambda^5 - 83748\sigma\lambda^4 + 83475\lambda^3 + 92610\sigma\lambda^2 + 22050\lambda + 1260\sigma), \quad (5.68)$$

$$c_{10}^{(\sigma)} = \frac{1}{5160960\lambda^{10}} (2222080\lambda^8 - 1756032\sigma\lambda^7 + 779152\lambda^6 - 203280\sigma\lambda^5 - 162582\lambda^4 + 487305\sigma\lambda^3 + 309960\lambda^2 + 55440\sigma\lambda + 2520). \quad (5.69)$$

Setting  $\sigma = \pm 1$  and  $\lambda = \log 2$ ,

$$\mathcal{I}_+(y) \sim X_+ - \frac{1}{4\lambda X_+} + \frac{1}{8\lambda^2 X_+^2} + \frac{2\lambda^2 - 3\lambda - 3}{48\lambda^3 X_+^3} + \frac{-4\lambda^2 + 15\lambda + 6}{192\lambda^4 X_+^4} + \dots, \quad (5.70)$$

$$\mathcal{I}_-(y) \sim X_- + \frac{1}{4\lambda X_-} + \frac{1}{8\lambda^2 X_-^2} + \frac{-2\lambda^2 - 3\lambda + 3}{48\lambda^3 X_-^3} + \frac{-4\lambda^2 - 15\lambda + 6}{192\lambda^4 X_-^4} + \dots. \quad (5.71)$$

The signs of the first corrections are those forced by proposition 5.35.

### 5.7.3 A differential form of the same coefficients

All three sources also give an operator form of the reversion, which is the convenient one for the Stokes analysis of section 5.9. We state it as a chapter-local lemma; it is a candidate for promotion, being independent of the swing factorial.

**Lemma 5.39** (Perturbed inverse in phase coordinates). *Let  $F_0$  be invertible near the point considered with  $F'_0 \neq 0$ , let  $g$  be a formal correction, and let  $X$  satisfy  $F_0(X) = z$ . Introduce*

$$\mathcal{D} := \frac{1}{F'_0(X)} \frac{d}{dX}. \quad (5.72)$$

*Then the solution  $x$  of  $F_0(x) + \eta g(x) = z$  with  $x|_{\eta=0} = X$  is, as a formal power series in the marker  $\eta$ ,*

$$x = X + \sum_{k \geq 1} \frac{(-\eta)^k}{k!} \mathcal{D}^{k-1} \left( \frac{g(X)^k}{F'_0(X)} \right). \quad (5.73)$$

*Proof.* This is theorem 1.18 under the dictionary  $F \mapsto F_0$ ,  $E \mapsto g$ ,  $\varepsilon \mapsto \eta$ ,  $\mu_0 \mapsto X$ ,  $y \mapsto z$  and  $\mathcal{D} \mapsto \mathcal{D}$ , which carries (1.32) to eq. (5.72) and (1.34) to eq. (5.73). Since  $g$  is here a formal and in general divergent series, no contour is available and the statement invoked is the coefficientwise one, corollary 1.19, whose only hypothesis is that  $F'_0(X)$  be invertible.  $\square$

**Proposition 5.40** (Differential formula for  $c_n^{(\sigma)}$ ). *Let  $\mathcal{Q}_\sigma(t) = \sigma \sum_{j \geq 1} d_j t^j$  and*

$$\mathcal{L}_\sigma := -\frac{t^2}{\lambda + \sigma t/2} \frac{d}{dt} \quad (t = X^{-1}), \quad (5.74)$$

*which is eq. (5.72) for  $F_{0,\sigma}(X) = \lambda X + \frac{\sigma}{2} \log X$  expressed in the variable  $t$ . Then, for every  $n \geq 1$ ,*

$$c_n^{(\sigma)} = [t^n] \sum_{k=1}^{\lfloor (n+1)/2 \rfloor} \frac{(-1)^k}{k!} \mathcal{L}_\sigma^{k-1} \left( \frac{\mathcal{Q}_\sigma(t)^k}{\lambda + \sigma t/2} \right), \quad (5.75)$$

*a finite sum requiring only  $d_1, \dots, d_n$ .*

*Proof.* Apply lemma 5.39 with  $F_0 = F_{0,\sigma}$ ,  $g = \sigma D$  and  $\eta = 1$ ; the substitution  $t = X^{-1}$  turns  $F'_{0,\sigma}(X) = \lambda + \sigma/(2X)$  into  $\lambda + \sigma t/2$  and  $d/dX$  into  $-t^2 d/dt$ . Setting the marker  $\eta$  to 1 is legitimate coefficientwise because  $\mathcal{Q}_\sigma(t)^k = O(t^k)$  while each application of  $\mathcal{L}_\sigma$  raises the order in  $t$  by one, so the  $k$ -th summand begins at  $t^{2k-1}$  and can contribute to  $[t^n]$  only when  $2k-1 \leq n$ .  $\square$

**Source claim.** S2 writes its operator expansion as an *equality*,  $\mathcal{I}_\sigma(y) = X + \sum_{m \geq 1} \frac{(-\sigma)^m}{m!} \mathcal{D}^{m-1} [D(X)^m / F'_0(X)]$ , after setting its coupling parameter to 1. **Defect.** With the marker removed there is no small parameter, the series over  $m$  does not converge, and the left-hand side is a number while the right-hand side is a divergent series; the statement as printed is false. **Repair.** The correct statement is the asymptotic one: the  $m$ -th summand is  $O(X^{-(2m-1)})$ , so for each fixed  $M$  the partial sum through  $m = M$  differs from  $\mathcal{I}_\sigma(y)$  by  $O(X^{-(2M+1)})$ , and regrouping by powers of  $X^{-1}$  reproduces eq. (5.75). S1 and S3 state the same formula correctly, as a coefficient identity.

### 5.7.4 The analytic statement, with a proof

**Source claim.** All three sources assert the Poincaré statement “ $\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n < M} c_n^{(\sigma)} X_\sigma(y)^{-n} + O(X_\sigma(y)^{-M})$ ”. **Defect.** S3 gives no proof at all (“Likewise, for each fixed  $M$ ,

..."); S1 gives a two-line sketch that appeals to the mean value theorem but never bounds the residual of the truncated ansatz, and in particular never uses any control on the forward remainder at the *perturbed* point; S2 proves the ingredients (a residual-to-index bound and derivative lower bounds) but does not assemble them. This is the one place in the three sources where a formal computation is upgraded to an analytic conclusion without an argument. **Repair.** Theorem 5.41 below supplies the proof, and makes the constant effective by using the enveloping remainder theorem 5.20.

**Theorem 5.41** (Inverse transseries, analytically). *Fix  $M \geq 1$  and  $\sigma \in \{-1, +1\}$ . Then, as  $y \rightarrow +\infty$ ,*

$$\boxed{\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n=1}^{M-1} \frac{c_n^{(\sigma)}}{X_\sigma(y)^n} + O(X_\sigma(y)^{-M})}, \quad (5.76)$$

with an effective implied constant depending only on  $M$  and  $\sigma$ . Since  $X_\sigma(y) = (\log y)/\lambda(1 + o(1))$ , the expansion is one in inverse powers of  $\log y$  once the dominant power–logarithmic phase has been inverted exactly.

*Proof.* Write  $X = X_\sigma(y)$  and  $x_M := X + \sum_{n=1}^{M-1} c_n^{(\sigma)} X^{-n}$ ; for  $y$  large,  $x_M = X(1 + O(X^{-2}))$ , so  $x_M > 1$  and  $x_M \asymp X$ . Consider the phase residual

$$\rho := \log \mathcal{S}_\sigma(x_M) - \log y = \lambda(x_M - X) + \frac{\sigma}{2} \log \frac{x_M}{X} + \sigma D(x_M),$$

where we used eq. (5.19) and  $\log y = \log C_\sigma + \lambda X + \frac{\sigma}{2} \log X$ . Put  $t = X^{-1}$  and  $U_{M-1}(t) = \sum_{n < M} c_n^{(\sigma)} t^n$ , so that  $x_M = X(1 + tU_{M-1})$  and

$$\rho = \lambda X \left[ \underbrace{tU_{M-1} + \frac{\sigma t}{2\lambda} \log(1 + tU_{M-1}) + \frac{\sigma t}{\lambda} D_{<M}(x_M)}_{=: t\mathcal{E}(t)} \right] + \sigma(D(x_M) - D_{<M}(x_M)),$$

where  $D_{<M}$  denotes the truncation of the series eq. (5.26) after the terms of order  $x^{-(M-1)}$ . Two estimates finish the proof.

(i) *The algebraic residual.* Substituting  $x_M = X(1 + tU_{M-1})$  and  $D_{<M}(x_M) = \mathcal{Q}_\sigma(t/(1 + tU_{M-1}))/\sigma$  truncated, the bracket  $\mathcal{E}(t)$  is exactly  $U_{M-1}(t) - \Psi_\sigma(t, U_{M-1}(t))$  modulo terms of order  $t^M$  coming from the truncation of  $\mathcal{Q}_\sigma$ . By theorem 5.38 the coefficients of  $t^1, \dots, t^{M-1}$  in  $U - \Psi_\sigma(t, U)$  vanish when  $U$  is replaced by  $U_{M-1}$  – this is the triangularity of the recurrence – so  $\mathcal{E}(t) = O(t^M)$  with an explicit rational-in- $\lambda$  constant obtained from  $c_1^{(\sigma)}, \dots, c_{M-1}^{(\sigma)}$  and  $d_1, \dots, d_M$ . Hence  $\lambda X \cdot t\mathcal{E}(t) = O(X^{-M+1} \cdot X^{-1}) \cdot X = O(X^{-M+1})$ ; more carefully,  $X t\mathcal{E}(t) = \mathcal{E}(t) = O(X^{-M})$ .

(ii) *The transcendental residual.* By theorem 5.20 with  $N = \lceil M/2 \rceil$ ,  $|D(x_M) - D_{<M}(x_M)| \leq |d_{2N+1}| x_M^{-(2N+1)} \leq c_M X^{-M}$  for an explicit  $c_M$ , since  $2N + 1 \geq M$  and  $x_M \asymp X$ .

Therefore  $|\rho| \leq C_M X^{-M}$  with  $C_M$  explicit. Finally,  $(\log \mathcal{S}_\sigma)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x) \geq \lambda - \frac{1}{2x} - \frac{1}{4x^2}$  by corollary 5.14, which exceeds  $\lambda/2$  once  $x \geq 2/\lambda$ . Theorem 1.42 (residual-to-root conversion) then gives

$$|x_M - \mathcal{I}_\sigma(y)| \leq \frac{|\rho|}{\lambda/2} \leq \frac{2C_M}{\lambda} X^{-M},$$

which is eq. (5.76). The final sentence follows from  $\lambda X + \frac{\sigma}{2} \log X = L_\sigma(y)$  and  $L_\sigma(y) = \log y + O(1)$ .  $\square$

**Remark 5.42** (An exact algebraic certificate). Step (i) of the proof is also the recommended *implementation* check. Given a candidate truncation  $U_{M-1}$ , substitute it into  $U - \Psi_\sigma(t, U)$  and expand: the coefficients of  $t^1, \dots, t^{M-1}$  must vanish identically in  $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$ . This is an exact algebraic certificate for every table of  $c_n^{(\sigma)}$ , and it is how the tables in eq. (5.62)–eq. (5.69) were verified here, independently of the recurrence that produced them.

### 5.7.5 The centred reversion

Applying theorem 1.25 to the centred normal form eq. (5.45) – that is, with  $a = 2\lambda$ ,  $b = \sigma/2$ ,  $C = 1/\sqrt{2\pi}$  and  $q_j = \sigma\nu_j$  ( $\nu_j = 0$  for odd  $j$ ) – gives the centred cores

$$\mathcal{U}_+(y) = \frac{W_0(8\pi\lambda y^2)}{4\lambda}, \quad \mathcal{U}_-(y) = -\frac{W_{-1}(-2\lambda/(\pi y^2))}{4\lambda}, \quad (5.77)$$

and the centred inverse expansion

$$\mathcal{I}_\sigma(y) \sim 2\mathcal{U}_\sigma(y) - \frac{1}{2} + 2 \sum_{n \geq 2} \frac{\hat{c}_n^{(\sigma)}}{\mathcal{U}_\sigma(y)^n}, \quad \hat{c}_1^{(\sigma)} = 0. \quad (5.78)$$

The vanishing of  $\hat{c}_1^{(\sigma)}$  is immediate from  $\nu_1 = 0$ . Recomputed in the present normalization,

$$\hat{c}_2^{(\sigma)} = -\frac{\sigma}{128\lambda}, \quad \hat{c}_3^{(\sigma)} = \frac{1}{512\lambda^2}, \quad \hat{c}_4^{(\sigma)} = \frac{\sigma(5\lambda^2 - 2)}{4096\lambda^3}, \quad (5.79)$$

$$\hat{c}_5^{(\sigma)} = -\frac{7\lambda^2 - 2}{16384\lambda^4}, \quad \hat{c}_6^{(\sigma)} = -\frac{\sigma(244\lambda^4 - 57\lambda^2 + 12)}{393216\lambda^5}, \quad \hat{c}_7^{(\sigma)} = \frac{334\lambda^4 - 75\lambda^2 + 12}{1572864\lambda^6}, \quad (5.80)$$

and

$$\hat{c}_8^{(\sigma)} = \frac{\sigma(4155\lambda^6 - 460\lambda^4 + 96\lambda^2 - 12)}{6291456\lambda^7}. \quad (5.81)$$

These agree with S1's centred table after the substitution  $\alpha = 2\lambda$  and the marker translation of remark 5.12; observe that  $\hat{c}_n$  is  $\sigma$ -even for odd  $n$  and  $\sigma$ -odd for even  $n$ , a reflection symmetry of eq. (5.44) that the uncentred coefficients do not have.

**Source claim.** S1 concludes that “the centred one is often the preferable computational normal form”. **Defect.** Its own table already contradicts the unqualified form of this claim: at  $m = 200$  the centred core beats the uncentred core by  $10^3$  ( $2.8 \times 10^{-7}$  against  $4.5 \times 10^{-4}$ ), but at  $R = 4$  the *uncentred* expansion is slightly better ( $7.3 \times 10^{-16}$  against  $1.1 \times 10^{-15}$ ). **Repair.** The correct statement, confirmed by the recomputation in table 5.5 and table 5.6, is: centring removes the order- $X^{-1}$  term entirely, so it improves the *bare core* by a factor  $\asymp X$  and is the right choice when no corrections are to be carried; once several corrections are carried, the two normalizations are comparable, with neither dominating uniformly.

## 5.8 Flattening the core into a polynomial–logarithmic transseries

### 5.8.1 The specialization

Set

$$H_\sigma(y) := \log \frac{L_\sigma(y)}{\lambda}, \quad (5.82)$$

and recall from theorem 1.33 that the inverse of the model definition 1.2 has the unique power–logarithmic form  $x \sim a^{-1} [L - bH + \sum_{n \geq 1} P_n(H)L^{-n}]$  with  $\deg P_n \leq n$  and leading term  $b^{n+1}H^n/n$ . With the dictionary eq. (5.21),

$$\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda} \left[ L_\sigma - \frac{\sigma}{2} H_\sigma + \sum_{n \geq 1} \frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n} \right], \quad (5.83)$$

where the  $P_n^{(\sigma)}$  are generated by theorem 1.33 from  $b = \sigma/2$ ,  $a = \lambda$ ,  $q_j = \sigma d_j$ ; the fixed-order meaning of eq. (5.83) is the analytic clause of theorem 1.33, namely an error  $O((1 + H_\sigma)^{N+1} L_\sigma^{-(N+1)})$  after  $N$

correction terms, whose hypotheses hold here by theorem 5.41. The recomputed polynomials are

$$P_1^{(\sigma)}(H) = \frac{H - \sigma\lambda}{4}, \quad (5.84)$$

$$P_2^{(\sigma)}(H) = \frac{\sigma H^2 - 2(\lambda + \sigma)H + 2\lambda}{16}, \quad (5.85)$$

$$P_3^{(\sigma)}(H) = \frac{1}{96} \left[ 2H^3 - (6\sigma\lambda + 9)H^2 + (18\sigma\lambda + 6)H + 4\sigma\lambda^3 - 6\lambda^2 - 6\sigma\lambda \right], \quad (5.86)$$

$$P_4^{(\sigma)}(H) = \frac{1}{384} \left[ 3\sigma H^4 - (12\lambda + 22\sigma)H^3 + (66\lambda + 36\sigma)H^2 + (24\lambda^3 - 36\sigma\lambda^2 - 72\lambda - 12\sigma)H - 8\lambda^3 + 30\sigma\lambda^2 + 12\lambda \right], \quad (5.87)$$

$$P_5^{(\sigma)}(H) = \frac{1}{3840} \left[ 12H^5 - (60\sigma\lambda + 125)H^4 + (500\sigma\lambda + 350)H^3 + (240\sigma\lambda^3 - 360\lambda^2 - 1050\sigma\lambda - 300)H^2 + (-280\sigma\lambda^3 + 780\lambda^2 + 600\sigma\lambda + 60)H - 192\sigma\lambda^5 + 160\lambda^4 - 80\sigma\lambda^3 - 270\lambda^2 - 60\sigma\lambda \right]. \quad (5.88)$$

These reproduce S2 exactly for  $1 \leq n \leq 5$ ; they were recomputed for  $\sigma = \pm 1$  separately and recombined in  $\mathcal{R}$ , and independently re-derived through  $n = 6$ . They also satisfy the cross-normalization identity  $P_n^{(\sigma)}(0) = \lambda^{n+1}c_n^{(\sigma)}$  of theorem 1.33: for instance  $P_1^{(\sigma)}(0) = -\sigma\lambda/4 = \lambda^2c_1^{(\sigma)}$  and  $P_2^{(\sigma)}(0) = \lambda/8 = \lambda^3c_2^{(\sigma)}$ , which ties eq. (5.84)–eq. (5.88) to eq. (5.62)–eq. (5.69) and catches any transcription error in either list.

### 5.8.2 The pure Lambert block in closed form

The pure block — the case  $Q \equiv 0$ , in which the whole correction series is suppressed and only the Lambert core survives — is theorem 1.34, proved there in the shortest of the three forms the sources use. Specializing its data to the swing branches,  $a = \lambda$  and  $b = \sigma/2$ , gives

$$P_n^{(0)}(H) = \left(\frac{\sigma}{2}\right)^{n+1} \sum_{m=1}^n (-1)^{m+1} \begin{bmatrix} n \\ m \end{bmatrix} \frac{H^{n-m+1}}{(n-m+1)!}, \quad (5.89)$$

with  $\begin{bmatrix} n \\ m \end{bmatrix}$  the unsigned Stirling numbers of the first kind. Reducing with  $\sigma^2 = 1$ , the two parities differ only by the overall sign  $\sigma^{n+1}$ : the even and odd pure blocks coincide for odd  $n$  and are opposite for even  $n$ .

Writing  $b = \sigma/2$ , the first few are

$$P_1^{(0)} = b^2H, \quad P_2^{(0)} = \frac{b^3}{2}(H^2 - 2H), \quad P_3^{(0)} = \frac{b^4}{6}(2H^3 - 9H^2 + 6H), \quad P_4^{(0)} = \frac{b^5}{12}(3H^4 - 22H^3 + 36H^2 - 12H),$$

and eq. (5.89) was recomputed here against the recurrence of theorem 1.33 for  $1 \leq n \leq 8$ , in exact arithmetic with  $b$  symbolic. In particular the leading coefficient is  $b^{n+1}/n$ , which is theorem 1.33's degree statement. Theorem 1.34 is the precise bridge between Lambert asymptotics and the first-kind Stirling calculus, and it is stated there rather than here because all four chapters use it.

### 5.8.3 The core series and its convergence

The flattening can also be organized as “expand the core, then compose”, which is S3's route and which is constructive at every order.

**Proposition 5.43** (All coefficients of the scaled core). *Let  $w$  solve  $w + \sigma \log w = \Lambda$  as in eq. (5.58), and put  $\ell = \log \Lambda$ . Define*

$$A_k(\ell) := \frac{1}{k+1} [z^k] (\ell + \log(1+z))^{k+1}, \quad k \geq 0. \quad (5.90)$$



Then

$$w = \Lambda + \sum_{k \geq 0} \frac{(-\sigma)^{k+1} A_k(\ell)}{\Lambda^k}, \quad (5.91)$$

and, with  $s(k, j)$  the signed Stirling numbers of the first kind,

$$A_0(\ell) = \ell, \quad A_k(\ell) = \frac{1}{(k+1)k!} \sum_{j=1}^k \binom{k+1}{j} j! s(k, j) \ell^{k+1-j} \quad (k \geq 1). \quad (5.92)$$

The first polynomials are

$$\begin{aligned} A_0 &= A_1 = \ell, & A_2 &= \frac{\ell(2-\ell)}{2}, & A_3 &= \frac{\ell(2\ell^2-9\ell+6)}{6}, & A_4 &= -\frac{\ell(3\ell^3-22\ell^2+36\ell-12)}{12}, \\ A_5 &= \frac{\ell(12\ell^4-125\ell^3+350\ell^2-300\ell+60)}{60}, & A_6 &= -\frac{\ell(20\ell^5-274\ell^4+1125\ell^3-1700\ell^2+900\ell-120)}{120}. \end{aligned}$$

*Proof.* Write  $w = \Lambda(1 + v)$  and  $z = \Lambda^{-1}$ . The equation becomes  $\Lambda v = -\sigma(\ell + \log(1 + v))$ , i.e.  $v = -\sigma z(\ell + \log(1 + v))$ . Lagrange inversion gives  $v = \sum_{k \geq 0} \frac{(-\sigma z)^{k+1}}{k+1} [v^k] (\ell + \log(1 + v))^{k+1}$ ; multiplying by  $\Lambda = z^{-1}$  yields eq. (5.91) with  $A_k$  as in eq. (5.90). Expanding the  $(k+1)$ -st power and using  $(\log(1 + z))^j = j! \sum_{k \geq j} s(k, j) z^k / k!$  gives eq. (5.92).  $\square$

The sign in eq. (5.91) matches S3 after  $\varepsilon = -\sigma$ ; explicitly  $w_- = \Lambda_- + \ell + \ell/\Lambda_- + \dots$  on the even sheet ( $W_{-1}$ ) and  $w_+ = \Lambda_+ - \ell + \ell/\Lambda_+ - \dots$  on the odd sheet ( $W_0$ ).

**Proposition 5.44** (An explicit convergence criterion). *Fix  $\rho \in (0, 1)$  and put  $M_\rho(\ell) = |\ell| + \log \frac{1}{1-\rho}$ . Then*

$$|A_k(\ell)| \leq \frac{M_\rho(\ell)^{k+1}}{(k+1)\rho^k}, \quad (5.93)$$

so the series in eq. (5.91) converges absolutely whenever  $M_\rho(\ell) < \rho|\Lambda|$  for some  $\rho \in (0, 1)$ , and its sum is then the solution branch  $w \sim \Lambda$ . In particular, for real  $\Lambda \geq 10$  the choice  $\rho = \frac{1}{2}$  already gives convergence.

*Proof.* On  $|z| = \rho$  one has  $|\log(1 + z)| \leq \sum_{k \geq 1} \rho^k / k = \log \frac{1}{1-\rho}$ , so  $|\ell + \log(1 + z)| \leq M_\rho(\ell)$  there; Cauchy's estimate applied to eq. (5.90) gives eq. (5.93). Summing the geometric-type majorant gives absolute convergence for  $M_\rho(\ell) < \rho|\Lambda|$ . The sum solves  $w + \sigma \log w = \Lambda$  because Lagrange's theorem identifies the series with the unique analytic solution  $v(z)$  of  $v = -\sigma z(\ell + \log(1 + v))$  vanishing at  $z = 0$ ; the branch with  $w \sim \Lambda$  is the one selected. For  $\Lambda = 10$ ,  $\ell = \log 10 = 2.3026$  and  $M_{1/2} = 2.3026 + 0.6931 = 2.9957 < 5$ .  $\square$

**Source claim.** S3 asserts that “the pure Lambert logarithmic expansion has a nonzero convergence domain for sufficiently large argument, as in the classical analysis of Lambert  $W$ ”, citing Corless et al. but giving no criterion. **Defect.** The cited classical result concerns the doubly-indexed series in  $(\log z)^{-1}$  and  $\log \log z$ ; the single series eq. (5.91), whose coefficients are polynomials in the slowly varying  $\ell$ , needs its own statement, since convergence depends on the ratio  $\ell/\Lambda$  and fails for  $\ell$  comparable to  $\Lambda$ . **Repair.** Proposition 5.44 supplies an explicit, proved criterion, which also delimits the regime where eq. (5.91) may be summed rather than truncated.

#### 5.8.4 Composition, and the equivalence of the two normalizations

**Theorem 5.45** (Constructive equivalence of the two inverse normalizations). *With  $v_j = (-\sigma)^j A_{j-1}(\ell)$  and*

$$R_r^{(m)}(\ell) = [z^r] \left( 1 + \sum_{j \geq 1} v_j z^j \right)^{-m} = \sum_{k=0}^r (-1)^k \binom{m+k-1}{k} \widehat{B}_{r,k}(v), \quad R_0^{(m)} = 1, \quad (5.94)$$

the Lambert-normalized expansion eq. (5.76) and the flattened expansion eq. (5.83) are term-by-term equivalent: in the scaled variables of eq. (5.58),

$$\mathcal{I}_\sigma(y) \sim \frac{\Lambda_\sigma}{2\lambda} - \frac{\sigma \ell_\sigma}{2\lambda} + \sum_{N \geq 1} \frac{\Pi_N^{(\sigma)}(\ell_\sigma)}{\Lambda_\sigma^N}, \quad \Pi_N^{(\sigma)}(\ell) = \frac{(-\sigma)^{N+1} A_N(\ell)}{2\lambda} + \sum_{m=1}^N c_m^{(\sigma)} (2\lambda)^m R_{N-m}^{(m)}(\ell), \quad (5.95)$$

with  $\deg \Pi_N^{(\sigma)} = N$  and leading coefficient  $(-1)^{N-1}(-\sigma)^{N+1}/(2\lambda N)$ .

*Proof.* Insert  $X_\sigma = w_\sigma/(2\lambda)$  into eq. (5.76), write  $w = \Lambda(1 + V)$  with  $V = \sum_j v_j z^j$  from eq. (5.91), and collect the coefficient of  $\Lambda^{-N}$ : the core contributes the first term of eq. (5.95) and the  $m$ -th correction contributes  $c_m^{(\sigma)}(2\lambda)^m [z^{N-m}] (1 + V)^{-m}$ . The Bell form of  $R_r^{(m)}$  is lemma 1.12. Degrees:  $A_N$  has degree  $N$  with leading coefficient  $(-1)^{N-1}/N$  by eq. (5.92), while every correction with  $m \geq 1$  contributes degree at most  $N - m$ .  $\square$

**Source claim.** S1 asserts the equivalence of its two inverse normalizations in prose: “expanding the Lambert carrier in the scale  $X^{-1}\mathbb{R}[\ell]$  and then substituting that expansion ... reproduces the recurrence coefficient by coefficient.” **Defect.** No proof and no formula; the assertion is exactly the sort of “successive matching” that the coefficient calculus is supposed to eliminate. **Repair.** Theorem 5.45, which follows S3, makes the equivalence a finite explicit composition with a proof, and identifies the degrees and leading coefficients.

*Remark 5.46* (Three flattening normalizations, and which one the volume uses). The three sources flatten in three different variables: S2 in  $L_\sigma = \log(y/C_\sigma)$  with  $H_\sigma = \log(L_\sigma/\lambda)$ ; S3 in  $\Lambda_\sigma = \log(\pi y^2) + \sigma \log(4\lambda)$  with  $\ell_\sigma = \log \Lambda_\sigma$ ; S1 in  $X_\epsilon = \log(y\alpha^{\beta_\epsilon}/K_\epsilon)$  in the half index. Since  $\Lambda_\sigma = 2L_\sigma + \sigma(\log(4\lambda) - \lambda) \neq 2L_\sigma$ , these are genuinely different variables, the three coefficient lists are not comparable termwise, and none is in error — all were recomputed and are correct in their own normalization. The volume adopts S2’s variables in eq. (5.83) (they are those of theorem 1.33) and records S3’s in eq. (5.95), where the composition proof is cleanest.

**Corollary 5.47** (Leading elementary forms). As  $y \rightarrow \infty$ ,

$$\mathcal{I}_\sigma(y) = \frac{1}{\lambda} \left( L_\sigma(y) - \frac{\sigma}{2} \log \frac{L_\sigma(y)}{\lambda} + O\left(\frac{\log L_\sigma(y)}{L_\sigma(y)}\right) \right). \quad (5.96)$$

The opposite signs of the  $\frac{1}{2} \log L$  term on the two sheets are the inverse-side signature of the forward powers  $n^{-1/2}$  and  $n^{+1/2}$ .

## 5.9 Asymptotic meaning, optimal truncation, and Stokes structure

### 5.9.1 What each expansion means at fixed order

For each fixed  $M$  the forward statement is eq. (5.36) with the effective constant of proposition 5.25, and the inverse statement is eq. (5.76). Both are Poincaré statements about divergent series: the coefficients  $d_j$  grow factorially (proposition 5.22), hence so do the  $A_n$  and, through the reversion, the  $c_n^{(\sigma)}$ . The divergence is not a defect but a datum: the Borel transform and the exact sum of the logarithmic series are known in closed form (theorem 5.13, proposition 5.16). By contrast the pure core series eq. (5.91) converges in the regime delimited by proposition 5.44; the full inverse is asymptotic only because of the gamma-ratio correction.

### 5.9.2 Optimal truncation and its image in the inverse variable

**Proposition 5.48** (Least-term truncation, forward and inverse). *Theorem 1.46 applied to eq. (5.30) with the singulant modulus  $\pi$  of proposition 5.16 gives the least-term index  $r \approx \pi x/2$  and the beyond-all-orders floor*

$$|R_N(x)|_{N \approx \pi x/2} = \frac{2\sqrt{2}}{\pi\sqrt{x}} e^{-\pi x} (1 + O(x^{-1})), \quad (5.97)$$

which by theorem 5.20 is an equality up to the stated factor and not merely an upper bound. Transporting it through the inverse by theorem 1.44 and  $(\log \mathcal{S}_\sigma)' = \lambda + O(x^{-1})$  gives the inverse floor

$$O\left(X_\sigma^{-1/2} e^{-\pi X_\sigma}\right) = O\left(y^{-\pi/\lambda} (\log y)^{-1/2 + \sigma\pi/(2\lambda)}\right), \quad \frac{\pi}{\lambda} = 4.5323601418 \dots \quad (5.98)$$

*Proof.* Equation (5.97) is proposition 5.22. For the image, use the core equation in the form  $e^{\lambda X} = (y/C_\sigma) X^{-\sigma/2}$ , so that  $e^{-\pi X} = (y/C_\sigma)^{-\pi/\lambda} X^{\sigma\pi/(2\lambda)}$ ; multiply by  $X^{-1/2}$  and use  $X \asymp \log y$ .  $\square$

**Remark 5.49.** Exponential smallness in the index becomes *algebraic* smallness in the level: an  $e^{-\pi x}$  forward sector maps to a  $y^{-\pi/\lambda}$  inverse sector. This is why hyperasymptotic corrections to the inverse live on a strictly finer grid than the algebraic powers of  $1/\log y$  produced by eq. (5.83) – they are invisible to that scale at every order.

### 5.9.3 Lateral sums and the aggregate Stokes multipliers

For a direction  $\theta$  let  $\mathcal{S}_{\theta\pm} D(x) = \int_0^{e^{i(\theta\pm 0)}\infty} e^{-xt} \widehat{D}(t) dt$  whenever the rotated Laplace integral converges, and let  $\text{Disc}_\theta$  denote the lower-minus-upper lateral difference.

**Proposition 5.50** (Bridge equations). *With the poles and residues of eq. (5.25),*

$$\text{Disc}_{\pi/2} D(x) = 2 \sum_{k \geq 0} \frac{e^{-i\pi(2k+1)x}}{2k+1} = 2 \operatorname{artanh}(e^{-i\pi x}) \quad (\Im x < 0), \quad (5.99)$$

$$\text{Disc}_{-\pi/2} D(x) = -2 \sum_{k \geq 0} \frac{e^{i\pi(2k+1)x}}{2k+1} = -2 \operatorname{artanh}(e^{i\pi x}) \quad (\Im x > 0), \quad (5.100)$$

the closed forms giving the analytic continuations beyond those half-planes. Consequently  $\text{Disc}_\theta \log \mathcal{S}_\sigma = \sigma \text{Disc}_\theta D$ , and the multiplicative Stokes factor of the forward branch across the ray  $\theta = \pm\pi/2$  is  $\exp(\sigma \text{Disc}_{\pm\pi/2} D(x))$ , that is

$$\mathfrak{M}_+^{(\sigma)}(x) = \left( \frac{1 + e^{-i\pi x}}{1 - e^{-i\pi x}} \right)^\sigma, \quad \mathfrak{M}_-^{(\sigma)}(x) = \left( \frac{1 - e^{i\pi x}}{1 + e^{i\pi x}} \right)^\sigma. \quad (5.101)$$

On the positive real axis no lateral choice arises, because  $\arg t = 0$  is a regular direction.

*Proof.* Rotating the Laplace contour past the ray through  $t_k = i\pi(2k+1)$  picks up  $2\pi i$  times the residue of  $e^{-xt} \widehat{D}(t)$  there, namely  $2\pi i e^{-t_k x}/t_k = 2e^{-i\pi(2k+1)x}/(2k+1)$ ; summing over  $k$  and using  $\operatorname{artanh} z = \sum_{k \geq 0} z^{2k+1}/(2k+1)$  gives eq. (5.99). The series converges for  $|e^{-i\pi x}| = e^{\pi \Im x} < 1$ , that is  $\Im x < 0$ . The lower ray is the conjugate computation. The last two displays follow from  $\log \mathcal{S}_\sigma = \log C_\sigma + \lambda x + \frac{\sigma}{2} \log x + \sigma D$ , in which only  $D$  is discontinuous, together with  $2 \operatorname{artanh} z = \log \frac{1+z}{1-z}$ .  $\square$

**Remark 5.51** (Convention dependence). S2 defines the discontinuity as lower minus upper lateral sum; S3 instead adopts the residue convention in which crossing a singular ray contributes  $+2\pi i$  times the enclosed residues, and works with  $Q = -D$ . The two sets of formulae differ by an overall sign, hence by a reciprocal in the multipliers, and both are internally consistent. A Stokes generator is only defined relative to a stated convention, so the volume states the convention explicitly in proposition 5.50 and does not declare either source's signs canonical.

### 5.9.4 Transport of an exponential sector through inversion

**Proposition 5.52** (Sector transport). *Let  $F$  be one lateral branch of  $\log \mathcal{S}_\sigma$  and let the other be  $F + \Delta$ . As a formal expansion in the sector  $\Delta$ ,*

$$x_{\text{new}} = x + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left( \frac{1}{F'(x)} \frac{d}{dx} \right)^{k-1} \left( \frac{\Delta(x)^k}{F'(x)} \right), \quad (5.102)$$

with  $\Delta = \sigma \operatorname{Disc}_\theta D$  and  $F'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$ . Quantitatively, and with no formal series at all: if  $|\Delta| \leq \delta$  on the interval joining  $x$  and  $x_{\text{new}}$  and  $F' \geq \mu > 0$  there, then

$$|x_{\text{new}} - x| \leq \frac{\delta}{\mu}, \quad x_{\text{new}} - x = -\frac{\Delta(x)}{F'(x)} + O(\delta^2), \quad (5.103)$$

the implied constant depending only on  $\sup |(F^{-1})''|$  on that interval.

*Proof.* Equation (5.102) is lemma 5.39 with  $F_0 = F$ ,  $g = \Delta$  and  $\eta = 1$ , read as an identity of formal series in the sector marker. The first inequality in eq. (5.103) is theorem 1.42; the second is Taylor's theorem for  $F^{-1}$  at the point  $F(x)$ .  $\square$

**Source claim.** S1 states its transport rule as a *theorem* (“Transseries-sector transport through inversion”) whose hypothesis is that “ $E$  is smaller than the algebraic scale retained in  $\Phi_{\text{alg}}$ ”, whose conclusion is prefaced by the word “formally”, and whose displayed first terms carry an unjustified error symbol  $O(E^3, E^2 E', E(E')^2)$ . **Defect.** The hypothesis is never used; the conclusion is a formal-series identity, not an analytic one; and the error term is asserted rather than proved. A statement that holds only formally must not stand in a theorem environment. **Repair.** Proposition 5.52 separates the two claims: the all-orders identity is formal (and is exactly lemma 5.39), while the quantitative first-order statement is proved by the mean value theorem under stated hypotheses. S2 and S3 already state their versions correctly, as formal expansions.

**Source claim.** S1's section on complex inverse sheets asserts that for every Lambert branch with  $|w_{\epsilon,k}| \rightarrow \infty$  in a suitable sector, “the same coefficients  $d_r^{(\epsilon)}$  give  $M_{\epsilon,k}(y) \sim w_{\epsilon,k}(y) + \sum_r d_r^{(\epsilon)} w_{\epsilon,k}(y)^{-r}$ ”. **Defect.** For  $k \neq 0, -1$  nothing has been said, let alone proved, about  $M_{\epsilon,k}$  being an inverse of anything: the existence of an analytic branch of  $\mathcal{S}_\sigma^{-1}$  along the corresponding path is a hypothesis, not a consequence, and it fails when the path meets a pole of the gamma quotient or a Stokes direction. **Repair.** The statement is demoted to remark 5.53, with the two missing hypotheses displayed. S2 states the same thing correctly, as a *formal* complex branch whose analytic validity is a separate requirement.

**Remark 5.53** (Complex branches, formally). For  $k \in \mathbb{Z}$  let  $X_{\sigma,k}$  be as in eq. (5.51) after fixing branches of the powers and logarithms in  $L_\sigma$ . Suppose that along a path with  $|X_{\sigma,k}(y)| \rightarrow \infty$  (i) the forward expansion eq. (5.36) is valid, which requires the path to stay in a sector free of the poles of  $\Gamma(x/2 + (3 - \sigma)/4)^{-2}$  and away from the Stokes directions of proposition 5.50, and (ii) an analytic branch  $\mathcal{I}_{\sigma,k}$  of the inverse exists along it. Then that branch satisfies  $\mathcal{I}_{\sigma,k}(y) \sim X_{\sigma,k}(y) + \sum_{n \geq 1} c_n^{(\sigma)} X_{\sigma,k}(y)^{-n}$  with the *same* coefficients, by the proof of theorem 5.41. The coefficient sequence does not depend on  $k$ : all branch dependence sits in  $W_k$  and in the chosen logarithm. Without hypotheses (i) and (ii) the display is a formal statement only.

## 5.10 Discrete inversion and index recovery

### 5.10.1 The three inverse objects, parity by parity

Definition 1.37 attaches three objects to a discrete sequence: the functional inverse of a chosen interpolation, the integer staircase, and the round-to-nearest generalized inverse.

**Warning 5.54** (None of the three exists for the unsplit sequence). By proposition 5.4 the set  $\{n : \text{sw}(n) \leq y\}$  is not an initial segment of  $\mathbb{N}_0$  – for instance  $\text{sw}(5) = 30 > 20 = \text{sw}(6)$  – so the staircase of  $\text{sw}$  is not monotone and does not invert it; the round-to-nearest inverse is ambiguous whenever two indices of opposite parity are nearly equidistant in value; and by proposition 5.9 no eventually monotone interpolation exists at all. All three objects exist, are unique, and obey theorem 1.39 *on each parity class separately*.

**Proposition 5.55** (Exact branchwise staircases). *For  $y \geq 1$  put*

$$N_{-}^{\leq}(y) = \max\{n \text{ even} : \text{sw}(n) \leq y\}, \quad N_{+}^{\leq}(y) = \max\{n \text{ odd} : \text{sw}(n) \leq y\}, \quad (5.104)$$

*both sets being non-empty because  $\text{sw}(0) = \text{sw}(1) = 1$ . Then*

$$N_{-}^{\leq}(y) = 2 \left\lfloor \frac{\mathcal{I}_{-}(y)}{2} \right\rfloor, \quad N_{+}^{\leq}(y) = 2 \left\lfloor \frac{\mathcal{I}_{+}(y) - 1}{2} \right\rfloor + 1, \quad (5.105)$$

*and the least index of each parity that reaches the level  $y$  is*

$$N_{-}^{\geq}(y) = 2 \left\lceil \frac{\mathcal{I}_{-}(y)}{2} \right\rceil, \quad N_{+}^{\geq}(y) = 2 \left\lceil \frac{\mathcal{I}_{+}(y) - 1}{2} \right\rceil + 1. \quad (5.106)$$

*These identities are exact, not asymptotic. The quantity eq. (5.106) is the staircase  $N_{*}$  of definition 1.37, and the displayed formula for it is theorem 1.39's exact-rounding identity  $N_{*} = \lceil \nu \rceil$ , applied to the bisection  $(\text{sw}(2m + \epsilon))_{m \geq 0}$  with its admissible interpolation  $m \mapsto \mathcal{S}_{\sigma}(2m + \epsilon)$ .*

*Proof.* By eq. (5.11) and proposition 5.8,  $\mathcal{S}_{-}$  is a strictly increasing bijection agreeing with  $\text{sw}$  at the even integers, so for even  $n$  we have  $\text{sw}(n) \leq y \iff \mathcal{S}_{-}(n) \leq y \iff n \leq \mathcal{I}_{-}(y)$ . The largest such even  $n$  is  $2 \lfloor \mathcal{I}_{-}(y)/2 \rfloor$  and the least even  $n \geq \mathcal{I}_{-}(y)$  is  $2 \lceil \mathcal{I}_{-}(y)/2 \rceil$ . The odd case is identical after the shift  $n \mapsto n - 1$ . The boundary case in which  $\mathcal{I}_{\sigma}(y)$  is an integer of the relevant parity is covered, because then  $y$  is exactly a sequence value and floor and ceiling return it.  $\square$

The round-to-nearest inverses on the two parity classes are

$$n_{-}^{\text{near}}(y) = 2 \left\lfloor \frac{\mathcal{I}_{-}(y)}{2} + \frac{1}{2} \right\rfloor, \quad n_{+}^{\text{near}}(y) = 2 \left\lfloor \frac{\mathcal{I}_{+}(y) - 1}{2} + \frac{1}{2} \right\rfloor + 1. \quad (5.107)$$

**Remark 5.56** (The rounding must be certified, not trusted). Equation (5.105)–eq. (5.107) are exact statements about the real numbers  $\mathcal{I}_{\sigma}(y)$ , whereas a truncated transseries supplies only an approximation to them. When  $y$  lies exponentially close to a sequence value the floor can flip. The remedy is cheap: use the transseries to produce a candidate integer, then decide the inequality  $\text{sw}(n) \leq y$  exactly, either in integer arithmetic or by comparing  $\log \mathcal{S}_{\sigma}(n)$  with  $\log y$  at the candidate and its same-parity neighbour, working through  $\log \Gamma$  to stay in range. Theorem 1.39 is the general separation condition and theorem 1.42 converts the logarithmic residual into a certified bracket for  $\mathcal{I}_{\sigma}(y)$ .

### 5.10.2 Which parity wins, and by how much

**Theorem 5.57** (The sheet gap). *Put  $Y = \log(y\sqrt{\pi})$  and  $h = \log(Y/\lambda)$ , so that  $L_{\sigma} = Y + \sigma\lambda/2$ . Then, as  $y \rightarrow \infty$ ,*

$$\mathcal{I}_{-}(y) - \mathcal{I}_{+}(y) = \frac{h}{\lambda} - 1 + \frac{1}{2Y} + O\left(\frac{h^2}{Y^2}\right) = \log_2\left(\frac{\log(y\sqrt{\pi})}{\log 2}\right) - 1 + o(1). \quad (5.108)$$

*In particular the gap tends to  $+\infty$ : every sufficiently large level is reached first on the odd sheet, about  $\log_2 \log y$  indices before the even sheet catches up.*

*Proof.* Subtract the two instances of eq. (5.83) truncated after  $n = 1$ . The leading terms contribute  $(L_- - L_+)/\lambda = -1$ . Next,  $H_\sigma = \log((Y + \sigma\lambda/2)/\lambda) = h + \sigma\lambda/(2Y) + O(Y^{-2})$ , and the two contributions  $-(\sigma/2\lambda)H_\sigma$  combine to  $(H_- + H_+)/(\lambda) = h/\lambda + O(Y^{-2})$ , the  $\sigma$ -odd parts cancelling. Finally, with  $P_1^{(\sigma)} = (H - \sigma\lambda)/4$  from eq. (5.84),

$$\frac{1}{\lambda} \left[ \frac{P_1^{(-1)}(H_-)}{L_-} - \frac{P_1^{(+1)}(H_+)}{L_+} \right] = \frac{1}{4\lambda} \left[ \frac{H_- + \lambda}{L_-} - \frac{H_+ - \lambda}{L_+} \right] = \frac{1}{4\lambda} \cdot \frac{2\lambda}{Y} + O\left(\frac{h}{Y^2}\right) = \frac{1}{2Y} + O\left(\frac{h}{Y^2}\right).$$

Collecting the three displays gives eq. (5.108); the second form follows from  $h/\lambda = \log_2(Y/\lambda)$  and  $1/(2Y) \rightarrow 0$ . The error term  $O(h^2/Y^2)$  absorbs the  $n = 2$  terms of eq. (5.83), whose polynomials have degree 2 in  $H_\sigma$ .  $\square$

**Remark 5.58** (Numerical confirmation, and agreement with S1). Recomputed with eight Lambert-normalized corrections on each sheet: for  $\log y = 10, 30, 100, 400$  the gap  $\mathcal{I}_-(y) - \mathcal{I}_+(y)$  equals 2.97806, 4.47863, 6.18565, 8.17591, against the prediction  $h/\lambda - 1 + 1/(2Y)$  equal to 2.97829, 4.47928, 6.18583, 8.17593. S1 states the same result in its half index as  $N_0 - N_1 = \frac{2}{\alpha} \log \log y + \frac{2}{\alpha} \log \frac{2}{\alpha} - 1 + o(1)$  with  $\alpha = 2\lambda$ ; since  $\frac{2}{\alpha} = \frac{1}{\lambda}$  and  $\log \frac{2}{\alpha} = -\log \lambda$ , this reads  $\frac{1}{\lambda}(\log \log y - \log \lambda) - 1 + o(1)$ , which is eq. (5.108). Directly from the sequence, the least index reaching  $y = 10, 10^3, 10^6, 10^{12}$  is 5, 11, 21, 39 on the odd sheet against 6, 14, 24, 44 on the even sheet: gaps 1, 3, 3, 5, consistent with the very slow  $\log_2 \log y$  growth.

**Corollary 5.59** (Counting function). *Let  $N(y) := \#\{n \in \mathbb{N}_0 : \text{sw}(n) \leq y\}$ . Then*

$$N(y) = \frac{\mathcal{I}_-(y) + \mathcal{I}_+(y)}{2} + O(1) = \log_2(y\sqrt{\pi}) + O(1). \quad (5.109)$$

*Proof.* By proposition 5.55 the even indices counted are  $0, 2, \dots, 2\lfloor \mathcal{I}_-(y)/2 \rfloor$ , that is  $\mathcal{I}_-(y)/2 + O(1)$  of them, and similarly there are  $\mathcal{I}_+(y)/2 + O(1)$  odd ones; this is the first equality. For the second, average the two instances of eq. (5.83): as in the proof of theorem 5.57 the  $\sigma$ -odd term  $-(\sigma/2\lambda)H_\sigma$  cancels in the average, leaving  $\frac{1}{2\lambda}(L_- + L_+) + O(\log \log y / \log y) = Y/\lambda + O(1)$ , and  $Y/\lambda = \log_2(y\sqrt{\pi})$ .  $\square$

## 5.11 Algorithms, certification, and validation

### 5.11.1 Generating the coefficients

**Algorithm 5.60** (Exact coefficient generation). Every coefficient family of this chapter is produced by a finite exact recurrence over  $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$ :

1.  $d_{2r+1}$  from eq. (5.26) (rational Bernoulli input;  $d_{2r} = 0$ );
2.  $A_n$  from the triangular recurrence eq. (5.35);
3.  $c_n^{(\sigma)}$  from eq. (5.61), or by solving  $U = \Psi_\sigma(t, U)$  coefficientwise, or from the differential formula eq. (5.75);
4.  $P_n^{(\sigma)}(H)$  from theorem 1.33 with the dictionary eq. (5.21), or from eq. (5.95) after proposition 5.43;
5. the exact certificate of remark 5.42 after every step.

Truncating every intermediate series at order  $n$  is exact for the  $n$ -th coefficient, because all three recurrences are triangular; at order  $N$  the cost is  $\tilde{O}(N^2)$  coefficient operations. No numerical fitting and no order-specific ansatz occurs anywhere.

**Remark 5.61** (How this chapter's tables were produced). The three sources supply three implementations of the above. For this chapter every coefficient was regenerated by a fourth, independent one: truncated power-series arithmetic over exact rationals with  $\lambda$  symbolic, run separately at  $\sigma = \pm 1$  and recombined in  $\mathcal{R}$ . Every displayed coefficient agreed with the source that printed it; the corrections recorded in this chapter concern statements, never numbers.

### 5.11.2 Certified numerical inversion

**Proposition 5.62** (Derivative bounds for certification). *For  $x \geq 1$ ,*

$$(\log \mathcal{S}_+)'(x) > \lambda + \frac{1}{2x} - \frac{1}{4x^2} > \lambda, \quad (\log \mathcal{S}_-)'(x) > \lambda - \frac{1}{2x}, \quad (5.110)$$

*the exact value being eq. (5.14). Hence if  $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$  and  $(\log \mathcal{S}_\sigma)' \geq \mu > 0$  on the interval joining  $x_*$  and  $\mathcal{I}_\sigma(y)$ , then  $|x_* - \mathcal{I}_\sigma(y)| \leq |r_*|/\mu$  by theorem 1.42.*

*Proof.*  $(\log \mathcal{S}_\sigma)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$  by eq. (5.19); apply  $0 < -D'(x) < 1/(4x^2)$  from corollary 5.14, and  $1/(2x) - 1/(4x^2) > 0$  for  $x > 1/2$ .  $\square$

**Algorithm 5.63** (Branchwise inversion of a level). Given a large  $y$  and a parity  $\sigma$ :

1. Evaluate the Lambert core:  $X_+ = W_0(4\pi\lambda y^2)/(2\lambda)$  or  $X_- = -W_{-1}(-4\lambda/(\pi y^2))/(2\lambda)$ . *The branch choice is not optional* (proposition 5.32).
2. Add corrections  $\sum_n c_n^{(\sigma)} X^{-n}$  by Horner nesting in  $1/X$ , stopping at or before the least term rather than summing indefinitely; alternatively use the centred form eq. (5.78), whose bare core is already accurate to  $O(\mathcal{U}^{-2})$ .
3. Certify: compute  $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$  through  $\log \Gamma$  and bound the error by proposition 5.62. The inequalities eq. (5.56) and eq. (5.57) give an independent one-sided check that costs nothing.
4. For higher accuracy run Newton's method on the logarithmic equation,

$$x_{j+1} = x_j - \frac{\log \mathcal{S}_\sigma(x_j) - \log y}{\lambda + \frac{\sigma}{2} [\psi(x_j/2 + 1) - \psi(x_j/2 + 1/2)]},$$

which is stable even when  $\mathcal{S}_\sigma(x)$  is far outside floating-point range. Proposition 5.8 guarantees a positive denominator, and the transseries seed lies in the basin for large  $y$ .

5. For a discrete answer, round by eq. (5.105) and verify the final inequality exactly (remark 5.56).

### 5.11.3 Validation

The tables below were recomputed at 50 significant digits against exact gamma-quotient reference values, with  $y$  generated as  $\mathcal{S}_\sigma(x_0)$  so that the exact inverse is  $x_0$ . They reproduce the corresponding tables of S2 and S3 to every printed digit, and, after the reparametrization of remark 5.12, those of S1 as well.

Table 5.4: Relative error of the forward expansion eq. (5.36) after the term  $\sigma^M A_M x^{-M}$ . Recomputed.

| $x$ | $\sigma$ | $M = 0$                 | $M = 2$                 | $M = 4$                  | $M = 6$                  | $M = 8$                  |
|-----|----------|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|
| 10  | -1       | $2.5273 \times 10^{-2}$ | $3.8527 \times 10^{-5}$ | $4.7159 \times 10^{-7}$  | $1.4186 \times 10^{-8}$  | $7.8542 \times 10^{-10}$ |
| 10  | +1       | $2.4650 \times 10^{-2}$ | $3.8626 \times 10^{-5}$ | $4.7375 \times 10^{-7}$  | $1.4233 \times 10^{-8}$  | $7.8709 \times 10^{-10}$ |
| 20  | -1       | $1.2573 \times 10^{-2}$ | $4.8642 \times 10^{-6}$ | $1.5087 \times 10^{-8}$  | $1.1546 \times 10^{-10}$ | $1.6336 \times 10^{-12}$ |
| 20  | +1       | $1.2417 \times 10^{-2}$ | $4.8704 \times 10^{-6}$ | $1.5121 \times 10^{-8}$  | $1.1565 \times 10^{-10}$ | $1.6353 \times 10^{-12}$ |
| 50  | -1       | $5.0122 \times 10^{-3}$ | $3.1226 \times 10^{-7}$ | $1.5560 \times 10^{-10}$ | $1.9152 \times 10^{-13}$ | $4.3651 \times 10^{-16}$ |
| 50  | +1       | $4.9872 \times 10^{-3}$ | $3.1242 \times 10^{-7}$ | $1.5573 \times 10^{-10}$ | $1.9164 \times 10^{-13}$ | $4.3668 \times 10^{-16}$ |

**Warning 5.64** (These tables are not convergence). The decreasing entries at fixed  $x_0$  do not indicate convergence as  $K \rightarrow \infty$ . Both the forward and the inverse series are divergent, with the beyond-all-orders floor eq. (5.98); the tables stop far short of the least term (at  $x = 20$  the least term of eq. (5.30) sits at  $r = 31$ , of size  $10^{-28}$ ). The non-monotone entries in the odd column at  $x_0 = 10$  are ordinary behaviour of an asymptotic series at small argument, not an error. Beyond the least term one should use the exact Laplace representation eq. (5.22), or a Newton step against  $\log \Gamma$ , or an exponentially improved representation transported by proposition 5.52.

Table 5.5: Absolute error  $|\mathcal{I}_\sigma^{[K]}(\mathcal{S}_\sigma(x_0)) - x_0|$  of the Lambert-normalized inverse eq. (5.76) with  $K$  corrections retained;  $K = 0$  is the bare core. Recomputed.

| $x_0$ | $\sigma$ | $K = 0$                 | $K = 2$                 | $K = 4$                  | $K = 6$                  | $K = 8$                  |
|-------|----------|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|
| 10    | -1       | $3.8813 \times 10^{-2}$ | $1.6744 \times 10^{-5}$ | $2.9057 \times 10^{-7}$  | $2.4840 \times 10^{-8}$  | $1.7249 \times 10^{-9}$  |
| 10    | +1       | $3.3589 \times 10^{-2}$ | $2.2677 \times 10^{-4}$ | $3.9282 \times 10^{-6}$  | $5.5615 \times 10^{-8}$  | $2.0051 \times 10^{-9}$  |
| 20    | -1       | $1.8701 \times 10^{-2}$ | $1.2039 \times 10^{-6}$ | $6.9374 \times 10^{-9}$  | $1.8267 \times 10^{-10}$ | $3.2855 \times 10^{-12}$ |
| 20    | +1       | $1.7399 \times 10^{-2}$ | $3.0220 \times 10^{-5}$ | $1.3176 \times 10^{-7}$  | $4.7142 \times 10^{-10}$ | $4.3075 \times 10^{-12}$ |
| 50    | -1       | $7.3186 \times 10^{-3}$ | $4.2970 \times 10^{-8}$ | $5.6559 \times 10^{-11}$ | $2.8529 \times 10^{-13}$ | $8.3851 \times 10^{-16}$ |
| 50    | +1       | $7.1104 \times 10^{-3}$ | $2.0095 \times 10^{-6}$ | $1.4043 \times 10^{-9}$  | $8.1104 \times 10^{-13}$ | $1.1816 \times 10^{-15}$ |
| 100   | -1       | $3.6329 \times 10^{-3}$ | $3.9487 \times 10^{-9}$ | $1.6091 \times 10^{-12}$ | $2.1882 \times 10^{-15}$ | $1.6191 \times 10^{-18}$ |
| 100   | +1       | $3.5808 \times 10^{-3}$ | $2.5440 \times 10^{-7}$ | $4.4455 \times 10^{-11}$ | $6.4401 \times 10^{-15}$ | $2.3379 \times 10^{-18}$ |

Table 5.6: The same errors for the centred inverse eq. (5.78). The bare core is better by a factor  $\asymp X$ , because  $\widehat{c}_1^{(\sigma)} = 0$ ; once corrections are retained the two normalizations are comparable. Recomputed.

| $x_0$ | $\sigma$ | $K = 0$                 | $K = 2$                 | $K = 4$                  | $K = 6$                  | $K = 8$                  |
|-------|----------|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|
| 10    | -1       | $8.7330 \times 10^{-4}$ | $5.5308 \times 10^{-5}$ | $1.1593 \times 10^{-7}$  | $2.7056 \times 10^{-9}$  | $4.4842 \times 10^{-11}$ |
| 10    | +1       | $7.6102 \times 10^{-4}$ | $5.6715 \times 10^{-5}$ | $2.3489 \times 10^{-7}$  | $7.9121 \times 10^{-9}$  | $4.0347 \times 10^{-10}$ |
| 20    | -1       | $2.2206 \times 10^{-4}$ | $7.4914 \times 10^{-6}$ | $5.2119 \times 10^{-9}$  | $3.7661 \times 10^{-11}$ | $3.3070 \times 10^{-13}$ |
| 20    | +1       | $2.0696 \times 10^{-4}$ | $7.5956 \times 10^{-6}$ | $7.4297 \times 10^{-9}$  | $6.3564 \times 10^{-11}$ | $8.0880 \times 10^{-13}$ |
| 50    | -1       | $3.5860 \times 10^{-5}$ | $5.0352 \times 10^{-7}$ | $6.5130 \times 10^{-11}$ | $8.3102 \times 10^{-14}$ | $1.4333 \times 10^{-16}$ |
| 50    | +1       | $3.4850 \times 10^{-5}$ | $5.0641 \times 10^{-7}$ | $7.5153 \times 10^{-11}$ | $1.0250 \times 10^{-13}$ | $2.0275 \times 10^{-16}$ |
| 100   | -1       | $8.9913 \times 10^{-6}$ | $6.3982 \times 10^{-8}$ | $2.1684 \times 10^{-12}$ | $7.1217 \times 10^{-16}$ | $3.2358 \times 10^{-19}$ |
| 100   | +1       | $8.8632 \times 10^{-6}$ | $6.4167 \times 10^{-8}$ | $2.3300 \times 10^{-12}$ | $7.9120 \times 10^{-16}$ | $3.8477 \times 10^{-19}$ |

## 5.12 Summary of the coefficient hierarchy

The whole calculation is a chain of constructive coefficient transformations, each with a closed extraction formula and a triangular recurrence:

$$\boxed{B_{2r+2} \xrightarrow{\text{gamma ratio}} d_{2r+1} \xrightarrow{\exp, B} A_n \xrightarrow{\text{Lagrange, } \widehat{B}} c_n^{(\sigma)} \xrightarrow{\text{unfold } W} P_n^{(\sigma)}(H),} \quad (5.111)$$

while the Borel kernel  $\tanh(t/2)/(2t)$  supplies, in one stroke, the exact sum of the first stage, the sign and size of every remainder (theorem 5.20), the singulant  $\pi$ , the least-term scale, and the complete Stokes data. Two facts are specific to the swing factorial and cannot be relegated to the general theory of Part 1:

1. *Parity is leading-order data.* The forward object is a period-two transseries eq. (5.40) whose discrete transmonomial  $\sigma_n$  switches the algebraic scale between  $2^n n^{-1/2}$  and  $2^n n^{+1/2}$ ; the inverse object is a *pair* of monotone branches, and proposition 5.9 shows that this is forced, not chosen.
2. *The two sheets need different Lambert branches.*  $W_{-1}$  on the even sheet,  $W_0$  on the odd sheet, the assignment being forced by sign  $b = \sigma$  in theorem 1.20. The principal branch on the even sheet does not merely lose accuracy: by proposition 5.32 it returns the root  $2/(\pi y^2)(1 + O(y^{-2})) \rightarrow 0$ , so that choice is fatal before any correction is computed.



## Chapter 6

# Gamma and Barnes $G$

### 6.1 What this chapter does

The four subjects of Chapters 2–5 are *sequences*; the next two are *special functions*, and the change matters in exactly one place. A sequence has three inverse objects (definition 1.37) and the staircase theorem is needed to pass between them; a function analytic and eventually strictly monotone on a real ray has one, and the inverse asked for is the honest compositional inverse of a restriction. Everything else — the exponential–power bookkeeping, the exact inversion of a dominant block, the all-orders reversion around it, the backward-error certificate — is the apparatus of Chapter 0 and is cited, not repeated.

This chapter consolidates three filed articles, listed in table 6.1. They were written independently and they agree: every coefficient, table and decimal any of them prints was recomputed here in exact arithmetic or to 60 digits, and all of it reproduced. What the three do *not* share is a normalization — one absorbs the Barnes constant  $\zeta'(-1)$  into the target and two do not, and the three use three different letters for the slope — so section 6.8 gives the dictionary and the rest of the chapter fixes one convention.

| Tag | Directory                           | Distinctive content                                                                                           |
|-----|-------------------------------------|---------------------------------------------------------------------------------------------------------------|
| S1  | inverse_gamma_barnesG_transseries   | Hahn supports; Newton valuation doubling; the $\sqrt{T^2 + 1/6}$ subsector                                    |
| S2  | inverse_gamma_barnes_transseries    | the pole branches of $\Gamma^{-1}$ ; universal $C_0, C_1, C_2$ for general $p$ ; $\zeta'(-1)$ left unabsorbed |
| S3  | Asymptotic_Inversion_Gamma_Barnes_G | the kernel coefficients $\kappa_n(p, v)$ ; the slope denominator bound; resonance; $b_3, b_4, b_5$            |

Table 6.1: Parameter dictionary for this chapter.

|                       | $\Gamma$                                  | $G$                                                 |
|-----------------------|-------------------------------------------|-----------------------------------------------------|
| inverted map          | $x \mapsto \Gamma(x)$ , increasing branch | $z \mapsto G(z + 1)$ , eventual branch              |
| centred variable      | $t = x - \frac{1}{2}$                     | $z = x - 1$                                         |
| core exponent $p$     | 1                                         | 2                                                   |
| core scale $a$        | 1                                         | $\frac{1}{2}$                                       |
| core shift $\beta$    | 1                                         | $\frac{3}{2}$                                       |
| absorbed constant $C$ | $\frac{1}{2} \log(2\pi)$                  | $\zeta'(-1)$                                        |
| perturbation support  | $2\mathbb{N}$                             | $\mathbb{N}$                                        |
| first perturbation    | $a_1 t^{-1} = -1/(24t)$                   | $\alpha z, \alpha = \frac{1}{2} \log(2\pi)$         |
| core                  | $T = e^\Lambda, \Lambda = w + 1$          | $U = e^{\lambda + \mathbb{I}}, \lambda = (v + 1)/2$ |
| inverse support       | $T^{-1}, T^{-3}, T^{-5}, \dots$           | $U^0, U^{-1}, U^{-2}, \dots$                        |

Two features distinguish this chapter from the first four.

The dominant block is not the one Chapter 0 inverts. Theorem 1.20 solves  $aX + b \log X = L$  exactly; here the unknown *multiplies* the logarithm, and the block to be inverted is  $ax^p(\log x - \beta)$ . That block is still solved exactly by  $W$ , but by a different substitution, and the resulting normalized slope is  $v + 1$  rather than  $a + b/X$ . Section 6.3 proves the core lemma in the generality the two subjects and the next chapter need, namely for  $ax^p(\log x - \beta)^r$ .

The forward phase is a Stirling series, not an exponential–power model. As in Part 3, the phase  $\frac{1}{2}z^2 \log z$  grows faster than  $z^\alpha$  for every  $\alpha \leq 1$  and definition 1.2 does not apply. What does apply is the axiomatized dominant core; we say precisely which axioms hold in remark 6.4 and do not force the model.

The chapter also proves two things none of the three sources states. The first is that the slope–denominator bound of proposition 6.15 is *attained*, with the deepest pole given in closed form (theorem 6.17); the second is that the two square roots the sources find —  $S1$ ’s and  $S3$ ’s  $\sqrt{U^2 + 1/6}$  for Barnes, and the previously unnoticed  $\sqrt{\Lambda^2 T^2 + 1/12}$  for Gamma — are the same phenomenon, the exactly solvable quadratic truncation of the normalized equation (lemma 6.16).

## 6.2 Forward normal forms

Bernoulli numbers and polynomials follow the conventions of Chapter 0:  $B_n = B_n(0)$  and  $ze^{xz}/(e^z - 1) = \sum_{n \geq 0} B_n(x) z^n / n!$ , so that  $B_1 = -\frac{1}{2}$ ,  $B_2 = \frac{1}{6}$ ,  $B_4 = -\frac{1}{30}$ , and

$$B_n\left(\frac{1}{2}\right) = (2^{1-n} - 1)B_n. \quad (6.1)$$

### 6.2.1 The half-shifted Gamma phase

**Lemma 6.1** (Symmetry-adapted Gamma normal form). *Put  $t = x - \frac{1}{2}$ . In every closed subsector of  $|\arg t| < \pi$ , as  $t \rightarrow \infty$ ,*

$$\log \Gamma\left(t + \frac{1}{2}\right) \sim t(\log t - 1) + \frac{1}{2} \log(2\pi) + \sum_{k \geq 1} \frac{a_k}{t^{2k-1}}, \quad (6.2)$$

where

$$a_k = \frac{B_{2k}(1/2)}{2k(2k-1)} = \frac{(2^{1-2k} - 1)B_{2k}}{2k(2k-1)}. \quad (6.3)$$

Only odd inverse powers of  $t$  occur. The first values are

$$a_1 = -\frac{1}{24}, \quad a_2 = \frac{7}{2880}, \quad a_3 = -\frac{31}{40320}, \quad a_4 = \frac{127}{215040}, \quad a_5 = -\frac{511}{608256}. \quad (6.4)$$

*Proof.* The shifted Stirling expansion is, for fixed  $h$  and in the stated sectors,

$$\log \Gamma(t + h) \sim \left(t + h - \frac{1}{2}\right) \log t - t + \frac{1}{2} \log(2\pi) + \sum_{k \geq 2} \frac{(-1)^k B_k(h)}{k(k-1)t^{k-1}}. \quad (6.5)$$

Set  $h = \frac{1}{2}$ . The coefficient of  $\log t$  collapses to  $t$  exactly, which is the entire point of the shift. By (6.1) and  $B_{2m+1} = 0$  for  $m \geq 1$ , we get  $B_{2m+1}(\frac{1}{2}) = 0$  for all  $m \geq 0$ , so only even  $k$  survives; writing  $k = 2n$  and noting  $(-1)^{2n} = 1$  gives (6.2)–(6.3). The five values follow from  $B_2 = \frac{1}{6}$ ,  $B_4 = -\frac{1}{30}$ ,  $B_6 = \frac{1}{42}$ ,  $B_8 = -\frac{1}{30}$ ,  $B_{10} = \frac{5}{66}$ .  $\square$

**Remark 6.2** (The shift is not cosmetic). Expanding in  $x^{-1}$  instead of  $t^{-1}$  leaves both parities in the forward phase, hence a perturbation support  $\mathbb{N}$  rather than  $2\mathbb{N}$ , hence — by corollary 6.13 — an inverse displacement with every integer power of  $T^{-1}$  instead of only the odd ones. The shift is the cheapest available instance of the normalization rule of remark 6.10: translate first, and only then declare what is small.

### 6.2.2 The Barnes phase in the centred variable

**Lemma 6.3** (Simplified Barnes normal form). *In every closed subsector of  $|\arg z| < \pi$ , as  $z \rightarrow \infty$ ,*

$$\log G(z+1) \sim \frac{1}{2}z^2 \log z - \frac{3}{4}z^2 + \alpha z - \frac{1}{12} \log z + \zeta'(-1) + \sum_{k \geq 1} \frac{c_k}{z^{2k}}, \quad (6.6)$$

where

$$\alpha = \frac{1}{2} \log(2\pi), \quad c_k = \frac{B_{2k+2}}{4k(k+1)}, \quad (6.7)$$

so that  $c_1 = -\frac{1}{240}$ ,  $c_2 = \frac{1}{1008}$ ,  $c_3 = -\frac{1}{1440}$ ,  $c_4 = \frac{1}{1056}$ .

*Proof.* Start from the classical form

$$\log G(z+1) \sim \frac{1}{4}z^2 + z \log \Gamma(z+1) - \left( \frac{1}{2}z(z+1) + \frac{1}{12} \right) \log z - \log A + \sum_{k \geq 1} \frac{B_{2k+2}}{2k(2k+1)(2k+2)z^{2k}},$$

$A$  being the Glaisher–Kinkelin constant, and substitute

$$\log \Gamma(z+1) \sim \left( z + \frac{1}{2} \right) \log z - z + \frac{1}{2} \log(2\pi) + \sum_{j \geq 1} \frac{B_{2j}}{2j(2j-1)z^{2j-1}}.$$

Collect by type. The  $z^2 \log z$  terms give  $z^2 \log z - \frac{1}{2}z^2 \log z = \frac{1}{2}z^2 \log z$ ; the  $z \log z$  terms cancel,  $\frac{1}{2}z \log z - \frac{1}{2}z \log z = 0$ ; the  $z^2$  terms give  $-z^2 + \frac{1}{4}z^2 = -\frac{3}{4}z^2$ ; the linear term is  $\alpha z$ ; and the lone  $\log z$  term is  $-\frac{1}{12} \log z$ .

Two constants remain. Multiplying the Gamma tail by  $z$  moves its  $j = 1$  term to  $B_2/2 = \frac{1}{12}$ , a constant; together with  $-\log A$  and  $\log A = \frac{1}{12} - \zeta'(-1)$  this is  $\frac{1}{12} - \frac{1}{12} + \zeta'(-1) = \zeta'(-1)$ .

At  $z^{-2k}$  two contributions meet: the Gamma tail with  $j = k+1$ , multiplied by  $z$ , contributes  $B_{2k+2}/((2k+2)(2k+1))$ , and the explicit Barnes tail contributes  $B_{2k+2}/(2k(2k+1)(2k+2))$ . Their sum is

$$\frac{B_{2k+2}}{(2k+1)(2k+2)} \left( 1 + \frac{1}{2k} \right) = \frac{B_{2k+2}}{(2k+2)(2k)} = \frac{B_{2k+2}}{4k(k+1)},$$

which is (6.7). □

**Remark 6.4** (Not an exponential–power model). Neither phase is of the form of definition 1.2. For  $\Gamma$  the phase is  $t \log t$  and for  $G$  it is  $\frac{1}{2}z^2 \log z$ ; in both cases  $\text{phase}/x^\alpha \rightarrow \infty$  for every  $\alpha \leq 1$ , by the argument already given for the double factorial in warning 3.14. What both phases do satisfy is the axiomatized dominant core: each is  $C^\infty$  and eventually strictly increasing on a real ray, its derivative is eventually positive and regularly varying, and the residual after removing the core admits a Poincaré expansion with a differentiable remainder in sectors. Those are the only properties used below, and they are what theorems 1.42 and 1.44 require.

**Remark 6.5** (Which inverse). On  $(0, \infty)$  the digamma function  $\psi$  is strictly increasing, because  $\psi'(x) = \sum_{n \geq 0} (n+x)^{-2} > 0$ , and it runs from  $-\infty$  to  $+\infty$ ; so it has exactly one zero and  $\Gamma' = \Gamma\psi$  changes sign exactly once. Every sufficiently large  $y > 0$  therefore has exactly two positive preimages. We write  $\Gamma_+^{-1}$  for the branch tending to  $+\infty$ , which is the subject of section 6.5, and  $\Gamma_0^{-1}$  for the branch tending to  $0^+$ , treated in section 6.6. These are genuinely different objects: the first has a divergent asymptotic transseries, the second a convergent power series in  $1/y$ .

For Barnes,  $(\log G(z+1))' = \frac{1}{2} + z + \frac{1}{2} \log(2\pi) - z\psi(z+1) + \dots$ ; rather than track that expression we use the cleaner statement of proposition 6.6, which gives an explicit ray on which  $z \mapsto G(z+1)$  is strictly increasing.

**Proposition 6.6** (An explicit ray of monotonicity). *For  $z > 0$ ,*

$$\frac{d}{dz} \log G(z+1) = \frac{1}{2} \log(2\pi) - \frac{1}{2} - z + z\psi(z+1) = \frac{1}{2} \log(2\pi) + \frac{1}{2} - z + z\psi(z). \quad (6.8)$$

*Consequently  $z \mapsto G(z+1)$  is strictly increasing on  $[3, \infty)$ , and*

$$\frac{d}{dz} \log G(z+1) \geq \frac{1}{2} z \log z \quad (z \geq 7). \quad (6.9)$$

*It therefore has a well-defined inverse  $\mathcal{B} : [G(4), \infty) \rightarrow [3, \infty)$ , and  $G_+^{-1}(y) = 1 + \mathcal{B}(y)$ .*

*Proof.* Write the Weierstrass form

$$\log G(1+z) = \frac{z}{2} \log(2\pi) - \frac{z}{2} - \frac{z^2(1+\gamma)}{2} + \sum_{k \geq 1} \left[ k \log \left( 1 + \frac{z}{k} \right) - z + \frac{z^2}{2k} \right],$$

which converges locally uniformly, so it may be differentiated termwise:

$$\frac{d}{dz} \log G(1+z) = \frac{1}{2} \log(2\pi) - \frac{1}{2} - z(1+\gamma) + \sum_{k \geq 1} \left[ \frac{k}{k+z} - 1 + \frac{z}{k} \right] = \frac{1}{2} \log(2\pi) - \frac{1}{2} - z(1+\gamma) + z \sum_{k \geq 1} \left[ \frac{1}{k} - \frac{1}{k+z} \right].$$

The last sum is  $\psi(z+1) + \gamma$ , which gives the first form in (6.8); the second follows from  $\psi(z+1) = \psi(z) + 1/z$ .

Now  $\psi(z+1) = \psi(z) + 1/z > \log z - 1/z + 1/z = \log z$ , using the standard bound  $\psi(z) > \log z - 1/z$ . Hence, with  $c := \frac{1}{2} \log(2\pi) - \frac{1}{2} = 0.41893\dots$ ,

$$\frac{d}{dz} \log G(z+1) > z(\log z - 1) + c \quad (z > 0). \quad (6.10)$$

The right-hand side has derivative  $\log z$ , so it increases on  $[1, \infty)$ ; at  $z = 3$  its value is  $3(\log 3 - 1) + c = 0.7147\dots > 0$ . This proves strict monotonicity on  $[3, \infty)$ .

For (6.9) it suffices, by (6.10), that  $z(\log z - 1) + c \geq \frac{1}{2} z \log z$ , i.e. that  $\frac{1}{2} \log z \geq 1 - c/z$ . The left side increases and the right side decreases, so it is enough to check  $z = 7$ :  $\frac{1}{2} \log 7 = 0.9729\dots$  and  $1 - c/7 = 0.9401\dots$   $\square$

*Remark 6.7* (How sharp the threshold is). The bound (6.10) is lossy. Solving (6.8) numerically, the derivative has its only positive zero at  $z = 1.55766\dots$ , so  $G(\cdot + 1)$  is in fact strictly increasing on  $[1.56, \infty)$ , and  $\frac{1}{2} z \log z$  is dominated from  $z = 5.5$  onward. The thresholds 3 and 7 are what the elementary argument gives; they are stated because a proof is wanted, not because they are best possible.

All three sources assert the existence of the eventual Barnes inverse with the words “ $(\log G)'(x) \sim x \log x$ , hence strictly increasing for all sufficiently large  $x$ ”. That is true but ineffective, and none of the three supplies a threshold; yet every numerical table in all three evaluates the inverse at arguments as small as  $z = 5$ , inside the range their own justification does not cover. Proposition 6.6 supplies a threshold, and  $3 < 5$ , so the tables are legitimate.

## 6.3 The power–logarithmic core

Theorem 1.20 inverts  $aX + b \log X$ . The block needed here is different and is inverted by a different substitution.

**Lemma 6.8** (Exact inversion of a power–logarithmic block). *Let  $a \neq 0$ ,  $p \neq 0$ ,  $r \neq 0$  and  $\beta \in \mathbb{C}$ , and put*

$$\Phi_0(x) = a x^p (\log x - \beta)^r. \quad (6.11)$$

*Fix  $R$  and compatible determinations of the logarithm, of the  $r$ -th root and of a Lambert branch  $W_k$ , and set*

$$v = W_k \left( \frac{p}{r} \left( \frac{R}{a e^{p\beta}} \right)^{1/r} \right), \quad u = \frac{r}{p} v, \quad X = e^{\beta+u}. \quad (6.12)$$

*Then  $\Phi_0(X) = R$ , and*

$$\log X - \beta = u, \quad \Phi'_0(X) = a X^{p-1} u^{r-1} (pu + r). \quad (6.13)$$

*In the case  $r = 1$  these read*

$$v = W_k \left( \frac{pR}{a} e^{-p\beta} \right), \quad X = e^{\beta+v/p} = \left( \frac{pR}{av} \right)^{1/p}, \quad \Phi'_0(X) = a X^{p-1} (v + 1). \quad (6.14)$$

*Proof.* By construction  $\log X - \beta = u$ , so  $\Phi_0(X) = a e^{p\beta} e^{pu} u^r$ . Hence  $\Phi_0(X) = R$  is equivalent to  $u^r e^{pu} = R/(a e^{p\beta})$ , and taking the  $r$ -th root, to  $u e^{pu/r} = (R/(a e^{p\beta}))^{1/r}$ . Multiplying by  $p/r$  turns this into  $(pu/r) e^{pu/r} = (p/r) (R/(a e^{p\beta}))^{1/r}$ , i.e.  $ve^v = (p/r) (\dots)^{1/r}$  with  $v = pu/r$ , which is (6.12).

For the slope, differentiate (6.11):

$$\Phi'_0(x) = a x^{p-1} (\log x - \beta)^{r-1} (p(\log x - \beta) + r),$$

and evaluate at  $x = X$ , where  $\log X - \beta = u$ . For  $r = 1$ ,  $u = v/p$  and  $pu + r = v + 1$ .  $\square$

**Remark 6.9** (The universal divisor). The factor  $v + 1$  that divides every inverse coefficient below is not an algebraic accident: by (6.14) it is the normalized slope  $\Phi'_0(X)/(aX^{p-1})$  of the exact core. Its vanishing at  $v = -1$  is the Lambert branch point, where the core map has a critical point and no inverse of this shape can exist. Every statement in this chapter that requires “ $v + 1$  invertible” is requiring distance from that turning point, and every error bound degrades as  $v + 1 \rightarrow 0$ .

**Remark 6.10** (What must be absorbed). A term of the *same* algebraic power as the core is not small and must be absorbed before anything is called a perturbation:  $ax^p \log x + cx^p = ax^p(\log x - \beta)$  with  $\beta = -c/a$ . Likewise an additive constant  $C$  in the forward phase is moved to the target,  $R := Y - C$ . Failing to absorb makes the normalized perturbation have valuation zero and destroys the hypothesis of theorem 6.12; failing to absorb the constant merely shifts the core, but by an amount that is not small compared with the corrections one is trying to compute. A phase  $x^p(a_r(\log x)^r + a_{r-1}(\log x)^{r-1} + \dots)$  with  $r \geq 2$  needs a genuine outer inversion in the logarithmic scale first; only its leading logarithmic power is covered by lemma 6.8.

## 6.4 Normalized displacement and triangular inversion

### 6.4.1 The universal kernel

Take  $r = 1$  from here on, and consider a phase

$$\Phi(x) = a x^p (\log x - \beta) + C + \Psi(x), \quad (6.15)$$

to be solved for  $\Phi(x) = Y$ . Put  $R = Y - C$ , construct  $v$  and  $X$  by (6.14), write  $q = X^{-1}$ , and substitute

$$x = X(1 + \epsilon). \quad (6.16)$$

Dividing  $\Phi(x) - Y$  by  $aX^p$  turns the equation into

$$\mathcal{K}_{p,v}(\epsilon) + \mathcal{R}(q, v, \epsilon) = 0, \quad (6.17)$$

where

$$\mathcal{K}_{p,v}(\epsilon) = (1 + \epsilon)^p \left( \frac{v}{p} + \log(1 + \epsilon) \right) - \frac{v}{p}, \quad \mathcal{R}(q, v, \epsilon) = \frac{\Psi(X(1 + \epsilon))}{aX^p}. \quad (6.18)$$

**Lemma 6.11** (Kernel coefficients). *Write  $\mathcal{K}_{p,v}(\epsilon) = \sum_{n \geq 1} \kappa_n(p, v) \epsilon^n$ . Then*

$$\kappa_n(p, v) = \frac{v}{p} \binom{p}{n} + \frac{\partial}{\partial p} \binom{p}{n}, \quad (6.19)$$

*a polynomial in  $p$  and  $v$ ; in particular*

$$\kappa_1(p, v) = v + 1. \quad (6.20)$$

*Where  $p, p-1, \dots, p-n+1$  are all nonzero one may write  $\partial_p \binom{p}{n} = \binom{p}{n} \sum_{j=0}^{n-1} (p-j)^{-1}$ , but (6.19) is the form valid at every  $p$ . For  $p=1$ ,*

$$\mathcal{K}_{1,v}(\epsilon) = v\epsilon + (1+\epsilon) \log(1+\epsilon) = (v+1)\epsilon + \sum_{m \geq 2} \frac{(-1)^m}{m(m-1)} \epsilon^m, \quad (6.21)$$

*and for  $p=2$ ,*

$$\mathcal{K}_{2,v}(\epsilon) = (v+1)\epsilon + \frac{v+3}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 - \frac{1}{12}\epsilon^4 + \frac{1}{30}\epsilon^5 - \dots \quad (6.22)$$

*Proof.* Since  $(1+\epsilon)^p \log(1+\epsilon) = \partial_p(1+\epsilon)^p$ , we have  $[\epsilon^n]((1+\epsilon)^p \log(1+\epsilon)) = \partial_p \binom{p}{n}$ , while  $[\epsilon^n]((v/p)(1+\epsilon)^p) = (v/p) \binom{p}{n}$ ; the constant  $-v/p$  affects only  $n=0$ . This is (6.19). For  $n=1$ ,  $\binom{p}{1} = p$  and  $\partial_p \binom{p}{1} = 1$ , so  $\kappa_1 = v+1$ . Both displayed specializations follow by direct expansion; note that the coefficient of  $\epsilon^2$  in (6.22) is  $(v/2) \binom{2}{2} \cdot 2 + \partial_p \binom{p}{2}|_{p=2} = \frac{v}{2} + \frac{3}{2} = \frac{v+3}{2}$ .  $\square$

### 6.4.2 Triangular inversion over a well-ordered support

**Theorem 6.12** (Triangular inversion on a Hahn grid). *Let  $\mathcal{C}$  be a field of characteristic zero containing  $p$ ,  $v$  and  $(v+1)^{-1}$ , let  $S \subset \mathbb{R}_{\geq 0}$  be a well-ordered additive submonoid, and let  $\mathcal{C}[[q^S]]$  be the corresponding Hahn ring. Assume*

$$\mathcal{R}(q, v, E) = \sum_{\rho \in S, \rho > 0} \sum_{m \geq 0} r_{\rho, m}(v) q^\rho E^m, \quad (6.23)$$

*with every monomial of positive  $q$ -exponent. Then (6.17) has a unique solution  $\epsilon$  of positive valuation in  $\mathcal{C}[[q^S]]$ , say  $\epsilon = \sum_{\gamma \in S, \gamma > 0} e_\gamma(v) q^\gamma$ , and writing  $\epsilon_{<\gamma}$  for its truncation below  $\gamma$ ,*

$$e_\gamma(v) = -\frac{1}{v+1} [q^\gamma] \left( \sum_{n \geq 2} \kappa_n(p, v) \epsilon_{<\gamma}^n + \mathcal{R}(q, v, \epsilon_{<\gamma}) \right). \quad (6.24)$$

*Proof.* Well-ordering is exactly what makes the coefficient extraction in (6.24) a finite sum. By Neumann's lemma, if  $A, B \subset \mathbb{R}$  are well ordered then  $A+B$  is well ordered and each of its elements has only finitely many representations  $a+b$  with  $a \in A, b \in B$ ; by induction the same holds for any finite number of summands. Since  $\text{val}(\epsilon) > 0$ , we get  $\text{val}(\epsilon^n) \geq n \text{val}(\epsilon) \rightarrow \infty$ , so at a prescribed exponent  $\gamma$  only finitely many  $n$  contribute, and for each such  $n$  only finitely many decompositions. The same applies to  $\mathcal{R}$  because every one of its monomials has  $\rho > 0$ .

Now fix  $\gamma$  and suppose  $e_\delta$  is known for all  $0 < \delta < \gamma$ . In the left-hand side of (6.17), the coefficient of  $q^\gamma$  receives  $\kappa_1 e_\gamma = (v+1)e_\gamma$  from the linear term, and from nothing else: a monomial of  $\epsilon^n$  with  $n \geq 2$  that involves  $e_\gamma$  has  $q$ -exponent at least  $\gamma + \text{val}(\epsilon) > \gamma$ , and a monomial  $q^\rho E^m$  of  $\mathcal{R}$  involving  $e_\gamma$  has  $q$ -exponent at least  $\rho + \gamma > \gamma$ . Solving the resulting linear equation gives (6.24); transfinite induction along the well-ordered support constructs  $\epsilon$ , and the same computation run on the difference of two solutions shows the first exponent at which they could differ satisfies  $(v+1)(e_\gamma - \tilde{e}_\gamma) = 0$ , so there is none.  $\square$

S1 states the same theorem under the hypothesis that the support semigroup is *locally finite*, meaning that every bounded interval meets it in a finite set. That is strictly stronger than well-ordering —  $\{1 - 1/n\} \cup \{1\}$  is well ordered and not locally finite — and it is more than the proof uses. What the

argument needs is precisely the finite-decomposition property, which Neumann's lemma supplies for every well-ordered set. The hypothesis is weakened accordingly here. For the two subjects of this chapter the distinction is invisible, since  $S = 2\mathbb{N}$  and  $S = \mathbb{N}$ ; it matters for the fractional and mixed grids of section 6.12.

**Corollary 6.13** (Support semigroup). *If the  $q$ -support of  $\mathcal{R}(q, v, 0)$  lies in  $A \subset \mathbb{R}_{>0}$  and no monomial of  $\mathcal{R}$  has a  $q$ -exponent outside the monoid generated by  $A$ , then  $\text{supp } \epsilon$  lies in the additive semigroup generated by  $A$ .*

*Proof.* Equation (6.24) produces  $e_\gamma$  from sums of products of earlier coefficients and of base exponents from  $A$ ; the set of exponents so reachable is closed under addition and contains  $A$ . Induction along the support.  $\square$

**Example 6.14** (The two supports). For  $\Gamma$  we have  $p = a = \beta = 1$ , and the perturbation  $a_k t^{-(2k-1)}$  normalized by  $aX^p = T$  becomes  $a_k q^{2k}$  up to a factor  $(1 + \epsilon)^{-(2k-1)}$ ; so  $A = \{2, 4, 6, \dots\}$ , the generated semigroup is  $2\mathbb{N}$ , and the displacement  $t - T = T\epsilon$  contains only the odd powers  $T^{-1}, T^{-3}, \dots$ . For  $G$  we have  $p = 2$ ,  $a = \frac{1}{2}$ ,  $\beta = \frac{3}{2}$ , and the linear forward term  $\alpha z$  normalized by  $\frac{1}{2}U^2$  becomes  $2\alpha q$ ; so  $1 \in A$ , the semigroup is all of  $\mathbb{N}$ , and every integer power occurs.

### 6.4.3 How deep the slope divisor goes

Every coefficient is produced by one division by  $v + 1$ , but earlier coefficients enter nonlinearly and carry their own divisions. The accumulation is bounded, and — this is new — the bound is attained.

**Proposition 6.15** (Slope-denominator bound). *Suppose  $S = \mathbb{N}$ , that the coefficients  $r_{\rho, m}$  of (6.23) lie in  $A[v]$  for a commutative ring  $A \supseteq \mathbb{Q}$ , and that none of them involves  $(v + 1)^{-1}$ . If  $\epsilon = \sum_{n \geq 1} e_n q^n$  is the solution of theorem 6.12, then*

$$(v + 1)^{2n-1} e_n \in A[v] \quad (n \geq 1). \quad (6.25)$$

*Proof.* Induction on  $n$ . For  $n = 1$ , (6.24) gives  $e_1 = -r_{1,0}/(v + 1)$ , and  $(v + 1)e_1 \in A[v]$ .

Let  $n \geq 2$  and assume the claim below  $n$ . In (6.24) the bracket is a sum of two kinds of terms. A term of  $\kappa_m(p, v)\epsilon^m$  with  $m \geq 2$  contributing to  $q^n$  is  $\kappa_m$  times a product  $e_{j_1} \cdots e_{j_m}$  with  $j_1 + \cdots + j_m = n$  and every  $j_i \geq 1$ ; since  $\kappa_m \in \mathbb{Q}[p, v]$  carries no divisor, the inductive hypothesis bounds its denominator exponent by

$$\sum_{i=1}^m (2j_i - 1) = 2n - m \leq 2n - 2.$$

A term  $r_{\rho, m} q^\rho E^m$  contributing to  $q^n$  has  $j_1 + \cdots + j_m = n - \rho$  with  $\rho \geq 1$ , so its denominator exponent is at most  $2(n - \rho) - m \leq 2n - 2$  as well (and at most 0 when  $m = 0$ ). Hence the bracket has denominator exponent at most  $2n - 2$ , and the final division by  $v + 1$  in (6.24) gives at most  $2n - 1$ .  $\square$

The next two statements are the chapter's first new contribution. They say that the bound of proposition 6.15 is exactly attained, and identify the coefficient of the deepest pole in closed form, in both subjects at once.

**Lemma 6.16** (The quadratic core). *Let  $A$  and  $c \neq 0$  be elements of a commutative  $\mathbb{Q}$ -algebra with  $A$  invertible, let  $\sigma$  be another element, and let  $\epsilon = \sum_{n \geq 1} \epsilon_n q^n$  be the unique solution of positive valuation of the quadratic model*

$$A\epsilon + c\epsilon^2 + \sigma q = 0. \quad (6.26)$$

*Then*

$$\epsilon = \frac{A}{2c} \left( \sqrt{1 - \frac{4c\sigma q}{A^2}} - 1 \right), \quad \epsilon_n = \frac{(-4c\sigma)^n}{2c} \binom{1/2}{n} A^{-(2n-1)}. \quad (6.27)$$

*In particular the denominator exponent of  $\epsilon_n$  in  $A$  is exactly  $2n - 1$  whenever  $c\sigma \neq 0$ .*

*Proof.* Solving the quadratic and choosing the root that vanishes at  $q = 0$  gives

$$\epsilon = \frac{-A + \sqrt{A^2 - 4c\sigma q}}{2c},$$

which is the first formula. Expanding  $\sqrt{1+w} = \sum_{n \geq 0} \binom{1/2}{n} w^n$  at  $w = -4c\sigma q/A^2$  and multiplying by  $A/(2c)$  gives the second.  $\square$

**Theorem 6.17** (The deepest pole is attained, with an explicit coefficient).

1. Gamma. For  $n \geq 1$  let  $d_{2n-1}(\Lambda)$  be the coefficients of theorem 6.20. Then

$$[\Lambda^{-(2n-1)}] d_{2n-1}(\Lambda) = \binom{1/2}{n} 12^{-n} \neq 0, \quad (6.28)$$

so the bound of proposition 6.15 is attained at every  $n$ , and the sum of the deepest-pole terms of the displacement is the convergent series

$$\sum_{n \geq 1} \binom{1/2}{n} 12^{-n} (\Lambda T)^{1-2n} = \sqrt{\Lambda^2 T^2 + \frac{1}{12}} - \Lambda T \quad (\Lambda^2 T^2 > \frac{1}{12}). \quad (6.29)$$

2. Barnes. For  $n \geq 1$  let  $e_n(\lambda)$  be the coefficients of theorem 6.28. Then

$$[\lambda^{-(2n-1)}] e_n(\lambda) = \binom{1/2}{n} (-2\alpha)^n \neq 0, \quad (6.30)$$

and the deepest-pole part of the displacement  $z - U$  sums to

$$\sum_{n \geq 1} \binom{1/2}{n} (-2\alpha)^n \lambda^{1-2n} U^{1-n} = \sqrt{\lambda^2 U^2 - 2\alpha U} - \lambda U \quad (\lambda^2 U > 2\alpha). \quad (6.31)$$

In both cases the deepest-pole family is the coefficient family of the quadratic model (6.26), with

$$(A, c, \sigma) = (\Lambda, \frac{1}{2}, a_1) \text{ for } \Gamma, \quad (A, c, \sigma) = (2\lambda, 1, 2\alpha) \text{ for } G. \quad (6.32)$$

*Proof.* Both parts have the same shape. Write  $\pi_n$  for the coefficient of the deepest possible pole,  $\pi_n := [A^{-(2n-1)}] e_n$ , which is well defined by proposition 6.15; the claim is that  $(\pi_n)$  obeys the recursion of the quadratic model. Everything turns on which terms of the normalized equation can reach pole order  $2n - 2$  after the coefficient  $e_n$  has been isolated – for the recurrence reads  $Ae_n = -(\text{bracket})$ , and  $A \cdot \pi_n A^{-(2n-1)} = \pi_n A^{-(2n-2)}$ .

*Gamma.* Here  $A = \Lambda = v + 1$  and, by (6.21), the normalized equation on the grid  $q^2$  is

$$\Lambda \epsilon + \frac{1}{2} \epsilon^2 - \frac{1}{6} \epsilon^3 + \cdots + \sum_{k \geq 1} a_k q^{2k} (1 + \epsilon)^{-(2k-1)} = 0. \quad (6.33)$$

Index by  $n$ , so that  $\epsilon = \sum_n e_n q^{2n}$  and  $e_n$  has pole order at most  $2n - 1$ . The term  $\frac{1}{2} \epsilon^2$  contributes at  $q^{2n}$  the sum  $\frac{1}{2} \sum_{j=1}^{n-1} e_j e_{n-j}$ , of pole order at most  $(2j - 1) + (2(n - j) - 1) = 2n - 2$  with top coefficient  $\frac{1}{2} \sum_{j=1}^{n-1} \pi_j \pi_{n-j}$ . A term  $\epsilon^m$  with  $m \geq 3$  has pole order at most  $2n - m \leq 2n - 3$ . In the Bernoulli sum, the summand  $k = n$  contributes the constant  $a_n$ , of pole order 0, which is  $2n - 2$  only when  $n = 1$ ; and a summand with  $1 \leq k \leq n - 1$  contributes  $a_k$  times a product of  $m \geq 1$  coefficients with indices summing to  $n - k$ , of pole order at most  $2(n - k) - m \leq 2n - 3$ . Hence for  $n \geq 2$

$$\pi_n = -\frac{1}{2} \sum_{j=1}^{n-1} \pi_j \pi_{n-j}, \quad \text{while} \quad \pi_1 = -a_1 = \frac{1}{24}.$$



Let  $g(w) = \sum_{n \geq 1} \pi_n w^n$ . The recursion says  $g = -a_1 w - \frac{1}{2} g^2$ , i.e.  $g^2 + 2g + 2a_1 w = 0$ , whence  $g = -1 + \sqrt{1 - 2a_1 w} = -1 + \sqrt{1 + w/12}$  and  $\pi_n = \binom{1/2}{n} 12^{-n}$ , which is (6.28); it is nonzero because  $\binom{1/2}{n} \neq 0$  for all  $n$ . This is exactly lemma 6.16 with  $(A, c, \sigma) = (\Lambda, \frac{1}{2}, a_1)$ : indeed  $(-4 \cdot \frac{1}{2} \cdot a_1)^n / (2 \cdot \frac{1}{2}) = (1/12)^n$ .

Since  $t - T = T\epsilon$ , the deepest-pole terms of the displacement are

$$\pi_n \Lambda^{-(2n-1)} T^{1-2n} = \binom{1/2}{n} 12^{-n} (\Lambda T)^{1-2n},$$

and summing the binomial series at  $w = 1/(12\Lambda^2 T^2)$  gives

$$\Lambda T \left( \sqrt{1 + \frac{1}{12\Lambda^2 T^2}} - 1 \right) = \sqrt{\Lambda^2 T^2 + \frac{1}{12}} - \Lambda T,$$

convergent for  $12\Lambda^2 T^2 > 1$ . This is (6.29).

*Barnes.* Here  $A = 2\lambda$  and, by (6.22) with  $v = 2\lambda - 1$  so that  $(v + 3)/2 = \lambda + 1$ , the normalized equation on the grid  $q$  is

$$2\lambda\epsilon + (\lambda + 1)\epsilon^2 + \frac{1}{3}\epsilon^3 - \cdots + 2\alpha q(1 + \epsilon) + 2bq^2(\lambda + 1 + \log(1 + \epsilon)) + 2 \sum_{k \geq 1} c_k q^{2k+2} (1 + \epsilon)^{-2k} = 0, \quad (6.34)$$

with  $b = -\frac{1}{12}$ . Pole orders are now taken in  $\lambda$ , and  $e_n$  has order at most  $2n - 1$ ; we must collect everything of order  $2n - 2$ .

The quadratic term  $(\lambda + 1)\epsilon^2$  splits. Its part  $1 \cdot \epsilon^2$  contributes  $\sum_{j=1}^{n-1} \pi_j \pi_{n-j}$  at order  $2n - 2$ . Its part  $\lambda\epsilon^2$  multiplies a quantity of order at most  $2n - 2$  by  $\lambda$ , giving order at most  $2n - 3$ ; it therefore does *not* reach  $2n - 2$ . Terms  $\epsilon^m$  with  $m \geq 3$  reach at most  $2n - 3$ .

In the perturbation,  $2\alpha q \cdot 1$  contributes only at  $n = 1$ , where its order 0 equals  $2n - 2 = 0$ ;  $2\alpha q\epsilon$  contributes  $2\alpha e_{n-1}$ , of order at most  $2n - 3$ . The term  $2b\lambda q^2$  has no pole at all, and  $2bq^2$  and  $2bq^2 \log(1 + \epsilon)$  have order at most  $2(n - 2) - 1$ , while multiplying the logarithm by  $\lambda$  lowers the order further; none reaches  $2n - 2$ . The same holds for every  $c_k$  term. Hence for  $n \geq 2$

$$2\pi_n = - \sum_{j=1}^{n-1} \pi_j \pi_{n-j}, \quad \text{while} \quad 2\pi_1 = -2\alpha, \quad \pi_1 = -\alpha.$$

With  $g(w) = \sum \pi_n w^n$  this reads  $2g = -2\alpha w - g^2$ , so  $g = -1 + \sqrt{1 - 2\alpha w}$  and  $\pi_n = \binom{1/2}{n} (-2\alpha)^n$ , which is (6.30) and agrees with lemma 6.16 at  $(A, c, \sigma) = (2\lambda, 1, 2\alpha)$ . Since  $z - U = U\epsilon$ , the deepest-pole terms are  $\pi_n \lambda^{-(2n-1)} U^{1-n}$ , and

$$\sum_{n \geq 1} \binom{1/2}{n} (-2\alpha)^n \lambda^{1-2n} U^{1-n} = \lambda U \sum_{n \geq 1} \binom{1/2}{n} \left( \frac{-2\alpha}{\lambda^2 U} \right)^n = \lambda U \left( \sqrt{1 - \frac{2\alpha}{\lambda^2 U}} - 1 \right),$$

which is (6.31). □

*Remark 6.18 (Two square roots, one mechanism).* S1 and S3 both isolate a square root in the Barnes problem, namely the coefficientwise limit  $\sqrt{U^2 + 1/6}$  of proposition 6.30 below, and both present it as a curiosity of Barnes  $G$  caused by the resonant  $-\frac{1}{12} \log z$  term. Theorem 6.17 shows it is not special to Barnes and not special to resonance. Every problem in this class has a distinguished exactly solvable truncation — keep the slope, the quadratic kernel coefficient, and the lowest forcing monomial, discard the rest — and that truncation is a quadratic, so it always resums to a square root. Which *extraction* of the coefficient family it computes depends on the subject: for Barnes, discarding by  $\lambda$ -degree gives the resonant limit (6.68); discarding by pole depth gives (6.31); for Gamma there is no resonance and only the second extraction exists. The three square roots

$$\sqrt{U^2 + \frac{1}{6}}, \quad \sqrt{\lambda^2 U^2 - 2\alpha U}, \quad \sqrt{\Lambda^2 T^2 + \frac{1}{12}}$$

are the same lemma applied three times.

*Remark 6.19* (What the deepest pole is good for). Two things. First, it is a sharp regression test: a symbolic implementation that produces  $d_{2n-1}$  or  $e_n$  can check the single rational number  $[A^{-(2n-1)}]$  against (6.28) or (6.30) at negligible cost, and that number is exactly the one most sensitive to an error committed early in the recursion, since it is the accumulation of all previous divisions. Second, it locates the breakdown: the expansions of this chapter are asymptotic in  $q$  but their coefficients degrade like  $(v+1)^{-(2n-1)}$ , so the useful range in  $n$  at a given distance from the branch point  $v = -1$  is governed by  $\Lambda T \gg 1$  resp.  $\lambda\sqrt{U} \gg 1$  – and by theorem 6.17 those are precisely the conditions under which the displayed square roots converge.

## 6.5 The increasing inverse of Gamma

### 6.5.1 Core variables

Let  $y \rightarrow +\infty$  and put

$$R = \log y - \frac{1}{2} \log(2\pi), \quad w = W_0(R/e), \quad T = \frac{R}{w} = e^{w+1}, \quad \Lambda = \log T = w + 1. \quad (6.35)$$

These are lemma 6.8 at  $p = a = \beta = r = 1$ : the core equation is

$$T(\log T - 1) = R, \quad (6.36)$$

its slope is  $\Lambda$ , and the three identities in (6.35) are equivalent to  $we^w = R/e$ . Write  $q = T^{-1}$  and  $t = T(1 + \epsilon)$ . Subtracting (6.36) from lemma 6.1 and dividing by  $T$  gives the exact normalized equation

$$\Lambda\epsilon + [(1 + \epsilon) \log(1 + \epsilon) - \epsilon] + \sum_{k \geq 1} a_k q^{2k} (1 + \epsilon)^{-(2k-1)} = 0. \quad (6.37)$$

### 6.5.2 The all-orders theorem

**Theorem 6.20** (All-orders expansion of  $\Gamma_+^{-1}$ ). *Let  $\Gamma_+^{-1}$  be the branch of remark 6.5 tending to  $+\infty$ , and let  $T, \Lambda$  be as in (6.35). There are uniquely determined  $d_{2n-1} \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$  such that, for every fixed  $N \geq 0$ ,*

$$\Gamma_+^{-1}(y) = \frac{1}{2} + T + \sum_{n=1}^N \frac{d_{2n-1}(\Lambda)}{T^{2n-1}} + \mathcal{O}_N\left(\frac{1}{\Lambda T^{2N+1}}\right), \quad y \rightarrow +\infty. \quad (6.38)$$

Writing  $\epsilon = \sum_{n \geq 1} d_{2n-1}(\Lambda) q^{2n}$  and  $E_{n-1} = \sum_{j < n} d_{2j-1} q^{2j}$ , they are computed by

$$d_{2n-1}(\Lambda) = -\frac{1}{\Lambda} [q^{2n}] \left[ (1 + E_{n-1}) \log(1 + E_{n-1}) - E_{n-1} + \sum_{k=1}^n a_k q^{2k} (1 + E_{n-1})^{-(2k-1)} \right], \quad (6.39)$$

and they satisfy

$$\deg_{\Lambda^{-1}} d_{2n-1} \leq 2n - 1, \quad d_{2n-1}(\Lambda) = -\frac{a_n}{\Lambda} + \mathcal{O}(\Lambda^{-2}). \quad (6.40)$$

Only the odd powers  $T^{-1}, T^{-3}, T^{-5}, \dots$  occur.

*Proof. Formal part.* Equation (6.37) is (6.17) with  $p = a = \beta = 1$ , support  $S = 2\mathbb{N}$  by example 6.14, and linear coefficient  $\kappa_1 = v + 1 = \Lambda$ ; so theorem 6.12 gives a unique formal solution and (6.39). The parity claim is corollary 6.13 together with  $t - T = T\epsilon$ : an exponent  $q^{2n}$  in  $\epsilon$  is an absolute power  $T^{1-2n}$ .

For (6.40), induct. The bracket in (6.39) has rational Taylor coefficients, so if every  $d_{2j-1}$  with  $j < n$  lies in  $\Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$  with degree at most  $2j - 1$ , the bound of proposition 6.15 applies verbatim and the bracket has degree at most  $2n - 2$ ; dividing by  $\Lambda$  gives degree at most  $2n - 1$  and no constant term. At order  $q^{2n}$  the only contribution not involving an earlier coefficient is  $a_n$ , whence  $d_{2n-1} = -a_n/\Lambda + \mathcal{O}(\Lambda^{-2})$ .

*Analytic part.* Fix  $N$  and set  $M = N + 1$ . We must know that (6.2) truncated after  $a_M t^{-(2M-1)}$  has remainder  $\mathcal{O}(t^{-(2M+1)})$  on the positive real axis, with a differentiable remainder. This follows from the Legendre duplication formula

$$\Gamma(t)\Gamma\left(t + \frac{1}{2}\right) = 2^{1-2t} \sqrt{\pi} \Gamma(2t), \quad \text{i.e.} \quad \log \Gamma\left(t + \frac{1}{2}\right) = \log \Gamma(2t) - \log \Gamma(t) + (1-2t) \log 2 + \frac{1}{2} \log \pi,$$

applied to two ordinary Stirling expansions on the positive real axis, each of which has the classical remainder bound with a remainder that may be differentiated once (Olver). Subtracting term by term reproduces (6.2) with a remainder of the stated order.

Now let  $x_N = \frac{1}{2} + T + \sum_{n \leq N} d_{2n-1} T^{1-2n}$  and  $f(x) = \log \Gamma(x)$ , and let  $\lambda = \log y$ . By construction the formal substitution cancels the normalized equation through  $q^{2N}$ , so the residual satisfies  $f(x_N) - \lambda = \mathcal{O}(T^{-(2N+1)})$ ; the forward truncation error just bounded is of the same order or smaller. On an interval around  $x_N$  of length  $\mathcal{O}(T^{-2N})$  we have  $f'(x) = \psi(x) \geq \frac{1}{2}\Lambda > 0$  for all large  $y$ , since  $\psi(x) = \log x + \mathcal{O}(1/x)$  and  $\log x_N = \Lambda + o(1)$ . Theorem 1.42 applied with  $m = \frac{1}{2}\Lambda$  converts the residual into  $|x_N - \Gamma_+^{-1}(y)| \leq 2|f(x_N) - \lambda|/\Lambda = \mathcal{O}(\Lambda^{-1}T^{-(2N+1)})$ , which is (6.38). The second half of theorem 1.42 also guarantees that a root exists within that distance, so no separate existence argument is needed.  $\square$

### 6.5.3 Explicit blocks

**Proposition 6.21** (The first five blocks). *With  $\Lambda = w + 1$ ,*

$$d_1 = \frac{1}{24\Lambda}, \tag{6.41}$$

$$d_3 = -\frac{14\Lambda^2 + 10\Lambda + 5}{5760\Lambda^3}, \tag{6.42}$$

$$d_5 = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{2903040\Lambda^5}, \tag{6.43}$$

$$d_7 = -\frac{822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 + 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625}{1393459200\Lambda^7}, \tag{6.44}$$

$$d_9 = \frac{309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 + 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 + 242550\Lambda + 40425}{367873228800\Lambda^9}. \tag{6.45}$$

*Equivalently, separated into powers of  $\Lambda^{-1}$ ,*

$$d_3 = -\frac{7}{2880\Lambda} - \frac{1}{576\Lambda^2} - \frac{1}{1152\Lambda^3}, \tag{6.46}$$

$$d_5 = \frac{31}{40320\Lambda} + \frac{7}{17280\Lambda^2} + \frac{17}{69120\Lambda^3} + \frac{5}{41472\Lambda^4} + \frac{1}{27648\Lambda^5}. \tag{6.47}$$

*Proof.* Run (6.39). The first two steps are short enough to display. Since  $(1 + \epsilon) \log(1 + \epsilon) - \epsilon = \frac{1}{2}\epsilon^2 - \frac{1}{6}\epsilon^3 + \dots$ , the coefficient of  $q^2$  is  $\Lambda d_1 + a_1$ , so  $d_1 = -a_1/\Lambda = 1/(24\Lambda)$ . At  $q^4$ ,

$$\Lambda d_3 + \frac{1}{2}d_1^2 - a_1 d_1 + a_2 = 0,$$

and substituting  $a_1 = -\frac{1}{24}$ ,  $a_2 = \frac{7}{2880}$ ,  $d_1 = \frac{1}{24\Lambda}$  gives (6.42). The remaining three were computed by truncated-series arithmetic over  $\mathbb{Q}(\Lambda)$  and are recorded in remark 6.22.  $\square$

*Remark 6.22* (Independent verification). Every rational number displayed in this chapter was recomputed for this volume, in exact arithmetic, by an implementation written from the normalized equations (6.37) and (6.55) and not from any source's coefficient list: the forward data  $a_k$  and  $c_k$  from Bernoulli numbers, then  $d_1, \dots, d_9$  over  $\mathbb{Q}(\Lambda)$  and  $b_0, \dots, b_5$  over  $\mathbb{Q}(\alpha, \lambda)$  by (6.39) and (6.58). All agreed with the three sources, in every digit of every numerator, including the 24-term numerator of  $b_5$ . The structural predictions were checked as well: the parity (6.59), the resonant limits (6.67) for  $m \leq 3$ , the deepest-pole formulas (6.28) and (6.30) for  $n \leq 6$ , and the two outer families of proposition 6.34. Separately, the nine high-precision roots that S2 prints were recomputed to 60 digits and agree in all 25 digits displayed.

*Remark 6.23* (Prior art). The block  $d_1$  is classical, and the expansion through  $d_3$  – in the separated form (6.46) – was proved by Gerhold, Hubalek and Tomovski. What the recurrence adds is that the expansion continues to all orders with an explicit triangular rule, a proof of the parity, and the coefficient ring. The leading Lambert approximation  $\Gamma_+^{-1}(y) \approx \frac{1}{2} + T$  itself goes back further; see Borwein–Corless and Jeffrey–Watt for its history and for the branch geometry of the complex inverse.

## 6.6 The branches of $\Gamma^{-1}$ at the poles

The Lambert transseries describes one branch. Every pole of  $\Gamma$  carries another, and those are convergent – a useful contrast, since it isolates exactly what makes the large branch divergent.

**Theorem 6.24** (The pole branches are convergent). *Fix  $n \in \{0, 1, 2, \dots\}$  and put*

$$\Phi_n(\varepsilon) = \frac{1}{\Gamma(1+\varepsilon)} \prod_{k=1}^n \left(1 - \frac{\varepsilon}{k}\right), \quad \Phi_n(0) = 1. \quad (6.48)$$

*Then  $1/\Gamma(-n+\varepsilon) = (-1)^n n! \varepsilon \Phi_n(\varepsilon)$ , and for all sufficiently large  $|y|$  the branch of  $\Gamma^{-1}$  tending to  $-n$  is given by the convergent series*

$$\Gamma_{[-n]}^{-1}(y) = -n + \sum_{m \geq 1} \frac{\tau^m}{m} [\varepsilon^{m-1}] \Phi_n(\varepsilon)^{-m}, \quad \tau = \frac{(-1)^n}{n! y}. \quad (6.49)$$

*With  $\psi_n = \psi(n+1) = H_n - \gamma$  and  $H_n^{(r)} = \sum_{k \leq n} k^{-r}$ , the first four terms are*

$$\Gamma_{[-n]}^{-1}(y) = -n + \tau + \psi_n \tau^2 + \frac{3\psi_n^2 + H_n^{(2)} + \zeta(2)}{2} \tau^3 + \frac{8\psi_n^3 + 6\psi_n(H_n^{(2)} + \zeta(2)) + H_n^{(3)} - \zeta(3)}{3} \tau^4 + \mathcal{O}(\tau^5). \quad (6.50)$$

*Proof.* From  $\Gamma(-n+\varepsilon) = \Gamma(1+\varepsilon)/(\varepsilon(\varepsilon-1)\cdots(\varepsilon-n))$  we get  $1/\Gamma(-n+\varepsilon) = \varepsilon \prod_{k=1}^n (\varepsilon-k)/\Gamma(1+\varepsilon) = (-1)^n n! \varepsilon \Phi_n(\varepsilon)$ . Thus  $\tau = \varepsilon \Phi_n(\varepsilon)$ , an analytic function of  $\varepsilon$  near 0 with derivative 1 there; ordinary Lagrange inversion (theorem 1.14, in its convergent analytic form) gives (6.49), convergent on a disc because  $\Phi_n$  is analytic and nonvanishing near 0.

For the coefficients, expand

$$\log \Phi_n(\varepsilon) = \sum_{k=1}^n \log \left(1 - \frac{\varepsilon}{k}\right) - \log \Gamma(1+\varepsilon) = -\psi_n \varepsilon - \frac{H_n^{(2)} + \zeta(2)}{2} \varepsilon^2 + \frac{\zeta(3) - H_n^{(3)}}{3} \varepsilon^3 - \dots,$$

using  $\log \Gamma(1+\varepsilon) = -\gamma \varepsilon + \sum_{j \geq 2} (-1)^j \zeta(j) \varepsilon^j / j$  and  $H_n - \gamma = \psi_n$ . Then  $\Phi_n^{-m} = \exp(-m \log \Phi_n)$  and  $[\varepsilon^{m-1}]$  of it gives, for  $m = 1, 2, 3, 4$ , the values  $1, 2\psi_n, \frac{9}{2}\psi_n^2 + \frac{3}{2}(H_n^{(2)} + \zeta(2))$  and  $\frac{32}{3}\psi_n^3 + 8\psi_n(H_n^{(2)} + \zeta(2)) - \frac{4}{3}(\zeta(3) - H_n^{(3)})$ ; dividing by  $m$  gives (6.50).  $\square$

**Corollary 6.25** (The branch tending to zero). *For  $n = 0, \tau = r := 1/y$  and, with  $\gamma$  Euler's constant,*

$$\begin{aligned} \Gamma_0^{-1}(y) = & r - \gamma r^2 + \frac{18\gamma^2 + \pi^2}{12} r^3 - \frac{8\gamma^3 + \pi^2\gamma + \zeta(3)}{3} r^4 \\ & + \frac{7500\gamma^4 + 1500\pi^2\gamma^2 + 2400\gamma\zeta(3) + 29\pi^4}{1440} r^5 \\ & - \left( \frac{54\gamma^5}{5} + 3\gamma^3\pi^2 + 6\gamma^2\zeta(3) + \frac{17\gamma\pi^4}{120} + \frac{\pi^2\zeta(3)}{6} + \frac{\zeta(5)}{5} \right) r^6 + \mathcal{O}(r^7), \end{aligned} \quad (6.51)$$

*a convergent expansion. Equivalently, in closed form,*

$$\Gamma_0^{-1}(y) = \sum_{m \geq 1} \frac{B_{m-1}(m\ell_1, \dots, m\ell_{m-1})}{m! y^m}, \quad \ell_1 = -\gamma, \quad \ell_j = (-1)^j (j-1)! \zeta(j) \quad (j \geq 2), \quad (6.52)$$

*where  $B_{m-1}$  is the complete exponential Bell polynomial of definition 1.10.*

*Proof.* For  $n = 0$  the product in (6.48) is empty, so  $\tau = r = \varepsilon/\Gamma(1 + \varepsilon)$  and  $\psi_0 = -\gamma$ ,  $H_0^{(r)} = 0$ . Substituting into (6.50) and using  $\zeta(2) = \pi^2/6$  gives the terms through  $r^4$ ; the  $r^5$  and  $r^6$  terms were computed the same way and use in addition  $\zeta(4) = \pi^4/90$ . For (6.52), Lagrange inversion of  $r = \varepsilon/\Gamma(1 + \varepsilon)$  gives  $[r^m] = \frac{1}{m!} (d/d\varepsilon)^{m-1} \Gamma(1 + \varepsilon)^m|_0$ , and  $\Gamma(1 + \varepsilon)^m = \exp(m \sum_j \ell_j \varepsilon^j / j!)$ , whose  $(m - 1)$ st derivative at 0 is the complete exponential Bell polynomial  $B_{m-1}(m\ell_1, \dots, m\ell_{m-1})$ .  $\square$

*Remark 6.26* (Why one branch converges and the other does not). The contrast is instructive and is the cleanest illustration in the volume of the distinction Chapter 0 insists on. Theorem 6.24 inverts an *analytic* map with nonvanishing derivative at a point, so its Lagrange series converges. Theorem 6.20 inverts an *asymptotic* series at infinity: the identity it establishes is an equality of formal objects plus a transported remainder bound at each fixed order, and the full series  $\sum_n d_{2n-1} T^{1-2n}$  diverges for every  $y$ , because the  $a_n$  grow factorially. No amount of care with the algebra can change that; what can be done is to truncate optimally, which is theorem 1.46 and section 6.11.

## 6.7 The eventual positive inverse of Barnes $G$

### 6.7.1 Core variables and the normalized equation

Let  $y \rightarrow +\infty$  and put

$$R = \log y - \zeta'(-1), \quad v = W_0\left(\frac{4R}{e^3}\right), \quad U = 2\sqrt{\frac{R}{v}} = e^{(v+3)/2}, \quad \lambda = \log U - 1 = \frac{v+1}{2}. \quad (6.53)$$

This is lemma 6.8 at  $p = 2$ ,  $a = \frac{1}{2}$ ,  $\beta = \frac{3}{2}$ ,  $r = 1$ ; the core equation is

$$\frac{1}{2}U^2 \left( \log U - \frac{3}{2} \right) = R, \quad (6.54)$$

and by (6.14) its slope is  $\frac{1}{2}U(v+1) = U\lambda$ . Write  $q = U^{-1}$  and  $z = U(1 + \epsilon)$ . Subtracting (6.54) from lemma 6.3 and dividing by  $\frac{1}{2}U^2$  gives

$$\mathcal{K}_\lambda(\epsilon) + 2\alpha q(1 + \epsilon) + 2bq^2(\lambda + 1 + \log(1 + \epsilon)) + 2 \sum_{k \geq 1} c_k q^{2k+2} (1 + \epsilon)^{-2k} = 0, \quad b = -\frac{1}{12}, \quad (6.55)$$

where, by (6.22) with  $v = 2\lambda - 1$ ,

$$\mathcal{K}_\lambda(\epsilon) = (1 + \epsilon)^2 \left( \lambda - \frac{1}{2} + \log(1 + \epsilon) \right) - \left( \lambda - \frac{1}{2} \right) = 2\lambda\epsilon + (\lambda + 1)\epsilon^2 + \frac{1}{3}\epsilon^3 - \frac{1}{12}\epsilon^4 + \frac{1}{30}\epsilon^5 - \dots \quad (6.56)$$

*Remark 6.27* (Absorbing  $\zeta'(-1)$ ). Taking  $R = \log y - \zeta'(-1)$  rather than  $R = \log y$  removes the constant from the perturbation, in accordance with remark 6.10. It is not mandatory — S2 does not do it — but it makes the coefficients shorter, and it is what makes the Glaisher constant disappear from them. Section 6.8 records the exact price of the other choice.

### 6.7.2 The all-orders theorem

**Theorem 6.28** (All-orders expansion of the Barnes inverse). *Let  $\mathcal{B}$  be the inverse of proposition 6.6, and let  $U, \lambda$  be as in (6.53). There are uniquely determined  $e_n \in \mathbb{Q}[\alpha, \lambda, \lambda^{-1}]$ ,  $n \geq 1$ , such that with  $b_n := e_{n+1}$  one has, for every fixed  $N \geq 0$ ,*

$$\mathcal{B}(y) = U + \sum_{n=0}^N \frac{b_n(\lambda)}{U^n} + \mathcal{O}_N(U^{-N-1}), \quad y \rightarrow +\infty, \quad (6.57)$$

and  $G_+^{-1}(y) = 1 + \mathcal{B}(y)$ . With  $E_{n-1} = \sum_{j < n} e_j q^j$  they are computed by

$$e_n(\lambda) = -\frac{1}{2\lambda} [q^n] \left[ \mathcal{K}_\lambda(E_{n-1}) - 2\lambda E_{n-1} + 2\alpha q(1 + E_{n-1}) + 2bq^2(\lambda + 1 + \log(1 + E_{n-1})) + 2 \sum_{2k+2 \leq n} c_k q^{2k+2} (1 + E_{n-1})^{-2k} \right], \quad (6.58)$$

they obey the parity rule

$$e_n(-\alpha) = (-1)^n e_n(\alpha), \quad (6.59)$$

and each is a bounded function of  $\lambda$  as  $\lambda \rightarrow +\infty$ .

*Proof. Formal part.* Equation (6.55) has support  $S = \mathbb{N}$  by example 6.14 and linear coefficient  $\kappa_1 = v + 1 = 2\lambda$ ; theorem 6.12 applies and yields (6.58) and uniqueness. The coefficient ring follows by induction, since the Taylor coefficients of  $\log(1 + \epsilon)$  and of the binomial powers are rational,  $\alpha$  enters only through  $2\alpha q$ , and the only division is by  $2\lambda$ .

For (6.59), observe that the substitution  $(q, \alpha, \epsilon) \mapsto (-q, -\alpha, \epsilon)$  leaves (6.55) invariant:  $\mathcal{K}_\lambda(\epsilon)$  has no  $q$ ;  $2\alpha q(1 + \epsilon)$  is invariant because both signs flip; and  $q^2, q^{2k+2}$  are even. Uniqueness therefore forces  $\epsilon(-q, -\alpha) = \epsilon(q, \alpha)$ , i.e.  $(-1)^n e_n(-\alpha) = e_n(\alpha)$ , which is (6.59).

Boundedness in  $\lambda$  is proposition 6.30 below, which exhibits the coefficientwise limit.

*Analytic part.* Let  $x_N = U + \sum_{n \leq N} b_n U^{-n}$  and  $f(z) = \log G(z + 1)$ ,  $\lambda_* = \log y$ . Retain the forward expansion (6.6) far enough that its remainder is  $\mathcal{O}(U^{-N-1})$  after division by the slope; this is legitimate because (6.6) is a Poincaré expansion with differentiable remainder on the positive axis, with explicit bounds available from Nemes. By construction the formal substitution cancels (6.55) through  $q^{N+1}$ , so the forward residual  $f(x_N) - \lambda_*$  is  $\mathcal{O}(\lambda U^{-N})$  — one factor  $\frac{1}{2}U^2$  is restored on undoing the normalization, and the first omitted normalized term is  $2\lambda e_{N+2} q^{N+2}$ . By proposition 6.6,  $f' \geq \frac{1}{2}z \log z$  on  $[7, \infty)$ , and along the relevant interval  $f' = U\lambda(1 + o(1))$ . Theorem 1.42 with  $m = \frac{1}{2}U\lambda$  turns the residual into  $|x_N - \mathcal{B}(y)| = \mathcal{O}(U^{-N-1})$ , which is (6.57), and again certifies that a root lies that close.  $\square$

### 6.7.3 Explicit blocks

**Proposition 6.29** (The first six blocks). *With  $\alpha = \frac{1}{2} \log(2\pi)$  and  $\lambda = (v + 1)/2$ ,*

$$b_0 = -\frac{\alpha}{\lambda}, \quad (6.60)$$

$$b_1 = \frac{6\alpha^2\lambda - 6\alpha^2 + \lambda^3 + \lambda^2}{12\lambda^3}, \quad (6.61)$$

$$b_2 = \frac{\alpha(8\alpha^2\lambda - 6\alpha^2 + \lambda^2)}{12\lambda^5}, \quad (6.62)$$

$$b_3 = -\frac{P_3(\lambda, \alpha)}{1440\lambda^7}, \quad b_4 = -\frac{\alpha P_4(\lambda, \alpha)}{1440\lambda^9}, \quad b_5 = \frac{P_5(\lambda, \alpha)}{362880\lambda^{11}}, \quad (6.63)$$

where

$$P_3 = 180\alpha^4\lambda^3 + 120\alpha^4\lambda^2 - 1500\alpha^4\lambda + 900\alpha^4 + 60\alpha^2\lambda^5 + 120\alpha^2\lambda^4 + 60\alpha^2\lambda^3 - 180\alpha^2\lambda^2 + 5\lambda^7 - \lambda^6 + 5\lambda^5 + 5\lambda^4, \quad (6.64)$$

$$P_4 = 576\alpha^4\lambda^3 + 480\alpha^4\lambda^2 - 2520\alpha^4\lambda + 1260\alpha^4 + 80\alpha^2\lambda^5 + 280\alpha^2\lambda^4 + 200\alpha^2\lambda^3 - 300\alpha^2\lambda^2 - 22\lambda^7 - 11\lambda^6 + 10\lambda^5 + 15\lambda^4, \quad (6.65)$$

$$P_5 = 22680\alpha^6\lambda^5 + 40824\alpha^6\lambda^4 - 349272\alpha^6\lambda^3 - 352800\alpha^6\lambda^2 + 1111320\alpha^6\lambda - 476280\alpha^6 + 11340\alpha^4\lambda^7 + 34020\alpha^4\lambda^6 + 3780\alpha^4\lambda^5 - 138600\alpha^4\lambda^4 - 132300\alpha^4\lambda^3 + 132300\alpha^4\lambda^2 + 1890\alpha^2\lambda^9 + 8568\alpha^2\lambda^8 + 15120\alpha^2\lambda^7 + 11718\alpha^2\lambda^6 - 3150\alpha^2\lambda^5 - 9450\alpha^2\lambda^4 + 105\lambda^{11} - 668\lambda^{10} - 378\lambda^9 - 21\lambda^8 + 175\lambda^7 + 105\lambda^6. \quad (6.66)$$

*Proof.* Run (6.58); see remark 6.22. The first step is  $2\lambda e_1 + 2\alpha = 0$ , giving  $b_0 = e_1 = -\alpha/\lambda$ . The parity (6.59) is visible:  $b_0, b_2, b_4$  are odd in  $\alpha$  and  $b_1, b_3, b_5$  even, since  $b_n = e_{n+1}$ . The denominators are  $\lambda^{2n-1}$  for  $e_n$ , in agreement with proposition 6.15, and the numerators' extreme coefficients  $-\alpha, -6\alpha^2/12, -6\alpha^3/12, -900\alpha^4/1440, -1260\alpha^5/1440, -476280\alpha^6/362880$  are  $-\alpha, -\frac{\alpha^2}{2}, -\frac{\alpha^3}{2}, -\frac{5}{8}\alpha^4, -\frac{7}{8}\alpha^5, -\frac{21}{16}\alpha^6$ , which is (6.30).  $\square$

### 6.7.4 Resonance: the surviving algebraic subseries

The forward term  $-\frac{1}{12} \log z$  is small relative to the core, but its *logarithm* is of the same size as the slope. It therefore survives the large-slope limit, and it is the only perturbation that does.

**Proposition 6.30** (Coefficientwise large-slope limit). *For each fixed  $n$ , the limit of  $e_n(\lambda)$  as  $\lambda \rightarrow +\infty$  exists and*

$$\lim_{\lambda \rightarrow \infty} e_{2m}(\lambda) = \binom{1/2}{m} 6^{-m}, \quad \lim_{\lambda \rightarrow \infty} e_{2m+1}(\lambda) = 0. \quad (6.67)$$

*Equivalently  $\lim b_{2m-1} = \binom{1/2}{m} 6^{-m}$  and  $\lim b_{2m} = 0$ ; the first values are  $b_1 \rightarrow \frac{1}{12}, b_3 \rightarrow -\frac{1}{288}, b_5 \rightarrow \frac{1}{3456}$ . The limiting coefficients are those of*

$$E_0(q) = \sqrt{1 + \frac{q^2}{6}} - 1, \quad \text{i.e.} \quad U(1 + E_0(U^{-1})) = \sqrt{U^2 + \frac{1}{6}}. \quad (6.68)$$

*Proof.* Divide (6.55) by  $2\lambda$  and let  $\lambda \rightarrow \infty$  with  $q$  formal. By (6.56),  $\mathcal{K}_\lambda(E)/(2\lambda) \rightarrow ((1+E)^2 - 1)/2$ , because the  $\lambda$ -free part of  $\mathcal{K}_\lambda$  is bounded coefficientwise. The term  $2\alpha q(1+E)/(2\lambda) \rightarrow 0$ ; each  $c_k$  term  $\rightarrow 0$ ; and

$$\frac{2bq^2(\lambda + 1 + \log(1+E))}{2\lambda} = bq^2 \left( 1 + \frac{1 + \log(1+E)}{\lambda} \right) \rightarrow bq^2.$$

So the limiting equation is  $((1+E_0)^2 - 1)/2 + bq^2 = 0$ , i.e.  $(1+E_0)^2 = 1 - 2bq^2 = 1 + q^2/6$ , whose branch with  $E_0(0) = 0$  is (6.68). That the coefficientwise limit may be taken inside the recursion is the triangular structure: at each  $n$  the coefficient equation has limiting linear divisor  $1 \neq 0$  and involves only finitely many lower coefficients, so induction commutes with the limit. Expanding  $\sqrt{1 + q^2/6}$  gives (6.67), and  $z = U(1 + \epsilon)$  turns  $E_0$  into  $\sqrt{U^2 + 1/6}$ .  $\square$

*Remark 6.31* (Two limits, not one). Proposition 6.30 is a statement about each coefficient separately. It does *not* say that  $\mathcal{B}(y) - \sqrt{U^2 + 1/6}$  is small: at any actual  $y$  the parameter  $\lambda$  is finite and the neglected terms, headed by  $b_0 = -\alpha/\lambda$ , are numerically the largest corrections in the whole expansion. S1 writes that the subsector “resums” to  $\sqrt{T^2 + 1/6}$  and then, in the following remark, correctly warns that it does not replace the expansion; the two statements sit awkwardly together. The accurate form is the one above: a limit of coefficients, useful as an exact structural check on a symbolic computation and as the identification of one coherent subseries, not an approximation to the inverse.

## 6.8 The three normalizations reconciled

The three sources use three coordinate systems. They agree, and the dictionary is worth recording because the disagreement is entirely in the choice of what to absorb.

**Proposition 6.32** (Dictionary). *Let  $C_n(D)$  denote the Barnes coefficients obtained from (6.55) when  $\zeta'(-1)$  is not absorbed, so that  $R = \log y$ ,  $Z = e^{(v+3)/2}$ ,  $D = v + 1$ , and the constant  $\kappa = \zeta'(-1) = \frac{1}{12} - \log A$  appears as the extra perturbation  $2\kappa q$ . Then the two families are related by*

$$b_n(\lambda, \alpha) = C_n(D) \Big|_{\log A \mapsto 1/12, \log(2\pi) \mapsto 2\alpha, D \mapsto 2\lambda}, \quad (6.69)$$

that is, the absorbed family is obtained from the unabsorbed one by setting  $\kappa = 0$  and halving the slope variable. In particular

$$C_1(D) = \frac{1}{12} + \frac{2 \log A}{D} + \frac{\log^2(2\pi)}{2D^2} - \frac{\log^2(2\pi)}{D^3}, \quad b_1(\lambda) = \frac{1}{12} + \frac{1}{12\lambda} + \frac{\alpha^2}{2\lambda^2} - \frac{\alpha^2}{2\lambda^3}, \quad (6.70)$$

and these agree under (6.69).

*Proof.* Both families solve the same normalized equation, the second with the extra term  $2\kappa q$  and with  $\lambda$  replaced by  $D/2$ . Since  $\kappa$  enters only through that term and only linearly, and since  $\kappa = \frac{1}{12} - \log A$  vanishes exactly when  $\log A = \frac{1}{12}$ , the substitution  $\log A \mapsto \frac{1}{12}$  turns the unabsorbed equation into the absorbed one; uniqueness of the solution does the rest. The two displayed forms of the first coefficient were checked directly, as recorded in remark 6.22.  $\square$

*Remark 6.33* (A cancellation worth seeing). In  $C_1$  the constant  $\log A$  appears and  $\frac{1}{6D}$  does not; in  $b_1$  the reverse. This is not a discrepancy. The forward term  $-\frac{1}{12} \log z$  contributes  $\frac{1}{12} + \frac{1}{12\lambda}$  to both, and the unabsorbed constant contributes  $-2\kappa/D = -\frac{1}{6D} + \frac{2 \log A}{D}$ , whose first half exactly cancels the  $\frac{1}{6D} = \frac{1}{12\lambda}$ . A reader comparing the two articles term by term without tracking the normalization will conclude that one of them has a sign error. Neither does.

## 6.9 Outer expansions in iterated logarithms

Retaining  $W$  exactly is both shorter and numerically better, but an answer purely in elementary logarithms is sometimes wanted. It is obtained by reverting the core equation directly, without first invoking a generic  $W$ -series.

**Proposition 6.34** (The two outer families).

1. Gamma. Put  $L = \log R$ ,  $\ell = \log L$ ,  $D_\Gamma = \ell + 1$ , and write  $T = (R/L)F$ . Then  $F$  is the unique solution with  $F(\infty) = 1$  of  $F(1 - D_\Gamma/L + L^{-1} \log F) = 1$ , and  $F = 1 + \sum_{n \geq 1} f_n L^{-n}$  with

$$\begin{aligned} f_1 &= D, & f_2 &= D(D-1), & f_3 &= \frac{1}{2}D(D-2)(2D-1), \\ f_4 &= \frac{1}{6}D(6D^3 - 26D^2 + 27D - 6), \\ f_5 &= \frac{1}{12}D(12D^4 - 77D^3 + 142D^2 - 84D + 12), \\ f_6 &= \frac{1}{30}D(30D^5 - 261D^4 + 725D^3 - 775D^2 + 300D - 30), \end{aligned} \quad (6.71)$$

abbreviating  $D = D_\Gamma$ .

2. Barnes. Put  $L = \log(4R)$ ,  $\ell = \log L$ ,  $D_G = \ell + 3$ , and write  $U = 2\sqrt{R/L}F$ . Then  $F$  solves  $F^2(1 - D_G/L + 2L^{-1} \log F) = 1$  and  $F = 1 + \sum_{n \geq 1} g_n L^{-n}$  with

$$\begin{aligned} g_1 &= \frac{1}{2}D, & g_2 &= \frac{1}{8}D(3D-4), & g_3 &= \frac{1}{16}D(5D^2 - 16D + 8), \\ g_4 &= \frac{1}{384}D(105D^3 - 568D^2 + 720D - 192), \\ g_5 &= \frac{1}{768}D(189D^4 - 1488D^3 + 3296D^2 - 2304D + 384), \\ g_6 &= \frac{1}{15360}D(3465D^5 - 36516D^4 + 120400D^3 - 150400D^2 + 67200D - 7680), \end{aligned} \quad (6.72)$$

abbreviating  $D = D_G$ .

*Proof.* For (i),  $T(\log T - 1) = R$  and  $\log T = L - \ell + \log F$  give  $(R/L)F(L - \ell - 1 + \log F) = R$ , i.e. the displayed equation. Substituting  $F = 1 + \sum f_n L^{-n}$  and equating coefficients of  $L^{-n}$  determines  $f_n$  uniquely from  $f_1, \dots, f_{n-1}$ , because the coefficient of  $f_n$  is 1. For (ii), write  $L = \log(4R)$ , so that



$\log(4R/L) = L - \ell$  and  $\log U = \frac{1}{2}(L - \ell) + \log F$ . Substituting  $U = 2\sqrt{R/L} F$  into  $\frac{1}{2}U^2(\log U - \frac{3}{2}) = R$  gives

$$\frac{2R}{L} F^2 \left( \frac{L - \ell}{2} - \frac{3}{2} + \log F \right) = R, \quad \text{i.e.} \quad F^2 \left( 1 - \frac{\ell + 3}{L} + \frac{2 \log F}{L} \right) = 1,$$

which is the displayed equation with  $D_G = \ell + 3$ . Both families were recomputed independently (remark 6.22).  $\square$

*Remark 6.35* (The outer variable is part of the answer). The polynomials  $g_n$  in (6.72) are attached to the choice  $L = \log(4R)$ ,  $D_G = \log L + 3$ . Choosing instead  $\tilde{L} = \log(4Re^{-3}) = L - 3$  and  $U = 2\sqrt{R/\tilde{L}} F$  leads to the *same* polynomials evaluated at  $D = \log \tilde{L}$  – the shift  $+3$  migrates from the argument into the variable. S1 uses the first convention and S3 the second, and both are correct; but the two lists of coefficients are only equal after the substitution is made, and comparing them term by term without it produces a spurious disagreement at every order. The same caution applies to the Gamma family, where the corresponding shift is  $+1$ .

*Remark 6.36* (The hierarchy, and why it must not be merged). For every fixed  $N$ ,  $T/L^N \rightarrow \infty$ ; so the constant  $\frac{1}{2}$  in theorem 6.20 is smaller than every term of the outer expansion of  $T$ , and the first Bernoulli block  $d_1/T$  is smaller again. The correct reading is lexicographic,

$$T \gg \frac{T}{L} \gg \frac{T}{L^2} \gg \cdots \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \cdots, \quad (6.73)$$

and for Barnes, with the extra intermediate level supplied by  $b_0$ ,

$$U \gg \frac{U}{L} \gg \cdots \gg 1 \gg \frac{1}{\lambda} \gg \frac{1}{U}. \quad (6.74)$$

An expansion that interleaves the outer  $L^{-1}$  series with the additive constants and the  $q$ -grid is not merely inelegant, it is wrong: no finite truncation of the outer series can reproduce even the first block  $d_1/T$ . This is the same phenomenon as the flattening caveat of theorem 1.33, and the reason the volume keeps  $W$  unexpanded by default.

## 6.10 Numerical verification

The tables below test the entire chain and involve no numerical inversion. An exact abscissa is chosen, the special function is evaluated at 80 digits, the core variables are reconstructed from that value alone, and the truncations are compared with the abscissa. Errors are signed, approximation minus exact.

Table 6.2:  $\Gamma_+^{-1}$ : signed error of  $\frac{1}{2} + T$  and of the truncations including  $d_1, \dots, d_9$ .

| $x$ | $\frac{1}{2} + T$        | $+d_1$                   | $+d_3$                    | $+d_5$                   | $+d_7$                    | $+d_9$                   |
|-----|--------------------------|--------------------------|---------------------------|--------------------------|---------------------------|--------------------------|
| 5   | $-6.1414 \times 10^{-3}$ | $2.8694 \times 10^{-5}$  | $-4.1862 \times 10^{-7}$  | $1.3452 \times 10^{-8}$  | $-8.2694 \times 10^{-10}$ | $8.4558 \times 10^{-11}$ |
| 10  | $-1.9470 \times 10^{-3}$ | $1.7430 \times 10^{-6}$  | $-5.7560 \times 10^{-9}$  | $4.4485 \times 10^{-11}$ | $-6.5578 \times 10^{-13}$ | $1.5870 \times 10^{-14}$ |
| 20  | $-7.1924 \times 10^{-4}$ | $1.4126 \times 10^{-7}$  | $-1.1183 \times 10^{-10}$ | $2.1113 \times 10^{-13}$ | $-7.5612 \times 10^{-16}$ | $4.4191 \times 10^{-18}$ |
| 50  | $-2.1572 \times 10^{-4}$ | $6.1957 \times 10^{-9}$  | $-7.6816 \times 10^{-13}$ | $2.2927 \times 10^{-16}$ | $-1.2910 \times 10^{-19}$ | $1.1809 \times 10^{-22}$ |
| 100 | $-9.1031 \times 10^{-5}$ | $6.2869 \times 10^{-10}$ | $-1.9388 \times 10^{-14}$ | $1.4440 \times 10^{-18}$ | $-2.0231 \times 10^{-22}$ | $4.5949 \times 10^{-26}$ |

Table 6.3:  $\mathcal{B}$ : signed error of  $U$  and of the truncations including  $b_0, \dots, b_5$ , at  $y = G(z + 1)$ .

| $z$ | $U$                     | $+b_0$                   | $+b_1$                   | $+b_2$                   | $+b_3$                    | $+b_4$                    | $+b_5$                    |
|-----|-------------------------|--------------------------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|
| 5   | 1.1187                  | $-1.3867 \times 10^{-2}$ | $-7.8343 \times 10^{-3}$ | $-1.5935 \times 10^{-3}$ | $-1.8345 \times 10^{-4}$  | $5.8515 \times 10^{-6}$   | $8.9807 \times 10^{-6}$   |
| 10  | $6.5273 \times 10^{-1}$ | $-2.0081 \times 10^{-2}$ | $-8.4026 \times 10^{-4}$ | $1.5249 \times 10^{-5}$  | $3.2299 \times 10^{-6}$   | $4.1936 \times 10^{-8}$   | $-1.3999 \times 10^{-8}$  |
| 20  | $4.4669 \times 10^{-1}$ | $-8.7205 \times 10^{-3}$ | $-6.6821 \times 10^{-5}$ | $2.3736 \times 10^{-6}$  | $3.4133 \times 10^{-8}$   | $-1.5564 \times 10^{-9}$  | $-1.9947 \times 10^{-11}$ |
| 100 | $2.5342 \times 10^{-1}$ | $-1.2959 \times 10^{-3}$ | $-3.9481 \times 10^{-7}$ | $8.1326 \times 10^{-9}$  | $-5.3660 \times 10^{-12}$ | $-8.9993 \times 10^{-14}$ | $6.0482 \times 10^{-16}$  |

Three things are visible. The Gamma expansion gains roughly two orders per block, as (6.38) predicts, and does so already at  $x = 5$ . The Barnes core error is much larger, because the linear forward term  $\alpha z$  produces the slowly decaying displacement  $b_0 = -\alpha/\lambda$ ; once that single block is included the improvement is again rapid. And at  $z = 5$  the last Barnes column is *worse* than the one before it ( $8.98 \times 10^{-6}$  against  $5.85 \times 10^{-6}$ ): the expansion is asymptotic, not convergent, and at a fixed moderate argument it has reached its useful limit. That is the behaviour theorem 1.46 describes and section 6.11 quantifies; it is not a defect of the coefficients, each of which was verified exactly.

*Remark 6.37* (What the tables do and do not establish). They check the algebra, the signs, the shifts and the branch choices, end to end from the special function to the inverse; a transcription error in any numerator would show immediately. They are not a substitute for the analytic remainder arguments in the proofs of theorems 6.20 and 6.28, and conversely those theorems, which control only the order of the remainder, would not catch a wrong rational number. The two kinds of evidence are complementary, and both were obtained.

## 6.11 Beyond all orders

The expansions above are complete relative to a Poincaré forward input. They do not contain the exponentially small sectors that the late Bernoulli terms encode.

**Proposition 6.38** (Transport of an exponentially small sector). *Suppose the forward phase splits as  $\Phi = \Phi_{\text{alg}} + E$ , where  $x_{\text{alg}}$  solves  $\Phi_{\text{alg}}(x) = R$  to the algebraic order considered and  $E$  is smaller than every term retained. If  $\Phi'_{\text{alg}}(x_{\text{alg}}) \neq 0$ , then the induced displacement of the root is*

$$\Delta x = -\frac{E(x_{\text{alg}})}{\Phi'_{\text{alg}}(x_{\text{alg}})} + \mathcal{O}\left(\frac{|E||E'|}{|\Phi'_{\text{alg}}|^2} + \frac{|E|^2|\Phi''_{\text{alg}}|}{|\Phi'_{\text{alg}}|^3}\right). \quad (6.75)$$

*Proof.* This is theorem 1.44 with  $f_0 = \Phi_{\text{alg}}$  and  $\varepsilon = E$ : part (ii) of that theorem gives exactly (6.75), the two error terms being its  $\theta$  and  $M_2$  contributions.  $\square$

**Proposition 6.39** (Least term and the inverse floor). *The half-shifted Stirling coefficients satisfy*

$$a_n = (-1)^n \frac{2(2n-2)!}{(2\pi)^{2n}} (1 + \mathcal{O}(4^{-n})), \quad (6.76)$$

*so the Gamma inverse blocks  $d_{2n-1} \sim -a_n/\Lambda$  decrease until  $n \approx \pi T$  and grow thereafter. Consequently the least term of (6.38) has size  $\exp(-2\pi T + \mathcal{O}(\log T))$ , and by theorem 1.42 the attainable accuracy of the inverse is*

$$\frac{\exp(-2\pi T + \mathcal{O}(\log T))}{\Lambda}. \quad (6.77)$$

*The Barnes analogue, from  $c_k = B_{2k+2}/(4k(k+1))$ , has least term near  $k \approx \pi U$  and floor  $\exp(-2\pi U + \mathcal{O}(\log U))/(U\lambda)$ .*

*Proof.* From  $B_{2n} = (-1)^{n+1} 2(2n)! \zeta(2n)/(2\pi)^{2n}$  and  $\zeta(2n) = 1 + \mathcal{O}(4^{-n})$ , together with  $a_n = (2^{1-2n} - 1)B_{2n}/(2n(2n-1))$  and  $2^{1-2n} - 1 = -1 + \mathcal{O}(4^{-n})$ , we get (6.76) using  $(2n)!/(2n(2n-1)) = (2n-2)!$ . The ratio of consecutive terms  $|a_{n+1}|T^{-2n-1}/(|a_n|T^{1-2n})$  is  $2n(2n-1)/((2\pi)^2 T^2)(1 + o(1))$ , which equals 1 at  $n \sim \pi T$ ; substituting that index into (6.76) and using Stirling gives the stated size, and dividing by the slope  $\Lambda$  transports it to the inverse. The Barnes computation is identical with  $B_{2k+2}$  in place of  $B_{2n}$ .  $\square$

*Remark 6.40* (What is not claimed). Proposition 6.39 locates the floor; it does not compute the Stokes multipliers, and this chapter does not. A genuine hyperasymptotic inverse must start from an exponentially improved *forward* expansion on the same branch and in the same sector — Nemes supplies these for both functions — and then apply (6.75). Appending terms of the shape  $e^{-2\pi T}$  to the real formula by pattern matching is not a branch-complete construction, and none of the three sources does more than indicate the first sector. This is recorded as open in section 6.13.

## 6.12 The general family

Nothing above used more about  $\Gamma$  and  $G$  than the shape of their phases. The following is the class the chapter actually treats, and it is what Part 7 will reuse.

**Theorem 6.41** (Inversion of a power–logarithmic phase). *Let*

$$\Phi(x) = ax^p(\log x - \beta) + C + \sum_{\rho \in A} x^{p-\rho} P_\rho(\log x) + \mathcal{E}(x), \quad (6.78)$$

where  $a \neq 0$ ,  $p > 0$ ,  $A \subset \mathbb{R}_{>0}$  generates a well-ordered monoid  $S$ , each  $P_\rho$  is a polynomial, and  $\mathcal{E}$  is smaller than every retained term. Put  $R = Y - C$  and let  $v, X, q$  be given by (6.14). If  $v + 1$  is invertible then the equation  $\Phi(x) = Y$  has a unique formal solution

$$x = X \left( 1 + \sum_{\mu \in S, \mu > 0} e_\mu(v) X^{-\mu} \right), \quad (6.79)$$

with  $e_\mu$  given by (6.24); the coefficients lie in  $K[v, (v+1)^{-1}]$  where  $K$  is any field containing the coefficients of the  $P_\rho$  and  $\beta$ ; and their denominators obey proposition 6.15 when  $S = \mathbb{N}$ . If in addition  $\Phi$  is real, eventually strictly increasing, with a differentiable Poincaré remainder at each truncation order, then each truncation of (6.79) approximates the actual inverse with the error obtained by dividing the forward residual by  $aX^{p-1}(v+1)$ .

*Proof.* Substituting  $x = X(1 + \epsilon)$  into (6.78) and dividing by  $aX^p$  gives (6.17) with

$$\mathcal{R}(q, v, \epsilon) = \frac{1}{a} \sum_{\rho \in A} q^\rho (1 + \epsilon)^{p-\rho} P_\rho \left( \beta + \frac{v}{p} + \log(1 + \epsilon) \right) + \frac{\mathcal{E}(X(1 + \epsilon))}{aX^p},$$

every monomial of which has  $q$ -exponent in  $S$  and positive. Powers of  $v$  in the coefficients are harmless:  $q^\rho v^M \rightarrow 0$  for fixed  $M$ , and the Hahn ring is graded by the  $q$ -exponent alone. Theorem 6.12 gives existence, uniqueness and the recurrence; corollary 6.13 the support; the coefficient ring follows because the only division performed is by  $v + 1$  and every other operation is a rational combination of the data. The analytic clause is theorem 1.42 applied with the slope  $\Phi'(X) = aX^{p-1}(v+1)(1 + o(1))$  from (6.14).  $\square$

**Example 6.42** (The two subjects, and a third).  $\Gamma$  is  $(a, p, \beta) = (1, 1, 1)$  with  $A = \{2, 4, 6, \dots\}$ ;  $G$  is  $(\frac{1}{2}, 2, \frac{3}{2})$  with  $A = \{1, 2, 4, 6, \dots\}$ , the exponent 1 coming from  $\alpha z$ . The hyperfactorial of Part 7 is the next member, and the higher multiple gamma functions continue the family with  $p = 3, 4, \dots$ , their perturbation supports read off from the drops in algebraic degree in the known forward expansions.

*Remark 6.43* (Where the closed form stops). Two boundaries deserve naming. A leading block  $ax^p(\log x - \beta)^r$  with  $r \neq 1$  is still exactly invertible, by lemma 6.8, with normalized slope  $(pu + r)/u$  in place of  $v + 1$ . A leading block  $ax^p(\log x)^r(\log \log x)^s$  with  $s \neq 0$  is not: there is no one-step elementary reduction, and the honest procedure is to invert the first two factors exactly, treat the third as a slow multiplier, and solve the residual slow equation by iteration before the  $q$ -grid theorem is applied. The method's real hypothesis is not “Lambert” but “the dominant transmonomial can be inverted exactly and its normalized slope is a unit”.

## 6.13 What is left open

1. *Closed forms for the coefficient families.* The recurrences compute  $d_{2n-1}$  and  $b_n$  to any order and theorem 6.17 gives the extreme coefficient of each in closed form, but no closed description of the full numerators is known. For Gamma the pattern  $d_{2n-1} = P_n(\Lambda)/(\gamma_n \Lambda^{2n-1})$  with  $P_n$  of degree  $2n - 2$  invites a Bell-polynomial formula decorated by Bernoulli values at  $\frac{1}{2}$ ; for Barnes the grading by powers of  $\alpha$  is a  $\mathbb{Z}/2$ -grading whose elementary proof is (6.59) but whose finer structure — which powers  $\alpha^j$  actually occur in  $b_n$ , and with what  $\lambda$ -degrees — is not determined here.

2. *Hyperasymptotics.* Remark 6.40.
3. *Uniformity near the branch point.* Everything degrades as  $v + 1 \rightarrow 0$ , and at  $v = -1$  the core map has a critical point where the correct local scale is a square root rather than a power series. A uniform approximation matching a branch-point expansion to the transseries developed here is not attempted. It is irrelevant for the positive real branches, which stay far from  $v = -1$ , and unavoidable for a global treatment of the complex sheets.
4. *Complex branches.* Replacing  $\log y$  by  $\text{Log } y + 2\pi im$  and  $W_0$  by  $W_k$  produces, in every sector where the forward expansion holds and the branch choices are continued consistently, a formal inverse sheet with the *same* coefficient functions. That is a local asymptotic chart, not a classification: distinct pairs  $(m, k)$  can describe continuations that meet, and the critical point of  $\Gamma$  creates branch geometry the leading Lambert model does not see. No global statement is made.
5. *Formalization.* The triangular recurrence is well suited to a proof assistant, since every requested coefficient depends on finitely many earlier ones; the natural decomposition separates the algebra of well-ordered supports, the formal implicit-function theorem (theorem 6.12), the exact rational certificates, and the analytic import of the Stirling remainder bounds. Nothing here is formalized.

# Chapter 7

## The hyperfactorial $K$ -function

### 7.1 What this chapter does

The  $K$ -function interpolates the hyperfactorial,  $K(n+1) = \prod_{j=1}^n j^j$ , and  $\log K(x) \sim \frac{1}{2}x^2 \log x - \frac{1}{4}x^2$ . Its dominant block is therefore of exactly the type inverted in lemma 6.8, with the same exponent  $p = 2$  as Barnes  $G$  and a different shift; and the whole apparatus of section 6.4 applies unchanged. What makes the subject worth a chapter of its own is a single feature: after the correct centring the entire perturbation of power gap two collapses to one term,

$$-\frac{1}{24} \log q, \quad (7.1)$$

and *that* term is resonant — its logarithm is of the same size as the core slope, so it does not shrink relative to the core the way an ordinary correction does. The three source articles deal with it in three different ways, and this chapter's main contribution is to prove that the three are one construction seen from three sides.

The three filed articles are listed in table 7.1. As in Part 6, they agree: every coefficient and constant any of them prints was recomputed for this volume in exact arithmetic and reproduced, including the eleven-digit numerators of the refined block  $b_5$ . Their difference is one of strategy, not of arithmetic.

| Tag | Directory                              | Strategy for (7.1)                                                                                  |
|-----|----------------------------------------|-----------------------------------------------------------------------------------------------------|
| T1  | inverse_k_function_transseries         | leave it in the perturbation, <i>and</i> absorb it into an algebraic shift $T = r^2 - \frac{1}{12}$ |
| T2  | K_Function_Inverse_Transseries         | absorb it into the core, using the $r$ -Lambert function; also the arbitrary-centre expansion       |
| T3  | K_function_inverse_transseries_article | leave it in the perturbation; slow-depth and ordered-block formulas, sectorial version              |

Table 7.1: The  $K$ -function in the notation of theorem 6.41.

|                             |                                                                 |
|-----------------------------|-----------------------------------------------------------------|
| inverted map                | $x \mapsto K(x)$ on $[2, \infty)$                               |
| centred variable            | $r = x - \frac{1}{2}$                                           |
| core $ax^p(\log x - \beta)$ | $a = \frac{1}{2}, p = 2, \beta = \frac{1}{2}$                   |
| absorbed constant           | $C = \log A - \frac{1}{8}$ , $A$ the Glaisher–Kinkelin constant |
| resonant term               | $-\frac{1}{24} \log r$                                          |
| perturbation support        | $2\mathbb{N}$                                                   |
| core                        | $\rho = e^L$ , $L = \frac{1}{2}(w+1)$ , $w = W_0(4Y/e)$         |
| inverse support             | $\rho^{-1}, \rho^{-3}, \rho^{-5}, \dots$                        |

Compare table 6.1. Barnes  $G$  has  $p = 2$ ,  $\beta = \frac{3}{2}$ , a linear term  $\alpha z$  generating the full integer support, and a resonant term  $-\frac{1}{12} \log z$ . The  $K$ -function has  $p = 2$ ,  $\beta = \frac{1}{2}$ , *no* linear term — so the support stays

even, as for  $\Gamma$  – and a resonant term  $-\frac{1}{24} \log r$ . It is, in a precise sense made in section 7.7, the cleanest member of the family: the only obstruction it has is the resonance.

## 7.2 The function and its increasing branch

**Lemma 7.1** (Exact logarithmic derivatives). *Write  $\kappa(x) = \log K(x)$ . For  $x > 0$ ,*

$$\kappa'(x) = \log \Gamma(x) + x - \frac{1}{2} - \frac{1}{2} \log(2\pi), \quad \kappa''(x) = \psi(x) + 1. \quad (7.2)$$

*Proof.* Use  $K(z) = \exp(\zeta'(-1, z) - \zeta'(-1))$  with the Hurwitz zeta function. Differentiating  $\partial_z \zeta(s, z) = -s\zeta(s+1, z)$  with respect to  $s$  gives  $\partial_z \zeta'(s, z) = -\zeta(s+1, z) - s\zeta'(s+1, z)$ , and at  $s = -1$ ,  $\partial_z \zeta'(-1, z) = -\zeta(0, z) + \zeta'(0, z)$ . Substituting  $\zeta(0, z) = \frac{1}{2} - z$  and  $\zeta'(0, z) = \log \Gamma(z) - \frac{1}{2} \log(2\pi)$  gives the first identity, and differentiating it again gives the second, since  $(\log \Gamma)' = \psi$  and the derivative of  $x - \frac{1}{2}$  is 1.  $\square$

**Proposition 7.2** (The increasing branch).  *$K$  is strictly increasing on  $[2, \infty)$ , with  $K(2) = 1$  and  $K(x) \rightarrow +\infty$ ; write  $K_+^{-1} : [1, \infty) \rightarrow [2, \infty)$  for the inverse. On  $(0, \infty)$ ,  $K$  has exactly two stationary points,*

$$x_{\max} = 0.29095698669918 \dots, \quad x_{\min} = 1.53768886373648 \dots, \quad (7.3)$$

*with  $K(x_{\max}) = 1.29867230693177 \dots$  and  $K(x_{\min}) = 0.87978684305939 \dots$ ; so below  $K(x_{\max})$  there are further real branches, none of which survives as  $y \rightarrow \infty$ .*

*Proof.* By lemma 7.1,  $\kappa'' = \psi + 1$ , and  $\psi$  is strictly increasing with  $\psi(x) \rightarrow -\infty$  as  $x \rightarrow 0^+$  and  $\psi(x) \rightarrow +\infty$ ; so  $\kappa''$  has exactly one zero and  $\kappa'$  is strictly convex with a unique minimum. Since  $\psi(2) = 1 - \gamma > 0$  we get  $\kappa'' > 0$  on  $[2, \infty)$ , and  $\kappa'(2) = \log \Gamma(2) + 2 - \frac{1}{2} - \frac{1}{2} \log(2\pi) = \frac{3}{2} - \frac{1}{2} \log(2\pi) = 0.5811 \dots > 0$ ; hence  $\kappa' > 0$  on  $[2, \infty)$  and  $K$  increases there.  $K(2) = 1$  is (7.4) at  $n = 1$ , and  $K \rightarrow \infty$  follows from theorem 7.4. Because  $\kappa'$  has a unique minimum and tends to  $+\infty$  at both ends of  $(0, \infty)$  while  $\kappa'(1) = 1 - \frac{1}{2} - \frac{1}{2} \log(2\pi) < 0$ , it has exactly two zeros; these are (7.3), located numerically.  $\square$

**Remark 7.3** (The notation is not cosmetic).  $K$  is not monotone on  $(0, \infty)$ , so “the inverse of  $K$ ” is ambiguous below  $K(x_{\max})$ . Everything in this chapter concerns  $K_+^{-1}$ , the branch tending to  $+\infty$ . The functional equation  $K(z+1) = z^z K(z)$  with  $K(1) = 1$  gives

$$K(n+1) = \prod_{j=1}^n j^j, \quad (7.4)$$

so  $K_+^{-1}$  restricted to the values  $\prod_{j \leq n} j^j$  inverts the hyperfactorial exactly.

## 7.3 Forward expansion, and why the centre is forced

The following puts every centring on the same footing; the distinguished one then identifies itself.

**Theorem 7.4** (Expansion about an arbitrary centre). *Fix  $\alpha \in \mathbb{C}$  and let  $x = r + \alpha$ . As  $r \rightarrow \infty$  in a closed subsector of  $|\arg r| < \pi$ ,*

$$\log K(r+\alpha) \sim \frac{r^2}{2} \log r - \frac{r^2}{4} + \left(\alpha - \frac{1}{2}\right) r \log r + \frac{B_2(\alpha)}{2} \log r + \log A + \frac{\alpha^2 - \alpha}{2} + \sum_{n \geq 1} \frac{(-1)^{n+1} B_{n+2}(\alpha)}{n(n+1)(n+2)r^n}. \quad (7.5)$$

*Truncation after  $r^{-N}$  leaves a remainder  $\mathcal{O}(r^{-N-1})$ .*

*Proof.* Start from the expansion at  $\alpha = 0$ , which follows by integrating (7.2): substituting Stirling's series into  $\kappa'(x) = \log \Gamma(x) + x - \frac{1}{2} - \frac{1}{2} \log(2\pi)$  gives

$$\kappa'(x) \sim \left(x - \frac{1}{2}\right) \log x - \frac{1}{2} + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)x^{2m-1}}, \quad (7.6)$$

and termwise integration gives

$$\log K(x) \sim \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\right) \log x - \frac{x^2}{4} + \log A - \sum_{m \geq 1} \frac{B_{2m+2}}{2m(2m+1)(2m+2)x^{2m}}, \quad (7.7)$$

the integration constant being  $\log A$  by the classical hyperfactorial asymptotic.

Now substitute  $x = r + \alpha$  and expand  $\log(r + \alpha) = \log r + \sum_{j \geq 1} (-1)^{j+1} \alpha^j / (j r^j)$  and  $(r + \alpha)^{-2m} = r^{-2m} \sum_{j \geq 0} (-1)^j \binom{2m+j-1}{j} \alpha^j r^{-j}$ . At each fixed inverse power only finitely many terms contribute. The polynomial and logarithmic part collects into the first line, using  $B_2(\alpha) = \alpha^2 - \alpha + \frac{1}{6}$  for the coefficient of  $\log r$ : the three sources of  $\log r$  are  $\frac{1}{2} r^2 \cdot 0$ , the term  $(\frac{\alpha^2}{2} - \frac{\alpha}{2} + \frac{1}{12}) \log r$  from the quadratic prefactor, which is  $\frac{1}{2} B_2(\alpha) \log r$ . For  $n \geq 1$  the coefficient of  $r^{-n}$  is a binomial convolution of Bernoulli numbers against powers of  $\alpha$ , and the identity  $B_{n+2}(\alpha) = \sum_j \binom{n+2}{j} B_j \alpha^{n+2-j}$  collapses it to  $(-1)^{n+1} B_{n+2}(\alpha) / (n(n+1)(n+2))$ . Translation by a fixed  $\alpha$  preserves closed proper subsectors, so the Poincaré remainder survives.  $\square$

**Corollary 7.5** (The midpoint is the unique good centre). *The coefficient of  $r \log r$  in (7.5) vanishes if and only if  $\alpha = \frac{1}{2}$ ; and at that same value  $B_{n+2}(\frac{1}{2}) = 0$  for every odd  $n$ , so the tail becomes even. With  $r = x - \frac{1}{2}$  and  $C = \log A - \frac{1}{8}$ ,*

$$\kappa\left(r + \frac{1}{2}\right) \sim \Phi(r) - \frac{1}{24} \log r + C + \sum_{m \geq 1} \frac{c_m}{r^{2m}}, \quad \Phi(r) = \frac{r^2}{2} \log r - \frac{r^2}{4}, \quad (7.8)$$

where

$$c_m = -\frac{B_{2m+2}(1/2)}{2m(2m+1)(2m+2)} = -\frac{(2^{-2m-1} - 1)B_{2m+2}}{2m(2m+1)(2m+2)}. \quad (7.9)$$

The first values are

$$c_1 = -\frac{7}{5760}, \quad c_2 = \frac{31}{161280}, \quad c_3 = -\frac{127}{1290240}, \quad c_4 = \frac{511}{4866048}, \quad c_5 = -\frac{1414477}{7380172800}. \quad (7.10)$$

*Proof.* The coefficient of  $r \log r$  is  $\alpha - \frac{1}{2}$ . At  $\alpha = \frac{1}{2}$  the constant is  $\log A + (\frac{1}{4} - \frac{1}{2})/2 = \log A - \frac{1}{8} = C$  and the coefficient of  $\log r$  is  $\frac{1}{2} B_2(\frac{1}{2}) = \frac{1}{2}(-\frac{1}{12}) = -\frac{1}{24}$ . Odd-index Bernoulli polynomials vanish at  $\frac{1}{2}$  by  $B_n(1 - \alpha) = (-1)^n B_n(\alpha)$ , so only even  $n = 2m$  survives, and  $B_{2m+2}(\frac{1}{2}) = (2^{-2m-1} - 1)B_{2m+2}$  gives (7.9).  $\square$

**Remark 7.6** (Two coincidences that are one). That the shift removing  $r \log r$  is also the shift killing the odd tail looks like luck. It is not: both are the reflection symmetry  $B_n(1 - \alpha) = (-1)^n B_n(\alpha)$  acting at its fixed point  $\alpha = \frac{1}{2}$ . The same remark applies to  $\Gamma$  in remark 6.2, where the half-shift does the same two jobs at once. Theorem 7.4 makes this visible by displaying every centre at the same time, instead of verifying the good one after the fact; this is the main reason it is stated in that generality.

## 7.4 The Lambert anchor and the canonical inverse

**Proposition 7.7** (Core variables). *Let  $Y = \log y$  and*

$$w = W_0\left(\frac{4Y}{e}\right), \quad \rho = e^{(w+1)/2} = 2\sqrt{\frac{Y}{w}}, \quad L = \log \rho = \frac{w+1}{2}. \quad (7.11)$$

*Then  $\Phi(\rho) = Y$  exactly, and  $\Phi'(\rho) = \rho \log \rho = \rho L$ .*

*Proof.* This is lemma 6.8 with  $a = \frac{1}{2}$ ,  $p = 2$ ,  $\beta = \frac{1}{2}$ ,  $r = 1$ : there  $v = W_k((pY/a)e^{-p\beta}) = W_0(4Y/e)$  and  $X = e^{\beta+v/p} = e^{(1+w)/2}$ , and the slope is  $aX^{p-1}(v+1) = \frac{1}{2}\rho(w+1) = \rho L$ .  $\square$

Write  $q = \rho^{-2}$ ,  $r = \rho(1+v)$ , and

$$\mathcal{D}_L(v) := \frac{1}{2}(1+v)^2(L + \log(1+v)) - \frac{1}{4}(1+v)^2 - \frac{L}{2} + \frac{1}{4} = Lv + \frac{L+1}{2}v^2 + \sum_{k \geq 3} \frac{(-1)^{k+1}}{k(k-1)(k-2)}v^k, \quad (7.12)$$

which is  $\mathcal{K}_{2,v}$  of (6.22) rescaled; its linear coefficient is  $L$  because the normalization here is by  $\rho^2$  rather than  $2\rho^2$ . Subtracting  $\Phi(\rho) = Y$  from (7.8) and dividing by  $\rho^2$  gives the exact normalized equation

$$\mathcal{D}_L(v) + q \left[ E - \frac{1}{24} \log(1+v) + \sum_{m \geq 1} c_m q^m (1+v)^{-2m} \right] = 0, \quad E := C - \frac{L}{24}. \quad (7.13)$$

**Theorem 7.8** (Canonical inverse transseries). *Equation (7.13) has a unique solution  $v = \sum_{n \geq 1} V_n(L)q^n$  with  $V_n \in \mathbb{Q}[C, L, L^{-1}]$ , given by*

$$V_n(L) = -\frac{1}{L}[q^n] \left\{ \mathcal{D}_L(v_{<n}) + q \left[ E - \frac{1}{24} \log(1+v_{<n}) + \sum_{m \geq 1} c_m q^m (1+v_{<n})^{-2m} \right] \right\}, \quad (7.14)$$

$v_{<n} = \sum_{j < n} V_j q^j$ ; only  $c_1, \dots, c_{n-1}$  can occur in  $V_n$ . For every fixed  $N \geq 1$ , as  $y \rightarrow +\infty$ ,

$$K_+^{-1}(y) = \frac{1}{2} + \rho + \sum_{n=1}^{N-1} V_n(L) \rho^{1-2n} + \mathcal{O}(\rho^{1-2N}). \quad (7.15)$$

The first coefficients are

$$V_1 = -\frac{E}{L} = \frac{1}{24} - \frac{C}{L}, \quad (7.16)$$

$$V_2 = -\frac{1}{L} \left( \frac{L+1}{2} V_1^2 - \frac{V_1}{24} + c_1 \right) = \frac{7L^2 - 240EL - 2880(L+1)E^2}{5760L^3}, \quad (7.17)$$

$$V_3 = -\frac{1}{L} \left( (L+1)V_1V_2 + \frac{V_1^3}{6} - \frac{1}{24} \left( V_2 - \frac{V_1^2}{2} \right) - 2c_1V_1 + c_2 \right), \quad (7.18)$$

$$V_4 = -\frac{1}{L} \left[ (L+1) \left( V_1V_3 + \frac{V_2^2}{2} \right) + \frac{V_1^2V_2}{2} - \frac{V_1^4}{24} - \frac{1}{24} \left( V_3 - V_1V_2 + \frac{V_1^3}{3} \right) + c_1(-2V_2 + 3V_1^2) - 4c_2V_1 + c_3 \right]. \quad (7.19)$$

*Proof.* The formal part is theorem 6.12 with  $S = 2\mathbb{N}$  and linear coefficient  $L$ : by (7.12) the coefficient of  $v$  in  $\mathcal{D}_L$  is  $L$ , which is invertible in  $\mathbb{Q}[C, L, L^{-1}]$ , and every monomial of the bracket in (7.13) carries at least one factor  $q$ . Hence a unique solution and (7.14). The displayed  $V_1, \dots, V_4$  follow by running it; they were recomputed independently (remark 7.9).

For the analytic statement, (7.8) is a Poincaré expansion with differentiable remainder on the positive ray, obtained by integrating the sectorial Stirling remainder in (7.6); retain it far enough that its own error is below the first omitted inverse term. The formal substitution kills the normalized equation through  $q^{N-1}$ , so the forward residual is  $\mathcal{O}(\rho^2 q^N) = \mathcal{O}(\rho^{2-2N})$ . By lemma 7.1 and (7.6),  $\kappa'(r + \frac{1}{2}) = r \log r + \mathcal{O}(r^{-1}) \asymp \rho L$  on the relevant interval, and this is bounded below by  $\frac{1}{2}\rho L > 0$  for large  $y$ . Theorem 1.42 converts the residual into an error  $\mathcal{O}(\rho^{2-2N}/(\rho L)) = \mathcal{O}(\rho^{1-2N}/L)$ , which is (7.15), and certifies that a root lies within it.  $\square$

*Remark 7.9* (Independent verification). As in remark 6.22, every rational number in this chapter was recomputed for this volume from the defining equations, in exact arithmetic, by an implementation



independent of the sources: the tail  $c_m$  from Bernoulli polynomials at  $\frac{1}{2}$  through  $m = 6$ ;  $V_1, \dots, V_4$  from (7.14), including the closed form (7.17) for  $V_2$ ; the accelerated forward blocks  $d_1, \dots, d_6$  both from the closed formula (7.27) and, independently, by direct expansion of the identity in the proof of theorem 7.12; the accelerated inverse blocks  $U_2, \dots, U_5$ ; and the refined blocks  $b_1, \dots, b_5$  together with the limits (7.44). All agreed with all three sources. The numerical tables of section 7.6 were computed afresh at 60 digits from  $K(x) = \exp(\zeta'(-1, x) - \zeta'(-1))$ , and reproduce  $K(5) = 27648 = 1^1 2^2 3^3 4^4$  exactly.

## 7.5 The resonance and its three resolutions

The term  $-\frac{1}{24} \log r$  enters (7.13) at  $q^1$  carrying the factor  $E = C - L/24$ , which grows with the slope. It is therefore resonant in the sense of proposition 6.30: it survives the large-slope limit, while every  $c_m$  and the constant  $C$  do not. The consequence is visible in  $V_1 = \frac{1}{24} - C/L$ , whose limit is not zero.

### 7.5.1 First resolution: leave it, and take the limit

**Proposition 7.10** (Coefficientwise limit). *For each fixed  $n$ ,*

$$\lim_{L \rightarrow \infty} V_n(L) = \binom{1/2}{n} 12^{-n}, \quad (7.20)$$

so  $V_1 \rightarrow \frac{1}{24}$ ,  $V_2 \rightarrow -\frac{1}{1152}$ ,  $V_3 \rightarrow \frac{1}{27648}$ . The limiting coefficients are those of

$$v_0(q) = \sqrt{1 + \frac{q}{12}} - 1, \quad \text{equivalently} \quad \rho(1 + v_0(\rho^{-2})) = \sqrt{\rho^2 + \frac{1}{12}}. \quad (7.21)$$

*Proof.* Divide (7.13) by  $L$  and let  $L \rightarrow \infty$  with  $q$  formal. By (7.12),  $\mathcal{D}_L(v) = L \cdot \frac{1}{2}((1+v)^2 - 1) +$  (terms free of  $L$ ), so  $\mathcal{D}_L(v)/L \rightarrow \frac{1}{2}((1+v)^2 - 1)$ . In the bracket,  $qE/L = q(C/L - \frac{1}{24}) \rightarrow -q/24$ , while  $q \log(1+v)/(24L) \rightarrow 0$  and every  $c_m$  term  $\rightarrow 0$ . The limiting equation is therefore

$$\frac{1}{2}((1+v_0)^2 - 1) - \frac{q}{24} = 0, \quad \text{i.e.} \quad (1+v_0)^2 = 1 + \frac{q}{12},$$

whose branch with  $v_0(0) = 0$  is (7.21). The limit may be taken inside the recursion because at each order the coefficient equation has limiting divisor 1 and involves only finitely many earlier coefficients. Expanding the square root gives (7.20), and  $r = \rho(1+v)$  turns  $v_0$  into  $\sqrt{\rho^2 + 1/12}$ .  $\square$

### 7.5.2 Second resolution: absorb it into an algebraic shift

**Lemma 7.11** (Absorbing a logarithmic correction). *Let  $a \neq 0$ ,  $p \neq 0$ , and suppose*

$$F(r) = ar^p(\log r - b) + \beta \log r + \gamma + \mathcal{O}(r^{-p}). \quad (7.22)$$

*Put  $T = r^p + \alpha$  and  $\Psi(T) = \frac{a}{p}T(\log T - pb)$ . Then*

$$\Psi(r^p + \alpha) = ar^p(\log r - b) + a\alpha \log r + a\alpha \left(\frac{1}{p} - b\right) + \mathcal{O}(r^{-p}), \quad (7.23)$$

*so the unique shift absorbing the term  $\beta \log r$  is*

$$\alpha = \frac{\beta}{a}. \quad (7.24)$$

*Proof.* Insert  $T = r^p + \alpha$  and use  $\log T = p \log r + \alpha r^{-p} + \mathcal{O}(r^{-2p})$ :

$$\Psi(T) = \frac{a}{p}(r^p + \alpha)(p \log r + \alpha r^{-p} - pb) + \mathcal{O}(r^{-p}) = ar^p \log r + \frac{a\alpha}{p} - abr^p + a\alpha \log r - ab\alpha + \mathcal{O}(r^{-p}),$$

which is (7.23). Matching the coefficient of  $\log r$  gives  $a\alpha = \beta$ .  $\square$

For the  $K$ -function  $a = \frac{1}{2}$ ,  $p = 2$ ,  $b = \frac{1}{2}$  and  $\beta = -\frac{1}{24}$ , so  $\alpha = -\frac{1}{12}$  and, remarkably, the constant correction  $a\alpha(1/p - b)$  vanishes because  $p^{-1} = b$ . The new coordinate is

$$T = r^2 - \frac{1}{12} = x^2 - x + \frac{1}{6}, \quad x = \frac{1}{2} + \sqrt{T + \frac{1}{12}}. \quad (7.25)$$

**Theorem 7.12** (Quadratic normal form). *With  $\Psi(T) = \frac{1}{4}T(\log T - 1)$  and  $T = x^2 - x + \frac{1}{6}$ , as  $T \rightarrow \infty$ ,*

$$\log K(x) \sim \Psi(T) + C + \sum_{n \geq 1} \frac{d_n}{T^n}, \quad (7.26)$$

where, writing  $a_0 = \frac{1}{12}$ ,

$$d_n = \frac{(-1)^n a_0^{n+1}}{4(n+1)} + \sum_{m=1}^n (-1)^{n-m} \binom{n-1}{m-1} a_0^{n-m} c_m. \quad (7.27)$$

The first values are

$$d_1 = -\frac{1}{480}, \quad d_2 = \frac{31}{90720}, \quad d_3 = -\frac{103}{725760}, \quad d_4 = \frac{179}{1330560}, \quad d_5 = -\frac{6480899}{28021593600}. \quad (7.28)$$

*Proof.* Since  $r^2 = T + a_0$  and  $\frac{1}{2}r^2 - \frac{1}{24} = \frac{T}{2}$ , the elementary part of (7.8) satisfies the *exact* identity

$$\left(\frac{r^2}{2} - \frac{1}{24}\right) \log r - \frac{r^2}{4} = \frac{T}{4} \log(T + a_0) - \frac{T}{4} - \frac{a_0}{4} = \Psi(T) + \frac{T}{4} \log\left(1 + \frac{a_0}{T}\right) - \frac{a_0}{4},$$

using  $\log r = \frac{1}{2} \log(T + a_0)$ . The first term of  $\frac{T}{4} \log(1 + a_0/T)$  is  $a_0/4$ , which cancels the constant  $-a_0/4$  exactly; this is why no constant is generated and  $C$  is unchanged. Expanding the remaining logarithm and  $(T + a_0)^{-m} = T^{-m}(1 + a_0/T)^{-m}$  and collecting the coefficient of  $T^{-n}$  gives (7.27).  $\square$

The gain is one full sector. In (7.8) the largest unresolved term was  $-\frac{1}{24} \log r$ , of size  $\log r$ ; in (7.26) it is  $d_1/T$ . Inverting exactly as before, with

$$\eta = Y - C, \quad \omega = W_0\left(\frac{4\eta}{e}\right), \quad \tau = \frac{4\eta}{\omega} = e^{\omega+1}, \quad S = \log \tau = \omega + 1, \quad (7.29)$$

so that  $\Psi(\tau) = \eta$  exactly, and writing  $T = \tau(1 + u)$ ,  $\nu = \tau^{-1}$ , the normalized equation becomes

$$\mathcal{E}_S(u) + 4 \sum_{m \geq 1} d_m \nu^{m+1} (1 + u)^{-m} = 0, \quad \mathcal{E}_S(u) = Su + \sum_{k \geq 2} \frac{(-1)^k}{k(k-1)} u^k. \quad (7.30)$$

**Theorem 7.13** (Accelerated inverse). *Equation (7.30) has a unique solution  $u = \sum_{n \geq 2} U_n(S) \nu^n$  in  $\nu^2 \mathbb{Q}[S, S^{-1}][[\nu]]$ , with*

$$U_n(S) = -\frac{1}{S}[\nu^n] \left\{ \mathcal{E}_S(u_{<n}) + 4 \sum_{m \geq 1} d_m \nu^{m+1} (1 + u_{<n})^{-m} \right\}, \quad (7.31)$$

and

$$U_2 = \frac{1}{120S}, \quad U_3 = -\frac{31}{22680S}, \quad U_4 = \frac{1030S^2 - 126S - 63}{1814400S^3}, \quad U_5 = -\frac{16110S^2 - 1023S - 341}{29937600S^3}. \quad (7.32)$$

Consequently, as  $y \rightarrow +\infty$ ,

$$K_+^{-1}(y) \sim \frac{1}{2} + \sqrt{\frac{1}{12} + \tau \left(1 + \sum_{n \geq 2} U_n(S) \tau^{-n}\right)}, \quad (7.33)$$

and truncating after  $U_N \tau^{-N}$  leaves an error  $\mathcal{O}(\tau^{-N-1/2})$  in  $x$ . Expanding the root,

$$K_+^{-1}(y) = \frac{1}{2} + R + \frac{1}{240S\tau R} - \frac{31}{45360S\tau^2 R} + \mathcal{O}(\tau^{-7/2}), \quad R = \sqrt{\tau + \frac{1}{12}}. \quad (7.34)$$

*Proof.*  $\mathcal{E}_S$  has linear coefficient  $S$ , a unit, and the forcing begins at  $\nu^2$  because  $d_m \nu^{m+1}$  starts at  $m = 1$ ; theorem 6.12 with  $S = \mathbb{N}$  gives uniqueness and (7.31), and  $u = \mathcal{O}(\nu^2)$ . The values (7.32) were computed and independently verified (remark 7.9). The analytic step is theorem 1.42 exactly as in theorem 7.8, with slope  $\Psi'(\tau) = \frac{1}{4}(\log \tau) = \frac{S}{4}$  in the  $T$  variable; converting to  $x$  through  $x = \frac{1}{2} + \sqrt{T + 1/12}$  costs the factor  $\frac{1}{2}T^{-1/2}$ , which is the source of the half-integer exponent.  $\square$

### 7.5.3 The first two resolutions are the same

**Theorem 7.14** (Acceleration is resummation). *The coefficientwise limit of proposition 7.10 and the algebraic substitution of lemma 7.11 are the same construction:*

$$\frac{1}{2} + \rho + \sum_{n \geq 1} \left( \lim_{L \rightarrow \infty} V_n(L) \right) \rho^{1-2n} = \frac{1}{2} + \sqrt{\rho^2 + \frac{1}{12}} = \frac{1}{2} + \sqrt{T + \frac{1}{12}} \Big|_{T=\rho^2-1/12}, \quad (7.35)$$

and the map  $T \mapsto \frac{1}{2} + \sqrt{T + \frac{1}{12}}$  appearing on the right is precisely the inverse of the accelerated coordinate (7.25). In other words, the leading term of the accelerated expansion (7.33) is the sum of the resonant subsector of the canonical expansion (7.15), and the residual series  $\sum U_n \tau^{-n}$  is what remains of the canonical coefficients after that subsector is removed.

*Proof.* By proposition 7.10 the limiting subsector is  $\sum_{n \geq 1} \binom{1/2}{n} 12^{-n} \rho^{1-2n} = \rho(\sqrt{1 + 1/(12\rho^2)} - 1) = \sqrt{\rho^2 + 1/12} - \rho$ , so adding  $\frac{1}{2} + \rho$  gives the middle expression in (7.35). The right-hand equality is (7.25) read backwards. For the last sentence: lemma 7.11 removes from the forward expansion exactly the term  $\beta \log r$  that is resonant, and by proposition 6.30 a perturbation is resonant precisely when it survives division by the slope in the large-slope limit; the remaining forward terms  $d_n T^{-n}$  have no  $\log T$  and are therefore non-resonant, so the accelerated coefficients  $U_n$  all tend to 0 — indeed  $U_n = \mathcal{O}(S^{-1})$  by (7.32) and the recurrence.  $\square$

*Remark 7.15* (The same statement for Barnes  $G$ ). Neither Barnes source performs this absorption, and it costs nothing. In lemma 6.3 the data are  $a = \frac{1}{2}$ ,  $p = 2$ ,  $b = \frac{3}{2}$ ,  $\beta = -\frac{1}{12}$ , so lemma 7.11 gives  $\alpha = -\frac{1}{6}$ , the coordinate  $T = z^2 - \frac{1}{6}$ , and the normal form

$$\log G(z+1) = \frac{1}{4}T(\log T - 3) + \alpha_* \sqrt{T + \frac{1}{6}} + \left( \zeta'(-1) - \frac{1}{12} \right) + \mathcal{O}(T^{-1}), \quad \alpha_* = \frac{1}{2} \log(2\pi), \quad (7.36)$$

the constant  $-\frac{1}{12}$  coming from  $a\alpha(1/p - b)$  in (7.23). The square root  $\sqrt{T + \frac{1}{6}}$  is exactly the resummed subsector  $\sqrt{U^2 + 1/6}$  that proposition 6.30 identifies as a coefficientwise limit — so remark 6.31, which warns that the limit is not an approximation, is answered here: it becomes one as soon as it is moved into the coordinate. The price for Barnes, and the reason its sources may have balked, is the half-integer power  $\sqrt{T + 1/6}$  forced by the linear term  $\alpha_* z$ , which the  $K$ -function does not have. That is the precise sense in which the  $K$ -function is the cleaner subject.

### 7.5.4 Third resolution: absorb it into the core

The shift of lemma 7.11 absorbs  $\beta \log r$  only modulo  $\mathcal{O}(r^{-p})$ . It can instead be absorbed *exactly*, at the cost of replacing  $W$  by a one-parameter relative.

**Definition 7.16** (The  $r$ -Lambert function). For fixed  $\varrho$ , let  $W^{(\varrho)}$  denote a branch of the inverse of  $s \mapsto se^s + \varrho s$ , so that

$$W^{(\varrho)}(z) e^{W^{(\varrho)}(z)} + \varrho W^{(\varrho)}(z) = z. \quad (7.37)$$

For  $\varrho = 0$  this is  $W$ .

**Theorem 7.17** (Exact inversion of an affine logarithmic power core). *Let*

$$\Psi_*(x) = ax^p \log x + b_* x^p + c_* \log x + d_*, \quad ap \neq 0. \quad (7.38)$$

*Fix compatible branches and set*

$$\varrho = \frac{c_*}{a} \exp\left(\frac{pb_*}{a}\right), \quad z = \frac{p}{a} \exp\left(\frac{pb_*}{a}\right) \left(Y - d_* + \frac{c_* b_*}{a}\right). \quad (7.39)$$

*If  $s = W^{(\varrho)}(z)$ , then*

$$x = \exp\left(\frac{s}{p} - \frac{b_*}{a}\right) \quad (7.40)$$

*solves  $\Psi_*(x) = Y$ , and every local branch arises from compatible choices of the logarithm and of a branch of  $W^{(\varrho)}$ .*

*Proof.* Put  $s = p \log x + pb_*/a$ , so that  $\log x = s/p - b_*/a$  and  $x^p = \exp(s - pb_*/a)$ . The two power terms collapse:

$$ax^p \log x + b_* x^p = x^p (a \log x + b_*) = \exp\left(s - \frac{pb_*}{a}\right) \cdot \frac{as}{p} = \frac{a}{p} \exp\left(-\frac{pb_*}{a}\right) s e^s,$$

where the middle step uses  $a \log x + b_* = a(s/p - b_*/a) + b_* = as/p$ . The logarithmic term is  $c_*(s/p - b_*/a)$ . Hence  $\Psi_*(x) = Y$  reads

$$\frac{a}{p} e^{-pb_*/a} s e^s + \frac{c_*}{p} s = Y - d_* + \frac{c_* b_*}{a},$$

and multiplying by  $(p/a)e^{pb_*/a}$  gives  $se^s + \varrho s = z$  with the data (7.39).  $\square$

**Corollary 7.18** (The refined  $K$  anchor). *Apply theorem 7.17 to  $\Psi_*(r) = \frac{1}{2}r^2 \log r - \frac{1}{4}r^2 - \frac{1}{24} \log r + C$ , that is to  $a = \frac{1}{2}$ ,  $p = 2$ ,  $b_* = -\frac{1}{4}$ ,  $c_* = -\frac{1}{24}$ ,  $d_* = C$ . Then*

$$\varrho = -\frac{1}{12e}, \quad z = \frac{48(Y - C) + 1}{12e}, \quad (7.41)$$

*and the resulting  $\rho_* = \exp(\frac{1}{2}W^{(\varrho)}(z) + \frac{1}{2})$  satisfies  $\Psi_*(\rho_*) = Y$  exactly, with slope  $\Psi'_*(\rho_*) = \rho_* \log \rho_* - \frac{1}{24\rho_*}$ . The residual perturbation is only the Bernoulli tail  $\sum_{m \geq 1} c_m \rho_*^{-2m}$ .*

*Proof.* Substitute the data into (7.39):  $pb_*/a = -1$ , so  $\varrho = (-\frac{1}{24}) \cdot 2 \cdot e^{-1} = -1/(12e)$  and  $z = 4e^{-1}(Y - C + (-\frac{1}{24})(-\frac{1}{4}) \cdot 2) = 4e^{-1}(Y - C + \frac{1}{48})$ , which is (7.41). The formula for  $\rho_*$  is (7.40) with  $p = 2$  and  $b_*/a = -\frac{1}{2}$ ; the slope is immediate.  $\square$

**Theorem 7.19** (Refined blocks). *With  $t = \rho_*^{-2}$ ,  $L_* = \log \rho_*$ ,  $r = \rho_* H$  and  $H = 1 + \sum_{n \geq 1} b_n(L_*) t^{n+1}$ , the normalized equation is*

$$H^2(2L_* - 1 + 2 \log H) - (2L_* - 1) - \frac{t}{6} \log H + 4 \sum_{m \geq 1} c_m t^{m+1} H^{-2m} = 0, \quad (7.42)$$

*its linearization at  $H = 1$ ,  $t = 0$  is  $4L_*$ , and*

$$b_n(L) = -\frac{1}{4L} [t^{n+1}] \text{ (left side of (7.42) at } H = H_{n-1}^*), \quad n \geq 1. \quad (7.43)$$

*Each  $b_n$  is rational in  $L$  with  $b_n = \mathcal{O}(L^{-1})$ , uses only  $c_1, \dots, c_n$ , and satisfies*

$$\lim_{L \rightarrow \infty} L b_n(L) = -c_n. \quad (7.44)$$

For every fixed  $N$ ,

$$K_+^{-1}(y) = \frac{1}{2} + \rho_* + \sum_{n=1}^N \frac{b_n(L_*)}{\rho_*^{2n+1}} + \mathcal{O}\left(\frac{\rho_*^{-2N-3}}{L_*}\right). \quad (7.45)$$

The fourth and fifth blocks are

$$b_4 = -\frac{12877200L^3 - 703428L^2 + 57552L + 539}{122624409600L^4}, \quad (7.46)$$

$$b_5 = \frac{1539856241280L^4 - 45140228640L^3 + 1666058823L^2 - 38145107L - 25162137}{8034351316992000L^5}. \quad (7.47)$$

*Proof.* The equation is (7.13) with the resonant term and the constant moved into the core, so only the tail remains; the linearization is  $4L_*$  because the normalization is by  $\frac{1}{4}\rho_*^2$ . Theorem 6.12 gives uniqueness and (7.43). Since the forcing begins at  $t^2$ , so does  $H - 1$ , which is why the ansatz uses  $t^{n+1}$ . For (7.44): at order  $t^{n+1}$  the only contribution independent of earlier blocks is  $4c_n t^{n+1}$ , so  $4Lb_n = -4c_n + \mathcal{O}(L^0 \cdot L^{-1})$ , giving  $Lb_n \rightarrow -c_n$ ; equivalently this is the linear response  $-S(\rho_*)/\Psi'_*(\rho_*)$ . The two displayed blocks were computed and independently verified (remark 7.9). The analytic clause is again theorem 1.42, with the slope  $\Psi'_*(\rho_*) \sim \rho_* L_*$ .  $\square$

*Remark 7.20* (Which resolution to use). All three are correct and they answer different questions. The canonical expansion is the one to state, because its coefficients live in  $\mathbb{Q}[C, L, L^{-1}]$  and it needs only  $W_0$ . The accelerated coordinate is the one to compute with: table 7.3 shows its leading term alone beating four canonical blocks at  $x = 5$ . The  $r$ -Lambert anchor is the one that explains: it shows that the obstruction is not deep, merely misplaced, since a single one-parameter special function removes it exactly and leaves nothing but the Bernoulli tail – which is the same tail that  $\Gamma$  and  $G$  leave, and which is beyond repair because it is divergent.

## 7.6 Numerical verification

The protocol is that of section 6.10: choose an exact abscissa  $x$ , evaluate  $K(x) = \exp(\zeta'(-1, x) - \zeta'(-1))$  to 60 digits, reconstruct the core variables from that value alone, and compare the truncations with  $x$ . Errors are signed.

Table 7.2: Canonical expansion (7.15): signed error of  $\frac{1}{2} + \rho$  and of the truncations including  $V_1, \dots, V_4$ .

| $x$ | $\frac{1}{2} + \rho$     | $+ V_1$                   | $+ V_2$                  | $+ V_3$                   | $+ V_4$                  |
|-----|--------------------------|---------------------------|--------------------------|---------------------------|--------------------------|
| 5   | $9.0012 \times 10^{-3}$  | $1.8437 \times 10^{-5}$   | $1.2665 \times 10^{-7}$  | $-5.9396 \times 10^{-10}$ | $6.4854 \times 10^{-11}$ |
| 10  | $1.3996 \times 10^{-3}$  | $-1.9257 \times 10^{-7}$  | $8.4998 \times 10^{-10}$ | $-5.1976 \times 10^{-12}$ | $6.4879 \times 10^{-14}$ |
| 20  | $-2.7600 \times 10^{-7}$ | $-5.5164 \times 10^{-8}$  | $2.0887 \times 10^{-11}$ | $-2.9637 \times 10^{-14}$ | $8.4762 \times 10^{-17}$ |
| 50  | $-2.0103 \times 10^{-4}$ | $-2.9310 \times 10^{-9}$  | $1.9068 \times 10^{-13}$ | $-3.7123 \times 10^{-17}$ | $1.5892 \times 10^{-20}$ |
| 100 | $-1.4839 \times 10^{-4}$ | $-2.6924 \times 10^{-10}$ | $5.4398 \times 10^{-15}$ | $-2.5162 \times 10^{-19}$ | $2.6077 \times 10^{-23}$ |

Table 7.3: Accelerated expansion (7.33): signed error of  $\frac{1}{2} + \sqrt{\frac{1}{12} + \tau}$  and of the truncations including  $U_2, \dots, U_5$ .

| $x$ | $\frac{1}{2} + \sqrt{\frac{1}{12} + \tau}$ | $+ U_2$                  | $+ U_3$                   | $+ U_4$                  | $+ U_5$                   |
|-----|--------------------------------------------|--------------------------|---------------------------|--------------------------|---------------------------|
| 5   | $-1.5162 \times 10^{-5}$                   | $1.2198 \times 10^{-7}$  | $-2.3300 \times 10^{-9}$  | $1.0850 \times 10^{-10}$ | $-9.0171 \times 10^{-12}$ |
| 10  | $-1.0786 \times 10^{-6}$                   | $1.9569 \times 10^{-9}$  | $-8.6886 \times 10^{-12}$ | $9.1990 \times 10^{-14}$ | $-1.7528 \times 10^{-15}$ |
| 20  | $-9.4572 \times 10^{-8}$                   | $4.0777 \times 10^{-11}$ | $-4.3490 \times 10^{-14}$ | $1.0944 \times 10^{-16}$ | $-4.9719 \times 10^{-19}$ |
| 50  | $-4.4020 \times 10^{-9}$                   | $2.9465 \times 10^{-13}$ | $-4.9102 \times 10^{-17}$ | $1.9149 \times 10^{-20}$ | $-1.3502 \times 10^{-23}$ |
| 100 | $-4.5974 \times 10^{-10}$                  | $7.6166 \times 10^{-15}$ | $-3.1503 \times 10^{-19}$ | $3.0371 \times 10^{-23}$ | $-5.2971 \times 10^{-27}$ |

The comparison is the point. At  $x = 5$  the accelerated *leading* term ( $1.5 \times 10^{-5}$ ) is already as accurate as the canonical expansion carried through its first block ( $1.8 \times 10^{-5}$ ), and the pattern persists down

the table: at every argument, accelerated column  $j$  tracks canonical column  $j + 1$ . This is theorem 7.14 in numbers. The accelerated leading term is not a better approximation of the same kind — it is the canonical expansion with an entire infinite subsector already summed, so it starts one block ahead and stays there. The accelerated form is also the cheaper of the two to evaluate, since  $U_2, \dots, U_5$  are shorter expressions than  $V_1, \dots, V_4$  and involve no  $C$ .

*Remark 7.21* (The anomalous row). The entry  $-2.76 \times 10^{-7}$  at  $x = 20$  in table 7.2 is far smaller than its neighbours at  $x = 10$  and  $x = 50$ , and the column changes sign there. This is not an accident of the tabulation and not a numerical artefact: by (7.16) the first correction vanishes exactly when

$$V_1(L) = \frac{1}{24} - \frac{C}{L} = 0, \quad \text{i.e.} \quad L = 24C = 2.97010\dots, \quad (7.48)$$

which corresponds to  $\rho = e^{24C} = 19.4944\dots$ , that is to  $x = \frac{1}{2} + \rho = 19.994\dots$ . The tabulated argument  $x = 20$  is within 0.006 of the zero of  $V_1$ , so at that one point the leading correction is almost exactly absent and the uncorrected core is anomalously good. The effect is entirely local; it says nothing about the quality of the expansion and everything about the numerical value of the Glaisher constant.

## 7.7 Comparison with $\Gamma$ and $G$

Table 7.4: The three power–logarithmic subjects of Parts 6 and 7 in the notation of theorem 6.41.

|               | $\Gamma(t + \frac{1}{2})$             | $K(r + \frac{1}{2})$                          | $G(z + 1)$                       |
|---------------|---------------------------------------|-----------------------------------------------|----------------------------------|
| $a$           | 1                                     | $\frac{1}{2}$                                 | $\frac{1}{2}$                    |
| $p$           | 1                                     | 2                                             | 2                                |
| $\beta$       | 1                                     | $\frac{1}{2}$                                 | $\frac{3}{2}$                    |
| linear term   | —                                     | —                                             | $\frac{1}{2} \log(2\pi) z$       |
| resonant term | —                                     | $-\frac{1}{24} \log r$                        | $-\frac{1}{12} \log z$           |
| support       | 2N                                    | 2N                                            | N                                |
| tail          | $a_k = \frac{B_{2k}(1/2)}{2k(2k-1)}$  | $c_m = \frac{-B_{2m+2}(1/2)}{2m(2m+1)(2m+2)}$ | $c_k = \frac{B_{2k+2}}{4k(k+1)}$ |
| square root   | $\sqrt{\Lambda^2 T^2 + \frac{1}{12}}$ | $\sqrt{\rho^2 + \frac{1}{12}}$                | $\sqrt{U^2 + \frac{1}{6}}$       |
| its source    | deepest pole                          | resonance                                     | resonance                        |

Three remarks are worth making from the merged view; none is in any of the six sources.

*Remark 7.22* (Where the  $\frac{1}{12}$  comes from, twice). The constant  $\frac{1}{12}$  appears in the last row of table 7.4 twice, for two apparently unrelated reasons: for  $\Gamma$  it is the deepest pole in the slope (theorem 6.17, where it arises as  $-2a_1$ ), and for  $K$  it is the resonant limit (proposition 7.10, where it arises as  $-24^{-1} \cdot (-2)$ ). The coincidence is not one. Both are  $-2 \cdot B_2(\frac{1}{2})/2 = \frac{1}{12}$ : for  $\Gamma$ ,  $a_1 = B_2(\frac{1}{2})/2 = -\frac{1}{24}$  is the first tail coefficient; for  $K$ ,  $\frac{1}{2} B_2(\frac{1}{2}) = -\frac{1}{24}$  is the coefficient of the resonant logarithm produced by corollary 7.5. The same Bernoulli value at the same reflection centre is doing both jobs, in one case as a tail coefficient and in the other as a logarithmic coefficient.

*Remark 7.23* (The square root is always the quadratic model). All three square roots in table 7.4, and the fourth one  $\sqrt{\lambda^2 U^2 - 2\alpha U}$  of (6.31), are instances of lemma 6.16: truncate the normalized equation to its linear and quadratic terms in the displacement together with the lowest forcing monomial, and solve. What differs between the cases is only which *extraction* of the coefficient family the truncation computes — the deepest pole in the slope, or the coefficientwise large-slope limit — and that in turn is decided by whether the forcing monomial carries a factor of the slope, i.e. by whether it is resonant. Resonance and pole depth are thus two readings of one quadratic.

*Remark 7.24* (What  $\Gamma$  and  $G$  are missing). The  $K$ -function sources supply two devices the  $\Gamma/G$  sources do not: the absorption lemma lemma 7.11 and the  $r$ -Lambert anchor theorem 7.17. Both apply to  $G$ . The absorption is carried out in remark 7.15. The  $r$ -Lambert anchor does *not* apply to  $G$  as it stands, because

theorem 7.17 requires the core to be  $ax^p \log x + b_*x^p + c_* \log x + d_*$  and the Barnes linear term  $\alpha_*z$  is of degree  $p - 1$ , outside that shape; absorbing it would need a two-parameter relative of  $W$  for which no closed theory is invoked here. Neither device applies to  $\Gamma$ , which has  $p = 1$  and no resonant term at all — for  $\Gamma$  the logarithmic coefficient  $\frac{1}{2}B_2(h)$  at  $h = \frac{1}{2}$  enters the *tail*, not the phase. So the three subjects are not three copies of one calculation: they are three different positions of a single obstruction, and the  $K$ -function is the one where it sits in the phase and can be removed.

## 7.8 What is left open

1. *A two-parameter anchor.* Remark 7.24 identifies the gap: a core of the shape  $ax^p \log x + b_*x^p + e_*x^{p-1} + c_* \log x + d_*$ , which would cover Barnes  $G$  exactly as  $W^{(e)}$  covers  $K$ . Whether its inverse is expressible through a named function, or whether the right statement is simply that the implicit equation defines one, is not settled here.
2. *Closed forms.* As in section 6.13, the recurrences compute  $V_n$ ,  $U_n$  and  $b_n$  to any order and their extreme coefficients are known — (7.20) and (7.44) — but no closed description of the full numerators is available. The refined blocks are the most promising, since (7.44) says their leading behaviour is exactly the forward tail with a sign, so the question is the structure of the correction to that.
3. *Hyperasymptotics.* Unchanged from remark 6.40. The  $K$ -function tail  $c_m$  inherits the factorial growth of  $B_{2m+2}$ , so the least term of (7.8) is near  $m \approx \pi r$  and the inverse floor is  $\exp(-2\pi\rho + \mathcal{O}(\log \rho))/(\rho L)$  by the argument of proposition 6.39. The Stokes multipliers are not computed.
4. *The other real branches.* Proposition 7.2 locates the two stationary points and hence the additional real branches below  $K(x_{\max})$ . Those branches are governed by the local behaviour at a critical point — a square-root, not a Lambert, scale — and are not treated.
5. *Formalization.* Unchanged from section 6.13: nothing in either chapter is machine-checked. The  $r$ -Lambert function would need its own development.

## Chapter 8

# The subfactorial

### 8.1 What this chapter does

The derangement numbers  $D_n = !n = n! \sum_{j \leq n} (-1)^j / j!$  are the first subject in this volume whose inverse is *not* governed by its own dominant block. The block is  $\Gamma(x+1)/e$ , whose inverse is the subject of section 6.5; everything that is specifically about derangements lives in a correction so small that it is beyond every fixed exponential scale relative to the inverse variable. The chapter is therefore organized around that separation, and its principal economy is that the carrier is *cited*, not recomputed.

All three sources rederive the inverse of the gamma envelope from scratch, each proving its own Lambert-centred reversion theorem and each printing the blocks

$$\frac{1}{24s}, \quad -\frac{14s^2 + 10s + 5}{5760s^3}, \quad \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5}, \quad \dots$$

These are *identical*, coefficient for coefficient, to  $d_1, d_3, d_5, d_7$  of proposition 6.21 — three more copies of a computation the volume already contains twice over. They are not repeated here. Proposition 8.9 states the dictionary and cites theorem 6.20; the general reversion theorem each source proves in order to get them is theorem 1.18, and is cited too.

The three filed articles are `inverse_subfactorial_transseries` (S1), `inverse_subfactorial_transseries-2` (S2) and `inverse_subfactorial_transseries-3` (S3). Their distinctive contributions are: S1, the exact core–tail decomposition, the median interpolation, and the threshold analysis of section 8.9; S2, the signed-remainder Bell expansion with derivatives and the a posteriori error bound; S3, the branch-indexed continuation, the single-frequency sector factorization of theorem 8.13, and the multi-frequency grid. They agree on everything they share; every constant and coefficient was recomputed for this volume and reproduced.

### 8.2 Which inverse?

$D_n$  is defined on  $\mathbb{N}_0$ , so definition 1.37 applies: there is an integer staircase, there are interpolated inverses — one for each admissible interpolation — and there is an asymptotic inverse attached to no interpolation. The subfactorial adds a fourth issue that the earlier chapters did not face: its natural analytic continuation is *branch-dependent*, and different branches disagree away from the integers while agreeing on them. Section 8.3 settles this before any inverse is taken.



### 8.3 The exact core–tail decomposition

**Theorem 8.1** (Branch splitting). *For  $\Re z > -1$  put*

$$F(z) = \frac{\Gamma(z+1)}{e}, \quad C(z) = \frac{1}{e} \int_0^1 t^z e^t dt = \frac{1}{e} \sum_{j \geq 0} \frac{1}{j! (z+j+1)}, \quad (8.1)$$

and for  $\kappa \in \mathbb{Z}$  set  $\omega_\kappa = (2\kappa+1)\pi i$ . Then the continuation of  $!z$  obtained by running the incomplete-gamma path from 0 to  $-1$  with argument  $(2\kappa+1)\pi$  is

$$\mathcal{S}_\kappa(z) = F(z) + e^{\omega_\kappa z} C(z), \quad (8.2)$$

and  $\mathcal{S}_\kappa(n) = D_n$  for every  $n \in \mathbb{N}_0$  and every  $\kappa$ .

*Proof.* Inclusion–exclusion gives  $D_n = n! \sum_{j \leq n} (-1)^j / j!$ , and the upper incomplete gamma identity  $\Gamma(n+1, w) = n! e^{-w} \sum_{j \leq n} w^j / j!$  at  $w = -1$  gives  $D_n = \Gamma(n+1, -1)/e$ . Continue: with  $s = z+1$  and the path  $t = \varrho e^{i(2\kappa+1)\pi}$ ,  $0 \leq \varrho \leq 1$ , on which  $e^{i(2\kappa+1)\pi} = -1$ ,

$$\gamma(s, -1) = e^{i(2\kappa+1)\pi s} \int_0^1 \varrho^{s-1} e^\varrho d\varrho,$$

so that  $\Gamma(s, -1) = \Gamma(s) - \gamma(s, -1)$  yields

$$\frac{\Gamma(z+1, -1)}{e} = F(z) - \frac{e^{i(2\kappa+1)\pi(z+1)}}{e} \int_0^1 \varrho^z e^\varrho d\varrho = F(z) + e^{\omega_\kappa z} C(z),$$

the sign coming from  $e^{i(2\kappa+1)\pi} = -1$ . Expanding  $e^t$  under the integral and integrating termwise gives the partial-fraction form in (8.1). At  $z = n$  the phase  $e^{\omega_\kappa n}$  equals  $(-1)^n$  for every  $\kappa$ , so all branches agree there and equal  $n!/e + (-1)^n C(n)$ , which is  $D_n$ .  $\square$

**Corollary 8.2** (The nearest-integer property). *For every  $n \geq 0$ ,  $D_n = n!/e + (-1)^n C(n)$ , and*

$$0 < C(x) < \frac{1}{x+1} \quad (x > -1). \quad (8.3)$$

Hence for  $n \geq 1$  the correction is less than  $\frac{1}{2}$  in absolute value and  $D_n$  is the nearest integer to  $n!/e$ .

*Proof.* The identity is theorem 8.1 at  $z = n$ . For the bound,  $C(x) = \int_0^1 e^{t-1} t^x dt$  and  $0 < e^{t-1} < 1$  on  $[0, 1]$ , so  $0 < C(x) < \int_0^1 t^x dt = 1/(x+1)$ . For  $n \geq 1$  this is at most  $\frac{1}{2}$ , with equality impossible by strictness.  $\square$

**Definition 8.3** (Two real interpolations). The branches  $\kappa = 0$  and  $\kappa = -1$  are complex conjugate on the real axis; their mean is the *median* (or symmetrized) interpolation

$$\mathcal{S}_{\text{med}}(x) = F(x) + C(x) \cos(\pi x), \quad (8.4)$$

which is real, agrees with  $D_n$  at every  $n \in \mathbb{N}_0$ , and is entire. The *parity envelopes* are

$$G_\sigma(x) = F(x) + \sigma C(x), \quad \sigma \in \{+1, -1\}, \quad (8.5)$$

so that  $G_+(n) = D_n$  for even  $n$  and  $G_-(n) = D_n$  for odd  $n$ .

**Remark 8.4** (The median is canonical, not unique). Any entire function vanishing on  $\mathbb{N}_0$  may be added to  $\mathcal{S}_{\text{med}}$  without disturbing the interpolation property, so (8.4) is not determined by interpolation alone. What distinguishes it is that it is the median of the two *nearest* lateral determinations of the same incomplete-gamma continuation, so it inherits their asymptotics rather than importing new ones. This is the same situation as Part 3, where no canonical real inverse of the double factorial exists until an interpolation is fixed; the resolution here is cleaner only because the two candidate branches are conjugates.

**Proposition 8.5** (Eventual monotonicity). *There is  $X$  such that  $\mathcal{S}_{\text{med}}$ ,  $G_+$ ,  $G_-$  and  $F$  are all strictly increasing on  $[X, \infty)$ .*

*Proof.*  $F'(x) = F(x)\psi(x+1)$ , and  $\psi(x+1) \sim \log x \rightarrow \infty$ , so  $F' > 0$  and  $F' \rightarrow \infty$  eventually. From (8.3) and  $C'(x) = \int_0^1 e^{t-1} t^x \log t \, dt$  we get  $C(x) = \mathcal{O}(x^{-1})$  and  $|C'(x)| = \mathcal{O}(x^{-2})$  – for the latter,  $|\log t| t^x$  integrates to  $(x+1)^{-2}$  on  $[0, 1]$ . Hence

$$\left| \frac{d}{dx} (C(x) \cos \pi x) \right| \leq |C'(x)| + \pi |C(x)| = \mathcal{O}(x^{-1}),$$

while  $F'(x) \rightarrow \infty$ ; the same bound covers  $\sigma C'$ . So the envelope derivative dominates.  $\square$

## 8.4 The Bell expansion of the tail

$C$  is a Stieltjes transform, and its moments are Bell numbers. This gives the tail expansion together with an exact remainder – one of the few places in this volume where a divergent expansion comes with a two-sided sign statement.

**Theorem 8.6** (Bell expansion with signed remainder). *Let  $\mathfrak{B}_m$  denote the Bell numbers,  $\mathfrak{B}_1 = 1$ ,  $\mathfrak{B}_2 = 2$ ,  $\mathfrak{B}_3 = 5$ ,  $\mathfrak{B}_4 = 15$ ,  $\mathfrak{B}_5 = 52$ ,  $\mathfrak{B}_6 = 203$ . For every real  $x > 0$  and integer  $M \geq 1$ ,*

$$C(x) = \sum_{m=1}^M (-1)^{m-1} \frac{\mathfrak{B}_m}{x^m} + R_M(x), \quad R_M(x) = \frac{(-1)^M}{e x^M} \sum_{j \geq 0} \frac{(j+1)^M}{j! (x+j+1)}. \quad (8.6)$$

*In particular  $(-1)^M R_M(x) > 0$  and*

$$|R_M(x)| \leq \frac{\mathfrak{B}_{M+1}}{x^{M+1}}, \quad (8.7)$$

*so the remainder has the sign of the first omitted term and is no larger than it. Consequently  $C(x) \sim \sum_{m \geq 1} (-1)^{m-1} \mathfrak{B}_m x^{-m}$ , an expansion that diverges for every  $x$ .*

*Proof.* For  $b > 0$  the finite geometric identity

$$\frac{1}{x+b} = \sum_{m=1}^M (-1)^{m-1} \frac{b^{m-1}}{x^m} + (-1)^M \frac{b^M}{x^M(x+b)}$$

is exact. Put  $b = j+1$ , divide by  $e j!$ , and sum over  $j \geq 0$ ; the partial-fraction form of  $C$  in (8.1) gives the left side. The  $m$ -th coefficient is

$$\frac{1}{e} \sum_{j \geq 0} \frac{(j+1)^{m-1}}{j!} = \mathfrak{B}_m, \quad (8.8)$$

which is Dobiński's formula combined with the Bell recurrence  $\mathfrak{B}_m = \sum_k \binom{m-1}{k} \mathfrak{B}_k$ : expanding  $(j+1)^{m-1}$  binomially and applying Dobiński's  $\mathfrak{B}_k = e^{-1} \sum_j j^k / j!$  to each power gives exactly that sum. The last term of the geometric identity gives  $R_M$ , whose summand is positive, so  $(-1)^M R_M > 0$ . Finally  $x+j+1 \geq x$  gives

$$|R_M(x)| \leq \frac{1}{e x^{M+1}} \sum_{j \geq 0} \frac{(j+1)^M}{j!} = \frac{\mathfrak{B}_{M+1}}{x^{M+1}},$$

again by (8.8). Divergence follows from  $\mathfrak{B}_{2m} \geq (2m-1)!!$ , so  $\mathfrak{B}_m^{1/m} \rightarrow \infty$ .  $\square$

**Corollary 8.7** (Derivatives). *For every  $r \geq 0$ ,*

$$C^{(r)}(x) = \frac{1}{e} \int_0^1 t^x (\log t)^r e^t dt, \quad (-1)^r C^{(r)}(x) > 0, \quad (8.9)$$

*and termwise differentiation of the expansion is legitimate at every fixed order:*

$$C^{(r)}(x) \sim \sum_{m \geq 1} (-1)^{m+r-1} (m)_r \mathfrak{B}_m x^{-m-r}, \quad (m)_r = m(m+1) \cdots (m+r-1). \quad (8.10)$$

*In particular  $C^{(r)}(x) = \mathcal{O}(x^{-r-1})$ .*

*Proof.* Differentiation under the integral sign is justified by dominated convergence on  $[0, 1]$ , and  $(\log t)^r$  has sign  $(-1)^r$  there. For (8.10), differentiate the exact identity (8.6)  $r$  times: the finite sum differentiates term by term, and  $R_M^{(r)}$  obeys the same kind of bound because differentiating the integral representation of  $R_M$  only inserts a bounded power of  $\log t$ .  $\square$

**Remark 8.8** (Numerical check). The moment identity (8.8) was verified symbolically for  $m \leq 6$ , giving  $\mathfrak{B}_1, \dots, \mathfrak{B}_6 = 1, 2, 5, 15, 52, 203$  as the coefficients in (8.6). The two representations of  $C$  in (8.1) were checked against each other to 30 digits at  $x = 5, 10, 20$ , and at each of those points the truncation after  $\mathfrak{B}_5$  was confirmed to have an error of the predicted sign and within the bound (8.7) — for instance  $|R_5(20)| = 2.616 \times 10^{-6}$  against the bound  $\mathfrak{B}_6/20^6 = 3.172 \times 10^{-6}$ . The general reversion coefficients of proposition 8.9 were recomputed and, specialized to the centred Stirling tail, reproduce  $d_1, d_3$  of proposition 6.21 exactly.

## 8.5 The carrier

**Proposition 8.9** (The carrier is Chapter 5's inverse). *Let  $u(Y)$  denote the eventual inverse of  $F(x) = \Gamma(x+1)/e$ , so that  $\Gamma(u+1) = eY$ . Then*

$$u(Y) = \Gamma_+^{-1}(eY) - 1, \quad (8.11)$$

*and therefore, by theorem 6.20, with*

$$R = 1 + \log Y - \frac{1}{2} \log(2\pi), \quad w = W_0(R/e), \quad Q = \frac{R}{w} = e^{w+1}, \quad s = w + 1, \quad (8.12)$$

*one has, for every fixed  $N$ ,*

$$u(Y) = -\frac{1}{2} + Q + \sum_{n=1}^N \frac{d_{2n-1}(s)}{Q^{2n-1}} + \mathcal{O}\left(\frac{1}{s Q^{2N+1}}\right), \quad (8.13)$$

*with  $d_1, d_3, d_5, d_7$  exactly as in proposition 6.21. Moreover  $u(Y) \asymp \log Y / \log \log Y$ .*

*Proof.*  $F(x) = Y$  means  $\Gamma(x+1) = eY$ , i.e.  $x+1 = \Gamma_+^{-1}(eY)$ , which is (8.11); the branch is the increasing one of remark 6.5, which exists eventually by proposition 8.5. Substituting  $y = eY$  into (6.35) gives  $R = \log(eY) - \frac{1}{2} \log(2\pi) = 1 + \log Y - \frac{1}{2} \log(2\pi)$ , and (6.38) then reads  $\Gamma_+^{-1}(eY) = \frac{1}{2} + Q + \sum d_{2n-1} Q^{1-2n} + \dots$ ; subtracting 1 gives (8.13). The order statement follows from  $Q = R/w$  with  $w = W_0(R/e) \sim \log R$ .  $\square$

**Remark 8.10** (The shift, again). The  $-\frac{1}{2}$  in (8.13) is the half-shift of remark 6.2 transported through (8.11):  $\frac{1}{2} - 1 = -\frac{1}{2}$ . It is what makes the correction series proceed in steps of  $Q^{-2}$ , and all three sources remark on it independently.

At  $x = u(Y)$  the derivatives of the carrier are needed; write

$$\lambda = \lambda(u) = \psi(u+1), \quad P_k(u) = \frac{F^{(k)}(u)}{F(u)}, \quad (8.14)$$

so that  $F^{(k)}(u) = Y P_k(u)$  and  $F'(u) = Y\lambda$ . Differentiating  $F' = F\lambda$  repeatedly gives

$$P_0 = 1, \quad P_{k+1} = P'_k + \lambda P_k, \quad (8.15)$$

so  $P_k = B_k(\lambda, \lambda', \dots, \lambda^{(k-1)})$  is the complete exponential Bell polynomial of definition 1.10; explicitly  $P_1 = \lambda$ ,  $P_2 = \lambda^2 + \lambda'$ ,  $P_3 = \lambda^3 + 3\lambda\lambda' + \lambda''$ . Since  $\lambda = \psi(u+1) \sim \log u$  and  $\lambda^{(j)} = \mathcal{O}(u^{-j})$  for  $j \geq 1$ , we have  $P_k = \mathcal{O}((\log u)^k)$ .

## 8.6 Inverting a perturbation of the carrier

Every inverse in this chapter solves  $F(x) + E(x) = Y$  for a correction  $E$  that is small compared with  $F'$ . That is exactly theorem 1.18, and no separate reversion theorem is needed.

**Proposition 8.11** (The additive reversion in carrier coordinates). *Let  $E$  be analytic near  $u = u(Y)$  with  $E = o(F')$  there. The solution of  $F(x) + E(x) = Y$  near  $u$  is*

$$x = u + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_u^{n-1} \left( \frac{E(u)^n}{F'(u)} \right), \quad \mathcal{D}_u = \frac{1}{F'(u)} \frac{d}{du} = \frac{1}{Y\lambda} \frac{d}{du}, \quad (8.16)$$

convergent for  $|E|$  small and, in the formal-jet sense, valid whenever  $F'(u)$  is invertible.

*Proof.* This is theorem 1.18 with  $F \mapsto F$ ,  $E \mapsto E$ ,  $\varepsilon \mapsto 1$ ,  $\mu_0 \mapsto u$ ,  $y \mapsto Y$  and  $\mathcal{D} \mapsto \mathcal{D}_u$ ; the operator identification uses  $F'(u) = Y\lambda$  from (8.14). The formal version is corollary 1.19.  $\square$

**Remark 8.12** (Why the amplitude series converges here). Theorem 1.18 gives convergence for  $|\varepsilon|$  below some  $\varepsilon_0$ , and one wants  $\varepsilon = 1$ . For the subfactorial that is legitimate, and for a stronger reason than in Part 4: in the coordinate  $v = F(x)$  the map  $v \mapsto Y - \varepsilon \tilde{E}(v)$ , with  $\tilde{E} = E \circ F^{-1}$ , has

$$\tilde{E}'(Y) = \frac{E'(u)}{F'(u)} = \mathcal{O}\left(\frac{1}{Y u \log u}\right), \quad (8.17)$$

by corollary 8.7 and  $F'(u) = Y\lambda$ . So for each sufficiently large fixed  $Y$  the map is a contraction on a small disc for all  $|\varepsilon| \leq 1$ , and the Lagrange series converges at  $\varepsilon = 1$ . What remains merely asymptotic is the subsequent expansion of the coefficients in powers of  $u^{-1}$  and  $\lambda^{-1}$ ; the sector expansion in  $Y^{-1}$  converges while the Bell and Stirling expansions inside each coefficient diverge. This is a genuinely different situation from every earlier chapter and is worth stating plainly.

## 8.7 Sector factorization

The correction is not merely small: it carries an exponential factor  $e^{\omega z}$ , and that factor organizes the whole inverse.

**Theorem 8.13** (Single-frequency sector factorization). *Let  $E(x) = e^{\omega x} C(x)$  with  $\omega$  fixed and  $C$  as in (8.1). Then the solution of  $F(x) + E(x) = Y$  has the formal expansion*

$$x \sim u(Y) + \sum_{n \geq 1} \frac{e^{n\omega u(Y)}}{Y^n} \Phi_n^{(\omega)}(u(Y)), \quad (8.18)$$

where, writing

$$\mathcal{L}_{n,r}^{(\omega)} H = \frac{H' + (n\omega - r\lambda)H}{\lambda}, \quad (8.19)$$

the coefficients are the finite operator products

$$\Phi_n^{(\omega)} = \frac{(-1)^n}{n!} (\mathcal{L}_{n,n-1}^{(\omega)} \circ \cdots \circ \mathcal{L}_{n,1}^{(\omega)}) \left( \frac{C^n}{\lambda} \right), \quad (8.20)$$

the empty product being the identity. Explicitly

$$\Phi_1^{(\omega)} = -\frac{C}{\lambda}, \quad (8.21)$$

$$\Phi_2^{(\omega)} = \frac{C D_\omega C}{\lambda^2} - \frac{C^2 P_2}{2\lambda^3}, \quad D_\omega := \frac{d}{du} + \omega, \quad (8.22)$$

$$\Phi_3^{(\omega)} = -\frac{C(D_\omega C)^2}{\lambda^3} - \frac{C^2 D_\omega^2 C}{2\lambda^3} + \frac{3C^2(D_\omega C)P_2}{2\lambda^4} + \frac{C^3 P_3}{6\lambda^4} - \frac{C^3 P_2^2}{2\lambda^5}. \quad (8.23)$$

Moreover

$$\Phi_n^{(\omega)}(u) = \mathcal{O}\left(\frac{u^{-n}}{\log u}\right), \quad (8.24)$$

so the grading by powers of  $Y^{-1}$  is asymptotically strict even when  $\omega$  is purely imaginary.

*Proof.* Start from (8.16). Its  $n$ -th summand begins with

$$\frac{E(u)^n}{F'(u)} = \frac{e^{n\omega u} C(u)^n}{Y \lambda(u)},$$

and the key observation is that  $\mathcal{D}_u$  maps objects of this shape to objects of the same shape. Let  $H$  be any function of  $u$  and  $r \geq 1$ . Along the differentiation  $Y$  is *not* constant – it is  $F(u)$  – so  $d(Y^{-r})/du = -rY^{-r-1}F'(u) = -rY^{-r}\lambda$ , and therefore

$$\mathcal{D}_u \left( \frac{e^{n\omega u} H(u)}{Y^r} \right) = \frac{1}{Y\lambda} \left( \frac{e^{n\omega u} (H' + n\omega H)}{Y^r} - \frac{r\lambda e^{n\omega u} H}{Y^r} \right) = \frac{e^{n\omega u}}{Y^{r+1}} \cdot \frac{H' + (n\omega - r\lambda)H}{\lambda} = \frac{e^{n\omega u}}{Y^{r+1}} \mathcal{L}_{n,r}^{(\omega)} H. \quad (8.25)$$

Applying this  $n-1$  times to  $H = C^n/\lambda$  starting at  $r=1$  gives (8.20), and the prefactor  $(-1)^n/n!$  is the one in (8.16). The displayed  $\Phi_1, \Phi_2, \Phi_3$  follow by expanding the products and using  $P_2 = \lambda^2 + \lambda'$ ,  $P_3 = \lambda^3 + 3\lambda\lambda' + \lambda''$ .

For (8.24): by corollary 8.7,  $C^{(j)} = \mathcal{O}(u^{-j-1})$ , so  $C^n = \mathcal{O}(u^{-n})$ ;  $\lambda \sim \log u$  and  $P_k = \mathcal{O}((\log u)^k)$ ; and each application of  $\mathcal{L}_{n,r}^{(\omega)}$  divides by  $\lambda$  while either differentiating (which lowers the power of  $u$  but not below  $u^{-n}$  overall, since the leading behaviour is set by  $C^n$ ) or multiplying by the bounded quantity  $n\omega - r\lambda$ , whose size  $\mathcal{O}(\log u)$  cancels the division. Induction gives (8.24).  $\square$

**Corollary 8.14** (Multi-frequency grids). *If  $E(x) = \sum_{\alpha=1}^M e^{\omega_\alpha x} C_\alpha(x)$ , then the inverse support lies on the grid  $\{Y^{-|r|} e^{(r \cdot \omega)u} : r \in \mathbb{N}^M \setminus \{0\}\}$  and*

$$x \sim u(Y) + \sum_{r \neq 0} Y^{-|r|} e^{(r \cdot \omega)u(Y)} \Phi_r(u(Y)), \quad (8.26)$$

ordered first by total degree  $|r|$ . Purely imaginary frequencies may give equal magnitudes or resonant phases, but the factor  $Y^{-|r|}$  keeps the total-degree grading well founded.

*Proof.* Expand  $E(u)^n$  in (8.16) by the multinomial theorem and apply (8.25) to each resulting monomial; the frequency  $r \cdot \omega$  and the power  $Y^{-|r|}$  are preserved by  $\mathcal{D}_u$ , as the computation there shows.  $\square$

## 8.8 The inverses

**Theorem 8.15** (Branch inverse). *Fix  $\kappa \in \mathbb{Z}$  and let  $z_\kappa(Y)$  be the solution of  $\mathcal{S}_\kappa(z) = Y$  with  $z_\kappa(Y) - u(Y) \rightarrow 0$ . Then, for every fixed  $N$ ,*

$$z_\kappa(Y) = u(Y) + \sum_{n=1}^N \frac{e^{n\omega_\kappa u(Y)}}{Y^n} \Phi_n^{(\omega_\kappa)}(u(Y)) + \mathcal{O}\left(\frac{Y^{-N-1}u^{-N-1}}{\log u}\right), \quad (8.27)$$

with  $\Phi_n$  from (8.20) and  $\omega_\kappa = (2\kappa + 1)\pi i$ . In particular the leading displacement is

$$z_\kappa(Y) - u(Y) = -\frac{e^{\omega_\kappa u} C(u)}{Y \lambda(u)} (1 + o(1)), \quad (8.28)$$

which is  $\mathcal{O}(1/(Y u \log u))$  in modulus.

*Proof.* Existence and uniqueness near  $u$ : by theorem 8.1,  $\mathcal{S}_\kappa(u) - Y = e^{\omega_\kappa u} C(u) = \mathcal{O}(u^{-1})$  while  $\mathcal{S}'_\kappa(u) = Y \lambda + \mathcal{O}(u^{-1})$ , so theorem 1.42 places a root within  $\mathcal{O}(1/(Y \lambda u))$  of  $u$  and remark 8.12 makes it unique in a disc of that size. The expansion is theorem 8.13 with  $\omega = \omega_\kappa$ , and the remainder is (8.24) applied to the first omitted sector together with the geometric domination of the convergent tail from (8.17).  $\square$

**Theorem 8.16** (The real median inverse). *Let  $\rho(Y) = \mathcal{S}_{\text{med}}^{-1}(Y)$  be the eventual real inverse of (8.4), and put  $H(u) = C(u) \cos(\pi u)$ . Then*

$$\rho(Y) \sim u(Y) + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_u^{n-1} \left( \frac{H(u)^n}{Y \lambda} \right), \quad (8.29)$$

whose  $n$ -th term contains the harmonics  $e^{im\pi u}$  with  $m = -n, -n+2, \dots, n$ . Explicitly,

$$\rho(Y) = u - \frac{H}{Y \lambda} + \frac{1}{Y^2} \left[ \frac{H H'}{\lambda^2} - \frac{H^2(\lambda^2 + \lambda')}{2\lambda^3} \right] + \mathcal{O}\left(\frac{Y^{-3}u^{-3}}{\log u}\right), \quad (8.30)$$

and the leading displacement is

$$\rho(Y) - u(Y) \sim -\frac{\cos(\pi u(Y))}{Y u(Y) \log u(Y)}. \quad (8.31)$$

At every derangement ordinate,  $\rho(D_n) = n$  exactly.

*Proof.* Apply proposition 8.11 with  $E = H$ ; (8.29) is (8.16). Writing  $\cos \pi u = \frac{1}{2}(e^{i\pi u} + e^{-i\pi u})$  and expanding  $H^n$  shows that the  $n$ -th term is a combination of  $e^{im\pi u}$  with  $m \equiv n \pmod{2}$  and  $|m| \leq n$ , which is corollary 8.14 for the two frequencies  $\pm \pi i$ . For (8.31),  $\Phi_1 = -H/\lambda$  with  $C(u) \sim 1/u$  from theorem 8.6 and  $\lambda \sim \log u$ . The interpolation statement is  $\mathcal{S}_{\text{med}}(n) = D_n$  from definition 8.3.  $\square$

**Corollary 8.17** (Parity envelopes). *For  $\sigma = \pm 1$  the eventual inverse of  $G_\sigma = F + \sigma C$  is*

$$x_\sigma(Y) \sim u(Y) + \sum_{n \geq 1} \frac{\sigma^n}{Y^n} \Phi_n^{(0)}(u(Y)), \quad (8.32)$$

which is theorem 8.13 at  $\omega = 0$  — no oscillation, and the coefficients are the same operator products with  $D_0 = d/du$ . In particular  $x_\sigma(Y) = u - \sigma C/(Y \lambda) + \mathcal{O}(Y^{-2}u^{-2}/\log u)$ .

*Proof.* Immediate from theorem 8.13 with  $\omega = 0$  and the amplitude  $\sigma$ , which enters as  $\sigma^n$  through  $E^n$ .  $\square$

## 8.9 The staircase and the thresholds

The discrete inverse is governed by theorem 1.39, and the only subject-specific input is where the jumps sit.

**Definition 8.18** (Thresholds). For  $n \geq 2$  let  $\alpha_n$  be defined by  $F(\alpha_n) = D_n$ , that is  $\alpha_n = u(D_n)$ . Since  $F$  is increasing, the generalized inverse of the sequence  $(D_n)$  is

$$N_*(y) = \min\{n \geq 2 : u(y) \leq \alpha_n\}. \quad (8.33)$$

**Theorem 8.19** (Threshold displacement). Write  $\varsigma_n = (-1)^n$ ,  $J_n = C(n)$  and  $\Pi_n = \psi(n+1)$ . Then

$$\alpha_n - n = \frac{\varsigma_n J_n}{F'(n)} - \frac{F''(n) J_n^2}{2F'(n)^3} + \frac{\varsigma_n (3F''(n)^2 - F'(n)F'''(n)) J_n^3}{6F'(n)^5} + \mathcal{O}\left(\frac{J_n^4}{F(n)^4 \Pi_n}\right), \quad (8.34)$$

and in particular

$$\alpha_n = n + (-1)^n \frac{e C(n)}{n! \psi(n+1)} + \mathcal{O}\left(\frac{1}{(n!)^2 n^2 \log n}\right) = n + (-1)^n \frac{e}{n! (H_n - \gamma)} \left(\frac{1}{n} - \frac{2}{n^2} + \frac{5}{n^3} - \dots\right). \quad (8.35)$$

Consequently  $|\alpha_n - n|$  is factorially small, and for every  $n \geq 2$  the separation condition (1.74) holds with room to spare.

*Proof.* By corollary 8.2,  $D_n = F(n) + \varsigma_n J_n$ , so  $\alpha_n$  is the value of the local inverse of  $F$  at  $F(n) + \varsigma_n J_n$ . Taylor expanding that inverse, with

$$(F^{-1})' = \frac{1}{F'}, \quad (F^{-1})'' = -\frac{F''}{(F')^3}, \quad (F^{-1})''' = \frac{3(F'')^2 - F'F'''}{(F')^5},$$

and substituting the increment  $\varsigma_n J_n$  gives (8.34). Now  $F(n) = n!/e$  and  $F'(n) = F(n)\Pi_n$ , so the first term is  $\varsigma_n J_n e / (n! \Pi_n)$ , which with  $\Pi_n = \psi(n+1) = H_n - \gamma$  and the Bell expansion of  $J_n = C(n)$  from theorem 8.6 gives (8.35); the second term is  $\mathcal{O}(J_n^2 / (F(n)^2 \Pi_n)) = \mathcal{O}((n!)^{-2} n^{-2} (\log n)^{-1})$  because  $F'' = F(\Pi^2 + \Pi')$  and  $J_n = \mathcal{O}(n^{-1})$ . Since  $|\alpha_n - n| \leq e / (n! (H_n - \gamma) n)$  is far below  $\frac{1}{2}$  for  $n \geq 2$ , theorem 1.39(ii) applies with any  $\eta$  up to  $\frac{1}{3}$ , say.  $\square$

**Remark 8.20** (What the staircase costs). Theorem 8.19 says the discrete inverse is easy: the thresholds sit factorially close to the integers, so rounding  $u(y)$  correctly needs only  $|u(y) - \nu(y)| < \frac{1}{2}$ , which (8.13) delivers already at  $N = 0$ . In this respect the subfactorial is far better behaved than the sequences of Parts 2 and 4, where the interpolated inverse and the staircase separate only slowly. The reason is structural: the correction that distinguishes  $D_n$  from  $n!/e$  is the same correction that displaces the threshold, and it is factorially small in both roles.

## 8.10 The four scales

**Proposition 8.21** (Hierarchy). As  $Y \rightarrow +\infty$ , with  $u = u(Y)$ , the inverse of any of the interpolations of definition 8.3 decomposes into four strictly ordered layers:

1. the Lambert–logarithmic layer, the carrier  $Q$  of (8.12), of size  $u$ ;
2. the Stirling layer, the blocks  $d_{2n-1}(s)Q^{1-2n}$  of (8.13), of sizes  $u^{-1}, u^{-3}, \dots$  divided by powers of  $s$ ;
3. the hyperasymptotic gamma layer, below the least term of the Stirling series, of size  $u^{-1/2}e^{-2\pi u}$  by proposition 6.39;
4. the Bell layer, the genuinely subfactorial correction, of size

$$\frac{1}{Y u \log u} = \exp(-u \log u + \mathcal{O}(u)). \quad (8.36)$$

Layer (iv) is smaller than  $e^{-cu}$  for every fixed  $c > 0$ .

*Proof.* Only the last statement needs an argument. By proposition 8.9,  $Y = F(u) = \Gamma(u+1)/e$ , so  $\log Y = u \log u - u + \mathcal{O}(\log u)$  by Stirling; hence  $1/(Yu \log u) = \exp(-u \log u + u + \mathcal{O}(\log u))$ , which is (8.36). Since  $u \log u - u \geq cu$  eventually for every fixed  $c$ , the claim follows. The ordering of (i)–(iii) is remark 6.36 transported by (8.11), and (iii) precedes (iv) because  $e^{-2\pi u}$  is larger than  $e^{-u \log u + \mathcal{O}(u)}$ .  $\square$

*Remark 8.22* (The consequence for practice). Proposition 8.21 says something uncomfortable and worth saying: for any  $Y$  at which one could actually compute, the subfactorial correction to the inverse is far below the accuracy that the Stirling series can deliver at all. Recovering it numerically requires either exact arithmetic on  $\Gamma(u+1)/e$  or a hyperasymptotic treatment of layer (iii). What the Bell sector is good for is exact statements at integers – corollary 8.2 and theorem 8.19 – where the correction is not competing with a divergent series but is the whole content.

## 8.11 What is left open

1. *Closed forms for  $\Phi_n$ .* Equation (8.20) is a finite operator product, hence computable, but its expansion in the differential ring generated by  $C$ ,  $\lambda$  and their derivatives has no closed description beyond  $n = 3$ . Because  $P_k$  is a complete Bell polynomial (8.15) and  $C$  has Bell-number coefficients, a double-Bell formula is plausible; it is not derived here.
2. *The hyperasymptotic gap.* Layer (iii) of proposition 8.21 separates the algebraic inverse from the Bell sector, and nothing in this chapter bridges it. A complete numerical theory of the real inverse below  $e^{-2\pi u}$  needs the exponentially improved inverse gamma of remark 6.40, which is itself open.
3. *Uniqueness of the real interpolation.* Remark 8.4: the median is canonical among the lateral determinations but not singled out by interpolation alone. A characterization – for instance by a growth or complete-monotonicity condition in the style of the Bohr–Møllerup theorem – is not attempted, and would be the right way to make “the” real inverse subfactorial well defined.
4. *Complex branches.* Theorem 8.15 is stated for fixed  $\kappa$  along the positive real  $Y$ -axis. How the sheets  $z_\kappa$  are connected, and what happens when  $\kappa$  grows with  $Y$ , is not addressed.



# Chapter 9

## A real-argument Fibonacci function

### 9.1 What this chapter does, and what it cannot use

This is the one subject in the volume with no Lambert core. Its dominant block is a pure exponential,  $\varphi^x$ , so the equation to be inverted is  $e^{\lambda x} \approx R$  and the leading inverse is an honest logarithm — no  $W$ , no slow coefficient field, no  $(v+1)^{-1}$ . What replaces the apparatus of section 6.3 is a different and, in one respect, much better structure: the inverse equation *factors exactly* into conjugate factors, the reversion is an ordinary analytic reversion at an ordinary point, and the resulting expansion *converges*.

The chapter is therefore organized around what carries over and what does not.

| Applies unchanged                                                                                           | Does not apply                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 1.18 and its formal form: the correction is an analytic perturbation of an exactly invertible core. | Definition 1.2: the phase is $\lambda x$ , not $ax^\alpha$ with a subdominant $b \log x$ ; the model's $\log x$ slot is empty and its $Q$ slot carries oscillation instead of a power series. |
| Theorem 1.14 and lemma 1.15, in their multivariate Lagrange–Good form (9.8).                                | Theorem 1.20 and lemma 6.8: nothing needs inverting beyond an exponential.                                                                                                                    |
| Theorem 1.42, for the residual certificate.                                                                 | Theorem 1.33: there is no $W$ to unfold, and the expansion is already elementary.                                                                                                             |
| Definition 1.37 and theorem 1.39: $F_n$ is a sequence.                                                      | Theorem 1.46: the expansion converges, so there is no least term and no beyond-all-orders floor.                                                                                              |

The three filed articles are `Fibonacci_Inverse_LogPeriodic_Transseries` (V1), `fibonacci_inverse_transseries_article` (V2) and `fibonacci_inverse_transseries_article-2` (V3). V1 supplies the conjugate factorization and the Lagrange–Good closed formula; V2 the structural laws for the real blocks, the residual certificate and the convergence radius; V3 the general block-reversion formula for products of exponentials. They agree, and every constant, coefficient and checkpoint was recomputed for this volume (remark 9.19).

### 9.2 The interpolation and its branches

**Definition 9.1** (The even Fibonacci modulus). Let

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \lambda = 2 \log \varphi, \quad \omega = \pi, \quad \rho = \frac{\omega}{\lambda}, \quad (9.1)$$

and define, for  $x \in \mathbb{R}$ ,

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(\lambda x) - \cos(\omega x)} \geq 0. \quad (9.2)$$

Numerically  $\lambda = 0.9624236501192068949955 \dots$  and  $\rho = 3.2642513026364969 \dots$

**Proposition 9.2** (Interpolation, evenness, and the modulus form).  *$\mathcal{F}$  is even, real, nonnegative, vanishes only at  $x = 0$ , and satisfies  $\mathcal{F}(n) = F_n$  for every integer  $n \geq 0$ , where  $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$ . Equivalently*

$$\mathcal{F}(x)^2 = \frac{1}{5}(e^{\lambda x} + e^{-\lambda x} - 2 \cos(\omega x)) \quad \text{and} \quad \mathcal{F}(x) = \frac{1}{\sqrt{5}} |\varphi^x - e^{i\pi x} \varphi^{-x}|. \quad (9.3)$$

Near the origin  $\mathcal{F}(x) = \sqrt{(\lambda^2 + \omega^2)/5} |x| + \mathcal{O}(|x|^3)$ , so the interpolation has a cusp there.

*Proof.* Evenness and nonnegativity are clear from (9.2), and  $\cosh(\lambda x) \geq 1 \geq \cos(\omega x)$  with equality only at  $x = 0$ . Expanding  $\cosh$  gives the first identity in (9.3), and since  $e^{\lambda x} = \varphi^{2x}$  that expression is  $\frac{1}{5} |\varphi^x - e^{i\pi x} \varphi^{-x}|^2$ , which is the second. For the interpolation, Binet's formula  $F_n = (\varphi^n - (-\varphi^{-1})^n)/\sqrt{5}$  gives

$$F_n^2 = \frac{1}{5}(\varphi^{2n} + \varphi^{-2n} - 2(-1)^n) = \frac{1}{5}(e^{\lambda n} + e^{-\lambda n} - 2 \cos(\pi n)) = \mathcal{F}(n)^2,$$

and  $F_n \geq 0$  for  $n \geq 0$ , so taking nonnegative roots gives  $\mathcal{F}(n) = F_n$ . The behaviour at the origin follows from  $\cosh(\lambda x) - \cos(\omega x) = \frac{1}{2}(\lambda^2 + \omega^2)x^2 + \mathcal{O}(x^4)$ .  $\square$

*Remark 9.3* (Which extension this is not).  $\mathcal{F}$  is even, so  $\mathcal{F}(-n) = F_n$ , whereas the usual signed convention has  $F_{-n} = (-1)^{n+1} F_n$ . The choice matters when the two outer inverse branches are named, and it is what makes the negative branch trivially  $-X_+$ .

**Proposition 9.4** (The outer branches). *Let*

$$x_* = \frac{1}{\lambda} \operatorname{arsinh}\left(\frac{\omega}{\lambda}\right) = 1.972992734136994 \dots \quad (9.4)$$

*Then  $\mathcal{F}'(x) > 0$  for every  $x > x_*$ ; in particular  $\mathcal{F}$  maps  $[2, \infty)$  strictly increasingly onto  $[1, \infty)$ . Write  $X_+(y)$  for the inverse of that restriction and  $X_-(y) = -X_+(y)$ . Every sufficiently large  $y$  has exactly the two real preimages  $X_{\pm}(y)$ .*

*Proof.* Differentiating (9.3),

$$\mathcal{F}'(x) = \frac{\lambda \sinh(\lambda x) + \omega \sin(\omega x)}{5 \mathcal{F}(x)}, \quad (9.5)$$

so the sign is that of the numerator. Since  $\sin(\omega x) \geq -1$ , the numerator is at least  $\lambda \sinh(\lambda x) - \omega$ , which is positive exactly for  $x > x_*$ . As  $x_* < 2$  and  $\mathcal{F}(2) = F_2 = 1$  with  $\mathcal{F} \rightarrow \infty$ , the stated bijection follows. Finally  $[0, 2]$  is compact, so  $\mathcal{F}$  is bounded there and every  $y$  above that bound has no preimage in  $[0, 2]$ ; with evenness this leaves exactly  $X_{\pm}(y)$ .  $\square$

*Remark 9.5* (Inner branches).  $\mathcal{F}$  is not monotone on  $(0, 2)$ : it has two critical points there, at  $x = 1.132379068 \dots$  and  $x = 1.719250886 \dots$ , with values  $1.013859885 \dots$  and  $0.911172684 \dots$ . So values slightly above 1 have extra preimages. These are governed by a simple critical point, where the local inverse is a Puiseux series in  $\sqrt{y - y_{\text{crit}}}$  rather than anything in this chapter's scale; V2 treats them and we do not. They are irrelevant to the large- $y$  problem.

### 9.3 Exact factorization

Everything in this chapter rests on one identity.

**Theorem 9.6** (Conjugate factorization). *Fix the outer branch, let  $y \geq 1$ , and put*

$$R = 5y^2, \quad u = e^{-\lambda x} \in (0, 1), \quad \alpha = 1 - i\rho, \quad \bar{\alpha} = 1 + i\rho. \quad (9.6)$$

*Then  $\mathcal{F}(x) = y$  is exactly equivalent to*

$$Ru = (1 - u^\alpha)(1 - u^{\bar{\alpha}}), \quad (9.7)$$

*where  $u^\alpha := e^{\alpha \log u}$  with the real logarithm of  $u \in (0, 1)$ .*

*Proof.* By (9.3),  $\mathcal{F}(x) = y$  is  $R = e^{\lambda x} + e^{-\lambda x} - 2 \cos(\omega x)$ . With  $u = e^{-\lambda x}$  we have  $\log u = -\lambda x$ , so  $e^{i\omega x} = e^{-i\rho \log u} = u^{-i\rho}$  and likewise  $e^{-i\omega x} = u^{i\rho}$ ; hence  $2 \cos(\omega x) = u^{i\rho} + u^{-i\rho}$  and

$$R = u^{-1} + u - u^{i\rho} - u^{-i\rho}.$$

Multiplying by  $u$  gives  $Ru = 1 + u^2 - u^{1+i\rho} - u^{1-i\rho}$ , and expanding the right side of (9.7),

$$(1 - u^{1-i\rho})(1 - u^{1+i\rho}) = 1 - u^{1-i\rho} - u^{1+i\rho} + u^2,$$

which is the same.  $\square$

*Remark 9.7* (Why this is the decisive step). Reverting (9.2) directly is possible but hides the structure: one is inverting a square root of a difference of two oscillating exponentials. After squaring and substituting  $u = e^{-\lambda x}$ , the problem becomes the reversion of an *ordinary analytic germ at the origin* in the two variables  $u^\alpha$  and  $u^{\bar{\alpha}}$ . That is why the answer converges and why a closed coefficient formula exists at all. It is also the precise sense in which this subject is easier than the other seven, not harder: the exponents are complex, but nothing is divergent.

## 9.4 The closed coefficient formula

**Theorem 9.8** (Lagrange–Good form of the outer inverse). *For all sufficiently large  $y$ , with  $R = 5y^2$ ,*

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} \frac{(-1)^{j+k+1}}{\lambda s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k} R^{-s_{jk}}, \quad s_{jk} = j + k - i\rho(j - k), \quad (9.8)$$

*the series converging absolutely. Truncating at  $j + k \leq N$  leaves an error  $\mathcal{O}(R^{-N-1}) = \mathcal{O}(y^{-2N-2})$ . The sum is real, because the terms  $(j, k)$  and  $(k, j)$  are complex conjugates.*

*Proof.* Write  $u = e^{-\lambda x}$ , so  $x = -\lambda^{-1} \log u$  and, by theorem 9.6,  $u$  is the small root of

$$u = \frac{(1 - u^\alpha)(1 - u^{\bar{\alpha}})}{R} =: \frac{1}{R} \Psi(u). \quad (9.9)$$

Set  $v = R^{-1}$ . Then  $u = v\Psi(u)$  with  $\Psi$  analytic at  $u = 0$  and  $\Psi(0) = 1$ ; so  $u = u(v)$  is the unique solution with  $u(0) = 0$ , analytic near  $v = 0$ , and Lagrange inversion (theorem 1.14 in its convergent analytic form) applies to the observable  $G(u) = -\lambda^{-1} \log u$ . Since  $G$  has a logarithmic singularity at 0 we use the standard variant: for  $n \geq 1$ ,

$$[v^n] (G(u(v)) - G_{\text{sing}}) = \frac{1}{n} [u^{n-1}] (G'(u) \Psi(u)^n) = -\frac{1}{\lambda n} [u^n] \Psi(u)^n,$$

using  $G'(u)u = -1/\lambda$ . Summing and adding the singular part  $-\lambda^{-1} \log v = \lambda^{-1} \log R$  gives

$$X_+(y) = \frac{\log R}{\lambda} - \frac{1}{\lambda} \sum_{n \geq 1} \frac{1}{n} [u^n] \Psi(u)^n R^{-n}. \quad (9.10)$$

It remains to extract  $[u^n] \Psi(u)^n$ . Here the exponents are not integers, and the clean way to say what “[ $u^n$ ]” means is to treat  $u^\alpha$  and  $u^{\bar{\alpha}}$  as independent variables – which is the content of Good’s multivariate form. Expanding

$$\Psi(u)^n = (1 - u^\alpha)^n (1 - u^{\bar{\alpha}})^n = \sum_{j,k \geq 0} (-1)^{j+k} \binom{n}{j} \binom{n}{k} u^{\alpha j + \bar{\alpha} k},$$

the monomial  $u^{\alpha j + \bar{\alpha} k} = u^{j+k-i\rho(j-k)} = u^{s_{jk}}$  contributes to the coefficient of  $u^n$  exactly when  $n = s_{jk}$ . Making that substitution in (9.10) – i.e. summing over the pairs  $(j, k)$  and setting  $n = s_{jk}$  throughout – gives

$$X_+(y) = \frac{\log R}{\lambda} - \frac{1}{\lambda} \sum_{j+k \geq 1} \frac{(-1)^{j+k}}{s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k} R^{-s_{jk}},$$

which is (9.8).

Convergence:  $|R^{-s_{jk}}| = R^{-(j+k)}$  because  $\Re s_{jk} = j+k$ , and  $|(s_{jk}^{(j)})(s_{jk}^{(k)})/s_{jk}|$  grows at most geometrically in  $j+k$  – each binomial is a product of  $j$  resp.  $k$  factors of modulus at most  $j+k+\rho|j-k|+1 \leq (1+\rho)(j+k)+1$ , divided by  $j!$  resp.  $k!$ , so the modulus is at most  $((1+\rho)(j+k)+1)^{j+k}/(j!k!)$ , and by  $j!k! \geq ((j+k)/2)!^2/(j^{j+k})$  and Stirling this is  $\mathcal{O}(c^{j+k})$  for a constant  $c$  depending only on  $\rho$ . Hence the series converges absolutely for  $R > c$ , and the tail beyond  $j+k=N$  is  $\mathcal{O}(R^{-N-1})$ .

Reality:  $\overline{s_{jk}} = s_{kj}$  and  $\overline{R^{-s_{jk}}} = R^{-s_{kj}}$  since  $R$  is real and positive, while  $\binom{s}{j} = \overline{\binom{s}{j}}$ ; so the  $(j, k)$  and  $(k, j)$  terms are conjugate and the total is real.  $\square$

*Remark 9.9* (On the coefficient extraction). The step “ $u^{s_{jk}}$  contributes to  $[u^n]$  when  $n = s_{jk}$ ” deserves the caution V1 gives it. The function  $\Psi$  is not a power series in  $u$  with integer exponents; it is a series in the two monomials  $u^\alpha, u^{\bar{\alpha}}$ , whose real exponents are  $j+k$  but whose imaginary parts are not zero. The correct formal setting is a Hahn series in the group generated by  $\alpha$  and  $\bar{\alpha}$ , ordered by real part, and Lagrange inversion in that setting is Good’s theorem applied to the two variables  $u^\alpha, u^{\bar{\alpha}}$  with the diagonal specialization. The reordering used to reach (9.8) is legitimate because the series converges absolutely, which the proof establishes independently.

## 9.5 The real form: log-periodic sectors

**Theorem 9.10** (Real sectors). *Put*

$$\xi = \frac{\log R}{\lambda} = \log_\varphi(\sqrt{5}y), \quad \theta = \rho \log R = \pi\xi, \quad q = R^{-1} = \frac{1}{5y^2}. \quad (9.11)$$

Then

$$X_+(y) = \xi + \sum_{n \geq 1} D_n(\theta) q^n, \quad (9.12)$$

where each  $D_n$  is a real trigonometric polynomial in  $\theta$  whose harmonics all have the same parity as  $n$  and whose degree is at most  $n$ . The first three are

$$D_1(\theta) = \frac{2}{\lambda} \cos \theta, \quad (9.13)$$

$$D_2(\theta) = -\frac{1}{\lambda} (2 + \cos 2\theta + 2\rho \sin 2\theta), \quad (9.14)$$

$$D_3(\theta) = \frac{1}{\lambda} \left[ (6 - \rho^2) \cos \theta + 5\rho \sin \theta + \left( \frac{2}{3} - 3\rho^2 \right) \cos 3\theta + 3\rho \sin 3\theta \right]. \quad (9.15)$$

In particular the leading correction is

$$X_+(y) = \log_\varphi(\sqrt{5}y) + \frac{\cos(\pi \log_\varphi(\sqrt{5}y))}{5(\log \varphi)y^2} + \mathcal{O}(y^{-4}). \quad (9.16)$$

*Proof.* Group the terms of (9.8) by  $n = j+k$ . Since  $R^{-s_{jk}} = R^{-n} e^{i\rho(j-k)\log R} = q^n e^{i(j-k)\theta}$ , the  $n$ -th group is  $q^n$  times a finite combination of  $e^{ih\theta}$  with  $h = j-k$  running over  $-n, -n+2, \dots, n$ ; so the harmonics have the parity of  $n$  and degree at most  $n$ , and by the conjugation of theorem 9.8 the combination is real. This is (9.12).

For  $n=1$  the pairs are  $(1,0)$  and  $(0,1)$  with  $s = 1 \mp i\rho$ ; since  $\binom{s}{1} = s$  and  $\binom{s}{0} = 1$ , each term is  $(-1)^2 \lambda^{-1} s^{-1} \cdot s \cdot R^{-s} = \lambda^{-1} R^{-s}$ , and the two add to  $\lambda^{-1} q (e^{i\theta} + e^{-i\theta}) = 2\lambda^{-1} q \cos \theta$ , which is (9.13). Substituting  $q = 1/(5y^2)$ ,  $\lambda = 2 \log \varphi$  and  $\theta = \pi\xi$  gives (9.16). The coefficients (9.14) and (9.15) follow from the pairs with  $j+k=2$  and  $j+k=3$  by the same computation; they were also obtained independently and checked against (9.8) numerically (remark 9.19).  $\square$

*Remark 9.11* (No limiting normalized coefficient). Every earlier chapter had a distinguished large-parameter limit of its coefficient family — the resonant limit of proposition 6.30 and proposition 7.10, or the deepest pole of theorem 6.17 — and in each case that limit summed to a square root. Here there is no such limit. The coefficients  $D_n(\theta)$  do not converge as  $y \rightarrow \infty$ :  $\theta = \pi\xi$  increases without bound and  $D_n$  is a nonconstant periodic function of it, so  $D_n(\theta)$  oscillates forever between fixed bounds. This is not a defect of the expansion — (9.12) converges, which none of the others do — but it does mean that the resummation technique of lemma 6.16 has no analogue here, and that statements like “the  $n$ -th coefficient tends to ...” must be replaced by statements about the mean or the extremes of  $D_n$  over a period.

## 9.6 Convergence, and a residual certificate

**Proposition 9.12** (Residual certificate). *Let  $\hat{x} \geq 2$  and  $y \geq 1$ . Then*

$$|X_+(y) - \hat{x}| \leq \frac{|\mathcal{F}(\hat{x}) - y|}{\min_{[\hat{x} \wedge X_+, \hat{x} \vee X_+]} \mathcal{F}'}, \quad \mathcal{F}'(x) \geq \frac{\lambda \sinh(\lambda x) - \omega}{5 \mathcal{F}(x)} > 0 \quad (x > x_*). \quad (9.17)$$

*In particular, for  $x \geq 2$  one may use  $\mathcal{F}'(x) \geq \lambda(\mathcal{F}(x) - \frac{1}{\mathcal{F}(x)})/2$  for  $\mathcal{F}(x)$  large, so the inverse is well conditioned: a residual of size  $\varepsilon$  certifies an error  $\mathcal{O}(\varepsilon/(\lambda y))$ .*

*Proof.* The first display is theorem 1.42 applied to  $f = \mathcal{F}$  on the interval joining  $\hat{x}$  to the root, together with the lower bound on  $\mathcal{F}'$  from (9.5) and  $\sin \geq -1$ , valid past  $x_*$  by proposition 9.4. For the conditioning,  $5\mathcal{F}^2 = e^{\lambda x} + e^{-\lambda x} - 2 \cos \omega x$  gives  $e^{\lambda x} = 5\mathcal{F}^2/2 + \mathcal{O}(1)$ , hence  $\lambda \sinh(\lambda x) = \lambda e^{\lambda x}/2 + \mathcal{O}(1) = \mathcal{O}(\lambda \mathcal{F}^2)$  and  $\mathcal{F}' \asymp \lambda \mathcal{F}$ ; dividing gives the stated error scale.  $\square$

*Remark 9.13* (What is different here). Theorem 9.8 is a *convergent* expansion of an analytic function, not a Poincaré asymptotic statement. There is no least term, no beyond-all-orders floor, and theorem 1.46 has nothing to say: one simply takes more terms. Every other chapter in this volume inverts a divergent Stirling- or Bernoulli-type series and must stop somewhere. The reason for the difference is visible in theorem 9.6: the forward function is, after squaring, an *entire* combination of exponentials, with no asymptotic series anywhere in it.

## 9.7 Is the log-periodicity intrinsic?

The synthesis chapter asks this of every subject with an unexpected feature, and here it has a clean answer in two halves.

**Proposition 9.14** (The oscillation is forced; its shape is not). *Let  $\mathcal{G}$  be any function that interpolates the Fibonacci numbers,  $\mathcal{G}(n) = F_n$  for  $n \geq 0$ , and is eventually strictly increasing with an inverse  $X_{\mathcal{G}}$ . Then*

$$X_{\mathcal{G}}(F_n) = n = \log_{\varphi}(\sqrt{5} F_n) + \frac{(-1)^n}{\sqrt{5} \varphi^n \log \varphi F_n} (1 + \mathcal{O}(\varphi^{-2n})), \quad (9.18)$$

*so along the sequence the displacement from the pure logarithm is necessarily of size  $F_n^{-2}$  and necessarily alternates in sign. By contrast, the values of  $X_{\mathcal{G}}$  between the  $F_n$ , and hence the harmonic content of the correction as a function of  $\log y$ , depend on  $\mathcal{G}$  and are not determined by the sequence.*

*Proof.* Binet gives  $\sqrt{5}F_n = \varphi^n - (-1)^n \varphi^{-n}$ , so

$$\log_{\varphi}(\sqrt{5}F_n) = n + \log_{\varphi}(1 - (-1)^n \varphi^{-2n}) = n - \frac{(-1)^n \varphi^{-2n}}{\log \varphi} + \mathcal{O}(\varphi^{-4n}).$$

Rearranging and using  $\varphi^{-2n} = (\sqrt{5}F_n \varphi^n)^{-1} (1 + \mathcal{O}(\varphi^{-2n}))$  gives (9.18). Since  $\mathcal{G}(n) = F_n$  forces  $X_{\mathcal{G}}(F_n) = n$  for every interpolation, the displayed relation is a statement about the *sequence*, independent

of  $\mathcal{G}$ . For the second half, let  $\chi$  be any nonzero entire function vanishing on  $\mathbb{N}_0$  and small enough that  $\mathcal{G} + \chi$  is still eventually increasing; it interpolates the same numbers and its inverse differs from  $X_{\mathcal{G}}$  between the integers.  $\square$

*Remark 9.15* (Reading the answer). So the log-periodicity is intrinsic in the only sense that a statement about a *sequence* can be: the second Binet term  $(-1)^n \varphi^{-n}$  is  $\varphi^{-2n}$  relative to the first, which is  $R^{-1}$ , and it alternates. Every interpolation must reproduce that. What  $\mathcal{F}$  contributes is the specific choice  $\cos(\pi x)$  – the minimal-growth real interpolation of  $(-1)^n$  – and *that* is what makes the correction a pure cosine at first order rather than some other function with the same values at integers. An article that says “the inverse Fibonacci function is log-periodic” is therefore making a statement half about Fibonacci and half about its own choice of continuation; the volume records both halves.

## 9.8 The general mechanism: inverting a product of exponentials

The factorization of theorem 9.6 is an instance of a general shape, and the closed formula of theorem 9.8 generalizes with it.

**Theorem 9.16** (Inverse of a finite exponential product). *Let  $\lambda > 0$ , let  $\alpha_1, \dots, \alpha_m$  have positive real part, and let  $c_1, \dots, c_m$  and  $\beta_1, \dots, \beta_m$  be complex. Suppose*

$$R = e^{\lambda x} \prod_{r=1}^m (1 - c_r e^{-\lambda \alpha_r x})^{\beta_r}, \quad (9.19)$$

*with the principal branch of each factor near 1. Then for  $R$  large the solution with  $x \rightarrow \infty$  is*

$$x = \frac{\log R}{\lambda} - \frac{1}{\lambda} \sum_{j \in \mathbb{N}^m \setminus \{0\}} \frac{1}{s_j} \prod_{r=1}^m (-c_r)^{j_r} \binom{s_j \beta_r}{j_r} R^{-s_j}, \quad s_j = \sum_{r=1}^m j_r \alpha_r, \quad (9.20)$$

*the series converging absolutely for  $R$  large.*

*Proof.* Put  $u = e^{-\lambda x}$ , so that (9.19) reads  $Ru = \Psi(u) := \prod_r (1 - c_r u^{\alpha_r})^{\beta_r}$ , i.e.  $u = v\Psi(u)$  with  $v = R^{-1}$  and  $\Psi(0) = 1$ . As in theorem 9.8, Lagrange inversion for the observable  $-\lambda^{-1} \log u$  gives

$$x = \frac{\log R}{\lambda} - \frac{1}{\lambda} \sum_{n \geq 1} \frac{1}{n} [u^n] \Psi(u)^n R^{-n},$$

and the generalized binomial theorem gives

$$\Psi(u)^n = \prod_r (1 - c_r u^{\alpha_r})^{n\beta_r} = \sum_{j \in \mathbb{N}^m} \prod_r (-c_r)^{j_r} \binom{n\beta_r}{j_r} u^{\sum_r j_r \alpha_r}.$$

The monomial  $u^{s_j}$  contributes at  $n = s_j$ , which gives (9.20). Absolute convergence follows as before:  $|R^{-s_j}| = R^{-\Re s_j}$  with  $\Re s_j \geq \delta |j|$  for  $\delta = \min_r \Re \alpha_r > 0$ , while the binomial products grow at most geometrically in  $|j|$ .  $\square$

**Corollary 9.17** (The Fibonacci case). *Taking  $m = 2$ ,  $\alpha_1 = \alpha = 1 - i\rho$ ,  $\alpha_2 = \bar{\alpha}$ ,  $c_1 = c_2 = \beta_1 = \beta_2 = 1$  in theorem 9.16 returns (9.8), with  $s_{(j,k)} = j\alpha + k\bar{\alpha} = j + k - i\rho(j - k) = s_{jk}$  and  $\prod_r (-c_r)^{j_r} = (-1)^{j+k}$ .*

*Proof.* Substitute those values into (9.20). The exponent is  $s_{(j,k)} = j\alpha + k\bar{\alpha} = j + k - i\rho(j - k) = s_{jk}$ ; each  $\binom{s_{j_r}}{j_r}$  becomes  $\binom{s_{jk}}{j}$  resp.  $\binom{s_{jk}}{k}$  because  $\beta_r = 1$ ; and  $\prod_r (-c_r)^{j_r} = (-1)^{j+k}$ , so the overall sign is  $-(-1)^{j+k} = (-1)^{j+k+1}$ , which is the sign in (9.8).  $\square$

*Remark 9.18* (Where the shape comes from, and where it stops). A forward function of the form (9.19) arises whenever a dominant exponential is multiplied by finitely many exponentially small corrections — which is exactly what a Binet-type formula with finitely many characteristic roots gives. Conjugate pairs  $\alpha, \bar{\alpha}$  produce the log-periodicity, since  $R^{-s}$  then carries  $e^{i\Im(s)\log R}$ . Two boundaries are worth naming. If the correction is an *infinite* product or a general analytic germ, the closed formula is replaced by a Lagrange–Good expansion whose coefficients are no longer products of two binomials; V1 and V3 both develop that case. And if the carrier is mixed,  $e^{\lambda x} x^\mu$ , then  $W$  reappears and one is back in the world of section 6.3 — which is the precise point at which this chapter’s subject rejoins the rest of the volume.

## 9.9 Numerical verification

*Remark 9.19* (What was checked). All of the following were recomputed for this volume at 50 digits, from the definitions and independently of the sources. The constants  $\lambda = 0.9624236501192068949955\dots$ ,  $\rho = 3.2642513026364969\dots$  and  $x_* = 1.972992734136994\dots$  of (9.1) and (9.4). The interpolation  $\mathcal{F}(n) = F_n$  for  $0 \leq n \leq 10$ , to 40 digits. The integer checkpoints  $X_+(2) = 3$ ,  $X_+(5) = 5$  and  $X_+(55) = 10$ , each to 40 digits — these test the branch convention as well as the formula. The exact factorization (9.7), at  $y = 3, 10, 100$ , to 35 digits. The closed formula (9.8): its imaginary part vanishes to 54 digits or better, and its truncations converge at the predicted rate. And the agreement between (9.8) truncated at  $j + k \leq 3$  and the trigonometric sectors (9.13)–(9.15), which matched to the full working precision at  $y = 100$  and  $y = 1000$ .

Table 9.1: Absolute error of (9.12) truncated after  $D_N$ , against the root of  $\mathcal{F}(x) = y$  computed to 50 digits.

| $y$  | $N = 0$ (i.e. $\xi$ )   | $N = 1$                  | $N = 2$                  | $N = 3$                  |
|------|-------------------------|--------------------------|--------------------------|--------------------------|
| 10   | $5.4520 \times 10^{-4}$ | $1.1350 \times 10^{-5}$  | $1.5430 \times 10^{-7}$  | $2.1470 \times 10^{-9}$  |
| 100  | $3.0100 \times 10^{-5}$ | $3.5620 \times 10^{-9}$  | $2.9720 \times 10^{-13}$ | $5.7170 \times 10^{-18}$ |
| 1000 | $4.1410 \times 10^{-7}$ | $1.7020 \times 10^{-13}$ | $2.5800 \times 10^{-19}$ | $4.2100 \times 10^{-25}$ |

Each additional block gains a factor  $R^{-1} = 1/(5y^2)$ , as theorem 9.8 predicts: from  $y = 100$  to  $y = 1000$  the  $N = 3$  error drops by about six orders of magnitude per block, and  $R^{N+1}$  times the error stays bounded across the table. Unlike every other error table in this volume, the columns would continue to improve indefinitely.

## 9.10 Comparison, and what is left open

*Remark 9.20* (Where this subject sits). Against the seven others: it is the only one whose inverse expansion converges; the only one with no Lambert core and no slow coefficient field; the only one whose coefficient family has no large-parameter limit (remark 9.11); and the only one where the interesting structure — oscillation — comes from a *pair of complex conjugate exponents* rather than from a logarithm. What it shares with the rest is the organizing principle: invert a dominant block exactly, and revert around it. That is theorem 1.18, and it is the only piece of Chapter 0 this chapter genuinely needs.

1. *Closed forms for  $D_n$ .* Theorem 9.10 gives the first three and a rule for the harmonics, but no closed formula for the coefficients of  $\cos h\theta$  and  $\sin h\theta$  inside  $D_n$ . They are finite explicit sums by (9.8); a generating function in  $\theta$  is not derived.
2. *The exact radius.* The proof of theorem 9.8 gives convergence for  $R$  larger than some constant depending on  $\rho$ , obtained from a crude bound on the binomials. The sharp radius — equivalently, the nearest singularity of  $v \mapsto u(v)$ , which is where  $\Psi'(u)v = 1$  — is not computed here; V1 states one and we did not verify it, so no value is quoted.

3. *Inner branches.* Remark 9.5: the preimages coming from the two critical points in  $(1, 2)$  are Puiseux, not log-periodic, and are not treated.
4. *Complex branches.* Replacing  $\log R$  by  $\log R + 2\pi im$  gives formal sheets, but  $\mathcal{F}$  is built from a modulus and is not analytic, so the continuation question is genuinely different from that of section 6.13(iv) and is not addressed.
5. *Other interpolations.* Proposition 9.14 shows the answer depends on the continuation. Which interpolation is canonical — by minimal growth, by complete monotonicity of some transform, or by agreement with the signed convention  $F_{-n} = (-1)^{n+1}F_n$  rather than with evenness — is open, and is the same question left open for the subfactorial in section 8.11(iii) and for the double factorial in Part 3.



## Chapter 10

# Synthesis: what the eight subjects share, and where they part

This chapter belongs to the consolidation rather than to any source. It compares the eight inversions carried out in Parts 2 to 9, states precisely how much of each is an instance of Part 1, and records what the twenty-four articles left open. Nothing here is asserted beyond what the earlier chapters prove.

### 10.1 One inversion, four dictionaries

The first four subjects of this volume were written about independently, and their *forward* analyses have almost nothing in common: a singularity analysis of a Pólya functional equation, a pair of gamma-quotient branches, a convergent Rademacher series over root-of-unity sectors, and a second, quite different pair of gamma quotients. Their *inverse* analyses are, to a degree that the individual articles could not see, the same computation.

The reason is definition 1.8. In each subject one first isolates a *dominant core*  $\Phi_0$  — an increasing function whose inverse is available in closed form — and writes the exact equation for the index  $x$  in terms of the observed value  $y$  as

$$\Phi_0(x) + (\text{a correction that is small relative to } \Phi'_0) = L(y).$$

The core is then inverted exactly by the Lambert  $W$  function, and the correction is removed to all orders by one triangular recurrence, theorem 1.29 (in the exponential–power case, theorem 1.25). The recurrence does not know which sequence it came from: it sees only the core data and the coefficients  $q_j$  of the correction.

Table 1.1 records the resulting dictionary. Three points in it deserve emphasis, because each was invisible from inside a single subject.

**The partition numbers are the same problem after one substitution.** Of the four, the partition numbers look least like the others: the Hardy–Ramanujan phase is  $\pi\sqrt{2N/3}$ , a *square root* of the variable, where the other three carry a phase linear in the index or in  $x \log x$ . The three partition articles accordingly build their inversion by hand around the equation they write as  $\rho - 2 \log \rho = L$ . By lemma 1.4 that equation *is* the linear-phase core  $a\xi + b \log \xi$  in the variable  $\xi = \sqrt{N}$ , with  $b/\alpha = -2$ ; the substitution costs nothing and removes the apparent exception. What the substitution does *not* do is commute with the arithmetic shift, and remark 1.6 shows why the order matters: re-expanding  $N = n - \frac{1}{24}$  in powers of  $n$  after the reduction manufactures an entire second tower of exponents  $n^{\alpha-k}$ , which is exactly the infinite algebraic array that one source reports and attributes, correctly, to the shift alone.

**The two “factorials” are not the same kind of object.** It is tempting — and this volume’s own brief made the mistake — to expect the double factorial and the swing factorial to sit side by side. They do not. The swing factorial  $n!/\lfloor n/2 \rfloor!^2$  has its factorials cancel to leave growth  $2^n n^{\pm 1/2}$ , so it is an exponential-power model with  $a = \log 2$  (Part 5). The double factorial does not: its logarithm is  $\frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$  with  $s = n + 1$ , a phase of factorial type, and since  $s \log s / s^\alpha \rightarrow \infty$  for every  $\alpha \leq 1$ , no choice of model data reproduces it (Part 3). What survives the difference is exactly the axiomatized core: both  $aX + b \log X$  and  $X(\kappa \log X + d)$  are admissible in the sense of definition 1.8, both are inverted exactly by  $W$  (theorem 1.20 and proposition 1.23), and the reversion above them is one theorem, theorem 1.29. The right statement is therefore not “four instances of one model” but “four instances of one *core calculus*, of which three share a model”.

**The Lambert branch is determined by a sign.** Each of the first twelve articles verifies by hand, for its own case, which branch of  $W$  produces the large root; several remark that the appearance of  $W_{-1}$  is essential and surprising. With the signed convention of remark 1.1 it is neither: by corollary 1.21 the branch is  $W_0$  when  $b > 0$  and  $W_{-1}$  when  $b < 0$ , with the fold at  $X_* = |b|/a$  and the threshold  $L_c = |b|(1 - \log(|b|/a))$ . Reading table 1.1 down the  $b$  column now predicts the branch in every chapter, including the two branches of Part 5, which differ only in the sign of  $\sigma$ .

## 10.2 Where the common frame stops

The shared apparatus covers the passage from an exact forward description to an all-orders inverse. It covers neither end of that passage, and it is worth being explicit about which parts of the four chapters are irreducibly their own.

- (a) *Getting the forward description at all.* The four forward analyses share no method. Part 2 needs a singularity analysis of  $T(z) = z \exp \sum_{k \geq 1} T(z^k)/k$ , the location and nature of the Otter singularity, and a transfer theorem; Parts 3 and 5 need Binet’s representation and Stirling’s series for gamma quotients; Part 4 needs the circle method, or rather its exact endpoint, Rademacher’s convergent series. Nothing in Part 1 produces any of these, and nothing in one of them helps with another.
- (b) *The arithmetic layer.* The sectors of Part 4 are indexed by a denominator  $k$  and carry Kloosterman-type amplitudes built from Dedekind sums. This structure has no analogue in the other three subjects, whose exponentially small corrections come from a competing singularity (Part 2) or from Borel geometry (Parts 3 and 5). The frame transports each sector through the inversion once the sector is given; it does not explain where the sectors come from.
- (c) *Parity.* Two of the four sequences are not the restriction of any single smooth interpolation, and in both cases this is forced, not accidental: the even and odd subsequences have different exponents ( $n^{\pm 1/2}$  for the swing factorial) or different gamma representations (the double factorial). Definition 1.37 and theorem 1.39(4) handle the consequence uniformly — the integer inverse is a minimum over residue classes — but the *fact* of the split is a property of the sequence.
- (d) *Which interpolation.* For a discrete sequence the asymptotic inverse is well defined only after an interpolation is chosen, and the choice is not innocent. Remark 1.38 separates the three inverse objects; Part 3 shows what changes when the OEIS periodic interpolation is used instead of the branchwise gamma continuation, namely a log-periodic oscillation in the inverse that is absent from either branch. One partition source records the sharper version of the same phenomenon: the algebraic part of the inverse is interpolation-independent, the nonperturbative part is not.

### 10.3 The three inverse objects, subject by subject

Definition 1.37 distinguishes the staircase inverse  $N_*$ , the inverse of a chosen interpolation, and the asymptotic inverse. The four chapters do not stand in the same relation to them, and the differences are not cosmetic.

| Chapter            | interpolation available     | parity split | exact staircase recovery |
|--------------------|-----------------------------|--------------|--------------------------|
| 2 rooted trees     | from the singular expansion | no           | along the range          |
| 3 double factorial | two branches, or OEIS       | yes          | per residue class        |
| 4 partitions       | from Rademacher             | no           | along the range          |
| 5 swing factorial  | two branches                | yes          | per residue class        |

Two cautions apply to the whole table. First, by theorem 1.39(5) the staircase  $N_*$  is *not* uniformly approximable by any continuous function:  $\sup_{y \geq Y} |N_*(y) - g(y)| \geq \frac{1}{2}$  for every continuous  $g$  and every  $Y$ . Statements of the form “ $N_*(y) = \nu(y) + o(1)$ ” are therefore false as uniform statements, however natural they look, and the volume asserts only the exact ceiling identity, recovery along the range  $y = A_n$ , and the residue-class refinement. Second, the error floor of the asymptotic inverse is not a defect of the reversion but is inherited: by theorem 1.46 the forward series has a least-term envelope, and theorem 1.42 converts it into a floor for the inverse. No amount of further reversion crosses it.

### 10.4 What this consolidation establishes

Several statements in this volume are not in any of its first twelve sources. They are listed here because a reader who knows the sources should be able to see quickly what has changed; the corrections themselves are boxed where they occur and collected in appendix B.

- (1) *A correct proof of the Lagrange fixed-point formula.* Five of the first twelve articles derive  $[t^n]U = \sum_{m \leq n} \frac{1}{m} [t^n u^{m-1}] \Psi^m$  by introducing a marker  $z$  and applying Lagrange–Bürmann in  $z$ . That application is not available: Lagrange–Bürmann needs  $\varphi(0)$  invertible, and here  $\varphi(0) = \Psi(t, 0) \in tR[[t]]$  is a non-unit. The identity is nonetheless true, and lemma 1.15 proves it by universality – it is a polynomial identity over  $\mathbb{Q}$  in the coefficients of  $\varphi$ , valid in a localization into which the relevant domain injects. Every reversion in this volume rests on this lemma, so the repair is not cosmetic.
- (2) *The branch rule* corollary 1.21, with the fold point and threshold, replacing four case analyses.
- (3) *The normalization constant of the flattened blocks.* The sources state  $P_n(0) = D_n$  in a case where  $a = 1$ ; the correct general statement is  $P_n(0) = a^{n+1}c_n$  (theorem 1.33(3)), and since  $a \neq 1$  in every chapter of this volume the factor is needed throughout.
- (4) *A quantitative obstruction for the staircase*, theorem 1.39(5), in place of an  $o(1)$  claim that is false uniformly.
- (5) *Backward error as a certificate.* Theorem 1.42 upgrades the sources’ conditional bound – which presupposes that a root exists – to an existence statement, by a sign change and the intermediate value theorem.
- (6) *One closed form where the sources had three.* Theorem 1.34 is proved equal to all three published forms of the pure Lambert block, and the equality was verified in exact rational arithmetic for  $1 \leq n \leq 8$ .
- (7) *A non-recursive formula for the factorial-core corrections.* Lemma 1.17 extends the fixed-point formula to  $\Psi = h + f$  with  $h \in u^2R[[u]]$ , which is what Part 3 needs; the source reaches the same coefficients only through an ad hoc compositional inverse.

## 10.5 The second four

Parts 6 to 9 add four subjects that are not sequences of the first kind, and they change the picture in three ways.

**A second core, and a second family.** Theorem 1.20 inverts  $aX + b \log X$ , in which the logarithm is *added*. Gamma, Barnes  $G$  and the  $K$ -function need  $ax^p(\log x - \beta)$ , in which the unknown *multiplies* the logarithm; that block is inverted exactly by  $W$  too, but by a different substitution, and its normalized slope is  $v + 1$  rather than  $a + b/X$ . Lemma 6.8 supplies it, in the generality  $ax^p(\log x - \beta)^r$ , and theorem 6.41 is the corresponding inversion theorem. Those three subjects form a family of their own, indexed by  $(a, p, \beta)$  and by where a resonant logarithm sits; the comparison is table 7.4.

**A subject that borrows its core from another chapter.** The subfactorial has no core of its own:  $D_n = n!/e + (-1)^n C(n)$ , and the inverse of the first term is the inverse of  $\Gamma$ , which is theorem 6.20. Everything specifically about derangements lives in a correction of size  $\exp(-u \log u + \mathcal{O}(u))$ , below every fixed exponential scale  $e^{-cu}$  — proposition 8.21. This is the first subject in the volume whose interesting content is *not* in its dominant block, and it is the reason Part 8 is the shortest chapter with the largest citation load.

**A subject the frame does not cover.** The real-argument Fibonacci function has a pure exponential core, no slow coefficient field, and no Lambert function anywhere. Its inverse expansion *converges*. Section 9.1 tabulates exactly which parts of Chapter 0 survive: essentially only theorem 1.18 and theorem 1.42. It is included because the purpose is the same — invert a rapidly growing function at infinity to all orders — and because the contrast is informative: it isolates which features of the other seven come from the *logarithm* in their phase rather than from inversion as such.

Table 10.1: The eight subjects. “Core” is the block inverted exactly; “slope” is the quantity every correction is divided by.

| Chapter            | Core                                   | Slope                        | Correction scale               |
|--------------------|----------------------------------------|------------------------------|--------------------------------|
| 2 rooted trees     | $aX + b \log X$                        | $a + b/X$                    | $X^{-1}, \alpha = 1$           |
| 3 double factorial | $X(\kappa \log X + d)$                 | $\kappa \log X + \kappa + d$ | $X^{-1}$                       |
| 4 partitions       | $aX + b \log X$                        | $a + b/X$                    | $X^{-1}, \alpha = \frac{1}{2}$ |
| 5 swing factorial  | $aX + b \log X$                        | $a + b/X$                    | $X^{-1}$                       |
| 6 $\Gamma$         | $t(\log t - 1)$                        | $\Lambda = w + 1$            | $T^{-2}$ (even)                |
| 6 Barnes $G$       | $\frac{1}{2}z^2(\log z - \frac{3}{2})$ | $U\lambda$                   | $U^{-1}$ (all)                 |
| 7 $K$              | $\frac{1}{2}r^2(\log r - \frac{1}{2})$ | $\rho L$                     | $\rho^{-2}$ (even)             |
| 8 subfactorial     | $\Gamma(x + 1)/e$                      | $Y\lambda$                   | $e^{\omega u}/Y$               |
| 9 Fibonacci        | $e^{\lambda x}$                        | $\lambda$                    | $R^{-1}$ , oscillatory         |

## 10.6 Three things only the merge shows

**Every subject with a logarithmic phase hides a square root.** Four apparently unrelated square roots appear in the second half of the volume:

$$\sqrt{\Lambda^2 T^2 + \frac{1}{12}}, \quad \sqrt{\lambda^2 U^2 - 2\alpha U}, \quad \sqrt{U^2 + \frac{1}{6}}, \quad \sqrt{\rho^2 + \frac{1}{12}}.$$

The first two are the deepest pole in the slope, for  $\Gamma$  and for  $G$  (theorem 6.17); the third and fourth are coefficientwise large-slope limits, for  $G$  and for  $K$  (propositions 6.30 and 7.10). All four are the same lemma: truncate the normalized equation to its linear and quadratic terms in the displacement

together with the lowest forcing monomial, and the truncation is a quadratic, hence solvable in closed form (lemma 6.16). Which *extraction* of the coefficient family the quadratic computes — deepest pole or large-slope limit — is decided by whether the forcing monomial carries a factor of the slope, that is by whether it is resonant. Resonance and pole depth are two readings of one quadratic, and no source saw either connection.

**Acceleration is resummation.** The  $K$ -function sources offer an “accelerated coordinate”  $T = x^2 - x + \frac{1}{6}$  that gains a full asymptotic sector, and present it as a change of variable. Theorem 7.14 proves it is the resummation of the resonant subsector of the unaccelerated expansion — the same square root, arrived at from the other side. The absorption lemma behind it (lemma 7.11) applies verbatim to Barnes  $G$ , which no Barnes source attempted; the resulting normal form is (7.36).

**One number, two jobs.** The constant  $\frac{1}{12}$  occurs as the  $\Gamma$  deepest pole and again as the  $K$  resonant limit, for reasons that look unrelated. Both are  $-2 \cdot B_2(\frac{1}{2})/2$ : for  $\Gamma$ ,  $B_2(\frac{1}{2})/2 = -\frac{1}{24}$  is the first Stirling tail coefficient; for  $K$ , the same value is the coefficient of the resonant logarithm. The identical Bernoulli value at the identical reflection centre is doing both jobs (remark 7.22). The same reflection symmetry is what makes the half-shift simultaneously remove  $r \log r$  and kill the odd tail, in remark 6.2 and remark 7.6.

## 10.7 What the second wave removed

The first twelve articles carried twelve copies of the Chapter 0 apparatus. The second twelve carry, in addition, copies of *each other*:

- The single block  $d_5 = (2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105)/(2903040\Lambda^5)$  of proposition 6.21 is printed, in one letter or another, in *seven* of the twelve second-wave sources: the three whose subject it is, all three subfactorial articles — which need the inverse of  $\Gamma(x+1)/e$  as their carrier and rederive it from scratch, each proving its own Lambert-centred reversion theorem to get there — and one  $K$ -function article. Four of those seven copies are removed here; proposition 8.9 cites theorem 6.20 instead.
- Eight of the twelve state a general triangular or formal-inversion theorem over a graded coefficient ring. They are all theorem 6.12, and one of them states it under a hypothesis strictly stronger than its own proof uses.
- Seven of the twelve state a named residual-to-error or a posteriori transfer result. They are all theorem 1.42, which the first wave had already needed.
- Seven of the twelve prove a Lagrange–Bürmann reversion adapted to their setting. In every case the content is theorem 1.18 or, where the perturbation is formal, corollary 1.19.

That the second wave’s sources duplicate the first wave’s is the strongest single piece of evidence that the consolidation was worth doing: the twelve new articles were written without knowledge of the twelve old ones, and converged on the same apparatus independently.

## 10.8 Open questions

- (i) *A Stokes-transport theorem.* The mechanism by which an exponentially small forward sector becomes an exponentially small inverse sector is uniform — it is the perturbation-operator form proposition 1.31 — but a theorem needs each subject’s Borel geometry as a hypothesis. The four chapters supply four separate instances; the general statement is not proved here.

- (ii) *The counting function.* One source states that the number of indices with  $A_n \leq y$  equals the average of the branch inverses plus  $\mathcal{O}(1)$ . The statement is plausible for every subject with a parity split, but it needs a per-subject monotonicity hypothesis and is not proved in this volume.
- (iii) *Rademacher-type sequences in general.* Part 4 gives a template for inverting a sequence presented as a convergent sum over arithmetic sectors. Whether the template applies to the other classical Rademacher-type expansions is untested here.
- (iv) *Multifactorials.* One double-factorial source sketches the  $m$ -fold generalization  $n!^{(m)}$ . The core is again of factorial type, so theorem 1.29 applies verbatim; what is not done is the residue-class analysis of the staircase for general  $m$ , which theorem 1.39(4) reduces to a finite computation but does not carry out.
- (v) *Relation to the formalized Lambert layer of this repository.* The surrounding development contains a Lean-verified treatment of both real branches of  $W$ , a lower-branch bracket with an explicit remainder rate, and the saddle expansions of the Fabius function. The Lambert core these four chapters are built on is therefore already formalized to a degree none of the sources assumes. Which of the reversion machinery above is within reach of that layer — theorem 1.20 and corollary 1.21 look closest — is an open and, in this repository, a natural question.

# Appendix A

## Provenance

This volume consolidates twenty-four independently written articles, three on each of its eight subjects. All twenty-four were filed in this repository on 3 September 2026 as a single quick-intake batch, at `drafts/series-and-transseries/special-function-inversion/`; the directories below are relative to that path. Nothing in the twenty-four was edited in place: this volume is a new document, and the sources remain available in git history.

| Source directory                                             | Source      | PDF           | Absorbed into    |
|--------------------------------------------------------------|-------------|---------------|------------------|
| <i>Chapter 2: Butcher–Pólya rooted trees</i>                 |             |               |                  |
| Butcher_Tree_Transseries                                     | 1 674 lines | 21 pp, A4     | Chapter 2        |
| Butcher_Tree_Transseries-2                                   | 2 853 lines | 62 pp, A4     | Chapter 2        |
| Butcher_Tree_Transseries-3                                   | 2 382 lines | 35 pp, A4     | Chapter 2        |
| <i>Chapter 3: the double factorial</i>                       |             |               |                  |
| Double_Factorial_Transseries                                 | 1 552 lines | 18 pp, A4     | Chapter 3        |
| Double_Factorial_Transseries-3                               | 2 287 lines | 28 pp, A4     | Chapter 3        |
| double_factorial_transseries-2                               | 2 138 lines | 44 pp, A4     | Chapter 3        |
| <i>Chapter 4: the partition numbers</i>                      |             |               |                  |
| Partition_Number_Transseries_and_Asymptotic_Inverse          | 1 901 lines | 32 pp, A4     | Chapter 4        |
| Partition_Number_Transseries_and_Inverse                     | 2 020 lines | 27 pp, A4     | Chapter 4        |
| Partition_Numbers_Transseries_and_Inverse                    | 1 905 lines | 33 pp, Letter | Chapter 4        |
| <i>Chapter 5: the swing factorial</i>                        |             |               |                  |
| Swing_Factorial_Transseries                                  | 1 885 lines | 28 pp, A4     | Chapter 5        |
| Swing_Factorial_Transseries_Article                          | 1 824 lines | 25 pp, A4     | Chapters 1 and 5 |
| Swing_Factorial_Transseries_and_Inverse                      | 1 389 lines | 22 pp, A4     | Chapter 5        |
| <i>Chapter 6: Gamma and Barnes <math>G</math></i>            |             |               |                  |
| Asymptotic_Inversion_Gamma_Barnes_G                          | 2 376 lines | 29 pp, A4     | Chapters 6 and 7 |
| inverse_gamma_barnesG_transseries                            | 1 655 lines | 28 pp, A4     | Chapter 6        |
| inverse_gamma_barnes_transseries                             | 1 827 lines | 25 pp, A4     | Chapter 6        |
| <i>Chapter 7: the hyperfactorial <math>K</math>-function</i> |             |               |                  |
| inverse_k_function_transseries                               | 2 259 lines | 29 pp, A4     | Chapter 7        |
| K_Function_Inverse_Transseries                               | 2 644 lines | 33 pp, A4     | Chapter 7        |
| K_function_inverse_transseries_article                       | 2 581 lines | 30 pp, A4     | Chapter 7        |
| <i>Chapter 8: the subfactorial</i>                           |             |               |                  |
| inverse_subfactorial_transseries                             | 2 631 lines | 38 pp, A4     | Chapter 8        |
| inverse_subfactorial_transseries-2                           | 2 188 lines | 27 pp, A4     | Chapter 8        |
| inverse_subfactorial_transseries-3                           | 1 878 lines | 35 pp, Letter | Chapter 8        |
| <i>Chapter 9: a real-argument Fibonacci function</i>         |             |               |                  |
| Fibonacci_Inverse_LogPeriodic_Transseries                    | 2 554 lines | 33 pp, A4     | Chapter 9        |
| fibonacci_inverse_transseries_article                        | 2 857 lines | 36 pp, A4     | Chapter 9        |
| fibonacci_inverse_transseries_article-2                      | 1 884 lines | 24 pp, A4     | Chapter 9        |

The shared apparatus of Chapter 1 is drawn from all twenty-four. Its organizing normal form, and the Lagrange fixed-point route to the reversion coefficients, follow the presentation in `Swing_Factorial_`

Transseries\_Article most closely; the  $\alpha$ -reduction that brings the partition numbers into the same frame is this volume's, since no source needed it.



# Appendix B

## Repairs to the sources

Every correction this volume makes to one of its twenty-four sources is boxed in the text where it occurs and listed here. A repair is recorded when a source states something false, states an asymptotic conclusion that its argument does not support, or asserts a named result without proof. Differences of convention are not repairs and are recorded in the text as remarks instead.

### B.1 Chapter 1: the common apparatus

**A proof that does not apply (five sources).** `Swing_Factorial_Transseries_Article`, `Partition_Number_Transseries_and_Inverse` (twice), `Partition_Numbers_Transseries_and_Inverse`, `Partition_Number_Transseries_and_Asymptotic_Inverse` and `double_factorial_transseries-2` all prove the Lagrange fixed-point formula  $[t^n] U = \sum_{m \leq n} \frac{1}{m} [t^n u^{m-1}] \Psi^m$  by introducing an auxiliary marker  $z$ , solving  $U = z\Psi(t, U)$ , and applying Lagrange–Bürmann in  $z$ . Lagrange–Bürmann requires the relevant  $\varphi(0)$  to be invertible; here  $\varphi(0) = \Psi(t, 0) \in tR[[t]]$  is a non-unit, and the residue proof genuinely breaks, since  $\psi = w/\varphi(w)$  is then not an element of  $wR[[t]][[w]]$ . The identity is true; lemma 1.15 proves it by universality instead. Every reversion in this volume depends on the lemma, so the repair is structural rather than cosmetic.

**An asymptotic claim that is false uniformly.** Several sources state, or leave the reader to infer, that the integer inverse satisfies  $N_*(y) = \nu(y) + o(1)$ . No continuous function approximates a staircase uniformly: theorem 1.39(5) shows  $\sup_{y \geq Y} |N_*(y) - g(y)| \geq \frac{1}{2}$  for every continuous  $g$  and every  $Y$ . The volume asserts only the exact ceiling identity, nearest-integer recovery *along the range*, and the residue-class refinement.

**A normalization constant.** The sources record  $P_n(0) = D_n$  for the flattened blocks, in a case where  $a = 1$ . The general statement is  $P_n(0) = a^{n+1}c_n$  (theorem 1.33(3)); since  $a \neq 1$  in every chapter of this volume the factor is needed throughout.

**Two truncation conventions in one article.** `Swing_Factorial_Transseries_Article` uses the exclusive convention in its forward error table and the inclusive one in its inverse table, so those two tables are not on a common scale. The volume is inclusive throughout (remark 1.1); tables inherited from that article’s forward section are re-indexed.

**A claim of this volume’s own, corrected.** The brief under which Part 1 was written asserted that the case  $\alpha = 1$  of definition 1.2 covers the double factorial. It does not:  $\log((x+1)!!) = \frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$  with  $s = x + 1$ , and  $s \log s / s^\alpha \rightarrow \infty$  for every  $\alpha \leq 1$ . The sources never claimed membership. Definition 1.8 axiomatizes the dominant core instead, and proposition 1.23 inverts the factorial phase exactly.

**Corrupt source text, not reproduced.** Swing\_Factorial\_Transseries\_Article equation eq:inverse-Stokes-leading and Butcher\_Tree\_Transseries-2 lines 2482–2483 contain corrupt control sequences. The affected statements are re-derived here rather than copied.

## B.2 Chapter 3: the double factorial

**Convergence asserted without a threshold.** The first source states that the periodic carrier “has the convergent local Lagrange–Bürmann expansion” for sufficiently large  $\Lambda$ , with no threshold and no proof of convergence as opposed to formal solvability; the second proves only a contraction estimate, from which convergence of the Lagrange series does not follow. Theorem 3.46 separates the two: existence and uniqueness of the real root by contraction for  $\Lambda > 2\pi a$ , and convergence of the Lagrange series under the explicit sufficient condition  $\mu = 2a/\Lambda \leq 1/5$ .

**Two assertions in place of proofs.** The deepest-pole coefficient (“only the quadratic kernel and the first Bernoulli coefficient can contribute”) is exactly the step needing the degree bookkeeping, and is supplied here by induction on the  $\Lambda^{-1}$ -degree. The coefficientwise sign law is argued in one source with “use a positive rooted-tree contribution ...; such a tree exists”, exhibiting no such object; theorem 3.37 keeps the correct positivity reduction and replaces the strictness step with an explicit attainability induction, verified symbolically for  $n \leq 7$ .

**A formal-to-analytic slide.** The third source presents an all-orders phase reversion as a theorem with an asymptotic conclusion, introduced by “setting  $\tau = 1$  as an asymptotic expansion”. The identity is formal; making it asymptotic needs estimates, uniform in the oscillatory phase, that are neither stated nor proved. It is demoted here to the formal proposition 3.50, with a proved remainder supplied separately.

**A silent three-way inconsistency.** One source centres the periodic inverse at the even-branch constant, the other two at the parity mean; the leading displacements consequently differ by  $2a/\Lambda$ . Both are correct for their own centring, but no source says so, and a reader comparing them would conclude that one is wrong. The volume adopts the mean centring and states the relation.

**A notation collision across three sources.** The letter  $C$  denotes the multiplicative amplitude, the logarithmic constant, and the branch prefactor in the three siblings respectively, and  $\alpha, \beta$  carry two incompatible meanings. A crosswalk table fixes the volume’s usage; the core parameters are never abbreviated, to avoid a fourth collision with  $\kappa$  in definition 1.8.

**A qualitative warning made quantitative.** All three sources warn that a different smooth interpolation “can change this sector” without exhibiting one. Theorem 3.8 exhibits the family  $\mathcal{D}(x)e^{\lambda \sin \pi x}$ , which interpolates  $n!!$  at every non-negative integer and changes the leading oscillatory amplitude from  $2a/\Lambda$  to  $2\sqrt{a^2 + \lambda^2}/\Lambda$  – a change larger than the entire Bernoulli grid. This is what makes the chapter’s organizing claim a theorem rather than a caution.

## B.3 Chapter 2: the rooted-tree numbers

**Digits printed beyond the accuracy of the computation.** One source prints the constants of the inherited  $-\sqrt{\rho}$  sector to forty decimal digits, but only about twenty-five of them are correct. Two independent recomputations here – including a from-scratch direct summation at eighty digits – give

$$r_0^- = 0.468636627924845548087202621931819314\dots,$$

and the volume prints only digits it has verified. This is the one place in the twenty-four articles where a printed numeral is wrong; it is a precision failure rather than a mistake in a formula.

**An audit appendix that contradicts its own main text.** The same source’s audit appendix reports  $D_3 = -3.582177326832123 \dots$ , disagreeing with the value its main text derives. The main text is correct; the appendix entry is not, and the volume uses the recomputed value.

**A symbol used for two different constants.** One source writes  $\alpha$  for the growth constant where the other two write  $\lambda$ , and the volume’s Chapter 1 uses  $\alpha$  for the phase exponent of definition 1.2. The chapter fixes  $\lambda$  for the growth constant throughout and never uses  $\alpha$  for it.

**An asymptotic statement with a formal proof.** A source states its inverse-coefficient theorem asymptotically but proves it only formally, and separately displays a global transseries with no hypothesis under which it could hold. The first is proved here in its asymptotic form; the second is replaced by a conditional theorem with a finite radius and an explicit hypothesis.

**A circular derivation of the singularity bounds.** The elementary bounds  $\frac{1}{4} \leq \rho \leq e^{-1}$  are used in the sources in a way that presupposes what they are used to prove. They are established here independently, from  $x \leq Te^{-T} \leq e^{-1}$  and the Catalan bound, and the critical characterization is then obtained non-circularly.

## B.4 Chapter 5: the swing factorial

**A divergent series written as an equality.** One source sets its bookkeeping coupling to 1 and prints the resulting operator expansion as an equality. There is then no small parameter, the sum diverges, and a number is equated to a divergent series; as printed the statement is false. The  $m$ -th summand is  $\mathcal{O}(X^{-(2m-1)})$ , so the correct statement is asymptotic, and regrouping gives the finite differential form used here. The other two sources state the same expansion correctly.

**A theorem whose hypothesis is never used.** A sector-transport result is labelled a theorem, its hypothesis is not used in the argument, and its conclusion is prefaced “formally”. It is split here into the formal all-orders identity, which is what the proof gives, and a quantitative first-order statement proved by the mean value theorem under stated hypotheses.

**An overclaim about complex branches.** One source asserts that every Lambert branch with  $|w| \rightarrow \infty$  yields an inverse branch with the same coefficients. For  $k \neq 0, -1$  the existence of an analytic branch of the inverse along the path is a hypothesis, not a consequence, and it fails at gamma poles and Stokes directions. The claim is demoted to a remark with the two hypotheses displayed.

**The analytic inverse statement, unproved in all three sources.** One source writes “likewise, for each fixed  $M, \dots$ ” with no proof; the second sketches an argument that never bounds the residual of the truncated ansatz nor uses forward control at the perturbed point; the third proves the ingredients but never assembles them. This is the one formal-to-analytic slide common to all three. Theorem 5.41 proves the statement in full, with an effective constant.

**A recommendation contradicted by its own table.** A remark asserts that the centred normalization is “often the preferable computational normal form”. The same article’s table shows the centred core better by a factor  $\approx 10^3$  at one end and the uncentred form slightly better at the other. The corrected statement is that centring kills the  $X^{-1}$  term and so improves the *bare core* by a factor  $\asymp X$ , while with several corrections retained the two are comparable and neither dominates uniformly.

**Two convergence claims without criteria.** An equivalence of the Lambert and logarithmic normalizations is asserted in prose with no formula and no proof; it is proved here as a finite explicit composition. A convergence claim appeals to “the classical analysis of Lambert  $W$ ”, but the classical result concerns a doubly indexed series, whereas the series at hand has coefficients polynomial in  $\ell$  and converges only when  $\ell/L$  is small; an explicit criterion is proved instead.

## B.5 Chapter 4: the partition numbers

**A definition that contradicts its own use.** The source abbreviated S2 in Part 4 defines the real Rademacher amplitude by a sum over  $1 \leq h \leq k$  with  $\gcd(h, k) = 1$ . For  $k \geq 2$  that is the intended index set, but at  $k = 1$  it returns  $\cos(2\pi\nu)$  rather than the constant 1, while the same article uses  $A_1 \equiv 1$  throughout. With the printed definition the interpolant would not interpolate  $p$ : its dominant term would oscillate off-lattice at full relative strength, and the article’s own eventual-monotonicity statement would be false. The volume takes  $h \in U_k$  with  $U_1 = \{0\}$ ; every later formula of that source is then correct as printed.

**An understated radius.** S1 states only that its universal sector-coefficient series “converges for all sufficiently small  $|t|$ ”. The radius is exactly  $\sqrt{24}$  in  $t = n^{-1/2}$ , the nearest singularities being the branch points of  $\sqrt{1 - t^2/24}$ ; convergence therefore holds for every real  $n > 1/24$ .

**A convergent series and an asymptotic total, conflated.** All three partition sources are loose about the distinction. The algebraic series  $\sum_m \omega_m n^{-m/2}$  converges, but its sum is the leading Rademacher block, *not*  $p(n)$ ; the Poincaré statement about  $p(n)$  carries an error that is the sum of a convergent algebraic tail and an exponentially small Rademacher remainder. Summing the algebraic series to convergence does not compute  $p(n)$ : the residual stalls at the  $e^{-\mu/2}$  floor.

**A false intermediate bound.** S1’s kernel estimate asserts  $g(z) \leq z^2/3$  on  $[0, 1]$ , which fails at  $z = 1$  where  $g(1) = e^{-1} = 0.3679\dots > 1/3$ . The chapter proves  $g(z) \leq (z^2/3) \cosh z$  instead, which is what the tail bound actually needs.

**An overclaim in this volume’s own Chapter 1.** Remark 1.3 originally said that the correction series  $Q$  is divergent of Gevrey type “in all four chapters”. It is not: for the partition numbers  $Q(t) = \log(1 - t)$  has radius of convergence 1. The remark now distinguishes the three divergent chapters from Part 4, where theorem 1.46 has no content and the error floor is set by the Rademacher remainder instead.

### The second wave: Chapters 6–9

**An existence claim that no source makes effective.** All six power–logarithmic sources justify the eventual inverse of Barnes  $G$  and of the  $K$ -function with the words “ $(\log G)'(x) \sim x \log x$ , hence strictly increasing for all sufficiently large  $x$ ”. That is true and ineffective: none of the six supplies a threshold, and yet every numerical table in all six evaluates the inverse at arguments as small as  $z = 5$ , inside the range their own justification does not cover. Proposition 6.6 proves that  $z \mapsto G(z + 1)$  is strictly increasing on  $[3, \infty)$  with slope at least  $\frac{1}{2}z \log z$  on  $[7, \infty)$ , and proposition 7.2 does the same for  $K$  on  $[2, \infty)$  with the two stationary points located. The tabulated arguments are then legitimate.

**A hypothesis stronger than the proof needs.** S1 of Part 6 states its triangular inversion theorem under the hypothesis that the support semigroup be *locally finite*, meaning that every bounded interval meets it in a finite set. That is strictly stronger than well-ordering —  $\{1 - 1/n\} \cup \{1\}$  is well ordered and not locally finite — and more than the argument uses, which is only the finite-decomposition property.

Neumann’s lemma supplies that for every well-ordered set, so theorem 6.12 is stated with the weaker hypothesis. For the two subjects of that chapter the distinction is invisible, since the supports are  $2\mathbb{N}$  and  $\mathbb{N}$ ; it matters for the fractional and mixed grids of section 6.12.

**A limit presented as a resummation.** S1 of Part 6 writes that the leading constant subsector of the Barnes coefficients “resums” to  $\sqrt{T^2 + 1/6}$ , and its next remark correctly warns that this does not replace the expansion; the two statements sit awkwardly together. The accurate form is proposition 6.30: a limit of coefficients, each taken separately as the slope grows, which is a structural identity and an exact check on a symbolic computation but not an approximation to the inverse — at any actual argument the neglected terms, headed by  $b_0 = -\alpha/\lambda$ , are the largest corrections present. Remark 6.31 states this, and remark 7.15 then shows what does turn the limit into an approximation: moving it into the coordinate.

**A carrier rederived three times.** All three subfactorial sources need the inverse of  $\Gamma(x + 1)/e$  as the carrier for everything else, and all three derive it from scratch, each proving its own Lambert-centred reversion theorem and each printing the blocks  $1/(24s)$ ,  $-(14s^2 + 10s + 5)/(5760s^3)$ ,  $(2232s^4 + \dots + 105)/(2903040s^5)$ . These are identical, coefficient for coefficient, to  $d_1, d_3, d_5$  of proposition 6.21. The block  $d_5$  is in fact printed in seven of the twelve second-wave sources. Proposition 8.9 states the dictionary  $u(Y) = \Gamma_+^{-1}(eY) - 1$  and cites theorem 6.20; the general reversion theorem the three sources prove to get there is theorem 1.18. This is not an error in any source — each is correct in isolation — but it is the largest single duplication the consolidation removes.

**An unverified radius, quoted by neither.** V1 of Part 9 states an exact radius of convergence for the Lagrange–Good series. The proof of theorem 9.8 establishes convergence for  $R$  beyond a constant depending only on  $\rho$ , by a crude bound on the binomial factors, and the sharp radius was not verified for this volume. No value is therefore quoted, and the question is recorded as open in section 9.10(ii). This is listed here because a reader comparing the volume with its source will find a number in one and not the other, and the reason is deliberate.